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Glenn Ledder

Mathematical Modeling for Epidemiology and Ecology

Second Edition



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Preface

Science is built up with facts, as a house is built with stones. But a collection of facts is no more a science than a heap of stones is a house.

Jules Henri Poincaré

Changes from the First Edition

The COVID-19 pandemic has had a greater effect on education than any natural event since the bubonic plague closed universities and forced Isaac Newton to study on his own. As we emerge from the pandemic with the understanding that this will not be the last novel disease humanity must face, it is natural for students of mathematics and biology to want to study mathematical epidemiology. The first edition of this book focused on ecology, with epidemiology generally appearing only in the problem sets. The primary motivation for a second edition has been to provide a significant coverage of mathematical epidemiology. Room for this additional coverage has been made by sharply tightening the focus of the book to mathematical modeling and dynamical systems. It is assumed that the reader has had a first course in calculus, but no prior study of more advanced topics, such as differential equations and linear algebra, is required.

A Focus on Modeling

As you read through this book, you will see that mathematical modeling goes far beyond the "application" problems that mathematics text authors include to make mathematics appear relevant. The problem is that what little modeling work appears in these problems is generally done by the author rather than the students. The experience of doing these problems only benefits science students if their science instructors are also good enough to do the modeling work for them.

This book is written from a modeling perspective rather than a mathematics or biology perspective. The lack of modeling content in the standard mathematics and science curricula means that the typical reader will have little or no modeling experience. For example, a basic knowledge of parameters, as provided in Sect. 1.1, will be unknown to many students, even

those who have had several mathematics courses beyond calculus. From a modeling perspective, this material should be a standard part of any secondary level advanced algebra course. Similarly, many students will be unused to being asked to interpret results of mathematical work in biological terms, rather than merely reporting mathematical results.

Part I of this edition is focused on mathematical modeling. Part II is a presentation of the theory and practice of dynamical systems analysis, but from a modeling perspective. Students are regularly asked to follow their mathematical work by addressing modeling questions.

One drawback of a modeling perspective is its unfamiliarity for mathematicians with no prior training in modeling. Some of my best personal experiences as a university professor were in teaching courses on subjects I had not previously studied. While preparation for these courses was certainly more work than preparation for familiar courses, the satisfaction of acquiring new expertise was a significant compensation. It is my hope that mathematicians who want to teach modeling, rather than "applicable mathematics," will have the same experience with this book.

Mathematical Epidemiology

Part of the reason for a second edition is the greatly heightened interest in mathematical epidemiology. The first edition focused on ecology. This second edition has some ecology, but there is also a focus on epidemiology in the text, the examples, the problems, and the projects. The treatment of epidemiology here is tailored toward issues arising from the COVID-19 pandemic. Whereas most epidemiology books concentrate on models on an endemic scale, where the focus is on finding equilibrium solutions and determining their stability, the presentation here pays equal attention to models on an epidemic time scale, where the focus must be on numerical simulations. Chapter 3 develops models in epidemiology in five sections: Sects. 3.1 and 3.2 discuss model components that determine the flow rates between classes, Sect. 3.3 develops the technique of compartment analysis with the SEIR epidemic model as an example, Sect. 3.5 presents two scenarios from the COVID-19 pandemic, and Sect. 3.9 presents a variety of endemic models, using different ways to incorporate demographics. Much of the analysis of one-component continuous models in Chap. 4 and multi-component continuous models in Chap. 6 focuses on a epidemiological models developed in Chap. 3.

Scientific Computation

This edition places a greater emphasis on scientific computation than the first edition. It is impossible to fully understand dynamical system behavior without looking at computer simulations as well as doing stability analysis,

and of course one cannot fit models to data without computation. A suite of 28 MATLAB programs accompanies this edition.

The choice of programming language to use for scientific computation requires an assessment of trade-offs. I used R in the first edition because it is free and is the most commonly used programming environment by biologists. I switched to MATLAB for this edition for three reasons: (1) MATLAB graphics are far superior to those of R, (2) It is easier to debug MATLAB programs because it is easier to print out intermediate results, (3) MATLAB programs can generally be run for free using Octave Online. I also saw that my primary biologist collaborator was able to switch from R to MATLAB with no significant difficulties.

It should also be mentioned that some mathematicians and biologists are now using Python. The advantages of Python are that it is free, it has a lot of web-based tutorials, its data manipulation features are superior to those of MATLAB, and it allows (but does not require) object-oriented programming. However, these advantages come with a price. Python is a more difficult language to master than MATLAB, and scientific computing tools have to be added as packages rather than being integrated into the main program.

Programming in any platform is difficult for many students, and it is likely that many of the readers of this book have little to no programming experience. This is a challenging pedagogical problem, but not an unsolvable one; the solution is for the author to write programs using a structure that makes them adaptable to new scenarios with a minimum of new coding. Many of the programs included with this book and others like them have been used successfully with students having no prior programming experience.

Appendix A presents a tutorial to MATLAB, focusing on explanation of every line in the two simplest programs, along with a description of what each of the 28 programs does. While one can find more comprehensive MATLAB tutorials online, the program suite for this book should be accessible with only the bare background provided in Appendix A.

I do not make any use of computer algebra systems to do routine algebra and calculus. The reader is welcome to use these tools to assist with the problems, but be forewarned that in many cases computer algebra systems are unable to do the nuanced algebra work required in this book without extensive direction. Algebra is a powerful tool when used with a high level of professional judgement. I have developed a number of useful guidelines for stability calculations; these are presented in Appendix G.

Description of Contents

Chapter 1 is a general introduction to the background needed for modeling and the ideas of modeling. It includes sections on parameters, biological data, randomness, and modeling concepts, along with a case study of an agent-based model.

Chapter 2 develops concepts and methods of empirical modeling, including three sections on least squares analysis and one on model selection using the Akaike Information Criterion, along with a case study that examines the different linearization techniques used to obtain parameter fits for the Michaelis–Menten model.

Chapter 3 develops concepts and methods of mechanistic modeling, beginning with sections on transition and interaction processes that serve as components of models, and followed by a section that introduces compartment analysis and develops the SEIR epidemic model, a section that presents some analysis for the SEIR model, and a case study section that introduces models for two scenarios from the COVID-19 pandemic. The chapter concludes with a section on equivalent forms, including dimensionless forms, and three additional case studies, each of which adds some important methods of mechanistic modeling. The lead poisoning case study shows how models can be simplified through empirical observation, the enzyme kinetics case study shows how models can be simplified through asymptotic approximation, and the endemic disease model case study provides a comparison of different ways to incorporate demographic assumptions into epidemic models.

Chapter 4 presents the concepts and techniques for the analysis of single variable dynamic models, both discrete and continuous. The chapter features a comparison of the discrete and continuous stability criteria in Sect. 4.4 and a novel improvement on the standard method for sketching phase lines in Sect. 4.5.

Chapter 5 presents the concepts and techniques for matrix projection models, beginning with an intuitive, biologically based, introduction in Sect. 5.1.

Chapter 6 presents the foundational graphical and analytical methods for nonlinear systems. The standard method of analysis using eigenvalues of the Jacobian appears in Sect. 6.2, but the key components of the chapter are the methods of nullcline analysis in Sect. 6.1 and stability via the Routh–Hurwitz conditions in Sect. 6.3.

The main text is accompanied by a set of 7 appendices, each consisting of material intended to help the reader without occupying space in the main text development.

1. Appendix A is a focused tutorial on MATLAB, presenting only what a reader needs to be able to use the program suite that accompanies the book.
2. Appendix B is a brief presentation of the idea of the derivative and methods of differentiation, including the partial derivative. This appendix provides a minimal reference for the calculus techniques required in Chaps. 3, 4, and 6.
3. Appendix C is a brief discussion of nonlinear optimization, which is needed as a reference for methods that appear in Chap. 2. In addition to the usual calculus method for use with known functions, the appendix includes two simple techniques for numerical optimization of a function of one variable that is defined as the output of a simulation rather than an explicit function.

4. Appendix D is a brief presentation of Runge–Kutta methods for numerical solution of differential equations, including Euler's method, the modified Euler method, and the explicit Runge–Kutta method of order 4, which is used in some of the differential equation programs needed for Chaps. 3, 4, and 6.
5. Appendix E is a presentation of the basic ideas of scaling that motivate some of the scaling choices that appear in Chaps. 3, 4, and 6.
6. Appendix F is a supplementary material that justifies the use of the Jacobian matrix for stability calculations by deriving the linearized system at an equilibrium point of a nonlinear system. This material may be of interest to instructors who want a more mathematically rigorous presentation in Chap. 6.
7. Appendix G is a compilation of best practices in the use of algebra for stability calculations. This material is essential for anyone doing the more challenging stability computations in Chap. 6.

Ways to Use This Book

This book was written with multiple goals in mind. In addition to serving as a textbook for a course, there are some sections that are sufficiently novel in approach or content that they could be advantageously used as a supplement to a different text that lacks these topics. There are also sections that present methods that are either novel or not thoroughly presented elsewhere.

A Textbook for a Course

Before listing possible course plans using the material in this book, it is important to start with a broad discussion of pacing. Books for some lower division courses are generally written under the assumption that each section will require 1 day of class. At this pace, it is not difficult to put more than 30 sections into a standard 3-credit course. Such a pace is reasonable if students are not expected to work on challenging problems or master complicated material. It will not work for a course with a focus on mathematical modeling because challenging problems and mastery are essential. Perhaps 20 sections from this book is a reasonable upper bound on how much can be included in a three-credit course. In counting sections, Sect. 6.1 should be counted as the equivalent of two sections; ideally this material should have been presented in two sections, but there was no sensible point at which to divide the material into two roughly equal parts. Another reason for limiting coverage to 20 sections is to free up time for students to work on projects. Some of the projects are open-ended research projects, while others are guided exploration of a problem of sufficient scope to have been an undergraduate research project if given without the guidance.

Four possible three-credit courses of 18 to 20 sections suggest themselves.

1. Mathematical Modeling
   Full coverage of Part I would make for a 19-section course on mathematical modeling that emphasizes epidemiological models. Some of the case studies could be omitted.
2. Dynamical Systems
   Full coverage of Part II, preceded by Sect. 1.1, would make for an 18-section course on dynamical systems (counting Sect. 6.1 as two sections).
3. Mechanistic Modeling and Continuous Dynamical Systems
   One way to make a course that incorporates both modeling and analysis would be to focus on mechanistic modeling and continuous dynamical systems. This would involve the equivalent of 20 sections, comprising Sects. 1.1 and 1.4, all of Chap. 3, Sects. 4.3 and 4.5, Sects. 5.3 and 5.4, and Sects. 6.1–6.4.
4. Mathematical Epidemiology
   An introductory course on mathematical epidemiology could be fashioned with a slight modification of the course on mechanistic modeling and continuous dynamical systems. This would involve the addition of Sect. 1.5 in place of Sects. 3.8 and 4.5.

A Supplementary Text for a Course

In addition to its potential use as a textbook for a mathematical biology course, this book contains a number of noteworthy topics that could serve as supplementary material for a course using a different book.

- A simple introduction to parameters appears in Sect. 1.1. Most mathematical presentations either use specific numerical values for parameters or assume a level of sophistication that not all readers have.
- A simple introduction to agent-based modeling appears in Sect. 1.5.
- Least squares analysis is gradually built up from the simple model $y = mx$ in Sects. 2.1–2.3, culminating in the study of "semilinear" models of the form $y = Af(x; p)$.
- An intuitive treatment of the Akaike Information Criterion appears in Sect. 2.4, followed by guidelines for how to use AIC to inform model selection.
- A computational experiment that compares different linearization methods for the Michaelis–Menten function appears in Sect. 2.5.
- A detailed look at transition processes in Sect. 3.1 includes a critique of the spontaneous transition assumption often used in epidemiology models and presentation of a multi-phase transition model. This section also presents the author's original vaccination models.

- A detailed look at interaction processes in Sect. 3.1 proceeds from the simple mass action interaction models to the Holling type 2 and 3 models used in ecology; these are derived from first principles.
- Section 3.3 uses the SEIR model as an example to develop the basic ideas of compartment analysis and mathematical epidemiology. These are extended in the two COVID-19 models of Sect. 3.5.
- A unique treatment of nondimensionalization and scaling within the broader context of equivalent forms of models appears in Sect. 3.6; in particular, Fig. 3.6.2 shows how dimensionless variables assist in the presentation of a model.
- The derivation of the Michaelis–Menten function from chemical kinetics using a differential equation model with asymptotic approximation appears in Sect. 3.8.
- A systematic discussion of how to add demographic processes to make an epidemic model into an endemic model appears in Sect. 3.9.
- A step-by-step description of the construction and interpretation of cobweb plots appears in Sect. 4.2.
- Section 4.4 presents a novel comparison of linearized stability criteria for discrete and continuous equations.
- The presentation of eigenvalues and eigenvectors is grounded in biology rather than in the usual context of geometric vectors, beginning with a treatment of long-term behavior and population structure in Sect. 5.1 that does not require any linear algebra background.
- The presentation of linearized stability analysis is enhanced by the use of Routh–Hurwitz (Sect. 6.3) and Jury (Sect. 6.5) conditions for two-dimensional and three-dimensional systems.

A Reference for Models and Research Techniques

There are also several topics that would be useful references for those doing mathematical biology research. Some of these are mathematical methods that are not well known, but should be.

- Section 4.5 shows how to use an imposed structure on an autonomous differential equation model to facilitate phase line analysis.
- The presentation of nullcline analysis in Sect. 6.1 includes a discussion of fast and slow dynamics and how to use this distinction to gain insight into the trajectories in a phase portrait.
- The use of the Routh–Hurwitz and Jury conditions in Sects. 6.3 and 6.5 for linearized stability analysis is difficult to locate elsewhere, but these methods are more powerful for many problems than the computation of eigenvalues.
- A brief set of guidelines for the thoughtful use of algebra in stability calculations appears in Appendix G.

Others are modeling ideas and techniques that I developed in my research work.

- The vaccination model in Sect. 3.1 and the COVID-19 models in Sect. 3.5 are taken from my own original research.
- The use of asymptotic approximation to reduce the number of components in models is illustrated for a lead poisoning model in Sect. 3.7 and a disease model in Sect. 6.4.
- A nuanced discussion of scaling appears in Appendix E, including material on how to choose dimensionless parameters (which I have not seen anywhere other than in my Bulletin of Mathematical Biology paper on which the appendix is based) and on how to rescale epidemiological models in which one or more dependent variables requires one scaling for transient behavior and a different one for long-term behavior.

Models and Problem Sets

It is natural to try to work a large number of problems as quickly as possible. However, this is not the best way to learn mathematics. A mathematician learning something new will work through a relatively small number of examples carefully rather than a large number of examples superficially. At a talk I heard on mathematics pedagogy, the speaker asked the audience, "Why do we ask our students to work problems? Is it because we want to know the answer?" Usually we don't care about the answer; we work problems to learn mathematics. Keep this in mind when you are working on a problem: your goal is to learn mathematics and mathematical modeling, not to get the answer to the problem.[1] Many of the problems are guided case studies and require quite a bit of time for a thorough understanding. Carefully working a small number of these will benefit the reader more than a cursory look at a larger number.

There are a large number of biological models that appear in this book as examples, case studies, projects, and problems. Many appear as a sequence of problems, each arising in the section where the relevant material is presented. The reader will get more out of the book by doing all of the problems related to some of these models than by indiscriminately choosing problems from each section. Here is a listing of models and where to find them.

abiotic resource	Problems 4.3.4, 4.4.13, 4.5.5
agent-based disease	Section 1.5, Projects 1A–D
aphid population growth	Project 5B
Beverton-Holt discrete population	Problems 4.1.4, 4.2.4, 4.4.4, 4.4.8
biotic resource	Problems 4.3.5, 4.4.14, 4.5.6
cheetah conservation	Project 5A
chemostat	Problems 3.6.13, 6.1.10, 6.2.8, 6.3.7

[1]Nevertheless, I have tried to make the problems as meaningful and interesting as possible, because nothing is as motivating as a desire to know the answer to a problem.

COVID-19	Section 3.5
discrete logistic growth	Section 4.1, Problems 4.1.2, 4.2.2, Project 4D
discrete resource consumption	Problems 4.1.3, 4.2.3, 4.4.7
disease transitions	Projects 1E, 3E
drug absorption	Problem 3.7.3
enzyme kinetics	Sections 3.8, 6.1
falcon conservation	Section 5.2, Problems 5.2.1–5.2.4, 5.4.9
flour beetles	Project 6F
fluorine at S pole	Problems 2.2.1, 2.2.6, 2.4.6
global temperatures	Problems 1.2.2, 1.2.3, 2.2.8–2.2.10, 2.4.8–2.4.10, Project 2
grape harvest dates	Problem 2.2.11, Project 2
HIV	Problems 3.3.9, 3.4.18, 3.6.11, 6.4.4, 6.4.5
immune system	Problems 3.3.10, 3.6.14, 6.1.15, 6.2.11, 6.3.9, Project 6D
insect pest populations (Hassell)	Problems 4.1.6, 4.2.6, 4.4.3, 4.4.10, Project 4A
lead poisoning	Sections 3.7, 6.2, 6.3, Problems 6.1.2, 6.2.10, 6.3.11
malaria	Problems 3.4.16, 3.4.17, 3.6.10, 3.8.7, 4.5.7, 4.5.8, 6.1.11, 6.3.8
onchocerciasis	Section 6.4, Problem 6.4.1
optimal harvesting	Problems 4.3.8, 4.3.9, 4.3.10
parasitoids	Problem 6.5.6, Project 6E
plankton	Problems 3.6.12, 6.1.8, 6.2.6, 6.3.5
polluted lakes	Project 4C
predator–prey systems	Section 6.3, Problems 1.4.1, 3.6.9, 6.1.4, 6.2.3, 6.3.2, 6.3.12, Project 6C
predation rate functions	Problems 1.2.8–1.2.10, 2.1.4, 2.2.3, 2.3.6, 2.4.4–2.4.5, Section 3.2
red blood cells	Problems 5.1.8, 5.3.6, 5.4.8
resource harvesting type 1	Problems 3.6.7, 4.3.3, 4.3.8, 4.3.10, 4.4.12
resource harvesting type 2	Problems 4.3.7, 4.3.9, 4.4.15, 4.5.9
resource harvesting type 3	Section 4.5
Ricker discrete population	Problems 4.1.5, 4.2.5, 4.4.2, 4.4.9
SEIR epidemic	Sections 3.3, 3.4, Problems 3.3.1, 3.3.7, 3.4.7–3.4.9
SEIR disease with fixed birth	Problems 6.2.12, 6.3.13
SEIR disease with extra features	Problems 3.3.3–3.3.7, 3.4.14, Projects 3A–D
SEIS disease with fixed population	Problems 6.1.9, 6.2.7, 6.3.6

self-limiting population	Problems 3.6.8, 4.3.6, 6.1.5
SIR epidemic	Problems 3.4.6, 3.4.13, 3.6.6, 6.1.3, Project 6B
SIR disease with logistic growth	Section 3.9, Problems 3.9.2, 3.9.3, Project 6A
SIR disease with fixed birth	Section 3.9, Problems 3.9.1, 6.1.7, 6.2.5, 6.3.4
SIR disease with temporary immunity	Problems 3.9.6, 3.9.7, 6.2.13, 6.3.14
SIS epidemic	Problem 3.2.10, Project 4B
SIS disease with fixed birth	Problems 3.9.4, 3.9.5, 6.1.6, 6.2.4, 6.3.3
SIS disease with logistic growth	Problems 3.9.8, 6.1.12, 6.2.9, 6.3.10
SIS disease with standard incidence	Problems 3.9.9, 3.9.10, 6.1.13, 6.1.14, 6.2.1, 6.2.2
teasel plant growth	Project 5C
vaccination	Section 3.1, Problems 3.1.2–3.1.5, 3.3.6, and 3.4.15, Project 3D

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The original version of this book has been revised. The incorrect URL links in the pages 17, 18, 28, 46, 60, 76, 81, 95, and 256 have now been corrected. The correction to this book is available at https://doi.org/10.1007/978-3-031-09454-5_7.

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Part I Mathematical Modeling

Colleagues who teach science and engineering courses often say that their students do not seem to be able to do the mathematics necessary for their subject. The real problem is not so much an inability to do mathematics but an inability to harness the power of mathematics in a scienti c context. More or different mathematics will not address this problem; it requires attention to mathematical modeling, which is largely absent from courses in either mathematics or science. Part I of this book is designed to arm the reader with an understanding of modeling and to teach the associated skills that are seldom taught in mathematics courses.

Mathematical modeling is the tendon that connects the muscle of mathematics to the bones of science. As such, it is distinct from mathematics, albeit with considerable overlap. In mathematics, we begin by specifying a set of theoretical conditions, and then we draw conclusions by applying rigorous methods. The focus is on demonstrating the certainty of the conclusions. In modeling, the starting point is a set of assumptions chosen to caricature a real-world scenario. The mathematical conclusions drawn from analysis of the model hold for the model with the certainty of mathematics; however, their applicability to the scenario that inspired the model is only as good as the assumptions used to build the caricature. Rigorous demonstration of conclusions is far less important than careful consideration of the assumptions and whether the conclusions match what is observed in the real setting. The material in this part is divided into three chapters. Chapter 1 offers an overview of modeling, including an introduction to parameters, a discussion of biological data, an attempt to provide a theoretical foundation for modeling, and a case study that uses an agent-based disease model to create biological data and build modeling intuition. Chapter 2 is a primer on empirical modeling, which is largely about using data to determine parameters for models and using statistical methods to choose among models. The treatment of this material differs from standard treatments in several ways: (1) we begin with the one-parameter model $y = mx$ rather than the more general model $y = b + mx$, (2) we introduce the class of *semilinear* models to facilitate fitting to data for models with two parameters, one of which is a scaling parameter, (3) model selection is based on the Akaike Information Criterion (AIC), a valuable tool that remains inexplicably absent from nearly all statistics books. Chapter 3 is a primer on mechanistic modeling, with a particular focus on the tools needed to build epidemiology models. We begin with two sections that look at modeling of transition and interaction processes before integrating these components into a general framework provided by compartment analysis. The remainder of the chapter consists of a section that looks at the analysis of the most fundamental epidemic model, a critically important section that presents scaling of models, and three case studies that present examples of mechanistic modeling.

Check for updates

1 Modeling in Biology

All mathematics texts include story problems. These are often associated with the term "modeling," which gives mathematics faculty and students the impression that they know what modeling is. However, most mathematics books are written by mathematicians rather than modelers, so the impression they give of modeling does not necessarily match what "modeling" means to modelers.

The story problems in mathematics books are usually of the sort that I call "applications," with characteristics that are different from true modeling:

- Applications are narrow in scope because they use fixed numbers rather than parameters and because the questions call for answers that are simply numbers. For example: "If a bacteria colony doubles every hour, how long does it take a single bacterium to become a population of one million?" Sometimes, the parameter values must be calculated indirectly, as in "A jar initially contains 1 g of a radioactive substance X. After 1 h, the jar contains only 0.9 g of X. How much more time is required before the jar contains only 0.01 g of X?"
- The mathematical setting in an application is implicitly assumed to be exactly equivalent to the real-world setting. Hence, the mathematical answers are unquestioningly accepted as the answers to the scientific questions.

Modeling as conceived by many modelers is quite different from the conception implied by these typical applications. Modeling deals with broad problems that use parameters rather than numbers and address questions that attempt to elucidate the fundamental behavior of the model. The mathematical setting is recognized to be a caricature of the real-world setting, raising questions about how well the mathematical answers apply to the biological questions.

This chapter contains some introductory material that modelers would like to see in precalculus mathematics courses, but which is not always present. The first section introduces the key concept of the *parameter*. Parameters allow a broad scope in modeling by introducing a small amount of abstraction to achieve generalization. The second section presents a brief discussion of biological data and addresses the dilemma of trying to make clear statements with only messy data in support. Section 1.3 expands on the theme of data in biology by presenting some important examples of probability distributions and the fundamental idea that means of samples approach a normal distribution as the number of samples increases. Section 1.4 introduces some essential concepts of modeling, including some that are not commonly articulated. Section 1.5 introduces agent-based models using a case study of a simple model of epidemic disease. Agent-based models are more computational than mathematical, but they provide simple settings for generating data and building intuition.

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The chapter concludes with five projects. Four of these ask students to add features to the agent-based model from the Sect. 1.5 case study. The final project develops the Erlang distribution. This distribution is known primarily in queueing theory rather than biology; however, it is a very valuable tool in disease modeling and worthy of study by biological modelers.

1.1 Working with Parameters

After studying this section, you should be able to:

- Perform algebraic manipulations on functions with parameters.
- Graph functions with parameters.
- Identify the mathematical significance of a parameter in a function.
- Interpret graphs of system properties in terms of a parameter.

We begin with an application problem typical of those found in a precalculus text.

▶ **Example 1.1.1** Jan takes two acetaminophen tablets (650 mg) for her headache. The amount y of acetaminophen in Jan's system is given by the function

$$y = 650e^{-0.3t}, \tag{1.1.1}$$

where t is the time in hours after the dose and y is given in milligrams. At what time will there be only 130 mg of acetaminophen in Jan's system?

We solve this problem by setting $y = 130$ in (1.1.1) and solving for t:

$$e^{-0.3t} = \frac{130}{650} = 0.2.$$

We can take a natural logarithm immediately, but it is easier to take the reciprocal of the equation first:

$$e^{0.3t} = 5.$$

Now, the natural logarithm yields

$$0.3t = \ln 5,$$

or

$$t = \frac{\ln 5}{0.3} \approx 5.36.$$

According to (1.1.1), the amount of acetaminophen in Jan's system will be 130 mg at about 5 h and 22 min (5.36 h) after the dose. □

Example 1.1.1 is fine as far as it goes, but that isn't very far. There is an underlying mathematical model, which we can write as

$$y = Ae^{-kt}, \qquad A, k > 0, \tag{1.1.2}$$

where the quantities A and k are *parameters*.

Definition 1.1.1

A **parameter** is a quantity in a mathematical model that can vary over some range, but takes a specific value in any instance of the model.

Equation (1.1.1) is only an *instance* of the model (1.1.2).[1] The model has been *used*, but not fully *utilized*, because the results cannot be extended to related or broader questions.

1. If we want to see what would happen with a different initial dose, or with ibuprofen instead of acetaminophen, we have to repeat the whole calculation.
2. We obtain quantitative results but no useful qualitative information.

Check Your Understanding 1.1.1:

Rework Example 1.1.1 for an initial dose of 390 mg and a rate constant $k = 0.2$ rather than 0.3.

In contrast, the same mathematical work can be done with the full model rather than an instance of the model, with advantages corresponding to the drawbacks of Example 1.1.1.

1. Work done in solving the full model need not be repeated when parameter values are changed.
2. Analysis of the full model can yield qualitative results that disclose model weaknesses or enhance scientific understanding.

Analysis of mathematical models requires facility in working with parameters.

1.1.1 Scaling Parameters

▶ **Example 1.1.2** Consider the one-parameter family of functions defined by $f(x) = mx$, where $m \in \mathbb{R}$.[2] How are we to understand this family of functions?

With modern technology, we can easily plot graphs and do computations with specific functions (that is, functions with no unspecified parameters). The presence of parameters makes the job more difficult. As a first try, we can graph several examples of the function by selecting a few values of the parameter. Figure 1.1.1 illustrates the family $y = mx$.[3] Graphically, the effect of the parameter m is to control the slope of the line $y = mx$. Algebraically, the effect of the parameter m is to modify all function values by the multiplicative constant m. This property of proportional change marks m as a *scaling* parameter. □

Scaling parameters have a very simple effect on the graph of a function. They magnify the function values by a fixed factor, so they cannot change the "shape" of the graph.

[1] We'll consider more general versions of the model in Examples 1.1.3–1.1.5.

[2] This notation means that the parameter m can be any real number.

[3] We use the word "illustrate" rather than "plot" or "graph" here because we can't simultaneously plot all lines $y = mx$. Instead, we plot several representative examples together and infer the properties of the whole family.

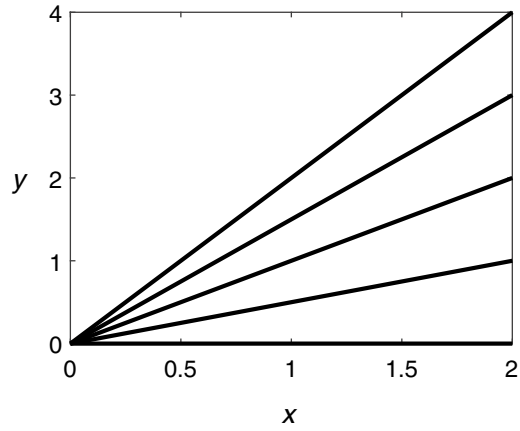


Fig. 1.1.1 The function $y = mx$, with $m = 0, 0.5, 1, 1.5, 2$

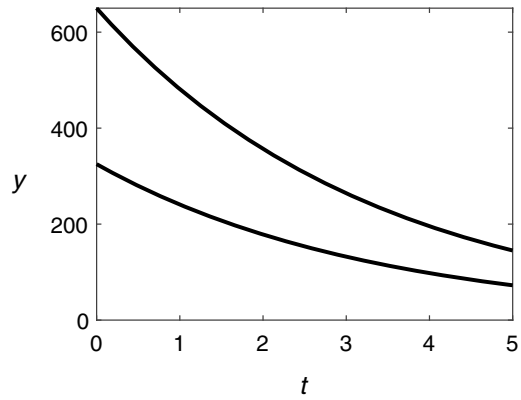


Fig. 1.1.2 The function $y = Ae^{-0.3t}$, with $A = 325, 650$

▶ **Example 1.1.3** Consider the family of functions

$$y = Ae^{-0.3t}.$$

The parameter A in this family, like the parameter m in the family $y = mx$, is a scaling parameter. In Example 1.1.1, where $A = 650$, the function value is reduced from 650 to 130 in 5.36 h. In the general case, the function value after 5.36 h is

$$y = Ae^{-(0.3)(5.36)} = 0.2A.$$

A time of 5.36 h reduces any dose to 20% of its initial value. Figure 1.1.2 illustrates the family of functions, using $A = 650$ and $A = 325$, corresponding to doses of one and two standard acetaminophen tablets. The scaling parameter no longer represents the slope of the graph, but it remains true that two instances with values of A differing by a factor of 2 have results at any given time that also differ by a factor of 2. □

We are now ready to rework Example 1.1.1 using the family $y = Ae^{-0.3t}$.

▶ **Example 1.1.4** Jan takes acetaminophen of initial dose A mg. The amount y of acetaminophen in Jan's system is given by the function

$$y = Ae^{-0.3t}, \tag{1.1.3}$$

where t is the time in hours after the dose and y is given in milligrams. At what time will there be only 130 mg of acetaminophen in Jan's system?

We must solve

$$130 = Ae^{-0.3t}$$

for t, which we can do following the same steps as in Example 1.1.1. Dividing by A yields

$$e^{-0.3t} = \frac{130}{A}.$$

Taking the reciprocal of the equation gives us

$$e^{0.3t} = \frac{A}{130},$$

and then the natural logarithm yields

$$0.3t = \ln\left(\frac{A}{130}\right) = \ln A - \ln 130.$$

Thus, we arrive at the answer

$$t = \frac{\ln A - \ln 130}{0.3}. \tag{1.1.4}$$

Because no numerical value has been specified for A, the answer retains A as a parameter. We can understand this answer by plotting a graph of time to 130 mg against A, shown in Fig. 1.1.3. Doses smaller than 650 mg take less time to be reduced to 130 mg than a 650-mg dose, but the curvature of the graph means that a dose of only half the size takes more than half as long. □

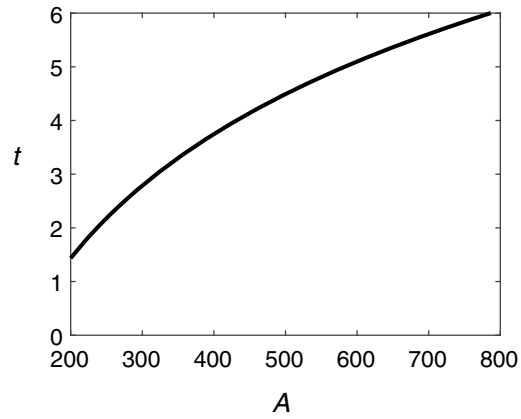


Fig. 1.1.3 The time (in hours) required for an acetaminophen dose of A mg to be reduced to 130 mg

Notice that A is *not* a scaling parameter in (1.1.4). It magnifies $y(t)$ by a fixed factor, but its effect on answers to more complicated questions is more subtle. Consequently, the result of Example 1.1.4 contains much more information than that of Example 1.1.1. We can now easily determine the required time for any size of dose, just as we can use Fig. 1.1.2 to determine the amount of medication remaining in the system at any given time. While application problems generally consist of simple numerical questions, mathematical models permit a large variety of questions, some answered simply with numerical values or functions of time and others answered by graphs and/or formulas involving parameters.

Scaling parameters can also simplify mathematical analysis, as we will see in Chap. 6.

1.1.2 Nonlinear Parameters

Most parameters are not scaling parameters. We can still study their effects by using thought experiments similar to those of Examples 1.1.3 and 1.1.4.

▶ **Example 1.1.5** Determine the key features of the one-parameter function defined by $f(t) = e^{-kt}$, where $k > 0$, on the interval $t > 0$.

All of these functions have $f(0) = 1$, all are decreasing, and all are positive. One way to see the effect of k is to find the times required for the function to decrease to one-half of its initial value. Let t_h be the desired time. Then

$$e^{-kt_h} = \frac{1}{2},$$

or

$$e^{kt_h} = 2.$$

Taking logarithms yields $kt_h = \ln 2$. Thus, e^{-kt} is reduced to one-half of its initial value at time

$$t_h = \frac{\ln 2}{k}. \tag{1.1.5}$$

The time t_h defined here is called the *half-life* for the decaying exponential function $y = e^{-kt}$. The half-life decreases as k increases, so larger k values make the function decrease faster. Figure 1.1.4 illustrates the function e^{-kt}. □

Decaying exponential functions of the form $y_0 e^{-kt}$ are used to model radioactive decay, drug clearance, and many other phenomena. The parameter y_0, which represents the initial value, is entirely dependent on context. The parameter k, which represents the decay rate, is a property of the specific quantity undergoing decay. Published sources generally provide the half-life t_h rather than the rate parameter k, but the relationship in (1.1.3) makes it easy to connect the desired parameter value to the known half-life.

Check Your Understanding 1.1.2:

Find the exact value and an approximate numerical value for the rate constant k if the half-life of a drug in the human body is 4 h.

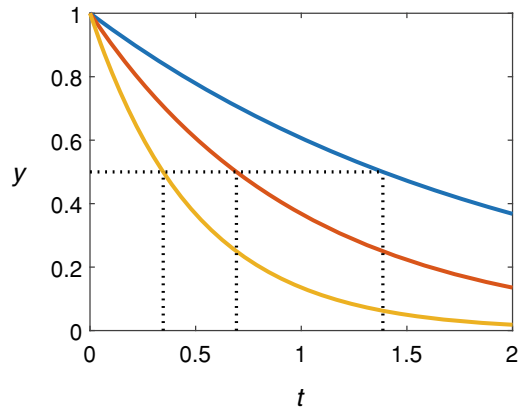


Fig. 1.1.4 The function $y = e^{-kt}$, with $k = 0.5$, $k = 1$, and $k = 2$ (top to bottom), showing the points $(t_h, 0.5)$

1.1.3 Bifurcations

In each of our examples so far, the parameter has had only a quantitative effect on the function. Nonlinear parameters can have a qualitative effect as well.

▶ **Example 1.1.6** Determine the nonnegative solutions of the equation $f(x) = x^3 - 2x^2 + bx = 0$, where b is any real number.

We can start by factoring out an x:

$$f(x) = x^3 - 2x^2 + bx = x(x^2 - 2x + b).$$

Thus, the function f always has a root at $x = 0$ and may have additional roots at points where $x^2 - 2x + b = 0$. Applying the quadratic formula to the latter equation, we have

$$x = \frac{2 \pm \sqrt{4 - 4b}}{2} = 1 \pm \sqrt{1 - b}.$$

There are two special cases. At $b = 1$, the quadratic formula yields only one root, $x = 1$; hence, the roots are $x = 0$ and $x = 1$. At $b = 0$, one of the roots from the quadratic formula is 0 and the other is 2, so the roots are $x = 0$ and $x = 2$. These special cases divide the range of b values into three intervals where the numbers of roots are different.

- For $b > 1$, there are no roots from the quadratic formula.
- For $0 < b < 1$, the square root quantity is between 0 and 1; hence, the quadratic formula yields two roots, one in the interval $0 < x < 1$ and one in the interval $1 < x < 2$.
- For $b < 0$, the square root yields a number larger than 1, so there is one positive root in the interval $x > 2$.

Figure 1.1.5 is a plot of the nonnegative roots as a function of b. The horizontal line indicates that $x = 0$ is a root for all values of b. The curve indicates roots that come from the quadratic formula, showing that there is one such root if $b < 0$, two such roots if $0 < b < 1$, and no roots (other than 0) for $b > 1$. □

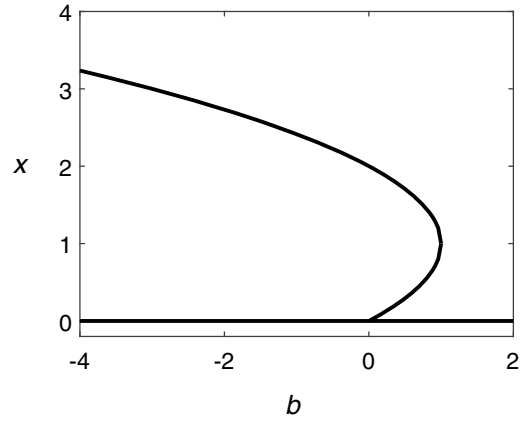


Fig. 1.1.5 The nonnegative roots of $x^3 - 2x^2 + bx$ as a function of b

The function $f(x) = x^3 - 2x^2 + bx$ of Example 1.1.6 is a case where the qualitative properties change as a parameter crosses a specific value. This kind of behavior is illustrative of a feature that is common in biological models.

Definition 1.1.2

A **bifurcation**[4] is a sudden change in the qualitative behavior of a system. The point at which it occurs is called a **bifurcation point**.

▶ **Example 1.1.7** Disease models have a parameter called the *basic reproduction number*, denoted $\mathcal{R}_0$ (and pronounced as "R-nought"), which measures the maximum average number of secondary infections that can be produced in a population by one primary infective. The system exhibits an epidemic outbreak if and only if this parameter is greater than one.[5] □

Check Your Understanding 1.1.3:

The rate of change of the infectious fraction of a population for a simple disease model is $di/dt = \mathcal{R}_0 i(1-i) - i$. Show that there is a positive value for i that makes $di/dt = 0$ if and only if $\mathcal{R}_0 > 1$.

Check Your Understanding Answers

1. $5 \ln 3 \approx 5.49$.
2. $k = \ln 2/4 \approx 0.173\,\text{hr}^{-1}$.
3. We get the solution $i = 1 - \mathcal{R}_0^{-1}$, which is positive only when $\mathcal{R}_0 > 1$.

Problems

1.1.1 The clearance rate constant k in (1.1.2) is slightly different for different individuals, as well as being considerably different for different drugs. Suppose k is the clearance rate for acetaminophen of

[4] BYE-fur-ca-shun.

[5] The best estimate of $\mathcal{R}_0$ for the original strain of COVID-19 is 5.7 [21], while that for the delta variant that was dominant in summer 2021 is about 10.

a patient who takes two acetaminophen tablets (650 mg). Determine the time t_1 at which a patient's system has only 130 mg of acetaminophen, in terms of the parameter k. Plot a graph of t_1 versus k. Interpret the graph in terms of the drug clearance process.

1.1.2* For the data of Example 1.1.1, find a formula for the time t_z at which the amount of acetaminophen in Jan's system is some unspecified amount z. Plot a graph of t_z versus the parameter z. Interpret the graph in terms of the drug clearance process.

1.1.3 Do an internet search to find the half-life of acetaminophen in a healthy body. Look at a variety of sites so that you will find a range of values. Determine the constant k for the largest and smallest half-lives and plot the functions $f(t) = Ae^{-kt}$ using these values together on the same axes. Use a standard dose for A.

1.1.4 Do an internet search to find the half-life of naproxen sodium in a healthy body. Look at a variety of sites so that you will find a range of values. Determine the constant k for the largest and smallest half-lives and plot the functions $f(t) = Ae^{-kt}$ using these values together on the same axes. Use a standard dose for A.

1.1.5 Determine the nonnegative roots of the polynomial $x^3 - bx^2 + x$ as a function of b. Plot these roots as in Fig. 1.1.5.

1.1.6*

(a) The one-parameter function $f(x) = x^2 - bx$, where b is any real number, has a graph that is a parabola. Find the vertex of the parabola in terms of the parameter b. (Hint: This is a simple problem using calculus. Without calculus, the problem requires algebraic manipulation. All quadratic equations can be written in the form $f = a(x - h)^2 + k$, where the vertex is at (h, k) and a is a parameter that represents the "broadness" of the parabola. By setting the formula given for the function equal to the desired form and requiring the polynomials to be identical, determine three equations that relate the desired parameters h, k, and a to the given parameter b.)

(b) Plot f with $b = -2, 0, 2, 4$. Verify that the vertices are in the location determined in part (a).

1.1.7 Suppose a disease reaches a closed population that has not previously experienced it. Of course, the outstanding example of that is COVID-19, but it also applied historically in instances where travelers brought a disease to a new area, such as when Europeans brought smallpox to the Americas. The fraction d of individuals who contract the disease can be estimated as a function of the basic reproduction number R_0, a parameter that represents the average number of secondary infections due to one infected individual in a population of susceptibles. Values of R_0 can vary from 0 to about 20,[6] with measles estimated at 12–18 and smallpox at 5–7 [3]. If $R_0 < 1$, then the disease can only infect a few individuals before it dies out. If $R_0 > 1$, then a disease epidemic occurs, and a model predicts that the fraction who get the disease satisfies the equation

$$R_0 d + \ln(1 - d) = 0.$$

(a) Solve the equation for R_0 and use a "guess-and-check" method to estimate the value of d for $R_0 = 5$. Independent of disease mortality, what does this suggest about the impact of smallpox on a Native American population newly exposed by explorers in the late fifteenth century?

[6] The mathematical epidemiologists Valerie Tweedle and Robert J. Smith? (The question mark in "Smith?" is part of the spelling. As of 2020, this author is now Stacey Smith?.) claimed that the most infectious "disease" was the social disease they called "Bieber fever," whose basic reproductive number they estimated to be about 27 [2].

(b) Plot d as a function of R_0 for the range $0 < R_0 < 5$. (Keep in mind that the given formula is not correct for all values of R_0. Note that tabular data for two related variables can be plotted with either variable on the horizontal axis.)

1.1.8 A resource x is consumed at the rate

$$f(x) = \frac{Ax}{B + x},$$

where A and B are positive parameters.

(a) What is the maximum consumption rate in terms of A and/or B, achieved in the limit as $x \to \infty$?
(b) Suppose the actual consumption rate is half of the maximum from part (a). What is the corresponding resource level x?

1.1.9 Suppose a quantity y decreases according to the formula $y = 1/(1 + at)$. Find the time at which the quantity reaches half of its starting value. Plot the result as a function of a.

1.1.10 In this problem, we consider the additional difficulties of modeling ibuprofen amounts in a patient, given that ibuprofen is slowly absorbed from the digestive system and relatively quickly eliminated from the body.

(a) Plot a concentration curve for ibuprofen using the model $y = Ae^{-kt}$. Assume a dose of 200 mg and a rate constant of $k = 0.35$.
(b) How long does it take for one-half of the drug to be eliminated? (Do this algebraically, not by trial and error.)
(c) A more sophisticated drug clearance model takes account of the need for an oral dose to be absorbed by the digestive system. This model is

$$y = \frac{Ab}{b - k}(e^{-kt} - e^{-bt}),$$

where A is the amount of the dose, k is the clearance rate for the bloodstream, and b is the clearance rate for the digestive system. Normally, $b > k$. Use this model to plot a drug versus time curve for a 200- mg dose of ibuprofen, assuming $k = 0.35$ and $b = 0.46$.
(d) Describe the important differences between the curves of part (a) and part (c). Consider qualitative features of the graphs as well as quantitative properties such as the peak value and the time at which the peak occurs.

1.1.11* Medication regimens for multiple doses are designed so that the minimum level (just before a dose) is high enough to be therapeutic and the maximum level (just after a dose) is not high enough to be toxic. We explore this idea in this problem.

(a) Suppose a person has been taking 650- mg doses of acetaminophen every 4 h for several days. This will result in a drug versus time curve that repeats over a period of 4 h. Let A be the unknown concentration immediately after taking a dose. Use the model for acetaminophen drug clearance, with $k = 0.3$, to obtain a formula for the amount of drug present after 4 h, immediately before the next dose.
(b) Use the result from part (a) to obtain a formula for the amount present immediately after the next dose.

(c) Given that the concentration curve is periodic, the amount present immediately after that next dose (from part (b)) must be A. Use this fact to calculate the value of A and to determine the minimum amount present during the 4-h period.
(d) Repeat parts (a)–(c) with doses of B mg of a drug with rate constant k taken every T hours. (Keep in mind that we are using A to represent the amount in the system at time 0, which includes both the new dose and whatever is remaining of previous doses.)
(e) Prepare a drug versus time curve showing the amount of acetaminophen from parts (a)–(c) over a 24-h period.[7]

1.1.12 A simple model for the optimal amount of territory for a bird to try to maintain was presented by R. McNeill Alexander [1]. Let A be the area of the territory and let p be the daily food availability per unit area, so that the total amount of food available per day is pA. If k is the amount of time (in days) per unit area needed to defend the territory, then $(1 - kA)$ is the amount of time (in days) available for collecting food. Assuming that the bird can collect food at the rate of q units per day, then the amount that can be collected in the time not spent defending the territory is $q(1 - kA)$. The optimal territory is that which has just enough food available so that the bird can collect all of it.

(a) Determine the optimal territory size in terms of the various parameters in the model.
(b) Let $Q = q/p$. Use this new quantity to obtain a formula for optimal territory size that depends on just two parameters.
(c) What is the biological meaning of the parameter Q?
(d) Suppose $k = 1$. This means that $A = 1$ represents the largest territory that can be defended. We can then interpret a specific value of A as the fraction of defendable territory that the bird should choose. Plot this optimal territory fraction as a function of the parameter Q.
(e) Interpret the graph biologically.

1.1.13 Under a given set of assumptions, we can obtain the model

$$r = \frac{\ln n}{a}$$

for the fractional yearly population growth (i.e., $r = 0.01$ corresponds to a growth rate of 1% per year) as a function of the average number of children per adult (n) and the average age of the mother when a child is born (a).

(a) Plot r as a function of n over the reasonable range $0.5 \leq n \leq 2$, with a values of 25, 30, and 35. These three curves should be plotted together on one set of axes. Use this figure to discuss the dependence of the growth rate on the number of children, focusing on how this dependence changes for different average reproduction ages.
(b) Plot r as a function of a over the reasonable range $25 \leq a \leq 35$, with n values of 1, 1.5, and 2. These three curves should be plotted together on one set of axes. Use this figure to discuss the dependence of the growth rate on the average reproductive age, focusing on how this dependence changes for different average numbers of children.

1.1.14 Consider a species that has a constant death rate of m individuals per capita per time unit.

(a) Given a constant per capita death rate, survival of individuals in a cohort born at the same time decreases exponentially. Use this fact to find a formula for the fraction $y(t)$ of individuals alive at age t. In particular, find the fraction s of individuals alive at age 1 as a function of m.

[7] Note that we are assuming the patient has been taking the drug for several days already. The results would be more complicated if our study began with the very first dose.

(b) Suppose individuals of our hypothetical species mature at age 1 and reproduce at constant rate n until age 2. It can be shown that the birth rate necessary for the overall population to stay at the same size is

$$n = \frac{m}{e^{-m} - e^{-2m}}.$$

Combine this formula with the result of part (a) to obtain an equation that determines the birth rate necessary for population maintenance as a function of the probability of surviving to maturity.

(c) Plot the resulting function $n(s)$. Choose a biologically reasonable range of s values for the plot. Discuss the biological significance of the result by comparing the birth rate requirements for animal species with different probabilities of survival to maturity.

1.2 Mathematics in Biology

After studying this section, you should be able to:

- Identify the role of mathematical modeling in science.
- Discuss the concept of demographic stochasticity and apply this concept to biological experiments.
- Generate questions about an experiment that could possibly be addressed with a mathematical model.

Before we can do mathematical modeling in biology, we first have to think about the basic question of where in biology is there even an opportunity to use mathematics. The connections between mathematics and biology are far less obvious than those between mathematics and physics; consequently, acceptance that mathematics has an important role in biology is not as universal as acceptance of the role of mathematics in physics.

In any modeling endeavor, the starting point has to be the real-world setting rather than the mathematics. Like any science, biology is a combination of theory and observation (whether of natural systems or purposeful experiments), with observation as an essential precursor to any meaningful theory. Since biological data can be different from physical science data in important ways, it makes sense to begin by gaining some experience with the kind of data that is common in biology.

1.2.1 Biological Data

Pick up a calculus or precalculus book and find a story problem with a scientific setting. Most likely, the problem you find has exact data. Real scientific data is not exact, and this difference must be understood before we can do mathematical modeling. We can explain the difference here, but you will understand it much better if you discover it yourself, through a combination of looking at data collected by others and collecting your own data.

The real world is not an easy setting for the collection of biological data. Even ignoring the difficulties in getting a good data set, there is the problem that data collection takes a lot of time and effort. This effort is necessary if we are going to practice real science, but it is a distraction if our purpose is to learn mathematical modeling. An alternative to collecting data from an experiment in the real world is to collect data from either a physical simulation, a practice that has been called "bean-bag biology" [12], or "a real experiment in a virtual world" [13]. Physical simulations can be done in a classroom or on a tabletop, while virtual worlds can be studied in a comfortable chair in front of a computer, without having to wait for events to occur in natural time. If a physical simulation or virtual world is carefully designed, what we learn from it might even be helpful in understanding the real world.

Table 1.2.1 Counts of undecayed particles from a virtual decay experiment

Day	0	1	2	3	4	5	6	7	8	9	10
y	100	87	83	75	67	58	51	46	44	40	35
Day		11	12	13	14	15	16	17	18	19	20
y		29	28	26	24	20	18	17	14	14	12
Day		21	22	23	24	25	26	27	28	29	30
y		11	11	10	10	7	7	7	6	6	6

Radioactive decay is modeled with a decaying exponential function; that is, given an initial amount A and half-life T, the amount remaining after time t is

$$y = A \cdot 2^{-t/T}, \tag{1.2.1}$$

more commonly written as

$$y = Ae^{-kt}, \qquad k = \frac{\ln 2}{T}. \tag{1.2.2}$$

This model is usually derived from physical principles, but it was almost certainly found originally from data—else how would one know the correct physical principles?

Most of us lack the facilities to set up a real experiment to measure radioactive decay. We can, however, produce real data from a virtual experiment based on the actual phenomena, although of course, we can't know that the virtual experiment authentically represents radioactive decay without comparing its results to that of a real experiment.

▶ **Example 1.2.1** The function program *decay.m* encodes a simulation of the decay of y_0 particles over a given amount of time, with the decay rate chosen so that the mean time for decay is μ days. The script *DecaySim.m* was set up to run one simulation using 100 particles with a mean decay time of 10 days. The results are shown in Table 1.2.1 and Fig. 1.2.1b. □

1.2.2 Deterministic Patterns in a Random World

Figure 1.2.1a shows the average land temperature on July 1 for years in the range of 1985 through 2015 [4]. The temperature data shows so much random variation from one year to another that is hard to see any meaning in the graph. Some of this variability comes from random events such as volcanic eruptions or variations in sunspot activity. Some of the unpredictability in real data such as this might come from minor flaws in experiment design or measurement and from our inability to completely control an experimental environment. Indeed, we should be somewhat skeptical that any real data is exactly correct.[8]

The data of Fig. 1.2.1b came from a computer simulation, so most sources of error in real data are absent here. Nevertheless, the individual data points do not exactly match our theoretical expectations. The virtual world was designed to have a 10% probability of decay for each particle in a given day. Thus, we "should" have had 10 decays in the first year, but instead, we had 13. The next year we had

[8] It has been observed that "nobody believes a model except the person who created it, while everybody believes data except the person who collected it."

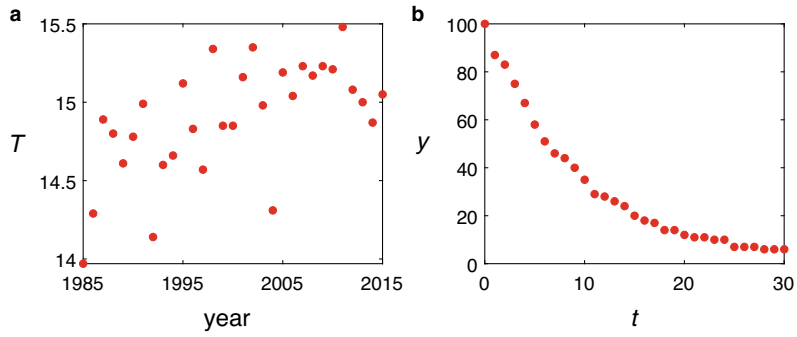


Fig. 1.2.1 **a** Average July 1 global land temperature in $°C$ [4]; **b** undecayed particle counts from Table 1.2.1

only 4 when we expected 8 or 9. Perhaps a different set of 100 particles would have had fewer decays on the first day and more on the second.

Unpredictability due to the randomness of a sample is called *demographic stochasticity*, and is present whenever the number of individuals in an experiment is small and differences in individual behavior are significant. This is typical in many areas of biology. In contrast, experiments in chemistry are free of demographic stochasticity because of the extremely large number of particles in even a small amount of material.[9] This beneficial averaging does not occur in experiments with single individuals and might be insufficient even with a sample of thousands.

At first thought, the inevitable randomness in biological data seems to suggest that mathematical modeling is pointless in biology. Mathematics is the most deterministic of disciplines, with many problems defined in such a way that there is a unique solution. If all biological events are affected by random factors, how can mathematics have any value in biology?

Obviously, the highly stochastic nature of biological events *limits* the possibilities for using mathematical methods in biology. It certainly *is* pointless to attempt to use mathematics to predict the number of days required for a particular person infected with COVID-19 to show symptoms. However, mathematics can be used to study the patterns that arise when experiments are repeated many times.

Biologists are often interested in identifying relationships between quantities in a system. At its most elementary level, this is the goal of ANOVA (analysis of variance), a statistical extension of descriptive statistics. However, ANOVA only seeks to determine the significance of the relationship between quantities. A more ambitious goal is to search for a quantitative description of a relationship. This enterprise rests on the view that real biological data can often be thought of as consisting of some random variation superimposed on a deterministic pattern.

A principal aim of mathematical modeling in biology is to develop methods for obtaining *deterministic* models for the *average* behavior of fundamentally *stochastic* biological processes.

It is sometimes necessary to present data in multiple ways to identify possible deterministic patterns. Figure 1.2.2 shows a plot of $\ln y$ against time from the data in Table 1.2.1. This plot suggests that if we averaged out the randomness of individual trials, we would obtain a set of points that lie on a straight line. We'll identify the best fit straight line in Sect. 2.2.

[9] Demographic stochasticity of virus particles is not an issue in a model that tries to predict quantities of these particles in a person suffering from a communicable disease; however, demographic stochasticity in a population of people could be quite significant.

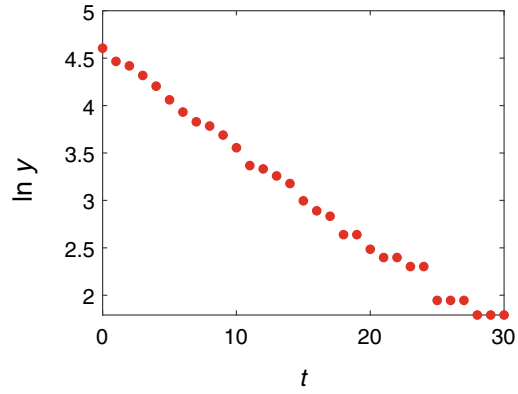


Fig. 1.2.2 Undecayed particle counts from Table 1.2.1, plotted as ln y rather than y

Problems

1.2.1 Suppose the underlying signal in the data of Fig. 1.2.1a is a straight line. Just by eye, sketch the straight line you think best fits the data.

(This problem is continued in Problem 2.2.8.)

1.2.2 [Global Temperature Change]
Use the data from 1985 through 2015 in the file *GlobalLandTemperatures_January.csv*[10] to create a plot similar to that of Fig. 1.2.1a. Just by eye, sketch the straight line you think best fits the data.

(This problem is continued in Problem 2.2.9.)

1.2.3 [Global Temperature Change]
Use the full data set in the file *GlobalTemperatures_July.csv*[11] to create a plot similar to that of Fig. 1.2.1a. The hypothesis of human-induced climate change suggests that the historical record should show no particular pattern before the beginning of a gradual temperature rise at a point corresponding roughly to the Industrial Revolution. Does a visual look at the plot seem to support this hypothesis? Discuss.

(This problem is continued in Problem 2.2.10 and Project 2A.)

1.2.4 Have someone carefully measure your height at regular intervals of 1 hour over the course of a day. Record the measurements to the nearest 1/8 inch or 2 mm. Plot a graph of the data. Describe your observations and suggest an explanation.

1.2.5 Do enough strenuous exercise to get yourself breathing hard, and then sit down. Record the number of your heartbeats over the course of the next minute. Restart the count so as to record the number of your heartbeats in the second minute. Continue to record the number of your heartbeats each minute for 15 min. Plot a graph of the data. Describe your observations and suggest an explanation.

1.2.6 A very simple epidemic model can be simulated using beans of two different colors and similar size and shape [12]. Start with a large cup containing 19 dark beans and one light bean. The dark beans represent people who do not have a disease, while the light beans represent people who do. Now perform a sequence of steps:

[10] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

[11] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

1. Randomly withdraw two beans from the cup without looking.
2. If the beans are of opposite colors, return the light bean to the cup and replace the dark one with a light one; otherwise just return both beans.
3. Record the number of light beans in the cup.

Repeat the sequence of steps until there are only four dark beans left.

(a) Plot a graph of the number of light beans versus time.

(b) Describe and explain the results.

(c) Identify specific features of the bean simulation that might not represent a real disease.

(d) Compare your graph with those of colleagues, or else repeat the experiment two more times to get three different graphs. What features do the graphs of different simulations have in common? Do any features vary significantly among the different simulations?

1.2.7 Use the program *DecaySim.m* to run several simulations using 100, 10,000, and 1,000,000 particles. How much variation is there in the graphs for each of these cases? Discuss the reason(s) for the results you observe.

Problems 1.2.8–1.2.11 require BUGBOX-predator.[12] Save the data sets from these problems, as they will be needed for problems in other sections.

BUGBOX-predator is a virtual world that creates an environment for collecting data on the amount of prey a predator captures in a given amount of time. It is based on a famous bean-bag biology experiment conducted by C. S. Holling in the late 1950s, before the capability of creating virtual worlds with computers [9]. Holling set up a virtual world consisting of sandpaper discs tacked onto a plywood board. The discs represented insects and a blindfolded student represented a predatory bird. In each experimental run, a student tapped the board with a finger at a steady pace, moving randomly around the board. Each time the student touched a sandpaper disk, (s)he removed it, placed it in a cup, and then returned to tapping. After 1 min, the student recorded the number of disks "eaten" in this manner. The data set consisted of pairs of numbers: disks available and disks "eaten." Holling used the data, along with his observations, to create the predation models that now bear his name. These models will be developed in Sect. 3.2.

The BUGBOX-predator world consists of a grid populated by x virtual aphids (the number x is chosen by the experimenter) and one virtual coccinellid (ladybird beetle). The predator moves randomly through the virtual-world environment, stopping to "consume" any virtual aphids in its path. The experiment outcome y is the number of prey animals eaten in 1 min.[13] There are two predator species, called *Predator speedius* and *Predator steadius*.

1.2.8 Collect a predation data set for *P. steadius*, using the default choice of no replacement. Use prey values of approximately 10, 20, and so on up to 140. Plot these data on a graph similar to Fig. 1.5.1.

(This problem is continued in Problems 2.1.4 and 2.2.3.)

1.2.9 Repeat Problem 1.2.8 for *P. speedius*.

(This problem is continued in Problem 2.3.6.)

[12] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

[13] Given unit time for the experiment and unit area for the environment, we can interpret y as the consumption rate per predator, in prey animals per unit time, and x as the prey density, in prey animals per unit area.

1.2.10 Repeat Problem 1.2.8, but with replacement.

(This problem is continued in Problems 2.1.4 and 2.2.3.)

1.2.11 From your experience in Problems 1.2.8 and 1.2.10, discuss the significance of the replacement option. Which option would be easier to implement in an experiment with real organisms? Which option allows for unambiguous reporting of data (think about possible differences between what x is supposed to mean and the way we measure it)? This example illustrates the difficulty of designing biological experiments, given the need for practical implementation and the importance of avoiding ambiguity.

1.3 Quantifying Randomness in Data

After studying this section, you should be able to:

- Identify the contexts in which the binomial and exponential distributions are appropriate.
- Explain the difference between distribution parameters and distribution outcomes
- Use the cumulative distribution function and survival function for exponential distribution calculations.
- Use the formulas for mean and standard deviation for the binomial and exponential distributions and distributions of sample means.
- Prepare histograms for distributions of means of samples of size n from a given underlying distribution.
- Discuss the properties of distributions of sample means as a function of sample size.

We've seen that individual measurements are not exactly predictable. Nevertheless, unpredictability itself can be quantified. This is the realm of probability and statistics, which are very large subjects and generally beyond the purview of a book on modeling. In this section, we summarize some results from probability theory that modelers need to know to understand biological data.

This section rests on a fundamental assumption—that there is no systematic bias in data. In reality, there can often be systematic bias through flawed experiment design, error in measuring equipment, or unforeseen complicating factors. Political polls, for example, include methods for selecting a sample of voters and assumptions about turnout probability, both of which can lead to a systematic difference between predicted and actual vote totals. Certain areas of biology are also prone to systematic errors, an example being that the number of confirmed COVID-19 cases in most practical contexts is a very poor estimate of the number of people infected. Here, we consider only unpredictability, assuming the measurements themselves are free of error.

1.3.1 Probability Distributions

Suppose you want to know your resting pulse rate. You can measure it once, but it might be unusually low or high at that moment. For a better estimate, you can measure it 10 times and compute the average. This would give you a "better" value if you have to report just one. You could compute your average from any number of measurements, however, and it is clear that doing so will change the average, if only slightly. No matter how many measurements we make, we are measuring only a finite *sample* of an infinite universe of possible measurements.

Modeling is about making approximations, and one approximation we can make is to think of the infinite universe of possible measurements as being fully known. This changes the subject from statistics, which deals with real measurements, to probability, which is a theoretical subject.

The simplest example is the probability distribution for the flip of a "fair" coin, which we expect to come up heads "exactly half of the time." If we flip a real coin 100 times and get heads 51 times, we will probably take this to be a natural random outcome rather than a systematic preference of that coin for heads. "Exactly half of the time" is a verbal statement of a mathematical *probability distribution*; in this case, the probability distribution for a fair coin is that both heads and tails have probabilities of exactly 0.5.

Definition 1.3.1

A **probability distribution** is a set of possible outcomes and a way of determining the probabilities for those theoretical outcomes. If the possible outcomes are continuous rather than discrete, then the distribution rule prescribes the probability that the outcome will fall within a range of the continuous values.

Probability distributions allow us to characterize random events using mathematical formulas that have only a small number of parameters. They can be defined in a number of different ways. Like other modeling settings, we must be careful to use a distribution only in a correct context. Here, we consider just two probability distributions: the binomial and exponential distributions. We introduce the family of Erlang distributions in the problem set.

The Binomial Distribution

A fair coin is a special case of a binary choice. In general, the probability of one choice could be any plausible value; for example, the probability of hitting a free throw in basketball could be anything from 0 for a child who lacks sufficient strength to nearly 1.0 for 2020 WNBA free throw percentage champion Tiffany Mitchell. Binary choices occur frequently in real-world situations, so there is a special mathematical structure to represent them.

Definition 1.3.2

A **Bernoulli**[14] **trial** is a random experiment with two possible outcomes called "success" and "failure", with probability $0 < p < 1$ of success and probability $q = 1 - p$ of failure.

▶ **Example 1.3.1** In a famous genetics experiment, the Austrian monk Gregor Mendel created a population of peas by carefully combining pollen and ovules from two different plants, one whose seeds were always smooth and one whose seeds were always wrinkled. He used these seeds, all of which were smooth, to collect a second generation of seeds from natural pollination. The second-generation seeds were either smooth or wrinkled, so the experiments of producing them were individual Bernoulli trials. If we arbitrarily define a wrinkled seed as a "success", Mendel's measurements suggested that the success rate for his experiment was $p = 0.25$; the explanation for this result is the foundation of Mendelian genetics. □

Individual Bernoulli trials are of value primarily as the building blocks of experiments consisting of a number of independent trials. The family of binomial distributions prescribes the theoretical probabilities for any given number of successes in multiple Bernoulli trials.

[14] Burr-NOO-lee.

Table 1.3.1 Probabilities for the number of wrinkled peas out of 24

k	0	1	2	3	4	5	6
$b(k; 24, 0.25)$	0.001	0.008	0.031	0.075	0.132	0.176	0.185
k	7	8	9	10	11	12	13
$b(k; 24, 0.25)$	0.159	0.112	0.067	0.033	0.014	0.005	0.002

Definition 1.3.3

The **binomial distribution** $b(k; n, p)$ represents the probability distribution of k successes in a set of n independent Bernoulli trials, each having success probability p.

The semicolon in our notation for the binomial distribution is one way of distinguishing the independent variable k from the parameters n and p.

The development of the binomial distribution formula and its properties is outside the scope of our treatment, but can be found in any standard probability or statistics text. Here, we simply present the results.

Theorem 1.3.1 (The Binomial Distribution)
The probability of k successes in n Bernoulli trials, each with the same success probability p, is given by the binomial distribution formula:

$$b(k; n, p) = \frac{n!}{k!(n-k)!}\left(\frac{p}{q}\right)^k q^n, \quad q = 1 - p. \tag{1.3.1}$$

The mean and variance of the binomial distribution with n trials and success probability p are

$$\mu = np, \qquad \sigma^2 = npq. \tag{1.3.2}$$

▶ **Example 1.3.2** To find the probabilities of getting k wrinkled seeds out of 24, we use the binomial distribution formula with $n = 24$ and $p = 0.25$. Mathematical software has built-in functions for the binomial distribution, so we don't need to do the calculations ourselves. Table 1.3.1 shows the probabilities of getting from 0 to 13 wrinkled seeds. From (1.3.2), we can calculate the mean to be 6, which is of course the number we would expect as 25% of 24. Note that we will get the expected value of 6 successes out of 24 trials less than 20% of the time—not because of any error or flaw, but simply because of unavoidable unpredictability of experiment results.[15] □

[15] Mendel published only a portion of his data, which the statistician and geneticist Ronald A. Fisher concluded had been adjusted to give better support to Mendel's theory than the full data set would have given. This sounds unethical to modern scientists, but careful control of data to minimize bias is a relatively recent innovation in science. A more complete discussion of Mendel's data and Fisher's analysis appears in [6].

The Exponential Distribution

Definition 1.3.4

The **exponential distribution** is the continuous probability distribution for a random variable T with outcomes $0 < t < \infty$ and defined by the *cumulative distribution function*

$$\mathrm{P}[T \leq t] = E(t; \lambda) \equiv 1 - e^{-\lambda t}, \tag{1.3.3}$$

where λ is a parameter.

Theorem 1.3.2 summarizes the key properties of the exponential distribution.

Theorem 1.3.2 (Exponential Distribution Properties)
The exponential distribution $E(t; \lambda)$ has mean and standard deviation

$$\mu = \frac{1}{\lambda}, \qquad \sigma = \frac{1}{\lambda}. \tag{1.3.4}$$

Definition 1.3.5

The survival function for decaying particles is the fraction remaining at time t, given a mean survival time of μ. From Theorem 1.3.2, this function is defined as[16]

$$S(t; \mu) = \mathrm{P}[T > t] = 1 - E(t; \lambda) = e^{-\lambda t}, \qquad \lambda = \frac{1}{\mu}. \tag{1.3.5}$$

▶ **Example 1.3.3** Suppose we start with a collection of A atoms whose decay times are exponentially distributed with mean time μ and that one half of the atoms remain undecayed at time t_h. Using the survival function, the number of undecayed particles is

$$\frac{A}{2} = A\mathrm{P}[T > t] = Ae^{-\lambda t_h}. \tag{1.3.6}$$

Solving for t_h yields

$$t_h = \frac{\ln 2}{\lambda}. \tag{1.3.7}$$

□

So far, we have merely asserted that the exponential distribution is appropriate for a natural decay process. To see why this is so, we need to consider the dependence of event times on history. Our human intuition suggests that events distributed over time become more likely as we wait longer for them to occur. This is often true; for example, a person who has had a cold for three days is much more likely to recover tomorrow than a person who only started having symptoms today. A sick person's internal

[16] We'll derive the survival function $e^{-\lambda t}$ from a mathematical model in Sect. 3.1.

state changes during their illness, and these changes are preparatory to recovery. This is not the case for natural decay processes. Decay of radioactive uranium atoms, for example, occurs spontaneously without any preparation; hence, the expected amount of time required for a particular atom to undergo decay should be independent of when we start the clock. This counterintuitive property is satisfied by the exponential distribution.[17]

▶ **Example 1.3.4** Epidemic models often assume that the time required to recover from a disease is exponentially distributed, even though time to recovery is often highly dependent on the amount of time since becoming sick. This sounds like a serious error, but in many cases, it makes little difference. At any given moment, there are individuals with various durations of illness. As long as the distribution of illness durations stays roughly the same over time, there is no harm in assuming that recovery times are exponentially distributed. □

The ahistoricity property makes the exponential distribution much easier to work with than other distributions of inter-event times; hence, it is common to use it even in cases where memory ought to matter. This issue will be further explored in Chap. 3.

1.3.2 Probability Distributions of Sample Means

In many scientific experiments, we want to determine the characteristics of some population but are unable to study all the individuals. We want to know how tall adult Americans are, but we cannot measure all American adults. We want to determine how many people have HIV, but we can get records only for people who have volunteered to be tested. We want to determine how much oxygen a particular Olympic swimmer uses in 200 m training swims, but we cannot test all of her sessions.[18]

In practice, we must determine the mean $\bar{x}$ and standard deviation s for a *sample* and then approximate the mean μ and standard deviation σ of the *population* by

$$\mu \approx \bar{x}, \qquad \sigma \approx s.$$

However, conclusions drawn from samples only apply to the full population if the sample is *representative*, which means that its characteristics match the population to a desired degree of accuracy. This poses a difficulty, because no procedure can guarantee that a sample is a representative. As an alternative, scientists try to collect a *random* sample, in which each member of the population has an equal chance of being selected. There are two issues that must be faced when doing this. First, a truly random sample is not necessarily representative. We saw in Example 1.3.2 that random samples can sometimes be noticeably different from the underlying population. This issue can be quantified, as we will see in the remainder of this section. Second, it is not always possible to obtain a sample that is truly random. Medical tests, for example, are conducted by institutions using subjects who are from a specific geographic region. This poses a difficulty: Can a random sample drawn from a subpopulation be representative of the population at large? We will not address this second question.

Properties of Sample Means

Because sampling is such a common occurrence, it is worth summarizing the conceptual framework for discussing samples. A total of n individual measurements are drawn from a base population having some distribution type with mean μ and standard deviation σ. The mean of the sample of measurements is a random variable X, which has its own distribution. It would be helpful to be able to connect the

[17] See Problem 1.3.8.

[18] The "population" in this case consists of the different 200 m swims for the single athlete.

properties of that distribution with those of the underlying distribution from which the samples have been drawn.

This sounds like a difficult task, but we already have an example of a distribution of sample means. The outcome of the binomial distribution is defined as a *sum* of successes in n Bernoulli trials. If we divide the outcome values k by the total number of trials n, the resulting outcomes are means of n values drawn from the simple distribution of the Bernoulli trial. The properties of the binomial distribution can be restated as the properties of the sample means. The resulting relationship between the probability p for a Bernoulli trial and the binomial distribution properties generalizes to a theorem about all distributions of sample means, which we state here without proof.[19]

Theorem 1.3.3 (Mean and Standard Deviation for a Distribution of Sample Means)
Let X be the mean of n independent random variables, each drawn from a distribution having mean μ and standard deviation σ. Then the mean μ_X and standard deviation σ_X for the random variable X are given by

$$\mu_X = \mu \,, \qquad \sigma_X = \frac{\sigma}{\sqrt{n}} \,. \tag{1.3.8}$$

Theorem 1.3.3 allows us to calculate theoretical means and standard deviations for sample means, given only that we know the mean and standard deviation for the population from which the samples are drawn.

▶ **Example 1.3.5** Suppose we draw a sample of four individuals from the exponential distribution with parameter $\lambda = 0.5$. From Theorem 1.3.2, the mean and standard deviation for individual measurements are $\sigma = \mu = 1/\lambda = 2$. Given a sample size of $n = 4$, Theorem 1.3.3 tells us that the sample means have a mean of $\mu_X = 2$ and standard deviation $\sigma_X = 2/\sqrt{4} = 1$. Similarly, samples of sizes 16 and 64 have means $\mu_X = 2$, with standard deviations $\sigma_X = 0.5$ and $\sigma_X = 0.25$, respectively. Figure 1.3.1 illustrates the underlying distribution and the distributions of sample means for sample sizes 4, 16, and 64. □

Check Your Understanding 1.3.1:
A die roll is a uniform distribution with possible values 1–6. It has a mean of 3.5 and standard deviation $\sqrt{35/12}$. Determine the mean and standard deviation for the average of 5 die rolls.

Distribution Type for Sample Means

We cannot make a general statement about the shape of distributions of sample means. However, there is an important pattern discernible from Fig. 1.3.1.

▶ **Example 1.3.6** In Example 1.3.5, we saw that the standard deviation of a distribution of sample means decreases by half each time the sample size quadruples. The change in distribution shape is dramatic. The original distribution (Fig. 1.3.1a) is highly skewed, with very small values more common than values near the mean. With a small sample of $n = 4$ (Fig. 1.3.1b), very small values are no longer common. They still arise in the underlying distribution, but most often in conjunction with three other values that are not very small. The first bin, with means less than 0.25, occurs only when the sum of

[19] See any standard probability and statistics book.

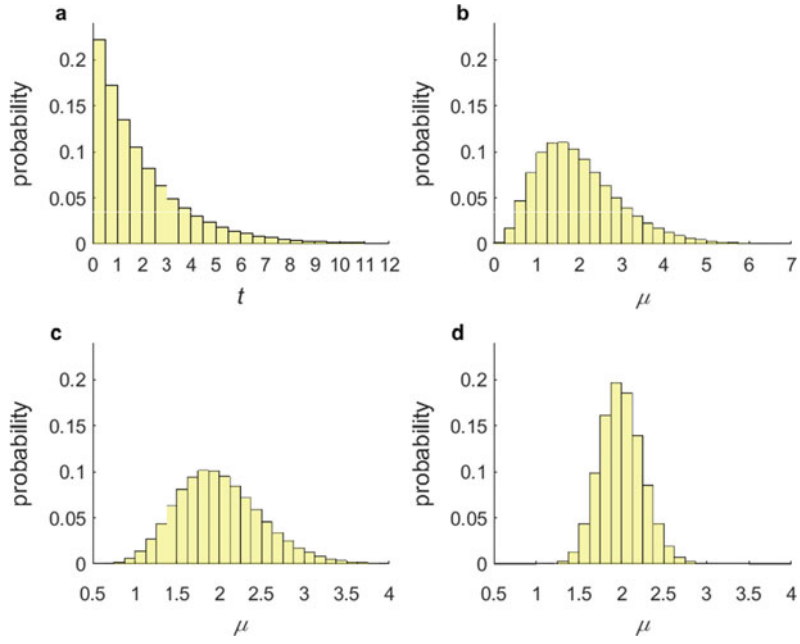


Fig. 1.3.1 Histograms for **a** the underlying distribution $E(t; 0.5)$ and the distributions of the means of samples of size **b** 4, **c** 16, and **d** 64

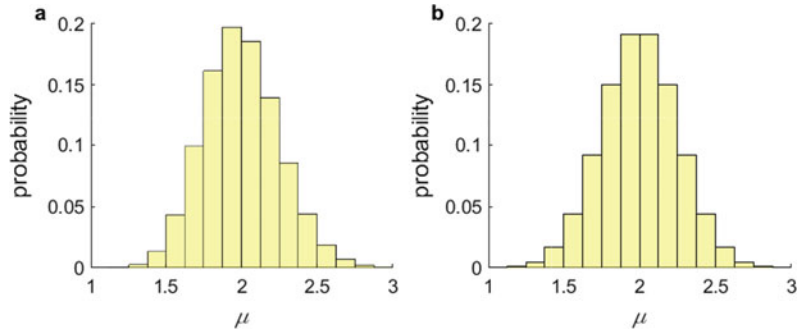


Fig. 1.3.2 A comparison of histograms: **a** the distribution of means of samples of size 64 from $E(t; 0.5)$ and **b** the normal distribution having the same mean $\mu = 2$ and standard deviation $\sigma = 0.25$

the four underlying values is less than 1, and this is rare. The distribution is still skewed; for example, a sample mean of 1.8 is much more common than a sample mean of 2.2, even though both are equidistant from the underlying distribution mean of 2.0. Proceeding through the panels, we see that larger samples produce distributions that are progressively less skewed, as well as narrower. □

The alert reader has perhaps noticed that the distribution for means of sample size 64 in Fig. 1.3.1d looks somewhat like the "bell-shaped curve" of a normal distribution. The similarity is striking when we compare directly with a normal distribution of matching mean $\mu = 2$ and standard deviation $\sigma = 0.5$, as shown in Fig. 1.3.2.

This observation is confirmed by a key theorem of probability, which we state here without proof.[20]

[20] See any standard probability and statistics book.

Theorem 1.3.4 (Central Limit Theorem)
Suppose X is a random variable obtained as a mean of n independent identically distributed random variables. In the limit as $n \to \infty$, the distribution of X approaches a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$, where μ and σ are the mean and standard deviation of the distribution from which the independent random variables are drawn.

Check Your Understanding Answers

1. $\mu_X = 3.5, \sigma_X = \sqrt{7/12}$

Problems

1.3.1* Like pea plants, tomato plants have some traits determined by just one pair of genes. These include the height (tall or dwarf) and leaf shape (potato leaf or cut leaf). A genetics experiment in 1931 was done with tall potato-leaf plants and dwarf cut-leaf plants. The second generation consisted of 926 tall cut-leaf plants, 293 dwarf cut-leaf plants, 288 tall potato-leaf plants, and 104 dwarf potato-leaf plants [16]. Determine the probabilities for each of these phenotypes. Based on the principles of Mendelian genetics, determine a genetic model that can account for these results. Support your choice by computing theoretical expected values for the probabilities.

1.3.2 Suppose the probability that a bird nest will be raided by a predator on any given day is 0.1. Assume that the fledglings survive if a nest is kept safe for 20 days. Determine the probability that the fledglings survive. Express this probability in terms of the binomial distribution.

1.3.3* Suppose a predator succeeds in 30% of its attacks on prey. How likely is it that it will need at least three tries to achieve its first success? What about five or more tries? State the probability of needing y or more tries in terms of the binomial distribution.

1.3.4 Suppose pairs of birds have a 25% chance of a successful nest. Given 10 pairs of birds, find the complete set of probabilities for the number of successful nests. Prepare a histogram and find the mean and standard deviation.

1.3.5 According to human genetics professor Daniel Geschwind in 2008, "Six out of the past 12 presidents [being left-handed] is statistically significant, and probably means something" [11, 17, 19].[21] Suppose the probability of left-handedness is 0.15.[22]

(a) Find the probability distribution for the number of left-handed people in a group of 12 randomly chosen people. Prepare a histogram and find the mean and standard deviation. How many standard deviations away from the mean is the outcome $X = 6$?
(b) The count of consecutive Presidents' left-handedness did not start at a random point in Presidential history. Geschwind could have started with any President, but he chose to start with a left-handed

[21] For a more scholarly discussion, see [15].

[22] This estimate is probably a little high, but not much. It is hard to know how many people are naturally left-handed, since some natural left-handers, such as the author's sister, were "trained" in school to be right-handed. Irrational prejudice against left-handers dates back many centuries; for example, the Latin word for "left-handed" is the original source of the English word "sinister."

Table 1.3.2 The frequency of N successes in 26,306 rolls of 12 dice, where a success means a roll of "5" or "6", from [18]

N	0	1	2	3	4	5	6	7	8	9	10	11	12
X	185	1,149	3,265	5,475	6,114	5,194	3,067	1,331	403	105	14	4	0

one. Hence, it seems clear that the first in the sequence shouldn't count. Repeat (a), but find the number of standard deviations away from the mean is the outcome $X = 5$ for a string of 11 randomly chosen people.

(c) To what extent do you think these calculations support Geschwind's view?[23]

1.3.6 The chi-square test in statistics was developed by Karl Pearson using data obtained by his colleague Walter Frank Raphael Weldon, who reported the results from 26,306 runs of an experiment consisting of rolls of a set of 12 dice [18].[24] The data is reproduced in Table 1.3.2.

(a) Determine the probability distribution for the number of successes in 12 die rolls, with "5" and "6" counted as successes. Prepare histograms of this distribution and Weldon's corresponding data. Do the dice appear to be fair?

(b) Use a stopwatch to estimate how long it takes to do 10 runs of the experiment (rolling 12 dice and recording the results as a list of tick marks). Estimate the time Weldon personally spent doing 19,300 experiment runs. Can you imagine a professional scientist rolling a set of dice 19,300 times for a scientific experiment?[25]

1.3.7* Suppose X is exponentially distributed with mean rate $\lambda = 4$. Find the probability that the next occurrence of the event occurs within 1 time unit.

1.3.8 Suppose X is exponentially distributed with mean rate λ.

(a) Calculate the probability that the next occurrence of the event requires more than one unit of time. Call that probability E.

(b) Calculate the probability that the next occurrence requires more than two units of time.

(c) Combine the answers of parts (a) and (b) to determine the probability that the clock reaches 2 time units after having already reached 1 time unit. Keep in mind that time unit one is only reached with probability E.

(d) Explain how the answers to parts (a) and (c) show that the probability of the next event occurring within 1 time unit from now does not depend on when the clock actually started.

1.3.9 Suppose X is exponentially distributed with mean rate λ.

(a) What is the probability that the value of X is greater than the mean of the distribution?

---

[23] Even if 5 of 11 is highly unusual, it does not mean that the result is significant. While any one coincidence discovered in a group of 12 people is unusual, there was no particular reason to look for the specific coincidence of left-handedness. The number of possible coincidences that could be discovered is probably quite large, so perhaps there is nothing unusual in discovering one. This point is most elegantly made by Tyler Vigen, a former law student and intelligence analyst, and the author of a book on spurious correlations [23]. The reader is encouraged to do a google search on "divorce rate in Maine and consumption of margarine" to find a graph on Vigen's web site of what is arguably the best example of a correlation devoid of significance.

[24] Why the dice were rolled 26,306 times, rather than some larger or smaller number, is lost to posterity. According to Weldon, 7,006 rolls were done by a clerk deemed "reliable and accurate," but Weldon did the other 19,300 rolls himself. There is no evidence that Weldon's experiments inspired the invention of the game "Yahtzee."

[25] Nowadays, a professional scientist would have his/her graduate student roll the dice 19,300 times. :-)

(b) What is the probability that the value of X is more than one standard deviation greater than the mean?
(c) What is the probability that the value of X is more than one standard deviation less than the mean?

1.3.10

(a) Prepare a histogram of the relative frequencies of data for the waiting times between firings of motor cortex neurons of an unstimulated monkey, recorded in the file WAITING.csv [8].[26] (Note that there are no headings in this file.) Superimpose the probability density function for the corresponding exponential distribution. Is the exponential distribution a reasonable model for this data set? What about the normal distribution?
(b) Repeat part (a) with the data for time intervals between successive pulses along a nerve fiber, recorded in the file NERVE.csv [8].

1.3.11* Consider samples drawn from an underlying exponential distribution $E(t; 0.2)$ and let X be the mean of a sample of size n.

(a) Determine the means and standard deviations of the random variable X for the cases where $n = 4$, $n = 16$, and $n = 64$.
(b) Draw 10,000 values of X for each of the three cases and plot histograms as in Fig. 1.3.1. Compare with the underlying distribution and with the corresponding normal distributions.
(c) Summarize the results of these experiments.

1.3.12 Approximately four million women gave birth in the United States in the year 2000, with single births, twins, and triplets occurring 98.51, 1.38, and 0.11% of the time, respectively, with a negligible probability of more than three births.

(a) Determine the mean and standard deviation for this probability distribution.
(b) Determine the means and standard deviations for the distribution of means for samples of size 100, 400, 2,500, and 10,000.
(c) Draw 10,000 values of X for each of the four distributions of means in (b) and plot histograms.
(d) Summarize the trend you see in the histograms.

1.3.13 Use Theorem 1.3.3 to derive the mean and variance formulas for the binomial distribution (1.3.2).

1.4 Basic Concepts of Modeling

After studying this section, you should be able to:

- Discuss the relationships between the real-world and mathematical models.
- Discuss the distinctions between mechanistic and empirical modeling.
- Discuss the concepts of parameterization, simulation, and characterization with mathematical models.
- Discuss the concepts of the narrow and broad view of mathematical models and the function of parameters in each view.

[26] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

Probably the best-known model in biology is the exponential growth model

$$P(t) = P_0 e^{kt}, \tag{1.4.1}$$

where $P(t)$ is the population of some group at time t, P_0 is the initial population, and $k > 0$ is a parameter called the *rate constant* or *proportionality constant*.[27]

From examining this formula, we can tease out some details about models and modeling.

1. Noninteger values are common; for example, with $P_0 = 1$ and $k = 1$, the population at time 1 is $P = e \approx 2.718$.
2. If you keep using larger and larger values of t, there is no limit to how large P can be.
3. The model can be rewritten in a different form by taking the logarithm of both sides to get $\ln P = \ln P_0 + kt$. This means that the graph of $\ln P$ versus t is a straight line and suggests a way to identify k from data.

Each of these statements illustrates a typical characteristic of mathematical models. First, models often give results that include all possible numbers in a range on a number line even when the quantities they represent can only take discrete values. While a value of 1079.6 is not technically correct for an integer population, there is no problem interpreting it as approximately 1080. Second, models can sometimes have qualitative properties that do not fit the intended scenario. Third, models can sometimes be rearranged or analyzed in ways that allow for convenient comparison with data.

Clearly, the exponential growth model is not "correct" or "true," but this does not mean that it has no value. Data from population growth experiments can adhere closely to the prediction of exponential growth until resource limits start to matter. The extreme example of this is the amount of data that can be put on a state-of-the-art computer chip, a quantity that has been growing exponentially since the 1960s and is only just starting to show signs of deviation [25].

None of our models are going to be "correct". They will be based on data that is not perfect or assumptions that are clearly oversimplifications. The models might have value by making general predictions about things like the impact of public health policies on the course of an epidemic. For example, a model can be used to address questions such as "How much of a difference does extensive contact tracing make?" Contact tracing requires significant infrastructure, so we'd like to have some idea of whether it will matter before we decide to do it. Models may not give us exact answers to questions such as these, but they are the best way we have to predict outcomes when we can't do real experiments.

Based on these considerations, we can make a tentative definition of the term *mathematical model*:

Definition 1.4.1

A **Mathematical model** is a self-contained collection of one or more variables together with a set of rules (usually formulas and equations) that prescribe the values of those variables. Models serve as an approximate quantitative description of some actual or hypothetical real-world scenario. They are created in the hope that the behavior they predict will capture enough of the features of that scenario to be useful.

[27] You may be familiar with this model using different symbols for some of the quantities, such as $y = Ae^{kt}$. The symbols *represent* the quantities, so the model is the same no matter what symbols are used. This will be dealt with in detail in Sect. 3.6.

Note the tentative language of the final sentence. The emphasis is on the uncertainty in the connection between the mathematical model and the real-world setting to which it is applied. This emphasis has two important consequences.

1. It allows us to ignore any minor deviations from reality that we see in the results.

A model is not "wrong" just because the results it gives fail to exactly match any actual data. A prediction of 1080 where the real data gives 1066 is pretty good in most circumstances. On the other hand, we should be bothered by a population of -5 or by a population fraction that is larger than 1, because these are qualitatively wrong.

2. It tells us that we must always question both the results a model gives and our interpretation of those results.

▶ **Example 1.4.1** The Lotka–Volterra model combines linear growth and predation models to describe the quantitative relationship between populations of predators and populations of prey (see [24], for example). It was developed to explain changes in Mediterranean fish populations that occurred during and after World War I, which it succeeded in doing. Subsequently, it has been used in some differential equations textbooks to "prove" that hunting coyotes (to keep them from eating farm animals) increases the population of the coyotes' natural prey *without decreasing the coyote population*. This claim is unsupported by any biological data and is obviously incorrect.[28] Nevertheless, most treatments of the Lotka–Volterra model in books and other references fail to mention this critical flaw. □

The Lotka–Volterra model seems at first thought to be appropriate for the coyote–rabbit setting because that setting involves a predator–prey system. However, it does not follow that just *any* predator–prey model will be appropriate. The correct approach is to think of the Lotka–Volterra model as only one *possible* model.[29] Instead of accepting a ridiculous result, such as the impossibility of eliminating predators, we should conclude that the model is inappropriate.

Mathematics has the benefit of certainty, as exemplified by proofs of theorems. This is of great value to mathematicians because it eliminates time and energy spent arguing about facts. Once a mathematical claim has been proven, everyone is obligated to accept it. However, this certainty only applies to mathematical claims about the *model*, not to mathematical claims about the *real-world setting* that inspired the model.

Claims about real-world behavior are only as good as the assumptions used to build the model. They cannot be tested mathematically, but must be addressed in other ways. The focus of the mathematical skills changes in modeling from proof and solution to characterization (understanding the broad range of possible behaviors) and simulation (visualizing the behavior in specific examples). The thinking you need for mathematical modeling is therefore somewhat different from the thinking associated with mathematics per se and more like the thinking associated with theoretical science, as illustrated in Example 1.4.1.

The value of a model depends on the setting to which it is applied and the questions it is used to address.

While we cannot hope our models will be "correct" for a real-world setting, we can aim to make them *valid*, in the sense of "giving meaningful results under a given set of real-world circumstances."

[28] See Problem 1.4.1.

[29] A model that exhibits oscillatory behavior in an appropriate way is the subject of Project 6D.

There are almost certainly quantitative differences between model results and real-world empirical results, and there may be important qualitative differences as well. If the differences are small enough in the given setting, we judge the model to be valid and use it with confidence. The model may work for somewhat different settings as well, but we must worry about its validity in the new setting. Where the validation is not satisfactory, we must revise the model and try again.

▶ **Example 1.4.2** The exponential decay model

$$y = y_0 e^{-kt}, \qquad k, y_0 > 0 \tag{1.4.2}$$

is valid for a macroscopic amount of a single radioactive substance. The model can also be applied to other settings where a quantity is decreasing to a fixed value, such as the clearance of medication from the bloodstream of an animal. The ultimate value of the quantity of interest could be nonzero, in which case we can interpret y as the difference between the current value and the ultimate value. Whatever the context, we have to be careful that the model is appropriate. In lead poisoning, a significant portion of the lead is deposited in the bones, so a more sophisticated model is needed to incorporate this physiological mechanism.[30] Time-release medications are slow to absorb from the digestive system and require a more sophisticated model as well. □

1.4.1 Mechanistic and Empirical Modeling

Mathematical models can be classified according to the method used to obtain them.

Definition 1.4.2

A **mechanistic model** is a collection of one or more variables, together with a self-contained set of rules that prescribe the values of those variables according to assumptions about the scientific principles that underlie the phenomena being modeled.

Definition 1.4.3

An **empirical model** is a mathematical model based on the examination of numerical data.

The distinction between the two types of models is sharpened by separating the "approximate quantitative description" in Definition 1.4.1 into two distinct processes: that of approximation and that of quantitative description. To clarify this point, it is helpful to introduce the idea of the *conceptual model*.

Definition 1.4.4

A **conceptual model** is an approximation of a real world scenario that serves as a verbal description of a mathematical model.

[30] See Sect. 3.7.

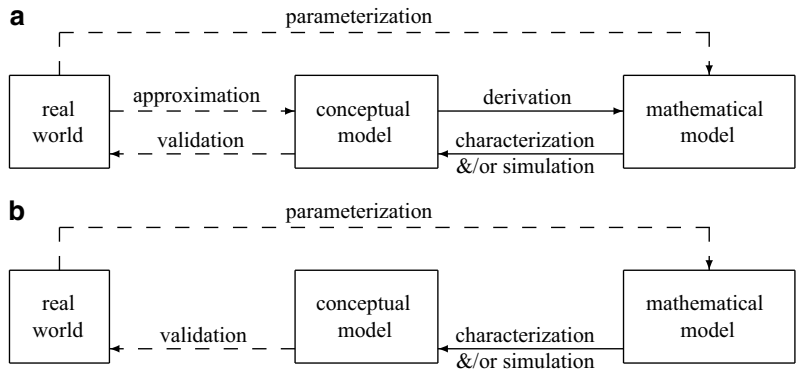


Fig. 1.4.1 Relationships between the real world, a conceptual model, and the corresponding mathematical model. *Solid arrows* indicate processes amenable to mathematical certainty, while *dashed arrows* indicate processes that must be viewed with scientific skepticism. **a** Mechanistic modeling. **b** Empirical modeling

Identifying the underlying conceptual model is necessary to understand biological literature that uses mathematics; however, these are not always made explicit in the presentation of a mathematical model.

Figure 1.4.1 illustrates the process flows of mechanistic and empirical modeling. These flows are not unidirectional. Each of the components feeds into the others, but it is important to note the lack of a direct connection from the mathematical model to the real world. It is this feature that distinguishes mathematical modeling from the "applications" of mathematics that appear in most textbooks.

Because of the lack of a clear direction in the flow, we describe these processes in alphabetical order before discussing some key issues in mathematical modeling.

Approximation: An intentional process of choosing which features to include in models, analogous to drawing a political cartoon.
Characterization: Obtaining general results about a model, using an explicit solution formula, graphical methods, or approximation. Characterization uses techniques of calculus as well as advanced techniques discussed in Chaps. 4–6.
Derivation: Constructing a mathematical model from a verbal description of assumptions and simplifying the model prior to analysis. Chapter 3 contains many examples of model derivation and simplification.
Model selection: Choosing a mathematical model from multiple options. In mechanistic modeling, construction of a conceptual model constitutes model selection. In empirical modeling, selection of a model is usually best done with the aid of the Akaike information criterion, a method for quantifying the statistical support a data set gives to a model. This topic is addressed in Sect. 2.4.
Parameterization: Using data to obtain values for the parameters in a model. This topic is addressed in Chap. 2.
Simulation: Using mathematics and computation to visualize model behavior for a given set of parameters. Chapter 3 includes simulation techniques for ordinary differential equation models.
Validation: Determining whether a model reproduces real-world results well enough to be useful. The criteria for validation depend on the purpose of the model.

1.4.2 Aims of Mathematical Modeling

Mathematical models can be used for different purposes, and the aim of the model plays a large role in determining the type of analysis and the criteria for validation. Sometimes the goal of modeling is

to predict the results of hypothetical experiments, as in the ATLSS simulation that models populations of animal and plant species in the Florida Everglades [22]. This model needs specific values for many parameters, such as mean daily temperatures and the average litter size and survival probability for Florida panther cubs. The parameters are estimated for the model because the goal is to predict populations in a hypothetical experiment for a real scenario. Given this goal, the criteria for model validity are quantitative. The model is valid if the results it predicts for experiments are within an acceptable tolerance of the actual experiment results. Demonstrating validity of a model used for quantitative prediction can be difficult. In a laboratory setting, the model simulation can be designed to match a specific experiment and the results can be directly compared. But we cannot conduct designed experiments for the Everglades. Instead, we look for historical events that can be thought of as experiments, in which case we can match the simulation to the historical event. If we know the effect of a historical housing development on the panther population, we can check to see that our model is quantitatively accurate for that specific case. If so, then we have evidence that our model will correctly predict the effect of a similar hypothetical event.

Mathematical models can also be used to address broad questions, such as that of how the equilibrium fraction of infectious individuals for an endemic disease like the common cold depends on the infectiousness of the disease. These investigations require mathematical characterization rather than numerical simulation. Model validation involves trying to confirm that the model behavior is qualitatively consistent with the behavior of the real biological system we are trying to model. Of course, we must specify what we mean by "consistent with the real behavior." For example, a model whose purpose is to study extinction risk for endangered species would need to be checked to ensure that it is actually capable of predicting extinction under some set of circumstances.

1.4.3 The Narrow and Broad Views of Mathematical Models

In any particular instance of a mathematical model, we have one or more dependent variables and one or more independent variables,[31] and a set of given values are assigned to the parameters. The focus of a simulation is on determining how the dependent variables depend on the independent variables. This is the *narrow view* of mathematical models.

In contrast, there is a *broad view* of mathematical models, in which the objective is to understand the effect of the parameter values on the model outcomes. Here is where we ask questions that we hope will be useful in interpreting the scenario that inspired the model rather than simply calculating results. In the broad view, we can study the effects of changes in the parameter values, asking questions such as "If two diseases have different incubation periods, how does that difference affect the outcomes of the model?" The outcomes can be whatever we are most interested in: for example, the maximum size of the infectious class, the day on which that occurs, and/or the total number of people who get the disease during the outbreak. Such information could be used to estimate the danger of running out of hospital space and the total number of deaths, for example.

The function concept is helpful in understanding how mathematical models work. In the narrow view, the dependent variables are (usually) functions of time. In the broad view, the outcomes are functions of the parameter values (Fig. 1.4.2).

Both the narrow view and broad view functions differ in an important way from the common functions that appear in most mathematics books. Functions in mathematics books are nearly always defined by explicit formulas, such as the formula for exponential growth (1.4.1). In mathematical modeling, functions are often defined in much more subtle ways; for example, as the solution of a mathematical problem rather than as a mathematical formula. Conceptually, however, the function

[31] Many models in biology are *dynamical systems*, in which time is the independent variable.

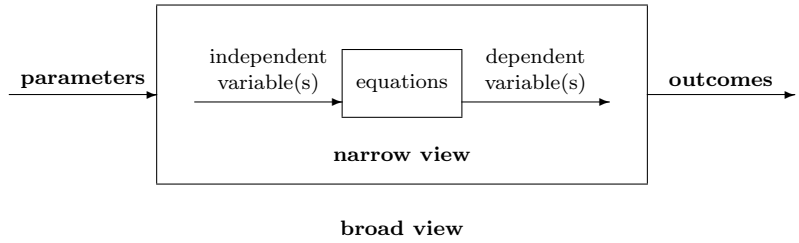


Fig. 1.4.2 Narrow and broad views of mathematical models

idea is the same regardless of whether the function is computed from a formula or as the result of a numerical procedure. It is best to think of functions in terms of their graphs rather than their formulas.

▶ **Example 1.4.3** Consider the family $y = \sin kt$, which is sometimes used as an empirical model for periodic data. When we choose a specific value for k and plot y as a function of t, we are working in the narrow view. Without choosing k, we can calculate the period to be the smallest time T such that $y(t + T) = y(t)$. Since the sine function repeats as the angle increases by 2π, the period is when $kT = 2\pi$, or $T = 2\pi/k$. If we plot T as a function of k, we are working in the broad view. □

As in Example 1.4.3, parameters function as constants in some aspects of model analysis and as variables in other aspects, corresponding to the narrow and broad views, respectively. This can be very confusing. Generally, we are working with the narrow view for simulations and the broad view for characterization. Both are important. We can make use of the full power of computers for simulations, but we can address deeper questions when we retain the broad view.

1.4.4 Accuracy, Precision, and Interpretation of Results

Most people make little distinction in ordinary language between the terms "accuracy" and "precision," but these terms have very distinct meanings in science. *Accuracy* is the extent to which results are correct, while *precision* is the extent to which results are reproducible. Precision is easier to measure than accuracy, but of course, it is accuracy that we really need. One cannot be confident of accuracy in the absence of precision, but results can be precise and yet inaccurate. Precision is limited in most areas of biology. Even where careful measurements are possible, results are not very reproducible. Have your blood pressure taken on five consecutive days, and you will see the point.

The lack of precision in most biological data has strong implications for how we interpret mathematical results. Computers give very precise results—divide 1.0 by 3 and you will not get 0.33, but 0.333333333333. This is alright if the numerator is certain to be very close to 1.0, but in biology, it could be that you measured 1.0 when the "correct" value is 0.9. If the data is off by 10%, then the additional digits beyond the second one are surely meaningless. Mathematics, of course, offers the possibility of infinite precision. In terms of biology, this does more harm than good. Apply infinitely precise methods to crude results and you get results that are infinitely precise in appearance without being reproducible. It is easy to take this apparent precision more seriously than it deserves. This is what I call the "measure it with your hand, mark it with a pencil, cut it with a laser" fallacy. The risk of this fallacy must be kept firmly in mind whenever we interpret results obtained from mathematical modeling applied to crude data.

Problems

1.4.1 Suppose we want to construct a realistic model for a predator–prey system. This model should allow for a variety of realistic results; in particular, it should predict three possible long-term results:

1. The predator and prey can coexist.
2. The prey can survive while the predator becomes locally extinct.
3. Both species can become locally extinct.

We look in a mathematical biology book and find the Lotka–Volterra model:

$$\frac{dx}{dt} = rx - qxy,$$

$$\frac{dy}{dt} = cqxy - my,$$

where x and y are the biomasses of the prey and predator, respectively,[32] r is the growth rate of the prey, m is the death rate of the predator, q measures the extent of predation, and c is a conversion factor for prey biomass into predator biomass. The model is presented as a predator–prey model, but we recognize the need to check that a model is appropriate for the setting we have in mind.

(a) Suppose the prey and predator populations stabilize to fixed biomasses $X \geq 0$ and $Y \geq 0$. If the biomasses at some time are $x = X$ and $y = Y$, then there should be no further change. This means that the fixed biomasses must make the right-hand sides of the differential equations be 0. Use this idea to find all possible pairs X and Y. Is this model capable of predicting all three possible long-term results? If not, which is missing?
(b) To help see what is wrong with the model,[33] write down the prey equation for the special case where the predator is absent. What does the model predict will happen?

1.4.2 Find an instance of a mathematical model in a biology book or research paper. Describe:

(a) The mathematical model itself,
(b) The conceptual model that corresponds to the mathematical model, and
(c) Features of the real-world setting that do not appear in the conceptual model.

1.4.3* Some genetic traits are determined by a single gene having two variants (alleles), with one (**A**) dominating the other (**a**). This means that individuals who have two dominant alleles (**AA**) and individuals who have one of each type (**Aa**) both exhibit the physical characteristics (phenotype) of the dominant trait, while the recessive phenotype is only found among individuals who have two recessive alleles (**aa**). It is sometimes helpful in genetics to model inheritance as a two-step process: first, all of the parents' genes are assembled into a gene pool; then, pairs of genes are randomly withdrawn from the gene pool for individuals in the next generation.

(a) Suppose q is the fraction of recessive genes in the gene pool. Based on the two-step conceptual model, what will be the fraction of individuals in the next generation who exhibit the recessive trait? What will be the fraction of individuals who have one dominant allele and one recessive

[32] Most descriptions of predator–prey models interpret the variables as the numbers of individuals, but the models are more realistic if the variables are viewed as being the total biomasses of the individuals.

[33] The point here is that using the Lotka–Volterra model to demonstrate that something can't happen in the real world is a logical fallacy when the model itself contains the assumption that the thing can't happen.

allele? What will be the fraction of individuals who do not have the recessive allele? (These results comprise the *Hardy–Weinberg principle*.)

(b) About 13% of the people of Scotland have red hair. Assuming that red hair is caused by a single recessive gene pair, what does the Hardy–Weinberg principle predict for the fraction of the recessive trait in the gene pool and the fraction of the population who do not have the recessive allele?

(c) Demographers estimate that 60% of the people of Scotland do not have the recessive allele for red hair. What flaws in the conceptual model might account for the difference between this estimate and the estimate you obtained from the Hardy–Weinberg principle?

1.4.4 The model $y = y_0 e^{-kt}$, with $k > 0$, is often used to model radioactive decay, where y is the amount of radioactive material remaining and y_0 is the initial amount of the material. This model is derived from a conceptual model in which the decay rate is k times the quantity of material. We can get some sense of what this means even without calculus. Consider the specific instance $y = e^{-2t}$. The average rate of decay over the interval $t_1 < t < t_1 + h$ is

$$r_h(t_1) = \frac{y(t_1 + h) - y(t_1)}{h}.$$

For the intervals $0 < t < 0.1$, $0.1 < t < 0.2$, and so on up to $0.9 < t < 1.0$, calculate the average rate of decay and compare it to the average of the quantities of radioactive material at the beginning and end of the time interval. Explain why the results are consistent with the conceptual model as described here.

1.4.5(a) Describe the qualitative predictions made by the exponential growth model $y = y_0 e^{kt}$. In particular, show that

$$G(t) = \frac{y(t + 1)}{y(t)}$$

does not actually depend on t.

(b) Describe an experiment that tests the prediction of part (a).

(c) Describe a physical setting in which this model for population growth is clearly not appropriate.

(d) Describe a physical setting in which this model for population growth might be appropriate.

1.4.6 Of Problems 1.4.4 and 1.4.5, one works primarily with the narrow view of a model and the other primarily with the broad view. Match these descriptions with the problems, explaining why that view is the focus of the problem.

1.4.7 Suppose individuals of group X and individuals of group Y interact randomly. If x and y are the numbers of individuals in the respective groups, it is reasonable to expect each member of group Y to have kx interactions with members of group X, where $k > 0$ is a parameter. (This says that doubling the membership of group X should double the contact rate with group X for a member of group Y.)

(a) Use the information about contact rates for individuals in group Y to find a model for the overall rate at which members of the two groups interact.

(b) Suppose p is the (fixed) fraction of encounters between individuals of the two groups that results in some particular event occurring between the individuals. Use this assumption to create a model for the rate R at which the events occur in the population.

(c) If the model of part (b) is used for the rate of infection of human populations with some communicable disease, what do the groups X and Y represent?

(d) The model of part (b) was used successfully to model an influenza outbreak in a small boarding school in rural England. Why do you think the model worked well in this case?

(e) The Centers for Disease Control in Atlanta did not use the model of part (b) to make predictions about the spread of the H1N1 virus in the United States in 2009. Explain why the model would not have been appropriate in this case.
(f) Describe some real-world settings other than epidemiology that could conceivably use this interaction model (Hint: This model finds common usage in chemistry and ecology as well as epidemiology.) How accurate do you expect the model to be in these different settings?

1.5 Case Study: An Agent-Based Epidemic Model

Mathematical models[34] often have continuous algebraic variables representing real discrete quantities like populations. They are often deterministic, which gives the illusion of short-term predictability, whereas real-world settings have a lot of messy randomness. These unrealistic features of standard models help make them useful by keeping them simple; however, they can give a misleading impression of the biological scenario. For this reason, it is helpful to gain some experience with a more intuitive model. *Agent-based models* (ABM) offer a nice introduction to modeling because of their intuitive nature [7, 20] and because they can be implemented as an activity, thereby providing students with valuable direct experience [12]. Observing the effect of randomness helps students obtain a healthy skepticism that will keep them from taking the results of their deterministic models literally.

1.5.1 Model Description and Physical Simulation

Definition 1.5.1

An **agent-based model** (also called an **individual-based model**) consists of a database of *individuals*, each identified by one or more *attributes* that can change over time, and a set of *rules* that update the attributes of each individual at each time step.

Agent-based models are usually implemented using simulations written with software such as Matlab or NetLogo. However, many can also be implemented as physical activities with actual people as the "individuals" in the model or with a table-top simulation in which one modeler manages all the "individuals."

As an example, we consider a model based on a physical activity created by G. Ledder and M. Homp to teach basic principles of epidemiology to a group of 5th graders and then adapted for use in a general education mathematics class [10, 14].

We assume a fixed population of N individuals, each of whom has a single attribute that indicates their current epidemiological status. There are four possible states, through which individuals move linearly:

1. "Healthy" individuals have not yet been infected;
2. "Pre-symptomatic" individuals have been infected and can transmit the disease, but do not show symptoms;
3. "Sick" individuals have been infected, can transmit the disease, and do show symptoms;
4. "Recovered" individuals are no longer sick, cannot transmit the disease, and cannot be reinfected.

We will sometimes refer to these states using the letters H, P, S, and R.[35]

[34] This section is adapted from [14].

[35] These letters are chosen to help students' intuition before we introduce formal models. In standard epidemiology models, the symbol S is used to represent susceptible individuals, but in the current context, it makes more sense to use S for sick individuals.

The state attribute can be identified in various ways. When enacting the model as a physical activity, the participants carry a set of four colored status cards—green for healthy, yellow for presymptomatic, red for sick, and blue for recovered—and place the appropriate card on top of their stack to show their status. In a computer implementation, the status can be a number: 1 for healthy, 2 for presymptomatic, and so on. The rules (below) that govern the changes in states have been carefully balanced; they work best if there are at least 16 individuals in the population, with a starting point of two presymptomatic individuals in a small initial population or 1–2% total infectious individuals ($P + S$) in a large one, and the rest initially healthy.

Once the initial states have been assigned to the individuals, the simulation consists of consecutive time steps, each divided into phases:

1. Individuals in the population are randomly assigned to pairs. This can be done using a fully random approach, such as dealing out a deck of cards (each representing a member of the population) in pairs or running a web-based app that creates pairs from a list, or using a partially random 'speed dating' structure that is more convenient for physical enactments because it takes less time [10]. In a fully automated implementation, only the healthy individuals need to be assigned a partner.
2. Healthy individuals who are paired with a presymptomatic or sick individual become infected with a fixed probability p; in a physical enactment, this is easily done with $p = 5/6$ by rolling a die and equating a die roll of one with avoiding the infection. Those who do become infected advance from healthy to presymptomatic.
3. After all pairs have been checked for disease spread, any individuals who were presymptomatic at the beginning of the time step become sick, while those who were sick at the beginning of the time step become recovered.
4. The numbers of individuals in each status category are recorded for subsequent analysis.

The simulation proceeds from one day to the next, each day starting with new partner assignments and continuing with status updates. Eventually, the simulation ends when all individuals are either healthy or recovered. Once there are no presymptomatic or sick individuals remaining, there can be no further status changes.

This simple agent-based model captures many of the features of real disease spread. Individuals interact randomly and possible transmission encounters may or may not result in actual transmission. Other features are not so realistic. In reality, the amount of time spent in a presymptomatic or sick state can vary from one person to another. Most diseases have an incubation period of noticeable duration, necessitating a latent stage between the healthy and presymptomatic stages. In real disease settings, individuals have large numbers of daily contacts, each with a low probability of transmission. The agent-based model obtains realistic results by reversing this pattern, with only one daily contact and a high probability of transmission.

The simplicity of the agent-based model also highlights the critical role assumptions play in modeling a phenomenon. In this context, the need for and outcomes of changing the model's assumptions can easily be understood, demonstrating the importance of consulting with experts to ensure that a model incorporates critical features and exhibits necessary behaviors. (Some of these assumptions will be considered in the exercises.)

1.5.2 Matlab Implementation

While physical simulations are great for building intuition, they are very inefficient for generating data. Implementations as computer programs are much more efficient. Three Matlab programs have been provided for this purpose:

1. *hpsr.m* contains the function *hpsr*, which runs one simulation using the standard HPSR rules.
2. *HPSR_onesim.m* is a script that uses the function *hpsr* to plot simulation results and report two key outcomes: the maximum infected fraction and the final healthy fraction.
3. *HPSR_avg.m* is a script that uses the function *hpsr* to compute means and standard deviations of the key outcomes and prepare histograms.

At its most basic level, individuals in the virtual population are classified as one of (H)ealthy, (I)nfectious, or (R)ecovered. Table 1.5.1 shows some data from one simulation of a virtual experiment that began with 50 presymptomatic individuals and no sick or recovered individuals out of a starting population of 10,000. The same data is plotted in Fig. 1.5.1. Like real data, the simulation data does not appear to fall on a perfectly smooth curve. Nor is it quantitatively reproducible. A different run of the simulation will yield a different final count of healthy individuals.

The simulation data provides clues that will facilitate analysis, but seeing these clues requires creative display of the data. Figure 1.5.2 displays the infectious class size as the logarithm of I rather

Table 1.5.1 Counts of healthy, infectious, and recovered individuals from an HPSR disease simulation starting with 50 presymptomatics out of a population of 10,000

Day	0	1	2	3	4	5	6	7	8	9	10	11
H	9950	9903	9821	9718	9563	9312	8877	8290	7420	6307	5059	3884
I	50	77	129	185	258	406	686	1022	1457	1983	2361	2423
R	0	20	50	97	179	282	437	688	1123	1710	2580	3693
Day	12	13	14	15	16	17	18	19	20	21	22	23
H	2943	2324	1969	1783	1676	1624	1598	1587	1582	1579	1579	1579
I	2116	1560	974	541	293	159	78	37	16	8	3	0
R	4941	6116	7057	7676	8031	8217	8324	8376	8402	8413	8418	8421

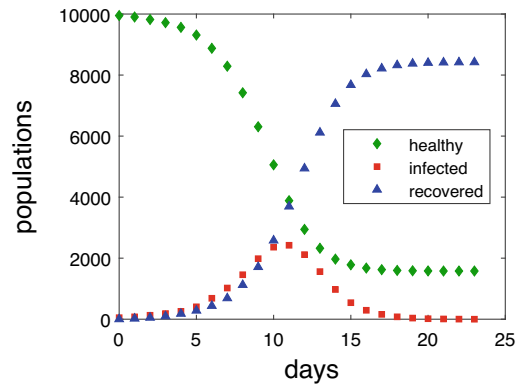


Fig. 1.5.1 Epidemic history for an HPSR simulation, from Table 1.5.1

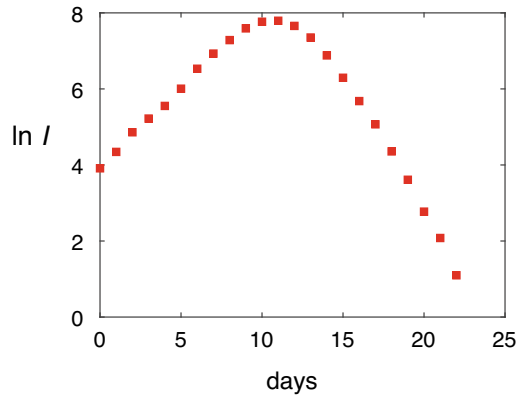


Fig. 1.5.2 The total infectious population for an HPSR simulation, from Table 1.5.1

than I itself. The plot strongly suggests that there is an underlying deterministic signal containing an initial phase with exponential growth followed by a period of waning infection.

Problems

Instructions for the HPSR physical simulation can be found at [10].

1.5.1 Run the HPSR simulation three times using either a physical simulation, with people playing the roles of the individuals, or a tabletop simulation. In each case, plot graphs of the daily class counts. Also record the number and identities of individuals who remained healthy throughout the simulation. Are those who don't get sick different in any meaningful way from those who do? In a real situation where an illness is spreading among a population, might there be a difference between those individuals who don't get sick and those who do? Discuss.

1.5.2 The agent-based HPSR model uses four different states, each marked by a different color in the physical simulation, to describe the illness progression.

(a) Using the rules as described, explain which, if any, of the color distinctions are needed and which are not.
(b) Now suppose the rules for the physical simulation are changed so that half of the individuals who are sick choose to isolate. How would you implement this change in the physical implementation of the model?
(c) How might the rules change if some of the individuals are vaccinated? How might this impact the meaning or number of the colored status cards?
(d) How might the rules change if individuals could contract the same illness more than once? How might this impact the meaning or number of the colored status cards?

1.5.3 The HPSR model assumes the duration of each phase of the illness is a single day. In reality, most illnesses last for at least several days. Visit the website [5] for information about the H1N1 virus, often referred to as "swine flu." Then develop a set of status cards and rules for an agent-based model of H1N1. Assume that the model would incorporate isolation of sick people. Would you still continue to use the same four states or would you need more? How many status cards would you use for each state? How many would you need for the full simulation? Would you need to incorporate additional

die rolls into the model? Fully justify any assumptions made about the states of the illness and the duration of each state.

1.5.4 Run the agent-based model program *HPSR_onesim.m* three times using parameter values that match your physical simulation in Problem 1.5.1. Compare the results. Do they give convincing evidence that the computer simulation and physical simulation actually implement the same model?

1.5.5 How does the final number of healthy individuals depend on the transmission probability parameter b? To address this question, run the program *HPSR_onesim.m* using values $0, 0.1, 0.2, \ldots, 1.0$ for b, with 50 presymptomatic individuals and the rest healthy out of a total population of 10,000. Plot the final number of healthy individuals versus b. Discuss the results.

1.5.6 [p] Address the same question as in Problem 1.5.5, except automate the calculations by writing a program that runs through each value of b in a loop and saves the final number of healthy individuals as part of a list. You can start with *HPSR_onesim.m*, removing any unnecessary statements and replacing the single function call with a loop that runs through a list of b values.

1.5.7 Run the agent-based model program *HPSR_avg.m* using parameter values that match your physical simulation in Exercise 1.5.1. Compare the results. Is this a good way to check that the computer simulation and physical simulation are actually implementing the same model?

1.5.8 [p] Use the programs *hpsr.m* and *HPSR_onesim.m* as templates to create an agent-based model for sharing of a secret. The population should be divided into two classes, (D)on't know and (K)now, with an initial population of 100 including 2 who know the secret. In each time step, the program needs to draw a random number to identify a partner value for each D (You need not worry about whether individual partners are called more than once.), determine how many of these partners are K's, draw a number for sharing probability, calculate the number of sharings, and update the numbers of D and K. You will need to change the condition that marks the end of the simulation. Most of the changes will be in *secret.m*, which contains the simulation of the model. The driver *Secret_onesim.m* will need only minimal changes to adapt to the changes in the simulation program.

(a) Use your programs to plot some simulation runs. A sharing probability corresponding to a die roll of 2–6 (as in *hpsr*) works well, as does a lower probability corresponding to a die roll of 3–6.
(b) Discuss the results.
(c) Explain why there are no classes analogous to the P and R classes in the disease model.

1.6 Projects

Projects A–D of this chapter ask students to modify the simple HPSR agent-based model of Sect. 1.5. In each, it is best to start by using the physical simulation to make sure that your model changes make sense. Then modify the Matlab program suite to collect data. Project E requires calculus and programming to develop the theory of Erlang distributions, which can be used for greater realism in epidemic modeling than the more commonly used exponential distribution.

Project 1A: Isolation

Use a modified version of the agent-based model to study the effect that isolation of the sick has on the course of the epidemic. Note that you will need a parameter to represent the probability that an individual isolates when sick. The effect of isolation will depend on the value of this parameter.

Project 1B: Isolation versus Vaccination

Use modified versions of the agent-based model to compare the relative benefits of isolation and vaccination in the absence of the other.

Project 1C: Longer Illness Duration

Suppose patients with the disease are sick for two days instead of one. Modify the model to account for that possibility. Try keeping the transmission probability the same and also adjusting it downward by some amount. Determine what transmission probability yields results most like those of the original model with the larger transmission probability and one day of sick time. Hint: You can break the sick class up into two subgroups.

Project 1D: Variable Illness Duration

As an extension of Project 1C, suppose a fraction p of patients with the disease are sick for two days, while the remainder is sick for just one day. Design experiments with this model and describe and explain the results. Adding isolation, as in Project 1A, will make this project even better.

Project 1E: Disease Transitions

Recovery from a disease is an example of a *transition*; that is, a process by which an individual changes from one state to another at some random time. It is common to use exponential functions to model transitions, but this is problematic. In this project, we develop a more sophisticated alternative called the Erlang distribution family. The Erlang distribution E_k represents a process that consists of a sequence of k identical exponentially distributed stages. The transition time is constructed by summing k random variables from a corresponding exponential distribution.

You will need some basic formulas from statistics, presented here without derivation.

1. Suppose a random variable X has outcomes x_k with probabilities p_k. The *expected value* of a function $g(X)$ is
$$Eg = \sum_k p_k \, g(x_k). \tag{1.6.1}$$
Note that the expected value of $g(X) = X$ is the mean of the distribution.
2. The expected value of a function of a continuous nonnegative random variable X is determined from
$$Eg = \int_0^\infty g(x) F'(x) \, dx, \tag{1.6.2}$$
where F' is the probability density function defined from the cumulative distribution function
$$F(x) = \mathrm{P}\,[X \le x]\,.$$
3. The standard deviation σ of a random variable X can be calculated as the square root of the variance, which is given as
$$\sigma^2 = EX^2 - (EX)^2. \tag{1.6.3}$$

We begin by identifying the need for an alternative to the exponential distribution. Then we use calculus to construct the survival function for E_2. Finally, we collect and examine data from several E_k distributions.

(a) What is there about the graph in Fig. 1.2.1b that is clearly wrong for a graph showing the number of people who are still sick after t days from a disease with a mean duration of 10 days? Discuss.

(b) The survival function for the Erlang distribution E_k is

$$S_k(t;\mu) = \mathrm{P}[T > t \mid ET = \mu] = 1 - \mathrm{P}[T \le t \mid ET = \mu], \tag{1.6.4}$$

where the probability notation means that we want the probability of a randomly selected time being larger (or smaller in the second case) than t, given total mean time μ. For S_2, we want the mean time for the sum of two exponentially distributed values to be μ. What value do we choose for the parameter λ in the associated exponential distribution?

(c) As a first step in calculating S_2, we need to express the probability of $T < t$ in terms of a sequence of two events. First, a random variable T_1 must be determined, and its value t_1 must be less than t; second, a random variable T_2 must be chosen, and its value must be small enough so that the total is less than t. Of course, the probability of this happening depends on t_1. Determine the correct formula for the probability $\mathrm{P}[T_2 < t - t_1]$ using the function $E(t;\lambda)$ that represents the exponential distribution with appropriate λ. We'll apply the answer to (b) at the end.

(d) The probability that $T_1 = t_1$ is arbitrarily small, but we can approximate it as a differential. Use linear approximation to obtain a formula for $\mathrm{P}[t_1 < T_1 < t_1 + dt_1]$ valid in the limit $dt_1 \to 0$. In order to get a valid approximation you must keep the largest term, which will have a factor dt_1 present. (In other words, don't actually take a limit as $dt_1 \to 0$, just discard terms that have more than one factor of dt_1.)

(e) Write down a formula for the probability that both conditions in (c) and (d) are true; that is, $t_1 < T_1 < t_1 + dt_1$ and $T_2 < t - t_1$. (As an analogy, the probability that the sum of two standard dice is 2 is (1/6)*(1/6) because each die roll has to be a 1.)

(f) The quantity t_1 ranges over all values from 0 to t; hence, the full probability for $T < t$ is an infinite sum of the infinitesimal quantities in part (e). Conclude that

$$\mathrm{P}[T < t] = \int_0^t \left(\lambda e^{-\lambda t_1} - \lambda e^{-\lambda t}\right) dt_1.$$

(g) Compute the integral in (f) to obtain $E_2(t;\lambda) = \mathrm{P}[T < t]$. Use this result and that of (b) to obtain the function $S_2(t;\mu)$.

(h) Use the expected value formula (1.6.2) to show that the mean of the survival distribution is μ and the standard deviation is $\mu/\sqrt{2}$.

(i) The file *TransitionSim.m* contains a Matlab script that simulates a recovery process whereby individuals move through k stages to recovery, each with a mean time of μ/k so that the total mean time is μ. Modify the script so that it plots simulation outputs using $k = 1$, $k = 3$, and $k = 6$ on a common set of axes. Discuss whether using more than one phase for a recovery process makes for more realistic results. (Note: The graph for $k = 1$ should be identical to the results from the *decay.m* function, subject to the usual random variation in individual simulations.)

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Empirical Modeling

2

Simulations require values for the model parameters, which raises the question of how parameter values should be determined. Occasionally, they can be measured directly, but more often they can only be inferred from their effects. This is done by collecting experimental data for the independent and dependent variables of a model and then using a mathematical procedure to determine the parameter values that give the best fit for the data. This is the subject of empirical modeling, which encompasses two questions:

1. Given a model and a set of data, how do we find the model parameters to fit the model to the data?
2. Given more than one model with best-fit parameters, how do we decide which model has more empirical support?

To these questions, we can provide a set of answers that work well in practice. As always in modeling, we must make some assumptions to get appropriate mathematics problems. The problem solutions are beyond doubt, but the assumptions are subject to some debate.

We begin in Sect. 2.1 by carefully developing the linear least squares method to fit the simple model $y = mx$. We extend this method in Sect. 2.2 to the more general linear model $y = b + mx$ and to linearizations for the exponential model $z = Ae^{kt}$ and the power function model $y = Ax^p$. Section 2.3 develops the notion of semilinear models, which generalize the exponential and power function models to any model of the form $y = Af(x; p)$, where p is a parameter in the nonlinear function $f(x)$. These three sections provide a practical guide to fitting models to data for the beginning empirical modeler.

Section 2.4 takes up the question of how to choose among models. This question admits a partial quantitative answer, thanks to the introduction of the Akaike Information Criterion in 1974 [1]. AIC has revolutionized model selection in biology, yet it has still not become a standard part of the mathematics or statistics curriculum. This glaring omission seems inexplicable, but I attribute it to an unwritten rule among mathematicians and statisticians that students should only learn things for which they can understand the theory.[1]

The chapter concludes with a case study devoted to righting an ongoing wrong in quantitative science procedure—the continuing use of the Lineweaver–Burk method to fit the Michaelis–Menten

[1] I like to say that a mathematician is someone who believes that you should not drive a car unless you have built one yourself.

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G. Ledder, *Mathematical Modeling for Epidemiology and Ecology*, Springer Undergraduate Texts
in Mathematics and Technology, https://doi.org/10.1007/978-3-031-09454-5_2

function to data in spite of its clear inferiority to a method from the same era and its having been definitively discredited in 1975 [2].

The chapter contains one project, in which students develop the method for fitting quadratic functions to data and use it with different sets of climate data to look for evidence of global warming.

2.1 The Basic Linear Least Squares Method ($y = mx$)

After studying this section, you should be able to:

- Use the linear least squares method to obtain the best-fit parameter values for the linear model $y = mx$.
- Discuss the assumptions made in claiming that the results of the linear least squares method are the best parameter values for the data.

Table 2.1.1 contains some predation data for *P. steadius* in the BUGBOX-predator virtual world.[2]

Because the data points (seen in Fig. 2.1.1) appear to lie roughly on a straight line through the origin, it makes sense to model the data using a linear function.

Table 2.1.1 Predation rate y for prey density x

x	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140
y	0	2	7	10	9	14	21	20	25	20	30	25	29	35	38

While the usual practice is to use the general linear function $y = b + mx$, it is better modeling practice in this case to use the simpler linear function $y = mx$. This assessment is based on three arguments.

1. Mathematically, it is much easier to find one parameter from data than two.
2. Biologically, the data point (0, 0) is theoretical as well as measured. While we could imagine getting more or less predation for a prey density of 10 than what the data shows, we cannot imagine any value of predation other than 0 when the prey density is 0.
3. Statistically, the model $y = mx$ may be more informative than the model $y = b + mx$ in the sense of having a more predictive and explanatory value.

In assessing the three arguments for using $y = mx$, we find a clear hierarchy. The biological argument is the strongest because it is about matching the model to the scenario. The mathematical argument is the weakest—in this particular case, we'll see in Sect. 2.2 that the mathematical convenience is minor.[3] The statistical argument is addressed in detail in Sect. 2.4.

[2] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

[3] There are other situations in which choices *are* made for mathematical convenience. In these cases, the modeler should carefully consider the trade-off between mathematical convenience and biological inaccuracy, something that is not done as often as it should be.

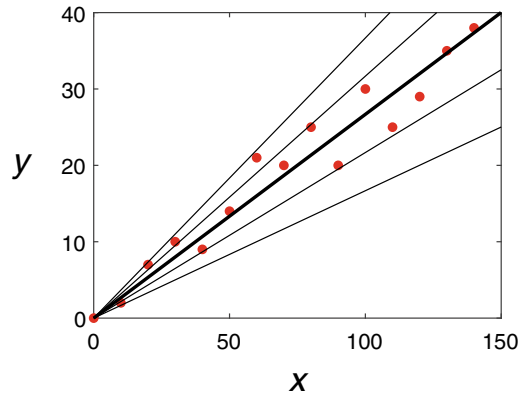


Fig. 2.1.1 Consumption rate y for prey density x from Table 2.1.1, showing several instances of the model $y = mx$; the heavy line is the instance that will emerge as the best fit

2.1.1 Overview of the Method

To fit a model to data, we must find and solve a mathematics problem to determine the parameter(s) that yield the best fit—in this case, a value of m for the given data set.

Obviously, there is no single value of m for which the model $y = mx$ fits the data exactly. For any given m, some or all of the data points lie off the graph of the model. Clearly, the slope m for the top line in Fig. 2.1.1 is too large and that for the bottom is too small. Somewhere between the top and the bottom is a "best" straight line, perhaps the thickest one in the plot.

To identify the best straight line, we must define and solve an optimization problem, which requires a combination of modeling and analysis:

1. Determine a function that expresses the quantity to be maximized or minimized in terms of one or more variable quantities and a permissible set of values for these variables.
2. Determine the permissible variable value that yields the maximum or minimum function value.

Optimization problems in empirical modeling are more conceptually difficult than optimization problems in calculus because of the different roles played by variables and parameters. In our current example, the variables x and y represent specific data points, while the parameter m is unknown. Hence, our usual thinking of x and y as variables and m as a constant is turned on its head.

> When parameterizing a mathematical model from data, the parameters in the model are the variables in the optimization problem, while the variables of the model appear in the optimization problem only as labels of values in the data set.

The imagery of the narrow and broad views of mathematical modeling (Fig. 1.4.2) is helpful in thinking about the problem of determining the best value of m. In step 1, we assume a fixed value of the parameter m, generate a set of "theoretical" (x, y) data points using the model $y = mx$, and calculate some quantitative measure of the total discrepancy between the actual data and the data obtained using the model. This step occurs within the narrow view because m is fixed. Once we have a formula for calculating the total discrepancy, we change our perspective. Now we think of the *data* as fixed and the total discrepancy for that fixed data set as a function $F(m)$. We then obtain the optimal value of m using methods of ordinary calculus. Treating m as a variable locates step 2 within the broad view.

2.1.2 Development of the Method

Quantifying total discrepancy (step 1) requires three choices. First, we have to decide how to determine the theoretical points to compare with the actual data points, then we have to decide how to measure the discrepancy between the corresponding pairs of points, and finally, we have to decide how to combine the pointwise discrepancies into a total.

There are several ways to determine the theoretical comparison points; the standard choice is to use the model to calculate a theoretical y value to go with each x value in terms of the model parameters. We'll revisit this choice at the end of the section. For the discrepancy between pairs of points, the only measurement that is ever used is the distance between the points in each pair, which will be Δy when the x values of the theoretical points are chosen to be the same as those of the data. Similarly, one could come up with different ways to sum up the discrepancies, but the only one ever used is to add up the squares.

Combining the choice of fixed x with distance squared, we have the following total discrepancy function:

$$F(m) = (\Delta y_1)^2 + (\Delta y_2)^2 + \cdots + (\Delta y_n)^2 , \tag{2.1.1}$$

where, for the model $y = mx$, the *residuals* Δy are given by

$$\Delta y_i = |mx_i - y_i| . \tag{2.1.2}$$

▶ **Example 2.1.1** Let $m = 0.3$. The data, model, and one of the residuals are shown in Fig. 2.1.2. The total discrepancy is $F(0.3) = 218$, calculated as the sum of the bottom row of Table 2.1.2. □

Table 2.1.2 Total discrepancy calculations for $y = 0.3x$ with the *P. steadius* data

x	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140
$0.3x$	0	3	6	9	12	15	18	21	24	27	30	33	36	39	42
y	0	2	7	10	9	14	21	20	25	20	30	25	29	35	38
Δy	0	1	−1	−1	3	1	−3	1	−1	7	0	8	7	4	4
$(\Delta y)^2$	0	1	1	1	9	1	9	1	1	49	0	64	49	16	16

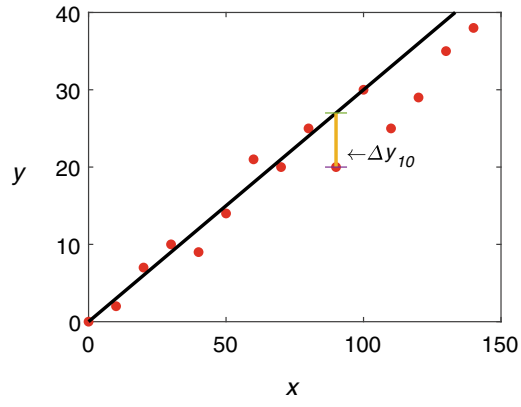


Fig. 2.1.2 Consumption rate y for prey density x, showing the model $y = 0.3x$ and the residual for $x = 90$

Check Your Understanding 2.1.1:

Repeat Example 2.1.2 for the model $y = 0.25x$. Is this model better or worse than $y = 0.3x$?

Total discrepancy calculations are easily automated with a spreadsheet; however, we can only do the calculation for one value of m at a time. To find the value of m that minimizes F, we would normally need to write a computer program to implement an optimization algorithm[4]; however, this particular function F is simple enough that hand calculation methods can determine the minimizing value of m.

Substituting (2.1.2) into (2.1.1) and expanding the squares, we have

$$F(m) = (m^2x_1^2 - 2mx_1y_1 + y_1^2) + \cdots + (m^2x_n^2 - 2mx_ny_n + y_n^2)\,,$$

which we can rearrange by combining terms with common powers of m:

$$\begin{aligned} F(m) = {} & (x_1^2 + x_2^2 + \cdots + x_n^2)m^2 \\ & - 2(x_1y_1 + x_2y_2 + \cdots + x_ny_n)m \\ & + (y_1^2 + y_2^2 + \cdots + y_n^2)\,. \end{aligned}$$

We can simplify this formula by assigning symbols to the coefficients of each power of m. For example, the coefficients of m^2 are

$$x_1^2 + \cdots + x_n^2 = \sum_{i=1}^{n} x_i^2.$$

This is a messy expression, but one which comes down in the end to a single number obtained by combining the data; hence, it makes sense to replace the messy notation with simpler symbols. To that end, we define

$$S_{xx} = \sum_{i=1}^{n} x_i^2, \qquad S_{xy} = \sum_{i=1}^{n} x_iy_i, \qquad S_{yy} = \sum_{i=1}^{n} y_i^2. \tag{2.1.3}$$

In the context of the curve fitting problem, these quantities are constants,[5] so the total discrepancy is a function of a single variable m,

$$F(m) = S_{xx}m^2 - 2S_{xy}m + S_{yy}\,. \tag{2.1.4}$$

This function is a simple parabola pointing upward, so we need only find the vertex of that parabola to obtain the important mathematical result[6]:

[4] See Appendix A.

[5] Context is crucial. As noted earlier, the parameter m functions as a *constant* in the model $y = mx$ (narrow view) but as a *variable* in the total discrepancy function F (broad view). Meanwhile, x and y are variables in the model, but the data points (x_i, y_i) function as parameters in the total discrepancy calculation because we have a fixed set of data.

[6] The proof of Theorem 2.1.1 is given as Problem 2.1.6.

Theorem 2.1.1 (Linear Least Squares Fit for $y = mx$)
Given a set of points (x_i, y_i) for $i = 1, 2, \cdots, n$, the value of m that minimizes the total discrepancy function for the model $y = mx$ is

$$m^* = \frac{S_{xy}}{S_{xx}} \tag{2.1.5}$$

and the corresponding residual sum of squares is

$$\text{RSS} = F(m^*) = S_{yy} - m^* S_{xy} \,. \tag{2.1.6}$$

The **residual sum of squares** is the total discrepancy for the model when the best value of m is used; that is, it is the minimum value of the function F. This will be needed for the semilinear data-fitting scheme of Sect. 2.3 and the model selection scheme of Sect. 2.4.

We now have the mathematical tools needed to find the optimal value of m for the *P. steadius* data set.

▶ **Example 2.1.2** For the data in Table 2.1.1, we obtain the results

$$S_{xx} = 101,500, \qquad S_{xy} = 27,080, \qquad S_{yy} = 7,331\,;$$

therefore, (2.1.5) and (2.1.6) yield the results $m^* \approx 0.267$ and RSS ≈ 106.1. The best-fit line is the heavy one in Fig. 2.1.1. □

Check Your Understanding 2.1.2:
Verify the values given in Example 2.1.2.

2.1.3 Implied Assumption of Least Squares

In choosing the point (x_i, mx_i) as the theoretical counterpart to the data point (x_i, y_i), we are making the implicit assumption that all of the measurement error or unpredictability in the data is in the value of y. This is often the case. In enzyme kinetics (see Sect. 2.5), the x values are concentrations of a chemical, which are easy to measure, while the y values are initial rates of reaction for a chemical reaction whose rate is gradually slowing, which are much harder to measure. So it is scientifically correct as well as mathematically convenient to assume all the discrepancy is in the y values.

In other contexts, the opposite might be true. In the main example of this section, it might be much easier to measure the consumption rate y than the prey density x, especially since the act of consumption changes the prey density. It might be better in such cases to use a total discrepancy function based on comparison points that have the same y values with the error assumed to be in the x values (Table 2.1.3).

Of course, the most difficult case is that in which we expect comparable uncertainty in the x and y values. In theory, we should then measure discrepancy from each data point to whichever theoretical point on a particular instance of a model is closest to the data point. This yields what is called the *total* least squares problem. In most cases, the final parameter values do not depend a lot on which

comparison point is used, so it is customary to use vertical discrepancy. This is an issue that should be considered in any practical problem.[7]

Check Your Understanding Answers

Table 2.1.3 Total discrepancy calculations for $y = 0.25x$ with the *P. steadius* data

x	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140
$0.25x$	0	2.5	5	7.5	10	12.5	15	17.5	20	22.5	25	27.5	30	32.5	35
y	0	2	7	10	9	14	21	20	25	20	30	25	29	35	38
Δy	0	0.5	2	2.5	1	1.5	6	2.5	5	2.5	5	2.5	1	2.5	3
$(\Delta y)^2$	0	0.25	4	6.25	1	2.25	36	6.25	25	6.25	25	6.25	1	6.25	9

1. The calculations appear in Table 2.1.3. The total discrepancy is $F(0.25) = 134.75$, which is less than $F(0.3) = 218$.

Problems

2.1.1 Use the standard least squares method to fit the function $y = mx$ to the data in Table 2.1.4, with calculations done on a hand calculator. Use *LeastSq_1var.m* to check your answer.

Table 2.1.4 Data for Problems 2.1.1 and 2.1.7.

x	0	1	2	3	4	5
y	0	1.0	2.4	3.4	4.0	5.0

2.1.2 Use the standard least squares method to fit the function $y = mx$ to the subset in Table 2.1.1 that includes points with $x = 0, 20, 40, \ldots 140$, with calculations done on a hand calculator. Plot the points and the best-fit straight line. Use *LeastSq_1var.m* to check your answer.

2.1.3 Roll dice to create a data set with six points. For each point (x_i, y_i), take $x_i = i$ and y_i the sum of a random roll of i dice.

(a) Identify the model that should be the theoretical best fit to the data. Is it linear? Should it pass through the origin? What should be the slope?
(b) Use the least squares formulas with a hand calculator to find the best-fit slope m^* and residual sum of squares for the model $y = mx$.
(c) Repeat part (c), but this time, obtain the y values by taking averages of four sets of die rolls. That is, to get y_3, roll a set of three dice four times and set y_3 as the average of the four point totals.
(d) Discuss the effect of the averages in part (c) on the results. Does the answer more closely conform to the theoretical result of part (a)? Is the residual sum of squares smaller? If so, by how much? Refer to Sect. 1.3 on distributions of sample means. Keep in mind that your die rolls were random, so your results may not be representative of what would happen on average.

[7] See [13] for a much more detailed look at the conceptual issues in fitting data as well as practical methods.

2.1.4 (Continued from Problems 1.2.8 and 1.2.10.)

(a) Fit the model $y = mx$ to your *P. steadius* data from Problem 1.2.8.
(b) Fit the model $y = mx$ to your *P. steadius* data from Problem 1.2.10.
(c) How much different (in percentage) are your results in (a) and (b) from the result in Example 2.1.2?
(d) Discuss whether or not replacement appears to be a significant source of differences in the results.

(This problem is continued in Problem 2.4.4.)

2.1.5* The data sets in Table 2.1.5 contain a parameter c that perturbs some of the data points away from the straight line $y = x$ while still maintaining an average y of 0. By examining the change in slope m as a function of c, we can measure the effect of measurement error on the result of the least squares procedure. To do this, plot the linear regression slope m as a function of the parameter c, where $0 \le c \le 1$, for each data set. How do measurement errors affect the least squares line? In particular, which errors are the least squares line more sensitive to?

Table 2.1.5 Two data sets for Problem 2.1.5

Set A						Set B					
x_1	−2	−1	0	1	2	x_2	−2	−1	0	1	2
y_1	−2+c	−1	0	1	2−c	y_2	−2	−1+c	0	1−c	2

2.1.6 Derive the results (2.1.5–2.1.6) of the linear least squares method for the model $y = mx$ by applying optimization methods from calculus[8] to the total discrepancy function

$$F(m) = S_{xx}m^2 - 2S_{xy}m + S_{yy} .$$

2.1.7 Derive the appropriate least squares formulas for the case of horizontal discrepancies. Use them, with calculations on a hand calculator, to fit the linear model to the data in Table 2.1.4. Plot the points and the best-fit straight line. Compare the results to Problem 2.1.1.

2.1.8* Use the formulas you derived in Problem 2.1.7 to fit the linear model to the data in Table 2.1.1. Plot the points and the best-fit straight line. Compare the results to Example 2.1.2.

2.2 Fitting Linear and Linearized Models to Data

After studying this section, you should be able to:

- Use the linear least squares method to obtain the best-fit parameter values for the linear model $y = b + mx$.
- Use the least squares method to fit the linearized versions of the models $z = Ae^{\pm kt}$ and $y = Ax^p$ to data, where A, k, and p are parameters.

[8] See Appendix C.

In Sect. 2.1, we developed the basic least squares method for the model $y = mx$. The method is easily adapted for the general linear model $y = b + mx$, the exponential model $z = Ae^{\pm kt}$, and the power function model $y = Ax^p$. It can be adapted to other nonlinear models, but it is important to study Sect. 2.5 before doing so.

2.2.1 Adapting the Method for $y = mx$ to the General Linear Model

Most straight lines in a plane do not pass through the origin. While there are theoretical reasons for insisting that the predation model passes through the origin, this is obviously not valid for *all* linear models; hence, Theorem 2.1.1 would seem to be of limited use. However, the following theorem reduces the problem of fitting the model $y = b + mx$ to that of fitting $y = mx$.[9]

Theorem 2.2.1
Given any set of points (x_i, y_i), with mean values $\bar{x}$ and $\bar{y}$, and any possible slope m, the best-fit straight line passes through the point $(\bar{x}, \bar{y})$.

We will assume that the best parameter pair (m^*, b^*) yields a line that passes through the point $(\bar{x}, \bar{y})$. By shifting coordinates, we can move the fixed point to the origin of an XY coordinate system, whereupon we can find the slope m by fitting the data to the model $Y = mX$. The same slope will be correct with the original coordinates. Theorem 2.2.1 tells us this procedure gives the right answer. Requiring the best line to pass through the mean point does not rule out any parameter pairs that might be better than the one we find.

We therefore have the following procedure.

Algorithm 2.2.1
Linear least squares fit for the general linear model $y = b + mx$

1. Convert the xy data to XY data using $X = x - \bar{x}$ and $Y = y - \bar{y}$, where $\bar{x}$ and $\bar{y}$ are the means of the x and y values, respectively.
2. Find the parameter m^* and the residual sum of squares for both the XY and xy models by applying Theorem 2.2.1 to the XY data:

$$m^* = \frac{S_{XY}}{S_{XX}}, \quad \text{RSS} = F(m^*) = S_{YY} - m^* S_{XY}, \tag{2.2.1}$$

where

$$S_{XX} = \sum_{i=1}^{n} X_i^2, \qquad S_{XY} = \sum_{i=1}^{n} X_i Y_i, \qquad S_{YY} = \sum_{i=1}^{n} Y_i^2.$$

3. The parameter b^* for the xy model is given by

$$b^* = \bar{y} - m^* \bar{x}. \tag{2.2.2}$$

[9] The derivation of this result is given as Problem 2.2.7.

▶ **Example 2.2.1** To fit the model $y = b + mx$ to the data in Table 2.1.1, we first compute the means $\bar{x} = 70$ and $\bar{y} = 19$. Then we subtract the means from the original data set to obtain a shifted data set, as shown in Table 2.2.1. Algorithm 2.2.1 then yields the results

$$m^* = 0.255\,, \qquad \text{RSS} = 100.4\,, \qquad b^* = 1.175\,. \qquad \square$$

Table 2.2.1 Consumption rate y for prey density x, along with the shifted data $X = x - \bar{x}$, $Y = y - \bar{y}$

x	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140
y	0	2	7	10	9	14	21	20	25	20	30	25	29	35	38
X	−70	−60	−50	−40	−30	−20	−10	0	10	20	30	40	50	60	70
Y	−19	−17	−12	−9	−10	−5	2	1	6	1	11	6	10	16	19

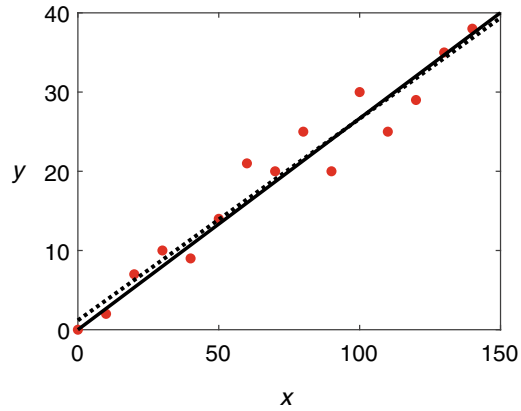


Fig. 2.2.1 Consumption rate y for prey density x, showing the linear least squares fits for the models $y = mx$ (*solid*) and $y = b + mx$ (*dotted*)

Check Your Understanding 2.2.1:
Verify the values given in Example 2.2.1.

We now have two best-fit results for the Table 2.1.1 data: the line $y = 0.267x$ from Example 2.1.2, with residual sum of squares 106.1, and the line $y = 1.175 + 0.255x$ from Example 2.2.1, with residual sum of squares 100.4. Figure 2.2.1 shows both of these lines along with the data. Does the slightly lower residual sum of squares mean that the two-parameter model is better than the one-parameter model? Not necessarily. The calculation of the residual sum of squares treats all data equally. However, the data point (0, 0) is free of experimental uncertainty, so perhaps we should be less tolerant of the discrepancy $\Delta y_1 = 1.175$ in the two-parameter model than the discrepancies at the other points. Perhaps we should insist that $y(0) = 0$ is a *requirement* for our model, even though doing so slightly increases the residual sum of squares. The greater complexity of the two-parameter model is also an issue, which we discuss in Sect. 2.4.

2.2.2 Fitting the Exponential Model by Linear Least Squares

Suppose we want to find the best parameters A and k to fit data on exponential growth to the model

$$z = Ae^{kt}. \tag{2.2.3}$$

Taking the natural logarithm changes the model equation to

$$\ln z = \ln A + kt\,. \tag{2.2.4}$$

Now suppose we define a new set of variables and parameters by the equations

$$y = \ln z, \qquad x = t\,, \qquad m = k\,, \qquad b = \ln A\,. \tag{2.2.5}$$

These definitions allow us to rewrite the logarithm model (2.2.4) as the standard linear model

$$y = b + mx\,. \tag{2.2.6}$$

The algebraic equivalence of the original model (2.2.3) and the linearized model (2.2.6) means that we can fit the exponential model to data by applying the linear least squares algorithm of Theorem 2.2.1.[10]

Algorithm 2.2.2
Linear least squares fit for the exponential model $z = Ae^{kt}$

1. Convert the tz data to xy data using $y = \ln z$ and $x = t$.
2. Convert the xy data to XY data using $X = x - \bar{x}$ and $Y = y - \bar{y}$, where $\bar{x}$ and $\bar{y}$ are the means of the x and y values, respectively.
3. Find the parameters m^* and b^* and the residual sum of squares for the xy model using the linear least squares formulas (2.2.1)–(2.2.2).
4. Calculate the parameters for the exponential model: $A^* = e^{b^*}$ and $k^* = m^*$.

Table 2.2.2 Data sets for the exponential model in Example 2.2.2

t	0	1	2	3	4	5	6	7	8
I	50	77	129	185	258	406	686	1022	1457
x	0	1	2	3	4	5	6	7	8
y	3.912	4.344	4.860	5.220	5.553	6.006	6.531	6.930	7.284
X	−4	−3	−2	−1	0	1	2	3	4
Y	−1.715	−1.283	−0.767	−0.406	−0.074	0.380	0.904	1.303	1.658

[10] Equivalent models are the subject of Sect. 3.6.

▶ **Example 2.2.2** Table 2.2.2 shows a portion of the data from the disease simulation in Sect. 1.5, including the number of infectious individuals I for each day from 0 to 8. The xy and XY data sets were calculated in steps 1 and 2 of Algorithm 2.2.2 (using I in the role of z), with $\bar{x} = 4$ and $\bar{y} = 5.627$. The linear least squares formulas (2.2.1)–(2.2.2) yield the results

$$m^* = 0.423\,, \qquad b^* = 3.94\,,$$

from which we obtain

$$k^* = 0.423\,, \qquad A^* = 51.2\,.$$

Figure 2.2.2 shows the data and best-fit model in both the tI and xy (t ln I) planes. □

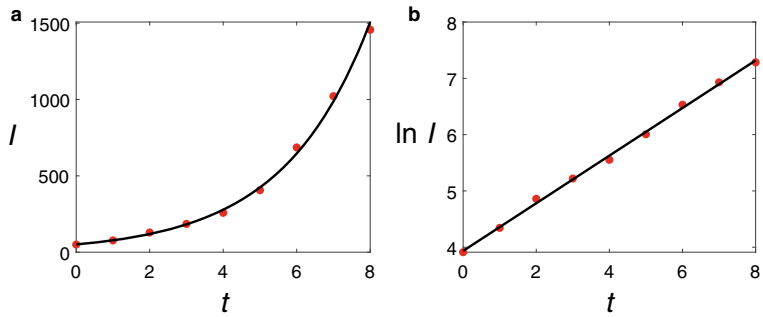


Fig. 2.2.2 The exponential model fit to the data in Table 2.2.2 using linear least squares: **a** $I = Ae^{kt}$; **b** $\ln I = b + mt$

2.2.3 Fitting the Power Function Model $y = Ax^p$ by Linear Least Squares

Table 2.2.3 presents predation data for *P. speedius* from the BUGBOX-predator virtual world. A plot of the data (Fig. 2.2.3a) appears to resemble a square root graph, suggesting a model of the form:

$$y = Ax^p\,, \qquad A,\, p > 0\,. \tag{2.2.7}$$

Table 2.2.3 Predation by *P. speedius* y for given prey density x

x	10	20	30	40	50	60	70	80	90	100	110	120	130	140
y	7	11	19	19	22	25	21	25	26	23	27	29	30	29

This model can be fit using the same linearization technique used for the exponential model. However, we have to be careful about notation. Often a particular symbol has different meanings in two or more formulas needed to solve a particular problem. Here, the symbol x represents the number of prey animals in the biological setting (and hence in the model (2.2.7)), but it also represents the generic independent variable in the models $y = b + mx$ and $y = mx$. This kind of duplication is unavoidable because many formulas have their own standard notation. One way to avoid error in these cases is to rewrite the generic formulas using different symbols while retaining the symbols as is in the formulas

that provide the context. In this case, let's use U, V, u, and v in place of X, Y, x, and y in the generic linear least squares formulation.

Taking a natural logarithm of (2.2.7) yields

$$\ln y = \ln A + p \ln x \,,$$

which is equivalent to the linear model $v = b + mu$ using the definitions

$$u = \ln x \,, \quad v = \ln y \,, \quad m = p \,, \quad b = \ln A \,.$$

We can then formulate an algorithm for fitting a power function model using linearized least squares.

Algorithm 2.2.3
Linear least squares fit for the power function model $y = Ax^p$

1. Convert the xy data to uv data using $u = \ln x$ and $v = \ln y$.
2. Convert the uv data to UV data using $U = u - \bar{u}$ and $V = v - \bar{v}$, where $\bar{u}$ and $\bar{v}$ are the means of the u and v values, respectively.
3. Find the parameters m and b for the uv model using the linear least squares formulas:

$$m^* = \frac{S_{UV}}{S_{UU}} \,, \qquad b^* = \bar{v} - m\bar{u} \,, \tag{2.2.8}$$

where

$$S_{UU} = \sum_{i=1}^{n} U_i^2, \qquad S_{UV} = \sum_{i=1}^{n} U_i V_i \,.$$

4. Calculate the parameters for the power function model: $A^* = e^{b^*}$ and $p^* = m^*$.

Notice that all the symbols used in Algorithm 2.2.3 are defined within the algorithm statement. The meaning of a biological symbol is seldom clear from the context alone, so it is good modeling practice to define all symbols in the statement of a model or algorithm.

▶ **Example 2.2.3** In fitting the model $y = Ax^p$ to the data from Table 2.2.3, we run into a problem. The change-of-variables formulas $u = \ln x$ and $v = \ln y$ do not work for the point $(0, 0)$. This is not a serious problem, because $(0, 0)$ satisfies $y = Ax^p$ exactly for any values of the parameters. Omitting $(0, 0)$, we obtain the linearized least squares result

$$y = 2.71x^{0.499} \,. \tag{2.2.9}$$

This model is plotted together with the data in Fig. 2.2.3. The first plot shows the data and model as a plot of y versus x, while the second plot shows the linearized data and model as a plot of $\ln y$ versus $\ln x$.

Check Your Understanding 2.2.2:

Construct a data table, similar to Table 2.2.2, to verify the results of Example 2.2.3. Note that the data point (0, 0) must be omitted.

Check Your Understanding Answers

1. See Table 2.2.4.

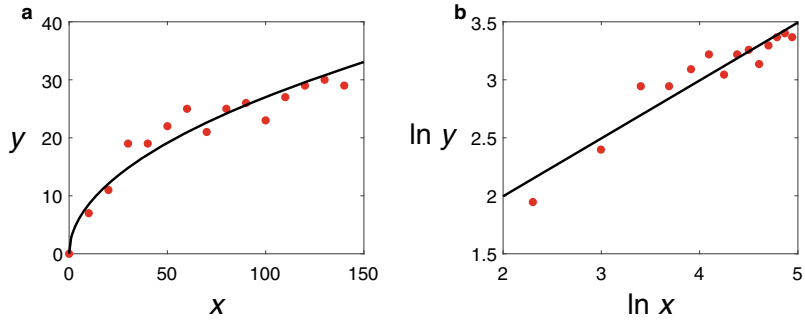


Fig. 2.2.3 The power function model fit to the data in Table 2.2.3 [without (0, 0)] using linear least squares

Table 2.2.4 Data sets for the power function model in Example 2.2.3

x	10	20	30	40	50	60	70	80	90	100	110	120	130	140
y	7	11	19	19	22	25	21	25	26	23	27	29	30	29
u	2.303	2.996	3.401	3.689	3.912	4.094	4.249	4.382	4.500	4.601	4.700	4.788	4.868	4.942
v	1.946	2.398	2.944	2.944	3.091	3.219	3.045	3.219	3.258	3.136	3.296	3.367	3.401	3.367
U	−1.799	−1.106	−0.701	−0.413	−0.190	−0.008	0.146	0.280	0.398	0.503	0.600	0.685	0.766	0.840
V	−1.099	−0.647	−0.101	−0.101	0.046	0.174	−0.001	0.174	0.213	0.090	0.251	0.322	0.356	0.322

Problems

2.2.1 [Fluorine at the South Pole]

(a) Fit a linear model to the data in Table 2.2.5 (using a calculator to do the computations). The data gives the concentration C, in parts per trillion, of the trace gas F-12 at the South Pole from 1976 to 1980.
(b) Plot the data and the best-fit line.
(c) Check your answer with the MATLAB script *LeastSq.m*.
(d) Discuss the quality of the fit of the model to the data.

(This problem is continued in Problems 2.2.6 and 2.4.6.)

Table 2.2.5 Concentration of F-12 at the South Pole by year, with 1976 as year 0 [14]

t	0	1	2	3	4
C	195	216	244	260	284

2.2.2*

(a) Fit a linear model to the data in Table 2.2.6 (using a calculator to do the computations).
(b) Plot the data and the best-fit line.
(c) Check your answer with the MATLAB script *LeastSq.m*.
(d) Discuss the quality of the fit of the model to the data.

(This problem is continued in Problem 2.4.7.)

Table 2.2.6 A data set for Problem 2.2.2

t	8.7	9	11	18	19	22	28
C	25	25	26	48	65	90	100

2.2.3 Repeat Problem 2.1.4 with the model $y = b + mx$ using *LeastSq.m*.

Table 2.2.7 Population of bacteria after t hours

t	0	1	2	3	4
N	6.0	9.0	13.0	21.0	29.0

2.2.4* The data in Table 2.2.7 gives the population of a bacteria colony as a function of time. Use a hand calculator rather than a computer program to find the exponential function that best fits the data set using the linearization method. Plot the linearized data with the best-fit line. Plot the original data with the exponential curve corresponding to the best-fit line.

2.2.5 Table 2.2.8 shows data from a radioactive decay simulation of 1,000 particles, each of which had an 8 % chance of decaying in any given time step. The original data consists of the time t and the number of remaining particles z for each time. Fit an exponential model to this data. Plot the data and model in the tz and $t \ln z$ planes.

(This problem is continued in Problem 2.3.3.)

Table 2.2.8 Data sets for the exponential model in Example 2.2.5

t	0	1	2	3	4	5	6	7	8	9
z	1,000	929	855	785	731	664	616	568	515	471

2.2.6 [Fluorine at the South Pole] (Continued from Problem 2.2.1.)

(a) Use the linearization method to fit the model $y = Ax^p$ to the data from Problem 2.2.1 and determine the residual sum of squares.
(b) What is clearly wrong with using the model $y = Ax^p$ for this data set?

(This problem is continued in Problem 2.4.6.)

2.2.7 Use the linear least squares results for the $y = mx$ case to derive the general linear least squares results of Theorem 2.2.1. Do this by solving the problem of minimizing the residual sum of squares for variable b with fixed m.[11] Show that the formula amounts to $b^* = \bar{y} - m\bar{x}$. Then show that the model produces $y = \bar{y}$ when $x = \bar{x}$.

Problems 2.2.8–2.2.10 use global temperature data [6] to look for evidence of global climate change.[12] Problem 2.2.11 addresses the same issues using data on grape harvest dates. These problems are extended in the Chap. 2 project.

2.2.8 (Continued from Problem 1.2.1.)
Use *LeastSq.m* to fit a linear model to the data from 1985 through 2015 in the file *GlobalLandTemperatures_July.csv*. Plot the data with the best-fit line. The hypothesis of human-induced climate change suggests that the historical record should show no particular pattern before the beginning of a gradual temperature rise at a point corresponding roughly to the Industrial Revolution. Does the temperature data support this hypothesis? Discuss, paying particular attention to the point that what we see is a combination of multiple trends of varying strengths.

(This problem is continued in Problem 2.4.8.)

2.2.9 (Continued from Problem 1.2.2.)
Repeat Problem 2.2.8, but with the file *GlobalLandTemperatures_January.csv*.

(This problem is continued in Problem 2.4.9.)

2.2.10 (Continued from Problem 1.2.3.)
Repeat Problem 2.2.8, but with the full data set in the file *GlobalTemperatures_July.csv*.

(This problem is continued in Problem 2.4.10.)

2.2.11^{p} The National Oceanographic and Atmospheric Administration (NOAA) has a data set on its website that gives the dates of the beginning of the grape harvests in Burgundy from 1370 to 2003 [5]. This data offers a crude, but long-term, look at global climate change.

(a) Fit a linear model using three different subsets of the data: (1) the years 1800–1950, (2) the years 1951–1977, and (3) the years 1979–2003. Of particular interest is the slope.
(b) On average, grape harvest dates have been getting earlier since 1800. By how many days did the expected grape harvest date change in each of the three periods?
(c) Plot all of the data from 1800 to 2003 as points on a graph. Explain why it is a mistake to connect the points to make a dot-to-dot graph.

[11] See Appendix C.

[12] The needed data sets are available at https://www.math.unl.edu/~gledder1/.

(d) Add the three linear regression lines to the plot, being careful to use only the appropriate time interval for each.
(e) What do the results appear to say about global climate change?
(f) Offer at least one possible explanation for the results that does not involve global climate change. (Hint: Think about possible biological explanations.)

2.3 Fitting Semilinear Models to Data

After studying this section, you should be able to:

- Use the semilinear least squares method to fit models of the form $y = Af(x; p)$ to data, where q and p are parameters.

In Example 2.2.3, we used linear least squares optimization to obtain the parameter values $A = 2.71$ and $p = 0.499$ for the data in Table 2.2.3. These values minimize the fitting error on the graph of $\ln y$ versus $\ln x$. But this is not the same thing as minimizing the fitting error on the graph of y versus x. It is tempting to accept the results as optimal, but if we really want to minimize the error in the model $y = Ax^p$ rather than the model $\ln y = \ln A + px$, then we need to use a graph of y versus x to define the discrepancies between the model and the data.

It is not particularly difficult to use a fully nonlinear method to minimize fitting error with a two-parameter model.[13] However, it is unnecessarily complicated. The function $y = Ax^p$ has a scaling parameter,[14] which means that it is what we might call a *semilinear* model.

Definition 2.3.1

A **semilinear model** is a mathematical model of the form:

$$y = Af(x; p)\,, \tag{2.3.1}$$

where p and A are parameters and f is a nonlinear function. The semicolon in the notation helps remind us that only x is an independent variable.

Both the exponential and power function models are semilinear. For exponential models, it is usually the case that the linearized graph is more representative of the function behavior than the graph of the original variables; hence, the linearization method, as illustrated in Sect. 2.2, is best. For most other semilinear models, it is best to fit the parameters on a graph of the original data rather than a linearized version.

The semilinear regression method for a model $y = Af(x; p)$ involves two distinct mathematics problems: first, we find the best A in terms of an arbitrary choice of p, and then we find the best p where the values of A are chosen to be the best for each p. We consider these problems in turn.

[13] For the standard approach, see any statistics book that includes nonlinear regression.
[14] Section 1.1.

2.3.1 Finding the Best A for Given p

The key to finding the best A for a given p is to modify the data set for that p so as to reconceptualize the model as a linear one.

▶ **Example 2.3.1** Suppose we assume $p = 0.5$ for the model $y = Ax^p$ and the data in Table 2.2.3. Then we can calculate the quantity x^p for each data point, since we know the x values. Defining $z = x^{0.5}$, we can rewrite the model $y = Ax^{0.5}$ as $y = Az$. This allows us to identify the best A by converting the original xy data into zy data and applying the linear least squares result of Sect. 2.1. The modified data set, given $p = 0.5$, appears in Table 2.3.1. With z playing the role of x and A playing the role of m, (2.1.5)–(2.1.6) yield the results

$$A^* = \frac{S_{zy}}{S_{zz}} \approx 2.67\,, \quad \text{RSS} = S_{yy} - A^* S_{zy} \approx 80.1\,.$$

□

Table 2.3.1 Predation by *P. speedius* y for given prey density x along with the zy data set using $z = x^{0.5}$

x	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140
y	0	7	11	19	19	22	25	21	25	26	23	27	29	30	29
z	0	3.16	4.47	5.48	6.33	7.07	7.75	8.37	8.94	9.49	10.00	10.49	10.95	11.40	11.83
y	0	7	11	19	19	22	25	21	25	26	23	27	29	30	29

Check Your Understanding 2.3.1:

Table 2.3.2 and Fig. 2.3.1 show the number of transmissions of a "secret" per day (T) as a function of the fraction of the population that does not already know the secret (y), from a run of the simulation in Problem 1.5.8. Find A^* and the corresponding residual sum of squares for the model $T = Ay(1 - y)$.

Table 2.3.2 Transmissions of a "secret" per day (T) as a function of the fraction of the population that does not already know the secret (y), from a run of the simulation in Problem 1.5.8

y	0.020	0.030	0.048	0.077	0.128	0.220	0.366	0.550	0.768	0.924
T	0.010	0.018	0.029	0.051	0.092	0.146	0.184	0.218	0.178	0.070

2.3.2 Finding the Best p

The residual sum of squares for $y = 2.71x^{0.499}$, the "best-fit" model we obtained by linearization in Example 2.2.3, is approximately 80.7. Thus, the new model $y = 2.67x^{0.5}$ from Example 2.3.1 is a little more accurate on a graph in the xy plane than the best fit obtained by linearization. This very slight improvement is not enough to justify the more complicated procedure for finding the parameter

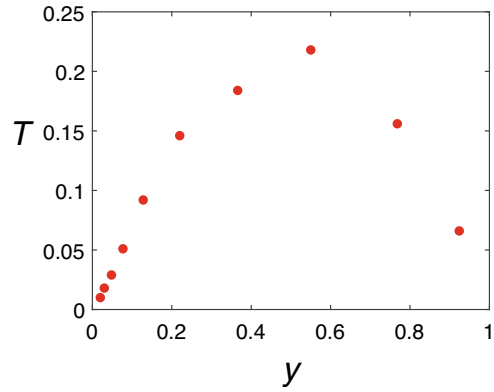


Fig. 2.3.1 Transmissions of a "secret" per day (T) as a function of the fraction of the population that does not already know the secret (y), from a run of the simulation in Problem 1.5.8

values. However, we only *guessed* the value $p = 0.5$; what we really need is a way to find the *best* p. We turn now to this mathematical problem.

When fitting a model $y = Af(x; p)$ to data, the goal is to choose the pair (p, A) that minimizes the residual sum of squares on a graph of y versus x. Optimization problems for two-parameter nonlinear models are relatively difficult, but in this case, we already know how to find the best value of A for any given choice of p. We can think of the method of Example 2.3.1 as defining the best A for each p as a function $A^*(p)$. If we choose these specific A values for each possible p, then the residual sum of squares ultimately depends only on p. We can formalize this idea, with F as the name given to the function that finds the best residual sum of squares for given p with best A for that p:

$$F(p) = \min_A (\text{RSS}(p, A)) = \text{RSS}(p, A^*(p)) . \tag{2.3.2}$$

▶ **Example 2.3.2** Suppose the data set is that of the top half of Table 2.3.1. In Example 2.3.1, we selected $p = 0.5$ and then obtained the results $A^*(0.5) = 2.67$ and RSS $= 80.1$. Using the notation of (2.3.2), we have $F(0.5) = 80.1$. We can repeat the entire calculation of that example using $p = 0.4$, with the result $F(0.4) = 63.1$. This result suggests that the optimal p is less than 0.5 and probably closer to 0.4. □

Check Your Understanding 2.3.2:

Repeat the calculation of Example 2.3.1 to obtain the result $F(0.4) = 63.1$.

For any given value of p, we have to create a modified data set and use the linear least squares formulas to get the corresponding F for that p. That is a lot of work, but it is a type of work for which computers are ideally suited. It takes only a simple program and the barest minimum of computer time to generate values of the function F for a given set of xy data. These can be used to estimate the minimizer as well as prepare a graph.

▶ **Example 2.3.3** Table 2.3.3 shows some values of $F(p)$ for the model $y = Ax^p$ using the *P. speedius* data from Table 2.3.1. From this table, we can see that the optimal p value is somewhere in the interval

$0.35 \le p \le 0.45$. We can then compute the values of $F(p)$ for $p = 0.350, 0.351, 0.352, \ldots 0.450$. These are plotted in Fig. 2.3.2. By examining the list of F values, we see that the optimal p value, to three significant figures, is 0.409, with a corresponding residual sum of squares of 62.9. We then obtain A by linear least squares, as in Example 2.3.1, leading to the result

$$y = 4.01x^{0.409}\,, \qquad \text{RSS} = 62.9\,. \tag{2.3.3}$$

This new model is plotted along with that of Example 2.2.3 in Fig. 2.3.3. □

Table 2.3.3 Some values of the residual sum of squares function $F(p)$ for the model $y = Ax^p$ using the *P. speedius* data from Table 2.3.1

p	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70
$F(p)$	92.3	71.2	63.1	66.6	80.1	102.4	132.3	168.7	210.7

2.3.3 The Semilinear Least Squares Method

The method described above works for any model of the general form $y = Af(x;\, p)$, where x is the independent variable, y is the dependent variable, and p and A are parameters.

To implement the method, we need to find a general formula for $\text{RSS}(p,\, A^*(p))$. In Example 2.3.1, we found

$$A^*(p) = \frac{S_{zy}}{S_{zz}}, \quad F(p) = \text{RSS}(p,\, A^*(p)) = S_{yy} - A^* S_{zy}\,,$$

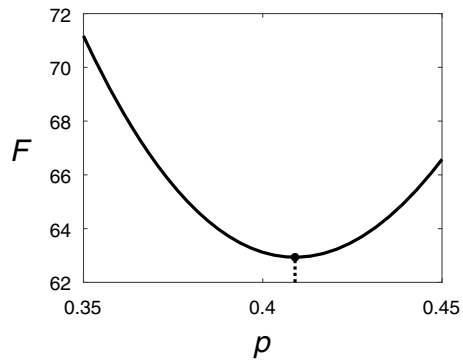


Fig. 2.3.2 The minimum residual sum of squares (F in (2.3.2)) for the *P. speedius* predation data with the power function model (Example 2.3.2)

with $z = x^p$ for the model $y = Ax^p$. In the general case, we replace z by $f(x;\, p)$. We can also eliminate A^* from the pair of equations to obtain the general result:

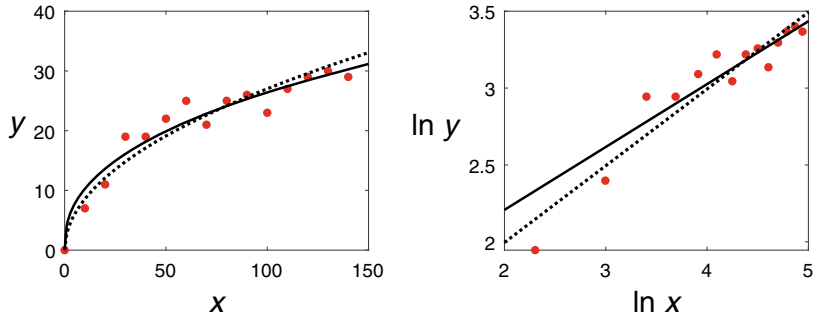


Fig. 2.3.3 The power function model fits to the *P. speedius* data (without (0, 0)) using linear least squares (*dashed*) and semilinear least squares (*solid*), shown in both the xy and $\ln x \ln y$ planes

$$F(p) = S_{yy} - \frac{(S_{yf})^2}{S_{ff}}, \tag{2.3.4}$$

where

$$S_{ff} = \sum_{i=1}^{n} [f(x_i; p)]^2, \qquad S_{yf} = \sum_{i=1}^{n} y_i f(x_i; p) \qquad S_{yy} = \sum_{i=1}^{n} y_i^2. \tag{2.3.5}$$

The full result is summarized in a theorem:

Theorem 2.3.1 (Semilinear Least Squares)
Given a model $y = Af(x; p)$ with data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, define a function F by

$$F(p) = \min_A (\text{RSS}(p, A)) = S_{yy} - \frac{(S_{yf})^2}{S_{ff}},$$

where S_{ff}, S_{yf}, and S_{yy} are defined in (2.3.5). The minimum residual sum of squares on the graph in the xy plane is achieved with the value p^* that minimizes F, along with

$$A^* = \frac{S_{yf}}{S_{ff}}.$$

Theorem 2.3.1 defines a function $F(p)$ whose minimizer is the desired result of the empirical modeling problem. The function F has a complicated formula but is easy to compute numerically for a given value of p. The standard method of optimization given in calculus books does not work well for problems like this, because finding a derivative formula for F is challenging and the resulting equation $F' = 0$ would have to be solved numerically. The brute force method of computing many function values, as we did in Example 2.3.2, works if we don't mind the amount of computer time needed to produce a graph. Alternatively, there are fully numerical methods that work well. One of

these is described in Appendix C and implemented as the author's *findmin.m*, which *Semilin.m* uses to determine best-fit parameter values for any semilinear model.

▶ **Example 2.3.4** A graph of transmissions per day of a secret versus individuals who already know the secret, from a numerical simulation, is presented in Fig. 2.3.1. Clearly, the number of transmissions is necessarily 0 when either everyone knows the secret or nobody knows it, so it makes sense to use a model that forces $T(0) = T(1) = 0$. It seems reasonable to assume there is a parameter A that gives the amplitude of the function family. Thus, $T = Ay(1 - y)$ is a reasonable choice for a minimal model, where y is the fraction of the population that knows the secret. Alternatively, we can seek a more complicated model that allows the maximum on the plot to be shifted away from 0.5, as the data seems to suggest. One way to achieve this is with the model

$$T = Ay(1 - y)(1 + py)\,. \tag{2.3.6}$$

If p ranges among nonnegative values, the maximum of T occurs in the range $[1/2, 2/3]$, which is enough of a range to accommodate the skewness of the graph. The MATLAB program *semilin.m* with the simulation data in Table 2.3.2 yields the best-fit curve seen in Fig. 2.3.4. □

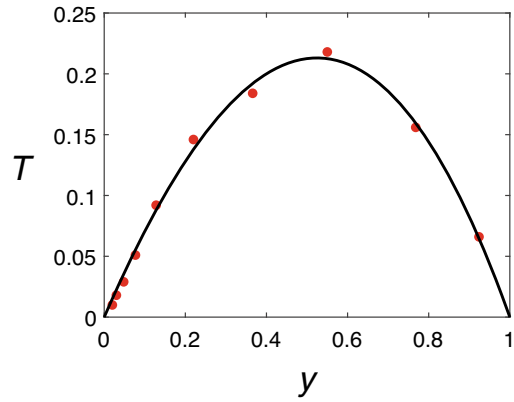


Fig. 2.3.4 The best fit of the model $T = Ay(1 - y)(1 + py)$ to the data of Fig. 2.3.1

Check Your Understanding 2.3.3:
Find the approximate parameters p^* and A^* and the corresponding residual sum of squares for Example 2.3.3.

2.3.4 To Linearize or Not?

When we rewrite the power function model in linear form, we are minimizing the total discrepancy on a graph of $\ln y$ versus $\ln x$ rather than a graph of y versus x. This makes a noticeable difference, as we see in the plots of Fig. 2.3.3. Each of the fitted models is best on the graph corresponding to the method used. The semilinear result seems almost to ignore the first point in the logarithmic plot;

however, the greater sensitivity to this specific data point exhibited by the linearized model exerts too much influence on its graph in the xy plane. The lesson is quite clear.

> Models must be fit to data using the variables that serve as the axes on the most meaningful graph of the data.

In the case of the power function model, the most meaningful graph depends on the context. Where the range of x values is small, as in our example, the most meaningful graph is generally y versus x. If the range of x values spans several orders of magnitude, then the graph of $\ln y$ versus $\ln x$ will likely be more meaningful. Linearization is usually preferable with exponential functions. Here, relative changes in y tend to be more meaningful than absolute changes, so it is almost always best to use the linearized form rather than the original form. For most other functions, the total discrepancy should be measured on the original graph. Section 2.5 offers a case study for one such function.

Check Your Understanding Answers

1. $A^* = 0.842$, RSS $= 0.000563$.
2. $p^* = 0.219$, $A^* = 0.766$, RSS $= 0.000298$.

Problems

The MATLAB file *Semilin.m* contains a script that uses the method of this section to fit a semilinear model to data.

2.3.1* Use semilinear least squares with the data in Problem 2.2.4 to fit the exponential model without linearization. Compare the parameter results from the two calculations. Plot the two resulting models as y versus t and again as $\ln y$ versus t. Discuss the results.

2.3.2 (Continued from Problem 2.2.2.)

(a) Use the linearization method to fit the model $y = Ae^{-kt}$ to the data from Problem 2.2.2.
(b) Use the semilinear method to fit the model and data from part (a).
(c) Find the residual sums of squares for the results of parts (a) and (b) on graphs of y versus t and on graphs of $\ln y$ versus t.
(d) Which is better, the linearization result or the semilinear result?

(This problem is continued in Problem 2.4.7.)

2.3.3 (Continued from Problem 2.2.5.)
Use semilinear least squares with the data in Table 2.2.8 to fit the exponential model without linearization. Compare the parameter results with Problem 2.2.5, which used the same data. Why are the results different? Plot the two resulting models as y versus t and again as $\ln y$ versus t. Draw a reasonable conclusion from your observations.

2.3.4* Use semilinear least squares to fit the model $y = Ax/(1 + px)$ to the data in Table 2.3.1. Compare the results with Example 2.3.2. Plot both models together as y versus x. Draw a reasonable conclusion from your observations.

2.3.5 Table 2.3.4 shows data for average lengths in centimeters of Atlantic croakers (a species of fish) caught off the coasts of three states. Use this data to fit the von Bertalanffy growth equation,

$$x(t) = x_\infty(1 - e^{-rt}),$$

where $x(t)$ is the length of the fish, x_∞ is the asymptotic maximum length, and r is a positive parameter.[15] How well does the model fit the data?

Table 2.3.4 Average length in centimeters of Atlantic croakers from New Jersey, Virginia, and North Carolina [4]

Age	1	2	3	4	5	6	7	8	9	10
NJ	30.3	31.1	32.4	34.2	35.0	34.8	37.4	36.6	36.1	37.4
VA	25.8	28.9	31.8	34.0	35.2	36.1	37.4	40.2	40.2	40.3
NC	24.6	27.3	29.7	33.1	35.2	37.2	37.8	38.4	37.7	38.1

2.3.6 (Continued from Problem 1.2.9.)

(a) Fit the model $y = Ax^p$ to your *P. speedius* data from Problem 1.2.9 using linearized least squares.
(b) Fit the model $y = Ax^p$ to the same data using the semilinear method.
(c) Plot the model from part (a) along with the data on a graph of $\ln y$ versus $\ln x$. Repeat for the model from part (b). Compare the visual appearances of the two plots.
(d) Plot the model from part (a) along with the data on a graph of y versus x. Repeat for the model from part (b). Compare the visual appearances of the two plots.
(e) Describe and explain any conclusions you can draw from this set of graphs.

(This problem is continued in Problem 2.4.5.)

2.4 Model Selection

After studying this section, you should be able to:

- Use the Akaike information criterion to compare the statistical validity of models as applied to a given set of data.
- Choose among models based on a variety of criteria.

It has long been understood that simplicity, as well as accuracy, must be figured into scientific judgment. This idea is enshrined in the scientific principle known as "Occam's Razor" in attribution to the fourteenth-century philosopher William of Ockham, in spite of its absence from any of Ockham's extant writings and its appearance in writings that predate Ockham. The form that has been passed down over time was written in 1639 by a British scientist named John Punch, translating from the Latin original as "Entities must not be multiplied beyond necessity." This is usually interpreted as a philosophical injunction to prefer simpler explanations over complex ones; however, we will see that

[15] It would be better to fit the data for individual lengths rather than averages; however, the raw data sets are quite large and not generally available.

it is possible to offer a mathematical interpretation that is simultaneously more correct as science, a more faithful rendering of Punch's statement, and (since 1974) capable of quantification. We'll return to this idea after further preparation.

2.4.1 Quantitative Accuracy

Quantitative accuracy is clearly an important criterion for choosing among possible models. It can easily be measured by the residual sum of squares, but comparisons can only be made when the residual sums of squares are measured on the same graph.

▶ **Example 2.4.1** In Examples 2.2.3 and 2.3.2, we fit a power function model using linear least squares on the linearized data and semilinear least squares on the original data, given in Table 2.3.1, with results shown in Fig. 2.3.3. Each set of parameter values wins on its "home field"; the semilinear result is clearly better when viewed on a plot of y versus x, while the linearized result is clearly better when viewed on a plot of $\ln y$ versus $\ln x$. The question of better fit in this case is not settled mathematically, but by the choice of which graph to use to measure errors. Unless we have a scientific reason for considering a graph of logarithms to be more meaningful than a graph of original values, we should use the semilinear method.[16] □

▶ **Example 2.4.2** The data in Table 2.1.1 was used to fit the model $y = mx$ in Example 2.1.2, with a residual sum of squares 106.1, and to the model $y = b + mx$, with residual sum of squares 100.4. The two-parameter linear model produced a lower total discrepancy than the one-parameter model; hence, it is more accurate. □

2.4.2 Complexity

We saw that the model $y = b + mx$ produced a more accurate result in Example 2.4.1 than the model $y = mx$. It could not have been otherwise because $y = mx$ is a special case of $y = b + mx$. If the optimal b for the latter model happens to be 0, then the models will be equally accurate; otherwise, the two-parameter model is bound to be better.

Before we conclude that model selection is usually a simple matter, we turn to the question "Shouldn't we *always* use the model that has the smallest residual sum of squares?" The answer to this question is an emphatic *NO*, as the following example makes clear.

Table 2.4.1 A small partial data set for consumption rate y as a function of prey density x

x	0	20	40	60	80
y	0	7	9	21	25

[16] For models that have more than one nonlinear parameter, one can use a fully nonlinear method. These can be found in any mathematical software package, such as MATLAB or R. When using other packages, such as spreadsheets, to fit exponential or power function models to data, it is important to look at the documentation to find out which method of evaluating total discrepancy is being used.

Table 2.4.2 A larger data set for consumption rate y as a function of prey density x, containing all the data in Table 2.4.1 along with additional data

x	0	10	20	30	40	50	60	70	80
y	0	2	7	10	9	14	21	20	25

▶ **Example 2.4.3** Suppose we collected the *P. steadius* data from Sect. 2.1 in stages. The initial data set, in Table 2.4.1, consists of every other point from the first part of the full data set. As noted in Example 2.4.1, we could get a lower residual sum of squares for $y = b + mx$ than for $y = mx$ simply because the second model is a special case of the first. If our goal is minimum residual sum of squares, we can just add more parameters. A fourth-degree polynomial has five parameters, which means that we can find one whose graph passes through all five data points, giving us a residual sum of squares of 0. The result is the model

$$y \approx 1.1375x - 0.0628x^2 + 0.00134x^3 - 0.00000859x^4 .$$

On the basis of quantitative accuracy for the data points used to obtain the parameters, this model is perfect. However, the graph of this model shows a problem. Figure 2.4.1a shows the five points used for the fit, the fourth-degree polynomial that passes through the points, and the $y = mx$ least squares line $y = 0.313x$. The sharp curves in the polynomial are suspicious.

Table 2.4.2 includes the full set of data. A useful model should remain reasonably good when more data is added; however, Fig. 2.4.1b shows that the fourth-degree polynomial has very little predictive value for the extra data in this case. If we fit a fourth-degree polynomial to the larger set of data, the graph of the best fit is significantly changed. Meanwhile, the model $y = 0.313x$ looks reasonably good with the larger set of data. □

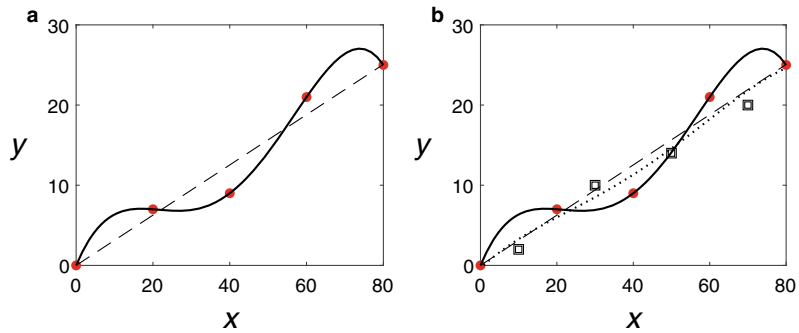


Fig. 2.4.1 The linear and fourth-degree polynomial models fit to the data in Table 2.4.1; **b** includes the additional data in Table 2.4.2 and a dotted line that shows the best-fit quartic polynomial for the full data set

Example 2.4.2 shows that the value of simplicity is more than philosophical; simpler models are more likely to be able to predict new data points than models that are focused too strongly on reproducing the given data. We can recast Punch's pithy statement of Occam's Razor in a way that is more mathematical:

Additional parameters should not be added to models unless the increased accuracy justifies the increase in complexity.

2.4.3 The Akaike Information Criterion

We have now established the idea that quantitative accuracy needs to be balanced against complexity. But how do we do that in practice? An ingenious solution to this problem was first presented in a 1974 paper by Hirotugu Akaike [1]. Akaike combined statistics and information theory to define what has become known as the Akaike Information Criterion, or AIC.[17] The theoretical underpinning of this result is very subtle, but the result itself is quite simple. AIC uses the number of parameters that have to be fit using statistics (K) as the quantitative measure of complexity and combines this with the residual sum of squares as

$$\text{AIC} = n \ \ln\left(\frac{\text{RSS}}{n}\right) + 2K \,,$$

where n is the number of points in the data set. The count for K must include the statistical variance along with all of the model parameters; for example, the simple linear model $y = mx$ has $K = 2$. To avoid confusion between statistical parameters and model parameters, it is more convenient to define AIC by

$$\text{AIC} = n \ \ln\left(\frac{\text{RSS}}{n}\right) + 2k + 2 \,, \tag{2.4.1}$$

where k is the number of parameters in the model. Whether to count model parameters or statistical parameters is a distinction without a difference. The actual values of AIC are unimportant. What matters is that smaller values represent a higher level of statistical support for the corresponding model. The difference between two AIC values is unchanged when both are augmented by the same constant.

▶ **Example 2.4.4** In Examples 2.1.2 and 2.2.1, we obtained two results from linear least squares for the data set in Table 2.1.1:

$$y = 0.267x \,,$$

$$y = 1.175 + 0.255x \,.$$

As an additional empirical option, we can also determine a best-fit parabola,[18]

$$y = 0.197 + 0.300x - 0.00032x^2 \,.$$

Finally, the best fit for a model called the Holling Type II model [10] (see Sect. 3.2) is

$$y = \frac{0.312x}{1 + 0.0016x} \,.$$

[17] A modification called the corrected AIC or AICc is also in common use. I recommend the original because of the work by Shane Richards [15]. There is also a Bayesian Information Criterion (BIC), which is only appropriate under the assumption that one of the possible models is the "true" one. BIC is most often used when trying to identify a probability distribution rather than a mathematical model.

[18] See Project 2.

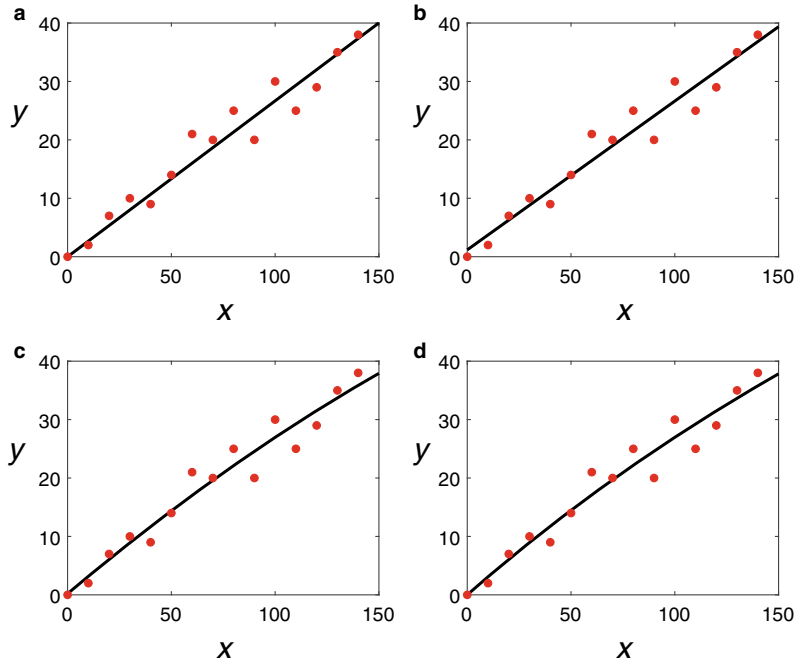


Fig. 2.4.2 The data and models of Example 2.4.3: **a** $y = mx$, **b** linear, **c** quadratic, **d** Holling type II

Table 2.4.3 summarizes the results and Fig. 2.4.2 displays the data and the various models. The model $y = mx$ has the smallest AIC value. The very slight improvements in accuracy for the other models do not quite compensate for the additional complexity.

Table 2.4.3 Comparison of four models for the data in Table 2.1.1 using AIC

Model	RSS	n	k	$n \ln\left(\frac{\text{RSS}}{n}\right)$	$2(k+1)$	AIC
$y = mx$	106.1	15	1	29.3	4	33.3
$y = Ax/(1+px)$	95.5	15	2	27.8	6	33.8
$y = b + mx$	100.4	15	2	28.5	6	34.5
$y = b + mx + ax^2$	96.1	15	3	27.9	8	35.9

Check Your Understanding 2.4.1:

In Sect. 2.3, we obtained residual sums of squares of 0.000563 for the model $T = Ay(1+y)$ and 0.000298 for the model $T = Ay(1+y)(1+py)$ using a data set with 10 points. Find the corresponding AIC values and determine which of the two models AIC recommends.

2.4.4 Choosing Among Models

Armed with the AIC, we are now ready to address the issue of model selection.

Choosing a model is a matter of informed judgment. The best choice using empirical criteria is determined by the AIC; however, AIC results are only as good as the data set being fit. When two models have very close AIC scores with a given data set, there is a possibility that the rankings would be reversed with a different data set. It is reasonable to ask how much of a difference in AIC should be considered definitive. We consider two separate cases.

When the Simpler of a Nested Pair has Lower AIC

A pair of models is said to be *nested* if one can be obtained from the other merely by setting one of its parameters to a constant, usually zero. For example, $y = mx$ and $y = b + mx$ are a nested pair. So is the pair of $y = mx$ and $y = Ax/(1 + px)$, because the latter model with $p = 0$ differs from the former model only in the symbol used for the parameter. For a nested pair, the best the simpler model can do is to achieve the same residual sum of squares as the more complicated one. In this case, the complexity penalty gives the simpler model an AIC advantage of 2. Thus, the required AIC difference needed to choose the simpler model must be some value less than 2. In practice, we should reject the more complicated model when its AIC difference compared to the simple model is more than some threshold. The following rule of thumb is a good choice:

> Model Selection Rule 1: If the simpler of two models in a nested pair has an AIC more than 1 unit less than the more complicated model, we are justified in rejecting the more complicated model.

▶ **Example 2.4.5** In Example 2.4.3, the lowest AIC was achieved by the simplest model, and each of the other models combine with the simplest as a nested pair. The AIC difference between $y = mx$ and $y = b + mx$ is larger than 1; hence, we can confidently reject the more complicated model. With an AIC difference of only 0.5 between $y = mx$ and $y = A/(1 + px)$, we cannot reject the more complicated model, and should consider non-statistical factors (see Example 2.4.4). The simpler model and the quadratic model are also a nested pair, with two extra parameters. While the residual sum of squares for the quadratic is lower (as it must be), the difference is nowhere near enough to justify adding two parameters. □

Other Cases

Suppose we have two models (not a nested pair) with an equal residual sum of squares, and model B has one more parameter than model A. In this case, there is no additional accuracy to trade-off for the extra complexity of model B. Clearly, we should reject model B. More generally, this suggests a rule of thumb:

> Rule 2: In cases where the model with the lower AIC is not the simpler model of a nested pair, we are fully justified in rejecting any model whose AIC is 2 or more units larger than that of the preferred model.

▶ **Example 2.4.6** In Example 2.3.2 and Problem 2.3.4, we obtained best fits for the power function and Holling type II model $y = Ax/(1 + px)$ for the data in Table 2.3.1. We can also try polynomials of various degrees. Table 2.4.4 summarizes the results. The AIC value for the Holling model is 2.2 units better than that for the second-place cubic model. Rule 2 suggests that we should reject all but the Holling model. The plots in Fig. 2.4.3 illustrate the differences between the two best models. The cubic model produces a lower residual sum of squares, but only by showing a very suspicious increase in slope at the end. □

Table 2.4.4 Comparison of four models for the data in Table 2.3.1 using AIC

Model	RSS	n	k	$n \ln\left(\frac{\text{RSS}}{n}\right)$	$2(k+1)$	AIC
$y = Ax/(1+px)$	43.2	15	2	15.9	6	21.9
$y = b + mx + ax^2 + cx^3$	38.5	15	4	14.1	10	24.1
$y = ax^p$	62.9	15	2	21.5	6	27.5
$y = b + mx + ax^2$	88.1	15	3	26.6	8	34.6

2.4.5 Some Recommendations

AIC has the advantage of being quantitative, but we should not overestimate its value. Nonquantitative criteria must often be considered. Here are some things to keep in mind.

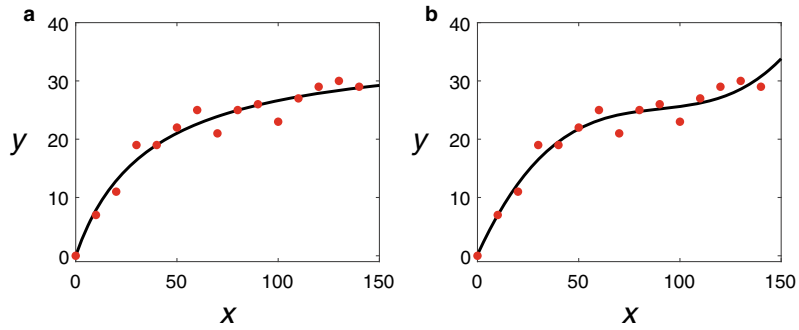


Fig. 2.4.3 The data and the two best models of Example 2.4.4: **a** Holling type II, **b** cubic

- Empirical models that fit data well over a given range may not work over a larger range. Empirical models should not be extrapolated.
- The ranking of models for a set of data depends on the location of the points on a graph, which can be expected to vary because of random factors. If we collect a new set of data from the same experiment, the model selected by AIC for the first data set may not be the best fit for the second set.
- Part of the amazing success of models in the physical sciences owes to their mechanistic derivation from physical principles. Empirical models, by definition, do not attempt to do this. We can have more confidence in a model that can be obtained mechanistically than in one that is strictly empirical.

We should usually prefer models that have a mechanistic justification, even if such models have AIC values that are not significantly better than the best empirical model.

▶ **Example 2.4.7** In Example 2.4.3, the lowest AIC score is for the model $y = mx$, but with only a slightly higher score for the Holling type II model. A different data set for the same experiment could yield a lower AIC score for the Holling model, or the difference could be greater than in the example. Both models have a mechanistic justification, which the other two models lack. Either is arguably the best choice. In practice, it is generally wise not to use a more complicated model than necessary, so the simple $y = mx$ is probably the better choice. □

Check Your Understanding Answers

1. The simpler model has an AIC of −93.8, while the more complicated model has −98.2. The actual values do not matter; the lower AIC gives support for the choice of the more complicated model in this case.

Problems

2.4.1 (a) Compute the residual sum of squares for the model

$$y \approx 1.1375x - 0.0628x^2 + 0.00134x^3 - 0.00000859x^4$$

with the full data set in Table 2.4.2.

(b) Compute the residual sum of squares for the model $y = 0.313x$ with the full data set in Table 2.4.2.
(c) Both of the models used in parts (a) and (b) were obtained by fitting to the partial data set in Table 2.4.1. Which has the better quantitative accuracy for the full data set?

2.4.2* The points corresponding to x values of 0, 10, 20, and 30 in Table 2.4.2 seem to lie close to the least squares line $y = 0.313x$. We can find a polynomial of the form $y = ax + bx^2 + cx^3$ that passes through the points $(0, 0)$, (h, y_1), $(2h, y_2)$, and $(3h, y_3)$ by the *method of successive differences*. The method yields simple formulas for the three coefficients:

$$c = \frac{y_3 - 3y_2 + 3y_1}{6h^3}, \qquad b = \frac{y_2 - 2y_1}{2h^2} - 3hc, \qquad a = \frac{y_1}{h} - hb - h^2c.$$

(a) Use these coefficient formulas to fit the data from Table 2.4.2 for the x values 10, 20, and 30.
(b) Plot the polynomial from part (a), along with the three data points, on a common graph.
(c) Does the cubic polynomial model work well for these data? Why or why not?

2.4.3 Using the *method of successive differences*, relatively simple formulas can be found to determine the coefficients of a polynomial of degree n that passes through $n + 1$ points having equally spaced x values. Suppose the points are (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4), and $x_4 - x_3 = x_3 - x_2 = x_2 - x_1 = h$. Then the polynomial $y = a + bx + cx^2 + dx^3$ has coefficients given by

$$d = \frac{y_4 - 3y_3 + 3y_2 - y_1}{6h^3}, \qquad c = \frac{y_3 - 2y_2 + y_1}{2h^2} - 3x_2 d,$$
$$b = \frac{y_2 - y_1}{h} - (x_1 + x_2)c - (3x_1x_2 + h^2)d, \qquad a = y_1 - x_1 b - x_1^2 c - x_1^3 d.$$

(a) Use these coefficient formulas to fit the data from Table 2.4.2 for the x values 10, 30, 50, and 70.
(b) Repeat part (a) using the data for the x values 20, 40, 60, and 80.
(c) Plot the polynomials from parts (a) and (b), along with all of the data, on a common graph.
(d) How good are these two models for the full data set?

2.4.4 (Continued from Problems 2.1.4 and 2.2.3.)

(a) Use your *P. steadius* data set from Problem 1.2.8 to fit the Holling type II model.
(b) Compute the AIC for the Holling type II model, the model $y = mx$ (Problem 2.1.4a), and the model $y = b + mx$ (Problem 2.2.3a).
(c) Which model gives the lowest AIC with your data? Combining these results with Example 2.4.3, which model do you recommend should be used for *P. steadius*? [Note: Comparison of AIC values is only meaningful when those values were obtained from one data set.]

2.4.5 (Continued from Problem 2.3.6.)

(a) Use your *P. speedius* data set from Problem 1.2.9 to fit the Holling type II model.
(b) Compute the AIC for the Holling type II model and the model $y = Ax^p$ (Problem 2.3.6b).
(c) Which model gives the lowest AIC with your data? Combining these results with Example 2.4.4, which model do you recommend should be used for *P. speedius*? [Note: Comparison of AIC values is only meaningful when those values were obtained from one data set.]

The remaining exercises of this section use the MATLAB program *PolyLS.m* to fit a polynomial of arbitrary degree to data.

2.4.6* (*Fluorine at the South Pole*) (Continued from Problems 2.2.1 and 2.2.6.)

(a) Use *PolyLS.m* to fit the models $y = b + mx + ax^2$ and $y = b + mx + ax^2 + cx^3$ to the data from Problem 2.2.1.
(b) Use the AIC to compare the results of part (a) with the linear model $y = b + mx$ and the results from Problem 2.2.6 for the model $y = Ax^p$. Plot the two models with the lowest AIC on a common graph along with the data.
(c) Are the AIC differences significant? Is the best model very good?

2.4.7 (Continued from Problems 2.2.2 and 2.3.2.)
Repeat the directions of Problem 2.4.6 with the data from Problem 2.2.2 and comparison to those earlier results.

Problems 2.4.8–2.4.10 use global temperature data [6] to look for evidence of global climate change.[19]

2.4.8 (*Global Temperature*) (Continued from Problem 2.2.8.)
Repeat the directions of Problem 2.4.6 with the 1985–2015 data in the file *GlobalLandTemperatures_July.csv*. Does the temperature data support the global warming hypothesis? Discuss, paying

[19] The needed data sets are available at https://www.math.unl.edu/~gledder1/.

particular attention to the point that what we see is a combination of multiple trends of varying strengths.

(This problem is continued in Project 2A.)

2.4.9 (*Global Temperature*) (Continued from Problem 2.2.9.)
Repeat the directions of Problem 2.4.6 with the 1985–2015 data in the file *GlobalLandTemperatures_January.csv*.

(This problem is continued in Project 2A.)

2.4.10 (*Global Temperature*) (Continued from Problem 2.2.10.)
Repeat the directions of Problem 2.4.6 with the full data set in the file *GlobalTemperatures_July.csv*.

(This problem is continued in Project 2A.)

2.5 Case Study: Michaelis–Menten Kinetics

With limited coverage in the mathematics or statistics curricula, model fitting is learned by scientists primarily from other scientists in their own field. This creates considerable inertia; once adopted, a method tends to remain in use long past what should be an appropriate shelf life.[20] One such example is the Lineweaver–Burk method for fitting the parameters in the Michaelis–Menten equation. This method was initially published in 1934 [11] and is still the most common method used by biochemists to determine the Michaelis–Menten parameters.[21] The poor performance of this method was noted in a 1965 paper by Dowd and Riggs, who found it to be inferior to two other linearization methods [7]. A more thorough statistical study by Atkins and Nimmo in 1975 concluded that the Lineweaver–Burk method was one of the least accurate of the seven methods known at that time [2]. The best method, nonlinear optimization (which is equivalent to our semilinear method), was computationally expensive in 1975, but modern computing has largely eliminated the need to prefer methods on grounds of extreme computational simplicity. Forty-five years after Atkins and Nimmo and roughly 35 years after the routine use of desktop computers, biochemists still learn the Lineweaver–Burk method from textbooks written by biochemists who reproduced what they learned from older biochemists.

In this case study, we repeat the experimental study of Atkins and Nimmo [2], facilitated by modern computing power, to do an updated numerical comparison of the three linearization schemes and the semilinear method.

2.5.1 The Michaelis–Menten Model and its Linearizations

Michaelis[22]–Menten reactions are enzyme-catalyzed reactions in biochemistry. The initial reaction rate (often called the *velocity* of the reaction) v depends on the concentration S of the principal reactant

[20] mi-KAY-lis.

[21] A better method (Hanes–Woolf, see below) had been published previously in 1932, but for some reason, it was Lineweaver–Burk rather than this better method that became standard practice.

[22] Marie Curie was not the only outstanding female scientist to overcome rampant discrimination in the first half of the twentieth century. Maud Menten earned a medical degree in Canada in 1911, but she wanted to do research in biochemistry. At the time, women were not allowed to do scientific research in Canada, so she went to Germany to work with Leonor Michaelis. Their joint work on what became known as Michaelis–Menten reactions was published in 1913 [12]. At the time, it was not uncommon for professors to take all the credit for work done by or with students. To his credit, Michaelis published their joint work under the names "L. Michaelis and Miss Maud L. Menten," giving Menten equal credit for their joint discovery.

(called the *substrate*), according to the model[23]

$$v = \frac{VS}{K+S}, \qquad S, V, K > 0\,,$$

where V is the maximum rate (corresponding to a very large substrate concentration) and K is the semisaturation parameter (the value of S for which the reaction velocity is half of its maximum). Since our aim is to focus on fitting data to the model, we'll improve readability by replacing the variables S and v with the generic variables x and y; thus,

$$y = \frac{Vx}{K+x}. \tag{2.5.1}$$

The Michaelis–Menten model is semilinear as defined in Sect. 2.3, which means that we can fit it to data using the relatively simple method developed in that section and implemented as a MATLAB function *semilinfit*, which is internal to the program *MMfit.m*.

Before the advent of the computer era, using the fully nonlinear method to fit the Michaelis–Menten model required a burdensome amount of hand computation (the semilinear method would have been better, but still burdensome), so it made sense to look for a linearization scheme. Three of these have been proposed.

A. The Lineweaver–Burk linearization scheme uses the reciprocal of the original [11]:

$$\frac{1}{y} = \frac{K+x}{Vx} = \frac{1}{V} + \frac{K}{V}\cdot\frac{1}{x}\,. \tag{2.5.2}$$

B. The Hanes–Woolf linearization scheme, developed independently by Charles Hanes [9] and Barnet Woolf [8], follows from multiplying the Lineweaver–Burk formula by x and rearranging terms [7]:

$$\frac{x}{y} = \frac{K}{V} + \frac{1}{V}\cdot x\,. \tag{2.5.3}$$

C. A third linearization scheme is a bit more challenging to derive, but its equivalence to the original model can easily be checked [7]:

$$y = V - K\cdot\frac{y}{x}\,. \tag{2.5.4}$$

2.5.2 Comparison of Methods

To test the various methods, we begin by constructing a model of real data. In the first step, we assume that the correct parameter values are $K=1$ and $V=1$.[24] We can then construct a data set using the x values $\{0.2, 0.4, 0.7, 1.0, 1.4, 2.0, 2.8, 3.8, 5.0\}$, which include three points to the left of the semisaturation value $x = 1$ and five points to the right, with spacing gradually increasing as the curve flattens out. The standard y values can then be calculated as $y = x/(1+x)$. We complete our construction of model data by assuming that each y value has some random measurement error given

[23] Michaelis and Menten used this model for empirical reasons, but a mechanistic derivation was subsequently discovered. We will see this derivation in Sect. 3.8.

[24] We'll see in Sect. 3.6 that there is no loss of generality in making this assumption.

as a percentage of the true value. Specifically, each y_i is multiplied by a random number drawn from a normal distribution with mean 1 and a standard deviation of 0.02. Thus, two thirds of the values have a percentage error less than 2%, while the rest have a larger error, but few more than 4%.

We then use each of the three linearization methods and the semilinear method to fit this randomly modified data set and collect the results for the fitted parameter values and the residual sum of squares. We automate this process to do 40000 runs and compute the means of the parameter values and the AIC difference between each model and the best model for that data set. Note that each model is minimizing fitting error on a different graph but that the residual sums of squares used to calculate AIC are based on the graph of y versus x.

Table 2.5.1 Comparison of three linearization methods for the Michaelis–Menten equation

Mean Result	Lineweaver–Burk	Hanes–Woolf	Method C	semilinear
ΔAIC	2.93	0.24	0.67	0
K	1.0009	1.0007	0.9976	1.0009
V	1.0001	1.0000	0.9989	1.0003

Table 2.5.1 shows the results of the computer experiment. The most important result by far is ΔAIC, which is the mean difference in AIC between the method identified in each column and whichever method gave the best results for that run.

Two of the ΔAIC entries in the table stand out. The entry for the semilinear method is 0, meaning that the semilinear method was always the best. This is guaranteed by the choice to measure the fitting error on a graph of y versus x, which is the only biologically appropriate choice.[25]

The second glaring result in the table is the horrible performance of the Lineweaver–Burk method. An AIC difference greater than 2 means that we would be justified in rejecting the model. A mean AIC difference of 2.93 is huge, considering that we are not looking at an incorrect model, but merely an incorrect choice of parameters. The AIC difference was greater than 2.0 46% of the time, and it was less than 0.5 only 25% of the time.

The horrible performance of the Lineweaver–Burk model is a definitive argument against its use, even in the pre-computer era. This is especially true when the Hanes–Woolf linearization yields results that are far better. Granted that AIC had not been discovered in 1965, the concept of residual sum of squares was well known at that time. A simple comparison of the three linear methods, as done by Dowd and Riggs [7], should have led to the uniform acceptance of the Hanes–Woolf linearization as the best of the three.

The results for the parameters K and V are also worth a brief mention. We would expect that the average of parameter values obtained from data sets that have unbiased errors should be very close to that of the error-free data. This is the case for three of the methods, but the deviations from the expected means show a slight bias in Method C. We can conjecture why this is so. The least squares method was implemented under the assumption that all of the error was in the vertical coordinate of each data point. The particular linearization for Method C has a horizontal coordinate that incorporates y; hence, the error imposed by the experiment has the effect of creating error in both coordinates for this one of the three methods.

[25] This is a theoretical statement. In practice, the computed RSS for the semilinear model can occasionally have a very slightly higher calculated value because of round-off error in computer calculations.

2.5.3 Conclusion

There can be no question that the semilinear least squares method produces the best fit on a plot of the original data, nor is there any reason in the world of fast computing to settle for a method that is not as good simply because it is faster for hand computation. In general, a good understanding of the theories of various methods for solving problems helps us to identify cases, such as this one, where older methods should be replaced by newer computer-intensive methods.

It is worth noting that in his original 1932 paper, Charles Hanes pointed out the flaw of linearization methods, which we have examined in Sect. 2.3 and here, namely that they fit the model on the wrong graph. Hanes was fully aware that he had found a method whose primary value was mathematical expediency, and he would have been very quick to abandon that method if modern computing power had suddenly become available.

Also worthy of note is the asymptotic behavior of the graphs for each of the linearization schemes. It is natural that there should be data that approaches the point $(0, 0)$. On the Lineweaver–Burk plot, both coordinates approach infinity for points where x is small, which is a problem. According to the model, $y/x \to (V/K)$ as $x, y \to 0$. Thus, points with small x approach the finite point $(0, K/V)$ on the Hanes–Woolf plot and the point $(V/K, 0)$ on plot C. This is why the errors for both of these methods are far less than those for the Lineweaver–Burk method.

While the semilinear method is clearly the correct one to use, the Hanes–Woolf linearization actually has an important use. It is always possible that one can make an error in writing a computer program or in using it with a given set of data. We must always be looking out for ways to confirm our results. The Hanes–Woolf linearization gives results that are very close to the best fit, and it has the conceptual value of being suitable for visualization. Plotting the data in the x/y versus $1/x$ plane gives a visual indication of how much measurement error is present, for example, there could be one point that is a clear outlier. One can draw an approximate best-fit line on a Hanes–Woolf plot by hand and use basic algebra to obtain the slope and intercept, yielding a crude estimate of the best-fit parameter values. These would then serve as a check on the results obtained from a computer program.

Problems

2.5.1 Modify *MMfit.m* by omitting the last point from the list of x values. Run the program and record the results. Then put the last point back in and remove the first point. Describe and explain the changes in the results compared to Table 2.5.1. (Hint: Think about where the points wind up on the linearized graphs.)

2.5.2* Table 2.5.2 contains a small data set from an experiment that used an enzyme from pig livers to study nicotine processing by mammals.

(a) Fit the Michaelis–Menten model to the data using the semilinear method. (Note that you can get a rough estimate of K from the data by recognizing that K is the value of S for which v is half of its maximum.) Plot the data and the model results on a common graph and record the residual sum of squares.
(b) Use the Lineweaver–Burk linearization to fit the model to the data. Plot the model results on the graph from (a) and determine the AIC difference.
(c) Repeat (b) using the Hanes–Woolf linearization.
(d) Discuss the results.

(These calculations can be done using *MMfit.m* with a few minor modifications: (1) set `T=1`, (2) add a line of data for `y`, and (3) comment out the line in the loop that calculates `y`.)

Table 2.5.2 Substrate concentration S and reaction velocity v for nicotinamide mononucleotide adenylyltransferase in pig livers [3]

S	0.138	0.220	0.291	0.560	0.766	1.46
v	0.148	0.171	0.234	0.324	0.390	0.493

2.5.3* Repeat Problem 2.5.2 using the data in Table 2.5.3. This is the data set from Hanes's 1932 paper that introduced the Hanes–Woolf linearization.

Table 2.5.3 Substrate concentration S and reaction velocity v for amylase hydrolysis of starch [9]

S	1.078	0.647	0.431	0.216	0.129	0.0863	0.0500	0.0400	0.0300
v	0.438	0.433	0.386	0.334	0.300	0.238	0.175	0.154	0.135

2.6 Project

This chapter's project looks at some large sets of climate data in an effort to try to obtain evidence for or against the hypothesis of human-induced global warming. Students need to understand that there is a difference between saying that a claim is true and saying that a certain body of evidence supports the claim. Similarly, there is a difference between saying that a body of evidence is insufficient to support and saying that the claim is untrue.

Project 2A: Global Climate Data Analysis

Problems 2.2.8–2.2.10 and 2.4.8–2.4.10 use global temperature data [6] to look for evidence of global climate change.[26] Problem 2.2.11 addresses the same issues using data on grape harvest dates. Explore one or more of the associated data sets in more depth by using the MATLAB program *PolyLS.m* to fit a quadratic polynomial in addition to the linear polynomials used earlier. Try modifying the ranges of years that you use. Robust results should not be especially sensitive to small differences, such as starting or finishing 10 years earlier or later. Draw what conclusions you think are justified, but do not overreach. In particular, if we want to look for evidence of global warming, what sort of data would be most useful? (Hint: If you have one glass of ice water in the refrigerator and another on the counter, how much of a temperature difference do you expect?)

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Mechanistic Modeling

3

In Sect. 2.2, we modeled radioactive decay with an exponential model. We were able to fit the data quite well, but an empirical justification such as this limits a model's explanatory value. Would an exponential model be a good fit with a different data set for the same substance? What about a data set that extends the total time of the experiment or a data set for a different radioactive substance? The same questions arise in the Michaelis–Menten model, which we saw in Sect. 2.5. Empirical modeling cannot answer these questions because empirical reasoning must *begin* with the data.

An alternative modeling approach is *mechanistic modeling*, in which we obtain a model from assumptions based on theoretical principles. Sometimes, a mechanistic justification can be found for a model we have already identified empirically, as we will see with our exponential model for radioactive decay and the Michaelis–Menten model. In these cases, the model gains explanatory value. In other cases, we may be able to discover a model not previously identified empirically.

Mathematical models are constructed from components at several levels of detail. Sections 3.1 and 3.2 address specific processes of transition and interaction, respectively. Section 3.3 introduces compartment analysis, which prescribes large-scale structure of some models. Because the COVID-19 pandemic has highlighted the importance of epidemiological models, the presentation in Sect. 3.3 focuses on the SEIR epidemic model, which is the best starting point for epidemic modeling.[1] We follow this model development with a section that characterizes key properties of the SEIR model, many of which are shared by more complicated epidemiology models. Section 3.5 is a case study on COVID-19 scenarios from March 2020 and January 2021.

Section 3.6 presents material on equivalent forms of models, progressing from mere variation in symbols through the crucial topic of scaling. The chapter text concludes with three additional case studies. A case study on lead poisoning illustrates how the number of variables in a model can sometimes be reduced without significantly changing the results. A case study on biochemical kinetics provides a mechanistic derivation of the Michaelis–Menten equation. The final case study presents a variety of endemic disease models.

The projects for this chapter include one that uses the SEIR epidemic model of Sect. 3.3, two that extend that model with additional features, one that explores the January 2021 COVID-19 scenario, and one that investigates how much error is caused by using the standard assumption of spontaneous transitions rather than a more realistic transition model.

[1] The better-known SIR model is generally used as an introduction to mathematical epidemiology; however, the omission of an incubation process means that the SIR model is too simple for most epidemic events.

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3.1 Transition Processes

After studying this section, you should be able to:

- Determine the dimensions of quantities that appear in formulas or equations.
- Discuss differential equation models of transition processes.
- Use the MATLAB program *Vaccination.m* to produce graphs for various vaccination scenarios.

The focus of this section is on transition processes, such as natural death or recovery from a disease. Before beginning this development, we need to take a brief look at a key modeling tool, dimensional analysis.

3.1.1 Dimensional Analysis

Dimensional analysis is based on a very simple idea: all models must be dimensionally consistent. This is a generalization of the colloquial saying, "You can't add apples to oranges." There are three requirements for dimensional consistency.[2]

Rules for Dimensional Consistency

1. Quantities can be added together or set equal to each other only if they have the same dimension.
2. The dimension of a product is the product of the dimensions.
3. The argument of a transcendental function, such as a trigonometric, exponential, or logarithmic function, must be dimensionless.

▶ **Example 3.1.1** A circle of radius r has circumference $2\pi r$. The ratio of circumference to radius is therefore 2π, which is, by definition, the radian measure of a full circle. The radian measure of any other angle is similarly defined as the ratio of the length of the corresponding circular arc to the radius of the circle. Because both circumference and radius are lengths, the number π and the radian measure of an angle are dimensionless. The rule that arguments of trigonometric functions must be dimensionless says that these should be given in radians rather than degrees. □

▶ **Example 3.1.2** In functions such as $\sin kt$, the argument does not necessarily correspond to a geometric angle, so the term "radians" does not really apply. Nevertheless, dimensional consistency requires the quantity k to have dimension 1/time. □

▶ **Example 3.1.3** The differential equation $\frac{dy}{dt} = -ky$ must be dimensionally consistent. The independent variable t is a time. If y is a mass, then the derivative dy/dt has dimension mass per time. The right side of the model must also be mass per time, and so the dimension of k must be 1/time. (Note that the dimension of y could have been anything, since y appears in the products on both sides.) □

[2] Note the distinction between dimensions, such as length, and the associated units of measurement, such as meters, feet, and light-years.

3.1.2 Spontaneous Transition

In Sect. 1.3, we used the argument that the decay probability for an individual particle should be independent of history to obtain the model $y = y_0 e^{-kt}$ for the number of radioactively decaying particles still undecayed at time t from a body of y_0 particles at time 0. Another mechanistic approach to this model is based on expressing the rate of a process as a function of the state of the system. Here, suppose we assume that the rate of decay is proportional to the number of particles present. This gives us the model

$$\frac{dy}{dt} = -ky, \qquad y(0) = y_0\,, \tag{3.1.1}$$

for some $k, y_0 > 0$, which leads to the same exponential function $y = y_0 e^{-kt}$ as the argument from probability. The constant k was shown earlier to be the reciprocal of the mean decay time; it can also be found by fitting the exponential model in linearized form to data.

We can also rewrite the model as

$$-\frac{1}{y}\frac{dy}{dt} = k\,; \tag{3.1.2}$$

thus, k is the decay rate $-dy/dt$ divided by the amount y, which is called the *relative rate of decay*. The conceptual model for radioactive decay from a process standpoint is that the relative rate of decay is constant. This idea generalizes to a rule of thumb for how to model continuous processes.

Conceptual models for continuous processes should be based on rates of change.

3.1.3 "Let the Buyer Beware"

As noted in Sect. 1.4, all mathematical models come with a disclaimer. Results obtained from their analysis are only guaranteed to be true for the corresponding conceptual models. Whether they are also true for the real-world setting depends on the quality of the approximation process, which is non-mathematical. It is common to use the decay model for all sorts of transitions, including ones that are clearly not spontaneous. For example, recovery from infection results from the action of the immune system, which is not spontaneous, yet the standard epidemiology models, including the one we will develop in Sect. 3.3, use spontaneous decay models. All modeling choices should be viewed with caution until they have been validated for the scenario of interest; hence, it behooves us to consider more sophisticated transition models.

3.1.4 A Model for Vaccination

Epidemiology models sometimes include a vaccination process, typically modeled as a single-phase transition.[3] Let $V(t)$ be the fraction of the population that has already been vaccinated and $W(t)$ the fraction that is waiting for vaccination. The vaccination rate is then $V'(t) = -W'(t)$, since everyone willing to be vaccinated is either vaccinated or waiting. If we are using a single-phase transition model, the transition is for leaving class W, which gives us the differential equation model

$$\frac{dW}{dt} = -\phi W\,, \qquad W(0) = a\,, \tag{3.1.3}$$

[3] This material is taken from [11].

where ϕ is a rate constant that represents the reciprocal of the mean time to be vaccinated and $a < 1$ is the fraction of the population that will accept the vaccine. The parameter a is limited by medical restrictions, such as no vaccines for young children, and vaccine refusal, which has been significant for the COVID-19 vaccine in most societies.

There are two things wrong with this model:

1. It fails to account for supply limitations in the early stages of vaccination, when there are not enough doses for everyone who wants to be vaccinated;
2. It fails to account for distribution limitation—even if supply is infinite, there are a limited number of doses that can be administered per day.

A suitable model must deal with both of these issues.

To correct the first issue, we could replace the constant ϕ with $\phi g(t)$, where g is a function that starts at 0 and increases to 1 at some time τ that marks the end of the restriction from limited doses. The simplest such model assumes that g is linear; thus,

$$\frac{dW}{dt} = -\phi g(t) W\,, \qquad W(0) = a\,, \tag{3.1.4}$$

where

$$g(t) = \min\left(\frac{t}{\tau}, 1\right)\,. \tag{3.1.5}$$

There is good reason to think that this second model will still be inadequate. While it accounts for limited supply, it fails to account for limited distribution capacity. To see this, consider what would happen in a real-world scenario, if supply were unlimited. Initially, we should expect the vaccination rate to be largely independent of the number of people waiting to be vaccinated, as it would be limited instead by the number of available vaccination appointments per day. Our current model has a rate that depends linearly on W, whereas a correct model should use a rate function that depends strongly on W only when W is small.

In terms of the process, it is reasonable to think of vaccination as analogous to enzyme kinetics, which was introduced in Sect. 2.5, and predation, which we will consider in Sect. 3.2. There is a large pool of patients (substrate/prey) and a finite number of vaccinators (enzyme molecules/predators). Initially, each vaccinator spends nearly all of its time processing patients. Only when the size of the patient pool becomes relatively small do the vaccinators have to spend time searching for patients. This description leads to the supply-adjusted Michaelis–Menten/Holling type 2 model,

$$\frac{dW}{dt} = -\frac{\phi g(t) W}{K + W}\,, \qquad W(0) = a\,, \tag{3.1.6}$$

with ϕ the theoretical maximum rate of vaccination ("achieved" as $W \to \infty$ and $g = 1$, except that in actuality $W \leq 1$), and K the value of W for which the rate is half of the theoretical maximum. We should also consider the Holling type 3 model,

$$\frac{dW}{dt} = -\frac{\phi g(t) W^2}{K^2 + W^2}\,, \qquad W(0) = a\,, \tag{3.1.7}$$

which accounts for the observation in some predator–prey systems that searching can be less efficient when the target population is very low.[4]

[4] The Holling models are developed in Sect. 3.2; for now, all that is important is the idea that rates of processes can be limited by processing capacity and that the models given here have that feature.

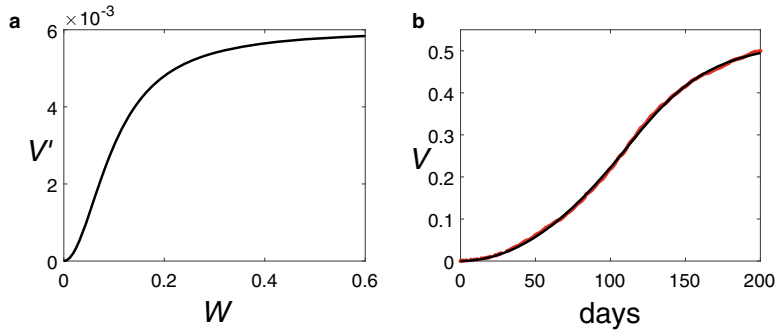


Fig. 3.1.1 **a** The vaccination rate $V' = -W'$ for (3.1.7) with $g = 1$ and best-fit parameters for ϕ, a, and K from CDC data; **b** The vaccinated population fraction V (black) with g from (3.1.5) along with CDC data (red), using best-fit parameters

Data from the United States Centers for Disease Control gives the cumulative number of vaccine doses administered, starting when the first doses were given on December 20, 2020 [3]. We can use this as data for the vaccinated fraction V by equating two administered doses with one person vaccinated; this is essentially equivalent to counting one dose as half of a vaccination, which is a reasonably good assumption based on dose efficacy. Fitting the models to data requires us to compute the best fits for the parameter set (ϕ, a, K, τ). The mathematics for this is outside the scope of this book, but the results are clear: the best model to fit the data is the supply-limited Holling type 3 model (3.1.7), with parameters

$$\phi = 0.0057\,, \quad r = 0.45\,, \quad K = 0.137\,, \quad \tau = 115\,. \tag{3.1.8}$$

Of course, the parameters will be different for different countries, as the supply and logistical difficulties were unique to each. We should expect the model to fit well for most countries, provided best-fit parameter values are computed from local data.

Figure 3.1.1a shows the dependence of vaccination rate on target population, assuming unlimited supply. As long as there are a lot of people waiting for vaccination, the process occurs at a rate close to the distribution-limited maximum (all appointment times are taken, with only a few no-shows); as the number of people wanting vaccination becomes small, it gets more difficult to locate the remaining people and deliver the vaccine to them.[5]

Figure 3.1.1b shows the predicted vaccination fraction over time compared to the data. The fit is unusually good for a model of biological processes.

3.1.5 Multi-Phase Transitions

Suppose a cohort of individuals are infected with a disease at the same time. If disease recovery were a spontaneous process, some of the individuals would recover immediately, while others might take an extremely long time. Instead, disease recovery is a result of changes in the immune system that accumulate over time. If it takes an average of four days to recover from a disease, very few people

[5] In the actual event of COVID-19 vaccination, this is exactly what happened. Initially, some people traveled long distances for the limited number of appointments, such as when a group of the author's acquaintances from Colorado drove to a small Texas town that had more doses than patients, but eventually, it became necessary for the health care system to send out vaccinators to people who were unable to provide either the transportation or the time for themselves, or who were willing but insufficiently motivated.

will recover on the first day; nevertheless, a natural decay model predicts that more people will recover on the first day than on each subsequent day.

We can build better models for recovery and other non-spontaneous transitions by thinking of the transition process as being divided into distinct spontaneous phases. In principle, each phase could have its own rate constant, but this adds parameters without making the model better; hence, we consider a k-phase process in which each phase has rate constant r. If the expected amount of time required for the full process is $1/\mu$, then each of the phases has expected time $1/k\,\mu$ and rate constant (which must have dimension 1/time) $r = k\mu$. In this way, a single parameter for the whole process (μ) determines all of the parameters needed in the model.

To keep things simple, we focus here on a process of two spontaneous phases. Let y be the number of individuals who have not completed the full process and let y_i be the number of individuals who are currently in phase i of the process. Since everyone is initially in phase 1, we have a model

$$\frac{dy_1}{dt} = -ry_1\,, \qquad y_1(0) = y_0, \tag{3.1.9}$$

$$\frac{dy_2}{dt} = ry_1 - ry_2\,, \qquad y_2(0) = 0\,, \tag{3.1.10}$$

$$y = y_1 + y_2\,, \tag{3.1.11}$$

where $r = 2\mu$. Note that there is a flow of individuals from phase 1 to phase 2, indicated by the presence of the term ry_1 in both equations, while only those individuals leaving phase 2 are actually leaving the whole system.

In most circumstances, we are not going to be able to solve the differential equations of mathematical models. The k-phase transition model for a cohort is one of the few exceptions. Equation (3.1.9) is linear and independent of the rest of the system. By inspection,[6] we can see that its solution is

$$y_1 = y_0 e^{-rt}. \tag{3.1.12}$$

This changes (3.1.10) to

$$\frac{dy_2}{dt} = -ry_2 + y_0 re^{-rt}. \tag{3.1.13}$$

This equation can be solved by the integrating factor method, which is part of the standard curriculum of a differential equations course; however, there is a simpler method based on a change of variables. Assume

$$y_2 = e^{-rt} z(t) \tag{3.1.14}$$

for some unknown function z. This assumption will be helpful if it leads to a problem for z that is easier than the problem for y_2.[7] Taking a derivative, we have

$$\frac{dy_2}{dt} = -re^{-rt} z + e^{-rt}\frac{dz}{dt} = -ry_2 + e^{-rt}\frac{dz}{dt}.$$

[6] This phrase means that no formal mathematical method is required, not that the result is obvious to everyone. The reader should simply check that the given formula satisfies both requirements in (3.1.9).

[7] The reader should be puzzled about why anyone would think this change of variables would be helpful. The answer is that it suggests itself based on experience with differential equations that the reader may not have.

After substituting this result into (3.1.13), the equation simplifies to

$$\frac{dz}{dt} = y_0 r. \tag{3.1.15}$$

Hence,

$$z = C + y_0 rt$$

for some unknown C. Substituting this result into (3.1.14) yields

$$y_2 = e^{-rt}(C + y_0 rt).$$

The initial condition for y_2 requires $C = 0$, so we obtain

$$y_2 = y_0 rte^{-rt}.$$

Substituting the results for y_1 and y_2 into (3.1.11) yields the solution

$$y = y_0 (1 + rt) e^{-rt}. \tag{3.1.16}$$

With more tedious calculation, the k-phase transition process can be fully characterized:

Theorem 3.1.1 (k-phase Transition)
The mathematical model for a transition comprised of k spontaneous phases, each with rate constant r, with a total mean time $1/\mu$, and with all individuals beginning as a cohort, consists of the differential equations

$$\frac{dy_1}{dt} = -ry_1, \quad \frac{dy_2}{dt} = ry_1 - ry_2, \quad \cdots \quad \frac{dy_k}{dt} = ry_{k-1} - ry_k,$$

initial conditions

$$y_1(0) = y_0, \qquad y_2(0) = \cdots = y_k(0) = 0,$$

and algebraic equations

$$y = y_1 + \cdots + y_k, \qquad r = k\mu.$$

The total quantity not fully transitioned is then given by

$$Y = \left(1 + T + \frac{1}{2}T^2 + \cdots + \frac{1}{(k-1)!}T^{k-1}\right) e^{-T},$$

where

$$Y = \frac{y}{y_0}, \qquad T = k\mu t.$$

▶ **Example 3.1.4** Figure 3.1.2 shows the fraction still infected from an initial cohort, given a process with k phases and a mean total recovery time of 5. The slope of the $k = 1$ curve is steepest at $t = 0$, indicating that recoveries are happening faster at the beginning than any later time. For a six-phase process, the slope is greatest at a time close to the mean, which is what we would expect. The question

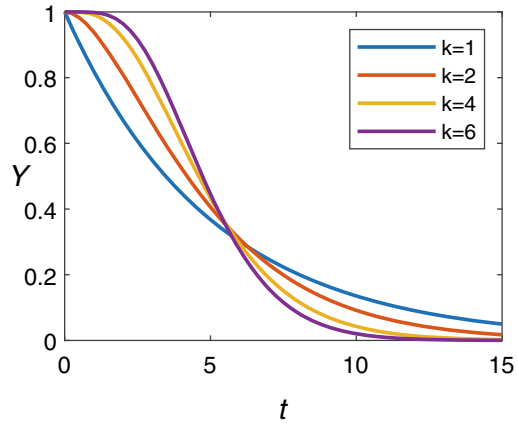


Fig. 3.1.2 Remaining fraction infected from an initial cohort for a k-phase recovery with mean total recovery time $1/\mu = 5$, from Theorem 3.1.1

of whether the number of phases makes a big difference in the results of an epidemic model is the subject of Project 3E. □

Problems

Problems marked with "p" require some programming beyond data entry. Problems marked with "c" require some calculus beyond ordinary differentiation.

3.1.1 Determine the dimensions of all the parameters in the model

$$\frac{dN}{dt} = RN\left(1 - \frac{N}{K}\right) - \frac{SN}{1 + HSN},$$

where $N(t)$ is the number of bacteria cells at time t and the other symbols are nonnegative parameters.

3.1.2 [Vaccination]
Suppose a population is divided into high-risk and low-risk subgroups, with vaccination given preferentially to those at high risk. Let W_1 and W_2 be the population fractions of low-risk and high-risk people waiting for vaccination, respectively. Assume that the vaccination model for the total population $W = W_1 + W_2$ is that of (3.1.7) and that the differential equation for W_2 is the same as that for W; that is,

$$\frac{dW_2}{dt} = -\frac{\phi g(t) W_2^2}{K^2 + W_2^2}, \tag{3.1.17}$$

with

$$g(t) = \min\left(\frac{t}{\tau}, 1\right).$$

(a) Use the equations for W and W_2 along with the relation $W = W_1 + W_2$ to derive the differential equation for W_1.
(b) If the two-class vaccination model is to be incorporated into an epidemiological model, the differential equation for W_1 will have to be written in the form:

$$\frac{dW_1}{dt} = -\Phi_1(W, W_2, t)W_1 .$$

Use the differential equation from (a) to determine the function Φ_1.

(c) Determine the initial conditions for W_1 and W_2, given an initial population fraction of S_0 susceptibles, with h, r, and r_2 as the fraction of the population at high risk, the fraction of vaccine refusers in the overall population, and the fraction of high-risk people who are vaccine refusers. We expect $r_2 \leq r$.

3.1.3 [Vaccination]
The MATLAB program *Vaccination.m* is set up to run a single-phase transition vaccination model $dW/dt = -\phi W$ with $W(0) = a$, where W is the fraction of the population waiting to be vaccinated and a is the fraction of the population that will be vaccinated.

(a) Use the program to plot the vaccination fraction $V = a - W$ using $a = 0.6$, which was approximately the average for the United States as of February 2022, and $a = 0.8$, which was a typical value for countries in western Europe. Use $\phi = 0.02$, corresponding to a mean vaccination time of 50 days and run the plot for 200 days.

(b) Modify the program to obtain similar graphs for the vaccination model of (3.1.7) using $\phi = 0.005$ and $K = 0.2$ with $g = 1$. Note: To compute $K^2/(K^2 + W^2)$ using MATLAB syntax, you must type `K^2./(K^2+W.^2)` or `K.^2./(K.^2+W.^2)` rather than `K^2/(K^2+W^2)`.[8]

(c) Discuss the two plots. Which is more realistic and why? Look at both the general shapes of the curves and the differences in the importance of a in the early stage of the process.[9]

(This problem is continued in Problems 3.3.6 and 3.4.15.)

3.1.4 [p] [Vaccination]
Do the plot of Problem 3.1.3a. Then modify *Vaccination.m* so that it uses the supply limitation given by (3.1.5) instead of unlimited supply. Describe the difference this makes in the results. Which is more realistic?

3.1.5 [p] [Vaccination]
Write a program that plots the low-risk and high-risk vaccination fractions $V_1 = a_1 - W_1(t)$ and $V_2 = a_2 - W_2(t)$ for the model of Problem 3.1.2. Assume a population that is 20% high-risk and that 80% of high-risk and 50% of low-risk patients want to be vaccinated. Assume constant supply rate coefficient $\phi = 0.005$ and semi-saturation parameter $K = 0.2$. Discuss the results.

3.1.6 [c] Assume a population of newly infectious people and let $y(t)$ be the fraction of this cohort that is still infectious at time t for a spontaneous transition model, so that

$$\frac{dy}{dt} = -\gamma y, \quad y(0) = 1.$$

(a) Use your knowledge of derivative formulas to identify the function $y(t)$ that satisfies these two requirements.

(b) Rewrite the result from (a) by solving it for t; this gives you the time at which the cohort has been reduced to a particular size.

(c) Use calculus to determine the average value of t over the interval $0 \leq y \leq 1$.

(d) Given the result of (c), how should we interpret the parameter γ?

[8] See Appendix A for a MATLAB primer.

[9] Think about the actual vaccination progress. How much difference would vaccine non-acceptance actually make at the beginning of vaccination?

3.1.7 [c] Mimic the solution of the problem (3.1.9)–(3.1.11) to obtain a result similar to (3.1.12) for the case $k = 3$. Use this result to verify Theorem 3.1.1 for that case.

3.1.8 [c] Verify Theorem 3.1.1 by differentiating the formula for Y to obtain the derivatives dy_j/dt and using the result to check the differential equations for y_1 and for y_j with $1 < j \leq k$.

3.1.9 [p] Write a function program that has input parameters k and μ and computes Y using Theorem 3.1.1. Test your program by using it to reproduce the curves in Fig. 3.1.2. Note that the input quantity t and output quantity Y are vectors, while the parameters k and μ are scalars.

3.2 Interaction Processes

After studying this section, you should be able to:

- Explain the conceptual model and derivations for disease transmission models.
- Discuss the conceptual models and derivations for the Holling predation functions.
- Run computer simulations of dynamical systems and interpret the results.

3.2.1 Person-to-Person Disease Transmission

Consider a population of size N that contains I infectious individuals (individuals who can transmit a disease) and S susceptible individuals (individuals who can be infected). We assume $S + I \leq N$ to allow for the possibility that the population contains other classes, such as individuals who have immunity.

Suppose each infectious person encounters a fraction c of the population per day. Assuming that all possible encounters are equally likely, we can expect that a fraction S/N of these encounters are with susceptible individuals. This means that each infectious person has cS encounters with susceptible individuals. If each encounter has a probability p of resulting in a transmission, then we can expect one infectious person to transmit the disease to an average of pcS individuals per day. If we have I infectious individuals rather than 1, the total rate of transmission is $pcSI$. We don't need both factors p and c, so we can write the transmission rate with a single parameter:

$$\text{transmission rate } = \beta SI. \tag{3.2.1}$$

It is always a good idea to check that the dimensions of a formula make sense. In this case, the dimension of c is not immediately clear, but we can work it out starting from what we know. The transmission rate must be in individuals per time, while S and I are in individuals. Hence, β must be 1/individual-time. We also know that $\beta = cp$ and p is dimensionless, being a pure probability. Therefore, c must also be 1/individual-time. If you reread the narrative, you will see that c is the fraction (dimensionless) of the population encountered per time *per infectious individual*.

Formula (3.2.1) can also be used if we are measuring the population classes as fractions of the total population rather than counts of individuals.

Check Your Understanding 3.2.1:

Determine the dimension of β if S and I are fractions of the population rather than counts of individuals.

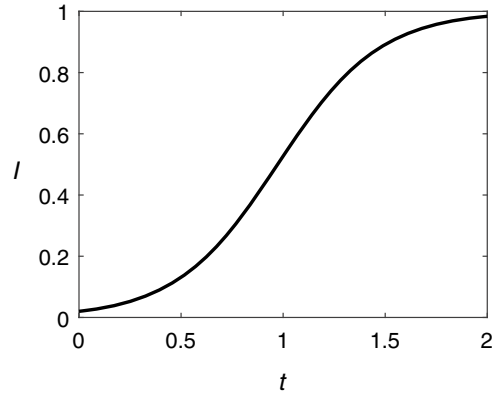


Fig. 3.2.1 The infectious population fraction from Example 3.2.1, with $\beta = 4$ and $I_0 = 0.02$

Since the dimension of β depends on whether populations are reported as individuals or as fractions, the numerical value of β for any particular disease is different in the two cases. As a consequence of this fact, we'll see in Sect. 3.3 that a more fundamental parameter is needed to quantify the infectiousness of a disease.

▶ **Example 3.2.1** Suppose a population is composed of infectious and susceptible subgroups, with fractions I and $S = 1 - I$. Then the rate of transmission is $\beta SI = \beta I(1 - I)$. If there are no other processes, such as recovery, then the infectious fraction is determined by the differential equation model

$$\frac{dI}{dt} = \beta I(1 - I), \qquad I(0) = I_0, \tag{3.2.2}$$

where $\beta > 0$ and $0 < I_0 < 1$. This problem can be solved exactly,[10] with solution

$$I = \frac{1}{1 + \left(I_0^{-1} - 1\right) e^{-\beta t}}. \tag{3.2.3}$$

Figure 3.2.1 illustrates the solution for one set of parameters. This example is somewhat unrealistic as a model for a disease; however, it is just right for a model of the sharing of secrets.[11] □

There are hidden assumptions in the simple formula (3.2.1) for the transmission rate. If you reread the explanation, you should notice that it doesn't distinguish individuals from each other. In reality, we live in a social network where we have frequent contacts with some people and no contact with nearly everyone else. Some of us are at the center of our network and have lots of contacts, while others have far fewer contacts. In the COVID-19 pandemic, some people deliberately reduced their frequency of contacts, while others did not. Additionally, infectious individuals vary in their level of infectivity and susceptibles vary in their level of susceptibility. While these are clearly flaws in the model from an individual standpoint, there is good reason to believe that they average out over a population in most circumstances.

[10] Problem 3.2.6.

[11] Presumably these cannot be forgotten; hence, there is no need for a "recovery" process.

The model also includes a fundamental assumption about how contacts occur. We assumed that the total number of contacts per person (cN) is proportional to the size of the population. This is called *density-dependent* or *mass action* incidence. We could instead have assumed that the total number of contacts per person is a fixed constant C. This is called *frequency-dependent* or *standard* incidence.

If we make this assumption, the transmission rate is $pCSI/N$. If the population is constant, we can define $\beta = pC/N$ and get the same formula (3.2.1). But some diseases increase the death rate of a population, while some models incorporate natural population growth. In these cases, the standard incidence transmission rate is

$$\text{transmission rate } = BS\frac{I}{N}, \qquad B = pC\,. \tag{3.2.4}$$

It is customary to use the same symbol β for the rate parameters in the standard incidence and mass action models; however, this is misleading on dimensional grounds, which is why we are using a different symbol for the standard incidence case.

Check Your Understanding 3.2.2:

What is the dimension of the parameter B in (3.2.4), assuming populations are given as numbers of individuals?

There are a variety of more complicated transmission models.[12]

3.2.2 Models for Consumption and Predation

Consider a situation in which an organism is placed in an environment with a fixed concentration x of some resource. Since the organism consumes the resource, it is therefore necessary for the resource to be replenished. We measure the rate y with which we have to replenish the resource; given that the amount of resource is fixed, this is equivalent to measuring the rate at which the resource is being consumed. Our goal is to construct a mathematical model that relates the intake rate y to the resource concentration x. Note that it doesn't make any difference how the organism obtains the resource. It could be a predator that feeds by hunting, an herbivore that feeds by grazing, or a single cell that feeds by absorbing nutrients through its surface.

To develop a conceptual model for this experiment, it helps to create a narrative version that appeals to human experience. Imagine yourself as the organism in the experiment. You live alone in a building with numerous hiding places that could contain servings of food. A total of x servings of food are distributed randomly throughout the building. Each time you eat a serving, a restaurant worker hides another serving in some randomly chosen location within the building, thereby keeping the food concentration at a constant level.[13] In this setting, our question is "How much do you eat per unit time in an environment with a constant food supply x per unit area?"

Linear Consumption/Predation Rate

For the simplest conceptual model, we imagine that you continually search the building, eating the food as you find it. It seems reasonable that you will search some fixed amount of space per unit of time. This search rate is a parameter, which we designate as s. The food density x could be measured

[12] See McCallum et al. [17] and Project 6C.

[13] Of course, this does not happen in a real feeding scenario; however, this is an assumption in the conceptual model. In practice, this discrepancy between the real experiment and the conceptual model can cause difficulties in the measurement of the parameters.

in servings per square meter, and the intake rate y could be measured in servings consumed per hour. The rate at which you locate food should be the product of the food density and the search rate; thus, we have the model

$$y = sx\,. \tag{3.2.5}$$

Note that the model also makes sense if we replace the quantities by their dimensions:

$$\frac{\text{food}}{\text{time}} = \frac{\text{area}}{\text{time}} \times \frac{\text{food}}{\text{area}}\,.$$

The Holling Models for Consumption/Predation

The model (3.2.5) is based on the assumption that you spend all of your time searching for food. This might be the case if food is extremely scarce. However, if you live in a buffet restaurant, where food is plentiful and easy to find, you spend only a tiny fraction of your time searching for it. You don't keep eating just because you can find more food! Instead, you spend nearly all of your time digesting and hardly any of it searching for more food. The time required for digestion is a feature of the actual scenario that our first conceptual model lacks. If you run a BUGBOX-predator trial[14] with *P. speedius* and a large prey density, you will notice that the predator spends only a small portion of the experiment time searching because it pauses whenever it locates prey. If the prey density is low, these pauses make little difference. They never make much of a difference for *P. steadius*, which moves slowly and digests quickly.

We need a conceptual model in which time is partitioned into two types: search time and "handling" time. With this distinction, our first model is no longer dimensionally consistent: the "time" in y is total time, while the "time" in s is search time. Our dimensional equation needs an additional factor to account for the distinction:

$$\frac{\text{food}}{\text{total time}} = \frac{\text{search time}}{\text{total time}} \times \frac{\text{area}}{\text{search time}} \times \frac{\text{food}}{\text{area}}.$$

Using f to denote the fraction of time spent searching, we have

$$y = fsx\,. \tag{3.2.6}$$

If f were a parameter, like s, then we would be done. However, this is not the case. Clearly, f is approximately 1 when the resource is scarce, but approximately 0 when it is plentiful. Thus, f is a second dependent variable. We therefore need a second equation. From our conceptual model,

$$\text{search time} + \text{handling time} = \text{total time}.$$

Dividing by total time, we have

$$\frac{\text{search time}}{\text{total time}} + \frac{\text{handling time}}{\text{total time}} = 1\,.$$

It seems reasonable that the amount of handling time should be proportional to the amount of food consumed, so we define a new parameter h to be the time required to handle one unit of the food. Now, we can think of the dimensional equation as

$$\frac{\text{search time}}{\text{total time}} + \left(\frac{\text{handling time}}{\text{food}} \times \frac{\text{food}}{\text{total time}}\right) = 1.$$

[14] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

In symbols, this is

$$f + hy = 1\,. \tag{3.2.7}$$

Equations (3.2.6) and (3.2.7) are a pair of algebraic equations for a pair of dependent variables. Thus, we have the right number of equations for a complete model. Substituting from (3.2.7) into (3.2.6), we have

$$y = (1 - hy)sx = sx - shxy\,,$$

or

$$y + shxy = sx\,.$$

Thus, we arrive at the model

$$y = \frac{sx}{1 + shx}\,. \tag{3.2.8}$$

The model (3.2.8) is known as the *Holling type 2* functional response model. It is named after C.S. Holling, who derived it in a seminal paper in 1959 [9]. The term "functional response" is used by ecologists to refer to what mathematical modelers would be more likely to call a predation rate. The linear model (3.2.5) is also known as Holling type 1.

Most ecological models use either Holling type 1 or type 2. The simplicity of the linear type 1 model makes it preferable for any biological system in which resources are scarce and consumers really do spend most of their time searching rather than processing. Such systems are relatively uncommon, as it is hard for organisms to survive if food is so scarce that they are almost constantly looking for it. Thus, the type 2 model is the one that is usually used in theoretical ecology. However, there is still a problem with the type 2 model that deserves attention. It is appropriate for specialist consumers, who are reliant on one specific resource. Generalist consumers have other options; as an example, bears eat a variety of foods and will decrease the amount of effort they put into fishing during periods when fish are scarce. The Holling type 3 model is designed for cases such as this. Instead of using a search rate s that is independent of the availability of the resource, suppose we make the search rate proportional to the resource level, with maximum rate S for a resource level at the carrying capacity K. This assumption reflects the idea that the consumer is dividing its time between different resources and will do more searching for a plentiful resource than a scarce one. Replacing s with Sx/K in the type 2 model (3.2.8) yields the type 3 model:

$$y = \frac{Sx^2}{K + Shx^2}\,. \tag{3.2.9}$$

Figure 3.2.2 compares the Holling type 2 and type 3 models. The type 3 model is initially concave up for small resource levels, which can lead to very different behavior when used in a resource management or predator–prey model.[15]

Check Your Understanding Answers

1. 1/time
2. 1/time

[15] See Sect. 4.5.

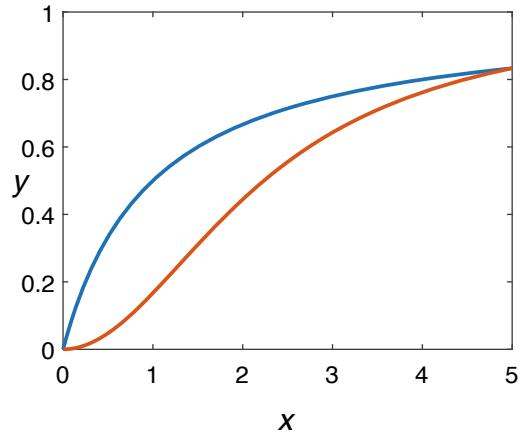


Fig. 3.2.2 The Holling type 2 (upper) and 3 (lower) consumption/predation models; x is the density of the resource relative to the carrying capacity of the environment and y is the rate of consumption for one consumer; parameters are $s = 1, h = 1, K = 5$

Problems

Problems marked with "p" require some programming beyond data entry. Problems marked with "c" require some calculus beyond ordinary differentiation.

3.2.1 Decide which of mass action incidence or standard incidence is a better fit for each of the following scenarios in which the population N is changing due to disease mortality or population growth. Explain your reasoning.

(a) Everyone lives in a community with a fixed number of dwellings. They don't move or build new dwellings in response to population change.
(b) Everyone lives in portable housing and transportation is expensive or time-consuming, so they move closer if the population decreases or they spread out if it increases.
(c) The disease in question is spread only through intimate sexual contact.

3.2.2 Suppose the real situation is somewhere between those of parts (a) and (b) of Problem 3.2.1. Suggest a model that might be better than either mass action or standard incidence for a scenario where a growing population results in a small amount of spreading out and a small increase in crowding.

3.2.3 Two possible models for the dynamics of a renewable resource (biotic or abiotic) are

$$\frac{dx}{dt} = 0.1 - \frac{xy}{1+x} \quad \text{and} \quad \frac{dx}{dt} = 0.1x - \frac{xy}{1+x},$$

where $x(t)$ is the amount of resource present at time t and y is the number of consumers. Which is the biotic resource and which is the abiotic one? Explain your reasoning.

3.2.4 The populations $x(t)$ and $y(t)$ of two interacting species are modeled using the equations

$$\frac{dx}{dt} = ax + bxy\,, \qquad \frac{dy}{dt} = cx + dxy\,,$$

where a, b, c, and d are parameters, not necessarily positive.

(a) Suppose the species are herbivores that compete for a common plant resource. Which of the four parameters should be positive and which negative? Explain.
(b) Repeat (a) for the case where x is an herbivore and y is a predator that eats the herbivore.

3.2.5 [c*] Use differentiation and algebra to show that the function defined in (3.2.3) satisfies the equations of (3.2.2).

3.2.6 [c] The usual way to solve (3.2.2) relies on differential equations theory and complicated integration. There is an easier way that involves repeated conversion of a hard problem into an easier one until you get to one whose answer you can simply identify. (You do have to thoroughly understand the chain rule and have solid algebra skills. If you don't have these, this is a great problem for developing them.)

(a) Define $y = I^{-1}$. Use the chain rule and algebra to obtain the formula $y' = -y^2 I'$, where $'$ indicates a time derivative.
(b) Substitute the equation for dI/dt into the formula from (a) and use $y = I^{-1}$ to obtain a differential equation that has only y', y, and β.
(c) Let $z = y - 1$. Use this definition with the equation from (b) to obtain the differential equation $z' = -\beta z$.
(d) Determine the correct value for $z(0)$, given $I(0) = I_0$.
(e) Use your knowledge of derivative formulas to identify the function that satisfies $z' = -\beta z$ and has the right value of $z(0)$.
(f) Use your solution for z to obtain y and then I.

3.2.7 Suppose a population grows by spreading into new territory. We can think of this process as an interaction between a population of occupied spaces (representing the organisms) and a population of available spaces.

(a) Write down a differential equation model for this scenario using $x(t)$ as the fraction of spaces that are currently available. For the initial condition, let y_0 be the (small) fraction of spaces that are initially occupied.
(b) To get a model for the population $y(t)$, define $y = 1 - x$ and replace x with $1 - y$ in the initial value problem of part (a).
(c) Discuss the connection between this model and the model (3.2.2).

Problems 3.2.8 and 3.2.9 provide some mathematical results needed in Problem 3.2.10.

3.2.8 [c] Show that the function

$$y(t) = \left(4 + 96e^{-t}\right)^{-1},$$

solves the initial value problem

$$\frac{dy}{dt} = f(y) = 4y(1 - y) - 3y = y - 4y^2, \quad y(0) = 0.01$$

3.2.9 [c] Show that the function

$$y(t) = \left[\frac{b}{a} + \left(\frac{1}{y_0} - \frac{b}{a}\right) e^{-at}\right]^{-1}$$

solves the initial value problem

$$\frac{dy}{dt} = ay - by^2, \quad y(0) = y_0.$$

(Hint: See Problem 3.2.8.)

3.2.10 [SIS epidemic model]
An SIS model has Susceptible and Infectious classes with the same transmission assumption as in Example 3.2.1. In addition, it has a spontaneous recovery process (take rate constant γ) that moves infectious individuals back into the susceptible class. Construct the model for this disease as a differential equation for I. Then use the result from Problem 3.2.9 to obtain a solution formula for $I(t)$. Plot the result with $\beta = 4$, $\gamma = 3$, and $I_0 = 0.01$.

3.2.11 [p] Consider a bacteria community whose population changes by two processes:

- Natural population increase with relative growth rate (rate per population) r.
- Predation by a single specialist predator, with the Holling type 2 predation function.

(a) Write down the mathematical model that corresponds to this conceptual model.
(b) Modify the MATLAB program *ODEsim.m* for the model of part (a) and use it to run a numerical simulation using parameter values $h = 1, s = 1, r = 1.1$. Try several initial bacteria counts. What happens if $r > s$?
(c) Repeat part (b), but with $r = 0.5$. You will need to try more than one initial bacteria count in order to see the full range of results.
(d) Is this a good model for predator-limited population growth? If not, is there something missing?

3.3 Compartment Analysis—The SEIR Epidemic Model

After studying this section,[16] you should be able to:

- Sketch compartment diagrams from verbal assumptions.
- Write down systems of differential equations from compartment diagrams.
- Explain the structure and properties of the SEIR epidemic model.
- Modify the SEIR epidemic model to embrace additional features of an epidemic disease.

Epidemiology is a rich area for mathematical modeling, thanks to the variety of diseases and the ongoing possibility that novel diseases, such as COVID-19, will arise. While the reader is undoubtedly most interested in COVID-19, a fully adequate model for that disease is too complicated to serve as a starting point. Instead, we begin with a well-known model called the SEIR model.

3.3.1 Classification of Epidemiological Models

Several features are needed to classify epidemiology models. We present these in rough order of importance.

1. Disease Type

   Infectious diseases can be divided into two subgroups, based on the transmission mechanism.

[16] This material is adapted from [13].

- **Person-to-person** transmission involves either direct transmission through physical contact or indirect transmission through the environment, such as droplets that enter the air through sneezing or coughing.
- **Vector-borne** transmission is required for diseases in which the pathogen has a complicated life cycle that requires multiple host species. An example is malaria, which is caused by a protozoan that lives part of its life in humans and part in mosquitoes.

Most diseases fit neatly into one of these two categories, but some are less clear. The plague that swept through much of the world in the medieval period had two forms: bubonic, which humans contracted from fleas, and pneumonic, which was passed from person to person via droplets in the air. For this section, we consider only person-to-person transmission.

2. Time Frame

- **Epidemic** models have no mechanism for replenishment of susceptible people, so the epidemic burns out when there are not enough susceptible people left to keep the fire going. Of course, such models are only valid for short-term scenarios.
- **Endemic** models are designed for the long term. They always include at least one mechanism for replenishment of susceptibles, typically birth of susceptible individuals.

We focus here on epidemic models and defer endemic models to Sect. 3.9.

3. Population Constancy

The total population can be fixed by omitting demographic processes or making sure birth and death rates are equal. This simplifies the model as compared to the more common case of variable-size populations and is typically done for all epidemic models as well as some endemic models.

4. Classes

Although the choice of classes is not more fundamental than the choice of time frame, it is traditional to name models according to the list of classes used in them. Our starting point is the SEIR model, where

- S is *Susceptible*, for individuals who are at risk of catching the disease.
- E is *Exposed*, for individuals who have been infected but are not yet infectious. This is a poor choice of term. In everyday language, we would say that a person has been "exposed" when they have contacted an infectious person, regardless of whether they have caught the infection, but in epidemiology, all members of the "Exposed" class have been infected. As a compromise between accuracy and consistency, we will retain the class symbol E but use the term *Latent* as the class name.
- I is *Infectious*, for individuals who can transmit the disease, even those who do not have symptoms.
- R is *Removed*, for individuals who are not currently infectious and are immune from further infection. This can include individuals who have recovered and individuals who are still sick but no longer infectious.

5. Processes

Epidemiological models need a list of processes that cause individuals to move from one class to another. The standard SEIR epidemic model has three processes:

a. a transmission process that moves susceptible individuals into the latent class;
b. an incubation process that moves latent individuals into the infectious class; and
c. a removal process that moves infectious individuals into the removed class.

6. Dynamical System Type

Discrete-time models are based on algebra, while continuous-time models are based on calculus. Discrete-time models seem more intuitive, but continuous dynamical systems have much better mathematical properties and are more appropriate for settings, such as diseases, where events can occur at any time. We will follow the more common practice of using continuous models in epidemiology.

3.3.2 Compartment Analysis

The idea of compartment analysis is the same as that of accounting. If we are keeping a budget and want to know how much money is in each account, we could keep track of changes and make regular updates to the total rather than counting the money every day. In continuous models, the accounting is done via rates.

Figure 3.3.1 is a compartment diagram for the SEIR model, showing the four classes as compartments and the three processes as arrows that connect compartments.

Notice what we might have included, but did not:

- It is customary in epidemic models to make no distinction between people who have died from the disease and people who have recovered. This undoubtedly seems horrible to the novice modeler, but it is actually very instructive. From a *human* perspective, we want to distinguish between healthy people, sick people, recovered people, people with ongoing deleterious effects, and people who have died. But our goal is to make an *epidemiological* model. There is no epidemiological distinction between infectious people who are sick and those who are not, or between removed people who are healthy, still suffering illness, or deceased. These human interest features can be added to the base epidemiological model during analysis.
- We have no birth or natural death processes. Obviously, births and natural deaths will occur during a disease outbreak. However, on an epidemic time scale of weeks or months, births and natural deaths only change the class counts to a limited extent. Remember that our model is not intended to exactly match reality. Too much detail makes models harder to understand without adding any predictive value.
- We have not included any modifications for public health measures, such as vaccination, isolation of infectious individuals, or quarantine of individuals through contact tracing. These can be added to the base model.

With the compartment diagram showing the process structure, it remains to quantify each of the processes, as discussed in Sects. 3.1 and 3.2. The standard SEIR model uses mass action incidence and spontaneous transitions.

It is customary to use lowercase Greek letters for the rate constants of each process. The specific symbol for a particular process varies from one author to another. Most authors use β for the

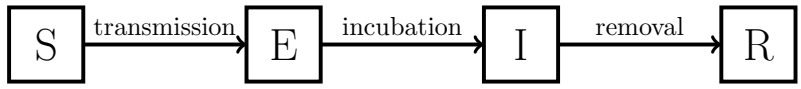


Fig. 3.3.1 The SEIR epidemic model in words

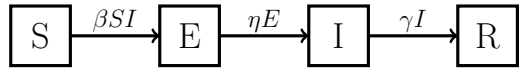


Fig. 3.3.2 The SEIR epidemic model in symbols

transmission rate constant, but the transition rate constant symbols vary widely. We'll use η for the incubation rate constant and γ for the removal rate constant. Filling in the details in the compartment diagram leads us to Fig. 3.3.2.[17]

Working directly from the diagram gives us the differential equations that describe the rates of change in terms of the state of the system; for example, the term ηE contributes a rate of decrease to the differential equation for E and a rate of increase to the differential equation for I. Full model specification also requires initial conditions. We use lowercase letters for these to avoid confusion between the symbol for the initial value of R and the symbol $\mathcal{R}_0$, defined below. We also assume that the class sizes are given as fractions of the total constant population. The final model is then

$$\frac{dS}{dt} = -\beta SI\,, \qquad S(0) = s_0 > 0\,; \tag{3.3.1}$$

$$\frac{dE}{dt} = \beta SI - \eta E\,, \qquad E(0) = e_0 \geq 0\,; \tag{3.3.2}$$

$$\frac{dI}{dt} = \eta E - \gamma I\,, \qquad I(0) = i_0 \geq 0\,; \tag{3.3.3}$$

$$\frac{dR}{dt} = \gamma I\,, \qquad R(0) = r_0 \geq 0\,; \tag{3.3.4}$$

where

$$s_0 + e_0 + i_0 + r_0 = 1, \qquad e_0 + i_0 > 0. \tag{3.3.5}$$

These last two requirements (3.3.5) ensure that the initial population total is 1 and that there are some infected people to get the outbreak started.

3.3.3 Model Behavior

We defer a careful analysis of the model to Sect. 3.4. For now, we simply examine a numerical simulation of the model, shown in Fig. 3.3.3. Note that the parameters η and γ are the reciprocals of the mean time for incubation and recovery; thus, these have been taken as 5 days and 10 days, respectively, roughly matching the data for COVID-19. The specific scenario has everyone susceptible at the beginning except for a very small number of latent individuals who somehow contracted the disease prior to the simulation start.

Several features common to epidemic models are visible in the graphs.

1. The epidemic gets off to a slow start whenever the initial number of infected individuals (latent plus infectious) is small, in this case just one out of 10,000. Eventually, the epidemic takes off in what appears to be exponential growth.

[17] It is more common to label the arrows with the rates per unit—that is, βI, η, and γ—rather than the rates themselves. I prefer using the full rates so that translation to the differential equations is purely mechanical.

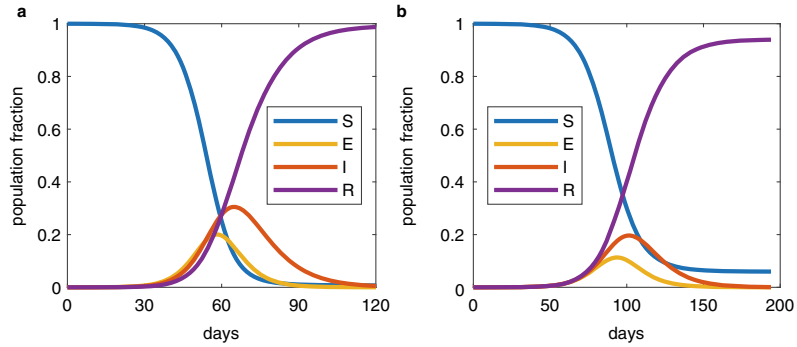


Fig. 3.3.3 Simulation results for the SEIR epidemic model with $\eta = 0.2, \gamma = 0.1, e_0 = 0.0001, i_0 = r_0 = 0$; **a**: $\beta = 0.5$; **b**: $\beta = 0.3$

2. The latent and infectious classes eventually peak, with the latent peak preceding the infectious peak. The latent peak is smaller in this example because the latent period is shorter than the infectious period.
3. The epidemic ends with some fraction of the population still susceptible. The size of that fraction depends strongly on the parameter values.

These features will be prominent in the analysis of Sect. 3.4.

3.3.4 Parameterization from Data

Sections 2.1–2.3 developed methods for fitting parameters in models to data, provided only one unknown parameter enters into a model in a nonlinear manner. A similar problem would be to obtain a best fit for the parameter β in the SEIR epidemic model, assuming η and γ are known. As in those problems, we can compute a residual sum of squares as a function of the unknown parameter by comparing dependent variable values from the data with the corresponding values from the model. There is, however, a fundamental difference in the two problems: the models in Chap. 2 are defined by explicit formulas, whereas the SEIR epidemic model is defined by a set of differential equations and evaluated numerically. We will be able to use the same basic mathematical ideas, but we will need a different method of implementing those ideas. One such method is the "quintsection" method described in Appendix C.

► **Example 3.3.1** The data for total infectious population from an agent-based SEIR simulation model[18] is shown as a set of points in Fig. 3.3.4. The durations of the latent and infectious stages for individuals were assumed to be normally distributed with a mean of 5 days for the latent period and 10 days for the infectious period, and standard deviations of 0.5 days and 1.5 days, respectively.

The SEIR epidemic model of this section was fit to that data by assuming the correct values for the parameters η and γ and fitting β. The actual value of β used in the simulation was 0.00005, and the best-fit value is 0.0000548. The results of the differential equation model with this value of β are shown as the solid curve in the plot.

Here are some important observations about the results:

[18] See *seirabm.m*.

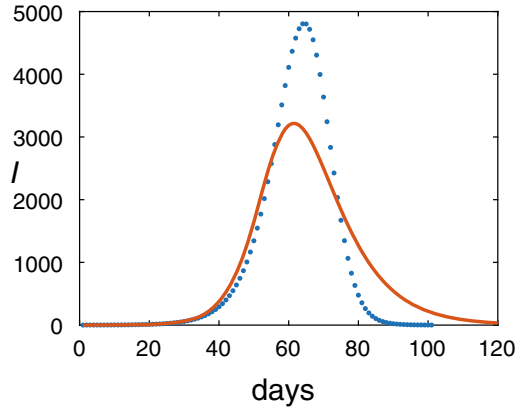


Fig. 3.3.4 Simulation results for an agent-based SEIR epidemic model with $N = 10000$, mean latent period of 5 days, mean infectious period of 10 days, $\beta = 0.00005$, and an initial population of 9999 susceptibles and 1 latent (dots), along with the differential equation SEIR epidemic model with best-fit parameter $\beta = 0.0000548$

1. The peak I count for the model is far less than the simulation results and the epidemic takes noticeably longer to run its course.
2. The model is able to approximate the peak time of the infection with a high accuracy.
3. The difference between the best-fit value of β and the actual value of β is about 10%, which is not too bad for an epidemiological model, especially given the larger errors in peak I and epidemic duration.

□

One lesson of Example 3.3.1 is that fitting a model to data does not necessarily mean that the model gives accurate results. If our primary interest is in the peak number of infectious individuals, the model performance is rather poor. The correct conclusion to draw is that the model is missing some important features of the disease dynamics. If you reread Sect. 3.1, you should be able to uncover the problem. Differential equation models assume that transition times are exponentially distributed, which makes for a much higher standard deviation than most distributions of actual data. A small fraction of infectious individuals in the differential equation model are infectious for a long duration, while the majority are only infectious for a short amount of time. This spreads out the transmission peak and delays the end of the epidemic. Project 3E attempts to correct this problem by using a multi-phase transition model.

Problems

3.3.1 Explain the SEIR model. Specifically,

(a) Why is the change in S proportional to both S and I?
(b) Why do the terms βSI, ηE, and γI appear in two different equations, and why are they positive in one instance and negative in the other?
(c) The model

$$\frac{dR}{dt} = kR$$

represents exponential growth of R. Why does

$$\frac{dR}{dt} = \gamma I$$

not represent exponential growth?

(d) Does the model clearly limit how large R can be?

3.3.2 Suppose a fraction p of infectious individuals self-isolate to reduce their contact rates to a factor of f times that for unisolated individuals. How does this change the SEIR model?

For each of Problems 3.3.3–3.3.7, sketch a compartment diagram and write down the appropriate differential equations. Each adds features to the base SEIR model.

3.3.3* Add a transition process where immunity is lost.
(This problem is continued in Problem 3.4.14.)

3.3.4 Add an asymptomatic stage (A) that occurs between the latent and symptomatic (I) stages. Assume that asymptomatic individuals are contagious.

3.3.5 Add an asymptomatic class (A) that is distinct from the symptomatic class (I). In other words, assume that exposed individuals become either asymptomatic (with probability p) or symptomatic, and that individuals in these classes recover rather than transitioning between A and I. Also assume that the infectiousness of asymptomatic individuals is only a fraction $f < 1$ of the infectiousness of symptomatic individuals.

3.3.6* [Vaccination]
(This problem is continued from Problem 3.1.3). If we want to add vaccination to the SEIR model, it is important to take vaccine non-acceptance into account. To do this, break the usual S class into two subclasses: (P)revaccinated and (U)nprotected. Individuals in both subclasses can be infected in the same way as individuals in class S of the SEIR model. However, prevaccinated individuals can also be vaccinated, which moves them to class R.[19]

(a) Sketch a compartment diagram for the PUEIR model.
(b) Write down the corresponding differential equations for the model, assuming the vaccination rate is $\Phi(W, t)$P, where W is the fraction of the population that will accept vaccination.
(c) To complete the model, it is necessary to use a vaccination model to identify the function Φ. Consider the model (3.1.7), which will be used to calculate W. The right side of the differential equation is the overall rate at which individuals leave the vaccination waiting class W. Since vaccination is offered to everyone, regardless of their epidemiological status, it is reasonable to expect that only a fraction P/W of vaccinations are administered to individuals in class P. Use this assumption along with (3.1.7) to determine Φ.

(This problem is continued in Problem 3.4.15.)
In a simulation, the initial population of susceptibles would be divided into classes P and S, with a fraction r of the initial susceptibles staying in class S as vaccine refusers.

[19] See Sect. 3.1.4.

3.3.7 Replace the single class E with classes E1 and E2, each with the same transition rate constant. Newly infected individuals are in class E1, pass to class E2 with a "partial incubation" process, and then pass on to class I with a second partial incubation process at the same rate.[20]

3.3.8 Construct a compartment diagram for the class and process structure that matches the individual-based model of Sect. 1.5.

3.3.9 [HIV]
The simplest model for viral infections of the immune system appears in an article by A.S. Perelson and colleagues that was published in the year 2000 [24]. The model has three components that interact in the bloodstream: healthy T cells (S),[21] infected T cells (I), and free virus particles (V). The dynamics of the system includes a number of processes:

1. Healthy T cells are produced by the body at a constant rate R.
2. Both healthy and infected T cells are lost through natural death at a rate proportional to the number of cells: DS for the healthy cells and DI for the infected cells.
3. Infected T cells can also die as a result of virus infection, with overall rate MI.
4. Healthy cells become infected through encounters with free virus particles. Since this is a chemical reaction, we assume that the rate is given by the law of mass action, which predicts that the rate is proportional to both S and V, with rate constant B.
5. Infected cells produce free virus particles at a constant rate, so the overall production rate of virus particles is proportional to the number of infected cells: PI.
6. The body removes free virus particles at a rate proportional to the concentration of such particles: CV.

(a) Sketch two compartment diagrams for the model, one for the T cells and one for the free virus. Use horizontal arrows for transmission processes that change individual cells from one state to another, vertical arrows entering the top of a box for processes that create T cells or virus particles, and vertical arrows that exit the bottom of the box for processes that kill cells or virus particles.
(b) Write down the corresponding system of differential equations.

(This problem is continued in Problem 3.4.18.)

3.3.10 [Immune system]
The immune system is an amazing piece of bioengineering, built and optimized over eons thanks to natural selection. Even a small improvement in the immune system would have offered a large fitness advantage for those lucky enough to have the right mutation. So where some components of human anatomy and physiology seem to be less than ideal (think of the prevalance of back pain, for example), we should expect the immune system to function well. Indeed, it is a rather complicated system that utilizes a variety of different components to combat infections, foreign bodies, and other dangers to the organism. A complete model of the immune system would be far beyond the scope of this book; however, we can study some components of the immune system in isolation, starting with a fairly simple three-component model created by Angela Reynolds and colleagues [21]. In addition to a pathogen, the model contains two different kinds of macrophages (white blood cells). There are generalist macrophages (M) that are omnipresent and serve as a rapid response force, and there are specialized macrophages (N) that have to be "trained" to recognize and attack a specific invader before they can respond. In its full-dimensional form, the model is

[20] See Sect. 3.1.5.
[21] T cells are a type of white blood cell.

$$\frac{dP}{dT} = RP\left(1 - \frac{P}{K}\right) - QMP - SNP\,, \tag{3.3.6}$$

$$\frac{dM}{dT} = L - DM - AMP\,, \tag{3.3.7}$$

$$\frac{dN}{dT} = \frac{CP}{H+P} - BN\,, \tag{3.3.8}$$

where we are using T for time rather than t in anticipation of scaling in Sect. 3.6.

(a) Explain the three terms in the pathogen equation (3.3.6). In particular, what assumptions does the model make about pathogen population growth and about interaction with macrophages? [Refer to Sects. 3.1 and 3.2.]
(b) Explain the three terms in the generalist macrophage equation (3.3.7). In particular, what are the processes by which the body maintains a population of these cells in the absence of infection, and what is the outcome on these cells of interaction with the pathogen?
(c) The first term in the specific macrophage equation (3.3.8) is simplified from a model that involves the training of cells from a reservoir of untrained cells. While it does not show all the features of that model, it does indicate how the rate at which these trained cells are created depends on the pathogen population. Explain this dependence using a graph of the production rate versus the pathogen population.
(d) Explain the biological process that removes trained cells from the body. In particular, what significant advantage is conferred by these cells as compared to the non-specific cells?

(This problem is continued in Problem 3.6.14.)

3.4 SEIR Model Analysis

After studying this section,[22] you should be able to:

- Explain the meaning and significance of the basic reproduction number.
- Use the *SEIR* suite of MATLAB programs to do virtual experiments with the SEIR model.
- Make simple modifications to the *SEIR* programs to adapt to other epidemic disease models.
- Explain the exponential growth phase in a disease outbreak.
- Explain how the final fraction of susceptibles is influenced by the basic reproductive number of the disease.

Before conducting a thorough analysis of the model (3.3.1)–(3.3.5), we first need to introduce the principal parameter that identifies the ease with which a disease spreads through a susceptible population. To begin, we assume that the mean durations t_L and t_I of the latent and infectious periods are known. Then the transition rate parameters are given as

$$\eta = \frac{1}{t_L}\,, \qquad \gamma = \frac{1}{t_I}\,. \tag{3.4.1}$$

3.4.1 The Basic Reproduction Number

The *basic reproduction number*, which is given the symbol $\mathcal{R}_0$ and read as "R-nought", is the fundamental measure of the infectiousness of a disease.

[22] This material is adapted from [13].

Definition 3.4.1

The **basic reproduction number** is the average number of secondary infections brought about by one infectious person in a population that is wholly susceptible.

This definition of $\mathcal{R}_0$ sounds complicated, but it is much simpler if we break it down into parts. The transmission rate formula βSI tells us the average number of secondary infections *per day* in a population of any composition. If that population is wholly susceptible, then we get an average of βNI secondary infections per day. That is the number produced by the whole infectious class; with $I = 1$, we see that "the average number of secondary infections *per day* brought about by one infectious person in a population that is wholly susceptible" is βN. To get the basic reproduction number, we just need to take into account that one infectious person has, on the average, t_I days in which to produce secondary infections. Total is rate times time, so the basic reproduction number is

$$\mathcal{R}_0 = \beta N t_I. \tag{3.4.2}$$

While we have calculated the basic reproduction number in terms of β, the actual use of this formula is often to determine β from $\mathcal{R}_0$. Of course, this means that we need to be able to determine $\mathcal{R}_0$ by some other means. You can look up values for well-known diseases. The issue of how to estimate $\mathcal{R}_0$ for a novel disease is critically important to accurate modeling results; this will be addressed later in the section.

Note that the value of $\mathcal{R}_0$ is independent of the units used for population class sizes. If we change N, it is the value of β that makes a corresponding change. Assuming we have chosen $N = 1$ by design, we can rearrange (3.4.2) and replace t_I with $1/\gamma$ to obtain

$$\beta = \gamma \mathcal{R}_0. \tag{3.4.3}$$

We can therefore specify a particular disease using t_L, t_I, and $\mathcal{R}_0$ as the three fundamental disease parameters and use (3.4.1) and (3.4.3) to calculate the parameters that appear in the model.

▶ **Example 3.4.1** Suppose the first generation consists of 10 infectious individuals. If $\mathcal{R}_0 = 3$, then each infectious person will generate an average of three new infections, for a total of 30 in the second generation. The epidemic will grow explosively as long as the population remains largely susceptible. Only when most of the population has been infected will we stop seeing more infections in the next generation than the previous one. In the end, nearly everyone will have got the disease. □

▶ **Example 3.4.2** In the scenario of Example 3.4.1, suppose instead that $\mathcal{R}_0$ is close to 1. If $\mathcal{R}_0 = 1.1$, there will be an average of 11 in the second generation. This is enough to keep the outbreak growing for a little while, but at a much slower rate than if $\mathcal{R}_0 = 3$, and a much smaller decrease in the susceptible population will be enough to stop the disease. Continuing with the comparison, if $\mathcal{R}_0 = 0.9$ then the second generation will average 9 individuals. Clearly the disease is unable to get a foothold. Thus, $\mathcal{R}_0 = 1$ is a critical value—a disease can only cause an epidemic outbreak if $\mathcal{R}_0 > 1$. □

We can now appreciate why COVID-19 is such a serious problem. The most common infectious diseases just prior to December 2019 were the common cold and influenza, with $\mathcal{R}_0$ values on the order of 1.5–3. It is not uncommon for a person to have known exposures to someone with the flu and not get the disease. In contrast, the standard twentieth-century childhood diseases of measles, chicken pox, and mumps have $\mathcal{R}_0$ values of 10 or more. Before the development of the vaccines for these diseases, virtually everyone who was exposed caught them. The best estimate for the original strain of COVID-19 is $\mathcal{R}_0 = 5.7$ [23], which is a very large value compared to influenza. The delta and

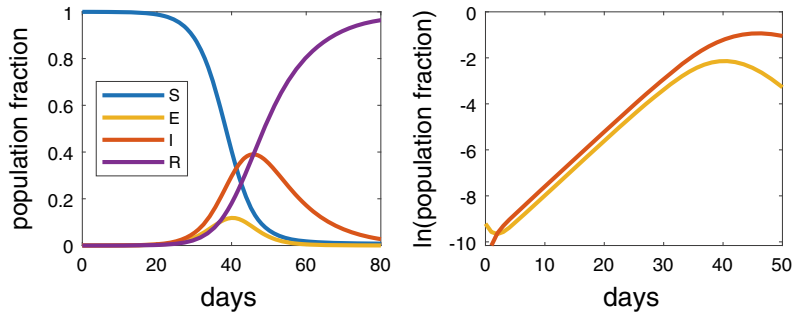


Fig. 3.4.1 Simulation results for the SEIR epidemic model with $R_0 = 5$, $t_L = 2$, $t_I = 10$, $e_0 = 0.0001$, $i_0 = r_0 = 0$

omicron strains are progressively more infectious. Initial estimates suggest that omicron has a basic reproduction number of at least 10, similar to that of mumps. Simulations show that in a society that completely ignores the threat, almost the entire population will get the disease in less than 2 months. (This is approximately the situation in Fig. 3.4.1.)

3.4.2 Goals of the Analysis

Analytical methods (calculus and algebra) and numerical methods (approximation, nowadays with computers) are complementary in many ways. One of these is that the behavior illustrated by numerical simulations can yield conjectures that can subsequently be confirmed by analysis.

Figure 3.4.1 shows the results of a simulation using an initial condition of no infectious or removed individuals and only one latent individual per 10K population. The plot on the left shows the typical pattern of an epidemic outbreak. It takes a while to get started, but then the infection grows rapidly. Both the latent and infectious classes reach a peak and then drop off to 0, with the latent peak occurring earlier in time than the infectious peak. The plot on the right shows some very important detail that can only be seen on a logarithm plot. There is a very fast initial phase during which the latent population is decreasing. This is because we started without any infectious individuals, so new transmissions had to wait until the first batch of latent individuals became infectious. Then there is a significant period of time during which the graphs of $\ln E$ and $\ln I$ are linear *and parallel*. Only as the latent population comes close to its peak do the graphs of $\ln E$ and $\ln I$ begin to curve downward. These same features appear with any realistic choices for the disease parameters, as long as the initial fractions of the exposed and infectious classes are small and $\mathcal{R}_0 > 1$. From this graph, we can make the following conjecture:

- After a short initial adjustment phase, there is a period in which the logarithms of the infected classes are linear with a common slope λ.

The model has six input parameters, the three that define the disease properties and three that define the starting point of the scenario. The goal of analysis is to study their impact on the model behavior. There are a number of possible outcomes we could be interested in. We focus on five of these:

1. The early-phase logarithmic slope parameter, λ, which tells us how rapidly the epidemic grows at the beginning;
2. The maximum infectious class size, I_{max}, which tells us how much impact the epidemic will have at the worst point in time;

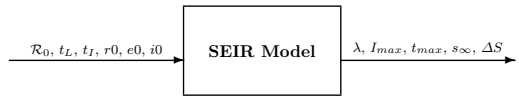


Fig. 3.4.2 Schematic diagram of the SEIR model as a function

3. The time at which the maximum infectious class size occurs, t_{max}, which tells us how much time there is to prepare for the peak;
4. The ending susceptible population, s_∞, which tells us how much of the population did not contract the illness and will be at risk in a subsequent scenario;
5. The total population infected during the scenario, $\Delta S = s_0 - s_\infty$. If we know what fraction of infected people die, we can use ΔS to estimate the total number of deaths in a given initial population.

Figure 3.4.2 frames the model analysis in the language of Sect. 1.4, with model outcomes as functions of model parameters. Our goals are to determine how these model outcomes depend on the parameters and also to use this information to develop a method for determining $\mathcal{R}_0$ for a novel disease.

The outcomes will need to be determined by a variety of methods. We can use analytical methods to compute λ and to derive an algebraic equation for s_∞. That equation cannot be solved using analytical methods, but it can be solved with numerical methods. Most epidemic model outcomes, such as I_{max} and t_{max} in the SEIR model, can only be determined by a fully numerical method.

3.4.3 Early-Phase Exponential Growth

Mathematical exploration of the insight from Fig. 3.4.1 leads to the following result[23]:

Theorem 3.4.1
Suppose the initial populations of the infected classes are small compared to that of the susceptible class. Then the SEIR model shows an extended exponential growth phase with

$$I \approx I_0 e^{\lambda t}, \quad E \approx \rho I_0 e^{\lambda t}, \quad S \approx s_0 \approx 1 - r_0, \tag{3.4.4}$$

where λ is the positive solution of the equation

$$(\lambda + \eta)(\lambda + \gamma) = \eta\gamma s_0 \mathcal{R}_0\,, \tag{3.4.5}$$

$$\rho = \frac{\lambda + \gamma}{\eta}\,, \tag{3.4.6}$$

and I_0 is a constant that represents the y-intercept of the straight line for the $\ln I$ plot.

Theorem 3.4.1 has two very important consequences. First, it gives us a way to estimate $\mathcal{R}_0$ from early data on the infectious class population. From data for $\ln I$, we can estimate the value of λ and then use (3.4.5) to estimate $\mathcal{R}_0$. This is the best way to estimate $\mathcal{R}_0$ for a novel disease, like COVID-19.[24]

[23] Problem 3.4.10.
[24] See Sect. 3.5.

The other important consequence of Theorem 3.4.1 is that it allows us to prescribe scenarios using only two initial conditions rather than three. For any scenario that starts with a small infected population, we can choose i_0 and then use $e_0 = \rho i_0$, where ρ is given by (3.4.6).

3.4.4 The End State

Dynamical systems often progress toward a fixed end state. These must be states for which all of the rates of change are 0. For endemic models, there are usually only one or two such states and there are straightforward methods to determine which is the end state.[25] The situation is much more complicated for epidemic models. To begin, note that a fixed end state must have no further changes in R. Since $dR/dt = \gamma I$, we can only have a fixed value of R if $I = 0$. We can similarly conclude that a fixed value $I = 0$ also requires $E = 0$. The chain stops here, however. There is no particular reason why a fixed end state should have $S = 0$ or any particular value of R. Based on the differential equation model (3.3.1–3.3.4), any state with values

$$S = s_\infty, \quad E = 0, \quad I = 0, \quad R = r_\infty = 1 - s_\infty, \tag{3.4.7}$$

with $0 \le s_\infty \le s_0$, could serve as the end state.

In the case of the SEIR epidemic model and a few other simple ones, the end state can be found using calculus. The idea is that the relationship between the variables S and R is determined by the differential equations for those two variables. If we assume that R is a function of S, rather than being an independent function of t, we can use the chain rule to obtain an expression for dR/dS. If that expression depends only on the variable S, as is the case here, then we can use integration techniques to find the one-parameter family of anti-derivatives for dR/dS. Only one of these anti-derivatives also satisfies the initial conditions for R and S. The result is as follows[26]:

Theorem 3.4.2
The initial and final values of the susceptible population are related by the equation

$$\ln s_0 - \ln s_\infty = \mathcal{R}_0(1 - r_0 - s_\infty). \tag{3.4.8}$$

Equation (3.4.8) is an analytical result that connects the initial values r_0 and s_0, the basic reproduction number $\mathcal{R}_0$, and the final value s_∞. It cannot be used to immediately calculate s_∞ for any set of input parameters because there is no way to solve the algebraic equation for s_∞. However, we can obtain several useful conclusions from the theorem:

1. The final state s_∞ depends only on the basic reproduction number and the initial conditions; it is unaffected by the values used for the rate constants η and γ. If these constants are relatively small for one scenario, then it will just take longer to reach the same final state.
2. The final state cannot have $s_\infty = 0$ because there are no values one can choose for the other parameters to satisfy (3.4.8).

[25] Analysis of long-term behavior is the subject of Chap. 6.
[26] Problem 3.4.12.

3. If we want to see how s_∞ depends on the infectiousness of the disease for a given initial scenario, we can use (3.4.8) to calculate $\mathcal{R}_0$ values from given values of s_∞ and then plot a graph of s_∞ versus $\mathcal{R}_0$, without ever solving the equation for s_∞.

If we want to be able to calculate s_∞ rather than approximating it from a graph, we need a numerical method, since (3.4.8) can't be solved for it using algebra. The equation can be manipulated in various ways to put it in the form $F(s_\infty) = 0$. There are a variety of well-documented numerical methods for solving such equations.[27] Scientific computing software, such as MATLAB, includes built-in functions for this task. These have been written by numerical analysis experts, and you can use them if you just want an answer. Some care is required, as the methods are harder to use in practice than in theory because of the requirement of having a good initial guess for some functions as compared to others. If you want better mathematical understanding, you have to write your own programs to test specific methods.

Problems

Problems marked with "p" require some programming beyond data entry. Problems marked with "c" require some calculus beyond ordinary differentiation.

3.4.1 Estimates of the basic reproductive number and durations for incubation and infectiousness can be found online for many illnesses. Use published information about H1N1 flu [5] to estimate reasonable parameter values for η, γ, and β. Be sure to justify all assumptions and cite all references.

3.4.2 Repeat Problem (3.4.1) using the infectious disease of your choice. Justify all assumptions and cite all references.

3.4.3 Find estimates for the incubation period, the infectious period, and the basic reproduction number for an infectious disease of your choice. Use this information to calculate η, γ, λ, and ρ for that disease. Also estimate s_∞, assuming $r_0 = 0$ and $s_0 \approx 1$, either by using a numerical solver with (3.4.8) or by using a plot of s_∞ versus $\mathcal{R}_0$ (see consequence 3 of Theorem 3.4.2).

3.4.4 Our best guesses for the original strain of COVID-19 are a basic reproduction number of 5.7 [23], an incubation period average of 5 days, and an infectious period average of 10 days. Assume an initial population that is entirely susceptible except for ten latent individuals per 100K. Run *SEIR_onesim.m* and describe what would have happened in a community that made no behavioral or public health adjustments.

3.4.5 Repeat Problem 3.4.4 using the disease you characterized in Problem 3.4.3. Make sure that the final susceptible fraction obtained by the computer simulation matches the one you estimated in that prob.

3.4.6 [Smallpox]
The Incan Empire had a population of over one million when it was conquered by 168 Spanish Conquistadores in 1525. The Spanish had gunpowder weapons and horses, but these advantages would not have been sufficient to defeat the huge Incan army. (It took about 2 min to reload a single-shot arquebus, during which time the number of conquistadores would have been significantly reduced.)

[27] These include Newton's method, the secant method, the bisection method, and other less well-known methods. Details of these methods can be found in any introductory numerical analysis book.

They also benefited by joining forces with peoples subjugated by the Incas, but that would not have happened without those peoples assessing the Spanish as having the advantage. Both historian William H. McNeill and natural scientist Jared Diamond have argued that the key factor in the Incan defeat was the European diseases the Spanish brought with them [6, 17]. To test this theory, set the basic reproduction number at 5, the incubation period at 12 days, and the infectious duration at 20 days, values that roughly match smallpox. Use a simulation to study the effect introduction of smallpox into Incan civilization would have had, even without considering the death toll of the disease. Discuss the implications of your findings.

3.4.7 For a more complete look at the effect of the basic reproduction number on epidemic progression, run *SEIR_comparison.m* using *R0* values 5, 3, 2, 1.5, and 1.25, with incubation period of 5 days, infectious duration of 10 days, and initial infectious fraction of 0.001.

(a) Discuss the graphs, explaining why the effects of *R0* are what you see.
(b) Suppose a disease with $R0 = 5$ is combated with social distancing and measures to decrease transmission probability for each contact. What effect do you expect these social policies to have and why?

3.4.8 Use *SEIR_paramstudy.m* to study the effect of the basic reproduction number on epidemic outcomes. Use *R0* values from 0 to 6 and the original default values for the other parameters. Describe and explain the results, paying particular attention to the behavior near $R0 = 1$.

3.4.9 Use *SEIR_paramstudy.m* to do a more thorough study of the effect of the disease duration on epidemic outcomes. Use $R0 = 2.5$ and t_I values from 4 to 12. Describe and explain the results. Pay particular attention to the axis limits.

3.4.10 [c] * Assume that $S \approx s_0$, $R \approx r_0$, $\ln I \approx \ln I_0 + \lambda t$, and $\ln E = \ln I + \ln \rho$ for some unknown values I_0 and ρ. Derive the results of Theorem 3.4.1 by substituting these assumptions into the differential equations (3.3.2)–(3.3.3).[28]

3.4.11 Assume that $\mathcal{R}_0$, η, γ, s_0, and i_0 are given, with i_0 small and e_0 unknown. Use the assumptions $S \approx s_0$ and $\ln E = \ln I + \ln \rho$ to derive a quadratic equation whose solution determines ρ. Find the positive solution(s) of this equation to obtain a formula that can be used to select an appropriate e_0 for a scenario that starts shortly after the beginning of an outbreak.[29]

3.4.12 [c] Assume R is a function of $S(t)$. Use this assumption and the model (3.3.1)–(3.3.4) to derive a formula for dR/dS. Then use calculus and algebra to derive the result of Theorem 3.4.2.

3.4.13 [c] * While there is no analytical method for finding the maximum size of the infectious class for the SEIR model, there is a method, similar to that of Problem 3.4.12, for the SIR model, which is

$$\begin{aligned} S' &= -\beta SI\,, \\ I' &= \beta SI - \gamma I\,, \\ R' &= \gamma I\,. \end{aligned}$$

[28] Figure 3.4.1b justifies the assumptions when e_0 and i_0 are small; however, it does not justify the additional assumption that I_0 is the same as i_0 because of the small transient at the left edge of the graph. This is why we take I_0 as an unknown to be determined.

[29] This means that we are defining "time 0" to be *after* the initial transient in Fig. 3.4.1b.

(a) Assume I is a function of $S(t)$. Use this assumption to derive a formula for dI/dS.
(b) Integrate the equation from (a) and apply the initial conditions $I(0) = I_0$, $S(0) = S_0$ to obtain the integration constant.
(c) Simplify the result by assuming $I_0 + S_0 = 1$ and $S_0 \approx 1$, which is appropriate for a novel disease with a small number of initial infectives. This should yield a formula for I as a function of S. Note that you can replace the parameter combination γ/β with $\mathcal{R}_0^{-1}$. This equation relates I and S for the full course of the outbreak.
(d) Explain why we know that the maximum value of I occurs at a point where $I' = 0$.
(e) Use the idea of (d) with the formula from (c) to obtain a formula for I_{max} in terms of $\mathcal{R}_0^{-1}$.
(f) Plot the result from (e) and discuss the effect of $\mathcal{R}_0$ on the maximum infectious fraction.

3.4.14 p (This problem is continued from Problem 3.3.3). Modify the SEIR program suite to incorporate a transition process where immunity is lost. Assume that the mean time over which immunity is lost is 100 days. Use the values from Fig. 3.4.1 for comparison with that simulation. You will need to run the simulation for 200 days or more.

3.4.15 p [Vaccination]
(This problem is continued from Problem 3.3.6). Modify the SEIR program suite to run the PUEIR model of Problem 3.3.6. You will need an additional differential equation to track the population waiting for vaccination (W); use the vaccination model of (3.1.7).

(a) Run the simulation with $a = 0$ as a test of the program. The plots for U, E, I, and R should be identical to the SEIR plots.
(b) Run the simulation with the parameters given in (3.1.8).
(c) Run the simulation again with $\tau = 30$, corresponding to a much faster increase in vaccine supply.
(d) Discuss the results of the simulations.

3.4.16* [Malaria]
One of the first examples of a mathematical epidemiology model was that published by Sir Ronald Ross[30] in 1911 on malaria [16, 22]. We can write this model as

$$\frac{dX}{dt} = bp\left(1 - \frac{X}{H}\right)Y - \gamma X\,,$$

$$\frac{dY}{dt} = bq\frac{X}{H}(V - Y) - \mu Y\,,$$

where X is the infectious human population out of a total population of H, Y is the infectious mosquito population out of a total population of V, b is the number of human bites per day per mosquito, p and q are the probabilities of transmission from human to mosquito and mosquito to human, respectively, γ is the rate constant for the human recovery process (after which humans are again susceptible), and μ is the rate constant for mosquito death (mosquitoes do not recover).

(a) Suppose we have a population in which the humans and the mosquitoes are almost all susceptible; that is, both X and Y are small compared to the population sizes. Determine the expected number of new human infections caused by one infectious mosquito during its lifetime.
(b) Similarly, determine the expected number of new mosquito infections caused by one infectious human during its infectious period.

[30] Prior to his mathematical modeling work, Ross did experimental studies of malaria. It was he who first demonstrated the complicated life cycle of the malaria parasite, thereby proving that people got malaria from mosquito bites. Ross was awarded the 1902 Nobel prize in medicine for this pioneering work.

(c) Combine the results of parts (a) and (b) to determine the basic reproduction number for the model.

(This problem is continued in Problem 3.6.10.)

3.4.17 P* [Malaria]
Modify *ODEsim.m* to run a simulation for the simplified Ross malaria model

$$\frac{dx}{dt} = \alpha(1-x)y - x\,,$$

$$\frac{dy}{dt} = m[\beta x(1-y) - y]\,,$$

using $m = 10$ and initial conditions $x = 0.001$, $y = 0$, corresponding to the introduction of malaria into a new region by people migrating from another region.

(a) Plot the infectious human population fraction x and the infectious mosquito population fraction y for $\alpha = 10$ and $\beta = 0.3$.
(b) The parameters α and β are both proportional to the mosquito biting rate. Repeat (a), but assume that public health measures are able to reduce the biting rate by 20%.
(c) Describe the differences a biting rate reduction makes in the outcome of the scenario.

(This problem is continued in Problems 3.8.7 and 6.1.11.)

3.4.18 P [HIV] (This problem is continued from Problem 3.3.9).

(a) Modify the MATLAB program *ODEsim.m* to run a simulation of the HIV model

$$\frac{dS}{dt} = R - DS - BVS\,,$$

$$\frac{dI}{dt} = BVS - DI - MI\,,$$

$$\frac{dV}{dt} = PI - CV\,,$$

where S and I are the concentrations of healthy and infected T cells relative to normal, and V is a measure of the number of free HIV virions in the bloodstream, relative to what might be considered an average initial viral load. Use initial conditions of $S(0) = 1$, $I(0) = 0$, and $V(0) = 1$. Plot separate graphs of the three variables over a time interval of 100 days. Reasonable parameter values are $D = 0.01$, $R = 0.01$, $M = 0.39$, $C = 3$, $B = 0.0000006$, and $P = 9000000$.
(b) Suppose an evil dictator wants to injure one of his critics by having him infected with HIV. Just to be sure it will work, let's say the initial dose is one million times the average initial viral load. Rerun the simulation and plot a different set of graphs for comparison.
(c) Describe the difference the initial viral load makes in the hypothetical scenario of (b).
(d) Compare the graphs of S, I, and V for the second scenario. Does something stand out in the graphs? (This will be explored further in Sect. 6.4.)
(e) Rerun either scenario for 300 days rather than 100 days. What seems to be happening?[31]

(This problem is continued in Problem 3.6.11.)

[31] This model is only intended for the initial stage of HIV infection, before AIDS develops.

3.5 Case Study: Two Scenarios from the COVID-19 Pandemic

As we have noted earlier, models need to be designed with a specific purpose in mind. Thus, there is no one "COVID-19" model.

For obvious reasons, the COVID-19 models that have generated the most interest are those with a goal of forecasting short-term trends such as ICU capacity and deaths. Models intended for this purpose require a large amount of detail, such as the age distribution of the population and the age-dependent frequencies of hospitalization and death. Such models require data not readily available and their complexity puts them outside the scope of this book.

Other models are intended to explore the possible impact of public health policies. These models do not require the same level of detail; indeed, extra detail may make it harder for the model to make meaningful predictions (see Sect. 2.4).

Because of their relative simplicity, models intended to predict the effect of policies need substantial revision when major events occur, such as the development of a vaccine or the change from one dominant variant to another. In this section, we consider two models the author created to explore the impact of public health policy at two critical moments in the progression of the pandemic: the beginning of testing and social distancing in March 2020 and the rise of the delta variant, which occurred at roughly the same time as the vaccine rollout in early 2021.

3.5.1 March 2020

COVID-19 was first identified in December 2019. By January 2020, the broad community of epidemiologists had identified the new disease as a danger. On March 11, the World Health Organization declared COVID-19 to be a pandemic. Within the next few weeks, most institutions and governments around the world had begun to impose public health policies such as social distancing and masking. The model presented here was originally created in late March 2020, at the beginning of interventions and public discourse about them. We consider a revised version created in January 2021 to better reflect what had been learned about the initial scenario.

Building the Model

The choice of model components is based on the natural history of the disease and the questions to be addressed. The logical starting point for COVID-19 is the SEIR model of Sect. 3.3. Several augmentations are required to model the March 2020 COVID-19 scenario.

1. We need an additional class for asymptomatic infectious patients (A). Compared to symptomatic patients, asymptomatic patients are less infectious, recover more quickly, and are less likely to be tested.
2. To assess the public health impact of the pandemic, we need to track either the number of hospitalized patients or the number of patients in ICUs, or both. For simplicity, our model includes an additional infectious class for hospitalized patients (H) and does not separately track ICU patients. To better account for the progression of the disease, we split the symptomatic infectious class into two groups: those who will not need hospitalization (I_1) and those who will (I_2).[32] We also add a separate class (D) for deceased individuals rather than incorporating these into the R class, as is done in the SEIR model. Thus, the model structure could be described as SEAIHRD.
3. To assess the impact of mitigation strategies on the pandemic, we need to build in testing, isolation of the sick, (possibly) quarantining of those known to be exposed, and the combined effect of

[32] Of course, no infectious individual knows whether (s)he is going to be hospitalized, but models only track class counts rather than individual results.

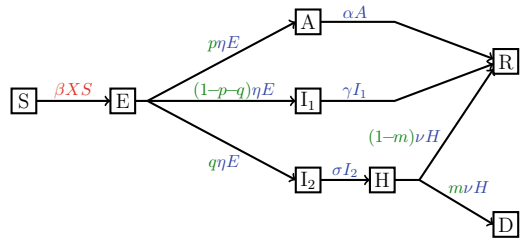


Fig. 3.5.1 Schematic diagram of the SEAIHRD model. Red indicates transmission processes, blue indicates transition processes, and green is for probabilities

social distancing and mask use. These factors do not change the class structure of the model, but they do change the formula for transmission.

Figure 3.5.1 displays the compartment diagram that represents our COVID-19 model. The diagram is based on the following assumptions about the processes that move individuals among the classes.

1. Susceptible individuals become infected at a rate proportional to the susceptible population count S and an "effective infectivity" count X. While the infectious class I is the only class capable of transmitting the disease in an SEIR model, the COVID-19 model has different categories of infectives with different levels of infectivity, which contribute to the effective infectivity in different ways (see below).
2. Latent individuals (class E) become infectious at rate ηE; a fraction p of these become asymptomatic, a fraction q become prehospitalized symptomatic, and the remainder becomes standard symptomatic.
3. Nonhospitalized infectives recover at rate γI_1 and asymptomatic individuals recover at rate αA. Prehospitalized infectives become hospitalized at rate σI_2.
4. Hospitalized individuals progress out of the hospital at rate νH; a fraction m of these die, while the rest recover.
5. Recovered individuals are immune for long enough that we can ignore possible loss of immunity.[33]
6. Deaths from unrelated causes and births are sufficiently small over the course of the epidemic that they can be ignored.

These assumptions lead to the differential equations

$$\frac{dS}{dt} = -\beta XS\,, \tag{3.5.1}$$

$$\frac{dE}{dt} = \beta XS - \eta E\,, \tag{3.5.2}$$

$$\frac{dA}{dt} = p\eta E - \alpha A\,, \tag{3.5.3}$$

$$\frac{dI_1}{dt} = (1-p-q)\eta E - \gamma I_1\,, \tag{3.5.4}$$

$$\frac{dI_2}{dt} = q\eta E - \sigma I_2\,, \tag{3.5.5}$$

$$\frac{dH}{dt} = \sigma I_2 - \nu H\,, \tag{3.5.6}$$

[33] Duration of immunity was at least several months for the original strain. Models for the omicron variant cannot neglect loss of immunity.

$$\frac{dR}{dt} = \alpha A + \gamma I_1 + (1-m)\nu H\,, \tag{3.5.7}$$

$$\frac{dD}{dt} = m\nu H. \tag{3.5.8}$$

To complete the model, we need to define the quantity X that represents the effective infectious population, which is the number of individuals of class I needed to match the total infectivity of the actual population distribution. It is here that the complexity of COVID-19 dynamics is seen. A significant number of additional assumptions are needed. (We use $I = I_1 + I_2$ for simplicity.)

1. A fraction c of class I are identified by a positive test.[34] These confirmed cases have decreased infectivity because they are put into isolation.[35]
2. The infectivity of each unconfirmed symptomatic infective is 1 (without loss of generality because there is the additional rate constant β in the transmission rate formula).
3. Asymptomatics, confirmed infectives, and hospitalized infectives have infectivities of f_a, f_c, and f_h (all less than 1) relative to that of unconfirmed symptomatic infectives. The overall contribution of hospitalized infectives to the pandemic is small enough that we take $f_h = 0$.[36]
4. There is a "contact factor" $\delta \leq 1$ that represents the level of risk from the average person's sum total of encounters, relative to normal. This parameter can be used to represent both physical distancing, which decreases the rate of encounters, and wearing of masks, which decreases the risk of each encounter. It is applied to unconfirmed infectives, both symptomatic and asymptomatic, but not to confirmed infectives (who are already in isolation).

With these assumptions, the effective number of infectives is

$$X = f_c c I + \delta[(1-c)I + f_a A]. \tag{3.5.9}$$

Parameterizing the Model

The model requires a large number of parameters, which we can classify by type:

1. Transition rate constants: η, α, γ, σ, and ν. These are calculated as reciprocals of the mean times for each transition process, for which we had some useful data by late spring 2020.
2. Probabilities: p, q, and m. Of these, p and m can be determined from data. The parameter q is difficult to determine directly from data. Instead, we use estimates for the fraction of confirmed cases (as of spring 2020) p_c and the fraction of confirmed cases that required hospitalization q_h and then calculate q as

$$q = \frac{q_h p_c}{1 - p_c}. \tag{3.5.10}$$

[34] In spring of 2020, testing in most countries was restricted to people with symptoms. For a scenario in which anyone can choose to be tested, we should use c_i and c_a for the fractions of confirmed cases in I and A, and the formula for X (3.5.9) would need to be modified.

[35] The word "quarantine" is incorrect here, as it refers to the isolation of individuals who have *not* tested positive, generally done because of known exposure—these individuals could be in any of classes S, E, A, I, or even R.

[36] Removing this process from earlier versions never made a visible change in any graphs. While it is undoubtedly true that some health care workers caught the disease from patients, a greater number of them probably caught the disease in the hospital cafeteria.

Table 3.5.1 Primary Parameter Values

Parameter	Meaning	Value	Reference
f_a	Relative infectivity of A	0.75	[2]
f_c	Relative infectivity with isolation	0.1	Estimate
m	Deaths per H	0.25	[4]
p	Asymptomatic fraction	0.4	[2]
p_c	Fraction of confirmed cases	0.09	[2]
q_h	Hospitalizations per confirmed case	0.12	[4]
t_2	Mean early doubling time of H (days)	3–5	[19, 23]
$t_a = 1/\alpha$	Mean infectious period for A (days)	8	[1]
$t_e = 1/\eta$	Mean incubation time (days)	5	[10]
$t_h = 1/\nu$	Mean hospitalization duration (days)	8	[7, 14]
$t_{i1} = 1/\gamma$	Mean infectious period for I_1 (days)	10	[8, 15]
$t_{i2} = 1/\sigma$	Mean transition time to H for I_2 (days)	6	[7]
c	Confirmed cases per I	0.1–0.8	
δ	Contact factor	0.1–1	

The value of p_c is at best a crude estimate, obtained by a study that tested everyone in a target population, including people with no known symptoms. The total number of deaths in a scenario will serve as a rough check on this parameter.

3. Infectivities: f_a and f_c. The first can be determined from data, while we can only estimate the second based on isolation behavior.
4. Transmission rate constant: β. This parameter is not easy to measure. As in Sect. 3.4.3, we will use a simplified early-phase model along with data on hospitalization counts (see below).
5. Public health parameters: c and δ: These will be specified as part of the scenario.
6. Initial conditions are the starting values for the state variables, which will be specified as part of the scenario. For some scenarios, the early-phase model and the initial hospitalization count determine the initial conditions for E, A, I_1, and I_2.

Table 3.5.1 contains the primary parameters taken directly from data or estimated.

Early-Phase Exponential Growth

For the first few weeks of the pandemic, the susceptible population fraction S can be approximated as a constant $S = 1$. This yields a linear model, which allows us to assume that all the infectious class counts are proportional to that of the latent class, with all growing exponentially with rate λ. The rate constant can be estimated from data using the doubling time equation (see (1.1.5)):

$$\lambda t_2 = \ln 2, \tag{3.5.11}$$

where t_2 is the doubling time of whichever class is easiest to measure. Given the difficulties of diagnosing COVID in the asymptomatic and mildly symptomatic, the best choice is the hospitalized class

H. The model resolves into a series of formulas to determine β. We summarize the result here, along with the formula for the basic reproductive number, leaving the derivations as an exercise.

Theorem 3.5.1
Suppose the initial population is almost entirely in the susceptible class. Then the SEAIHRD model (3.5.1)–(3.5.10) shows an extended exponential growth phase with

$$E \propto e^{\lambda t}, \quad A = aE, \quad I_1 = iE, \quad I_2 = jE, \quad H = hE, \quad S \approx 1, \tag{3.5.12}$$

where λ is determined from the hospitalization doubling time t_2 by

$$\lambda = \frac{\ln 2}{t_2}$$

and

$$a = \frac{p\eta}{\lambda + \alpha}, \quad i = \frac{(1-p-q)\eta}{\lambda + \gamma}, \quad j = \frac{q\eta}{\lambda + \sigma}, \quad h = \frac{\sigma j}{\lambda + \nu}.$$

Furthermore, the parameter β is given by

$$\beta = \frac{\lambda + \eta}{i + j + f_a a}$$

and the basic reproductive number is

$$R_0 = \beta \left[\frac{f_a p}{\eta} + \frac{(1-p-q)}{\gamma} + \frac{q}{\sigma} \right]. \tag{3.5.13}$$

The values for p_c and q_h in Table 3.5.1 yield $q = 0.018$. Doubling times from 3 days to 5 days ultimately yield $\mathcal{R}_0$ values from 6.8 to 3.9. In late March of 2020, the "accepted" value of $\mathcal{R}_0$ for COVID-19, obtained by statistical analysis of known transmissions, was 2.6. This low value corresponds to a doubling time of 8 days, which is clearly too large in comparison with known data. The problem with the statistical analysis result is that the method misses most asymptomatic cases. The best estimate we have for the original strain, $\mathcal{R}_0 \approx 5.7$, was obtained using a method similar to ours and published in July 2020 [23]. Our method gives this value using a doubling time of 3.5 days, which is consistent with the known data.[37] While we do not have adequate data to reliably use the same method to estimate $\mathcal{R}_0$ for subsequent variants, we do have a rough estimate of a 2-day doubling time for omicron, with which our model (along with updates for the transition rate parameters) suggests a basic reproduction number of about 10.

[37] It is possible to match data reasonably well with an unrealistically low value for $\mathcal{R}_0$, but doing so requires that the value of mitigation strategies such as masking and testing be underestimated. If the goal is to make predictions rather than merely to match data, it is important that each of the parameter values be realistic; therefore, modelers should use $\mathcal{R}_0$ values derived from epidemiological outcomes rather than statistical analysis of case data.

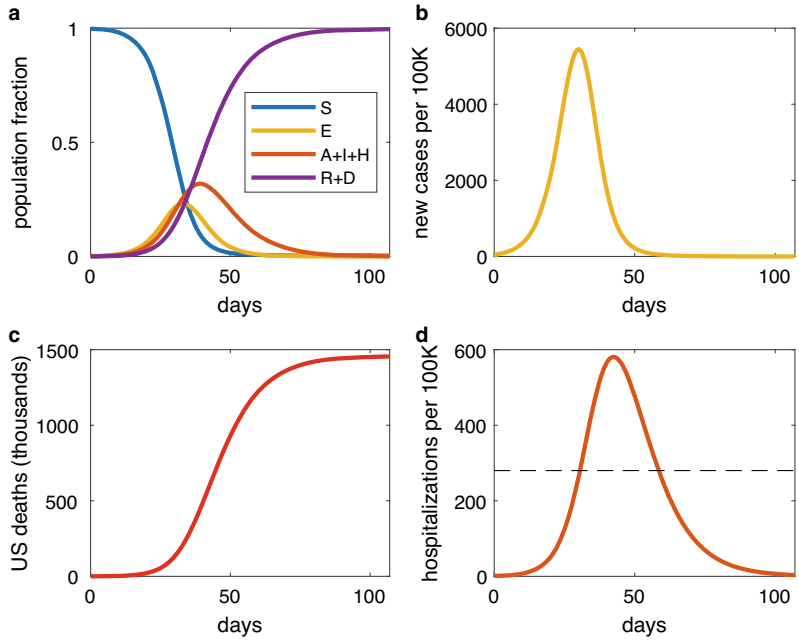


Fig. 3.5.2 The SEAIHRD COVID-19 model default scenario—no testing or distancing

Model Investigation

Figure 3.5.2 shows the default scenario for March 2020: a basic reproductive number of 5.7, no testing, masking, or social distancing, and a starting hospitalization level of one patient per 100K total population. Panel **a** shows the usual plot of susceptible, latent, total infectious, and removed, and we can see that the pattern is similar to an SEIR model with the same basic reproduction number. Note that the pandemic runs its course in about three months, with nearly everyone infected by the 2-month mark. Panel **b** shows a peak of nearly 6000 new cases per day per 100K people occurring about one month from the beginning of the scenario, but of course, this consists largely of unconfirmed cases. Panels **c** and **d** show the projected death toll for the United States and the projected hospitalization count per 100K. The death toll in this scenario is 1.5 million, which seems consistent with what actually happened, given that there was some mitigation. Similarly, the actual hospitalization count was high enough to exceed the capacity for some regions (the dashed line in the plot is the average capacity in the US), and the dire prediction in the plot seems consistent with what would have happened without any mitigation.

The purpose of the model is to test different mitigation strategies. There are a lot of experiments we could do; here we show just two of them. Figure 3.5.3 shows the effect of testing by comparing simulation plots similar to those of Fig. 3.5.2 but with different values of the confirmed case fraction c, keeping all other parameters the same. The blue curves are the default scenario. Modest amounts of testing (50% or less) have only a small impact on the course of the pandemic. With $c = 0.75$, there is a noticeable decrease in the severity—for example, maximum hospitalization is reduced from about 6 per thousand to just under 4 per thousand—but it is still high enough to overwhelm the health care system. The total number of deaths is not reduced much by this change in testing. Only when testing becomes nearly universal do we see a large benefit to the health care system, but the total death count is still above one million for the United States. Of course, the exact numbers are affected by our estimates for the parameters. The best conclusion to draw is that universal testing reduces the death toll by about 20%. Testing clearly has some value, but not enough without other mitigation strategies.

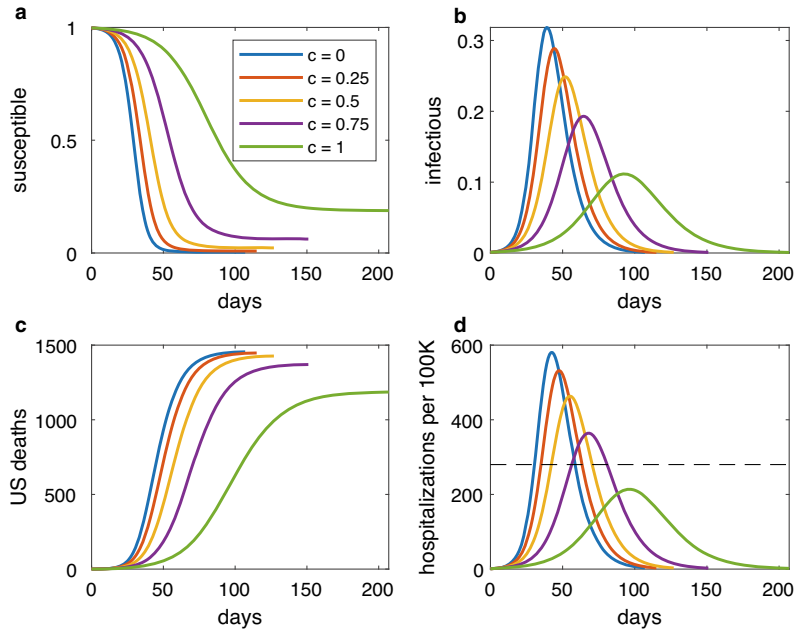


Fig. 3.5.3 The SEAIHRD COVID-19 model with different levels of testing; other parameters are the default values

Figure 3.5.4 shows the results of an experiment that tests the importance of the contact factor. The results depend strongly on how δ compares to a critical value of roughly 0.2. This value reduces the effective basic reproduction number to 1. It is slightly more than $1/\mathcal{R}_0$ because lowering contact rates only affects the unisolated. Below the critical value of δ, we have the pandemic under control, but this is only an expedient while waiting for a treatment or a vaccine. The scenario ends quickly, but with nearly the entire population still susceptible. If we were to end our vigilance at this point, the pandemic would come roaring back like a wildfire that is only partially contained when the fire crews go home. As δ decreases from 1, we don't see much improvement in the total death count, but we immediately see improvement in the maximum hospitalization count. With the given parameter estimates, we need δ to be about 0.25 in order to decrease the U.S. death toll to the roughly 800,000 that we had when the vaccine rollout occurred in January 2021. This seems to be a reasonable figure. Many people decreased their transmission risk by much more than a factor of 4, while other people largely ignored public health recommendations.

3.5.2 January 2021

Two events significantly altered the trajectory of the pandemic in the U.S. in early 2021. The mRNA vaccines became available in late December 2020, but only in very limited supply. And the delta variant began its rise to dominance in late January or early February. While these events were not quite simultaneous, it is reasonable from a historical modeling perspective to treat them as if they were.

The change from the original strain to the delta strain does not require a significant change to the model. The natural histories of the two strains are the same; the only difference is in the parameter values. For simplicity, we assume that the only significant parameter value changes are the basic reproduction number and the hospitalization fraction, which were both higher for delta. We don't have good data for either, but 8.0 seems to be a reasonable estimate for $\mathcal{R}_0$ and 0.03 for q.

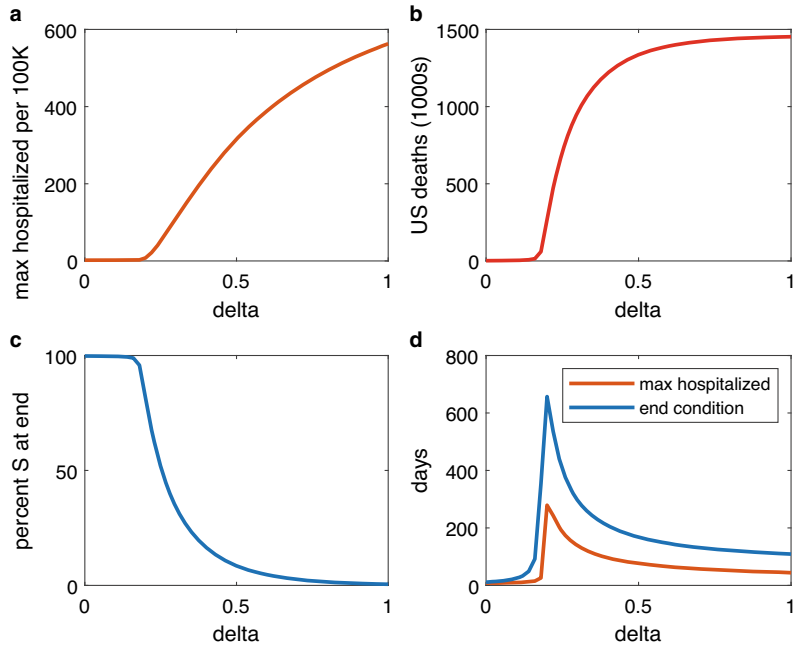


Fig. 3.5.4 The effect of social distancing and masking on the SEAIHRD COVID-19 model outcomes, assuming a low level of testing ($c = 0.1$)

The primary difference we need to model the January 2021 scenario is the addition of vaccination. The usual way to add vaccination to an SEIR model is with a transition process that moves individuals directly from class S to class R. This is inadequate for our COVID-19 scenario for several reasons:

1. There was (and has continued to be) significant resistance to vaccination in addition to those unable to be vaccinated.
2. The standard way of adding vaccination to a model assumes that distribution of vaccine is not limiting. In reality, people who wanted to be vaccinated had to wait weeks or months for an appointment.
3. The standard vaccination model assumes that supply is not limiting. In reality, it took time for manufacturers to ramp up production.
4. Vaccination does not necessarily eliminate infection; for some patients, it allows breakthrough infections of lower severity.
5. Vaccination was administered preferentially to patients deemed to be at high risk.

Each of these features must be addressed in a COVID-19 model for January 2021.

1. The susceptible class is divided into a prevaccinated class for those waiting their turn for the vaccine and an unprotected class for those who are unable or unwilling to be vaccinated.
2. The vaccination component of the model is based on the limited-distribution vaccination model from Sect. 3.1.
3. The vaccination rate coefficient is taken as an increasing function of time rather than a constant, as in Sect. 3.1. In addition, the model will need to account for vaccine doses administered to people who are already in the removed class.

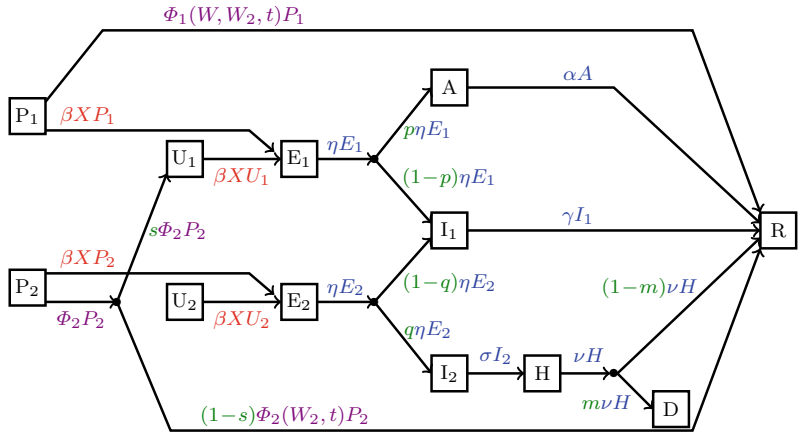


Fig. 3.5.5 Schematic diagram of the PSEAIHRD model. Red indicates transmission processes, blue indicates transition processes, violet is for vaccination processes, and green is for probabilities

4. The prevaccinated, susceptible, and latent (E) classes are divided into low-risk and high-risk subgroups, with vaccination conferring immunity to low-risk people and moving high-risk people into the low-risk category when not conferring immunity.[38]

Building the Model

Figure 3.5.5 displays a schematic diagram for the PUEAIHRD model that adds vaccination to the earlier scenario.[39] The additional prevaccinated (P) class allows us to draw a distinction between individuals who are susceptible in the usual way and individuals who are currently susceptible, but in line for vaccination. The model is largely the same as before, but with these additional assumptions.

1. At the beginning of the scenario, susceptible individuals are divided into low- and high-risk prevaccinated classes P_1 and P_2 and corresponding unprotected classes U_j, with a fraction h of high risk and vaccine non-acceptance fractions r_j.
2. Individuals in classes P and U are infected with COVID-19 in the same manner as in the March 2020 model.
3. Individuals leave classes P_2 and P_1 through vaccination at rates $\Phi_2(W_2, t)P_2$ and $\Phi_1(W, W_2, t)P_1$ that will have to be determined by a vaccine submodel. We assume that vaccination always moves low-risk individuals from class P_1 to class R. High-risk individuals move from class P_2 to S_1 with probability s and class R otherwise.[40]

We retain the assumption that removed individuals remain immune to further infection. This gradually became less true, but the extra complication of building in loss of immunity is not necessary in a model intended to address short-term questions.

[38] There is no clear definition of high risk, so it is reasonable to define high risk as people who are at risk of breakthrough infections and low risk as people who have a good chance of being asymptomatic.

[39] The model presented here is from [11].

[40] Note that we are neglecting outright failure of the vaccine, which was not common.

From Fig. 3.5.5, the model equations are

$$P_1' = -\beta X P_1 - \Phi_1(W, W_2, t) P_1, \tag{3.5.14}$$
$$P_2' = -\beta X P_2 - \Phi_2(W_2, t) P_2, \tag{3.5.15}$$
$$U_1' = s\Phi_2(W_2, t) P_2 - \beta X U_1, \tag{3.5.16}$$
$$U_2' = -\beta X U_2, \tag{3.5.17}$$
$$E_1' = \beta X S_1 - \eta E_1, \tag{3.5.18}$$
$$E_2' = \beta X S_2 - \eta E_2, \tag{3.5.19}$$
$$A' = p\eta E_1 - \alpha A, \tag{3.5.20}$$
$$I_1' = (1-p)\eta E_1 + (1-q)\eta E_2 - \gamma I_1, \tag{3.5.21}$$
$$I_2' = q\eta E_2 - \sigma I_2, \tag{3.5.22}$$
$$H' = \sigma I_2 - \nu H, \tag{3.5.23}$$
$$R' = \Phi_1(W, W_2, t) P_1 + (1-s)\Phi_2(W_2, t) P_2 + \alpha A + \gamma I_1 + (1-m)\nu H, \tag{3.5.24}$$
$$D' = m\nu H, \tag{3.5.25}$$

where $S = P + U$ and the prime symbol is the derivative with respect to time t.

The initial conditions for P_j and U_j depend on the initial susceptible population S_0, the high risk fraction h, and the vaccine non-acceptance fractions r_j:

$$P_1(0) = (1-r_1)(1-h)S_0\,, \quad P_2(0) = (1-r_2)hS_0\,, \tag{3.5.26}$$

$$U_1(0) = r_1(1-h)S_0\,, \quad U_2(0) = r_2 hS_0\,. \tag{3.5.27}$$

The Vaccination Submodel

The vaccination submodel is based on the vaccination model of Sect. 3.1, modified to account for the preferential delivery of vaccination to those deemed to be of high risk.[41] Let W, W_1, and W_2 be the population fractions of all people waiting to be vaccinated, low-risk people waiting to be vaccinated, and high-risk people waiting to be vaccinated. Using the model (3.1.7) for both W and W_2, we have

$$\frac{dW}{dt} = -\Phi_2(W, t) W\,, \qquad W(0) = 1 - r\,; \tag{3.5.28}$$

$$\frac{dW_2}{dt} = -\Phi_2(W_2, t) W_2\,, \qquad W_2(0) = (1-r_2)h\,; \tag{3.5.29}$$

where

$$\Phi_2(W, t) = \frac{\phi g(t) W}{K^2 + W^2}\,, \qquad g(t) = \min\left(\frac{t}{\tau}, 1\right)\,. \tag{3.5.30}$$

It remains to determine Φ_1, which is defined by

$$\Phi_1(W, W_2, t) = -\frac{W_1'}{W_1} = -\frac{W' - W_2'}{W_1}. \tag{3.5.31}$$

[41] See Problem 3.1.2.

After some algebra, we obtain the result

$$\Phi_1(W, W_2, t) = \frac{\phi K^2 g(t)(W + W_2)}{(K^2 + W^2)(K^2 + W_2^2)}. \tag{3.5.32}$$

The model is now complete, with equations (3.5.14)–(3.5.27) augmented by (3.5.28)–(3.5.29), and with appropriate initial conditions for E_1, E_2, A, I_1, I_2, H, and R.

Model Investigation

The key questions to ask for the January 2021 scenario are about the impact of the gradual vaccine rollout, particularly if it is accompanied by a decrease in the use of prior mitigation strategies. It is reasonable to guess that the benefit from the introduction of vaccine may possibly be less important than the simultaneous relaxing of mask use and social distancing. In the event, this turned out to be the case.

We defer specific investigations to Project 3D.

Problems

Problems marked with "p" require some programming beyond data entry.

3.5.1 Use *SEAIHRD_onesim.m* to reproduce Fig. 3.5.2.

3.5.2 Use *SEAIHRD_onesim.m* to redo Fig. 3.5.2, but with a contact factor of 0.5 instead of 1.0. Discuss the significance of a 50% reduction in contact rates.

3.5.3 Use *SEAIHRD_comparison.m* to reproduce Fig. 3.5.3.

3.5.4 Explore the impact of the contact factor by using *SEAIHRD_comparison.m* to prepare a graph similar Fig. 3.5.3, using contact factors of 1, 0.8, 0.6, and 0.4. Assume $c = 0.25$, which is a reasonable guess for the early stages of testing. Discuss the impact of the contact factor in this range.

3.5.5 Repeat Problem 3.5.4 using contact factors of 0.24, 0.22, 0.2, 0.18, and 0.16. Discuss the impact of the contact factor in this range.

3.5.6 Use *SEAIHRD_paramstudy.m* to reproduce Fig. 3.5.4.

3.5.7 Repeat Problem 3.5.6, but with testing fractions $c = 0.5$ and $c = 0.75$. Discuss how different levels of testing affect the impact of the contact factor.

3.5.8 Some U.S. officials claimed in the early stages of the pandemic that the main effect of more testing is a higher reported case count, while public health experts argued that testing was an essential component of public health policy. These contrasting views can be explored by varying the testing rate in a scenario that otherwise matches the early stages of the pandemic. Modify *SEAIHRD_paramstudy.m* to prepare a graph similar to Fig. 3.5.4, but with c as the independent variable. Assume $\delta = 0.3$, which is a reasonable guess for the early stages of public health intervention. Discuss the results.

3.5.9 On April 1, 2020, Dr. Anthony Fauci, head of the U.S. National Institute for Allergy and Infectious Diseases, warned that even with aggressive measures, the total number of deaths from

COVID-19 in the United States during the time required to control the initial outbreak could be 100,000–200,000. (This figure was considered outrageously high by many at the time. In the event, 200,000 deaths had occurred by late September.) He also suggested that the most aggressive measures could reduce these numbers noticeably. Assess Dr. Fauci's claims, assuming a testing rate of $c = 0.3$.

(a) Use trial and error with *SEAIHRD_comparison.m* to identify the range of values for δ that yield deaths between 100K and 200K. A few experiments should suffice to get an estimate to the nearest 0.001 for each level. Remember that you can use multiple values with *SEAIHRD_comparison.m*—but not too many.
(b) For a more in-depth look at the effect of contact factor on death counts, run *SEAIHRD_paramstudy.m* with a range of contact factors and examine the graph of total deaths. Try the ranges $0 \leq \delta \leq 1$ to get a picture for the full range and $0.1 \leq \delta \leq 0.2$ to get a picture that focuses on the critical range.
(c) Given that masks at that time reduced transmission by about a factor of 4 and physical distancing by about a factor of 2, does the model support the claim that more aggressive measures would have significantly impacted the death rate? Also, comment on how the epidemic duration and final susceptible percentage changes and what these results mean for public health.

3.5.10 [p] In an interview on March 18, 2020, a U.S. government official said "If we can get all America to pitch in for the next 15 days, we can flatten the curve." This suggests that a 15-day lockdown would have had a permanent benefit. In mid-April, other government officials recommended 45 days.

To address the implied claim of a permanent benefit from a limited lockdown, we need to modify the base function *seaihrd* so that `delta` can change in the middle of a scenario from a low value (the lockdown) to 1 (post-lockdown) and modify *SEAIHRD_comparison.m* to utilize the new function.

The changes to *SEAIHRD_comparison.m* just add a new parameter called `lockdays`:

1. Change line 81 so that `xvals` is assigned to the name `lockdays`.
2. Change the function call in line 83 by making the function name `seaihrd2` and adding an additional item `lockdays` at the beginning of the argument list.

The changes to *seaihrd.m* are more substantial. Start by changing the name of the function to `seaihrd2` and saving the file as *seaihrd2.m*. Then add `lockdays` at the beginning of the argument list in the `function` statement. The COMPUTATION section needs some significant changes:

1. Some code needs to be inserted that sets `delta` to 1, either because there is no lockdown or because the lockdown has elapsed. These are two different triggers, so they must be implemented separately.
   a. At the beginning of the COMPUTATION section, add an `if` block that sets `delta = 1` whenever `lockdays==0`.[42]
   b. After the statement that saves the results at time t+1, add an `if` block that sets `delta = 1` and `summ = sum(Y(2:6))` whenever `t==lockdays`.
2. Code needs to be inserted so that the check for the end condition (the `if-else` construction at the end of the `for` loop) is only checked after the lockdown. To do this, put the `if-else` block at then end of the loop inside an `if` statement that triggers when `t>lockdays`.

In addition to the program changes, set the default scenario data to $\delta = 0.1$ and $c = 0.1$, along with the usual default values for the other parameters. Note that the post-lockdown value of $\delta = 1$ is built into the new function *seaihrd2*.

[42] Don't forget the semicolon at the end of the assignment statement.

(a) Run the modified program *SEAIHRD2_comparison.m* using lockdown times of 0 days, 15 days, and 45 days as `xvals`.[43] Describe the effect these limited lockdowns have on the course of the epidemic. Also, explain why the results come out this way. How accurate were the claims that a lockdown of limited duration would have had a permanent benefit?

(b) We set the total infectivity for unconfirmed symptomatics (δ) to match our assumed infectivity of confirmed symptomatics ($f_c = 0.1$). This is a reasonable guess for how much we can reduce transmission risk with maximum effort. However, we can also test the most optimistic assumption possible by rerunning the experiment with $f_c = 0$ and $\delta = 0.05$, reflecting a 20-fold reduction in transmission risk. This would eliminate all transmission during the limited lockdown. Does this help? Why or why not?

3.6 Equivalent Forms

After studying this section, you should be able to:

- Identify equivalent forms of a mathematical model.
- Convert a mathematical model to dimensionless form, given appropriate choices for reference quantities.
- Explain why equivalent forms do not always produce equivalent parameter values when fit to a data set.

In physics, everyone chooses the same way to write Newton's Second Law of Motion: $F = ma$. The model could be written in other ways, but this form has become standard. This uniformity makes it easy for a reader to compare material written by different authors. In contrast, uniformity of model appearance is rare in biology. Different authors inevitably choose different ways to write the same model. Hence, the reader who wants to understand mathematical models in biology needs to develop the skill of discriminating between different models and different forms of the same model. This is a skill that mathematicians take for granted, but it is problematic for most students. In this section, we consider different ways of writing the same model, working our way up from forms that differ only in notation to forms with more substantive differences.

3.6.1 Notation

Any given model can be written with a variety of notations. Sometimes these differences are due to the different settings in which the model can arise; other times, they are due simply to the lack of standardization.

▶ **Example 3.6.1** Many important biochemical reactions are of the Michaelis–Menten type. The initial rate of reaction depends on the concentration of the principal reactant, called the substrate. The model is generally written as

$$v = \frac{V_M[S]}{K_M + [S]}, \qquad [S], V_M, K_M > 0,$$

where v is the initial rate of the chemical reaction, $[S]$ is the concentration of the substrate, V_M is the initial rate of chemical reaction for infinite $[S]$, and K_M is called the semisaturation parameter.

[43] Examine the plots carefully as a way of checking that your program is doing what it is supposed to do. The plots for a lockdown time of 0 should be identical to what you get in *SEAIHRD_onesim.m* with $\delta = 1$ and the right value of c. The other plots should start out looking like a simulation with a smaller value of δ.

The Monod growth function in microbiology is used to model the rate of nutrient uptake by a microorganism as a function of the concentration of nutrients in the environment. The Monod model, in one of several common notations, is

$$r = \frac{qS}{A+S}, \qquad S, q, A > 0,$$

where r is the rate of nutrient uptake, S is the concentration of the nutrient, q is the maximum uptake rate, and A is the semisaturation parameter.

The Michaelis–Menten and Monod models were designed for different contexts. Although the interpretation of each model is based on its own context, the models are mathematically identical, differing only in notation. □

The notation for the Michaelis–Menten equation is almost standard, while the Monod growth function has almost as many systems of notation as the number of authors who have written about it. Although the two formulas use different symbols, they are identical in mathematical form. Both formulas say that the dependent variable is a rational function of the independent variable, with the numerator of the form "parameter times independent variable" and the denominator of the form "parameter plus independent variable." If we focus on the *symbols*, we are misled by the apparent differences. If we focus on the *roles* of the symbols, we see that the formulas are identical. Few symbols in mathematics have a fixed meaning. Even π, which is the universal symbol for the ratio of circle circumference to circle diameter, is occasionally used for other purposes. This lack of standardization makes it necessary to provide the meaning of each symbol for any given context.

3.6.2 Algebraic Equivalence

Occasionally, it is possible to write the same model in two different ways by performing algebraic operations. This is most common with formulas that include a logarithmic or exponential function.

▶ **Example 3.6.2** In fitting an exponential model by linearization in Sect. 2.2, we converted the model

$$z = Ae^{kt}$$

to the form

$$\ln z = \ln A + kt.$$

These forms are equivalent in the sense that they both produce the same z values for any given t value. □

▶ **Example 3.6.3** A certain problem in plant physiology eventually reduces to the algebraic equation

$$Pe^{-P} = e^{-R},$$

where R is the independent variable and P is the dependent variable. Hence, the formula defines P implicitly in terms of R. Various algebraic operations can be performed to change this equation. For example, we can multiply by e^P and e^R to obtain the equation

$$Pe^{R} = e^{P}.$$

We can then take the natural logarithm on both sides, obtaining

$$\ln P + R = P\,.$$

Further rearrangement yields

$$P - \ln P = R\,.$$

All of these forms are algebraically equivalent, although they appear different at first glance. □

None of the formulas in Example 3.6.3 allow us to solve algebraically for P. The choice among them is a matter of taste. I prefer the last one, because that form allows us to graph the relationship by calculating values of R for given values of P.

3.6.3 Different Parameters

More subtle than differences in algebraic presentation are differences resulting from parameter choices. Sometimes it requires some algebraic manipulation to see that two similar models are actually equivalent.

▶ **Example 3.6.4** In Sect. 3.2, we developed the Holling type 2 model

$$y(x) = \frac{sx}{1 + hsx}\,, \qquad x, s, h > 0\,, \tag{3.6.1}$$

for the relationship between the prey biomass eaten by a predator of unit size in a unit amount of time (y) and the total prey biomass available in a specific region (x), where s is the amount of habitat that a predator can search per unit time and h is the time required for the predator to process one unit of prey biomass unit. The same model can also be written as

$$y(x) = \frac{qx}{a + x}\,, \qquad x, q, a > 0\,, \tag{3.6.2}$$

where the parameters q and a are different from the original s and h. The *functions* in (3.6.1) and (3.6.2) are mathematically different, so the models are not *identical*. However, the *models* represented by the functions are *equivalent* if we define the parameters in (3.6.2) by $q = 1/h$ and $a = 1/(hs)$. Note that the Holling type 2 model is also equivalent to the Michaelis–Menten and Monod models, with yet another context. □

Check Your Understanding 3.6.1:
Substitute $q = 1/h$ and $a = 1/(hs)$ into (3.6.2) and simplify the result to obtain (3.6.1).

Each of the equivalent forms (3.6.1) and (3.6.2) is preferable from a particular point of view. Form (3.6.2) is preferable from a graphical point of view, because the parameters q and a indicate properties directly visible on the graph.[44] It has the additional advantage of being semilinear,[45] so we can fit it to empirical data by the semilinear least squares method.[46] Form (3.6.1) is better from a biological point of view, because it allows us to study the effects of search speed and handling time separately.

[44] Problem 1.1.8.
[45] Section 2.3.
[46] Problem 2.3.4.

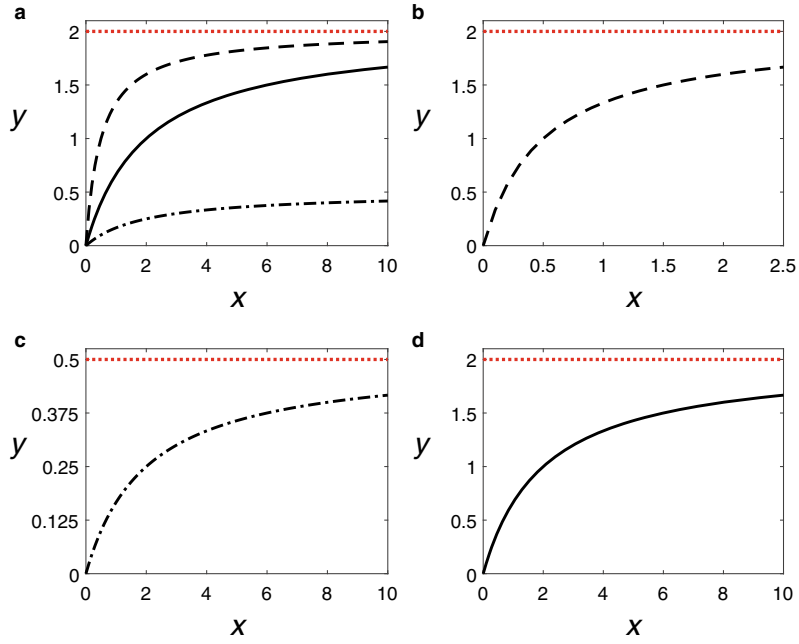


Fig. 3.6.1 The model $y(x) = qx/(a + x)$, with (q, a) values of (2, 0.5) (*dashed*), (2, 2) (*solid*), (0.5, 2) (*dash-dotted*)

3.6.4 Visualizing Models with Graphs

The appearance of a graph depends on the ranges chosen for the variables. In choosing the ranges, it is important to have a purpose in mind. Consider three instances of the predation model (3.6.2):

$$y_1 = \frac{2x}{0.5 + x}, \qquad y_2 = \frac{2x}{2 + x}, \qquad y_3 = \frac{0.5x}{2 + x}.$$

If our purpose is to see what effect the different choices of q and a have on the graph, we should plot the three models together, as in Fig. 3.6.1a. But suppose instead that we want to plot each curve separately in a way that best shows the behavior of the function. The ranges chosen for Fig. 3.6.1a look best for y_2, so we plot this curve with the same ranges (Fig. 3.6.1d). The function y_3 has a much lower maximum value than y_2, so we might choose a narrower range for the vertical axis (Fig. 3.6.1c). On the other hand, most of the variation in y_1 occurs on the left side of Fig. 3.6.1a, so we might choose a narrower range for the horizontal axis (Fig. 3.6.1b). Notice that the three individual graphs are now identical, except for the numbers on the axes.

3.6.5 Dimensionless Variables

Look again at Fig. 3.6.1. Notice that the vertical axis range is $0 \le y \le q$ for each of the single-curve plots. Similarly, the horizontal axis range for each of the curves is $0 \le x \le 5a$. We can make use of this observation to produce one plot with axis values that are correct for these three cases, and all others as well. One way is to incorporate the parameters q and a into the axis values (Fig. 3.6.2a). Alternatively, we could incorporate the parameter values into the variables themselves, as in Fig. 3.6.2b.

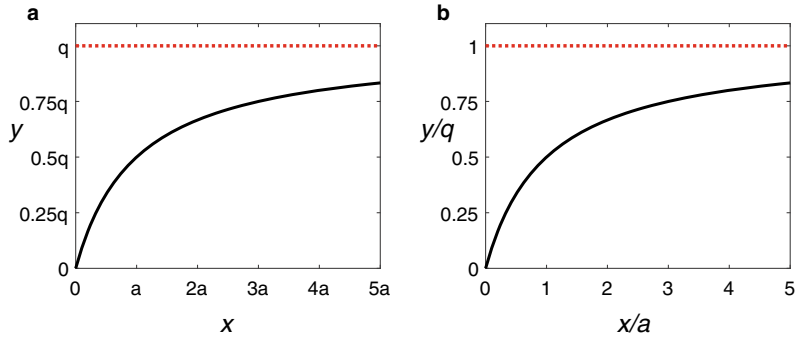


Fig. 3.6.2 The model $y(x) = qx/(a + x)$, using two different labeling schemes: **a** the factors a and q are in the axis values; **b** the factors a and q are in the axis labels

The quantities y/q and x/a are dimensionless versions of the original quantities y and x. This means that they represent the same things (food consumption rate and food density, respectively), but are measured differently. Where y is the food consumption rate measured in terms of some convenient but arbitrary unit of measurement (prey animals per week or grams per day, for example), y/q is the food consumption rate measured as a fraction of the maximum food consumption rate q. Where x is the prey concentration in a convenient but arbitrary unit, x/a is the prey concentration relative to the semisaturation concentration a.

3.6.6 Dimensionless Forms

The quantities y/q and x/a have an algebraic benefit as well as a graphical benefit. To see this benefit, we define dimensionless variables Y and X by[47]

$$Y = \frac{y}{q}, \qquad X = \frac{x}{a}. \tag{3.6.3}$$

Rearranging these definitions yields substitution formulas:

$$y = qY, \qquad x = aX. \tag{3.6.4}$$

Now we replace y and x in (3.6.2) using the formulas of (3.6.4). This yields the equation

$$qY = \frac{qaX}{a + aX}. \tag{3.6.5}$$

Removing common factors yields the dimensionless form of the predation model:

$$Y = \frac{X}{1 + X}. \tag{3.6.6}$$

[47] In this text, we will usually adopt the practice of using one case for all of the original variables in a model and the opposite case for the corresponding dimensionless variables. Other systems, such as those that add accent marks for either the dimensional or the dimensionless quantities, have greater flexibility but other disadvantages. Distinguishing by case has the advantage of easy identification of corresponding quantities without the clumsiness of accent marks.

The graph of the dimensionless model is the same as Fig. 3.6.2b, except that the axis labels are X and Y instead of x/a and y/q. The dimensionless variables measure quantities in terms of units that are intrinsic to the physical setting. The statement $y = 0.5$ cannot be understood without a unit of measurement or a value of q for comparison. The statement $Y = 0.5$ can immediately be understood as a predation rate that is half of the maximum predation rate.

Back in the Michaelis–Menten case study,[48] we said that we could assume the parameters V and K were both unity without loss of generality. This was because doing so is equivalent to working with the dimensionless form for any pair of parameter values.

3.6.7 Scaling of Differential Equation Models

Differential equation models can also be nondimensionalized. This requires substitution formulas for the independent variable of derivatives.

▶ **Example 3.6.5** Consider the dimensional differential equation model

$$\frac{dY}{dT} = -kY\,, \qquad Y(0) = Y_0,$$

where we are using T for dimensional time in order to reserve t for dimensionless time. We can define new variables y and t by

$$y = \frac{Y}{Y_0}\,, \qquad t = kT\,.$$

These formulas correspond to the substitution formulas $Y = Y_0\, y$ and $1/T = k/t$, so we have the substitution formulas

$$\frac{d}{dT} = k\frac{d}{dt}\,, \qquad \frac{dY}{dT} = kY_0\frac{dy}{dt}\,.$$

The final result is

$$\frac{dy}{dt} = -y, \qquad y(0) = 1\,. \tag{3.6.7}$$

□

Dimensionless models are not always parameter-free like (3.6.6) and (3.6.7), but they always have fewer parameters than the original dimensional model. That alone makes them preferable for model characterization. Sometimes making a model dimensionless provides valuable insight into the behavior of the model, even before any analysis is performed.[49]

Problems

3.6.1 Derive (3.6.2) from (3.6.1). In so doing, find the algebraic formulas that define h and s in terms of the parameters q and a.

[48] Section 2.5.

[49] This will be a recurring theme that we will see in the case studies of this chapter and throughout Chap. 6.

3.6.2 Explain the graphical significance of the parameter a in the dimensional model

$$y = \frac{qx}{a+x}.$$

(Hint: What is y if you set $x = a$?)

3.6.3* Rewrite the model $y = y_0 e^{-kt}$ in dimensionless form by choosing appropriate dimensionless quantities to replace y and t.

3.6.4 Create a set of four plots similar to Fig. 3.6.1 using the functions

$$y_1 = \frac{4x}{2+x}, \qquad y_2 = \frac{2x}{2+x}, \qquad y_3 = \frac{2x}{4+x}.$$

Your plots of the individual functions should be identical to those of Fig. 3.6.1, except for the numerical values on the axes.

3.6.5 Suppose the model $Q = aX^2 - bX^3$ represents the surplus of energy available to an organism of length X.

(a) Explain why it makes sense for the ingestion term to be proportional to X^2, while the metabolism term is proportional to X^3.
(b) Nondimensionalize the model using a reference length corresponding to the length at which the organism has no surplus energy. For the reference energy level, note that there is one combination of a and b that has the same dimension as Q.
(c) Use calculus to determine the dimensionless length at which the surplus energy is maximum.
(d) How large do you expect an organism to grow relative to the maximum size it could possibly achieve? Explain your answer.

3.6.6 [SIR epidemic model]
Consider the SIR model of Problem 3.4.13, but use T for dimensional time.

(a) Replace the dimensional variables in the model with dimensionless variables by using the total population N as the scales for S, I, and R, and the mean infectious duration $t_I = 1/\gamma$ as the scale for time.
(b) The dimensional parameters N, β, and γ have now sorted themselves into a single dimensionless grouping. Write the fully dimensionless form of the model using this one dimensionless parameter.
(c) In Sects. 3.3 and 3.4, we focused our analysis of epidemic models on the effect of just one parameter on the progress of an outbreak. Explain why this was appropriate.

(This problem is continued in Problem 6.1.3.)

3.6.7 [Resource harvesting]
The Schaefer model,

$$\frac{dX}{dT} = RX\left(1 - \frac{X}{K}\right) - QX, \qquad R, K, Q > 0,$$

is often used to model the impact of harvesting on a fish population, where X is either the number of fish or the biomass of fish and T is dimensional time.

(a) Explain the conceptual model that is represented by the Schaefer model. Your explanation should include both the overall compartment structure of the model and the specific forms of the two terms.
(b) Derive the dimensionless form of the Schaefer model by using $x = X/K$ and $t = RT$ for the dimensionless population and time and using an appropriate definition for E. The symbol E is commonly used in this context to suggest that it represents the harvesting "effort."

(This problem is continued in Problem 4.3.3.)

3.6.8 [Self-limiting population]
Many populations are self-limiting in the sense that their waste products decrease the population directly or decrease the capacity of the environment to support them. An example of such a model is [5]

$$\frac{dW}{dT} = AP\,,$$
$$\frac{dP}{dT} = RP\left(1 - \frac{P}{K}\right) - BPW.$$

(a) Explain the assumptions implicit in the differential equations.
(b) Nondimensionalize the model using reference quantities K for p, R/B, for W, $1/R$ for T, and parameter $a = ABK/R^2$.
(c) Explain the significance of the reference value R/B for W.

(This problem is continued in Problem 4.3.6.)

3.6.9* [Predator—prey and consumer–resource models]
Rosenzweig–MacArthur population models have the general form

$$\frac{dV}{dT} = RV\left(1 - \frac{V}{K}\right) - Cf(V)\,,$$
$$\frac{dC}{dT} = ECf(V) - MC\,,$$

where all parameters are positive and f has the properties

$$f(0) = 0\,, \qquad f' > 0\,, \qquad f'' \le 0\,.$$

These models can be used for consumer–resource systems, where V is the biomass of resource and C the biomass of consumer, or for predator—prey systems, where V and C are biomasses of prey species and predator species, respectively.

(a) Explain the assumptions inherent in the model.
(b) The simplest Rosenzweig–MacArthur model has a linear predation term, $f(V) = SV$. Nondimensionalize this version of the model using the scales K, R/S, and $1/R$ for V, C, and T, with parameters $m = M/R$ and $h = ESK/M$.
(c) What is the biological significance of the reference value R/S for the consumer population?

(This problem is continued in Problem 6.1.4.)

3.6.10 [Malaria] (Continued from Problem 3.4.16.)
Scale the Ross malaria model,

$$\frac{dX}{dT} = bp\left(1 - \frac{X}{H}\right)Y - \gamma X\,,$$

$$\frac{dY}{dT} = bq\frac{X}{H}(V - Y) - \mu Y\,,$$

using H and V as the scales for X and Y and $T_i = 1/\gamma$ as the scale for the dimensional time T. Use $\beta = bq/\mu$ as one of the parameters.

3.6.11 [HIV] (Continued from Problem 3.4.18.)

(a) Scale the HIV model,

$$\frac{dS}{dT} = R - DS - BVS\,,$$

$$\frac{dI}{dT} = BVS - DI - MI\,,$$

$$\frac{dV}{dT} = PI - CV\,,$$

where S and I are the concentrations of healthy and infected T cells relative to normal, and V is a measure of the number of free HIV virions in the bloodstream, relative to a typical initial viral load, using R/D as the scale for S, $R/(M + D)$ as the scale for I, $PR/C(M + D)$ as the scale for V, and $T_i = 1/D$ as the scale for the dimensional time T. Use parameters

$$b = \frac{BRP}{CD(M + D)}\,, \qquad \epsilon = \frac{D}{C}\,, \qquad \delta = \frac{D}{M + D}\,.$$

(b) Explain the choices for the scales in biological terms.
(c) Use the data in Problem 3.4.18 to calculate the values of the dimensionless parameters.
(d) Explain the meanings of ϵ and δ in biological terms, including what they represent and the significance of their values.

(This problem is continued in Problem 6.4.5.)

3.6.12 [Plankton]
The health of oceans depends on having a healthy population of plankton. While a full model of plankton populations that incorporates all important biological and environmental factors is outside the scope of this book, we can examine an elementary model that looks only at the balance of free nitrogen F, phytoplankton (microscopic plants) P, and zooplankton (microscopic animals) Z. The quantities of plankton could be expressed in various units, but we choose to express them in terms of their nitrogen content. In this way, each unit of nitrogen in the system counts as one unit of either F, P, or Z.

(a) Suppose the following mechanisms account for nitrogen transfers:

1. Phytoplankton die, returning their nitrogen to the water, at a rate proportional to the population size, with rate constant a.
2. Zooplankton also die and return their nitrogen to the water at a rate proportional to the population size, this time with rate constant b.
3. Free nitrogen is consumed by phytoplankton at a rate proportional to both the nitrogen concentration and the phytoplankton population, with rate constant c.
4. Phytoplankton are consumed by zooplankton at a rate proportional to both populations, with rate constant d.

Write down the appropriate differential equations for the functions $F(T)$, $P(T)$, and $Z(T)$, using T for time.

(b) Show that $N = F + P + Z$ is constant.

(c) Use (b) to eliminate F from the differential equation for P.

(d) Nondimensionalize the P and Z equations using the reference quantities N for phytoplankton, $cN/(c+d)$ for zooplankton, and $1/cN$ for time.[50] Choose dimensionless parameters α, β, and δ so that the resulting dimensionless model is

$$p' = p(1 - \alpha - p - z)\,,$$
$$z' = \delta z(p - \beta)\,.$$

(e) Explain the biological significance of the parameters α and β. [Hint: Think of cN as representing the quantity cF.]

(This problem is continued in Problem 6.1.8.)

3.6.13 [Chemostat] A chemostat is a device consisting of a container filled with water, bacteria, and nutrients, set up so that water and nutrients enter the container at some rate and contents are drained from the container at the same rate, thereby maintaining a constant volume. A simple model for a chemostat is

$$\frac{dR}{dT} = Q(R_0 - R) - \frac{SRC}{A+R}\,,$$
$$\frac{dC}{dT} = \frac{ESRC}{A+R} - QC\,,$$

where $R(T)$ and $C(T)$ are the amounts of resource and consumer in the container, Q is the flow rate in container volumes per unit time, R_0 is the concentration of the resource in the input stream, measured in resource units per container volume, S is a consumption speed, and E is the amount of consumer biomass that is added through consumption of one unit of resource.

(a) Explain each of the five terms in the model (taking QR_0 and QR as distinct terms). [Hint: Look at the signs of the terms, whether they are transitions, interactions, or otherwise, and consider the specific details if nonlinear.]

(b) Use the scales A, QR_0/S, and $1/ES$ for R, C, and T to obtain the dimensionless form

$$r' = qr_0\left(1 - \frac{r}{r_0} - \frac{rc}{1+r}\right)\,,$$
$$c' = c\left(\frac{r}{1+r} - q\right)\,.$$

You will need to find the appropriate definitions for the dimensionless parameters q and r_0.

(c) Explain why QR_0/S is a good choice of reference quantity for the consumer biomass. [Hint: Use the resource equation to interpret the quantities QR_0 and SC.]

(d) Explain the biological meaning of the parameters q and r_0. Note that these parameters are largely under control of the experimenter.

[50] The choice of reference quantity for Z is somewhat unusual. Given that the total nitrogen amount is A, it is natural to scale Z by A. The extra factors of c and $c+d$ make the dimensionless model simpler by having only two parameters that determine the equilibria rather than three. The drawback of this choice is that one unit of zooplankton no longer corresponds to one unit of phytoplankton.

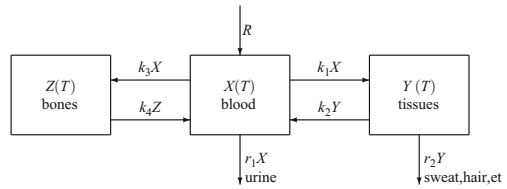


Fig. 3.7.1 Lead transport in a vertebrate body

(This problem is continued in Problem 6.1.10.)

3.6.14 [Immune system] (Continued from Problem 3.3.10.)

(a) Scale the immune system model (3.3.6)–(3.3.8) using the reference quantities K, L/M, C/B, and $1/R$ for P, M, N, and T. Choose dimensionless parameters so that the scaled model is

$$p' = p(1 - p - qm - sn)\,,$$
$$m' = a[\delta(1 - m) - mp]\,,$$
$$n' = b\left(\frac{p}{h + p} - n\right)\,.$$

(b) The parameter δ is always small. What is the biological meaning of this fact?
(c) The parameter h is always small. What is the biological meaning of this fact?

(This problem is continued in Problems 6.1.15 and 6.2.11.)

3.7 Case Study: Lead Poisoning

Pharmacokinetics is the study of the rates of drug interactions in living organisms.[51] While elementary models track a single quantity, it is usually necessary to consider the amounts of the drug stored in distinct components of the organism. Since the various rates at which the drug moves between components or exits the system depend on the amounts of the drug that are present in those different components, pharmacokinetic models appear as systems of differential equations.

Models for pharmacokinetics usually ignore the chemical reactions of the drug, but focus instead on its movement between different components of the organism. If a chemical reaction occurs in one component, it enters the model by changing the rate at which the drug can move out of that component.

As an example, we consider a model for lead poisoning. When lead is ingested, it is first absorbed by the digestive system into the bloodstream. Some is then filtered out of the blood by the kidneys and discarded in the urine. Unfortunately, lead is also absorbed into the body's tissues, and it tends to bind with bone. Simple models for lead poisoning can ignore the digestion, which happens very quickly, but they must track the amounts of lead in the blood, bones, and other tissues, and account for the various processes by which it moves among these three areas.

Figure 3.7.1 is a compartment diagram that depicts the bones, blood, and tissues as three compartments containing quantities $Z(T)$, $X(T)$, and $Y(T)$ of lead, respectively, where T is time.[52] The

[51] We use the word *drug* to mean any biochemically reactive substance introduced into an organism from external sources.

[52] T is used for dimensional time to reserve t for dimensionless time. Using capitals for the original model has the slight advantage that most of the work will then be done on the lowercase version.

arrows indicate rates at which lead enters the body, leaves the body, or moves from one compartment to another. Each of these rates includes a parameter. For example, lead enters the blood compartment from outside the system at the constant rate R and from the Y and Z compartments at the rates $k_2 Y$ and $k_4 Z$, respectively. Lead leaves the blood compartment by three different processes, with the combined rate $(k_1 + k_3 + r_1)X$. The differential equations for the compartments[53] are

$$\frac{dX}{dT} = R - (k_1 + k_3 + r_1)X + k_2 Y + k_4 Z\,, \tag{3.7.1}$$

$$\frac{dY}{dT} = k_1 X - (k_2 + r_2)Y\,, \tag{3.7.2}$$

$$\frac{dZ}{dT} = k_3 X - k_4 Z\,. \tag{3.7.3}$$

The differential equations based on the compartment diagram must be supplemented by initial conditions. Two scenarios are of primary interest:

1. Development of lead poisoning, with $R > 0$, $X(0) = Y(0) = Z(0) = 0$.
2. Clearance of lead from the body, with $R = 0$ and $X(0)$, $Y(0)$, and $Z(0)$ determined by the outcome of the first scenario.

Estimates of the parameters are supplied by Rabinowitz et al. [20], who collected data from a controlled study of an otherwise-healthy volunteer, with R in micrograms/day and the various rate constants in days^{-1}:

$$R = 49.3\,, \quad r_1 = 0.0211\,, \quad r_2 = 0.0162\,, \tag{3.7.4}$$

$$k_1 = 0.0111\,, \quad k_2 = 0.0124\,, \quad k_3 = 0.0039\,, \quad k_4 = 0.000035\,. \tag{3.7.5}$$

The compartment model of (3.7.1)–(3.7.3) can be used for simulations, which we consider here, as well as analysis, which we defer to Sect. 6.3.

▶ **Example 3.7.1** Suppose a person ingests 49.3 μg of lead per day for 2 years and then stops ingesting any more lead. To model this scenario, we start by running a simulation for 2 years with the model (3.7.1)–(3.7.3) using 0 initial conditions and $R = 49.3$. Then we use the values of the state variables at the 2-year mark as initial conditions for a second phase with $R = 0$. The results in Fig. 3.7.2 were obtained with the built-in MATLAB function *ode45*. Other software implementations should yield similar results. Observe what happens when the lead dose changes at times 0 and 2. The lead amount in the blood changes very rapidly and approaches a constant value. The amount in the tissues behaves similarly, although the rate of change is not as rapid. The bones accumulate lead at a rate comparable to the tissues during the ingestion phase, but their capacity for storing lead is apparently much larger than that of the blood and tissues. When the lead ingestion ends, the bones yield their store of lead very slowly.[54] □

[53] See Sect. 3.3.

[54] Future President Andrew Jackson received a bullet in his chest in a duel in 1806. The bullet could not be removed. Many historians believe that Jackson's death in 1845 was due at least in part to the long-term effect of lead poisoning from the bullet.

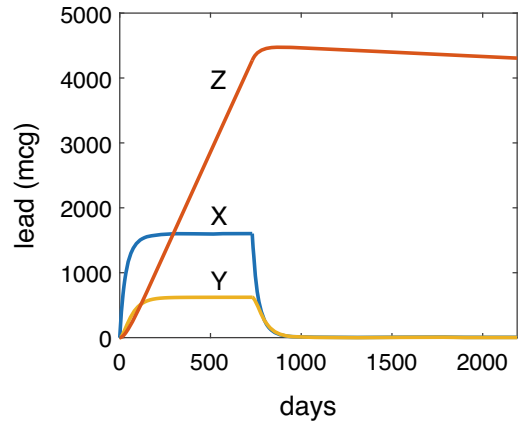


Fig. 3.7.2 The lead poisoning model, with steady infusion for 2 years, followed by 4 years with no further infusion

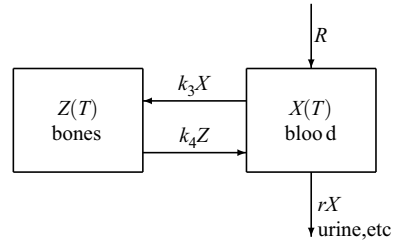


Fig. 3.7.3 A two-compartment model for lead transport in a vertebrate body

3.7.1 A Simplified Model

Sometimes the purpose of a model is to create simulations that are as accurate as possible. Other times, the goal is to better understand the scientific principles in the scenario being modeled. In this latter case, it can be beneficial to search for a simplified model that still captures the important features of the setting. The simulation result for the three-compartment lead poisoning model suggests a possible simplification. Note that the graphs of the lead amounts in the blood (X) and the tissues (Y) are very similar. Except for the first few days after a change in the external input rate R, the amount in the tissues seems to be merely a constant multiple of the amount in the blood. If we make the assumption $Y = cX$, where c is a positive parameter, then we can use this simple algebraic equation to replace the differential equation for Y. After algebraic simplification to eliminate Y from the original model, we would have a system that corresponds to the simplified compartment diagram of Fig. 3.7.3. The parameters R, k_3, and k_4 are unchanged, but the new parameter r must somehow incorporate both r_1 and the parameters previously associated with the tissue compartment (k_1, k_2, and r_2). The calculation of r is addressed in Problem 3.7.1.

Following the same procedure as for the full model, the compartment diagram of Fig. 3.7.3 corresponds to the model

$$\frac{dX}{dT} = R - (k_3 + r)X + k_4 Z\,, \tag{3.7.6}$$

$$\frac{dZ}{dT} = k_3 X - k_4 Z\,. \tag{3.7.7}$$

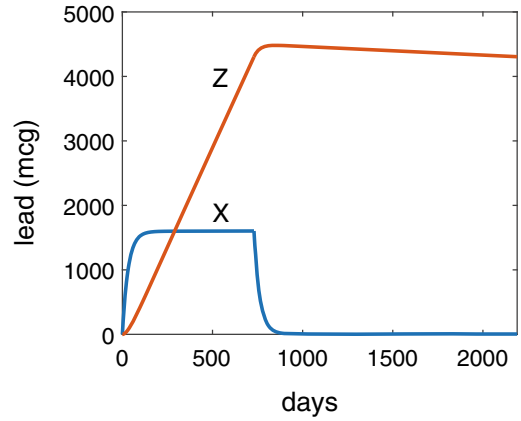


Fig. 3.7.4 The two-compartment lead poisoning model, with steady infusion for 2 years, followed by 4 years with no further infusion

Figure 3.7.4 displays the results of the same scenario as Fig. 3.7.2 using the parameter value $r = 0.0277$. The graphs for X and Z are indistinguishable from those of Fig. 3.7.2.

The comparison of Figs. 3.7.2 and 3.7.4 illustrates a key principle of mathematical modeling, which we first saw in Sect. 2.4:

> A simpler model can be preferable to a more complex model if the more complex model is only slightly more accurate.

The two-compartment model has three rate parameters (k_3, k_4, and r), compared to six rate parameters in the three-compartment model. If real data were available, we could use the Akaike information criterion[55] to determine which model has greater statistical support. Given uncertainty in parameters and the tiny difference between the results of the two models, the simpler model would almost certainly prove to be the better choice.

3.7.2 The Dimensionless Model

If we want to determine the general behavior of the model over a range of possible parameter values, we can obtain further simplification by scaling.[56]

Using $1/k_4$ as the reference time and R/r and $(k_3/k_4) * R/r$ as the reference lead masses for blood and bones,[57] we obtain the model

$$\epsilon \frac{dx}{dt} = iq - (1+q)x + z\,, \tag{3.7.8}$$

$$\frac{dz}{dt} = x - z\,. \tag{3.7.9}$$

[55] Section 2.4.

[56] Section 3.6.

[57] These choices are not obvious. The quantities R/r and $(k_3/k_4) * R/r$ are the amounts of lead that will be in the blood and bones after a long period of ingestion at the rate R. The time $1/k_4$ is the expected amount of time that a lead atom spends in the bones.

where

$$q = \frac{r}{k_3} \approx 7.1\,, \qquad \epsilon = \frac{k_4}{k_3} \approx 0.0090\,, \tag{3.7.10}$$

and i takes the value 1 during the accumulation phase and 0 during the clearance phase.[58] This model is analyzed in detail in Sect. 6.2.

Problems

Problems marked with "c" require some calculus beyond ordinary differentiation.

3.7.1* The combined parameter r in the two-component lead poisoning model can be calculated from the data used for the three-component model. The idea is that the approximation that Y reaches equilibrium quickly corresponds to replacing the Y equation in the original model with the algebraic equation

$$k_1 X - (k_2 + r_2)Y = 0\,.$$

Derive the two-component model by solving this equation for Y and substituting the result into the original X equation

$$\frac{dX}{dT} = R - (k_1 + k_3 + r_1)X + k_2 Y + k_4 Z$$

to obtain a new X equation. This new equation reduces to the form

$$\frac{dX}{dT} = R - (k_3 + r)X + k_4 Z\,,$$

where the quantity inside the parentheses appears as a sum of k_3 and several other terms. Redefine r to be the sum of these other terms and confirm that $r = 0.0277$ is consistent with the given data.

3.7.2 Obtain the dimensionless model (3.7.8)–(3.7.10) from the original model (3.7.6)–(3.7.7). What does the parameter q represent?

3.7.3 [c] [Drug absorption]
Figure 3.7.5 shows a very simple two-compartment model for a drug that is absorbed in the digestive tract and then circulated in the blood compartment (including the liver and kidneys).

(a) Write down the appropriate differential equations for the model. Note that the equations must be supplemented by initial conditions $W(0) = W_0$ and $X(0) = X_0$.
(b) Notice that the equations are decoupled, meaning that the W equation does not contain X. Solve the W equation by finding the correct anti-derivative from the table of derivatives in Appendix B. Note that the parameter A must be chosen so that $W(0) = W_0$.
(c) The X equation can be simplified with a clever trick. Let $Y(T) = e^{rT}X(T)$. Use the product rule to obtain Y' and then substitute from the differential equation for X to derive the resulting equation

$$Y' = ke^{rT}W(T)\,.$$

(d) Use the result for W and integration to obtain a one-parameter family of functions for Y.

[58] The Greek letter ϵ is commonly used in mathematical modeling to represent a dimensionless parameter whose value is typically much less than 1.

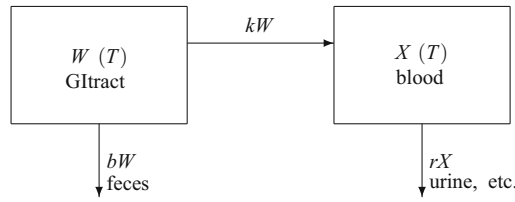


Fig. 3.7.5 A compartment diagram for drug absorption

(e) Use the definition of Y to obtain a family of functions for X. Use the initial condition $X(0) = X_0$ to complete the solution for X. In writing your solution, it is advantageous to replace $r - k - b$ by $-(k + b - r)$; under most circumstances, $k > r$.
(f) To study the effect of k, set $b = 0$, $X_0 = 0$, $r = 1$, and $W_0 = 1$ and obtain a simplified formula for X. Now plot $X(T)$ on a single graph using $k = 4$, $k = 3$, and $k = 2$. Discuss the effect of k on the functioning of the drug.
(g) Plot the function X from part (f) with $k = 0.5$. What happens when drug absorption is very slow?
(h) Suppose drug manufacturers can control the value of k without changing the efficacy of the drug. What would be the relative advantages of a larger k and a smaller k?

3.8 Case Study: Enzyme Kinetics

The Michaelis–Menten equation for the reaction rate of enzyme-catalyzed biochemical reactions is one of the best known models in biology. It was originally developed as an empirical model, but its derivation from a dynamical system model is very instructive, combining together the modeling technique of compartment analysis with an approximation technique from asymptotic analysis.

Michaelis–Menten reactions are biochemical reactions in which an enzyme decomposes a substrate into products. Typically the substrate is a complicated molecule, such as sugar, while the products generally include multiple chemical species, such as water and carbon dioxide. Of course, there is a great variety in the details, but all can be written in a generic form as

$$\mathrm{S} + \mathrm{E} \rightleftharpoons \mathrm{C} \rightarrow \mathrm{P} + \mathrm{E}\,. \tag{3.8.1}$$

Unlike in actual chemical formulas, where each symbol refers to a single chemical species, the symbol P in this generic form refers abstractly to the whole collection of products. This needs to be made explicit, else a person with experience in chemistry but not biochemistry would think the overall reaction S $\rightarrow$ P means that S and P are the same thing. The symbol C refers to an *activated complex*, which is an unstable chemical species formed from the union of S and E that can either revert to its original components or progress to the collection of products along with the restored enzyme. The critical feature of these reactions is that the enzyme is conserved in the sense that every molecule of enzyme is either present in its native form or as part of the complex.

As indicated by the three arrows in the generic reaction equation, the mechanism consists of three separate steps:

1. A forward reaction in which the substrate and enzyme produce the complex.
2. A backward reaction in which the complex decomposes into substrate and enzyme.
3. A completion reaction in which the complex decomposes into products and enzyme.

Mathematical models for chemical kinetics follow from simple mechanistic rules that determine the rate of change of each chemical species in terms of the concentrations of all the species. Let S be the concentration of substrate, Y the concentration of enzyme, Z the concentration of complex, and P the concentration of product. We need to determine the rates of the three reactions in terms of these variables.

The backward and completion steps occur spontaneously because of the instability of the complex; thus, it is reasonable to assume that the rates of these reactions are simply proportional to the amount of complex present. Specifically, we assume rates $k_2 Z$ for the backward reaction and $k_3 Z$ for the completion reaction. The forward reaction requires molecules of substrate and enzyme to encounter each other in the mixture. Assuming that the molecules move randomly, the rate is proportional to the concentrations of both[59]; thus, we write the rate of the forward reaction as $k_1 SY$.

Armed with expressions for the reaction rates, we can write down differential equations for the rates of change of each of the quantities in the reaction. Each unit of the forward reaction decreases the concentration of substrate by one unit, while each unit of the backward reaction increases the substrate concentration by one unit. Since the rate of the forward reaction is $k_1 SY$, the rate of increase of substrate via the forward reaction is also $k_1 SY$; similarly, the rate of decrease of substrate via the backward reaction is the same as the reaction rate $k_2 Z$. The overall rate of change of substrate is the sum of the changes caused by the two reactions; therefore, we have[60]

$$\frac{dS}{dT} = -k_1 SY + k_2 Z\,. \tag{3.8.2}$$

We can apply similar reasoning to obtain the differential equations for the enzyme, complex, and product:

$$\frac{dY}{dT} = -k_1 SY + k_2 Z + k_3 Z\,, \tag{3.8.3}$$

$$\frac{dZ}{dT} = k_1 SY - k_2 Z - k_3 Z\,, \tag{3.8.4}$$

$$\frac{dP}{dT} = k_3 Z\,. \tag{3.8.5}$$

Because the right-hand sides of the equations for Y and Z are identical except for the sign, we can add them together to obtain the simple equation

$$\frac{d(Y+Z)}{dT} = 0\,,$$

which means that the sum of enzyme and complex concentrations is invariant. At any given moment, each enzyme molecule is either free, in which case it counts toward Y, or bound, in which case it counts toward Z. Assuming that the initial amounts of these chemicals are Y_0 and 0, we have

$$Y + Z = Y_0\,.$$

[59] This is the *law of mass action*.

[60] We are using T for time to preserve the symbol t for dimensionless time.

We can discard (3.8.3) and substitute $Y = Y_0 - Z$ into (3.8.2) and (3.8.4), obtaining the system

$$\frac{dS}{dT} = -k_1 S(Y_0 - Z) + k_2 Z\,, \qquad S(0) = S_0\,, \tag{3.8.6}$$

$$\frac{dZ}{dT} = k_1 S(Y_0 - Z) - k_2 Z - k_3 Z\,, \qquad Z(0) = 0\,. \tag{3.8.7}$$

$$\frac{dP}{dT} = V = k_3 Z\,, \qquad P(0) = 0\,. \tag{3.8.8}$$

Technically, we don't need the extra equation for P either, because the quantity $S + Z + P$ is also invariant; however, we retain this equation because P is the quantity most commonly measured. The rate of change of P is generally called the *reaction velocity*, but it is really the rate of product formation. In keeping with standard notation, we label this quantity V.

3.8.1 Scaling

As we saw in Sect. 3.6, there are often advantages to scaling a model. Here, we choose reference quantities S_0 for S and P, Y_0 for Z, $k_1 S_0 Y_0$ for V, and $1/(k_1 Y_0)$ for T. Given that the original variables are all uppercase, we use the corresponding lowercase letters for the dimensionless variables. Thus, we apply the substitution formulas:

$$S = S_0 s\,, \quad Z = Y_0 z\,, \quad P = S_0 p\,, \quad V = k_1 S_0 Y_0 v\,, \quad \frac{d}{dT} = k_1 Y_0 \frac{d}{dt}\,. \tag{3.8.9}$$

The reference quantities $k_1 S_0 Y_0$ for reaction velocity and $1/k_1 Y_0$ for time are not obvious and should not greatly concern the reader.[61]

With the given substitutions, the system (3.8.6)–(3.8.8) becomes

$$\frac{ds}{dt} = -s(1 - z) + hz\,, \qquad s(0) = 1\,, \tag{3.8.10}$$

$$\epsilon \frac{dz}{dt} = s(1 - z) - hz - rz\,, \qquad z(0) = 0\,, \tag{3.8.11}$$

$$\frac{dp}{dt} = v = rz\,, \qquad p(0) = 0\,, \tag{3.8.12}$$

where

$$h = \frac{k_2}{k_1 S_0}\,, \qquad r = \frac{k_3}{k_1 S_0}\,, \qquad \epsilon = \frac{Y_0}{S_0} \ll 1\,. \tag{3.8.13}$$

[61] The choice for reaction velocity is based on the insight that the forward reaction is the one that drives the overall reaction, and its initial rate is $k_1 S_0 Y_0$. We can approximate the amount of time that corresponds to a unit change of substrate by substituting $\Delta S \approx -S_0$ in the equation

$$\frac{\Delta S}{\Delta T} \approx -k_1 S_0 Y_0$$

and solving for ΔT, with the result $\Delta T \approx 1/k_1 Y_0$. Note that we had other choices. Using the same procedure, we could have found the amounts of time corresponding to a unit change of complex from the forward reaction, a unit change of either substrate or complex from the backward reaction, or a unit change of complex from the completion reaction. Intuitively, the given choice is best, because the substrate is the principal reactant and the forward reaction is the one that uses up substrate.

The final model has dependent variables s, z, and v, which represent the concentration of substrate, concentration of complex, and rate of product formation, respectively. There are three dimensionless parameters: h, r, and ϵ. There was some flexibility in defining the dimensionless parameters, but there are some advantages to the form used here. The parameter h represents the strength of the backward reaction relative to the forward reaction. Larger values of h should make the overall reaction slower because molecules of the complex will tend to decompose more quickly than they are formed. Similarly, the parameter r represents the strength of the completion reaction relative to the forward reaction. Larger values of r relative to h mean that the complex is more likely to produce product than to decompose into reactants.

The parameter ϵ plays a special role in the analysis because it appears only as a factor on the left side of a differential equation. While h and r represent the relative strengths of the component reactions, ϵ represents the relative magnitude of the rates of change determined by the set of component reactions. The notation $\epsilon \ll 1$ means that we expect ϵ to be very small,[62] which follows from the assumption that the enzyme is usually in short supply relative to the substrate. Insights such as this can simplify the analysis of a model, as we will see later.

3.8.2 Simulation

Our principal question for this model is "How does the reaction velocity $v(t)$ depend on the parameters h, r, and ϵ?" This question requires analysis because it assumes that these parameters are allowed to vary. Nevertheless, we can obtain some insights by considering simulations with fixed parameter values.

There are no methods that can be used to determine the functions $s(t)$ and $z(t)$ from the differential equations (3.8.10) and (3.8.11). However, numerical methods can be used for satisfactory simulations. In our current problem, it is important to recognize that the system has a small parameter in front of one of the differential equations. This means that rates of change of that variable will sometimes be large. Having very different rates of change marks a problem as *stiff*, which means one must be careful using standard numerical methods. MATLAB has several differential equation solvers intended for stiff problems. In practice, the best results are often obtained by using the standard *ode45* solver, reducing the error tolerance if necessary. In our examples, we will choose more modest values of ϵ than normally occurs, both to decrease the stiffness and to make the behavior more apparent in graphs.

Figure 3.8.1 shows the results of a simulation using the system (3.8.10)–(3.8.12), with parameters $h = 1$, $r = 2$, and $\epsilon = 0.2$. The curves show a striking feature of Michaelis–Menten reactions: the concentration of the complex (and the reaction velocity) changes rapidly from an initial value of 0 to a maximum value, after which it gradually decreases. With a more realistic (smaller) choice of ϵ, we would not even notice the very short phase in which the reaction velocity rises to its peak. It is this peak that biochemists are particularly interested in, and which will be the focus of our analysis.

3.8.3 Asymptotic Approximation

So far, we haven't seen anything that looks like the Michaelis–Menten function. This will emerge from application of an asymptotic approximation to the dimensionless system. The key to developing this approximation is understanding that the behavior observed in the graph is caused by the small factor ϵ in (3.8.11). We can rewrite this equation as

$$\frac{dz}{dt} = \frac{s - (s + h + r)z}{\epsilon}.$$

[62] In oral communications, this symbol is generally read as "much less than."

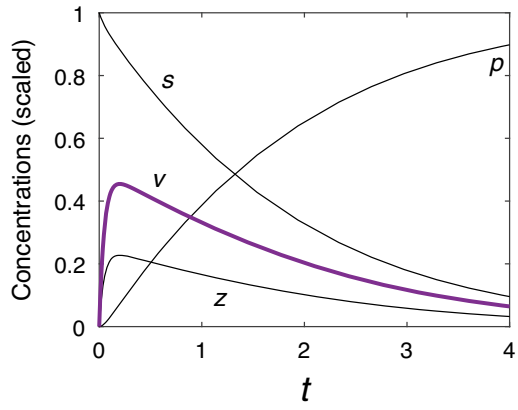


Fig. 3.8.1 The Michaelis–Menten system ((3.8.10)–(3.8.12)), with $h = 1$, $r = 2$, $\epsilon = 0.2$

At the beginning of the reaction, $s = 1$ and $z = 0$, so $dz/dt = 1/\epsilon \gg 1$. Thus, dz/dt is initially large because ϵ is small. The complex increases rapidly, but only for a short interval of time. There can be only one way for the graph of z to flatten out, and that is for the quantity $s - (s + h + r)z$ to be very small. This occurs at approximately $t = 0.1$ in this example, but will occur much sooner with a realistically smaller ϵ. From this time on, dz/dt is never large, so we must have $s \approx (s + h + r)z$. Solving for z, we have

$$z \approx \frac{s}{s + h + r} \,. \tag{3.8.14}$$

This is a short-time asymptotic approximation of the differential equation (3.8.11). We can ignore the initial transient in all but the first instant of the reaction, so we can use the quasi-steady equilibrium equation (3.8.14) in place of the differential equation (3.8.11). Substituting this approximation into (3.8.10) and (3.8.12) yields the simplified system[63]

$$\frac{ds}{dt} = -\frac{rs}{s + h + r} \,, \qquad v = \frac{rs}{s + h + r} \,. \tag{3.8.15}$$

We could at this point have a computer plot the solution of the differential equation for s and use it to obtain a graph of v versus t; this would look just like the graph of Fig. 3.8.1 except that it would have its peak at exactly time 0. Instead, the approximate system (3.8.15) is used primarily to estimate the maximum of the reaction velocity curve. This is determined by substituting the maximum value of s ($s = 1$) into the v equation to get

$$v_0 = \frac{r}{1 + h + r} \,,$$

where v_0 is the initial rate of product formation, with "initial" understood to mean "after the brief transient period during which the concentration of the complex reaches equilibrium with the substrate."

Replacing the dimensionless quantities v_0, r, and h in the maximum reaction rate equation with the corresponding dimensional quantities, we obtain[64]

$$V_0 = \frac{V_{max} S_0}{S_0 + K_m} \,, \tag{3.8.16}$$

[63] Problem 3.8.3.

[64] Problem 3.8.4.

where

$$V_{max} = k_3 Y_0 \,, \quad K_m = \frac{k_2 + k_3}{k_1} \,. \tag{3.8.17}$$

The parameters V_{max} and K_m are usually fit from data for V_0 versus S_0, without attempting to evaluate the k_i or Y_0 independently.[65] However, note that only the parameter K_m is a fundamental property of the biochemical system. The parameter V_{max} is a function of the enzyme concentration as well as the biochemical system, so it could be different in different experiments for a particular system.

Problems

Problems marked with "p" require some programming beyond data entry.

3.8.1 Obtain the dimensionless Michaelis–Menten model in the text using the given definitions of s, z, p, and v.

3.8.2 Nondimensionalize the Michaelis–Menten model using the same definitions of s, z, p, and v as in the text, but using $t = k_1 S_0 T$ for the time. Use the same parameters as in the text. What is the difference in the resulting model?

3.8.3 Derive (3.8.15).

3.8.4 Derive (3.8.16).

3.8.5 p Use a computer simulation to study the effect of the backward rate constant h on the Michaelis–Menten model.

(a) Modify the program *ODEsim.m* to run simulations for (3.8.10)–(3.8.12). Check your program by running it for the scenario in Fig. 3.8.1.
(b) Hold $r = 2$ and $\epsilon = 0.2$, and plot the simulation results for two values of h larger than 1 and two values of h smaller than 1. Choose values that adequately illustrate the effect of h.
(c) Explain the effect of h in words by reference to the Michaelis–Menten reaction: (1) What does h quantify in the reaction? (2) Given the meaning of h, how should the reaction progress change as h increases? Keep in mind that the graph of z, while interesting, is not important to biochemists. The purpose of the reaction is either to produce product or to break down the substrate.

3.8.6 Use a computer simulation to study the effect of ϵ on the Michaelis–Menten model. Hold $r = 2$ and $h = 1$ and try $\epsilon = 0.1, 0.2, 0.4, 0.8$. Plot the simulation results and explain the effect of ϵ. (Hint: You may need to try different time horizons.)

3.8.7* [Malaria] (Continued from Problem 3.4.17.)
Normally, a model for a vector-borne disease, such as malaria, needs to consider both the human population and the vector population, in this case the mosquitoes. Occasionally, when the pathogen life cycle involves longer periods of time in the human host than in the vector, it is possible to use an approximate model that looks like an infectious disease model with an unusual transmission term.[66] The method is the same asymptotic argument we used to derive the Michaelis–Menten equation. Here, we try this plan with the Ross malaria model, given in dimensionless form as

[65] See Sects. 2.3 and 2.5.
[66] See also Sect. 6.4.

$$\frac{dx}{dt} = \alpha(1-x)y - x\,,$$

$$\frac{dy}{dt} = m[\beta x(1-y) - y]\,.$$

Given a typical value of $m = 10$, there is some hope that the approximation of treating m as a large parameter will be reasonably accurate.

(a) Derive a single-component model for the human infectious fraction x by assuming that m is large and y' is not large.
(b) Run a simulation with $\alpha = 10$, $\beta = 0.3$, and $x(0) = 0.001$.
(c) Compare the plots with that of Problem 3.4.17.
(d) Repeat (b) and (c) with $m = 100$.
(e) Discuss the results. Does the asymptotic simplification work as well here as it did with the Michaelis–Menten reaction?

(This problem is continued in Problem 4.5.7.)

3.9 Case Study: Adding Demographics to Make an Endemic Disease Model

Sections 3.3 and 3.4 presented a derivation and analysis for an epidemic disease model. Epidemiology models can be roughly categorized as epidemic or endemic: epidemic if there is no replenishment of the susceptible class and endemic if there is. One way to build replenishment of susceptibles into a model is to assume that recovered individuals are not immune. In its simplest form, this is the SIS (susceptible-infectious-susceptible) model. We defer consideration of loss of immunity for the exercises. Here we consider models obtained by adding demographic features, such as birth, death, and immigration. For simplicity, we use the SIR model, rather than the SEIR model, as the base for these additional features.

Different authors use different assumptions to add demographics to an SIR model. These have varying degrees of realism, and there is not always a clear narrative description of the assumptions. In these cases, we can construct a narrative by reverse engineering the differential equations. Regardless of whether a set of assumptions was clearly stated or only implied in the equations, it is important to think about how realistic the assumptions are.

3.9.1 A Generic SIR Model with Demographics

The different versions of SIR endemic models are easier to compare if we embed them in a common framework by defining some generic functions, as shown in Fig. 3.9.1. In any particular model, some of these will reduce to the simple version that appears in the SIR epidemic model.

1. $B(N)$ is the birth rate, which could depend on the total population $N(t)$. By making the birth rate a function of the total population only, we are assuming that epidemiological status does not affect one's ability to have children. This is a reasonable assumption, as we will see that the infectious population is almost always small enough to have little effect on the birth rate.
2. $M(X, N)$ is the natural death rate for any population class X. Usually we will assume the death rate coefficient to be independent of population density, so we'll have μS as the death rate for susceptibles, μI as the death rate for infectives, and so on. If the death rate does depend on population density, then N will appear in the formula as well as X.

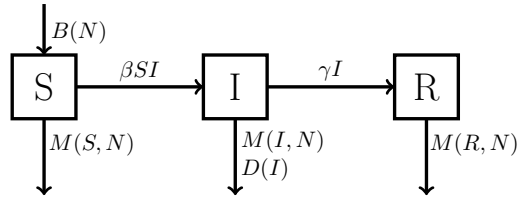


Fig. 3.9.1 The generic SIR model with population demographics

3. $D(I)$ is the disease-related death rate for class I. Where present, this will almost always be taken as a spontaneous transition process; that is, $D(I) = \alpha I$.

Our generic model assumes mass action incidence for the transmission process. One could instead assume standard incidence.[67] This may be appropriate for some disease scenarios, but we omit this option here because it has no bearing on the general question of how to add demographics to an epidemic model.

From the compartment diagram, the generic model is

$$S' = B(N) - \beta SI - M(S, N)\,, \tag{3.9.1}$$

$$I' = \beta SI - \gamma I - D(I) - M(I, N)\,, \tag{3.9.2}$$

$$R' = \gamma I - M(R, N)\,. \tag{3.9.3}$$

Adding these yields a total population equation, which will be used in some versions in place of one of the state equations:

$$N' = B(N) - M(N, N) - D(I)\,. \tag{3.9.4}$$

Constant Population

The simplest way to add demographics to the SIR epidemic model is to assume a natural death process without density dependence, $M(X, N) = \mu X$, no disease-induced deaths, and a birth rate process that exactly balances the death rate, $B(N) = \mu N$. The population is then constant, as seen in (3.9.4). The equation for R is not needed, so we have a 2-component model:

$$S' = \mu N - \beta SI - \mu S\,, \tag{3.9.5}$$

$$I' = \beta SI - \gamma I - \mu I\,. \tag{3.9.6}$$

3.9.2 Several Approaches to a Variable Population Version

If we want to include disease-related deaths in an endemic disease model, we need to have a mechanism that allows either a higher birth rate or a lower natural death rate to compensate for the extra deaths, or else we need to have a net positive immigration rate. Since the rate of additional deaths depends on I rather than N, we can expect the population to be variable rather than fixed. This usually means that a 3-component model is required. While such a model could be written using R in addition to the usual S and I, it is often more convenient to use the total population N as the third component.

[67] See Sect. 3.2.

Fixed Birth Rate

The simplest way to incorporate disease-related death is to use a fixed birth rate Λ in place of a birth rate that balances the natural death rate, resulting in the model

$$N' = \Lambda - \mu N - \alpha I \,. \tag{3.9.7}$$

$$S' = \Lambda - \beta SI - \mu S \,, \tag{3.9.8}$$

$$I' = \beta SI - \gamma I - \alpha I - \mu I \,, \tag{3.9.9}$$

A population greatly decreased by disease-induced death is going to have a lower birth rate than an unaffected population, with a risk of extinction if the death rate is too high. The fixed birth rate model is unrealistic in such a scenario. The death rate μN declines with population but the birth rate Λ does not; hence, the population cannot go extinct. This limits the use of the model to scenarios where disease-induced deaths are not a threat to overall population survival.[68]

This model decouples into a two-dimensional SI system along with an additional equation for N, allowing for two-dimensional methods of analysis. We'll see in Chap. 6 that the difficulty of model analysis increases significantly as the number of components in the system increases. It would therefore be mathematically beneficial to use the relatively simple but unrealistic fixed birth rate model whenever its results are comparable to those of models that are more realistic but less simple. We can do this for diseases with low mortality.

Compensating Disease-Induced Death with Density-Dependent Birth

Rather than compensating for disease-induced deaths by assuming a larger fixed birth rate, we could assume that the natural birth rate increases when the population is low. This is the most realistic assumption for most scenarios, but it requires some care to derive. Rather than making an assumption about the birth rate directly, it is better to make an assumption about the population dynamics in the absence of disease and use that to get the birth rate.

We start by assuming that the population grows logistically in the absence of disease:

$$N' = rN\left(1 - \frac{N}{K}\right) \,.$$

This form does not directly carry over to a disease model because it doesn't track births and deaths separately, but we can combine it with a formula for either the birth or death rate to derive the other. If we assume that the density dependence is in the birth rate only, then we can take the death rate to be

$$M(X, N) = \mu X \,. \tag{3.9.10}$$

Since logistic growth implicitly incorporates both birth and death, we can define the birth rate as the growth rate plus the death rate:

$$B(N) = rN\left(1 - \frac{N}{K}\right) + \mu N \,. \tag{3.9.11}$$

[68] This is a good time to remind the reader of the importance of matching models to scenarios rather than automatically using a model created for a different scenario.

We therefore get the equations

$$N' = rN\left(1 - \frac{N}{K}\right) - \alpha I\,. \tag{3.9.12}$$

$$S' = rN\left(1 - \frac{N}{K}\right) + \mu N - \beta SI - \mu S\,, \tag{3.9.13}$$

$$I' = \beta SI - (\gamma + \alpha + \mu)I\,. \tag{3.9.14}$$

Alternatively, one could make the death rate density-dependent, rather than the birth rate. However, this makes the model noticeably more complicated without appreciably changing the results. Given that there are other assumptions that are more problematic, there is no value to adding this extra complexity.

3.9.3 Scaling

If we want to simulate a specific disease scenario, we can use the dimensional versions of the models presented here. However, if we want to do analysis, it is much better to scale the models, partly to cut down the number of parameters and partly to facilitate the use of asymptotic methods to simplify the analysis. Here we focus on the model with fixed birth rate: (3.9.7)–(3.9.9). The issues involved in scaling any of the other models are similar.[69]

The first step in scaling an epidemiological model is to choose a reference population and a reference time. This is a difficult task to manage without significant experience. The interested reader can find details in Appendix E, where we explain the reasons for choosing population and time scales

$$K = \frac{\Lambda}{\mu}\,, \qquad T_i = \frac{1}{\gamma + \alpha + \mu}\,, \tag{3.9.15}$$

and dimensionless parameters

$$\mathcal{R}_0 = \frac{\beta K}{\gamma + \alpha + \mu}\,, \qquad \epsilon = \frac{\mu}{\gamma + \alpha + \mu} \ll 1\,, \qquad d = \frac{\alpha}{\gamma + \alpha + \mu}\,, \tag{3.9.16}$$

which represent the basic reproductive number, the ratio of the natural death rate to the overall rate of removal from class I, and the fraction of infectious individuals who die from the disease, respectively.

The notation $\ll$ is used in asymptotic analysis to indicate a parameter that is assumed to be arbitrarily small at such time as this information is useful. This assumption is well justified for the parameter ϵ. Given a life expectancy of about 70 years, a disease duration of 3.5 weeks would yield a value $\epsilon = 0.001$. Given that many diseases have shorter durations, ϵ will often be even smaller than this. In many cases, the presence of a small parameter helps simplify the computations for linearized stability analysis.[70]

The substitutions

$$X = Kx\,, \qquad \frac{d}{dT} = (\gamma + \alpha + \mu)\frac{d}{dt}\,,$$

where T is being used for dimensional time to reserve t for dimensionless time, yield the dimensionless model

[69] See [12] for a detailed presentation of scaling for models in epidemiology.

[70] Section 6.3.

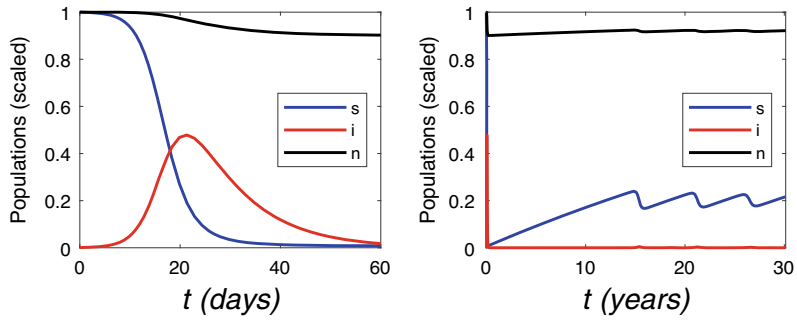


Fig. 3.9.2 The development of a classic childhood disease starting with the initial epidemic, from Example 3.9.1

$$n' = \epsilon(1 - n) - di. \tag{3.9.17}$$

$$s' = \epsilon(1 - s) - \mathcal{R}_0 si, \tag{3.9.18}$$

$$i' = \mathcal{R}_0 si - i. \tag{3.9.19}$$

3.9.4 Simulations

Simulations with disease models can be tricky because the very small value of ϵ means that any system with disease processes and demographics is stiff.[71] When setting up a simulation, one can either use the original dimensional model or a dimensionless version such as (3.9.17)–(3.9.19). The dimensional version will give graphs with easily interpreted time coordinates, but will also require more parameters to be specified.

For convenience in interpreting dimensionless time, we can define several fixed times and time coordinates. The critical fixed times are

$$T_i = \frac{1}{\gamma + \alpha + \mu}, \qquad T_\mu = \frac{1}{\mu}, \tag{3.9.20}$$

which are the expected duration of infectiousness and the expected lifespan in the absence of disease. These are the two parameters that are easiest to estimate for most disease scenarios.[72] For convenience, we assume that these are given in days and years, respectively. Once we have completed a simulation, we will want the graphs to show dimensional time in either days (t_d) or years (t_y). We need formulas to calculate ϵ in terms of T_i and T_μ and both dimensional times in terms of the dimensionless time t. These are

$$\epsilon = \frac{T_i}{365 T_\mu}, \qquad t_d = T_i t, \qquad t_y = \frac{t_d}{365}. \tag{3.9.21}$$

▶ **Example 3.9.1** Figure 3.9.2 shows the early and long-term course of an endemic disease with a disease duration of $T_i = 10$ days, a mortality probability of 10%, a basic reproduction number of 5,

[71] See the discussion in Sect. 3.8.

[72] It is much easier to estimate T_i from data than to measure its component rates γ and α.

a time scale ratio $\epsilon = 0.0005$, corresponding to a life expectancy of roughly 55 years, and an initial state with 0.1% infectious. The plots show the combination of fast and slow processes. There is a massive first wave in which roughly half of the population is simultaneously infectious within a few weeks and by two months the epidemic has run its course, with virtually nobody remaining in the susceptible class. Slow demographic processes allow the susceptible class to recover gradually while the disease continues to impact a tiny fraction of the population. After about 14 years, the susceptible population reaches roughly $s = 1/\mathcal{R}_0$, which is the level needed for herd immunity. Further births of susceptibles ignites a second epidemic. With such a small fraction of the population in the susceptible class, this second epidemic is small, with barely a blip in the graph of class i. Thereafter, new epidemics occur roughly every 7 years, each with slightly less intensity than the previous one. Over a period of several centuries, the system will approach a steady state at a population that is roughly $1 - d$ times the 'normal' population and with a susceptible fraction of roughly $1/\mathcal{R}_0$, all children. This scenario roughly describes the process by which measles and mumps became childhood diseases.[73] □

3.9.5 Rescaling

The system (3.9.17)–(3.9.19) is conveniently scaled for simulations, but a rescaling makes the problem better suited for analysis. The issue is that the infectious populations tend toward levels on the order of ϵ over time. Thus, long-time analysis occurs in the regime where i is small. We can make that smallness explicit by defining a new infectious variable y that will be $O(1)$ when $i = O(\epsilon)$. With the substitution

$$i = \epsilon y\,, \tag{3.9.22}$$

the system becomes

$$n' = \epsilon(1 - n - dy). \tag{3.9.23}$$

$$s' = \epsilon(1 - s - \mathcal{R}_0 sy), \tag{3.9.24}$$

$$y' = \mathcal{R}_0 sy - y, \tag{3.9.25}$$

See Appendix E for details.

Problems

3.9.1* [SIR disease with fixed population]
Scale the base SIR model with fixed population (3.9.5)–(3.9.6). Compare the result with the fixed birth rate version in the text. What results will be the same for both models, and what results will be different? In particular, address the issue of how the disease impacts total population.

(This problem is continued in Problem 6.1.7.)

[73] It would have been more dramatic for these diseases than for the example, as $\mathcal{R}_0 \approx 10$ for mumps and is higher for measles.

3.9.2 [SIR disease with logistic growth]
Scale the SIR model with logistic growth (3.9.12)–(3.9.14). Use the same scales and parameters as in the fixed birth rate version, but you will need an additional parameter

$$\rho = \frac{r}{\mu}.$$

Explain what this parameter represents.

(This problem is analyzed in Project 6A.)

3.9.3 [SIR disease with logistic growth]
Modify the program *ODEsim.m* for the SIR model with logistic growth from Problem 3.9.2. Run two simulations, using the same parameters as in Example 3.9.1, with $\rho = 0.25$ for one simulation and $\rho = 4$ for the other. Compare the results with each other and with Fig. 3.9.2.

3.9.4* [SIS disease with fixed birth rate]

(a) Modify the model (3.9.7)–(3.9.9) to change from an SIR disease, where recovery confers immunity, to an SIS disease, where recovered patients are once again susceptible.
(b) Scale the model, as in the text. Explain why you do not need the s equation.
(c) Modify *ODEsim.m* to run a simulation, using the same parameters as in Example 3.9.1.
(d) Compare the results with those of Fig. 3.9.2.

(This problem is continued in Problems 3.9.5 and 6.1.6.)

3.9.5* [SIS disease with fixed birth rate] (Continued from Problem 3.9.4.)
Consider the scaled SIS model with fixed birth rate, from Problem 3.9.4(b).

(a) Replace d by a parameter $\delta = \mu/\alpha = \epsilon/d$. Then set both derivatives equal to 0 to get algebraic equations for n and i. When $\mathcal{R}_0 > 1$, these will be the long-term stable (equilibrium) values of n and i for the population.[74] The change of parameters means that these algebraic equations will depend only on two parameters instead of three.
(b) Solve the algebraic equations to determine the equilibrium values for n and i as functions of $\mathcal{R}_0$ and δ. We anticipate δ to be small, so write your answers using δ rather than δ^{-1}.
(c) Compare your results with the graphs from Problem 3.9.4. They should be consistent.
(d) Define x to be the ratio of i at equilibrium to n at equilibrium and determine it as a function of $\mathcal{R}_0$ and δ. Explain why this result has more biological significance than i separately.
(e) Explain the impact of $\mathcal{R}_0$ and δ on n and x and the biological meanings of the results. (It will help to get a simple approximation by keeping in mind that $\delta = \mu/\alpha$ is likely to be small.) In particular, address the question of whether a disease with only modest mortality can severely decrease a population. Think particularly in terms of animal populations in nature.

3.9.6 [SIR disease with fixed birth rate and loss of immunity]

(a) Modify the model (3.9.7)–(3.9.9) by adding a loss of immunity term σR to the S equation.
(b) Scale the model, as in the text, with $\phi = \sigma/\mu$ as an additional parameter.
(c) Modify *ODEsim.m* to run two simulations, using the same parameters as in Example 3.9.1, along with $\phi = 0.5$ and $\phi = 5$.

[74] See Chap. 6.

(d) Compare the results with those of Fig. 3.9.2.

(This problem is continued in Problems 3.9.7 and 6.2.13.)

3.9.7 [SIR disease with fixed birth rate and loss of immunity] (Continued from Problem 3.9.6.) Consider the scaled SIS model with fixed birth rate, from Problem 3.9.6(b).

(a) Rescale the problem using $i = \epsilon y$.
(b) Obtain two equations relating the equilibrium values of n and y by setting the derivatives to 0. When $\mathcal{R}_0 > 1$, these will be the long-term stable values of n and y for the population.[75]
(c) Solve the equilibrium equations to determine the equilibrium values for n and y as a function of $\mathcal{R}_0$, d, and ϕ.
(d) Define x to be the ratio of y at equilibrium to n at equilibrium and determine it as a function of $\mathcal{R}_0$, d, and ϕ. Explain why this result has more biological significance than y separately.
(e) Fix $\mathcal{R}_0 = 4$ and plot n versus ϕ for $d = 0$. Use $0 \le \phi \le 5$. Add similar curves for $d = 0.1, d = 0.2$ to the same plot.
(f) Repeat (e) but with x instead of n.
(g) Discuss what we learn from the plots of (e) and (f). (Keep in mind that $x = y/n$, not $x = i/n$.)

3.9.8 [SIS disease with logistic growth]

(a) Modify the model (3.9.12)–(3.9.14) to change from an SIR disease, where recovery confers immunity, to an SIS disease, where recovered patients are once again susceptible.
(b) Scale the model using the same scales and parameters as in the text example. You will need an additional parameter, $\rho = r/\mu$. You will not need the S equation; however, you will need to replace S in the I equation prior to doing the scaling.
(c) Modify *ODEsim.m* to run a simulation, using the same parameters as in Example 3.9.1 and $\rho = 1$
(d) Compare the results with those of Fig. 3.9.2 and Problem 3.9.1.

(This problem is continued in Problem 6.1.12.)

3.9.9 [SIS disease with logistic growth and standard incidence]
Consider an SIS model in which population demographics is determined by a natural carrying capacity along with disease-related mortality. With the additional assumption that the contact rate is independent of the population size (appropriate for a population that expands or contracts its range to maintain constant density[76]), we have the model

$$\frac{dN}{dT} = rN\left(1 - \frac{N}{K}\right) - \alpha I\,,$$
$$\frac{dI}{dT} = B\frac{I}{N}S - (\gamma + \alpha + \mu)I\,,$$

where N is the total population, I is the infectious population, $S = N - I$ is the susceptible population, γ is the mean recovery time, μ is the life expectancy in the absence of the disease, α is an additional rate constant for disease-induced death, B is the transmission coefficient, K is the natural carrying capacity in the absence of the disease, and r is the relative growth rate of small populations.

[75] See Chap. 6.

[76] See the discussion of standard incidence in Sect. 3.2.

(a) Using K as the population scale and $1/(\gamma+\alpha+\mu)$ as the time scale, derive the scaled model

$$n' = \delta[n(1-n) - wi],$$
$$i' = \mathcal{R}_0 i\left[1 - \mathcal{R}_0^{-1} - \frac{i}{n}\right].$$

You will need to identify the correct combinations of parameters for δ and w.
(b) Explain why we can expect δ to be very small.
(c) Explain the biological significance of the parameter w.

(This problem is continued in Problems 3.9.10 and 6.1.13.)

3.9.10 [SIS disease with logistic growth and standard incidence]
The model of Problem 3.9.9 is difficult to analyze because of the term i/n in the i equation. We can recast the problem to make it much easier by replacing i with a new variable $x = i/n$.

(a) Explain the biological meaning of the variable x and give a reason why we might be more interested in x than i.
(b) Use the product rule to differentiate the equation $i(t) = n(t)x(t)$. Solve the resulting equation for nx' and substitute in the formulas for n' and i'. Eliminate i from the resulting equation to get the differential equation for x.
(c) Some of the terms in the x' equation contain a factor of δ, while others do not. Explain why it is a good modeling decision to omit the δ terms. This requires answers for two questions: "Why can we be confident that the results will not change noticeably?" and "What benefit do we gain from the change in the equation?"
(d) Compare the nx and ni systems. Identify features that make the former likely to be easier to analyze than the latter. [One particular feature is especially important.]

(This problem is continued in Problem 6.1.14.)

3.10 Projects

Projects 3A–3D involve modifications of the SEIR model. Project 3E investigates the impact of using a multi-phase transition rather than the standard spontaneous transition.

Project 3A: Herd Immunity

At a U.S. Senate hearing on September 22, 2020, one senator claimed that current COVID-19 case loads were low in New York (less than 50 per 100K) because the state had earlier suffered a sufficient number of cases to achieve herd immunity. In reply, Dr. Anthony Fauci cited a current estimate that about 22% of New Yorkers were immune because of past infection and that no infectious disease expert or epidemiologist considered 22% to be even close to the level required for herd immunity. In this project, we investigate the specific claim that 22% immunity is sufficient for herd immunity and the more general claim that any high level of immunity is by itself sufficient.

(a) Run *SEAIHRD_comparison.m* to compare the results using initial immunity levels of 22%, 50%, 60%, and 70%, paying particular attention to the hospitalization count, which is the simplest measure of the seriousness of a COVID-19 outbreak. The senator's claim was that behavioral measures made little difference, so set $\delta = 0.8$, indicating minimal distancing and mask use, and choose the optimistic (at that time) confirmation fraction $c = 0.5$ to bias the results toward lower infection levels. Under these circumstances, how much initial immunity would be needed to produce the low hospitalization numbers experienced in New York in late September? Do our model results support the senator, Dr. Fauci, or neither?

(b) Herd immunity is a very subtle concept that relatively few people understand completely. To develop this understanding, run *SEAIHRD_onesim.m* with $\delta = 1$, $c = 0.7$, and 80% initial immunity, and then repeat with 0% initial immunity.

   a. What is the final percent susceptible when initial immunity is 80%? Did herd immunity offer significant protection in this case?
   b. When initial immunity is 0%, we eventually reach a point where the total removed population is 80%, and yet the final susceptible percentage is far lower than when we assumed initial immunity of 80%. According to the naive view, herd immunity should have worked as soon as we reached one of these thresholds. Why didn't it?

(c) The claim that herd immunity was already having an influence rests on a more fundamental claim that herd immunity occurs as soon as an adequate level of population immunity is achieved. How should Dr. Fauci have responded to that claim?

(d) Write a short paper that explains herd immunity. Use what you learned from your experiments, but write the paper for a nontechnical audience. In other words, base your arguments on a verbal description of the herd immunity phenomenon, but do not present any mathematical work or simulation results. You may want to create a graphic to assist your explanation; the graphic should be factually correct, but non-technical, as you might expect to present at a Senate hearing.

Project 3B: Isolation

One way to incorporate isolation of symptomatic infectious patients into a disease model is to divide the infectious class into two classes, a (P)resymptomatic class and an (I)nfectious class with symptoms, leading to an SEPIR model.

1. Building the Model

   a. Prepare a compartment diagram and differential equations for the SEPIR model by incorporating the following assumptions into the SEIR model of Sect. 3.3:

      i. Classes P and I are equally effective in transmitting the disease.
      ii. Presymptomatic individuals develop symptoms at rate ϕP, while symptomatic individuals are removed at rate σI.
      iii. A fraction q of presymptomatic individuals are isolated when they develop symptoms, which means they move directly to class R, while the remaining fraction $p = 1 - q$ enter class I.

   b. Suppose the mean total duration of infectiousness is T_I, with a fraction ρ of that time spent as presymptomatic. Determine ϕ and σ in terms of these parameters.

2. Analysis

   a. Let $\mathcal{R}_0$ be the basic reproduction number for the case where there is no isolation of the symptomatic and let $\mathcal{R}_q$ be the basic reproduction number for the more general case. Find a formula that determines $\mathcal{R}_q$ in terms of $\mathcal{R}_0$, q, and ρ. (Hint: There are two different kinds of patients who create secondary infections, those who eventually isolate and those who don't. For each group, you know the expected transmission rate, the time available for transmission, and the fraction of individuals who are in the group.)
   b. Determine the isolation fraction q necessary to prevent the epidemic in terms of $\mathcal{R}_0$ and ρ.
   c. Discuss the results of (c), using one or more graphs to aid in the explanation.

3. Simulation
   Run two simulations to illustrate the effect of isolation on the model, using $\beta = 0.4$, $T_I = 10$, $T_E = 5$ (mean time in class E), and $T_P = 2$, with values of 0.4 and 0.8 for q. Match the initial conditions to Fig. 3.3.3, with $P(0) = 0$. Compare the results with that figure.

Project 3C: Masks and Distancing

Build and study a model that incorporates social distancing and/or mask usage to decrease transmission. Start with the SEIR model, but make it more realistic by dividing the population into a compliant subgroup and a noncompliant subgroup. Keep in mind that individuals in the two groups still interact with each other. See Sect. 3.5 for information about how to model settings where different groups have different levels of infectivity.

Project 3D: Vaccine Impact

Modify the SEAIHRD program suite so that it fits the January 2021 COVID-19 scenario of Sect. 3.5. Assume a basic reproduction number of 8 for the delta strain. Assume that 20% of people are at high risk, that the vaccine refusal fractions are 36% for all people and 12% for high-risk people, that 20% of vaccinated high-risk people become low-risk susceptibles, that 2.5% of infected high-risk people are pre-hospitalized, that 50% of infected low-risk people are asymptomatic, and that 50% of asymptomatic and 90% of symptomatic patients are tested. For other disease parameters, use the same values as for the March 2020 scenario. For the vaccination parameters, use the best-fit values from Sect. 3.1.4 except when experimenting with shorter values of τ.

The principle questions to be addressed are how the progress of the pandemic in this scenario is impacted by the contact factor δ and by the time τ required for vaccine manufacture to reach its maximum, with a goal of understanding the relative importance of maintaining mitigation strategies during vaccine rollout.

Project 3E: Multi-Phase versus Spontaneous Transitions

Modify the standard SEIR model of Sects. 3.3 and 3.4 by replacing the single-phase incubation and removal transitions with 2-phase transitions having the same mean times, as in Sect. 3.1.5.

Run simulations using an appropriately modified version of the *SEIR* program suite. Compare the maximum of class I and the final susceptible fraction obtained from this new SEEIIR model with those for the original SEIR model, using $\mathcal{R}_0 = 5$, $t_L = 5$, $t_I = 10$. Also compare the final susceptible fraction obtained analytically (Sect. 3.4.4). Then modify the program *R0fit.m* to find the best-fit value

of $\mathcal{R}_0$ from data produced by *seirabm.m* (which you do not need to change) and plot the corresponding model results with the data. Use your findings to discuss the issue of whether there is a significant error caused by assuming spontaneous transitions.

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Part II Dynamical Systems

Dynamical systems analysis focuses on determining the long-term behavior of autonomous systems—that is, systems in which the changes or rates of change of the state variables depend only on the state of the system. In principle, this can be accomplished through solution formulas; in practice, we can seldom solve the dynamical system equations analytically. Instead, we have a variety of graphical, analytical, and numerical methods at our disposal. Graphical methods include cobweb analysis for one-component discrete-time systems, phase line analysis for one-component continuous-time systems, and nullcline analysis for two-component continuous-time systems. The analytical method in all cases is linearized stability analysis, which for systems of more than one component involves either the determination of eigenvalues or the application of inequality criteria based directly on the entries in the matrix that represents the system. Numerical methods include numerical computation of eigenvalues and simulation. Analytical methods can be difficult to employ, but they can often produce results that indicate how outcomes depend on parameters, whereas numerical methods require full specification of parameters.

The importance of proper scaling of models is a recurring theme in the examples, case studies, and problems. At minimum, nondimensionalization reduces the number of parameters requiring estimated values for simulation or study in analysis. Beyond that, it can sometimes be used to reduce the number of essential components in a model. As will be seen in Chap. 6 in particular, analysis of models becomes more difficult as the number of components increases, and graphical methods are generally limited to one-component discrete models and two-component continuous models. The case study of onchocerciasis in Sect. 6.4 illustrates how asymptotic approximation can simplify a problem by making use of small dimensionless parameters that typically arise in biological models.

Other themes of our treatment of dynamical systems are the casting of problems into structures that simplify analysis, as exemplified by Sect. 4.5, and careful use of algebra to simplify stability calculations, as seen in Sect. 6.3.

Our treatment of dynamical systems begins with one-variable discrete and continuous equations in Chap. 4, progresses to discrete linear systems in Chap. 5, and concludes with nonlinear systems in Chap. 6. It is easy to see how to interpret discrete systems and use them for simulations. Their advantages end there; the remaining advantages lie with continuous systems, which have superior graphical methods and simpler mathematical properties. These advantages, which will become apparent in Chap. 4, more than offset the initial advantages of discrete models. In general, one should only use discrete models when the synchronicity of events dictates discrete time.



Dynamics of Single Populations

4

In this chapter, we use ecological scenarios as settings in which to develop and study models for the change of populations over time. We restrict ourselves for now to models that require careful monitoring of only one population.[1] There are two main categories of dynamic models, discrete and continuous, differing in the assumption made about how to mark time. *Discrete dynamic models* assume that time can be broken up into distinct uniform intervals. The length of the interval depends on the life history of the organism being modeled. Salmon have yearly spawning periods, so a time interval of 1 year is chosen for a discrete salmon model. *Continuous dynamic models* assume that time flows continuously from one moment to the next. The assumption of continuity in time is relative to the overall duration of the population. For example, it is common to ignore the diurnal variation of temperature and sunlight in a model that tracks a population of plants over a complete growing season.

The primary goal of this chapter is to empower the reader with tools for the analysis of both discrete and continuous models of a single dynamic variable. A secondary goal is to help the reader understand how to interpret discrete and continuous models and how to judge which kind of model is more suitable for a given biological setting.

Section 4.1 introduces discrete models and is followed by a section that presents the graphical method of analysis for such models. Section 4.3 introduces continuous models and the corresponding graphical analysis. Section 4.4 presents linearized stability analysis, a particularly useful application of the derivative that works for both discrete and continuous models, although in different ways. The chapter concludes with a case study of a model that may help explain why some renewable resources, such as whales, suffered significant population declines in a very short period of time and have been difficult to restore. While the material here is presented in the context of a case study, the section should also be considered as an important source of valuable modeling skills; in particular, the phase line analysis tool is greatly strengthened by the technique introduced in this section.

The chapter concludes with four projects. Project 4A considers a discrete-time model of insect pest control. Project 4B examines the effect of a limited treatment protocol on a simple disease model. Project 4C examines a continuous-time model of lake eutrophication. Project 4D uncovers some of the detailed behavior of the discrete logistic model.

4.1 Discrete Population Models

After studying this section, you should be able to

- Build a discrete population model from a set of assumptions about processes that contribute to each year's population.

[1] Stage-structured and interacting populations are examined in Chaps. 5 and 6, respectively.

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- Run simulations with discrete population models.
- Find fixed points of discrete population models.

Suppose we want to model the growth of a fish population introduced in a newly renovated lake. How do we construct such a model? The model cannot be very complicated because we have only a limited knowledge of the relevant facts. Presumably we have estimates of the vital rates of the population—birth, survival, and growth—but only limited knowledge of details. We need a conceptual model that incorporates the essential biological processes, such as survival of adults from year to year, birth and survival to the following year, and perhaps also processes that change the population from outside, such as natural migration, stocking, accidental release, or harvesting, but we also want to ignore relatively minor effects.

4.1.1 A General Seasonal Population Model

Population models are created by careful accounting. Suppose we posit a population census that occurs at a particular point in the yearly cycle. With this yearly cycle in mind, we need mathematical formulas to predict the contribution of each relevant biological process to the year $t+1$ population. Even in the absence of a real yearly census, the model could have qualitative value; however, it will be most useful if the census can actually be conducted to check on the model predictions. Most likely, the census will need to rely on statistical methods, such as the mark-and-recapture method.

Let N_t be the population in year t. For simplicity, we assume that the census takes place just before the annual birth pulse. We assume that the following processes determine the population in year $t+1$:

1. The year t population produces an average of b eggs per individual in the population.
2. A fraction p of the eggs hatch into offspring that survive to become adults at the next census.
3. A fraction S of the adults from the year t census survive to be counted as adults in the year $t+1$ census.

With these assumptions, we have two sources of individuals for the year $t+1$ census: survivors from the year t population and *recruitment* from the eggs laid at the beginning of year t.[2] In words,

$$\begin{aligned}\text{next year's population} = {}& \text{survivors from this year's population} \\ & + \text{recruitment from this year's eggs.}\end{aligned}$$

The number of adult survivors is SN_t, while the number of recruits is bpN_t; thus, we have the model

$$N_{t+1} = (bp + S)N_t\,, \tag{4.1.1}$$

where

$$b > 0\,, \quad 0 < p < 1\,, \quad 0 < S < 1\,.$$

In general, any of the parameters b, p, and S could be *density dependent*, that is, they could be functions of the population N_t. As a practical matter, all are probably functions of population, but p should be particularly population dependent, as larger populations increase competition for food, which decreases survival of the young. If the survival probabilities p and S are functions of N, they should be decreasing

[2] "Recruitment" is a term biologists use to describe the combined processes of birth and survival to the next census.

functions, as higher populations decrease survival fractions. The parameter b is probably a decreasing function of N in most cases, but occasionally it could be decreasing for very small N.

4.1.2 Discrete Exponential Growth

For a first approximation, we assume that all parameters are fixed. Only when the overall population is low is it reasonable to assume fixed parameter values; hence, we should not expect the model results to remain meaningful as the population grows.

For reasons that will only become clear in hindsight, it is often better to think of the model as a formula for the change in the population rather than the population itself. The model in this form is

$$N_{t+1} - N_t = (bp + S - 1)N_t.$$

For convenience, let $r = bp + S - 1$. We then arrive at the simplest mathematical model for population growth,

$$N_{t+1} = (1 + r)N_t\,, \qquad r > -1\,. \tag{4.1.2}$$

The reader should be wondering what makes the introduction of the parameter r "convenient." We could instead have defined a parameter $R = bP + S$, and obtained the model $N_{t+1} = RN_t$. This is a matter of taste. We will see later that r is the relative growth rate, which means that the discrete Eq. (4.1.2) is analogous to the continuous model $dy/dt = ry$, whereas the equation $N_{t+1} = RN_t$ is not.

Some additional insight into the model follows from rewriting the condition $r > 0$ as $bp > 1 - S$. In this inequality, bp is the number of recruits per the previous year's adults, and $1 - S$ is the fraction of the previous year's adults that die. The condition $r > 0$ means that the number of recruits is more than sufficient to replace the adults that have died.

If the initial population N_0 and the net relative growth rate r are known, we can use (4.1.2) to determine each year's population from the previous year's. This equation is simple enough that we can also obtain an explicit solution formula for the population in year t. Given N_0, we have $N_1 = (1 + r)N_0$, $N_2 = (1 + r)N_1 = (1 + r)^2 N_0$, $N_3 = \cdots = (1 + r)^3 N_0$, and so on. The pattern suggests the general formula

$$N_t = (1 + r)^t N_0, \tag{4.1.3}$$

a conjecture that is easily confirmed. Since the explicit solution is an exponential function of the independent variable t, the model is called the *discrete exponential growth model*.

We now have two equations for the discrete exponential growth model. The original model (4.1.2) defines the sequence recursively, meaning that N_{t+1} is given in terms of N_t rather than in terms of t directly. Mathematicians use the term *difference equation* for this type of definition to emphasize that it is a problem to be solved rather than a solution formula. The sequence of populations is defined explicitly by the solution (4.1.3) of the difference equation. The ready determination of an explicit formula for a discrete model is a luxury reserved for a few of the simplest models. Most models worth studying are ones for which we have only the original recursive definition, so we need methods for studying such equations without a solution formula.

The solution formula reveals an unrealistic property of the discrete exponential growth model, which follows from the unrealistic assumption that all three parameters b, p, and S are independent of population size. Observe that

$$\lim_{t\to\infty} N_t = \infty.$$

Eventually, there will be so many fish that the model lake will have no room for water, but the model population will be growing faster than ever.[3] Real populations do not behave this way, and of course we should prefer models that don't do so either. The discrete exponential model can be used successfully for a short time only—during the period of growth when space and resources are plentiful. Eventually, the growth of the population leads to crowding and limitation of resources. A better model should include a mechanism for slowing the net relative growth rate for larger populations.

Before we turn to more realistic models, we note one other insight that can be gained from manipulating the appearance of the model. If we rearrange the original model to isolate r, we obtain the form

$$\frac{N_{t+1} - N_t}{N_t} = r. \tag{4.1.4}$$

The quantity $N_{t+1} - N_t$ is the change in population from time t to time $t + 1$. Since one time unit has elapsed, we can also think of $N_{t+1} - N_t$ as the rate of change of population per unit time. Dividing by N_t yields the *relative rate of change* of the population, which is the constant r. Equation (4.1.4) provides a clear interpretation of the model, which is that the relative rate of change of the population is constant. Given $r > 0$, the model predicts that the population growth rate will be unabated as the population grows. While this conclusion is confirmed by the solution formula (4.1.3), the point is that we could have obtained the same conclusion without actually knowing the formula.

4.1.3 The Discrete Logistic Model

Mathematical models in physics and chemistry are almost always derived from first principles. However, it is seldom possible to derive ecological models from first principles, leading us to accept heuristic models instead. A *heuristic* model is one in which the specific details of the formulas that appear in the model are chosen because they have the right qualitative behavior [5]. We can develop a heuristic fishery model that improves on the discrete exponential growth model by modifying (4.1.4). We want the relative rate of growth to fall to 0 at some large value that represents the capacity of the environment. The easiest way to do that is to make the relative rate of growth linear with a negative slope. Using the parameters specified in Fig. 4.1.1, we obtain the *discrete logistic model*,

$$\frac{N_{t+1} - N_t}{N_t} = r\left(1 - \frac{N_t}{K}\right). \tag{4.1.5}$$

The relative rate of change in the logistic model is the same as in the exponential model for $N_t \approx 0$, but it decreases as the population grows. Eventually, the relative rate of change becomes 0 when $N_t = K$.

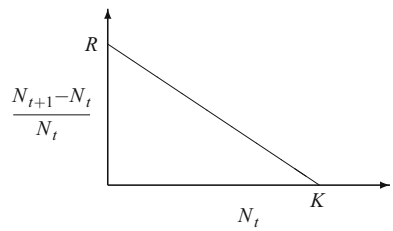


Fig. 4.1.1 The relative rate of change for the discrete logistic model

[3] For an amusing illustration of this principle, see the short story *Pigs is Pigs* [2].

The discrete logistic model is the simplest population model that can account for the effect of a finite amount of space and resources. More realistic models include the Beverton–Holt model and the Ricker model, which appear in the problems. Both of these models can be derived from first principles.

For the purpose of simulations, we can rewrite the discrete logistic model as a formula for N_{t+1} in terms of N_t:

$$N_{t+1} = N_t + rN_t\left(1 - \frac{N_t}{K}\right). \tag{4.1.6}$$

4.1.4 Simulations

If the parameters in a model are assigned values, we can run simulations by choosing different starting populations and calculating the results.

▶ **Example 4.1.1** Consider the model

$$N_{t+1} = N_t + 0.2N_t\left(1 - \frac{N_t}{1,000}\right), \qquad N_0 = 100\,.$$

This is the discrete logistic model with $r = 0.2$ and $K = 1,000$ and an initial population of 100, which might be realistic for our fishery scenario. From (4.1.5), we find the initial relative growth rate to be $0.2(1 - 100/1{,}000) = 0.18$, which is only slightly different from the initial growth rate in the discrete exponential model. As the population increases, this relative growth rate decreases, so it must be recalculated at each step. Nevertheless, it is a simple matter to use the model to project the population to any future time. For example,

$$N_1 = 100 + 20(1 - 100/1,000) = 118\,,$$

$$N_2 = 118 + 23.6(1 - 118/1,000) = 138.8\,.$$

The solution is illustrated in Fig. 4.1.2 along with the populations obtained with initial values of 0, 10, 500, and 1,000. Note that the model generally yields fractional values. This may seem to be a problem, but it really isn't. At best, the results of a mathematical model only approximate reality. Most models include errors that are far more significant than that of allowing fractional fish. □

Check Your Understanding 4.1.1:

Find N_1, N_2, and N_3 for the model of Example 4.1.1, given $N_0 = 500$.

4.1.5 Fixed Points

▶ **Example 4.1.2** Consider the model

$$N_{t+1} = N_t + 0.2N_t\left(1 - \frac{N_t}{1{,}000}\right), \quad N_0 = 1{,}000.$$

Using the model to determine population values for times greater than 0, we begin with $N_1 = 1{,}000 + 200(1 - 1{,}000/1{,}000) = 1{,}000$. Thus, $N_1 = N_0$. Similarly, we have $N_{t+1} = N_t$ for all t, that is, the population in the model will be 1,000 in all subsequent years. This is just what we should have expected,

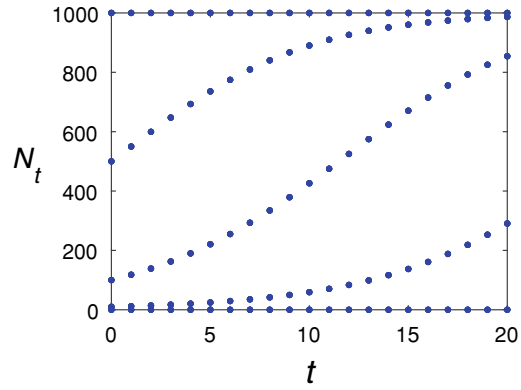


Fig. 4.1.2 Population results for Example 4.1.1.

since we set the initial population N_0 to be the same as the environmental capacity K. We also have $N_1 = N_0$ and $N_{t+1} = N_t$ for $N_0 = 0$. Since there is no migration process, there is no mechanism to start population growth if there is no initial population. □

Definition 4.1.1

A value of N for which $N_{t+1} = N_t$ is called a **fixed point**.

Fixed points play a special role in the study of discrete dynamic models; hence, determination of fixed points is a key part of the analysis of any such model. In general, fixed points can be determined by simple algebra. Suppose $N_t = N$, where N is a fixed point. Then by definition, $N_{t+1} = N$ also. The substitutions $N_{t+1} = N_t = N$ into the discrete logistic model (4.1.6) yield the equation

$$N = N + rN\left(1 - \frac{N}{K}\right).$$

This equation defines fixed points in terms of the model parameters r and K. After simplifying, we obtain

$$N\left(1 - \frac{N}{K}\right) = 0.$$

There are two fixed points, $N = K$ and $N = 0$. You can quickly check that these are fixed points by computing $N_1 = 0$ from $N_0 = 0$ and $N_1 = K$ from $N_0 = K$.

▶ **Example 4.1.3** Consider the model

$$N_{t+1} = \frac{RN_t^2}{1 + N_t^2},$$

where R is a positive constant. To find the fixed points of this model, we set $N_{t+1} = N$ and $N_t = N$ and obtain the equation

$$N = \frac{RN^2}{1 + N^2}.$$

As with most population models, $N = 0$ is a fixed point.[4] If $N \neq 0$, the equation becomes

$$1 = \frac{RN}{1 + N^2},$$

which we can rewrite as

$$1 + N^2 = RN.$$

This is a quadratic equation for N. Applying the quadratic formula to the form $N^2 - RN + 1 = 0$, we obtain

$$N = \frac{R \pm \sqrt{R^2 - 4}}{2}.$$

These fixed points exist only if $R > 2$. It can be shown that $R < 2$ means that the population at time $t + 1$ is always less than the population at time t; hence, $R > 2$ is a requirement for viability.[5] Assuming this requirement is met, the model then has three fixed points:[6]

$$N = 0, \quad N = \frac{R - \sqrt{R^2 - 4}}{2}, \quad N = \frac{R + \sqrt{R^2 - 4}}{2}.$$

□

Check Your Understanding 4.1.2:

Find the nonnegative fixed points of the model

$$N_{t+1} = \frac{AN_t}{B + N_t}$$

in terms of the positive parameters A and B.

Fixed points can be categorized according to the behavior of solutions in approaching them or receding from them.

Definition 4.1.2

A fixed point N^* is **locally asymptotically stable** if $\lim_{t \to \infty} N = N^*$ for any N_0 close enough to N^*; the stability is said to be **global** if no restriction on N_0 is required. A fixed point is **unstable** if sequences that begin arbitrarily close to it move away from it.

▶ **Example 4.1.4** Figure 4.1.2 appears to show that the fixed point $N^* = 1000$ is at least locally asymptotically stable, while the fixed point $N^* = 0$ is unstable. These claims can be confirmed by cobweb analysis (Sect. 4.2). □

We often use the simple term "stable" to mean "locally asymptotically stable," as the full phrase is rather a mouthful. Note that there is a boundary case, sometimes referred to as "neutrally stable,"

[4] This is always true if there is no migration from outside.

[5] See Problem 4.1.1.

[6] Only nonnegative fixed points have biological meaning. Given $R > 2$, all three fixed points are nonnegative.

between unstable and asymptotically stable. These arise in mathematical examples but have limited biological importance.

Discrete models do not necessarily converge to a fixed value, as will be seen in the problems. Nevertheless, those that do must converge to a stable fixed point; hence, the determination of stable fixed points is always a key component of model analysis.

Check Your Understanding Answers

1. $N_1 = 550$, $N_2 = 599.5$, $N_3 = 647.52$.
2. $N_0^* = 0$ and $N_1^* = A - B$, with the latter meaningful only if $A > B$.

Problems

4.1.1 Show that $R < 2$ means that the sequence defined in Example 4.1.3 is always decreasing, as claimed in the example.

4.1.2* [Discrete Logistic Model]
The discrete logistic model has some interesting mathematical properties. Among these are the possibilities of periodic or chaotic solutions. To explore these possibilities, we consider the model (4.1.6) with $K = 1000$, as in Example 4.1.1, but with larger values of r. In each of the simulations indicated below, determine the ultimate fate of the sequence. Start with a total time of 20 steps. If that is not enough to see a pattern, try 40 steps or 80 steps.

(a) $r = 1.5$ and $N_0 = 20$.
(b) $r = 1.8$ and $N_0 = 20$.
(c) $r = 2.2$ and $N_0 = 20$. Then switch to $N_0 = 980$.
(d) $r = 2.5$ and $N_0 = 980$.
(e) $r = 2.56$ and $N_0 = 770$.
(f) $r = 3.0$ and $N_0 = 550$.
(g) $r = 3.0$ and $N_0 = 551$.
(h) Discuss the simulation results, emphasizing what happens as r increases.

(This problem is continued in Problem 4.2.2.)

4.1.3 [Discrete resource consumption]
The model

$$N_{t+1} = N_t + RN_t\left(1 - \frac{N_t}{K}\right) - \frac{CN_t}{A + N_t}$$

represents a population that undergoes logistic growth along with consumption by a population of C consumers. We explore this model with $K = 1000$, $r = 2.2$, and $A = 200$. In each of the simulations indicated below, start with a total time of 20 steps. If that is not enough to see a pattern, try 40 steps or 80 steps.

(a) $C = 0$ and $N_0 = 980$ (This is the same scenario as Problem 4.1.2(c).)
(b) $C = 100$ and $N_0 = 1000$
(c) $C = 300$ and $N_0 = 1000$.
(d) Discuss the simulation results, emphasizing what happens as C increases.

(This problem is continued in Problem 4.2.3.)

4.1.4* [Beverton–Holt model]
One specific fishery model is the Beverton–Holt model, which can be written in dimensionless form as

$$N_{t+1} = \left(S + \frac{A}{1+N_t}\right) N_t; \quad A > 0\,, \quad 0 \le S < 1.$$

(a) Find the relative rate of change for the model.
(b) What is the biological significance of the assumption $S = 0$?
(c) Find a general formula for the non-negative fixed points of the model. Note that some fixed points may only exist for certain ranges of parameter values.
(d) Plot (on one set of axes) graphs of the fixed points in the SN-plane, given $A = 0.5$ and $A = 2$.
(e) Run a simulation with the model using $A = 0.8$, $S = 0$, and $N(0) = 0.2$. Discuss the simulation results, emphasizing the connection to the results of part (c).
(f) Repeat part (e), but with $A = 2$.
(g) Repeat part (e), but with $S = 0.5$.

(This problem is continued in Problem 4.2.4.)

4.1.5 [Ricker model]
Another fishery model is the Ricker model, which can be written in dimensionless form as

$$N_{t+1} = \left[S + Ae^{-N_t}\right] N_t.$$

(a) Find the relative rate of change for the model.
(b) What is the biological significance of the assumption $S = 0$?
(c) Find a general formula for the non-negative fixed points of the model. Note that some fixed points may only exist for certain ranges of parameter values.
(d) Plot (on one set of axes) graphs of the fixed points in the SN-plane, given $A = 0.5$ and $A = 2$.
(e) Run a simulation with the model using $S = 0$, $A = 2$, and $N(0) = 0.2$. Discuss the simulation results, emphasizing the connection to the results of part (c).
(f) Repeat part (e), but with $A = 8$.
(g) Repeat part (e), but with $A = 14$.

(This problem is continued in Problem 4.2.5.)

4.1.6 [Hassell insect model]
The Hassell model, which is sometimes used for population dynamics of insect pests, can be written in dimensionless form as

$$N_{t+1} = \left(\frac{A}{1+N_t}\right)^b N_t, \qquad A, b > 0.$$

(a) Find a general formula for the non-negative fixed points of the model. Note that some fixed points may only exist for certain ranges of parameter values.
(b) Run a simulation with the model using $b = 0.5$ and $A = 10$. Discuss the simulation results, emphasizing the connection to the results of part (a).
(c) Repeat part (b) using $b = 2$, $A = 6$, which roughly matches a laboratory population of a species of weevils.

(d) Repeat part (b) using $b = 3$, $A = 4$, which roughly matches a field population of Colorado potato beetles.[7]

(This problem is continued in Problem 4.2.6.)

4.2 Cobweb Analysis

After studying this section, you should be able to

- Construct cobweb plots.
- Use cobweb plots to run simulations.
- Use cobweb plots to find fixed points and determine their stability.

Cobweb plots are graphical representations of single-component discrete dynamic models and are used to draw conclusions regarding long-term behavior. The model must have the general form

$$N_{t+1} = g(N_t)\,, \tag{4.2.1}$$

for some function g. Note that the function g depends only on the current state of the system N_t, and not explicitly on the time t.[8] This requirement rules out models that account for yearly changes in the environment.[9] Fortunately, there are a lot of useful models that are of the required form, including the Beverton–Holt and Ricker models introduced in Sect. 4.1 problems.

4.2.1 Cobweb Plots

Figure 4.2.1 shows a cobweb plot and time history for the discrete exponential growth model

$$N_{t+1} = 1.5N_t.$$

At first glance, cobweb plots seem to be a jumble of lines going in different directions. Interpreting them is a skill that takes practice. We focus first on the construction of the plot and then fill in details to help with the interpretation.

The basis of a cobweb plot is a graph that displays $y = g(N)$ and $y = N$ on the same set of axes. To minimize confusion, it is best to label the horizontal axis as N and either leave the vertical axis unlabeled or use the generic label "y." In the cobweb plot of Fig. 4.2.1a, the heavy blue line is $y = g(N) = 1.5N$ and the medium black line is $y = N$.

The light red lines with a stair-step shape are a graphical rendition of a simulation. The procedure for creating these lines on a plot that already contains $y = g(N)$ and $y = N$ is specified in Algorithm 4.2.1.

[7] Parts (b)–(d) use parameter values adapted from [1].

[8] Processes that depend only on the state of a system, and not directly on the time, are called *autonomous*. Much of the analysis we can do with population models only works for autonomous models.

[9] This does not mean that we cannot study the effects of global climate change. We can use analysis to study its effects on fixed points and stability. If we want to study the short-term effects of climate change, however, we need to run simulations.

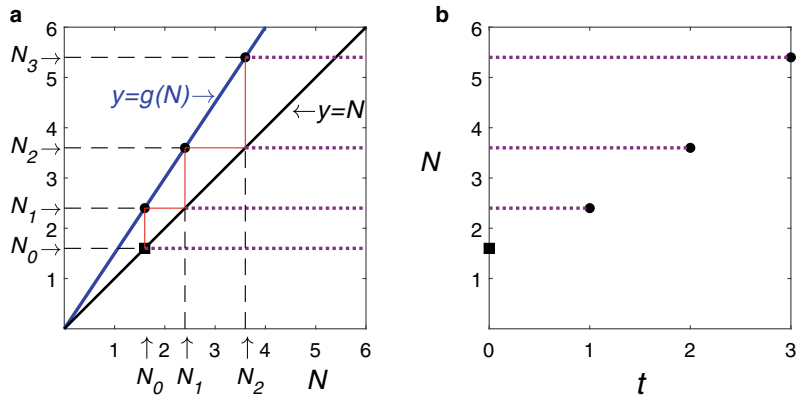


Fig. 4.2.1 Cobweb plot (left) and time history (right) for $N_{t+1} = 1.5N_t$ with $N(0) = 1.6$

Algorithm 4.2.1
Cobweb plot for $N_{t+1} = g(N_t)$

1. Mark the point on the $y = N$ line corresponding to a chosen initial population N_0.
2. Draw a vertical line segment from the point marked in step 1 to the curve $y = g(N)$. Mark the intersection point. Note that this line segment could go either up or down, depending on the relative locations of the curve $y = g(N)$ and the line $y = N$.
3. Draw a horizontal line segment from the point marked in step 2 to the line $y = N$. This line segment could go either to the left or to the right.
4. Continue extending the plot by alternating vertical and horizontal line segments, always moving up or down to $y = g(N)$ and left or right to $y = N$, and marking each point that is an intersection of a cobweb segment with the curve $y = g(N)$.
5. For a time history plot, as in Fig. 4.2.1b, extend horizontal lines from the marked points in the cobweb plot to another set of axes. These horizontal lines should end at consecutive time values, starting with $t = 0$ for the horizontal line corresponding to the first marked point.

Just as in a simulation by calculation, all of the parameters must be chosen before constructing the cobweb plot, including the initial value N_0. Once the parameters are chosen, the process of constructing the cobweb plot is mechanical. Just plot the two functions and draw the line segments in the prescribed manner. Think of the construction and interpretation as separate steps, and the construction is easy.

We begin the interpretation of the cobweb plot with the first marked point. The horizontal coordinate of this point is N_0 and the point is on the line $y = N$; hence its vertical coordinate is also N_0. The dotted horizontal line transfers this vertical coordinate to the time history plot, where it is marked at time 0.

The vertical line segment in step 2 has the horizontal coordinate N_0 and intersects with the line $y = g(N)$. Thus, its vertical coordinate is $g(N_0)$, which is N_1. The dotted horizontal line transfers this coordinate to the point $(1, N_1)$ on the time history plot.

The horizontal line segment of step 3 has the y-coordinate N_1. It reaches the line $y = N$ at the point (N_1, N_1). The dashed vertical line from this point transfers the coordinate N_1 to the horizontal axis. This dashed line is not necessary, but it helps understand the overall plot.

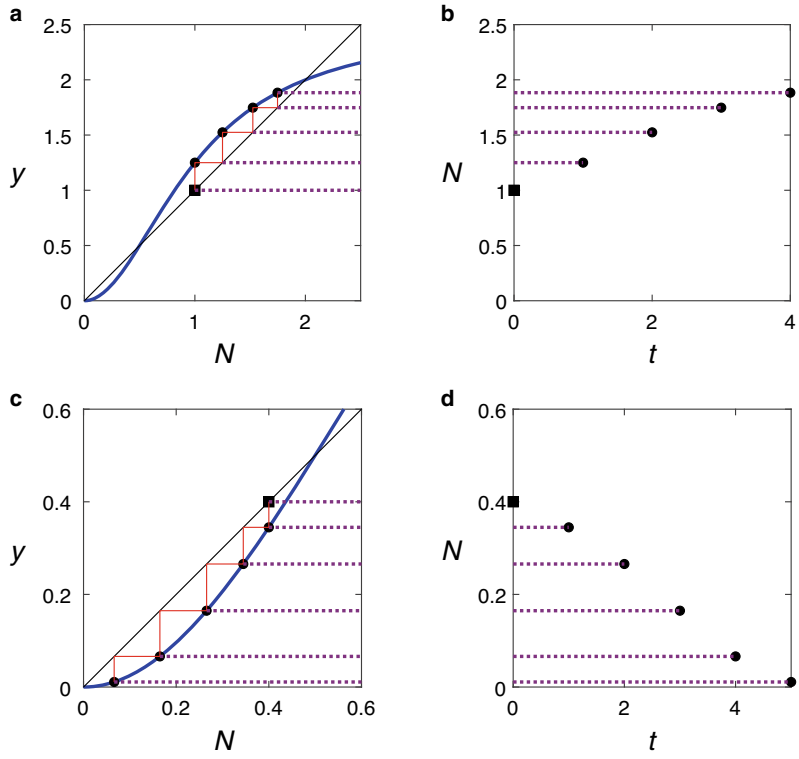


Fig. 4.2.2 Cobweb plots and time histories for $N_{t+1} = \frac{2.5N_t^2}{1+N_t^2}$ with different initial conditions: **a** $N_0 = 1.6$; **b** $N_0 = 0.4$

The next vertical line segment has horizontal coordinate N_1. It ends at the point $(N_1, g(N_1))$, which is (N_1, N_2). The dotted horizontal line transfers this coordinate to the time history plot.

And so the process continues in pairs of steps: a horizontal step to transfer the coordinate N_t from the vertical axis to the horizontal, followed by a vertical step to identify the coordinate N_{t+1} on the vertical axes of both the cobweb and time history plots.

Once you understand the reason for the alternating vertical and horizontal lines, the cobweb plot loses its mystery and becomes a simple tool.

4.2.2 Stability Analysis

Up to this point, we have focused on explaining the process of the cobweb plot and its interpretation as a simulation. Now we turn to the use of the plot for analysis. What information does it contain that is not found in the corresponding time history?

In a time history, we can only see the current point and past points. In a cobweb plot, we can see the future as well, that is, we can predict the trend of future populations from the relative positions of the graphs of $y = g(N)$ and $y = N$. For the model of Fig. 4.2.1, the future is that the sequence will continue to increase as the plot is extended to larger values of N and y. Every horizontal line segment will go to the right, so each value of N will be larger than the preceding value.

▶ **Example 4.2.1** Figure 4.2.2 shows two cobweb plots for

$$N_{t+1} = \frac{2.5N_t^2}{1 + N_t^2}.$$

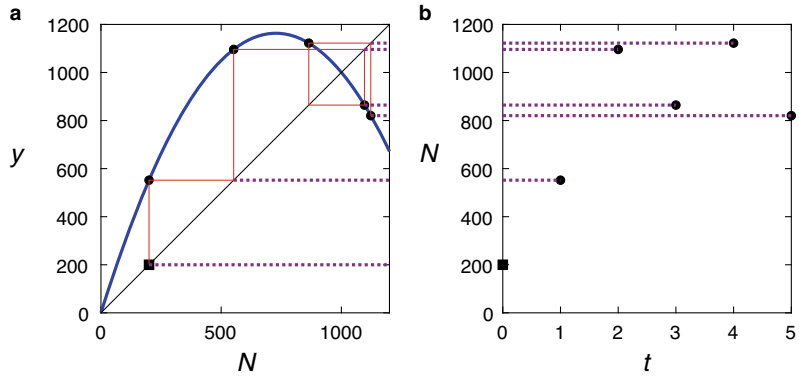


Fig. 4.2.3 Cobweb plot and time history for $N_{t+1} = N_t + 2.2N_t(1 - N_t/1000)$ with $N_0 = 200$

These plots tell a different story from that of Fig. 4.2.1. The graphs of $y = g(N)$ and $y = N$ cross at the fixed point $N = 2$. As the simulation in panels **a** and **b** progresses, the horizontal line segments are all moving to the right and the vertical line segments are all moving up, but they must move in such a way that the fixed point is never crossed. The solution must approach the fixed point $N = 2$ over time; hence, this fixed point is asymptotically stable.[10] Panel **c** shows a cobweb plot for the same model, but this time with an initial value $N_0 = 0.4$. This time, the horizontal line segments all move to the left, so the marked points approach the fixed point $N = 0$. Thus, $N = 0$ is a second asymptotically stable fixed point for the model. The point $N = 0.5$ is an unstable fixed point that marks the dividing line between initial conditions that lead to $N = 2$ and those that lead to $N = 0$. □

The model of Example 4.2.1 is a typical well-behaved discrete model. The stable fixed points (at 0 and 2) serve as sequence limits for ranges of initial values delimited by the unstable fixed point(s) (at 0.5 in the example).

▶ **Example 4.2.2** Figure 4.2.3 shows a cobweb plot and time history for the discrete logistic model with $r = 2.2$ and $K = 1000$. If r is small enough, the fixed point $N = 1000$ is stable. With $r = 2.2$, $N = 1000$ is still a fixed point, but it is no longer stable. Starting from a small initial value, the population increases to a point larger than the fixed point, but then it settles into a pattern that appears to be periodic, with values alternating between approximately 800 and 1150. The cobweb plot shows why this happens. Each vertical and horizontal move goes beyond the fixed point before the appropriate curve ($y = g(N)$ or $y = N$) is reached. □

Eventually, the problem of Example 4.2.2 settles into a pattern of alternation between the two values 746.3 and 1162.8. Alternation between two values is called a *2-cycle*. Many other patterns can arise from the discrete logistic equation.[11] The problems in this section include some preliminary explorations, and Project 4D considers the phenomena in much more detail.

[10] Section 4.1.

[11] See [8] for some very unusual patterns in a two-component discrete model.

Problems[12]

4.2.1* Prepare a cobweb plot for the model

$$N_{t+1} = \frac{N_t^2}{1 + N_t}, \qquad N_0 = 1.$$

Describe the long-term behavior of the model.

4.2.2 (Continued from Problem 4.1.2.)
Prepare a cobweb plot for the discrete logistic model

$$N_{t+1} = N_t + 2.5N_t\left(1 - \frac{N_t}{1000}\right)$$

using $N_0 = 550$. [This will be best if you use five time steps and no time history plot.] Discuss the results with reference to those of Problem 4.1.2.

4.2.3 [Discrete resource consumption] (Continued from Problem 4.1.3.)
Prepare cobweb plots for each of the cases indicated below for the model

$$N_{t+1} = N_t + 2.2N_t\left(1 - \frac{N_t}{1000}\right) - \frac{CN_t}{200 + N_t}.$$

[Try six time steps and adjust as needed.]

(a) $C = 0$ and $N_0 = 980$.
(b) $C = 100$ and $N_0 = 1000$.
(c) $C = 300$ and $N_0 = 1000$.
(d) Discuss the simulation results, emphasizing what happens as C increases.

(This problem is continued in Problem 4.4.7.)

4.2.4* [Beverton–Holt model] (Continued from Problem 4.1.4.)
Prepare cobweb plots for the Beverton–Holt model

$$N_{t+1} = \left(S + \frac{A}{1 + N_t}\right)N_t, \quad A > 0, \quad 0 \le S < 1,$$

with $N_0 = 0.2$, using

(a) $A = 0.8$, $S = 0$.
(b) $A = 2$, $S = 0$.
(c)* $A = 0.8$, $S = 0.5$.
(d) Discuss the results with reference to those of Problem 4.1.4.

(This problem is continued in Problem 4.4.4.)

4.2.5 [Ricker model] (Continued from Problem 4.1.5.)
Prepare cobweb plots for the Ricker model

[12] The author's MATLAB program *CobwebPlotter.m* provides a convenient way to produce cobweb plots similar to the figures in this section.

$$N_{t+1} = AN_t e^{-N_t}$$

with $N_0 = 0.2$, using

(a) $A = 2$.
(b) $A = 8$.
(c) $A = 14$.
(d) Discuss the results with reference to those of Problem 4.1.5.

(This problem is continued in Problem 4.4.2.)

4.2.6 [Hassell insect model] (Continued from Problem 4.1.6.)
Prepare cobweb plots for the Hassell model

$$N_{t+1} = \left(\frac{A}{1+N_t}\right)^b N_t, \qquad A, b > 0,$$

using

(a) $b = 0.5$, $A = 10$, $N_0 = 1$.
(b) $b = 2$, $A = 6$, $N_0 = 2$.
(c) $b = 3$, $A = 4$, $N_0 = 2.5$.
(d) Discuss the results with reference to those of Problem 4.1.6.

(This problem is continued in Problem 4.4.10.)

4.3 Continuous Dynamics

After studying this section, you should be able to

- Build continuous dynamic models from assumptions about rates of change.
- Use computer simulations to study continuous dynamic models.
- Find equilibria of continuous dynamic models.
- Plot a phase line for a continuous dynamic model and use it to determine stability of equilibria.

We have already seen differential equation models in Sects. 3.3 and 3.4. The focus there was simply on defining models and examining computer-generated results. Here we turn to the methods of analysis for models of continuous change of single variables. We briefly consider analytical solutions and numerical simulation before turning our focus to graphical methods.

4.3.1 Exponential Growth

The simplest population model is the exponential growth model, which is characterized by a constant relative growth rate. To derive this model, we note that the derivative dy/dt of a function $y(t)$ of continuous time is the *absolute rate of change*. The *relative rate of change* is the ratio of the absolute rate of change to the magnitude $|y|$; since $y > 0$, a constant relative rate of change corresponds to the equation

$$\frac{1}{y}\frac{dy}{dt} = r\,. \tag{4.3.1}$$

We can rewrite the equation in the standard form as

$$\frac{dy}{dt} = ry\,. \tag{4.3.2}$$

The model (4.3.2) is usually described as representing the number of individuals in a population; however, it is often more appropriate to think of the variable y as representing the biomass of the population. Consider a population of plants, for example. Two small plants make the same contribution to the overall population size and resource requirements as a single plant that is twice as large; hence, it makes more sense to measure a plant population in terms of its biomass rather than the number of individuals. Though less obvious, this statement is equally true for animals. One large animal probably eats about the same amount of food as two animals half its size. Another advantage of using biomass to represent populations is that it changes on a continuous basis, unlike the count of individuals, which only changes as a result of birth events.

Although solution methods for differential equations are beyond the scope of this book, (4.3.2) can be solved by inspection. We are looking for a function whose derivative is a constant times the function. A quick review of a table of derivatives[13] reveals that the exponential function has this property; specifically, the function $y(t) = y_0 e^{rt}$ has the required property and also satisfies the initial condition $y(0) = y_0$.

Check Your Understanding 4.3.1:

Determine an appropriate mathematical relationship between the discrete and continuous relative growth rates r_d and r_c by comparing the solution of the continuous model $dy/dt = r_c y$, $y(0) = 1$, with that of the corresponding discrete model $N_{t+1} - N_t = r_d N_t$, $N_0 = 1$.

4.3.2 Logistic Growth

The exponential model is only useful while resources are abundant. The unbounded solution of (4.3.2) makes the model unsuitable for any situations in which growth should be restricted. A more realistic model is the *logistic growth model*. This model is usually chosen on heuristic grounds as the simplest model that transitions from a growth rate of r for small populations to 0 for a population of size K, as was done in the discrete case in Fig. 4.1.1, but with the continuous relative rate of change,

$$\frac{1}{y}\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right),$$

which we can rewrite as

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right). \tag{4.3.3}$$

Equation (4.3.3) can be solved explicitly; however, it is not particularly advantageous to do so. All important analytical results can be obtained directly from the differential equation, as we shall see in the remainder of this chapter. Simulations can be conducted with any degree of accuracy using numerical methods, which work by computing the solutions to discrete approximations of the differential equation.

▶ **Example 4.3.1** Analogous to Example 4.1.1, consider the model

$$\frac{dy}{dt} = 0.2y\left(1 - \frac{y}{1,000}\right), \quad y(0) = 100.$$

[13] See Appendix B.

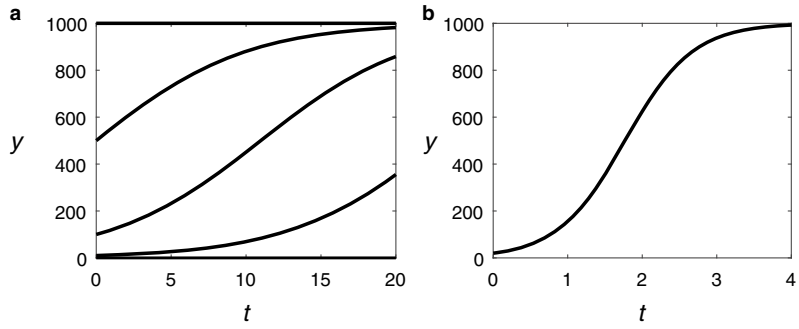


Fig. 4.3.1 Population results for the logistic growth model $dy/dt = ry(1 - y/1000)$: **a** $r = 0.2$, **b** $r = 2.2$

A collection of numerical methods for solving differential equations is presented in Appendix D; here, we summarize the simplest one, which is called Euler's method. We choose a sequence of time values t_j, for which we seek corresponding approximate population values y_j. With $t_0 = 0$, we have a starting value of $y_0 = 100$. Numerical methods work by using a discrete approximation to the differential equation to calculate each successive function value. Euler's method uses the forward difference approximation

$$\frac{dy}{dt}(t_j) \approx \frac{y_{j+1} - y_j}{t_{j+1} - t_j}. \tag{4.3.4}$$

Substituting this approximation into the differential equation yields the discrete dynamic model

$$y_{j+1} = y_j + 0.2hy_j\left(1 - \frac{y_j}{1,000}\right), \quad y_0 = 100,$$

where we have assumed time intervals of equal length $h = t_{j+1} - t_j$.

With sufficiently small time steps, graphs of the simulation result are indistinguishable from the exact solution, as shown in Fig. 4.3.1a. As we might expect, the behavior of the continuous logistic model in this example is similar to that of the corresponding discrete model seen in Fig. 4.1.2; however, the complicated behavior seen in the discrete model with larger growth rate parameter (Fig. 4.2.3) is not replicated in the continuous version. □

4.3.3 Dynamical Systems

In a calculus text, the derivative of a function $y(t)$ is nearly always given as a function of the independent variable t. In mathematical modeling, the derivative is nearly always given as a function of the dependent variables.

Definition 4.3.1

An **autonomous dynamical system** is a system of one or more quantities whose rates of change in time are given as functions of the quantities themselves.

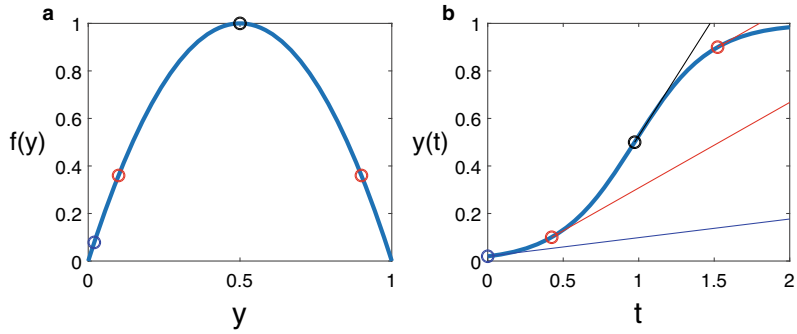


Fig. 4.3.2 **a** The function $f(y) = 4y(1-y)$; **b** the solution of $dy/dt = f(y)$ with $y(0) = 0.01$. The slopes in panel **b** correspond to the $f(y)$ values in panel **a**

▶ **Example 4.3.2** In Example 3.2.1, we developed a mechanistic model for the sharing of a secret:

$$\frac{dy}{dt} = f(y) = \beta y(1-y)\,, \quad y(0) = y_0\,, \qquad 0 < y_0 < 1\,, \tag{4.3.5}$$

where y is the fraction of a population that knows the secret and β is a parameter that represents how quickly the secret is shared. Note that this is just the logistic model built around a different narrative. □

In functional terms, any model $dy/dt = f(y)$ says that the rate of change of the variable y depends directly on its current value and only indirectly on the time t. The rate may be fast or slow at any particular time, depending only on the current state of the system at that time.

How should we try to understand a dynamical system $dy/dt = f(y)$? The simplest way is to use the information shown by a graph of the known function $f(y)$ for a range of meaningful values (from $y = 0$ to $y = 1$ if y is a fraction of some maximum value, as in Example 4.3.2), not to find a formula for the function $y(t)$, but to check slopes on a sketch of the graph of y versus t. If we know the current value of y at any point in time, then we can use the graph of f to identify dy/dt, which is the slope of the graph of $y(t)$ at that particular time.

▶ **Example 4.3.3** Figure 4.3.2 shows two graphs, one of the function $f(y) = 4y(1-y)$ and one of the function $y(t)$ that solves the equation $dy/dt = 4y(1-y)$. The first graph gives us information we can use to verify the second graph. Suppose the value of y is initially 0.01. From the left panel of Fig. 4.3.2, we can see that the value of f, and hence of dy/dt, is small and positive. So y is increasing, but with a fairly flat rise. The first marked point in the right panel shows the same value of y and a tangent line that has the slope found from the f-axis in the left panel. As y increases, the graph of f tells us that y will continue to rise, with increasing slope; in other words, it will curve upward as it increases. The second marked point is at $y = 0.1$ and shows the value of dy/dt to be a little less than 0.4. The corresponding point in the panel on the right has $y = 0.1$ and shows the tangent line with the correct slope. This behavior starts to change when we reach $y = 0.5$, since that is where f achieves its maximum value. This point corresponds to the steepest slope on the graph of $y(t)$. After that, the function f is still positive, but its value decreases as y decreases. So the graph of $y(t)$ will continue to increase, but with a flattening or downward curvature. The fourth marked point has the same value of f as the second, which means that the slopes of the graph of y at those two points are the same. Note that the graph of $y(t)$ can never cross the threshold value $y = 1$ because f is 0 when $y = 1$, meaning that y is no longer increasing. □

The graph of y in Fig. 4.3.2b was created using a solution formula obtained by calculus; however, it could have been obtained from a numerical simulation. Without either of these methods, we would still have been able to get the right slopes for each value of y, which give us the correct overall shape for the graph. We just would not have been able to get the times right for the four marked points.

To summarize, a differential equation of the form $dy/dt = f(y)$ specifies the slope of the graph of y as a function of the value of y. This information, along with an initial condition, is sufficient to define the graph, which we can roughly sketch with nothing more than logical statements about how f changes as y changes.

Check Your Understanding 4.3.2:

Let $f(y) = -y^2$. Observe that the function $y = 1/(1+t)$ has derivative $dy/dt = -1/(1+t)^2 = -y^2 = f(y)$. Plot graphs of $f(y)$ and $y(t)$ as shown in Fig. 4.3.2. Then mark by hand the points on both graphs where y =0, 0.1, 0.2, and 0.3; add tangent lines to the graph of $y(t)$; and compare the slopes of those tangents to the marked values of $f(y)$.

4.3.4 Equilibrium Points and Stability

Analogous to the fixed points of discrete models are *equilibrium points* of continuous models.

Definition 4.3.2

A value of y for which $f(y) = 0$ is called an **equilibrium point** of the differential equation $dy/dt = f(y)$.

Equilibria can be classified according to whether solutions approach them or recede from them over time. The definitions regarding stability in Definition 4.1.2 carry over to continuous dynamical systems, but with the phrase "solutions curves" in place of "sequences."

▶ **Example 4.3.4** Setting $dy/dt = 0$ in the differential equation

$$\frac{dy}{dt} = 4y(1-y)$$

yields the algebraic equation

$$4y(1-y) = 0\,.$$

The solutions $y = 0$ and $y = 1$ are the equilibrium points for the differential equation. The solution curve in Fig. 4.3.2b shows that $y = 0$ is unstable. An additional solution curve with initial condition $y_0 > 1$ would confirm that $y = 1$ is asymptotically stable. □

4.3.5 The Phase Line

The function values in Fig. 4.3.2a tell us the slopes of solution curves, which we can use to obtain a sketch of those curves. In many cases, our primary interest is in the stability of the equilibria rather than a fully detailed graph. For this purpose, we do not need to know the values of $f(y)$, but merely the algebraic signs of f for ranges of y. This information can be obtained from the graph of f and used to prepare a *phase line* plot rather than a plot of solution curves.

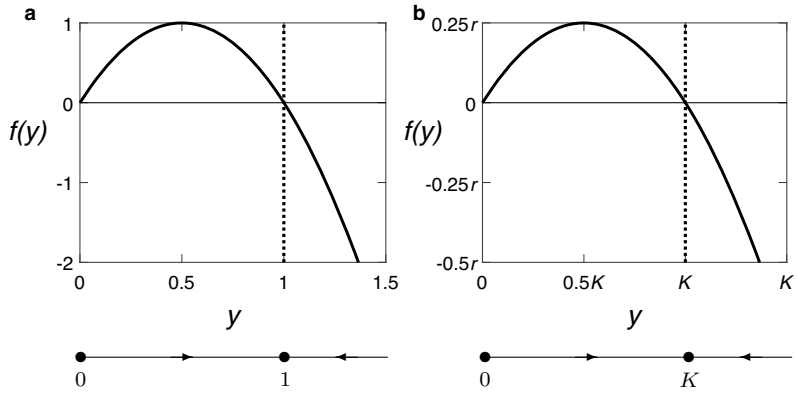


Fig. 4.3.3 **a** The function $f(y) = 4y(1-y)$ and the phase line for $y' = 4y(1-y)$; **b** The function $f(y) = ry(1-y/K)$ and the phase line for $y' = ry(1-y/K)$

A phase line plot consists of a single axis for the dependent variable. This axis may be displayed as horizontal or vertical. The advantage of vertical orientation is that it corresponds to plots of solution curves; the advantages of horizontal orientation are that it is easier to obtain the phase line plot from the graph of f and that it makes the plot analogous to the phase plane plots we will use to study systems of two dynamic variables in Chap. 6.

Regardless of orientation, the phase line consists of an axis, points to mark the equilibria, and arrows to indicate whether the solution curves are increasing or decreasing in the regions marked out by the equilibria.

A simple procedure is all that is needed to create a phase line plot.

Algorithm 4.3.1
Phase line representation for $dy/dt = f(y)$

1. Find the equilibrium points ($f(y) = 0$) and mark these points on the phase line.
2. The equilibrium points partition the interval $[0, \infty)$ into regions.[14] Each of these regions needs an arrow. The arrowhead points to the right/up for regions in which $f(y) > 0$ and to the left/down in regions where $f(y) < 0$.

▶ **Example 4.3.5** Figure 4.3.3a shows a graph of the function $f(y) = 4y(1-y)$ along with the (horizontal) phase line representation of the logistic equation

$$\frac{dy}{dt} = 4y(1-y)\,.$$

□

The phase line representation has a generality that a plot of solution curves lacks. Initial conditions need not be specified, as the phase line shows the behavior for all possible cases. Parameters can often

[14] This assumes that the dependent variable cannot be negative in the model, as is the case when the dependent variable represents a population. If the model makes sense for negative values of the dependent variable, then use the interval $(-\infty, \infty)$.

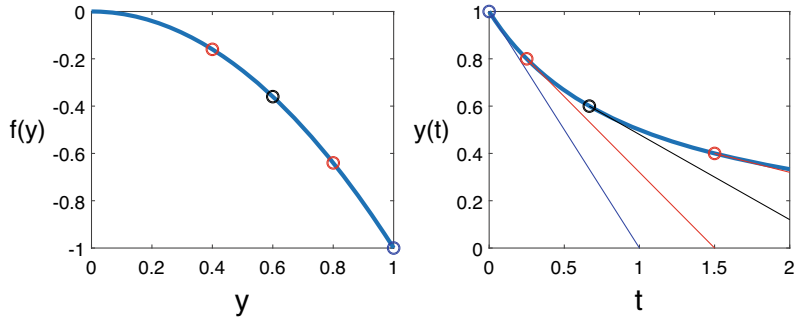


Fig. 4.3.4 **a** The function $f(y) = -y^2$; **b** the function $y = 1/(1+t)$, which is the solution of $dy/dt = f(y)$ with $y(0) = 1$

be left unspecified as well, since the method requires only a rough sketch of the graph of f. The generality of the phase line sketch allows us to determine global as well as local stability.

▶ **Example 4.3.6** Figure 4.3.3b shows a graph of the function $f(y) = ry(1 - y/K)$ along with the phase line representation of the logistic equation

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right).$$

The phase line shows that the equilibrium $y = K$ is globally asymptotically stable and that the equilibrium $y = 0$ is unstable. □

Check Your Understanding Answers

1. $1 + r_d = e^{r_c}$
2. See Fig. 4.3.4.

Problems

4.3.1* [Vaccine Implementation]
Suppose a new vaccine for a novel disease becomes available at time $t = 0$. Because of production limitations, it is expected to take t_v weeks to produce enough vaccine for the whole population. Given the difficulty of locating all people for vaccine administration, we can assume a Holling type 2 function[15] for the actual vaccination rate as a function of the unvaccinated population size. These considerations lead us to the model

$$\frac{du}{dt} = -\frac{Vu}{w+u}, \qquad u(0) = u_0,$$

where u is the unvaccinated fraction of the population, u_0 is the fraction of the population that is willing to be vaccinated, $V = 1/t_v$ is the maximum vaccination rate in population fraction per week, and w is the population fraction at which vaccination occurs at only half of the maximum rate.

(a) Plot a graph of the function that represents du/dt versus u, assuming that the total production time is 20 weeks and that vaccination occurs at half of the maximum rate when 10% of the population is still unvaccinated. Your graph should be similar to Fig. 4.3.2a, except of course that the function is different.

[15] Sections 3.2 and 3.6.

(b) Use your graph from (a) to sketch a possible graph of the solution $u(t)$, similar to Fig. 4.3.2b. Do this by hand, simply by looking at the graph from (a). For simplicity, assume everyone is willing to be vaccinated.
(c) Run a numerical simulation of the model by modifying the MATLAB function *odesim.m*. Run it (by modifying *ODEsim_test.m*) using the parameter values from (a).
(d) If there were no delays caused by difficulty in locating people, everyone would be vaccinated in 20 weeks. What fraction of the population is still unvaccinated at the 20-week mark of this scenario?
(e) Compare your hand-generated and computer-generated graphs of u. They should be qualitatively similar. If not, reconsider both your hand graph and your program to identify any errors.

4.3.2* Use phase line analysis to determine the stability of the equilibria for the model

$$\frac{dN}{dt} = -N\left(1 - \frac{N}{T}\right)\left(1 - \frac{N}{K}\right), \qquad 0 < T < K.$$

The parameter K has the same biological meaning that it has in the logistic model. Assuming that N is a population, explain the biological meaning of T.

(This problem is continued in Problem 4.4.11.)

4.3.3* [Resource harvesting] (Continued from Problem 3.6.7.)
The dimensionless Schaefer model,

$$x' = x(1 - x) - Ex, \qquad x, E \geq 0,$$

is often used to model the impact of harvesting on a fish population, where x is either the number of fish or the biomass of fish as a fraction of the environmental carrying capacity and E is a measure of the effort put into fishing.

(a) Determine the equilibrium points for the model. In particular, what restriction is there for E if there is to be a positive equilibrium population level?
(b) Use phase line analysis to determine the stability of the equilibria. You will need to consider two cases, for different ranges of E. (You do not need to consider the boundary case.)

(This problem is continued in Problem 4.4.12.)

4.3.4 [Abiotic resource]
The (scaled) model

$$\frac{dx}{dt} = R - \frac{x}{1 + x}, \qquad x, R \geq 0$$

is a possible model for the environment level of an abiotic resource that is created at a fixed rate and consumed at a rate given by Holling type 2 dynamics.

(a) Find all meaningful equilibrium points, noting the requirements on R for their existence.
(b) Sketch the function $f(x) = dx/dt$ and use the graph to prepare a phase line plot for the case $R = 2$. Determine the prediction the model makes for the long-term resource level.
(c) Repeat (b) with $R = 0.5$.

(d) Is this a reasonable model for an abiotic resource, if only for some circumstances? Explain your answer.

(This problem is continued in Problems 4.4.13 and 4.5.5.)

4.3.5 [Biotic resource]
The (scaled) model

$$\frac{dx}{dt} = rx - \frac{x}{1+x}, \qquad x, r \geq 0$$

is a possible model for the biomass of a biotic resource that grows exponentially but is consumed at a rate given by Holling type 2 dynamics. Follow the directions for Problem 4.3.4 with this model, using $r = 2$ and $r = 0.5$.

(This problem is continued in Problems 4.4.14 and 4.5.6.)

4.3.6 [Self-limiting population]
(This problem is continued from Problem 3.6.8.)
The dimensionless model

$$\begin{aligned} w' &= ap\ , \\ p' &= p(1-p-w) \end{aligned}$$

represents a self-limiting population, such as yeast in bread or beer, with p the population, w the concentration of toxic waste, and $a > 0$ a parameter that represents the rate of waste production.

(a) Define a new variable $z = p + w$. Obtain a differential equation for z that has the form $z' = pf(z)$.
(b) The equation for z in part (a) contains the unknown population $p(t)$; however, it can still be studied with phase line analysis. Explain why this statement is true.
(c) Use phase line analysis to determine a value that the sum $p + w$ cannot exceed.
(d) Note that w must increase as long as $p > 0$. Explain the consequence of this fact in conjunction with the result of the phase line analysis.
(e) The simple conclusion of this study is only as good as the model from which it was obtained. It serves to explain what happens to yeast in the rising of dough or the brewing of beer, but it obviously does not apply to fish in a pond. What important feature of the pond environment would need to be added to the model to make it realistic for fish?

(This problem is continued in Problem 6.1.5.)

4.3.7 [Resource harvesting]
The model

$$v' = v\left(1 - \frac{v}{k}\right) - \frac{cv}{1+v}, \qquad k, c > 0$$

represents the resource level of vegetation, where k is the amount of vegetation that the environment can carry and c is a measure of the number of consumers.

(a) Explain the difference in modeling assumptions between this model and the Schaefer model of Problem 4.3.3. The extra parameter k is not important; focus on the difference in the functions used for the negative terms. Refer to Sect. 3.2.
(b) Sketch a graph that shows the positive equilibrium points v^* as a function of c for the case $k = 8$. Note that there are ranges of c for which there are 0, 1, or 2 positive solutions, along with $v_0^* = 0$.

(Hint: Solving the equation $v' = 0$ is unnecessarily complicated. Instead, solve the equilibrium equation for c, calculate c for a range of solutions v^*, and then graph v^* vs c. Don't forget that one must always make a separate check to see if $v^* = 0$ is an equilibrium value.)

(c) Use phase line analysis to determine the stability of the equilibria for the case $k = 8$, $c = 1$. You do not need to determine the exact value(s) of the equilibria. (Note that your graph from (b) will give you the number and approximate values of the equilibria.)

(d) Repeat (c) for the cases $c = 2$ and $c = 3$.

(e) Suppose there is initially a small number of consumers ($c < 1$) and a large amount of vegetation. Explain what the model predicts will happen during the following sequence of events (assume that the vegetation does not fully disappear):

1. The number of consumers is increased to $c = 2$.
2. The number of consumers is further increased to $c = 3$.
3. The number of consumers is decreased back to $c = 2$.

(This problem is continued in Problems 4.4.15 and 4.5.9.)

4.3.8 [Optimal harvesting]
The term Ex in the differential equation of Problem 4.3.3 represents the yield rate of the fishery. If the fishery is managed, the goal might be to maximize the sustained yield $y = Ex^*(E)$, where $x^*(E)$ is the equilibrium population for any particular level of effort. Find the effort E^* that achieves the maximum sustainable yield and the corresponding maximum yield.

4.3.9 [Optimal harvesting]
Determine the maximum sustainable yield

$$y = \frac{cv^*}{1 + v^*}$$

and the corresponding vegetation level v^* and consumer level c for the resource consumption model

$$v' = v\left(1 - \frac{v}{k} - \frac{c}{1 + v}\right).$$

To do this, you need to be able to write y as a function of either c or v^* by using the equilibrium equation to replace one of those quantities.

4.3.10* Problems 4.3.8 and 4.3.9 are concerned with harvesting in which the goal is to produce the maximum sustainable yield. In some cases, it is more realistic to use a bioeconomic approach. Suppose there is a cost C associated with the harvesting effort. As an example, consider the Schaefer model of Problem 4.3.3 with cost function

$$C(E) = A\sqrt{E}$$

for some $A > 0$. It is reasonable to assume that the optimal effort will be the one that maximizes the revenue, which we can think of as the sustained yield Ex^* minus the cost.

(a) The goal is to maximize revenue $R(E) = Ex^*(E) - C(E)$, where $x^*(E)$ is the equilibrium solution of the Schaefer equation. Use the formula for the equilibrium to eliminate x^* from the formula for R.

(b) Obtain an equation for the critical points E^* of $R(E)$. Ideally, we would like to solve this equation for the optimal effort as a function of the parameter A. Instead, solve the equation for A and plot the result in the AE^*-plane.

(c) Explain how the optimal effort changes as the cost coefficient A increases. (In particular, what is the best harvesting strategy for A larger than some critical value?) How does this change the value of x^* corresponding to the maximum effort?

4.4 Linearized Stability Analysis

After studying this section, you should be able to:

- Use the derivative to determine the stability of fixed points of discrete dynamic variables.
- Use the derivative to determine the stability of equilibria of continuous dynamic variables.

In Sects. 4.2 and 4.3, we learned how to analyze first-order autonomous population models graphically, with cobweb plots for discrete models $N_{t+1} = g(N_t)$ and phase line analysis for continuous models $N'(t) = f(N)$. Now we consider a method that is based on calculation. Asymptotic stability is a *local* property, which means that it depends only on the properties of the model very close to the point of interest. Local properties can be analyzed using calculus.

4.4.1 Stability Analysis for Discrete Models: A Motivating Example

▶ **Example 4.4.1** Note that $N = 2$ is a fixed point for the model

$$N_{t+1} = \frac{2.5N_t^2}{1 + N_t^2}.$$

To analyze the stability of this fixed point, we need to know the local behavior of the function

$$g(N) = \frac{2.5N^2}{1 + N^2}$$

at the point $N = 2$. We have

$$g'(N) = \frac{(5N)(1 + N^2) - (2.5N^2)(2N)}{(1 + N^2)^2} = \frac{5N}{(1 + N^2)^2}.$$

Specifically, $g'(2) = 0.4$, so the linear approximation of g at $N = 2$ is

$$g(N) \approx g(2) + g'(2)(N - 2) = 2 + 0.4(N - 2).$$

In the vicinity of $N = 2$, the original model is therefore approximated by the linear model

$$N_{t+1} = 2 + 0.4(N_t - 2).$$

We can simplify this model by defining the population perturbation $x = N - 2$. Systematically replacing N by $2 + x$ yields the linearized model

$$x_{t+1} = 0.4x_t.$$

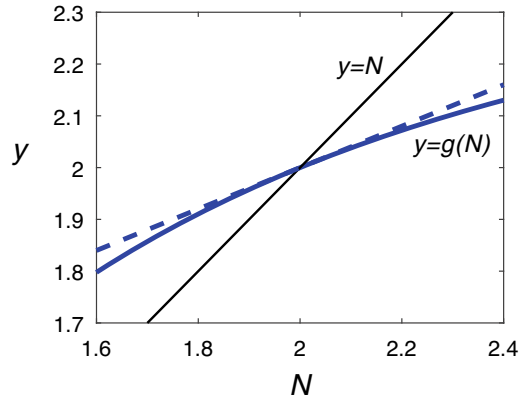


Fig. 4.4.1 $y = g(N)$ (*heavy*), $y = N$ (*medium*), and $y = 2 + 0.4(N - 2)$ (*dashed*) for $N_{t+1} = \dfrac{2.5N_t^2}{1 + N_t^2}$

This is the exponential model,[16] whose solution we can write as

$$x_t = 0.4^t x_0.$$

From this result, we can determine the long-term behavior of the model:

$$\lim_{t\to\infty} x_t = 0.$$

Since x_t is the approximate difference between N_t and 2, we also have

$$\lim_{t\to\infty} N_t = 2,$$

which marks the fixed point $N = 2$ as locally asymptotically stable. □

To see why the method of Example 4.4.1 works, consider the stability question using a cobweb plot. Figure 4.4.1 shows the basic elements of the cobweb plot, including both the actual curve $y = g(N)$ and the linear approximation $y = 2 + g'(2)(N - 2)$. As we zoom in on the fixed point, the curve and the dashed line come closer together, so the simulation lines in the cobweb plots for the linear and nonlinear models will be indistinguishable, provided we choose an initial value close enough to the fixed point. The advantage of using the linearized model is that we don't need to analyze it with the cobweb plot because it can be solved explicitly. In general, this is how analytical stability methods work. Instead of using a graphical method for the original nonlinear model, we replace the model with a linear approximation that can be analyzed with simple calculations.

4.4.2 Stability Analysis for Discrete Models: The General Case

The power of mathematics is in its ability to obtain general results from motivating examples. In Example 4.4.1, the conclusion of asymptotic stability for $N = 2$ derived from the solution formula $x_t = 0.4^t x_0$, which has a limit of 0 because $|0.4| < 1$. The number 0.4 originally came from the calculation of $g'(2)$, which is the slope of the linear approximation of $g(N)$ at 2. If we change the problem, the details will be different, but the general result will be the same. Given a fixed point N^* for the equation $N_{t+1} = g(N_t)$, the linearized model at that fixed point is

[16] Section 4.1.

$$x_{t+1} = g'(N^*)x_t, \qquad x = N - N^*.$$

Thus, the difference between N and N^* is approximately

$$x_t = [g'(N^*)]^t x_0.$$

This quantity vanishes as $t \to \infty$ whenever $|g'(N^*)| < 1$, which serves as a sufficient condition for asymptotic stability. Every example will work the same way, with only superficial differences owing to different g functions. Instead of repeating the full calculation every time, we can summarize the result in a theorem, which contains some additional detail.

Theorem 4.4.1 (Stability of Discrete Dynamic Variables)
Let N^* be a fixed point for the sequence defined by $N_{t+1} = g(N_t)$, where g is a differentiable function.

1. The fixed point N^* is asymptotically stable if $|g'(N^*)| < 1$ and unstable if $|g'(N^*)| > 1$;
2. $N_t - N^*$ alternates in sign whenever $g'(N^*) < 0$ and retains a uniform sign whenever $g'(N^*) > 0$.

Theorem 4.4.1 allows for a very efficient determination of stability, as long as $g'(N^*) \neq \pm 1$. With a combination of linearized stability analysis and cobweb plots, we can often obtain a complete picture of long-term behavior for a discrete model with an arbitrary initial point.

▶ **Example 4.4.2** Consider again the model

$$N_{t+1} = \frac{2.5N_t^2}{1 + N_t^2}.$$

The fixed points are the solutions of

$$N = \frac{2.5N^2}{1 + N^2};$$

hence, either $N = 0$ or $1 = 2.5N/(1 + N^2)$. The latter simplifies to the quadratic equation $N^2 - 2.5N + 1 = 0$, which yields the solutions $N = 0.5$ and $N = 2$. The fixed points are thus $N_0^* = 0$, $N_1^* = 0.5$, and $N_2^* = 2$. To determine the stability of each fixed point, we first need to determine g'. From Example 4.4.1, we have

$$g'(N) = \frac{5N}{(1 + N^2)^2}.$$

Thus,

$$g'(0) = 0\,, \quad g'(0.5) = 1.6\,, \quad g'(2) = 0.4\,.$$

By Theorem 4.4.1, 0 and 2 are asymptotically stable and 0.5 is unstable. Note that questions about global stability cannot be answered with linearized stability analysis, which only shows what happens if the initial condition is arbitrarily close to the fixed point. Combined with the cobweb plot of Fig. 4.2.2, we can conclude that the population always approaches 2 if $N_0 > 0.5$ and always approaches 0 if $N_0 < 0.5$. This is an example of a model in which extinction can occur when the population is below some threshold value, a phenomenon biologists call the *Allee effect*. □

So far, we have used the linearization technique only for a model with fixed parameters. It can also be applied without fixing the values of parameters.

▶ **Example 4.4.3** Consider the discrete logistic model

$$N_{t+1} = N_t + rN_t\left(1 - \frac{N_t}{K}\right), \qquad r > -1\,.$$

The fixed points are $N_0^* = 0$ and $N_1^* = K$. We have

$$g(N) = N + rN - \frac{rN^2}{K};$$

thus,

$$g'(N) = 1 + r - \frac{2rN}{K}\,.$$

Hence, $g'(0) = 1 + r > 0$. The stability requirement for $N_0^* = 0$ is $-1 < 1 + r < 1$. The first inequality is always true, so the fixed point is asymptotically stable when $r < 0$ and unstable when $r > 0$. The sign of $g'(0)$ means that solutions do not alternate in sign. □

Check Your Understanding 4.4.1:
Apply Theorem 4.4.1 to the fixed point $N_1^* = K$ for Example 4.4.3.

Note that local stability analysis, while easy to perform, leaves some questions about the discrete logistic equation unanswered.

1. Are the fixed points globally asymptotically stable when they are locally stable?
2. What happens when $r > 2$ (since there are no stable fixed points)?

The first of these questions can be addressed with cobweb analysis. As we've seen, graphical analysis is global, rather than local, and examination of cobweb plots answers the first question in the affirmative. The second question has already been answered in part through cobweb analysis, as we have seen a stable 2-cycles in Example 4.2.2 and can also find stable 4-cycles, 8-cycles, and so on, chaotic behavior, and even oddities such as a stable 3-cycle. The local analysis method is adapted to determine the stability of 2-cycles in Project 4D.

4.4.3 Stability Analysis for Continuous Models

Stability analysis for continuous models is also based on examination of a linearized model. The idea is exactly the same as for discrete models, but with a different critical condition.

Let y^* be an equilibrium point for the problem $dy/dt = f(y)$. As with the discrete case, we can replace f near $y = y^*$ by a linear approximation

$$f(y) \approx f(y^*) + f'(y^*)(y - y^*) = f'(y^*)(y - y^*)\,,$$

where we have used the fact that $f(y^*) = 0$ because y^* is an equilibrium point. Thus, we have

$$\frac{dy}{dt} \approx f'(y^*)(y - y^*).$$

With $x(t) = y(t) - y^*$, we have the linearized model

$$\frac{dx}{dt} = f'(y^*)x.$$

This is the exponential growth equation, and it has the solution

$$x = x(0)e^{f'(y^*)t}.$$

Thus, $x \to 0$ whenever $f'(y^*) < 0$. This gives us a theorem analogous to Theorem 4.4.1.

Theorem 4.4.2 (Stability of Continuous Dynamic Variables)
Let y^* be an equilibrium point for the differential equation $y' = f(y)$, where f is a differentiable function. Then the equilibrium solution $y = y^*$ is asymptotically stable if $f'(y^*) < 0$ and unstable if $f'(y^*) > 0$. Solutions are never oscillatory.

▶ **Example 4.4.4** The model

$$y' = f(y) = ry\left(1 - \frac{y}{K}\right)$$

has equilibrium points $y_0^* = 0$ and $y_1^* = K$. We have

$$f' = r\left(1 - \frac{y}{K}\right) + ry\left(-\frac{1}{K}\right) = r\left(1 - \frac{y}{K}\right) - r\left(\frac{y}{K}\right).$$

Thus, $f'(0) = r > 0$ and $f'(K) = -r < 0$. By Theorem 4.4.2, 0 is unstable and K is asymptotically stable. Note that we simplified the analysis by not multiplying out products or combining terms unless necessary. Using the product rule without further simplification gave us a derivative formula in which each term was 0 at one of the equilibria. □

4.4.4 Comparison of Discrete and Continuous Dynamics

The standard forms for discrete and continuous models are $N_{t+1} = g(N_t)$ and $N' = f(N)$. The functions g and f do not represent corresponding quantities; one is the new population, while the other is the rate of change. This means that discrete results stated in terms of g and continuous results stated in terms of f look considerably different. If we want to understand the similarities and differences between discrete and continuous models, we must first rewrite the standard form for discrete models to focus on the rate of change. As discussed earlier,[17] the rate of change in a discrete model is the same as the difference between consecutive values; this corresponds to a general form

$$N_{t+1} - N_t = F(N_t), \tag{4.4.1}$$

for some function F. In standard form, this would be

$$N_{t+1} = N_t + F(N_t).$$

[17] Section 4.1.

Thus, the function F that represents the relative rate of change in a discrete model is related to the function g that represents the new population value by the equation

$$g(N) = N + F(N).$$

This identification allows us to recast Theorem 4.4.1 in terms of the rate of change.[18]

Theorem 4.4.3 (Stability of Discrete Dynamic Variables)
A fixed point N^* for the discrete model $N_{t+1} - N_t = F(N_t)$, with F a differentiable function, is asymptotically stable if and only if $-2 < F'(N^*) < 0$.

Compare Theorems 4.4.2 for the continuous case and 4.4.3 for the discrete case. The requirements $f'(y^*) < 0$ and $F'(N^*) < 0$ are equivalent. This requirement is enough for the continuous case, but not for the discrete case. While y^* is stable in the continuous case no matter how negative $f'(y^*)$ is, stability is lost in two stages in the discrete case if $F'(N^*)$ is too negative, that is, the solution becomes oscillatory when $F'(N^*)$ is less than -1 and then unstable when it is less than -2. This difference in model behaviors has important consequences for biology, modeling, and analysis:

1. Populations whose dynamics are governed by synchronous processes, such as fish that reproduce in one short period out of the year, are more likely to be unstable than those whose dynamics are governed by asynchronous processes.
2. Using a discrete model when all processes are asynchronous can lead to fundamental qualitative errors in results.
3. When using a discretization method to simulate a continuous model, one must be careful that the discretization does not introduce behavior that is not inherent in the model itself. Such errors can usually be fixed by using smaller time intervals.

Generally, we should prefer discrete models when similar events happen simultaneously, such as in populations that have a specific season for births. Continuous models are preferable when similar events are spread over time and occur simultaneously with other events. In most disease situations, for example, some members of the population are just getting sick while others are recovering, and the sick individuals are at different stages of the disease; hence, we should use continuous models.

Compared to linearized stability analysis, graphical analysis has the advantage of being global rather than local. This difference makes the phase line method preferable to linearized stability analysis for continuous dynamics. The same cannot be said for cobweb analysis. The difference is that the phase line method can be applied with arbitrary parameters, while cobweb analysis requires parameter values to be chosen. Study of discrete dynamics should combine graphical and analytical methods, while study of continuous dynamics can be done with graphical methods alone. We will see in Chap. 6 that this latter conclusion does not apply to systems of more than one dynamic variable.

Check Your Understanding Answers

1. We have $g'(K) = 1 - r$, so the stability requirement for $N_1^* = K$ is $-1 < 1 - r < 1$. Multiplying by -1 and reversing the inequalities yield $1 > r - 1 > -1$, which reduces to $0 < r < 2$. The fixed point is unstable if $r < 0$ or $r > 2$. When the fixed point is stable, the approach will be monotone when $r < 1$ and oscillatory when $r > 1$.

[18] The derivation of this result is in Problem 4.4.16.

Problems

4.4.1 Find the fixed points and their stability for the model

$$N_t = \frac{5.2N_t^2}{1 + N_t^2}.$$

Compare with Example 4.4.2.
(This problem is continued in Problems 4.4.5 and 4.4.6.)

4.4.2* [Ricker model] (Continued from Problems 4.1.5 and 4.2.5.)

(a) Determine the stability of the fixed points for the Ricker model

$$N_{t+1} = 8N_t e^{-N_t}.$$

(b) Discuss the results with reference to Problems 4.1.5 and 4.2.5.

(This problem is continued in Problem 4.4.9.)

4.4.3 [Hassell insect model] (Continued from Problems 4.1.6 and 4.2.6.)

(a) Determine the stability of the fixed points for the Hassell model

$$N_{t+1} = \frac{64N}{(1 + N_t)^3}.$$

(b) Discuss the results with reference to Problems 4.1.6 and 4.2.6.

(This problem is continued in Problem 4.4.10.)

4.4.4 [Beverton–Holt model] (Continued from Problems 4.1.4 and 4.2.4.)

(a) Determine the stability of the fixed points for the Beverton–Holt model

$$N_{t+1} = \left(0.5 + \frac{0.8}{1 + N_t}\right) N_t.$$

(b) Discuss the results with reference to Problems 4.1.4 and 4.2.4.

(This problem is continued in Problem 4.4.8.)

4.4.5 Show that the fixed point $N_0^* = 0$ is locally asymptotically stable for the model

$$N_{t+1} = \frac{RN_t^2}{1 + N_t^2}$$

for any finite value of R, no matter how large.

4.4.6 (This problem generalizes Example 4.4.2 and Problem 4.4.1.)
Consider the model

$$N_t = \frac{RN_t^2}{1 + N_t^2}\,.$$

(a) Find the formulas for the two positive fixed points as functions of R. Note that there is a minimum value of R necessary for these fixed points to exist.
(b) Use Theorem 4.4.1 to show that positive fixed points are asymptotically stable if and only if

$$N^* > \frac{2}{R}.$$

[Hint: Use the fixed point equation to simplify the formula for $g'(N^*)$.]
(c) Use the results from (a) and (b) to explain why the larger of the two fixed points is always stable. [This can be done with almost no calculations by finding a constant that is guaranteed to be smaller than N^* and yet at least as large as $2/R$.]
(d) Show that the smaller of the two fixed points is never stable. [The easiest way to do this is to assume that the smaller point satisfies the requirement of (b) and then do algebraic simplification until you find a contradiction.]

4.4.7* [Discrete resource consumption] (Continued from Problem 4.2.3.)

(a) Find the fixed points for the growth/consumption model

$$N_{t+1} = N_t + 2.2N_t\left(1 - \frac{N_t}{1000}\right) - \frac{100N_t}{200 + N_t}\,.$$

[Keep in mind that only meaningful fixed points count.]
(b) Use Theorem 4.4.3 to determine the stability of the fixed points.
(c) Discuss the results in conjunction with the results of Problem 4.2.3.

4.4.8* [Beverton–Holt model] (Continued from Problems 4.1.4, 4.2.4, and 4.4.4.)

(a) Determine the stability of the fixed points for the Beverton–Holt model

$$N_{t+1} = \left(S + \frac{A}{1 + N_t}\right)N_t\,, \quad A > 0\,;\ \ 0 \le S < 1\,.$$

(b) Discuss the results with reference to Problems 4.1.4, 4.2.4, and 4.4.4.

4.4.9 [Ricker model] (Continued from Problems 4.1.5, 4.2.5, and 4.4.2.)

(a) Determine the stability of the fixed points for the Ricker model

$$N_{t+1} = \left[S + Ae^{-N_t}\right]N_t\,.$$

(b) Discuss the results with reference to Problems 4.1.5, 4.2.5, and 4.4.2.

4.4.10 [Hassell insect model] (Continued from Problems 4.1.6, 4.2.6, and 4.4.3.)

(a) Determine the stability of the fixed points for the Hassell model

$$N_{t+1} = \left(\frac{A}{1+N_t}\right)^b N_t, \qquad A, b > 0,$$

for the cases $b \leq 1$, $b = 2$, and $b = 3$.
(b) Determine whether the stable solutions found in part (a) oscillate as they approach the fixed point.
(c) Discuss the results with reference to Problems 4.1.6, 4.2.6, and 4.4.3.

4.4.11* (Continued from Problem 4.3.2.)
Use Theorem 4.4.3 to determine the stability of the equilibria for the model

$$\frac{dN}{dt} = -N\left(1-\frac{N}{T}\right)\left(1-\frac{N}{K}\right), \qquad 0 < T < K.$$

Discuss the results with reference to Problem 4.3.2.

4.4.12 [Resource harvesting] (Continued from Problem 4.3.3.)
Determine the stability of all equilibria for the Schaefer model

$$x' = x(1-x) - Ex, \qquad E > 0.$$

Discuss the results with reference to Problem 4.3.3.

4.4.13 [Abiotic resource] (Continued from Problem 4.3.4.)
Determine the stability of all equilibria for the proposed abiotic resource model

$$x' = R - \frac{x}{1+x}.$$

Discuss the results with reference to Problem 4.3.4.
(This problem is continued in Problem 4.5.5.)

4.4.14* [Biotic resource] (Continued from Problem 4.3.5.)
Determine the stability of all equilibria for the proposed biotic resource model

$$x' = rx - \frac{x}{1+x}.$$

Discuss the results with reference to Problem 4.3.5.
(This problem is continued in Problem 4.5.6.)

4.4.15* [Resource harvesting] (Continued from Problem 4.3.7.)
Determine the stability of all equilibria for the Holling type II resource consumption model

$$v' = v\left(1 - \frac{v}{k} - \frac{c}{1+v}\right), \qquad k = 8, \quad c > 0.$$

Discuss the results with reference to Problem 4.3.7.
(This problem is continued in Problem 4.5.9.)

4.4.16 Derive the result of Theorem 4.4.3 by defining the appropriate function g for the model $N_{t+1} - N_t = F(N_t)$ and applying Theorem 4.4.1.

4.4.17 Consider a model for a population that runs over a 2-year cycle:

$$Y_{t+1} = g(A_t), \qquad A_{t+1} = f(Y_t),$$

where

$$f(0) = g(0) = 0, \qquad f', g' > 0, \qquad f'', g'' < 0, \qquad f(Y) < Y, \qquad g(A) < g_m.$$

(a) Sketch graphs of f and g.
(b) Derive a model that tracks adults only with a 2-year census interval.
(c) Determine the restriction necessary on the functions f and g for the population to persist (in other words, for the fixed point with $A^* = 0$ to be unstable).

4.5 Case Study: A Mathematical Model of Resource Conservation

Some of the exercises in Sects. 4.3 and 4.4 introduced a variety of possible models for renewable resource harvesting. In this section, we consider a systematic approach to creation of renewable resource models and then focus on one that is capable of predicting a variety of behaviors.[19]

Biotic resource levels change according to two processes:

1. A growth process works to bring the resource to the carrying capacity of the environment in the absence of consumption.
2. A harvesting process depresses the resource level.

The simplest renewable resource models follow from the assumptions that the number of consumers is constant and that the amount of resource consumption by each individual does not depend on the total number of consumers. These assumptions yield the generic model

$$\frac{dX}{dT} = G(X) - CH(X), \tag{4.5.1}$$

where $X(T)$ is the biomass of the resource,[20] C is the number of consumers, $G(X)$ is a function that represents the growth process, and $H(X)$ is the harvest rate per consumer.

We assume that there is no growth when the resource level is 0 but that there is positive growth for resource levels sufficiently small; thus, $G(0) = 0$ and $G'(0) > 0$. The harvesting rate should be an increasing function of resource level, but 0 if there is no resource; thus, $H(0) = 0$ and $H' \geq 0$.

4.5.1 Growth and Harvesting Functions

The simplest and most common example of a consumer-resource model is the Lotka–Volterra model, obtained by choosing linear functions $G(X) = RX$ and $H(X) = SX$. This widely used model is actually a very poor choice for predator–prey or consumer–resource interactions.[21] The proposed

[19] This section is adapted from [7].

[20] The usual lowercase t has been replaced by the uppercase T to allow for systematic use of upper and lower cases in nondimensionalization.

[21] Problem 1.4.1.

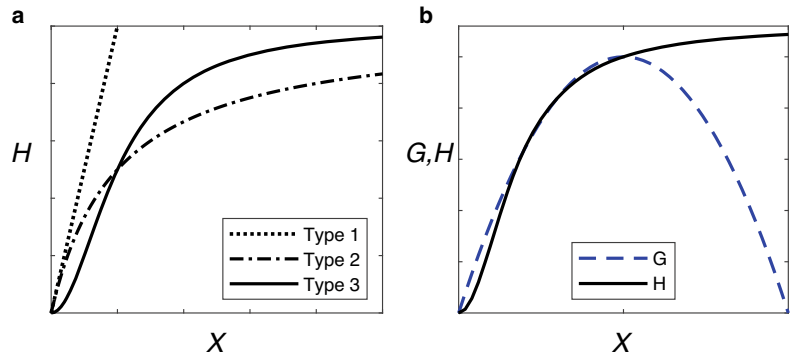


Fig. 4.5.1 **a** The Holling functions; **b** A comparison of the logistic growth and Holling type 3 harvesting functions

biotic resource model from Problem 4.3.5 with Holling type 2 harvesting and linear growth turned out to be even worse. The failures in both cases are because the linear growth model produces unrealistic results in the absence of harvesting.

Instead, the natural choice for a growth function is

$$G(X) = RX\left(1 - \frac{X}{K}\right), \tag{4.5.2}$$

which prescribes logistic growth with maximum rate R and carrying capacity K. Our resource will then exhibit logistic growth in the absence of consumers.

A variety of choices are possible for the harvesting function $H(X)$, including the three Holling functions,[22] presented in Fig. 4.5.1a. Type 1 is used in the Schaefer model.[23] This model is used sometimes, but it is a little too simple because it fails to account for the significant effect of harvesting time. Type 2 is best for settings in which the consumer has no alternative resources, while type 3 is best for settings where there are alternatives. The latter is common; for example, ranchers can feed their cattle something other than the native grass and whaling crews can alter their plans to hunt different animals. With whale populations in mind for this study, we choose the type 3 function,

$$H(X) = \frac{SX^2}{K + BSX^2}, \tag{4.5.3}$$

where S is the maximum search speed, K is the carrying capacity from the growth function, and B is a measure of the time required for harvesting one unit of resource.

Figure 4.5.1b compares the growth and type 3 harvesting functions. With the given parameter values, the G and H curves cross at three points, resulting in different system behaviors for different ranges of X.

4.5.2 Scaling

Combining (4.5.1)–(4.5.3) yields the dynamic equation

$$\frac{dX}{dT} = RX\left(1 - \frac{X}{K}\right) - C\frac{SX^2}{K + BSX^2}. \tag{4.5.4}$$

[22] Section 3.2.

[23] Problems 4.3.3 and 4.3.8.

This equation has five parameters (although it could be thought as just 4 parameters by using CS and BS instead of C, B, and S). This is not an issue if our interest is limited to studying the model with one or two sets of known parameter values, but it is far from ideal if our goal is to characterize the full range of model behaviors. We can address this issue by scaling the model.[24]

The choice of scales for a model is often tricky, but this is not the case here. We are thinking of scenarios in which the consumption parameters C, S, and B change gradually over time, so it is best to use scales that come from the growth term rather than the harvesting term. This means we should choose K for the reference resource level and $1/R$ for the reference time. Thus, we factor the variables X and T in terms of scale factors K and $1/R$ and dimensionless variables x and t as

$$X = Kx\,, \qquad \frac{d}{dT} = R\frac{d}{dt}\,. \tag{4.5.5}$$

Making these substitutions and rearranging factors yield

$$x' = x\,(1-x) - \frac{CS}{R}\,\frac{x^2}{1+BSKx^2}\,,$$

with the prime symbol indicating the derivative with respect to dimensionless time. The five-dimensional parameters have naturally sorted themselves into two dimensionless groupings, allowing us to rewrite the model in terms of two dimensionless parameters. There are multiple ways to do this [6], and the best choice is often clear only after doing the analysis. Our choice here is to factor BSK out of the denominator to get

$$x' = x\,(1-x) - \frac{cx^2}{p+x^2}\,, \qquad p = \frac{1}{BSK}\,, \quad c = \frac{C}{BRK}\,. \tag{4.5.6}$$

It is helpful to examine the specific dimensionless parameters to identify a biological interpretation. The quantity $1/B$ is the rate at which harvested resource can be processed, while the quantity SK is the rate of resource discovery when the resource is at its carrying capacity; thus, the parameter p can be interpreted as the "processing-to-discovery" ratio. We will focus on small values of this ratio, corresponding to large natural populations. The maximum growth rate of the resource is $RK/2$, which is enough production to fully support $BRK/2$ units of consumers, given that each consumer requires B units of time to process one unit of resource. A value $c = 0.5$ therefore means that the maximum productivity is just adequate to support the number of consumers present. Clearly values of c at or above 0.5 will deplete the resource. Of course the actual results will be more subtle than this, but it is helpful to have some sense of what to expect prior to doing the analysis.

4.5.3 Plan for Analysis

Our model has two parameters, one that represents the consumption effort (c) and one that combines the biological characteristics of the growth and hunting processes (p). Suppose we want to understand how resources become overexploited and then recover through conservation. This requires us to look at changes in parameter values over the course of historical time. In general, technological change means that search speed increases while handling time decreases. These trends tend to result in less long-term change in the product SB that appears in the definition of p than the change we should expect for c. It is therefore not unreasonable to make the assumption that values of p are largely unchanged over time, but of course the value will be dependent on the resource. We therefore ask the following question:

[24] Section 3.6.

- For any given value of p, how does the pattern of stable and unstable equilibria depend on the parameter c?

For now, we assume a specific value $p = 0.01$. This value is chosen in hindsight because it is in the range where the most interesting behavior occurs. We'll address the influence of p after a full analysis of the $p = 0.01$ case.

There are four types of analysis that we could consider doing with a continuous dynamic equation: finding an analytical solution, using the phase line to determine equilibria and stability, using linearized stability analysis to determine stability, and running simulations. Simulations can be run using any numerical differential equation solver, but one can only do examples rather than general cases. Analytical solutions have little advantage, as it is usually harder to determine solution behavior from an analytical solution than from the original model and harder to calculate numerical values from an analytical solution than from a simulation. Linearized stability analysis is not as good as phase line analysis, both because it is more work[25] and because phase line analysis yields global as well as local stability. These considerations suggest we focus on phase line analysis.

4.5.4 A Structured Approach to Phase Line Analysis

The naive way to do phase line analysis for (4.5.6) is to use a graph of x' versus x to determine when x is increasing, as in Algorithm 4.3.1. However, the graph of the function

$$x\,(1-x) - \frac{cx^2}{p+x^2}$$

depends on the parameters p and c in complicated ways. Even with a single fixed value of p, the analysis will be entirely procedural because we can only obtain the graph for specific values of c.

As an alternative to the naive approach, we can use an approach that simplifies the analysis by imposing a structure on the function in the differential equation. The plan is to write the formula for x' using the structure

$$x' = w(x)[g(x) - h(x)], \qquad w(0) = 0, \quad w' > 0, \tag{4.5.7}$$

where wg is the growth rate and wh is the harvesting rate. These requirements do not yield a unique factorization; for example, we could choose

$$x' = x\left[(1-x) - \frac{cx}{p+x^2}\right] \tag{4.5.8}$$

or

$$x' = \frac{x}{p+x^2}\left[(1-x)(p+x^2) - cx\right]. \tag{4.5.9}$$

Before choosing which factorization to use, it helps to understand the properties of a dynamic equation in the factored form of (4.5.7).

Theorem 4.5.1 (Properties of the Equation $x' = w(g-h)$)
For any factorization

[25] This is assuming that the phase line analysis is done using the method presented in this section.

$$x' = w(x)[g(x) - h(x)]\,, \qquad w(0) = 0\,, \quad w' > 0:$$

1. $x = 0$ is an equilibrium point.
2. All equilibria other than $x = 0$ are points where the graphs of g and h intersect.
3. The state variable x is increasing whenever the graph of g is above the graph of h and decreasing whenever the graph of g is below that of h.

Check Your Understanding 4.5.1:
Explain the three points in Theorem 4.5.1.

If possible, the choice of factoring should be made so that the graphs of g and h can be sketched by hand. Neither of the options works for both g and h for either function, but option (4.5.9) is more attractive because our principal parameter c appears only in the simpler function. For any fixed p, the nonlinear function g needs to be graphed just once and we can superimpose multiple plots of the linear function h.

Figure 4.5.2a shows the graph of the nonlinear function $g(x) = (1 - x)(p + x^2)$ from (4.5.9), with $p = 0.01$, along with lines $h(x) = cx$ of three different slopes. While the graphs of g and h are not the growth and harvesting functions, they nevertheless *represent* these functions in a relative sense, that being given by conclusion 3 of the theorem.

When $c = 0.12$ (panel **b**), there is one positive equilibrium, at a value not much less than the environmental capacity 1. This equilibrium is identified by the intersection of the curve and the line and marked as a disk on the x-axis. The point $x = 0$ is also marked as an equilibrium because this is built in to the structure (4.5.7). For populations between these two equilibrium values, the curve g is above the line h. This means that growth outstrips consumption and the population increases, as marked by the arrow pointing to the right. To the right of the positive equilibrium, the graphs are reversed, so the population decreases. The arrows show that the positive equilibrium is globally asymptotically stable, while the extinction equilibrium is unstable. Given $p = 0.01$, the specific case $c = 0.12$ is representative of a range of "small" values of c. Consumption is relatively low and the stable equilibrium population is relatively high.

When $c = 0.36$, as seen in panel **d**, the situation is similar, except that the consumption level is sufficiently large that the single positive equilibrium is at a very small value of the resource. It is still true that the curve is above the line for values of x between the two equilibria and below the line when x is to the right of the positive equilibrium. As before, the positive equilibrium is globally asymptotically stable and the extinction equilibrium is unstable. The difference is that a moderate value, like $x = 0.4$, is now in the "large x" region where the curve is below the line. The number of consumers is high, but the actual amount of consumption is not, since the consumption is given as wh and both w and h are small.

Panel **c** shows the more interesting case of an intermediate value of c. Here there are three positive equilibria, which partition the x-axis into four regions. In the first and third of these, the curve is above the line, so growth exceeds consumption and the population increases. Similarly, the population decreases in the second and fourth regions because the curve is below the line. The result of this analysis is that the largest and smallest of the positive equilibria are locally asymptotically stable, while the middle one and the extinction equilibrium are unstable. The positive unstable equilibrium has special significance, as illustrated by the arrows: it marks the boundary between the domains of attraction of the two stable equilibria. Thus, the ultimate fate of the system for moderate c depends on whether it starts above or below this critical value. This feature will play an important role in the application

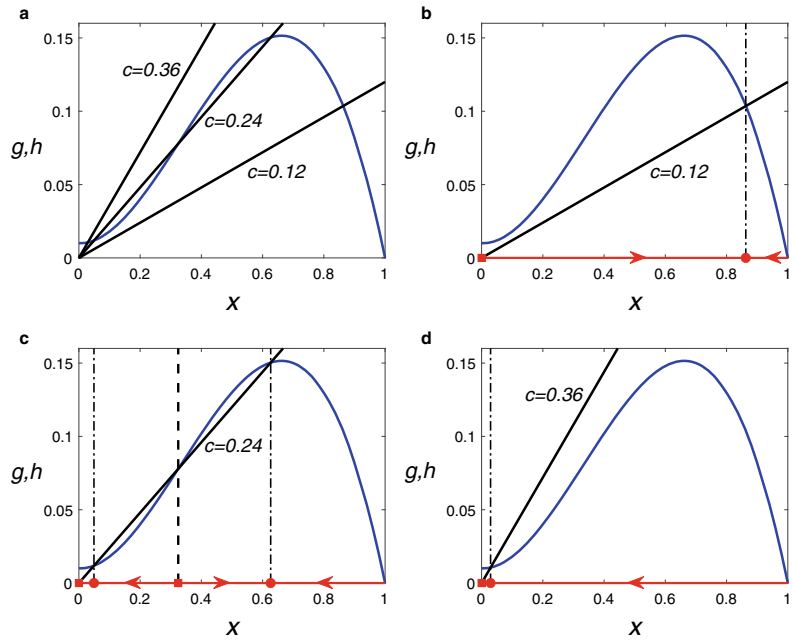


Fig. 4.5.2 Plots of g and h from (4.5.9) with $p = 0.01$. **a**: Three cases for c, showing a single large positive equilibrium for $c = 0.12$, three positive equilibria for $c = 0.24$, and a single small positive equilibrium for $c = 0.36$. The other panels show the cases individually, with the phase line drawn on the x-axis, using disks for stable equilibria, squares for unstable equilibria, and arrows showing the direction of change, which is always to the right when the curve (g) is above the line (h) and to the left otherwise. The unstable equilibrium at $x = 0$ is where $w = 0$

of these results to the history of resources that became depleted very quickly and have made only a modest recovery, such as some species of whales.

4.5.5 A Reconstructed History of Whale Populations

Figure 4.5.3 shows a possible history of a resource that has suffered a sudden depletion and been difficult to restore, such as many populations of whales. All phases assume $p = 0.01$; hence, the g curve is unchanged. The parameter c represents the capacity of humans to harvest whales rather than just the number of consumers. Thus, it increases naturally if unregulated, through a combination of human population growth and technological change. We assume that changes in c occur slowly enough that the system is able to continually adjust to the new stable equilibrium value, provided that equilibrium depends continuously on c. (This is not entirely realistic, as technological change can be sudden.) The history assumes that c increases naturally, represented by a steepening of the straight line, until some point at which it becomes controlled by public policy. Changes in population occur on two different time scales: both the increase in c and natural population growth occur on a scale of tens of years, while the depletion from unsustainable consumption can be much faster.

Phase 1: Depletion

Prior to the introduction of advanced technology, humans functioned as natural predators of whales. A small value of c, as indicated by the dotted line in the phase 1 plot, resulted in an equilibrium population only slightly below the environmental carrying capacity. Over a long period of time, the

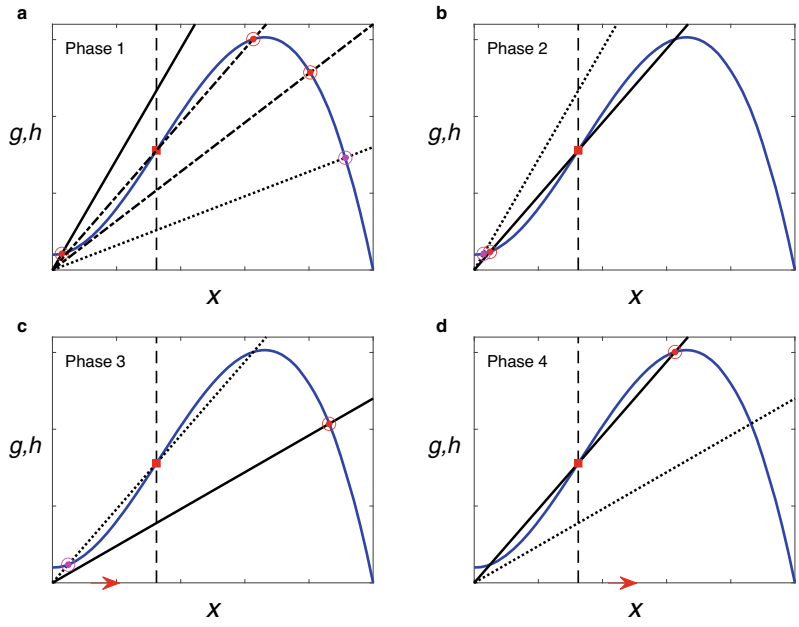


Fig. 4.5.3 A reconstructed history of whale resource levels using $p = 0.01$. Each phase begins with a value of c corresponding to the dotted line and proceeds to the value of c corresponding to the solid line, with any intermediate values shown as dash-dot. The bull's-eye markers show the corresponding stable equilibria and the square markers show unstable positive equilibria. The arrows on the x-axis in phases 3 and 4 indicate the direction of change toward a stable equilibrium

value of c rose steadily, gradually decreasing the equilibrium x, but slowly enough as to be unnoticeable over a human lifespan. Eventually, the value of c rose above the critical value that separates cases 1 and 2. At that point, seen in the second dash-dot line, there would have been two stable equilibria. The existence of a stable low-level equilibrium would have had no effect on the population, as the initial state for each successive consumption level would have been on the positive side of the unstable equilibrium. Only when c exceeded the critical value that separates cases 2 and 3 would there have been a drastic change. Once that critical value was exceeded, the high-level stable equilibrium disappeared, leaving only the heretofore unobserved low-level equilibrium. At that point, the initial condition was far from the final equilibrium value, so the decrease in population would have occurred on a time scale corresponding to the harvesting, whereas previously the population changes occurred on the much slower time scale corresponding to consumption increase. There is no question that some populations of whales, among other resources, have experienced a population crash at some point without a large contemporary change in consumption. This is different from instances where the consumption rate changed suddenly, such as the depletion of British forests during the rise of industry.

Phase 2: Inadequate Correction

Once a whale population became depleted, hunting them stopped being commercially viable. This would have caused a significant decrease in whale hunting efforts, which would have decreased the consumption capacity. Between that and the initiation of whale conservation efforts, c might perhaps have been lowered to the solid curve in the plot. The system is back in case 2; unfortunately, this time the initial condition is at the prior low-level equilibrium, so it is the high-level equilibrium that is unachievable. As with phase 1, the phase 2 pattern is well documented for whales and other resources. This is what seems to have happened after the beginning of conservation in 1961. It was difficult to

sustain politically because the payoff was so low. Some stocks of whales are undoubtedly still in this phase [10].

Phase 3: Strict Conservation and Restoration

As the environmental movement grew, international policy shifted toward further decrease in consumption, helped along by environmental activists such as Greenpeace. In phase 3, the decrease in c continues until the system is back in case 1. At that point, the low-level equilibrium is lost and the system begins to move toward the high-level equilibrium. Here again, the improvement in some whale populations is well documented and explainable using our simple model. Unfortunately, the time scale for restoration is much slower than the time scale for the depletion that occurred at the end of phase 1. In the depletion event, consumption outpaced population growth and drove the population decrease. In the restoration phase, the population increase occurs on the slower time scale of natural population growth.

Phase 4: Restoration and Sustainable Management

With a harvesting level corresponding to the dotted line in the phase 4 plot (which is the same as the solid line in the phase 3 plot), the whale population is recovering. If we maintain strict conservation for a while and then return to case 2, the outcome depends on whether the amount of restoration has raised the population above the unstable equilibrium value for the case 2 consumption rate. If so, as represented by the arrow on the x-axis, we will have moved the population into the domain of attraction for the high-level equilibrium and the population will continue to grow in spite of an increase in consumption capacity. If not, we will still be in the domain of attraction for the low-level equilibrium and the population will begin to decline again. Of course the description of our current situation using the phase 3 and phase 4 plots is qualitative at best. We don't know the parameter values and there are flaws in the model that, while not serious enough to falsify our narrative, are serious enough to make prediction uncertain. The prudent approach would be to maintain strict conservation until the population recovery rate is large and then increase consumption capacity only gradually. By monitoring the whale population and insisting on further growth, we can make sure that we have entered case 2 on the correct side of the unstable equilibrium. The point, though, is that it is not necessary to maintain strict conservation forever. A moderate consumption level that was not small enough for recovery in phase 3 is small enough for sustainability in phase 4. Based on recorded population data, we are clearly in phase 4 for some whale stocks, such as the west South Atlantic humpback and New Zealand right whales [11].

4.5.6 Bifurcation Analysis

In our analysis of the model, we saw that the sensitivity of the equilibrium solutions to the parameter c is small at times and large at others, sometimes so large as to be discontinuous. The dependence of a solution on parameter values is of both mathematical and biological interest. It can be seen in a 1-parameter system by plotting a curve of the equilibrium value against the parameter. In a 2-parameter system, we can plot multiple such curves, using several different values for the second parameter. Figure 4.5.4a shows such a plot for the model (4.5.6). The consumption parameter c is taken as the independent variable on the graph, while the process-to-discovery ratio parameter p is set at multiple values. The $p = 0.01$ case we have been studying is the second curve from the left and shows that case 2 requires c values roughly in the range 0.18–0.27. The bull's-eye marker indicates what we might call a critical point, as it marks the critical value p^* that is the boundary between smaller p values, for which bifurcation occurs at some values of c, and larger p values that show no bifurcation in c. The plot shows that the corresponding value c^* also marks the largest value of c among bifurcation points.

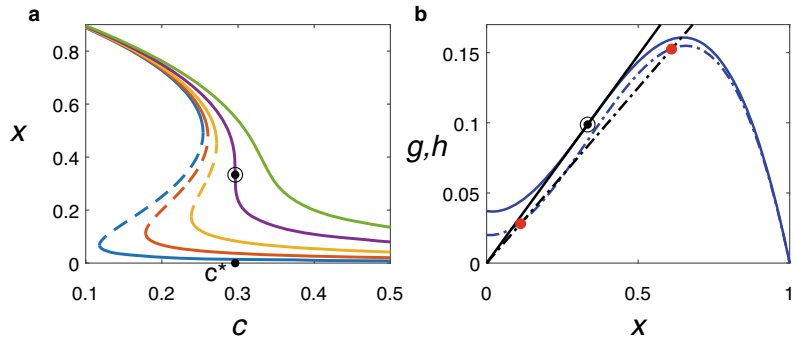


Fig. 4.5.4 **a** Equilibrium resource levels as a function of the consumer parameter c, with $p = 0.004$, 0.01, 0.02, 1/27, and 0.06, from left to right. Equilibria on the solid curves are stable, while those on the dashed curves are unstable. The critical value $p^* = 1/27$ is that for which the bifurcation curve has a vertical tangent, at the point with a bull's-eye marker. **b** Multiple positive equilibria are possible if and only if $p < p^*$ and c is in a narrow range below c^*. The solid curve and solid line are for p^* and c^*. The dash-dot curve shows a smaller value of p and the dash-dot line shows a value $c < c^*$ corresponding to case 2, with disks marking the stable equilibria

The critical parameter values can be determined analytically using a combination of calculus and algebra. The key to doing this is to identify the properties of the critical case on a plot of g and h from (4.5.7, 4.5.9), shown in Fig. 4.5.4b. The solid line that marks the critical case is tangent to the graph of g at the inflection point at the bull's-eye marker. The dash-dot curve in the plot shows that multiple equilibria occur for $p < p^*$, provided c is smaller than c^*, but not so much smaller as to be in case 3 of Fig. 4.5.2.

The critical point is found by solving the set of equations for the properties that the critical case must satisfy, as shown in Fig. 4.5.4b:

$$g(x^*) = h(x^*), \quad g'(x^*) = h'(x^*) = c^*, \quad g''(x^*) = 0. \tag{4.5.10}$$

Setting $g'' = 0$ yields the result $x^* = 1/3$. Then $g'(x^*) = c^*$ yields $c^* + p^* = 1/3$, and finally $g(x^*) = h(x^*)$ then yields $p^* = 1/27$. The discontinuous time histories of Sect. 4.5.5 only occur when natural resource stocks are large enough for the process-to-discovery parameter p to be sufficiently small.

Problems

4.5.1* Find the equilibrium points for the following cases:

(a) $p = 0.01, c = 0.12$.
(b) $p = 0.01, c = 0.24$.
(c) $p = 0.01, c = 0.36$.

Compare to Fig. 4.5.2.

4.5.2 Run computer simulations for the following cases:

(a) $p = 0.01, c = .12, x(0) = 0.8$.
(b) $p = 0.01, c = .24, x(0) = 0.8$.
(c) $p = 0.01, c = .36, x(0) = 0.8$.
(d) $p = 0.01, c = .24, x(0) = 0.4$.

(e) $p = 0.01, c = .24, x(0) = 0.2$.
(f) Describe the behaviors that can be seen in the model, using these simulations as examples. Compare to figures in the text.

4.5.3 Suppose $p = 0.004$. Determine (approximately) all equilibrium solutions for the cases $c = 0.1$, $c = 0.2$, and $c = 0.3$.

4.5.4(a) Use Theorem 4.4.2 to show that a positive equilibrium of the renewable resource model (4.5.6) is stable if

$$2x + \frac{2pcx}{(p + x^2)^2} > 1\,.$$

(b) Use the equation for positive equilibria to eliminate c from the condition of part (a) and rearrange the result to get the form

$$P(x) + p > 0\,,$$

where P is a simple polynomial.
(c) Use the result from part (b) to show that any equilibrium that satisfies $x \geq 0.5$ must be asymptotically stable.[26]

4.5.5 [Abiotic resource] (Continued from Problem 4.3.4.)
Use Theorem 4.5.1 with $w = 1/(1 + x)$ to determine the stability of all equilibria for the proposed abiotic resource model

$$x' = R - \frac{x}{1 - x}$$

for the cases $R < 1$ and $R > 1$. Compare the work required to do this with that required in Problem 4.3.4.

4.5.6 [Biotic resource] (Continued from Problem 4.3.5.)
Use Theorem 4.5.1 with $w = x/(1 + x)$ to determine the stability of all equilibria for the proposed biotic resource model

$$x' = rx - \frac{x}{1 + x}$$

for the cases $r < 1$ and $r > 1$. Compare the work required to do this with that required in Problem 4.3.5.

4.5.7 [Malaria] (Continued from Problem 3.8.7.)
The scaled single-variable malaria model is

$$\frac{dx}{dt} = \frac{\alpha\beta x}{1 + \beta x}(1 - x) - x,$$

where x is the fraction of the human population that has malaria, α is a measure of the transmission from mosquitoes to humans, and β is a measure of the transmission from humans to mosquitoes.

(a) Write the model using the structure (4.5.7) with $w = x/(1 + \beta x)$ and $g(x)$ a constant.
(b) Find the equilibria of the model using the structure in (a).

[26] This is an example of a useful algebra calculation that could not reasonably be done with a computer algebra system. Of course, one could get a CAS to do the calculation by giving it the sequence of steps in the calculation, but not by expecting the CAS to do algebra with human ingenuity.

(c) Do a thorough phase line analysis for this model using Theorem 4.5.1. You will need to do multiple cases depending on the values of α and β.

4.5.8 [Malaria]
The structure defined in Problem 4.5.7 also helps with linearized stability analysis.

(a) Differentiate the function $f(x) = w(x)[g - h(x)]$ using the product rule, without using the formulas for w, g, and h (but with the observation that g is a constant).
(b) Determine the stability of $x_0^* = 0$ by evaluating f' from (a) when $w = 0$. Given $w' > 0$, use the result to determine the stability for $x_0^* = 0$.
(c) Determine the stability of the positive equilibrium x_1^* by evaluating f' from (a) when $g = h$. The stability result follows almost immediately.
(d) Confirm that the results of the linearized stability analysis are consistent with those of the phase line analysis of Problem 4.5.7.

4.5.9 [Resource harvesting] (Continued from Problem 4.3.7.)

(a) Use Theorem 4.5.1 to determine the stability of all equilibria for the Holling type II resource consumption model

$$v' = v\left(1 - \frac{v}{k} - \frac{c}{1+v}\right)$$

with $k = 8$ and $c = 0.36$.
(b) Repeat with $c = 0.12$.
(c) Repeat with $c = 0.24$.
(d) Compare the work required to do this with that required in Problem 4.3.7.

4.6 Projects

Each of the projects in this chapter requires a significant amount of work, but each offers an opportunity to study a challenging biological or mathematical problem. Projects 4A and 4D are on discrete models, with 4A focusing on the biological issue of using a non-toxic method of pest control and 4D focusing on exploring some of the interesting mathematics of limit cycles in discrete dynamics. Projects 4B and 4C deal with an epidemiological question and an environmental question. Both of these projects rely heavily on the material from the case study of Sect. 4.5.

Project 4A: Pest Control

One method for controlling insect pests is to release sterile males into a field population. This reduces the future population because females that mate with sterile males do not contribute to the next generation.

(a) Suppose a field population of insects in year t consists of N_t females and the same number of males. In addition, a population of S sterile males is artificially established by release into the environment. Derive the model

$$N_{t+1} = \left(\frac{A}{1+N_t}\right)^b \frac{N_t}{N_t + S}\, N_t\,, \qquad S > 0$$

by modifying the Hassell model of Problem 4.1.6. Carefully explain your assumptions.

One of the most important steps in a mathematical modeling investigation is the development of questions. These should include some general questions as well as specific ones. Here we might consider these, arranged with the specific questions first and then the general questions from easiest to hardest.

1. What happens to a Colorado potato beetle population ($b = 3$, $A = 4$) as S increases? (We could ask the same question using a different pair of b and A values.)
2. For given b and A, how much does $S > 0$ suppress an insect pest population that starts at the $S = 0$ fixed point of $N = A - 1$?[27]
3. What circumstances guarantee that the fixed point $N_0^* = 0$ is locally asymptotically stable?
4. What is the local stability of positive fixed points?
5. What circumstances guarantee that the fixed point $N_0^* = 0$ is globally asymptotically stable? (The biological equivalent of global asymptotic stability of $N_0^* = 0$ is extinction of the insect pest population.)

1. Potato beetle examples ($b = 3$, $A = 4$)

(b) The fixed point equation is difficult to solve for N, but we can pick a value of N and solve for S in order to get an example. Find the value of S needed to have the fixed point $N^* = 1$. Using that value of S, plot $(1 + N)^3(N + S)$ and $64N$ on a common graph and approximately identify any other fixed points for that S.

(c) Run simulations and obtain cobweb plots using the value of S you found in (b) with initial values $N_0 = 1.2$, $N_0 = 0.5$, and $N_0 = 0.1$. Describe the results and identify the stable fixed points.

(d) Repeat (c), but with $S = 10$ and $N_0 = 1.2$.

(e) Repeat (c), but with $S = 0.5$ and $N_0 = 1.2$. Try some other initial conditions as well.

(f) Use your results from (b)–(e) to conjecture the behavior of the case $b = 3$. Be careful to identify multiple ranges of S values that show different behaviors.

2. Positive Fixed Points ($b = 3$)

(g) In part (b), we found a point on the curve of N^* vs S for the parameters $b = 3$ and $A = 4$. Use the same procedure to plot graphs of N^* versus S for $A = 2$, $A = 3$, $A = 4$, and $A = 5$, all with $b = 2$. Plot the four curves on a common set of axes. Make sure you only plot realistic values of the variables.

3. Local Stability of $N_0^* = 0$

(h) Use Theorem 4.4.1 to show that the fixed point $N_0^* = 0$ is always locally asymptotically stable (for any values of b, A, and S).

(i) Compare the result of (h) with what you wrote for (f). Did you get this point right? If not, explain why it can sometimes be difficult to identify general results from looking at examples.

4. Local Stability of $N^* > 0$

(j) Use Theorem 4.4.1 to derive the stability requirements

[27] If $A < 1$, then the insect population is not viable on its own, so the questions need not be asked unless $A > 1$.

$$\frac{S}{S+N} < \frac{bN}{1+N}, \qquad \frac{bN}{1+N} < 2 + \frac{S}{S+N}.$$

This requires nuanced use of algebra. You already have a messy formula for $g'(N)$ from part (h). If you remove common factors from the terms in the formula for $g'(N^*)$, you can then use the fixed point equation to replace a messy product with the constant 1.

(k) Show that the first stability criterion is equivalent to $dS/dN^* < 0$, where the derivative comes from the formula for the fixed point after solving for S. You can make the same algebraic simplification as in (j) by using the fixed point equation to eliminate the factor $A^b(1+N)^{-b}$. Apply the result to the plot in (g) to show that the smaller positive N^* is always unstable.

(l) Show that the second stability criterion simplifies to $N < \hat{N}(S) = \sqrt{1+3S}$ for the special case $b = 3$. Use this result to write the stability criterion in the form $A < \hat{A}$, where $\hat{A}$ is a fairly simple function of S and $\hat{N}(S)$. Plot this result as S versus $\hat{A}$ to get a graph that shows how large S must be to stabilize the (larger) fixed point.

5. Global Stability of $N_0^* = 0$

(m) To establish global stability of the extinction fixed point, it is sufficient to show that $N_{t+1} < N_t$ for any N_t. Define $G(N)$ so that $N_{t+1} = N_t G(N_t)$. Show that the maximum value of G occurs when

$$S = \frac{bN^2}{1+N-bN}.$$

(n) Use the result of (m) to show that the maximum value of G is less than 1 whenever

$$A^b < \frac{(1+N)^{b+1}}{1+N-bN}.$$

(o) Use N as a parameter and the functions in (m) and (n) to plot S versus A for $b = 3$. (Note that you have to be careful to limit the values of N that you use.) If you put this plot on the same graph as the plot in (l), it shows how large S must be to stabilize the pest population and how much larger it needs to be to eliminate the pest population.

Project 4B: A Disease Model with Treatment[28]

The simplest disease model is the SIS epidemic model with no mortality,[29] in which the infectious population fraction i satisfies the equation

$$i' = \mathcal{R}_0 i(1-i) - i,$$

where $\mathcal{R}_0$ is the basic reproductive number and time has been scaled so that $t = 1$ is the average recovery time. Rearrangement shows this model to be equivalent to the logistic growth model, with final infectious fraction $1 - \mathcal{R}_0^{-1}$.

[28] This project is adapted from a model by Maia Martcheva [9].

[29] Problem 3.2.10.

Suppose there is a treatment for our disease that enhances recovery time by a factor $1+\phi$. This would make the recovery rate $(1+\phi)\gamma I$ in the dimensional model and $(1+\phi)i$ in the dimensionless model.[30]

If the treatment rate were uniformly $1+\phi$, we could have stuck with the standard SIS model and just used a larger value for γ. Instead, we assume that the enhanced recovery rate is only universally available when the infectious population is small. As the infectious population grows, distribution problems arise, so some people get the enhanced rate and others don't. The treatment rate in the model should be a population-wide average rate, which should have a multiplicative factor that is a decreasing function of i with value $1+\phi$ at $i=0$. To keep things simple, let's assume that the recovery rate is not enhanced at all for $i=1$. We could use a linear function, but instead let's use a function that is concave up.

The above considerations lead to the model

$$i' = \mathcal{R}_0 i(1-i) - \frac{1+\phi}{1+\phi i} i \equiv f(i)\,.$$

Note that this model reduces to the original if we take $\phi = 0$. It is good modeling practice to add enhancements in a way that the enhanced model reduces to the original just by setting a parameter to 0.

The analysis of the model should focus on identifying equilibria, including the conditions required for their existence, and determining the stability of the equilibria. Following the analysis, we will want to tell a story about what happens if the treatment rate ϕ increases. This particular model is interesting because the effect of enhanced treatment depends on how infectious the disease is, that is, we will see different versions of the story for different values of $\mathcal{R}_0$.

1. Preparation for Analysis
   As in Sect. 4.5, this complicated differential equation is easier to analyze if we utilize a specific structure rather than taking f as a unit. Write the model in the form $i' = w(i)[g(i) - C]$, where

$$w(i) = \frac{\mathcal{R}_0 i}{1+\phi i}.$$

   Note that $w' > 0$ and that $w = 0$ if and only if $i = 0$. Thus, equilibria have either $w = 0$ or $g = C$.

2. Equilibrium Analysis

   a. Sketch $g(i)$ for the case $\phi < 1$. There will be two cases, one for $C > 1$ and one for $C < 1$. Add dashed horizontal lines to the figure using a value of C in each of these ranges. Sketch the phase line for both cases and determine which equilibria are stable.

   b. For the case $\phi > 1$, find the value of the function g at the maximum point. We'll call this value g_m.

   c. Repeat (a) for $\phi > 1$. Note that there will be three cases this time.

   d. For each of the five cases, note the corresponding range of ϕ values, as functions of $\mathcal{R}_0$, keeping in mind that some of the cases are only possible for some ranges of $\mathcal{R}_0$ values.

   e. Sketch a graph in the $\mathcal{R}_0\phi$-plane that shows the regions of the five cases. In each region, identify whether $i^* = 0$ is stable and whether there is a stable equilibrium with $i^* > 0$.

   f. Use linearized stability analysis to confirm the results obtained from phase line analysis. Note that the structure $f = w(g - C)$ is helpful.

[30] Note that we've incorporated the additional feature into the model by using a dimensionless parameter rather than a dimensional one. We could instead have made the improved recovery rate be $\gamma + \kappa$, but then κ would need to be scaled.

3. The Impact of Treatment
   a. For the case $\mathcal{R}_0 = 2$, plot a graph of the stable equilibrium value(s) as a function of ϕ. Describe any possible ways the disease could be eliminated.
   b. Repeat (a) for $\mathcal{R}_0 = 3$.
   c. Repeat (a) for $\mathcal{R}_0 = 4$.

Project 4C: Lake Eutrophication

If you visit a lot of lakes of different sizes in different parts of the world, you will see that there are two common physical states. Some lakes are relatively clean and fresh-smelling—these are called *oligotrophic*. In contrast, *eutrophic* lakes are overgrown with algae and have a rank smell. The physical difference is due to differences in phosphorus content, with that of eutrophic lakes being many times higher than that of oligotrophic lakes. Biologists have long known that there does not appear to be a gradation of intermediate states; indeed, rapid eutrophication of formerly oligotrophic lakes is an environmental problem that can be caused by runoff of fertilizer from farms. Why there are no intermediate states, how eutrophication occurs, and the possibility of restoring a eutrophic lake are all questions that can be explored with a simple mathematical model due to a 1999 paper by Carpenter et al. [3]:

$$\frac{dP}{dT} = B - SP + R\frac{P^q}{M^q + P^q}, \qquad B, S, R, M > 0, \quad q \geq 2. \tag{4.6.1}$$

This model incorporates three mechanisms for change in phosphorus content in a lake. The first term represents the influx of phosphorus from the environment at constant rate B; this can include artificial sources such as farm runoff as well as natural sources such as decomposition of plants.[31] The second term represents the combined processes of sedimentation, outflow, and absorption by plants, all of which remove phosphorus from the water through spontaneous chemical reactions. The last term represents the recycling of phosphorus from sediments. This term plays a significant role in the physical system, because the large values typical of q (from 2 for a cold deep lake to as much as 20 for a warm shallow lake) mean that the recycling rate is roughly R when $P > M$ and very small when $P < M$. Thus, eutrophic lakes, which are high in phosphorus, have large recycling rates that keep the phosphorus concentration high; oligotrophic lakes, in contrast, have very little recycling of sedimentary phosphorus.

In this project, we conduct a case study similar to that of Sect. 4.5, consisting of scaling, phase line analysis for a specific range of parameters, examination of a multi-phase scenario, and a bifurcation study.

1. Scaling
   Obtain a dimensionless form of the model using M as the scale for P and $1/S$ as the scale for T, along with the parameters $b = B/SM$ and $r = R/SM$ in addition to q. We will generally consider q and r to be fixed by environmental factors, with b ranging from an environmentally determined minimum value $b = b_0$ upward. Scaling has reduced the number of parameters from five to three, which makes it easier to analyze the model and reduces the number of uncertain parameters that must be estimated for simulations. Three is still a lot. We'll further reduce the number of parameters by fixing q at an intermediate value of 8, leaving the influx parameter b and the recycling parameter r as study parameters.

[31] The standard use of phosphates in laundry and dishwashing detergents was linked to lake eutrophication in the late 1960s and spawned one of the early conflicts between the environmental movement and industry. Phosphates are still used in some detergents, but smaller amounts and better treatment of wastewater have significantly reduced their contribution to eutrophication.

2. Phase Line Analysis

   a. Rearrange the differential equation from step 1 so that it takes the form

$$\frac{dp}{dt} = r[N(p) - L(p)],$$

   where N is a nonlinear function that contains q, but not r or b, and L is a linear function that contains r and b. Explain why this is a useful form for phase line analysis in which q is fixed and r and b are allowed to vary.

   b. Plot the nonlinear function in part (a) using $q = 8$ and the linear function using $r = 2$ along with the b values 0.2 and 0.8. Use the graph to determine approximate values for the equilibrium points. (These parameter values illustrate typical model behavior and are plausible for a variety of lakes.)[32]

   c. Discuss the pattern of equilibrium solutions, that is, note the presence or absence of large, small, and intermediate values of p at equilibrium for the given values of b and the ranges these values represent. For an example, see Sect. 4.5.3.

   d. Determine the approximate values of the equilibria for the lake eutrophication model with $q = 8$, $r = 2$, and $b = 0.8$. Use phase line analysis to determine the stability of the equilibria.

   e. Repeat part (d) with $b = 0.2$.

3. History of a Polluted Lake
   Explain the model predictions for a lake that experiences the following sequence of events, assuming that the lake is initially at equilibrium with $b = 0.2$ and a small amount of phosphorus.

   a. Flooding raises b to 0.8, where it remains for a long period of time.

   b. Conditions eventually reduce b back down to 0.2, where it remains for a long period of time.

4. Bifurcation
   Another approach to understanding the equilibria of the model is to plot the equilibria as a function of the parameters.

   a. With $q = 8$ and $r = 2$, plot the equilibria as a function of b on the interval $0 \le b \le 1$. (Hint: The desired plot should be in the bp-plane, but points on the plot can conveniently be found by calculating b values from selected p values. The plot from step 2(b) is helpful for determining a suitable range of p to try. Restrict the plot to realistic values of b.) Repeat with $r = 1$ on the same graph.

   b. Assume that there are measures that can be taken to temporarily change the recycling rate for a lake. Describe a procedure that could be used to reverse eutrophication in that case.

   The plots of parts 2b and 4a raise additional questions.

   c. For what range of r values is it possible to have only one small p-equilibrium for small b? To address this question, note that $r = 2$ does not meet this requirement because the large p^* portion of the curve in the plot of p^* versus b extends all the way to the b-axis, while $r = 1$ does meet this requirement because the large p^* portion of the curve ends at a vertical tangent point at approximately $b = 0.38$. We need to find the unique pair of r and p values that put that vertical tangent point on the b-axis. This means that the derivative db/dp^* is 0 at $b = 0$ along with the usual equilibrium relation $N(p^*) = L(p^*)$. Use these requirements to obtain a formula for the

[32] Carpenter, Ludwig, and Brock estimated $q = 7.8$ and $r = 7.7$ for Lake Mendota, which is adjacent to the campus of the University of Wisconsin at Madison and is probably the most thoroughly studied lake in the world [3].

critical value of r in terms of q. Check your result by plotting p^* versus b with that value of r and $q = 8$.

d. From the plot in step 2b, it is clear that $b > 1$ guarantees that there cannot be a small p-equilibrium. The same is true for values a little less than 1, but clearly not for $b = 0.2$. There must be some critical value of b for which no small equilibrium is possible. To address the question, consider that we are interested in specific cases where the graphs of N and L are tangent. The requirement for equilibrium and the requirement for tangency comprise two algebraic equations relating p^*, r, and b. Eliminate r from these equations to obtain the formula

$$b = p - \frac{p(1 + p^q)}{q}.$$

Given $q = 8$, use calculus to find the maximum value of b that can be obtained from this relationship. A larger value of b means that equilibria with small p are impossible.

Project 4D: 2-Cycles in the Discrete Logistic Model

We've already seen some of the interesting properties of the discrete logistic model. This project addresses one of the questions about these properties:

- For what values of R is there an asymptotically stable 2-cycle?

If you have not already done Problems 4.1.2 and 4.2.2, you should do these first.

Without loss of generality, we need to consider only the scaled version

$$x_{t+1} = x_t + Rx_t(1 - x_t)\,,$$

equivalent to setting $K = 1$. As you work on the problem, keep in mind that we have already learned that the fixed point $x = 1$ is asymptotically stable when $0 < R < 2$ and that the fixed point $x = 0$ is unstable for all values of R (that is, $R > 0$, since negative values do not make biological sense).[33]

Our plan is based on the idea that a stable 2-cycle corresponds to a pair of stable fixed points for a model that has a census every 2 years.

1. Use the model to derive the formula

$$x_2 = x_0 + Rx_0(1 - x_0)H(x_0), \qquad H(x) = 1 + (1 - Rx)(R + 1 - Rx)\,.$$

Use this formula to create a 2-year version of the model by identifying y_t with x_0 and y_{t+1} with x_2.

2. Suppose Y is a fixed point for the equation of step 1 that is *not* also a fixed point of the x equation. What equation must Y satisfy? [It is best not to substitute in the formula for the function H until absolutely necessary.]
3. The equation from step 2 can be simplified by replacing Y and R with more convenient substitutes. Define

$$Z = RY\,, \qquad B = R + 2$$

and substitute into the equation for Y. The result should be a simple quadratic equation for Z. Find the general solution formula for this equation. Substitute for B to obtain Z as a function of R.

4. Check your results of step 3 by calculating Y for the particular cases $R = 2.2$ and $R = 2.5$ and comparing the results with simulations from Problem 4.1.2.

[33] Example 4.4.3.

5. Use the formulas from step 3 to calculate the quantities $2Z - B$ and $RZ - Z^2$, both as functions of R. Note that you do not need to compute Z^2 because you have an equation that can be used to express Z^2 in terms of something simpler to calculate. Also note that you have to be careful with formulas that include $\pm$. Either do the calculations twice, once with each sign, or make sure that you use the fact that the negative of $\pm$ is $\mp$.
6. Stability of Y depends on $F'(Y)$, where F is defined so that $y_{t+1} = y_t + F(y_t)$. Take the derivative of F using the product rule and without substituting for H. When you evaluate F' at $y = Y$, some of the terms will disappear. Simplify the formula. (If you do this right, the quantities you calculated in step 5 will appear; after you have used the formulas from that step, your final result for $F'(Y)$ should be a polynomial in R that has no $\pm$ terms or square roots.)
7. Use Theorem 4.4.3 to find the range of R values for which the 2-cycle is stable.
8. Confirm that the results you got in step 7 are consistent with what you have seen in Problem 4.1.2, Fig. 4.2.3, and Problem 4.2.2.

References

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Discrete Linear Systems

5

In Chap. 4, we considered the dynamics of single quantities changing in either discrete or continuous time. Here we consider the dynamics of systems of several related quantities changing in discrete time. This chapter deals exclusively with linear systems, which are used to represent dynamics of structured populations divided into classes by age, size, or developmental stage. Discrete linear systems are a major tool in conservation biology modeling, where the primary goal is to determine the effects of parameters on the growth rate of a population. Nonlinear discrete systems appear in Sect. 6.5.

We begin in Sect. 5.1 with an introduction to the dynamics of structured populations using scalar notation. The models we obtain are analogous to exponential growth models for single quantities, except that the possibility of various distributions of population among the classes makes the exponential growth rate difficult to determine. However, we can demonstrate that such systems do eventually tend toward exponential growth (or decay); for problems with a limited number of classes, we can prescribe an intuitive method for determining both the eventual growth rate and the stable distribution of the population. The concepts and methods of this scalar approach are extended to an example from conservation biology in the peregrine falcon case study of Sect. 5.2.

It is often true that problems that can be solved by intuition are more easily solved by a formal mathematical procedure based on prior conceptual development. The analysis of discrete linear dynamical systems is an outstanding example of this phenomenon. In Sect. 5.3, we develop some of the basic mathematical theory of matrix algebra, which we then apply in Sect. 5.4 to the problem of determining the eventual growth rate and stable population distribution for structured models. This material is applied in Sect. 5.5 to the conservation of loggerhead sea turtles, work based on a classic paper published in 1994.

The matrix algebra theory of Sects. 5.3 and 5.4 is essential background for the analysis of continuous systems in Chap. 6. Indeed, this material is essential for much of the mathematical analysis performed in biology, and the reader is advised to aim for mastery.

5.1 Discrete Linear Systems

After studying this section, you should be able to:

- Construct a structured discrete linear population model from a narrative description.
- Identify a narrative description from a structured discrete linear population model.
- Describe the general behavior of discrete linear population models in terms of the growth rate and stable population ratios.

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- Determine the long-term growth rate and stable population ratios for two and three-component discrete linear population models.

In this section, we extend the basic ideas of the single-component linear model to analogous models of populations with structure. We begin with a brief summary of the earlier model, adapted from Sect. 4.1. We consider the model in the form

$$N_{t+1} = \lambda N_t \,, \tag{5.1.1}$$

which can be rearranged as

$$\lambda = \frac{N_{t+1}}{N_t} \,. \tag{5.1.2}$$

This version reveals that λ is the constant factor by which the population is augmented at each time step.

The model (5.1.1) has the explicit solution

$$N_t = \lambda^t N_0 \,, \tag{5.1.3}$$

which indicates that the population grows without bound if $\lambda > 1$, stays constant if $\lambda = 1$, and shrinks toward 0 if $\lambda < 1$. Thus, the unique fixed point $N = 0$ is asymptotically stable if $\lambda < 1$, neutrally stable if $\lambda = 1$, and unstable if $\lambda > 1$.

Multi-component models exhibit similar behavior.

5.1.1 Simple Structured Models

Simple models of single (unstructured) populations track only the total population size. For reasons that will become clear through the examples, such models can sometimes fail to capture important aspects of population behavior. Better models may be obtained by adding structure to a population model.

Definition 5.1.1

A **structured population model** is a population model in which individuals are categorized according to some discrete or continuous property, usually age or life stage.

We restrict consideration to models in which the individuals are divided into discrete classes. Models that have continuous structure but are discrete in time are called integral projection models and are beyond the scope of this text.

▶ **Example 5.1.1** A population consists of juveniles and adults and changes from year to year through survival and reproduction:

- Ten percent of the juveniles survive to become adults.
- All adults die after 1 year.
- Adults produce an average of 20 juveniles in their single season of life, through reproduction.
- Juveniles also reproduce, with an average of one juvenile offspring each.

For simplicity, we assume the population has only one sex.

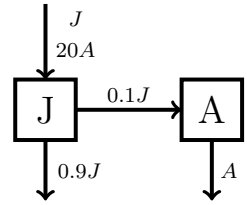


Fig. 5.1.1 Life history diagram for Examples 5.1.1–5.1.3 and 5.1.5. Each year, 10% of juveniles become adults and 90% die, while all adults die. The $t+1$ population of J consists exclusively of the total births $J_t + 20A_t$, while the $t+1$ population of A consists exclusively of the juveniles that survived the year to become adults

From this narrative, we obtain the schematic diagram of Fig. 5.1.1 and a set of equations that compute J_{t+1} and A_{t+1} in terms of the time t populations:

$$J_{t+1} = J_t + 20A_t\,, \qquad A_{t+1} = 0.1J_t\,. \tag{5.1.4}$$

□

As with one-component dynamic models, we can study the model (5.1.4) with simulations, provided we prescribe initial population values.

▶ **Example 5.1.2** Suppose we have an initial population of 200 juveniles and 10 adults for the model of Example 5.1.1; that is,

$$J_0 = 200\,, \qquad A_0 = 10\,.$$

The populations at time 1 are then found using (5.1.4) with $t = 0$:

$$J_1 = J_0 + 20A_0 = 400\,, \qquad A_1 = 0.1J_0 = 20\,.$$

Similarly, we find

$$J_2 = J_1 + 20A_1 = 800\,, \qquad A_2 = 0.1J_1 = 40\,.$$

In this simulation, the populations of both classes double at each time step. In other words,

$$\frac{J_t}{J_{t-1}} = 2\,, \qquad \frac{A_t}{A_{t-1}} = 2\,.$$

Alternatively, we may write an exact solution as

$$J_t = 2^t J_0\,, \qquad A_t = 2^t A_0\,.$$

□

In Example 5.1.2, the structured model (5.1.4) displays exactly the same behavior as the unstructured model (5.1.1). However, we should be careful not to draw too strong a conclusion from this. The simulation results will generally be more complicated if we start with different initial populations.

▶ **Example 5.1.3** Consider the model (5.1.4) with an initial population of 100 juveniles and 10 adults. Table 5.1.1 and Fig. 5.1.2 show the population growth over several time steps. The population does

Table 5.1.1 Populations for (5.1.4) with $J_0 = 100$, $A_0 = 10$

Time	0	1	2	3	4	5	6	7	8
J	100	300	500	1,100	2,100	4,300	8500	17100	34100
A	10	10	30	50	110	210	430	850	1710
J_t/J_{t-1}		3	1.67	2.2	1.91	2.05	1.98	2.01	1.99
A_t/A_{t-1}		1	3	1.67	2.2	1.91	2.05	1.98	2.01
J/A	10	30	16.7	22	19.1	20.5	19.8	20.1	19.9

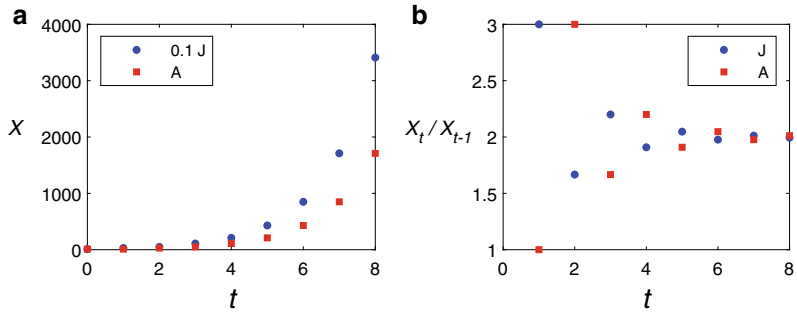


Fig. 5.1.2 **a** Populations and **b** population ratios for Example 5.1.3

not double at each time step, which would require $J_t/J_{t-1} = 2$ and $A_t/A_{t-1} = 2$ for each t. However, these ratios of successive populations do approach 2 gradually as time increases. □

Check Your Understanding 5.1.1:
Verify the data in Table 5.1.1.

The comparison between Examples 5.1.2 and 5.1.3 is instructive. In both cases, the model predicts a final growth rate of 2; however, the simulation produces a growth rate of exactly 2 in the first instance and only approaches 2 gradually in the second. We can gain crucial insight into the behavior of structured models by examining the ratio of J to A. In Example 5.1.2, this ratio is initially 20:1; because both population classes double at each time step, the ratio of 20:1 is maintained. However, in Example 5.1.3, the ratio of J to A is initially 10:1. Over the course of the simulation, the ratio changes because the growth rates of the two component populations are not exactly the same. However, the ratio appears to approach 20:1 as the growth rate approaches 2.

Examples 5.1.2 and 5.1.3 illustrate the basic behavior of discrete linear systems. There is a special ratio of initial populations, 20 to 1 in this case, for which the model shows growth at a constant rate $\lambda > 0$.[1] If a simulation starts with a different ratio of initial populations, then the model populations will only gradually settle into a pattern with the characteristic growth rate and proportions.

What about a system with more than two components?

▶ **Example 5.1.4** A population consists of larvae, young adults, and older adults. One percent of larvae grow into young adults each year, and 30% of young adults survive to become older adults. Young

[1] Of course there could be "growth" at a rate less than 1 for some models.

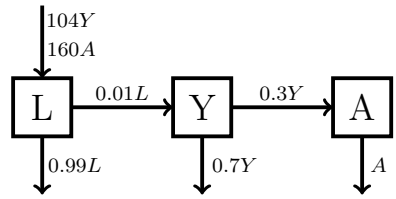


Fig. 5.1.3 Life history diagram for Examples 5.1.4 and 5.1.6

adults have an average of 104 offspring each year, while older adults have an average of 160 offspring. Characterize the growth of this population.

From the description, we obtain the schematic diagram of Fig. 5.1.3 and the system

$$L_{t+1} = 104Y_t + 160A_t, \tag{5.1.5}$$
$$Y_{t+1} = 0.01L_t, \tag{5.1.6}$$
$$A_{t+1} = 0.3Y_t\,. \tag{5.1.7}$$

Table 5.1.2 and Fig. 5.1.4 show the results of a simulation starting with 1000 larvae, 50 young adults, and 5 older adults. There is not a clear pattern at the beginning; for example, the older adult population triples in the first year before dropping to 3 and then rising back up to 18. Eventually, however, the population dynamics settles into a pattern in which each component is growing by 20% per year. During the first 8 years or so, the proportions of L to A and Y to A gradually adjust from initial values of 200 and 10 to stable values of 480 and 4. □

Table 5.1.2 Populations for (5.1.5–5.1.7) with $L_0 = 1000$, $Y_0 = 50$, and $A_0 = 5$

Time	0	1	2	3	4	5	6	7	8	9	10
L	1,000	6,000	3,440	6,720	6,458	8,640	9,942	12,085	14,486	17,341	20,867
Y	50	10	60	34	67	65	86	99	121	145	173
A	5	15	3	18	10	20	19	26	30	36	43
L_t/L_{t-1}		6.00	0.57	1.95	0.96	1.34	1.15	1.22	1.20	1.20	1.20
Y_t/Y_{t-1}		0.20	6.00	0.57	1.95	0.96	1.34	1.15	1.22	1.20	1.20
A_t/A_{t-1}		3.00	0.20	6.00	0.57	1.95	0.96	1.34	1.15	1.22	1.20
L/A	200	400	1,147	373	626	429	513	466	486	478	480
Y/A	10.0	0.7	20.0	1.9	6.5	3.2	4.5	3.8	4.1	4.0	4.0

Note that plots of populations show the trend of accelerating growth, while plots of population ratios give more detailed information about the manner of growth.

5.1.2 Finding the Growth Rate and Stable Stage Distribution

We've seen that discrete linear systems seem to have a characteristic growth rate and population ratio that solutions approach regardless of the initial population values. Is there some method that can be

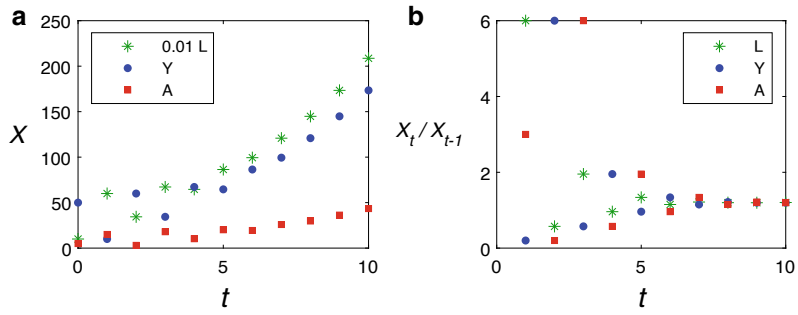


Fig. 5.1.4 **a** Populations and **b** population ratios for Example 5.1.4

used to find these characteristic features without having to resort to simulations? The answer is "yes," but some theory of matrix algebra is needed before the method can be fully developed; this is addressed in the next two sections. Meanwhile, it is instructive to solve the problem for small models, such as those of Examples 5.1.1 and 5.1.4, using basic principles rather than a sophisticated mathematical technique.

▶ **Example 5.1.5** To find the characteristic growth rate and population ratios for (5.1.4), we mimic the scenario of Example 5.1.2, except that we don't know what initial conditions are needed. We do know that only the ratio matters, so we can arbitrarily set $A_0 = 1$ as long as we allow all possible values for J_0; that is, $J_0 = j$, where j is a variable whose value must be determined. In the scenario of Example 5.1.2, the population ratio is $J/A = j$ for all time, with both populations increasing by a factor λ each time step. This can only work if we get the right value of j; hence, we have to determine j as well as λ. We know from our simulations that the results are $\lambda = 2$ and $j = 20$. Our current goal is to determine these values without recourse to simulations.

The key idea is that we can calculate J_1 and A_1 in two different ways by utilizing all of the given information.

- The model equations allow us to compute J_1 and A_1 for any initial values. Thus,

$$J_1 = J_0 + 20A_0 = j + 20\,, \qquad A_1 = 0.1J_0 = 0.1j\,. \tag{5.1.8}$$

- If we choose j correctly, then both component populations will change by a factor of λ in each time step; hence,

$$J_1 = \lambda J_0 = \lambda j\,, \qquad A_1 = \lambda A_0 = \lambda\,. \tag{5.1.9}$$

These two calculations must yield the same answers, which gives us two equations, one for each of the component populations J_1 and A_1:

$$j + 20 = \lambda j\,, \qquad 0.1j = \lambda\,. \tag{5.1.10}$$

To solve this system, we can use the second equation to eliminate one of the variables from the first equation. In more complicated problems, it will be better to eliminate the ratio variable(s) to get a single equation for λ, so we adopt that practice now. The second equation becomes $j = 10\lambda$; substituting into the first equation then yields

$$10\lambda + 20 = 10\lambda^2\,,$$

or

$$0 = 10\lambda^2 - 10\lambda - 20 = 10(\lambda - 2)(\lambda + 1)\,.$$

Curiously, we get more than one answer for λ. One is negative, which is clearly wrong. The other is the value we got from our simulation: $\lambda = 2$, with its corresponding ratio value $j = 10\lambda = 20$. □

Check Your Understanding 5.1.2:
Find the long-term growth rate and component ratio for the model

$$J_{t+1} = 0.8J_t + 6A_t\,, \qquad A_{t+1} = 0.08J_t\,.$$

5.1.3 General Properties of Discrete Linear Models

Example 5.1.5 suggests three important mathematical questions.

1. In general, what sort of equation do we get for λ?
2. How many solutions does the equation for λ have?
3. If there is more than one solution for λ, how do we know which one is correct?

These issues will be explored in more depth in the next two sections; in short, here is what we will discover:

Theorem 5.1.1 (Long-term Growth).
Discrete linear population models with n components always have a long-term growth rate λ, which is the largest positive root of a polynomial of degree n.

▶ **Example 5.1.6** The method we used in Example 5.1.5 to find the long-term growth rate and population ratios also works for the model of Example 5.1.4. We can arbitrarily set the initial number of adults at 1 and the other initial populations at unknown values ℓ and y. Using the model and the initial conditions, we obtain

$$L_1 = 104y + 160\,, \qquad Y_1 = 0.01\ell\,, \qquad A_1 = 0.3y\,. \tag{5.1.11}$$

The assumption that each component grows at rate λ from the starting values $(\ell, y, 1)$ yields

$$L_1 = \lambda\ell\,, \qquad Y_1 = \lambda y\,, \qquad A_1 = \lambda\,. \tag{5.1.12}$$

Combining these two sets of equations yields three equations for λ, l, and y:

$$\begin{aligned} 104y + 160 &= \lambda\ell\,,\\ 0.01l &= \lambda y\,,\\ 0.3y &= \lambda\,. \end{aligned}$$

Solving these three equations is more complicated than solving two equations in Example 5.1.5. We'll have a more sophisticated method later in the chapter. For now, we proceed by working from bottom to top to obtain an equation that has only λ.

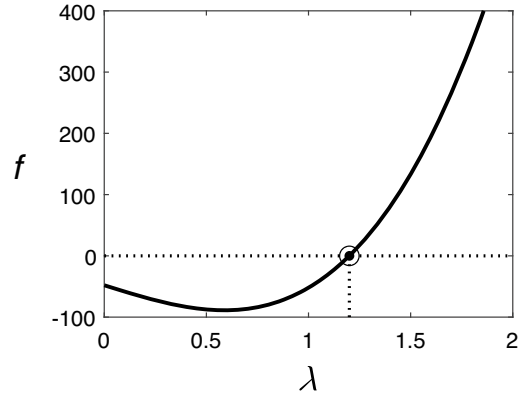


Fig. 5.1.5 The polynomial $f(\lambda)$ for Example 5.1.5

The third equation yields a substitution formula for y:

$$y = \frac{1}{0.3}\lambda\,.$$

Then the second equation yields a substitution formula for ℓ:

$$\ell = 100\lambda y = \frac{100}{0.3}\lambda^2\,.$$

Substituting these results into the first equation yields

$$\frac{104}{0.3}\lambda + 160 = \frac{100}{0.3}\lambda^3\,,$$

or

$$104\lambda + 48 = 100\lambda^3\,.$$

Hence, we need to find the largest positive root of the third degree polynomial equation

$$f(\lambda) = 100\lambda^3 - 104\lambda - 48 = 0\,. \tag{5.1.13}$$

There is no convenient solution formula for third degree polynomial equations. Sometimes one can guess a root and then factor the equation, but this is unlikely unless the parameters have been chosen for arithmetic convenience. Nevertheless, there is no difficulty in getting an approximate solution by graphing. From the graph of f in Fig. 5.1.5, we see that the largest positive solution is approximately $\lambda = 1.2$. In fact, this value solves the equation exactly.

Once λ is known, we can quickly recover the other variables from the substitution formulas:

$$\ell = \frac{100}{0.3}\lambda^2 = 480\,, \qquad y = \frac{1}{0.3}\lambda = 4\,.$$

The stable age distribution has 480 larvae and 4 young adults for every adult. As expected, the growth rate and population ratios match the results we obtained using a simulation. □

It may seem at first that there is no reason to find the growth rate and population ratios using messy algebra when we can find them from simulations. However, a simulation merely shows the behavior that follows from a particular initial condition. Could there be multiple sets of stable growth rates and population ratios, and which one you get depends on the starting point? This is not a question that can be definitively answered by examples. A formal calculation, such as that of Example 5.1.6, settles the question completely.

Check Your Understanding Answers

2. $\lambda = 1.2, \quad j = 15$

Problems

5.1.1* Consider a population with the same life history as that of Example 5.1.1, but with only 5% survival of juveniles and average reproduction rates of 11 per year for adults and 0.6 per year for juveniles.

(a) Run a computer simulation for 10 time steps, assuming an initial population of 10 adults.
(b) Plot the ratios J_t/J_{t-1}, and A_t/A_{t-1} together on a common set of axes, starting at $t = 1$.
(c) Plot the ratio J_t/A_t.
(d) Based on the plots of parts (b) and (c), describe the long-term behavior that the model predicts.
(e) Use the method of Example 5.1.5 to determine the long-term growth rate and population ratios.
(f) Discuss the question of whether the long-term behavior described by the model is biologically realistic. Could it be realistic in the short term?

5.1.2 Repeat Problem 5.1.1, but with adult reproduction at 5.4 rather than 11.

5.1.3 Consider a population of juveniles and adults for which 5% of juveniles survive to become adults, 60% of adults survive in any given year, and adults have an average of 11 offspring per year.

(a) Write down the equations for the model based on the given assumptions.
(b) Run a computer simulation for 10 time steps, assuming an initial population of 10 adults.
(c) Plot the ratios J_t/J_{t-1} and A_t/A_{t-1} together on a common set of axes, starting at $t = 1$.
(d) Plot the ratio J_t/A_t.
(e) Based on the plots of parts (c) and (d), describe the long-term behavior that the model predicts.
(f) Use the method of Example 5.1.5 to determine the long-term growth rate and population ratios.
(g) Discuss the question of whether the long-term behavior described by the model is biologically realistic. Could it be realistic in the short term?

5.1.4 Consider a population with the same life history as that of Example 5.1.5, but with 4% survival of larvae, 60% survival of young adults, and average reproduction rates of 11 per year for young adults and 50 per year for older adults.

(a) Run a computer simulation for 10 time steps, assuming an initial population of 10 for each group.
(b) Plot the ratios L_t/L_{t-1}, Y_t/Y_{t-1}, and A_t/A_{t-1} together on a common set of axes, starting at $t = 1$.
(c) Plot the ratios L_t/A_t and Y_t/A_t together on a common set of axes.
(d) Based on the plots of parts (b) and (c), describe the long-term behavior that the model predicts.
(e) Use the method of Example 5.1.5 to determine the long-term growth rate and population ratios.
(f) Discuss the question of whether the long-term behavior described by the model is biologically realistic. Could it be realistic in the short term?

5.1.5* Consider a population with the same life history as that of Example 5.1.1, but with r for the survival probability of juveniles and average reproduction rates of f per year for adults and s per year for juveniles. Determine the relationship that the parameters must satisfy to achieve a long-term growth rate of exactly 1.

5.1.6 Consider a population with the same life history as that of Problem 5.1.3, but with r for the survival probability of juveniles, b for the survival probability of adults, and an average reproduction rate of f per year for adults.

(a) Determine the relationship that the parameters must satisfy to achieve a long-term growth rate greater than 1.
(b) Suggest a biological interpretation of the requirement in part (a). (Hint: Write the requirement in the form $g(b, r, f) \geq 1$ and determine the biological meanings of the terms in the function g.)

(This problem is continued in Problem 5.3.5.)

5.1.7 Consider a population with the same life history as that of Problem 5.1.4, but with r for the survival probability of juveniles, p for the survival probability of young adults, and reproduction rates of f_1 per year for younger adults and f_2 per year for older adults.

(a) Determine the relationship that the parameters must satisfy to achieve a long-term growth rate greater than 1.
(b) Suggest a biological interpretation of the requirement in part (a). (Hint: Write the requirement in the form $g(p, r, f_1, f_2) \geq 1$ and determine the biological meanings of the terms in the function g.)

5.1.8 (Red blood cells)
Let R_t be the number of red blood cells in circulation on day t and let M_t be the number produced by the bone marrow on day t. Assume that a fraction f of red blood cells are removed from the circulation by the spleen each day and that all red blood cells produced by the marrow become part of the circulation on the following day. Also assume that the number produced by the marrow on a given day is γ times the number that were removed from circulation on the previous day [7].

(a) Write down the equations for the model based on the given assumptions.
(b) Suppose the number of red blood cells is approximately constant over long periods of time. Determine the value of γ necessary for this result.
(c) Find a solution formula for λ in terms of f and γ.
(d) What is obviously unrealistic about this model? Explain your answer.

(This problem is continued in Problem 5.3.6.)

5.2 Case Study: Peregrine Falcons

Peregrine falcons (*Falcoperegrinus anatum*) were placed on the endangered species list in 1970 due to a combination of poisoning from the pesticide DDT, habitat loss, and hunting. The population recovered because of bans on DDT and hunting along with fostering of baby falcons to improve survival rates. By 1999, there were more than 2,000 breeding pairs in the United States, which was deemed sufficient to remove peregrine falcons from the endangered list. In 2001, the U.S. Fish and Wildlife Service established a regulation permitting the harvesting of up to 5% of newborn peregrine

falcons for use by falconers.[2] Several environmental groups objected to the harvesting permits and unsuccessfully appealed the regulation in 2005. Shortly thereafter, the issue was taken up by a group of undergraduate mathematics students and their mentors at the University of Nebraska–Lincoln REU site, who constructed a mathematical model to assess the viability of the falcon population with and without harvesting [5].[3]

The falcon population model considers the population to consist of three classes: fledglings (B), juveniles (J), and adults (A), with only females considered. Changes in these populations are assumed to be the result of a small number of processes:

1. A fraction s_0 of female fledglings survive each year to become juveniles;
2. A fraction s_1 of female juveniles survive each year to become adults;
3. A fraction s_2 of female adults survive each year and remain in the same adult class;
4. Each year's surviving female adults produce an average of f female fledglings.

From these assumptions, we obtain the discrete system

$$B_{t+1} = f s_2 A_t \,, \tag{5.2.1}$$

$$J_{t+1} = s_0 B_t \,, \tag{5.2.2}$$

$$A_{t+1} = s_1 J_t + s_2 A_t \,. \tag{5.2.3}$$

Parameter value estimates from a population in Colorado are

$$s_0 = 0.544, \quad s_1 = 0.670, \quad s_2 = 0.800, \quad f = 0.830. \tag{5.2.4}$$

Harvesting of fledglings effectively reduces the parameter f; hence, with 5% harvesting we have $f = 0.7885$.

The broad *ecological* question to be addressed is whether the population will remain viable with 5% harvesting. To address this, we need to choose relevant *mathematical* questions.[4] Here are some possibilities:

1. At what rate will the model population grow without harvesting, assuming the measured parameter values?
2. How much will 5% harvesting change the growth rate, again assuming the measured parameter values?
3. What relationship must the parameters satisfy so that the long-term growth rate is at least 1?
4. Assuming the measured parameter values, what percentage of harvesting could be permitted while still maintaining the population?
5. Assuming 5% harvesting, how much of an error in the measured values can be tolerated while still maintaining the population?

[2] In fact, much of the fostering and general population increase was due to the efforts of falconers, who then felt that their investment of time and money should be rewarded by being permitted to harvest baby falcons for their sport. This case serves as an example of the difficult issues involved in trying to create conservation policies that satisfy diverse interests.

[3] The work presented in the paper focuses on the reliability of population projections when parameter values are uncertain, which is beyond the scope of what we can do here. Nevertheless, our methods allow us to obtain population growth rate predictions for a variety of circumstances and to determine critical parameter values for population viability.

[4] In any scientific experiment or field study, the quality of the research depends on there being a thoughtful set of questions to be addressed. Similarly, the most important part of a modeling investigation is the development of meaningful questions. The outcome of a modeling investigation is not a set of formulas or theorems, but a story about what it all means.

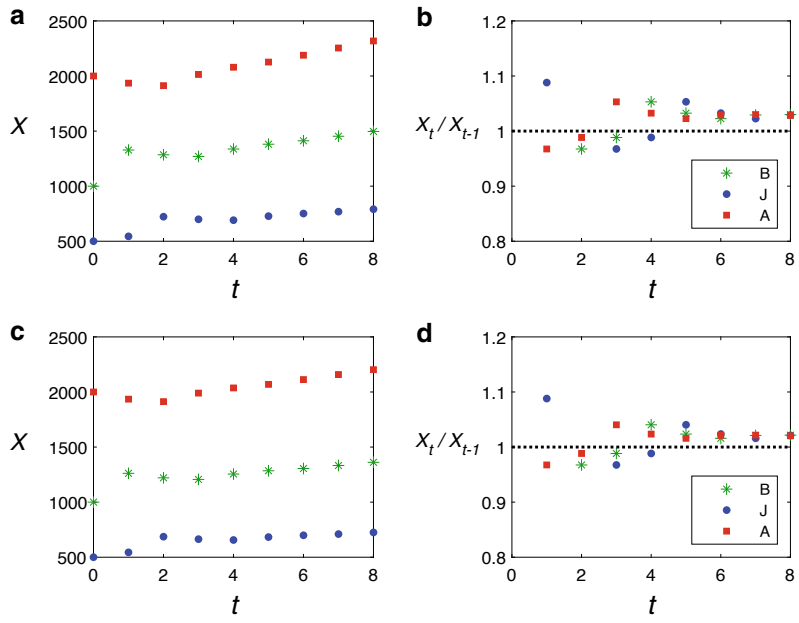


Fig. 5.2.1 Populations **a** and **c** and population ratios **b** and **d** for the peregrine falcon model. The upper panels are with no harvesting and the lower panels are with 5% harvesting

Note that these questions are of two types, corresponding to the distinction between narrow and broad views made in Sect. 2.2. The first two questions are in the narrow view because they assume the specific parameter values reported for a single population. Question 3 is in the broad view because it does not make any assumptions about parameter values. Questions 4 and 5 are broad view questions even though they are not fully general because they ask how the results depend on at least one parameter value.

As a first look at the model predictions, we consider an 8-year simulation with an initial population consisting of 2000 adults, 1000 fledglings, and 500 juveniles, which are rough estimates of the populations in 2001. Figure 5.2.1 shows the results for the cases with and without harvesting.

From these plots, it is clear that the model predicts healthy population growth, even with 5% harvesting. We can see that the equilibrium ratio estimate of 0.5:0.25:1 with which the scenario began is off a little, because the first 2 years show a decrease in adults and an increase in juveniles. The equilibrium ratio of juveniles to adults seems to be closer to 1/3 than 1/4. These values could be ascertained from the data itself, but it is easier to do a long-term growth rate analysis.

5.2.1 Mathematical Analysis

Suppose the initial populations are

$$B_0 = b\,, \quad J_0 = j\,, \quad A_0 = 1\,. \tag{5.2.5}$$

Then the model yields year 1 populations of

$$B_1 = f s_2\,, \quad J_1 = s_0 b\,, \quad A_1 = s_1 j + s2\,. \tag{5.2.6}$$

Alternatively, growth at rate λ yields

$$B_1 = \lambda b\,, \quad J_1 = \lambda j\,, \quad A_1 = \lambda\,. \tag{5.2.7}$$

Setting these population equal yields three equations for b, j, and λ:

$$fs_2 = \lambda b\,, \quad s_0 b = \lambda j\,, \quad s_1 j + s2 = \lambda\,. \tag{5.2.8}$$

From last to first, we can solve these and substitute into the previous equation to get

$$j = \frac{\lambda - s_2}{s_1}\,, \tag{5.2.9}$$

$$b = \frac{\lambda(\lambda - s_2)}{s_0 s_1}\,, \tag{5.2.10}$$

$$\lambda^2(\lambda - s_2) = fs_0 s_1 s_2\,. \tag{5.2.11}$$

Our primary interest is in the growth rate λ, so we can conveniently restate (5.2.11) as

$$P(\lambda) \equiv \lambda^2(\lambda - s_2) = rs_2\,, \qquad r = fs_0 s_1\,. \tag{5.2.12}$$

On the average, an adult that survives the year will produce f fledglings; a fraction s_0 of these will survive to become juveniles and a fraction s_1 of those will survive to become new adults. Hence, r is the expected number of new adults produced from the offspring of a surviving adult; we can interpret r as representing the rate of recruitment.[5]

While it is natural to write the cubic polynomial equation for λ in the standard form, it is instructive to leave it in the form (5.2.12). The function P is an increasing function of λ that achieves the value 0 at $\lambda = s_2$ and approaches infinity as λ increases. Thus, it achieves all possible positive values once on the interval (s_2, ∞). This means that there is a unique solution of the equation with $\lambda > s_2$ for any value of the parameter r.[6] Figure 5.2.2 shows a plot of P using the value $s_2 = 0.8$, along with horizontal lines that indicate the values of rs_2 with and without harvesting and vertical lines that mark the corresponding solutions for λ. We see that 5% harvesting reduces the model's predicted growth rate from about 3% per year to about 2% per year.

The model suggests that 5% annual harvesting is sustainable. In our report to whatever agency might have asked us to do this study, we should be careful to point out that the conclusions are only as good as the parameter estimates, and that these can be very different from one year or location to another. The actual harvesting policy should be informed by an annual census, with harvesting suspended if the population appears to be declining.[7]

5.2.2 General Analysis Questions

So far, we have focused on the narrower problems based on the specific parameter values we have for one population. With only two parameters, we can study more general questions. In particular, we might want to know what combinations of recruitment and adult survival are necessary for sustained population growth. The threshold for growth is $\lambda = 1$, so we can examine the critical case where this value is the solution of (5.2.12), that is,

$$r = \frac{1 - s_2}{s_2}. \tag{5.2.13}$$

[5] See Sect. 4.1.

[6] Mathematicians will recognize this as an application of the Intermediate Value Theorem.

[7] In the actual event, the attempts to stop the harvesting lost in court, and the harvesting was implemented with no significant impact on the falcon population recovery.

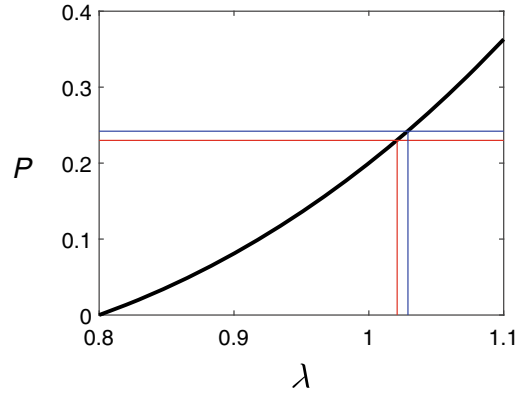


Fig. 5.2.2 The function $P(\lambda)$ from (5.2.12) along with horizontal lines marking rs_2 for the cases with and without harvesting. The corresponding vertical lines indicate the resulting growth rates of $\lambda = 1.021$ and $\lambda = 1.029$

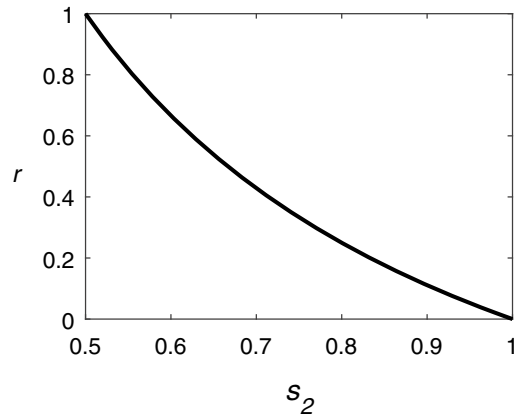


Fig. 5.2.3 The critical value of the recruitment parameter r needed to maintain a population with given adult survival probability s_2

Growth occurs if the recruitment value is larger than this critical value. Figure 5.2.3 shows the result for the case where the adult survival probability is at least 50%.

Other more general questions relating to the specific parameter data appear in the exercises.

Problems

5.2.1* Assuming the measured parameter values, determine what percentage of harvesting could be permitted while still maintaining the population.

5.2.2 Assuming 5% harvesting, determine how much of an error in each of the measured parameter values (taken one at a time) can be tolerated while still maintaining the population.

5.2.3 Plot the ratios B_t/A_t and J_t/A_t for the case without harvesting.

5.2.4 Calculate the long-term ratios b and j for the cases with and without 5% harvesting.

5.3 A Matrix Algebra Primer

After studying this section, you should be able to:

- Multiply matrices and vectors.
- Write discrete linear systems in matrix-vector form.
- Compute determinants of 2×2 and 3×3 matrices.
- Determine whether an equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions for a given nonzero matrix $\mathbf{A}$.
- Compute nonzero solutions of an equation $\mathbf{Ax} = \mathbf{0}$ when they exist.

A working knowledge of stage-structured models in biology requires some understanding of topics in matrix algebra. Therefore, we take a break from modeling to develop the concepts, beginning with the mathematical definitions needed so that discrete linear models can be written in matrix-vector form and continuing with what for us is the central problem of matrix algebra:

- Given a nonzero matrix $\mathbf{A}$, find (if possible) a nonzero vector $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$.

The reason why this problem is central will only become clear in the next section.

5.3.1 Matrices and Vectors

In Sect. 5.1, we considered models that had two or more related dynamic quantities, each with a separate symbol and each computed by its own formula. These quantities can be advantageously combined into a single multicomponent quantity called a vector,[8] in which the individual (or scalar) quantities are arranged in a column.[9]

▶ **Example 5.3.1** In Example 5.1.3, we considered a model of a population with classes J and A and initial populations $J_0 = 100$ and $A_0 = 10$. We can define a two-dimensional vector $\mathbf{x}$ whose components are J and A. Thus,

$$\mathbf{x} = \begin{pmatrix} J \\ A \end{pmatrix}, \qquad \mathbf{x}_0 = \begin{pmatrix} J_0 \\ A_0 \end{pmatrix} = \begin{pmatrix} 100 \\ 10 \end{pmatrix}.$$

□

A system of linear equations involving the components of a vector contains several coefficients in addition to the variables. These are grouped together into matrices, with one row for each equation and one column for each variable. Matrices have a dimension $m \times n$, where m is the number of rows and n the number of columns. In a population biology context, we always have equal numbers of equations and variables; $n \times n$ matrices are said to be *square*.

[8] The reader may remember using the term "vector" for quantities in physics that have magnitude and direction, and can therefore be represented by arrows with components for each cardinal direction. The mathematics of physical vectors and abstract biological vectors is identical, except that the concept of length makes more sense in a geometric setting than a biological one. In the abstract biological setting, the number of components in a vector depends on the biological model rather than geometric considerations.

[9] The advantage of the formalism of vectors will only become apparent after we have defined arithmetic operations that allow for vector calculations to faithfully reproduce the corresponding scalar calculations.

▶ **Example 5.3.2** The model of Example 5.1.3 consists of two scalar equations, which we can write as

$$\begin{aligned} J_{t+1} &= 1J_t + 20A_t \\ A_{t+1} &= 0.1J_t + 0A_t \end{aligned},$$

to emphasize that there is a coefficient for each possible term in the linear functions on the right sides of the equations. The four coefficients can be arranged as a 2×2 matrix:

$$\mathbf{M} = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix}.$$

□

In Example 5.3.2, we constructed the matrix $\mathbf{M}$ by systematically putting the coefficient in equation i for variable j into the matrix in row i and column j. This allows us to use matrices to represent the corresponding systems of equations. Before we can do so, we need a few more definitions. In general, we use the notation m_{ij} to refer to the entry in row i and column j of the matrix $\mathbf{M}$.

Definition 5.3.1

The **main diagonal** of a matrix is the set of entries whose row and column numbers are the same; that is, entries of the form m_{kk}. The $n \times n$ **identity matrix** is the matrix of appropriate size in which all entries on the main diagonal are 1 and all other entries are 0; for example, the 3×3 identity matrix is

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The identity matrix is particularly helpful when we want to subtract a common value λ from each entry on the main diagonal of a matrix $\mathbf{M}$.[10]

▶ **Example 5.3.3** Let $\mathbf{M}$ be the matrix in Example 5.3.2. Let λ be an unknown number. Define a matrix $\mathbf{A} = \mathbf{M} - \lambda\mathbf{I}$. Then

$$\mathbf{A} = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1-\lambda & 20 \\ 0.1 & -\lambda \end{pmatrix}.$$

Notice that $\mathbf{A}$ is actually a family of matrices with parameter λ. □

So far, our matrices and vectors have merely served to combine quantities into a data structure. The power of these structures becomes apparent only after we have endowed them with mathematical operations. The addition operation is defined for pairs of vectors or matrices of the same size. We compute a vector $\mathbf{u} + \mathbf{v}$ by adding the corresponding components of the individual vectors, and matrix addition works the same way. Multiplication is somewhat more complicated. For the moment, we consider only multiplication of a vector of size n by a matrix of size $n \times n$, with the matrix on the left.

[10] The reason for wanting to do this will become clear in Sect. 5.4.

Definition 5.3.2

The **matrix product** of an $n \times n$ square matrix $\mathbf{A}$ and an n-vector $\mathbf{x}$ is the vector

$$\mathbf{Ax} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{pmatrix}. \tag{5.3.1}$$

▶ **Example 5.3.4** Let $\mathbf{M}$ be the matrix of Examples 5.3.2 and 5.3.3 and let $\mathbf{x}_0$ be as in Example 5.3.1. We can multiply $\mathbf{x}_0$ on the left by $\mathbf{M}$:

$$\mathbf{M}x_0 = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix} \begin{pmatrix} 100 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \cdot 100 + 20 \cdot 10 \\ 0.1 \cdot 100 + 0 \cdot 10 \end{pmatrix} = \begin{pmatrix} 300 \\ 10 \end{pmatrix}.$$

□

Keep in mind that the product $\mathbf{Ax}$ of a square matrix $\mathbf{A}$ and a vector $\mathbf{x}$ is defined only when the matrix is on the left and both matrix and vector have the same size n. The product is a vector of size n.

5.3.2 Population Models in Matrix Notation

The alert reader may have noticed the correspondence between the preceding examples and Example 5.1.3. Using the matrix $\mathbf{M}$ and vector $\mathbf{x}$, we observe the relationship

$$\mathbf{M}\mathbf{x}_t = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix} \begin{pmatrix} J_t \\ A_t \end{pmatrix} = \begin{pmatrix} J_t + 20A_t \\ 0.1J_t \end{pmatrix} = \begin{pmatrix} J_{t+1} \\ A_{t+1} \end{pmatrix} = \mathbf{x}_{t+1}.$$

Hence, the structured population model can be written in a very simple form using matrix-vector notation.

All discrete linear population models can be written in the form

$$\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t.$$

The matrix $\mathbf{M}$ is called a **population projection matrix**.

▶ **Example 5.3.5** In Example 5.1.4, we encountered the model

$$\begin{aligned} L_{t+1} &= 104Y_t + 160A_t, \\ Y_{t+1} &= 0.01L_t, \\ A_{t+1} &= 0.3Y_t. \end{aligned}$$

This model takes the form $\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$, with

$$\mathbf{x} = \begin{pmatrix} L \\ Y \\ A \end{pmatrix}, \qquad \mathbf{M} = \begin{pmatrix} 0 & 104 & 160 \\ 0.01 & 0 & 0 \\ 0 & 0.3 & 0 \end{pmatrix}.$$

If we start with 1,000 larvae, 50 young adults, and 5 old adults, we can compute the populations at time 1 by matrix multiplication:

$$\begin{pmatrix} L_1 \\ Y_1 \\ A_1 \end{pmatrix} = \mathbf{M} \begin{pmatrix} L_0 \\ Y_0 \\ A_0 \end{pmatrix} = \begin{pmatrix} 0 & 104 & 160 \\ 0.01 & 0 & 0 \\ 0 & 0.3 & 0 \end{pmatrix} \begin{pmatrix} 1,000 \\ 50 \\ 5 \end{pmatrix} = \begin{pmatrix} 6,000 \\ 10 \\ 15 \end{pmatrix}.$$

□

5.3.3 The Central Problem of Matrix Algebra

Let $\mathbf{A}$ be an $n \times n$ matrix with at least one nonzero entry and let $\mathbf{0}$ be the n-vector whose entries are all 0. For any given matrix $\mathbf{A}$, we are interested in finding nonzero solutions $\mathbf{x}$ for the equation

$$\mathbf{Ax} = \mathbf{0},$$

if there are any.

It is instructive to examine the corresponding scalar problem. If $a \neq 0$, then the only solution of $ax = 0$ is $x = 0$. This is not necessarily the case for the matrix equation $\mathbf{Ax} = \mathbf{0}$.

▶ **Example 5.3.6** Let $\mathbf{A}$ and $\mathbf{x}$ be given by

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then

$$\mathbf{Ax} = \mathbf{0}.$$

This means that it is possible for an equation of the form $\mathbf{Ax} = \mathbf{0}$ to have nonzero solutions for $\mathbf{x}$ without requiring $\mathbf{A} = \mathbf{0}$. □

▶ **Example 5.3.7** Are there any nonzero solutions to $\mathbf{Ix} = \mathbf{0}$, where $\mathbf{I}$ is the 2×2 identity matrix? To answer this, suppose

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

with x_1 and x_2 to be determined. Then

$$\mathbf{Ix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The equation $\mathbf{Ix} = \mathbf{0}$ corresponds to the equation pair $x_1 = 0$, $x_2 = 0$. Clearly, these equations have no nonzero solutions. □

Examples 5.3.6 and 5.3.7 demonstrate that nonzero solutions exist for some matrices, but not for others. Is there some way to predict whether a given matrix has nonzero solutions without trying to find them first? Before we can answer this crucial question, we need to develop the computational tool called the *determinant*.

5.3.4 The Determinant

The determinant of a matrix is a number that is calculated from its entries using a complicated formula. Here we consider only the determinants of 2×2 and 3×3 matrices, which are relatively simple.

Definition 5.3.3

The **determinant of a** 2×2 **matrix**

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is the quantity

$$\det(\mathbf{A}) = ad - bc\,.$$

The **determinant of a** 3×3 **matrix**

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

is the quantity

$$\det(\mathbf{A}) = (aei + bfg + cdh) - (ceg + bdi + afh)\,.$$

These formulas may seem almost random at first, but there is a pattern. Each of the positive terms is a product of elements aligned diagonally from top left to bottom right, and each of the negative terms is a product of elements aligned diagonally from top right to bottom left.[11]

▶ **Example 5.3.8** Let

$$\mathbf{A} = \begin{pmatrix} -1.2 & 104 & 160 \\ 0.01 & -1.2 & 0 \\ 0 & 0.3 & -1.2 \end{pmatrix}.$$

Then

$$\begin{aligned}\det(\mathbf{A}) = {} & (-1.2)(-1.2)(-1.2)+(104)(0)(0)+(160)(0.01)(0.3)-(160)(-1.2)(0) \\ & -(104)(0.01)(-1.2)-(-1.2)(0)(0.3) = -1.728+0.48+1.248 = 0.\end{aligned}$$

□

[11] Lightly copying the first two columns of the 3×3 determinant to the right of the matrix, giving the appearance of a 3×5 matrix, will help you to see this. These patterns do NOT hold in higher dimensional determinants. The reader who wants to work with higher dimensional matrices should consult a linear algebra book for a complete definition of the determinant as well as computational schemes involving cofactors.

Check Your Understanding 5.3.1:
Find the determinant of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 \\ 3 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}.$$

5.3.5 The Equation $\mathbf{Ax} = \mathbf{0}$

The determinant is an efficient way to identify matrices for which $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions. We state the principal result without proof.

Theorem 5.3.1 (Singular Matrices)
The equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions for $\mathbf{x}$ if and only if $\det(\mathbf{A}) = 0$. Such a matrix is said to be **singular**.

Suppose we want to find the nonzero solutions $\mathbf{x}$ for a singular matrix $\mathbf{A}$. The general procedure is complicated, but in most biological problems we need only consider cases where *all* of the components of $\mathbf{x}$ are nonzero. This allows us to use a simple variation of the usual procedure.[12]

▶ **Example 5.3.9** Let $\mathbf{A}$ be the matrix from Example 5.3.8. This matrix is singular, so we know the equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions For now, we assume all components of those solutions are nonzero and look specifically for a solution having $x_3 = 1$. Then $\mathbf{Ax} = \mathbf{0}$ is equivalent to the scalar equations

$$-1.2x_1 + 104x_2 + 160 = 0, \quad 0.01x_1 - 1.2x_2 = 0, \quad 0.3x_2 - 1.2 = 0\,.$$

We can solve the third equation to get $x_2 = 4$ and then solve the second equation to get $x_1 = 480$. We have apparently found a solution

$$\mathbf{x} = \begin{pmatrix} 480 \\ 4 \\ 1 \end{pmatrix}.$$

Note that we never used the first of the three equations. This equation is available as a check. Substituting the results into the first equation yields

$$(-1.2)(480) + (104)(4) + (160)(1) = 0\,,$$

which confirms that our solution is indeed correct. Note that this is the same procedure we used in Example 5.1.5, that is, we reverted to a scalar method once we knew the matrix is singular. □

Check Your Understanding 5.3.2:
Find a nonzero solution to the equation $\mathbf{Ax} = \mathbf{0}$ where $\mathbf{A}$ is the matrix from Check Your Understanding 5.3.1.

Was the choice of $x_3 = 1$ in Example 5.3.9 special? No, it was convenient, but not necessary. We could have chosen any nonzero value for any of the three variables. We would have obtained a different

[12] See any linear algebra book for the general procedure.

answer, but all the solutions of $\mathbf{Ax} = \mathbf{0}$ are simple multiples of each other. For example, had we started with $x_2 = 1$, we would have found $x_3 = 0.25$ and $x_1 = 120$. The ratio $x_1 : x_2 : x_3$ is $480 : 4 : 1$ in all cases.

In general, the procedure of Example 5.3.9 could have failed only if the correct solutions required $x_3 = 0$. Had that been the case, we would not have found a solution, and we could have tried again with $x_3 = 0$ rather than $x_3 = 1$.

A Warning

By setting one of the components of the solution vector in Example 5.3.9 to an arbitrary value, we obtained a solution that was otherwise unique. This does not happen with all matrices, but it is guaranteed for matrices that arise in population models. A full development of matrix algebra is beyond the scope of our treatment.

Check Your Understanding Answers

1. $\det \mathbf{A} = 0$
2. One solution is $x_1 = 1, x_2 = 3, x_3 = 1$. Any nonzero multiple of that vector is also a solution.

Problems

In Problems 5.3.1–5.3.4, compute the determinant of the indicated matrix.

5.3.1* $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 0 & 1 \end{pmatrix}$

5.3.2 $\begin{pmatrix} 2 & 3 & -1 \\ 0 & 5 & 3 \\ -4 & -6 & 2 \end{pmatrix}$

5.3.3 $\begin{pmatrix} a & b & 0 \\ 0 & a & b \\ a & 0 & b \end{pmatrix}$

5.3.4 $\begin{pmatrix} a & 0 & 2a \\ 0 & b & 3b \\ 3c & c & 2c \end{pmatrix}$

5.3.5* (Continued from Problem 5.1.6.)
Write the model of Problem 5.1.6 in matrix-vector form. Then find the determinant of the matrix. (This problem is continued in Problem 5.4.7.)

5.3.6 [Red blood cells] (Continued from Problem 5.1.8.)
Write the model of Problem 5.1.8 in matrix-vector form. Then find the determinant of the matrix. (This problem is continued in Problem 5.4.8.)

5.3.7* Let $\mathbf{A} = \begin{pmatrix} -\lambda & 1 \\ 3 & 2-\lambda \end{pmatrix}$.

(a) Find all values of λ for which the equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions.
(b) Find one nonzero solution for each λ in part (a).

5.3.8 Let $\mathbf{A} = \begin{pmatrix} 3-\lambda & 2 \\ 1 & 2-\lambda \end{pmatrix}$.

(a) Find all values of λ for which the equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions.
(b) Find one nonzero solution for each λ in part (a).

5.3.9 Let $\mathbf{A} = \begin{pmatrix} 2 & 0 & c \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix}$.

(a) Find all values of c for which the equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions.
(b) Find one nonzero solution for each c in part (a).

5.3.10 Let $\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ c & 1 & 0 \\ 1 & 3 & 2 \end{pmatrix}$.

(a) Find all values of c for which the equation $\mathbf{Ax} = \mathbf{0}$ has nonzero solutions.
(b) Find one nonzero solution for each c in part (a).

5.4 Long-Term Behavior of Linear Models

After studying this section, you should be able to:

- Explain the biological significance of eigenvalues and eigenvectors.
- Compute (real-valued) eigenvalues and their associated eigenvectors.
- Determine the dominant eigenvalue of a matrix.
- Describe the long-term behavior of discrete linear models.

In Sect. 5.3, we developed the basic theory of matrix algebra, with two important results.

1. Discrete linear population models can be written as $\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$, where $\mathbf{x}$ is a vector of n component populations and $\mathbf{M}$ is an $n \times n$ matrix.
2. The equation $\mathbf{Ax} = \mathbf{0}$, where $\mathbf{A}$ is a nonzero $n \times n$ matrix, has nonzero solutions if and only if $\det(\mathbf{A}) = 0$.

In this section, we combine these two results to develop an efficient mathematical procedure for determining the long-term behavior of discrete linear systems.

5.4.1 Eigenvalues and Eigenvectors

As we saw in Sect. 5.1, discrete linear population models exhibit growth at a uniform constant rate when the initial conditions are just right. We obtained a set of equations for the growth rate and population ratios by analyzing what happens in the first time step when the initial conditions have the right proportions. This same method can be applied to models written in matrix notation. Suppose we have a discrete linear model $\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$ and an initial condition $\mathbf{x}_0 = \mathbf{v}$ for which growth occurs at a constant rate λ. Then we can calculate $\mathbf{x}_1$ in two ways. From the mathematical model, we get

$$\mathbf{x}_1 = \mathbf{M}\mathbf{x}_0 = \mathbf{M}\mathbf{v}\,.$$

The exponential growth property, given the assumptions that the initial proportions are just right, gives us

$$\mathbf{x}_1 = \lambda\mathbf{x}_0 = \lambda\mathbf{v}\,.$$

Combining these equations yields a matrix algebra equation,

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v}\,, \tag{5.4.1}$$

in which both the scalar λ and the vector $\mathbf{v}$ are unknown. This is the *eigenvalue problem*[13] of matrix algebra:

Definition 5.4.1

The **eigenvalue problem** for a nonzero $n \times n$ matrix $\mathbf{M}$ is the problem of finding solutions to $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. A value of λ for which non-trivial solutions exist is called an **eigenvalue** of the matrix and any corresponding solution $\mathbf{v}$ is called an **eigenvector** of the matrix corresponding to the given eigenvalue.

To solve (5.4.1), we must recast it in a more convenient form. Using the identity matrix $\mathbf{I}$,[14] we have $\mathbf{v} = \mathbf{I}\mathbf{v}$, which allows us to rewrite $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ as

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} = \lambda\mathbf{I}\mathbf{v}\,,$$

or

$$\mathbf{M}\mathbf{v} - \lambda\mathbf{I}\mathbf{v} = \mathbf{0}\,.$$

We can now use the distributive property of matrix multiplication (factoring out the $\mathbf{v}$) to get the equivalent equation

$$(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}\,. \tag{5.4.2}$$

Now observe that (5.4.2) is of the form $\mathbf{A}\mathbf{x} = \mathbf{0}$. The key result follows from Theorem 5.3.1.

Theorem 5.4.1 (Eigenvalues)

The eigenvalues of a square matrix $\mathbf{M}$ are the solutions of the polynomial equation

$$\det(\mathbf{M} - \lambda\mathbf{I}) = 0\,. \tag{5.4.3}$$

The polynomial $\det(\mathbf{M} - \lambda\mathbf{I})$ is called the *characteristic polynomial* of the matrix $\mathbf{M}$.

▶ **Example 5.4.1** Let $\mathbf{M}$ be the matrix from Example 5.3.2. We have

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} 1 & 20 \\ 0.1 & 0 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 20 \\ 0.1 & -\lambda \end{pmatrix}.$$

[13] "EYE-gen-value," with a hard g as in "get."

[14] Section 5.3.

Thus,

$$0 = \det \begin{pmatrix} 1-\lambda & 20 \\ 0.1 & -\lambda \end{pmatrix} = (1-\lambda)(-\lambda) - (20)(0.1)$$
$$= \lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1)\,.$$

The matrix $\mathbf{M}$ has eigenvalues 2 and -1. □

Once the eigenvalues are known, we can determine eigenvectors from (5.4.2). This is best done by rewriting the equation in scalar form and applying the procedure of Example 5.3.9. For the purpose of determining long-term behavior, we need only the eigenvector for the largest eigenvalue.

▶ **Example 5.4.2** With $\mathbf{M}$ from Example 5.4.1 and $\lambda = 2$, we have

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} -1 & 20 \\ 0.1 & -2 \end{pmatrix}.$$

The system $(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ is equivalent to the scalar equations

$$-J + 20A = 0, \qquad 0.1J - 2A = 0\,.$$

These equations are redundant, so any solution of one is a solution of the other. (This must be the case, else $\lambda = 2$ was not actually an eigenvalue.) Arbitrarily taking $A = 1$, the first equation yields $J = 20$, and this solution also satisfies the second equation. Thus, any vector of the form

$$\mathbf{v} = c_1 \begin{pmatrix} 20 \\ 1 \end{pmatrix}$$

is an eigenvector corresponding to $\lambda = 2$. □

Check Your Understanding 5.4.1:
Find a formula for all of the eigenvectors of the matrix $\mathbf{M}$ from Example 5.4.2 corresponding to the eigenvalue $\lambda = -1$.

▶ **Example 5.4.3** Let

$$\mathbf{M} = \begin{pmatrix} 0 & 104 & 160 \\ 0.01 & 0 & 0 \\ 0 & 0.3 & 0 \end{pmatrix},$$

as in Example 5.3.5. We have

$$0 = \det \begin{pmatrix} -\lambda & 104 & 160 \\ 0.01 & -\lambda & 0 \\ 0 & 0.3 & -\lambda \end{pmatrix} = [-\lambda^3 + (160)(0.01)(0.3)] - [-(104)(0.1)(\lambda)]$$
$$= -\lambda^3 + 10.4\lambda + 0.48\,.$$

In Example 5.1.6, we derived this equation using scalar methods, and we found the root $\lambda = 1.2$ by graphing. To find the corresponding eigenvector, we note that the equation $(\mathbf{M} - 1.2\mathbf{I})\mathbf{v} = \mathbf{0}$ is equivalent to the scalar equations

$$-1.2J + 104Y + 160A = 0\,, \qquad 0.01J - 1.2Y = 0\,, \qquad 0.3Y - 1.2A = 0\,.$$

Taking $A = 1$, the third and second equations yield $Y = 4$ and $J = 480$, respectively. Substituting these values into the first equation confirms that the answers are correct. Note that the method for obtaining the eigenvalue equation has been improved by matrix notation, whereas the method for finding the eigenvector components is identical to the scalar method used in Sect. 5.1. □

Algorithm 5.4.1 summarizes the procedure for finding eigenvalues and eigenvectors.

Algorithm 5.4.1
To find eigenvalues and eigenvectors of a matrix $\mathbf{M}$:

1. Find the characteristic polynomial, defined by
$$P(\lambda) = \det(\mathbf{M} - \lambda\mathbf{I})\,.$$
2. Solve the characteristic equation $P(\lambda) = 0$ (usually graphically or numerically).
3. Substitute an eigenvalue λ into the equation
$$(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}\,;$$
then rewrite this equation as a set of scalar equations.
4. Set one of the scalar unknowns to 1 and solve for the remaining unknowns.[15] This leaves one unused scalar equation to check the solutions.
5. If the last scalar equation does not check, there is an error; most likely the value of λ that you used is not actually an eigenvalue.

5.4.2 Eigenvalue Decoupling

Occasionally we can identify eigenvalues with less work than by following Algorithm 5.4.1.

▶ **Example 5.4.4** For the matrix
$$\mathbf{M} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\,,$$
the characteristic polynomial is
$$P(\lambda) = (\lambda - a)(\lambda - d)\,;$$
hence the eigenvalues are a and d. □

[15] This won't work if you choose a scalar unknown whose value needs to be 0, but you can try again with that scalar unknown set to 0 instead. This never happens in stage-structured population models, because the stable structure of a viable population must have positive numbers for each stage. The corresponding mathematical property is guaranteed by the Perron–Frobenius theorem. See [8] or other advanced books on matrix theory.

The matrix $\mathbf{M}$ in Example 5.4.4 has an entry of 0 in the row 2, column 1 position, which is the only one below the main diagonal. A matrix that has all 0's below (or above) the main diagonal is said to be *triangular*. Example 5.4.4 generalizes to triangular matrices of any size; that is, the eigenvalues of triangular matrices are given by the entries on the main diagonal.

A matrix does not have to be triangular for it to have an eigenvalue that can be found by inspection.

▶ **Example 5.4.5** For the matrix

$$\mathbf{M} = \begin{pmatrix} a & b & 0 \\ d & e & 0 \\ g & h & i \end{pmatrix},$$

the characteristic polynomial has a factor $\lambda - i$; hence, $\lambda = i$ is one of the eigenvalues. The other eigenvalues are those of the submatrix

$$\mathbf{M_3} = \begin{pmatrix} a & b \\ d & e \end{pmatrix},$$

which was formed by removing row 3 and column 3 from $\mathbf{M}$. □

We can generalize the result of Example 5.4.5 as a theorem:

Theorem 5.4.2 (Decoupling of Eigenvalues)
Let m_{ij} be the entry in row i and column j of an $n \times n$ matrix $\mathbf{M}$. Suppose one of these conditions is met for some value of $k \in \{1, n\}$:

1. All entries of the form $m_{k,j}$ with $j \neq k$ are 0,
2. All entries of the form $m_{i,k}$ with $i \neq k$ are 0.

Then $m_{k,k}$ is an eigenvalue of $\mathbf{M}$ and the remaining $n - 1$ eigenvalues come from the $(n - 1) \times (n - 1)$ submatrix from which row k and column k have been omitted.

▶ **Example 5.4.6** The matrix

$$\mathbf{M} = \begin{pmatrix} a & b & c \\ d & 0 & 0 \\ 0 & 0 & i \end{pmatrix}$$

satisfies condition 1 of Theorem 5.4.2 with $k = 3$. Therefore, one of the eigenvalues is $\lambda = i$ and the other two come from the submatrix

$$\mathbf{M_3} = \begin{pmatrix} a & b \\ d & 0 \end{pmatrix}.$$

□

Check Your Understanding 5.4.2:
Execute steps 3–5 in Algorithm 5.4.1 to find an eigenvector for the eigenvalue $\lambda = 1$ for the matrix of Example 5.4.6 if $a = 3, b = -3, c = 1, d = 2$, and $i = 1$. It is most convenient to choose $x_1 = 1$ in step 4.

5.4.3 Long-Term Behavior

The eigenvalues and eigenvectors of a matrix $\mathbf{M}$ can be used to determine the set of all solutions to the matrix equation $\mathbf{x_{t+1}} = \mathbf{M}\mathbf{x_t}$. The interested reader can find the method in any book on difference equations. In ecological modeling, we are usually concerned only with the general characteristics of models, rather than the detailed results of specific simulations. Among the important general characteristics is the long-term behavior, which is the term in the solution formula that is most important as $t \to \infty$. The results are summarized in a theorem.

Theorem 5.4.3 (Long-Term Behavior of Discrete Linear Population Models)
Given a matrix $\mathbf{M}$ with no negative entries, solutions of $\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$ eventually approach

$$\mathbf{x}_t = c_1 \lambda_1^t \mathbf{v}$$

for some value of c_1, where λ_1 is the largest positive eigenvalue of $\mathbf{M}$ and $\mathbf{v}$ is the corresponding eigenvector. This eigenvalue determines the long-term growth rate and the corresponding eigenvector determines the long-term solution ratios.[16]

▶ **Example 5.4.7** Let

$$\mathbf{M} = \begin{pmatrix} 0 & 104 & 160 \\ 0.01 & 0 & 0 \\ 0 & 0.3 & 0 \end{pmatrix}.$$

The dominant eigenvalue $\lambda = 1.2$ and the corresponding eigenvector were found in Example 5.4.3. From Theorem 5.4.3, we have the qualitative result: the long-term behavior of the model is growth at a rate of 20 % with the stable age distribution of 480 : 4 : 1. □

Check Your Understanding Answers

1. $\mathbf{v} = c_2 \begin{pmatrix} -10 \\ 1 \end{pmatrix}$. This is not the only possible answer. Any 2-vector for which the first component is -10 times the second component works equally well.
2. The vector $(1, 2, 4)$ is an eigenvector.

Problems

Find the eigenvalue of largest magnitude and a corresponding eigenvector for each of the matrices in Problems 5.4.1–5.4.6.

5.4.1*

$$\mathbf{M} = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$

[16] Some of the conclusions of the theorem must be generalized for matrices with negative entries. In particular, there could be complex eigenvalues, in which case it is the eigenvalue or eigenvalue pair with largest real part that determines the long-term behavior. It is also possible to have eigenvectors with different component ratios for the same eigenvalue, which can occur for an eigenvalue λ_1 when the factored characteristic polynomial has factor $(\lambda - \lambda_1)^n$ for $n > 1$. Details can be found in any book on difference equations.

5.4.2

$$\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}$$

5.4.3*

$$\mathbf{M} = \begin{pmatrix} 1 & a & b \\ 0 & 2 & c \\ 0 & 0 & 3 \end{pmatrix}$$

5.4.4

$$\mathbf{M} = \begin{pmatrix} 4 & b & -1 \\ 0 & 1 & 0 \\ 6 & c & -1 \end{pmatrix}$$

5.4.5

$$\mathbf{M} = \begin{pmatrix} 0 & 9 & 12 \\ 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix}$$

5.4.6

$$\mathbf{M} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & 0 \\ 2 & 2 & 0 \end{pmatrix}$$

5.4.7 (Continued from Problem 5.3.5.)

(a) Derive the polynomial equation for the eigenvalues of the matrix model of Problem 5.3.5.
(b) Determine the inequality that the parameters must satisfy for a growth rate of at least 1.

5.4.8 (Continued from Problem 5.3.6.)

(a) Derive the polynomial equation for the eigenvalues of the matrix model of Problem 5.3.6.
(b) Determine the inequality that the parameters must satisfy for a growth rate of at least 1.

5.4.9 [Peregrine falcons]

(a) Write the model of Sect. 5.2 in matrix-vector form.
(b) Derive the polynomial equation for the eigenvalues of the matrix in part (a).
(c) Determine the inequality that the parameters must satisfy for a growth rate of at least 1.
(d) Compare your answers with Eqs. (5.2.12) and (5.2.13).

5.5 Case Study: Loggerhead Turtles

One of the best-known matrix population projection models in conservation biology is a study of loggerhead sea turtles (*Caretta caretta*) by Larry B. Crowder, Deborah T. Krause, Selina S. Heppell, and Thomas H. Martin, published in 1994 [4]. The compartment diagram for the model is in Fig. 5.5.1. It shows a model with five stages: hatchlings, small juveniles, large juveniles, subadults, and adults. Each stage has parameters r_j for recruitment to the next stage, s_j for survival without recruitment, and m_j for mortality, with $s_1 = r_5 = 0$. Note that all individuals fit into one of these streams,[17] so

[17] The return streams with parameters s_j could have been omitted from the diagram, but it is advantageous to include them because those terms appear explicitly in the model equation.

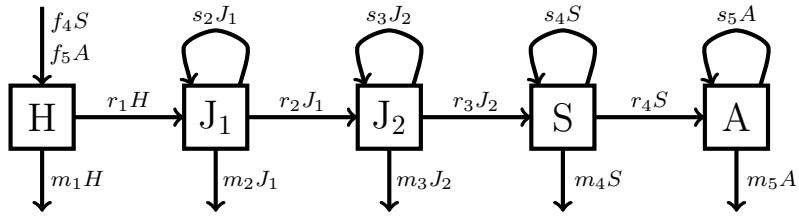


Fig. 5.5.1 Loggerhead sea turtle life history. The f parameters represent production of new hatchlings, the r parameters represent recruitment from one stage to the next, the s parameters represent survival in a stage without moving on to the next stage, and the m parameters represent mortality

$$r_j + s_j + m_j = 1\,. \tag{5.5.1}$$

From the diagram, we can identify the matrix for the stage-structured model as

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & f_4 & f_5 \\ r_1 & s_2 & 0 & 0 & 0 \\ 0 & r_2 & s_3 & 0 & 0 \\ 0 & 0 & r_3 & s_4 & 0 \\ 0 & 0 & 0 & r_4 & s_5 \end{pmatrix}. \tag{5.5.2}$$

The m_j do not appear explicitly in the model, but we will think of these as fundamental parameters from which r_j and s_j are calculated; this will allow us to easily modify the model parameters to account for alternate assumptions about mortality. We will also want to define the ratios of recruitment to survival, or "advancement" fractions, as fundamental parameters. These are given by

$$a_j = \frac{r_j}{r_j + s_j}\,. \tag{5.5.3}$$

If m_j and a_j are known, then we can obtain formulas for r_j and s_j from (5.5.1), (5.5.3):

$$r_j = a_j(1 - m_j)\,, \qquad s_j = (1 - a_j)(1 - m_j)\,. \tag{5.5.4}$$

5.5.1 Status Quo for South Carolina Loggerheads in 1994

As of 1994, estimates of vital rates for loggerhead turtles in South Carolina [4] yielded the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 4.665 & 61.896 \\ 0.675 & 0.703 & 0 & 0 & 0 \\ 0 & 0.047 & 0.657 & 0 & 0 \\ 0 & 0 & 0.019 & 0.682 & 0 \\ 0 & 0 & 0 & 0.061 & 0.809 \end{pmatrix}.$$

The numbers give us a sense of the challenge for turtle populations. About 2/3 of hatchlings survive to become small juveniles. Each year, about 3/4 of the small juveniles survive, but only a tiny fraction of these become large juveniles. The rest of the survivors return as small juveniles, and again experience a loss of about 1/4 with only a small number advancing to the next stage. Thus, the overall probability of survival to adulthood is very low. In exchange for that, adults have an 80% survival probability and an average of 62 offspring per year of survival. The question is whether this is enough to compensate

for the low number of adults. Given the numerical values of the parameters, the answer is no. The dominant eigenvalue is $\lambda_1 = 0.95$, indicating a population decline of about 5% each year.

5.5.2 A Model that Accounts for Trawler Mortality

Crowder and colleagues were interested in identifying possible ways to stabilize the sea turtle population. They collected field data that showed that approximately half of the mortality of the large juvenile, subadult, and adult stages was due to incidental capture and drowning in shrimp trawls [4]. This observation suggested that the turtle population could be saved by requiring shrimp trawlers to install turtle excluder devices. Of course the shrimp industry did not want such a regulation because of the cost. From a public good perspective, the question came down in part to how much of a difference turtle excluder devices could make. Crowder and colleagues showed that a two-thirds reduction in trawling mortality would be sufficient to restore and stabilize the turtle population.

To assess Crowder's claim and consider nuanced differences, we need to think of m_j and a_j as the primary scenario parameters and use them to calculate the model parameters r_j and s_j from (5.5.3). We introduce some additional formal notation to make the logic easier to follow.

1. Let T_j be the fraction of the total mortality m_j that is due to trawlers in the absence of turtle excluder devices. Following Crowder et al, we'll assume $T_j = 0.5$ for $j = 3, 4, 5$ and $T_j = 0$ for $j = 1, 2$ for most of our experiments, but the model will allow for these parameters to take other values for different scenarios.
2. Let x_j be the fractional decrease in trawler mortality due to modifications in the shrimp industry. These are the parameters we will use to define scenarios. Note that we only need an x_j when $T_j > 0$; hence, there are three study parameters in the base scenario.
3. For each parameter m_j, r_j, s_j, we add a "hat" accent mark to indicate the values from Crowder's data. The versions of these quantities without hats will be functions of x_j. We assume that a_j is independent of mortality changes, so the hats are not needed for them.

From the published data, we have

$$\hat{\mathbf{r}} = [0.675 \;\; 0.047 \;\; 0.019 \;\; 0.061 \;\; 0]\,,$$
$$\hat{\mathbf{s}} = [0 \;\; 0.703 \;\; 0.657 \;\; 0.682 \;\; 0.809]\,.$$

Then from (5.5.1) and (5.5.3), we have

$$\hat{\mathbf{m}} = [0.325 \;\; 0.250 \;\; 0.324 \;\; 0.257 \;\; 0.191]\,, \tag{5.5.5}$$
$$\mathbf{a} = [1 \;\; 0.0627 \;\; 0.0281 \;\; 0.0821 \;\; 0]\,. \tag{5.5.6}$$

With all of the numerical data in hand, it remains to establish the connections between the variable parameters r_j and s_j that are needed for the model with the study parameters x_j. To that end, we note that the mortality for any stage is divided into two portions: the trawler mortality in the base case is $T_j\hat{m}_j$ and the natural mortality is $(1 - T_j)\hat{m}_j$. Adding these together gives the total mortality $\hat{m}_j$ when $x_j = 0$. If we are able to reduce trawler mortality by a fraction x_j of its base value, then the new trawler mortality is $(1 - x_j)T_j\hat{m}_j$. Adding this new trawler mortality to the natural mortality gives us a total mortality of

$$m_j = (1 - T_j)\hat{m}_j + (1 - x_j)T_j\hat{m}_j = (1 - x_jT_j)\hat{m}_j\,. \tag{5.5.7}$$

The model parameters r_j and s_j can now be calculated by combining (5.5.7) with (5.5.4).

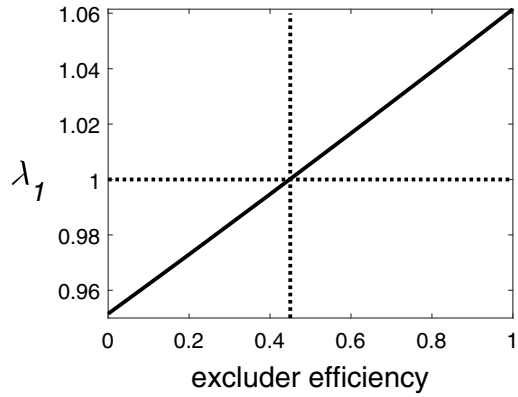


Fig. 5.5.2 The dependence of the long-term sea turtle growth rate on turtle excluder efficiency

While we have done more work than was minimally necessary, we have built a more general model that can make the default trawler loss and the value of turtle excluders depend more fully on the life stage. More generally, we could apply the model to any stage-structured setting in which mortality can be divided into a portion that can be controlled and a portion that can't.

5.5.3 A Simple Experiment to Test the Value of Turtle Excluder Devices

For a simple experiment, suppose we assume that there is a common value of x that applies to all stages that are affected by trawling, so that we have a model in which the single study parameter x determines the outcome, along with the fixed values of T_j, $\hat{m}_j$, and a_j. The growth rate λ_1 is then a function of the turtle excluder efficacy x. This function can be evaluated with a MATLAB program that creates the matrix in terms of the parameters and then calculates the dominant eigenvalue. Figure 5.5.2 shows the results. Although the dependence of the matrix entries on the parameter x is not linear, the effect of x on the growth rate is approximately linear. The turtle excluders are enough to stabilize the turtle population if they reduce trawler mortality by about 45%. A two-thirds reduction is enough to allow for modest growth of a little better than 2% per year.

A slightly more complicated experiment is considered in Problem 5.5.1.

Problems

Problems marked with "p" require some programming beyond data entry.

5.5.1 p

(a) Write a program to calculate the dominant eigenvalue as a function of the turtle excluder efficiency. The program *LinSys.m* is a good starting point, as it contains the necessary MATLAB instructions to find eigenvalues.
(b) Check your program by reproducing the results of Fig. 5.5.2.
(c) Modify the program to account for the possibility that turtle-excluding devices only work for small and full-size adults; in other words, that $x_3 = 0$.
(d) Write a brief summary report that combines the findings in the case study with your findings in (c).

5.5.2 Look up literature on loggerhead sea turtle populations after 1994. Have turtle excluder devices been implemented? In either case, to what extent have the conclusions of Crowder and colleagues turned out to be correct?

5.6 Case Study: Phylogenetic Distance

Based on obvious physical differences between humans and the other great apes, taxonomists once thought that humans diverged from that family relatively early, prior to the division between chimpanzees and gorillas. We now know that a combined human-chimpanzee lineage split from the gorilla lineage before humans split from chimpanzees. This conclusion is based on the estimation of *phylogenetic distance*, which is a measure of the accumulated DNA differences between species.

While it is easy to say that the human and chimpanzee genomes have greater differences with the gorilla genome than they have with each other, any attempt to determine the dates at which lineages diverged requires a mathematical model. The simplest such models use equations of the form

$$\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$$

to track dynamic changes in probabilities. In this case study, we consider the problem of how to use the number of DNA differences we can measure to estimate the number of DNA changes that have occurred in the divergence of two species from a common ancestor.

5.6.1 Some Scientific Background

DNA is a polymer, which means that it is a chain of indeterminate length composed of individual units. Each of these individual units of DNA incorporates one of four functional groups, called **nucleotides**: adenine (A), guanine (G), cytosine (C), and thymine (T). The nucleotides function as letters in an alphabet; that is, they have no individual meaning but combine together into three-letter "words" called **codons**. The codons follow one after another in a long run-on sentence, called a **chromosome**. The full collection of chromosomes for an individual constitutes that individual's **genome**. The codon "dictionary" appears in Table 5.6.1.

Because the codons are run together in chromosomes, the DNA "language" requires "punctuation" to indicate the start and end of a functional unit, which we can think of as a gene. The beginning of a functional unit is marked by a special pattern that is independent of the codon dictionary, while the end of a functional unit is marked by special stop codons.

With 4 nucleotides taken in sets of 3, there are $4^3 = 64$ distinct codons. These represent only 21 units of meaning, consisting of the 20 amino acids and the stop marker. Hence, redundancy is inevitable. This redundancy of codons is not uniform; for example, there are six different ways to "spell" leucine, arginine, and serine, but only one way to "spell" methionine and tryptophan. The first two nucleotides in a codon are more likely to matter than the third one. Out of the 16 possible combinations of the first two nucleotides, there are 8 cases (TC, CT, CC, CG, AC, GT, GC, GG) in which the same amino acid is obtained no matter what the third nucleotide is. This redundancy means that there is not a direct connection between nucleotide change and change in organism function; some nucleotide changes merely change the "spelling" of the same "word." A second reason why the connection is not direct is that there is a one in four chance that two consecutive changes in a nucleotide will only serve to restore the original.

Table 5.6.1 The 64 DNA codons

Codon	Meaning	Codon	Meaning	Codon	Meaning	Codon	Meaning
TTT	Phenylalanine	TCT	Serine	TAT	Tyrosine	TGT	Cysteine
TTC	Phenylalanine	TCC	Serine	TAC	Tyrosine	TGC	Cysteine
TTA	Leucine	TCA	Serine	TAA	Stop	TGA	Stop
TTG	Leucine	TCG	Serine	TAG	Stop	TGG	Tryptophan
CTT	Leucine	CCT	Proline	CAT	Histodine	CGT	Arginine
CTC	Leucine	CCC	Proline	CAC	Histodine	CGC	Arginine
CTA	Leucine	CCA	Proline	CAA	Glutamine	CGA	Arginine
CTG	Leucine	CCG	Proline	CAG	Glutamine	CGG	Arginine
ATT	Isoleucine	ACT	Threonine	AAT	Asparagine	AGT	Serine
ATC	Isoleucine	ACC	Threonine	AAC	Asparagine	AGC	Serine
ATA	Isoleucine	ACA	Threonine	AAA	Lysine	AGA	Arginine
ATG	Methionine	ACG	Threonine	AAG	Lysine	AGG	Arginine
GTT	Valine	GCT	Alanine	GAT	Aspartate	GGT	Glycine
GTC	Valine	GCC	Alanine	GAC	Aspartate	GGC	Glycine
GTA	Valine	GCA	Alanine	GAA	Glutamate	GGA	Glycine
GTG	Valine	GCG	Alanine	GAG	Glutamate	GGG	Glycine

We can think of the DNA in a genome as falling broadly into three categories:

1. *Essential* DNA is in the form of genes that are crucial to species survival, such as the genes that determine the network of blood vessels or the function of organs. These genes are largely resistant to change because such changes from the norm tend to be harmful. The corresponding DNA may be different between species, but will likely be almost the same for individuals of the same species.
2. Some DNA is in the form of genes that play at best a small role in species survival, such as the genes that determine hair color. The corresponding DNA shows significant variation within a population. This DNA is useful for the identification of individuals in a species.
3. There is also *non-essential* DNA, which does not affect the characteristics of the organism but is merely a residue of the evolutionary past. At one time it was thought that most DNA is non-essential, but scientists now estimate that this category typically encompasses about 20 % of a genome.

The genome of a species can be thought of as being defined by the combination of its essential and non-essential DNA. Although these portions are largely inherited intact from one's parents, there are two important processes that cause them to change over time: natural selection and mutation. Essential DNA is subject to natural selection. If there are individual variations in a portion of this DNA, then some individuals will be more successful at survival and reproduction than others; over time, the population will be dominated by those individuals who have the more successful variation.[18] In contrast, non-essential DNA is not subject to natural selection.

Natural selection must, of course, have individual variation to work with. For organisms that reproduce sexually, this variation results from genetic mutations that occur in the production of sperm and egg cells. Mutations are rare events, and those that alter the individual's fitness are either removed by natural selection or gradually replace earlier versions. Successful mutations and mutations in non-

[18] An example of this is the delta variant of the SARS-CoV-2 virus that causes COVID-19. This variant arose over time through a sequence of mutations and came to outcompete the original variant and other competing variants because it was better at infecting new hosts. Subsequently, the delta variant was replaced by the omicron variant, which had the selection advantage of being resistant to both the vaccines and immunity acquired from older variants.

essential DNA can accumulate over evolutionary time. This is what allows us to create a concept of phylogenetic distance based on differences in similar regions of DNA between species. While there is not a simple linear relationship between number of mutations and evolutionary time, it seems reasonable that more evolutionary time should result in more mutations. Thus, a larger difference between species A and B than between species B and C indicates a more recent common ancestor for B and C. The full story is actually much more complicated.

1. Substitutions appear to account for only 35–50 % of mutations [6]. There are a number of other types of mutations, the most common being insertions and deletions, in which a small bit of DNA is inserted between two formerly adjacent nucleotides, or a small bit is lost from a section of DNA. These mutations are much harder to identify over long periods of evolutionary time and harder to quantify.[19] Methods that consider only substitutions can only be used on portions of a chromosome in which any insertions or deletions are known.
2. Non-essential DNA is subject to mutation without natural selection, which raises the question of how mutations in non-essential DNA could be identical for individuals of the same species.
3. Natural selection occurs at the level of genome function, not genome structure. Changes in individual nucleotides do not change the function in cases where both the new and old codons make the same protein.
4. Natural selection is based on the preferential survival of some mutations over others. The accumulation rate of a mutation depends on the frequency of mutations and the amount of survival difference the mutation makes. We need a new vaccine for influenza every year, but the measles vaccine is the same now as it was when first created.[20] Hence, the molecular clock that connects mutations with time does not tick at a constant rate across species or even within species. The molecular clock is close to constant for species that are closely related and for genome portions that have the same or no function.

5.6.2 A Model for DNA Change

Suppose we leave the details of what portion of a genome to study to the molecular biologists. Assume that there are J nucleotides in a strand that has had no insertions or deletions and let N be the unknown number of generations that have passed between the ancestral strand and the contemporary strand. For any position in the sequence, the nucleotide must be either A, G, C, or T. By comparing the ancestral and contemporary strands, we can measure the fraction of DNA sites that are different between the two. This value, commonly called β, is a measure of the difference between genomes.

Now let α be the probability of a mutation in one site over one generation. Over N generations, we expect the total number of mutations to be αN for each site, yielding a total of $\alpha N J$ for the strand. The product $d = \alpha N$ is the number of mutations per site. This is the phylogenetic distance, which we tentatively assume to be proportional to evolutionary time. Our goal is to infer d from β.

At first thought, this sounds easy. The total number of differences between the strands is βJ and the total number of mutations is $\alpha N J = d J$. These should be equal, so $d = \beta$. However, this reasoning is flawed. If a site starts as A, mutates to G, and then mutates back to A, with no further changes, then both of these mutations are counted toward dJ. However, the two strands are identical because the

[19] One extreme case similar to insertion and deletion can be seen in a comparison of the human and chimpanzee genomes. Humans have 23 pairs of chromosomes, while chimpanzees have 24, which seems to refute the claim that the two species are closely related. However, a careful study of human chromosome 2 shows evidence that it consists of two formerly distinct chromosomes that joined together, with each portion corresponding to one of two chimpanzee chromosomes.

[20] SARS-CoV-2 has a mutation rate that is higher than most viruses, but lower than influenza, suggesting that we will probably need annual COVID-19 vaccination boosters but that these will be more effective than the annual influenza vaccines.

second mutation reversed the first one, so neither of them contributes to βJ. Thus, $d > \beta$, because some mutations actually *decrease* the number of differences between the strands. We need a nuanced mathematical model to connect the unknown phylogenetic distance with the known fraction of sequence differences.

Let $p_A(n)$, $p_G(n)$, $p_C(n)$, and $p_T(n)$ be the probabilities of having each given nucleotide at a particular site in the nth generation. Mutations from one generation to the next change these probabilities, and we must quantify these changes. The simplest assumption is that all possible changes are equally likely. Since α is the probability of change, and each nucleotide has three possible changes, the probability of any particular change is $\alpha/3$. Of course the probability of no change is $1 - \alpha$. With these assumptions, the probability that a site will contain the nucleotide A at time $n + 1$ is the sum of the probabilities of starting with A and not changing plus the probabilities of starting with one of the others and then changing to A:

$$p_A(n+1) = (1-\alpha)p_A(n) + \frac{\alpha}{3}p_G(n) + \frac{\alpha}{3}p_C(n) + \frac{\alpha}{3}p_T(n). \tag{5.6.1}$$

Similar equations can be written for the other probabilities in generation $n + 1$, and the four equations can be combined into a single matrix equation of the form

$$\mathbf{x}_{n+1} = \mathbf{M}\mathbf{x}_n. \tag{5.6.2}$$

Definition 5.6.1

A **Markov chain** is a model in which a vector of probabilities is modified over time through multiplication by a transition matrix $\mathbf{M}$.[21]

Each row in the matrix corresponds to the coefficients in one of the four equations. Since we have arbitrarily chosen the order A, G, C, T, the coefficients of (5.6.1) are the first row of the transition matrix $\mathbf{M}$. The full probability vector $\mathbf{x}$ and the matrix $\mathbf{M}$ are given by

$$\mathbf{x} = \begin{pmatrix} p_A \\ p_G \\ p_C \\ p_T \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} 1-\alpha & \alpha/3 & \alpha/3 & \alpha/3 \\ \alpha/3 & 1-\alpha & \alpha/3 & \alpha/3 \\ \alpha/3 & \alpha/3 & 1-\alpha & \alpha/3 \\ \alpha/3 & \alpha/3 & \alpha/3 & 1-\alpha \end{pmatrix}. \tag{5.6.3}$$

The specific model we are examining, defined by the assumption that the transition probabilities are all the same, is called the *Jukes–Cantor* model.[22]

[21] Given that we have always done matrix-vector multiplication with the matrix on the left and the vector on the right, as in (5.6.2), this is the natural way to proceed. Unfortunately, most of the literature on Markov chains makes the opposite choice. In our matrix $\mathbf{M}$, the entry in row i and column j represents the probability of a transition from state j to state i. In the more common representation of Markov chains, the matrix is written so that the entry in row i and column j represents the probability of a transition from state i to state j. This sounds more natural, but it means that the probability vectors must be written as rows rather than columns and the matrix multiplication must have the vector on the left. This necessitates changes in the definition of eigenvectors, which is an unfortunate complication that we choose to avoid with our choice of mathematical structure.

[22] Other models make more sophisticated assumptions about the relative probabilities of specific substitutions. The Jukes–Cantor model illustrates the important features of Markov chain models and phylogenetic distance while keeping complications to a minimum.

5.6.3 Equilibrium Analysis of Markov Chain Models

There are some fundamental mathematical similarities and differences between the matrices obtained in structured population models and those obtained in Markov chain models. These lead to some differences in the features of the corresponding dynamical systems.

1. Structured population models have matrices in which the ij entry represents the contribution of population component j at time n to population component i at time $n+1$. Thus, none of the entries can be negative. Nonnegative matrices have three special properties: (1) the eigenvalue of largest magnitude is always positive, (2) there is a one-parameter family of eigenvectors corresponding to this dominant eigenvalue, and (3) the eigenvector corresponding to the dominant eigenvalue is positive. These properties guarantee that solutions will approach an asymptotic growth rate that corresponds to a stable distribution of component populations.
2. Markov models also have nonnegative entries, so they inherit the properties in item 1. They have additional structure due to the fact that the ij entry represents the probability of being in state i at time $n+1$ after having been in state j at time n. Thus, each entry is between 0 and 1. Moreover, there is always exactly one state at the end of each time step, so the total of the probabilities for any time step must be 1. This means that the sum of entries in each column is 1. This additional structure guarantees that the dominant eigenvalue is $\lambda = 1$.[23] This means that Markov models generally have an equilibrium solution that represents a stable distribution of probabilities.

▶ **Example 5.6.1** Let $\mathbf{M}$ be the matrix of (5.6.3). Finding eigenvalues of a general matrix this size is outside the scope of our presentation; however, we can start with the assumption that $\lambda = 1$ is an eigenvalue. If $\mathbf{x}$ is an eigenvector corresponding to $\lambda = 1$, then it satisfies the equation

$$(\mathbf{M}-\mathbf{I})\mathbf{x} = \mathbf{0}\,, \qquad \mathbf{M}-\mathbf{I} = \begin{pmatrix} -\alpha & \alpha/3 & \alpha/3 & \alpha/3 \\ \alpha/3 & -\alpha & \alpha/3 & \alpha/3 \\ \alpha/3 & \alpha/3 & -\alpha & \alpha/3 \\ \alpha/3 & \alpha/3 & \alpha/3 & -\alpha \end{pmatrix}.$$

The components of $\mathbf{x}$ must satisfy a system of four equations, each corresponding to a row of $\mathbf{M}-\mathbf{I}$. Such a system would normally be difficult to solve, but here we can observe that the entries in each row sum to 0. If all four components of the vector are the same, then the products of coefficients and components will also sum to 0. The stable distribution of probabilities has to be an eigenvector, and as a set of probabilities it also has to sum to 1, which means that each probability is 1/4. This should not be surprising, as the symmetry in the rule that determines the probability of each possible mutation represents a process in which none of the nucleotides is favored over the others. □

5.6.4 Analysis of the DNA Change Model

The initial goal of our analysis is to connect the measured value of β with the phylogenetic distance $d = \alpha N$. The equilibrium distribution discovered in Example 5.6.1 is of no help in accomplishing this goal; by definition, this is the distribution we expect to see as $N \to \infty$. Instead, we proceed by a method that follows the strategy of calculating a quantity in two different ways, one involving β and the other involving α and N. The method requires us to use another eigenvector in addition to the one for $\lambda = 1$. This calculation is beyond the scope of our treatment, so we simply present the result.

[23] There are some additional requirements that guarantee these properties; further discussion of this topic is outside the scope of this presentation.

The vectors

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{v_2} = \begin{pmatrix} 3 \\ -1 \\ -1 \\ -1 \end{pmatrix}$$

are eigenvectors of the matrix $\mathbf{M}$ of (5.6.3) corresponding to the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 1 - \frac{4}{3}\alpha$.

Check Your Understanding 5.6.1:

Verify that the vector $\mathbf{v_2}$ is an eigenvector of $\mathbf{M}$ corresponding to the eigenvalue $\lambda_2(\alpha) = 1 - \frac{4}{3}\alpha$.

Define the vector $\mathbf{u}$ by[24]

$$\mathbf{u} = \mathbf{M}^N(\mathbf{v_1} + \mathbf{v_2}). \tag{5.6.4}$$

We now proceed to calculate $\mathbf{u}$ by two different methods, taking advantage of two facts:

1. $\mathbf{v_1}$ and $\mathbf{v_2}$ are eigenvectors, which means that multiplication by $\mathbf{M}$ yields a simple result.
2. The sum $\mathbf{v_1} + \mathbf{v_2}$ is also very simple.

Calculating u in Terms of N and α

The calculation of $\mathbf{u}$ is somewhat tedious, so we leave much of it as a problem. The essential idea is that repeated use of the eigenvector equation $\mathbf{Mv} = \lambda\mathbf{v}$ leads to a more general result,

$$\mathbf{M}^N\mathbf{v} = \lambda^N\mathbf{v}, \tag{5.6.5}$$

with which we eventually obtain the answer[25]

$$\mathbf{u} = \begin{pmatrix} 1 + 3\lambda_2^N \\ 1 - \lambda_2^N \\ 1 - \lambda_2^N \\ 1 - \lambda_2^N \end{pmatrix}, \quad \lambda_2 = 1 - \frac{4}{3}\alpha. \tag{5.6.6}$$

Estimating u in Terms of β

The matrix $\mathbf{M}^N$ represents the overall transition probabilities for N successive generations. We can't calculate this matrix directly, but we can estimate it. Given that β is the measured fraction of sites that have changed nucleotides over N generations, we can approximate $\mathbf{M}^N$ by

$$\mathbf{M}^N = \begin{pmatrix} 1-\beta & \beta/3 & \beta/3 & \beta/3 \\ \beta/3 & 1-\beta & \beta/3 & \beta/3 \\ \beta/3 & \beta/3 & 1-\beta & \beta/3 \\ \beta/3 & \beta/3 & \beta/3 & 1-\beta \end{pmatrix}. \tag{5.6.7}$$

[24] There is no obvious reason why this should be helpful. It is always more satisfying when methods have a clear conceptual motivation, but occasionally mathematicians must resort to methods that appear simply as clever tricks.

[25] Problem 5.6.1a.

This is not entirely correct, as it assumes both that the fraction of changed sites is the same, no matter what the original nucleotide, and that the changed sites are equally distributed among the three possible nucleotides. These assumptions are no worse than the basic Jukes–Cantor assumption about the structure of **M**, however. Combining (5.6.4) and (5.6.7) yields the result

$$\mathbf{u} = \frac{4}{3}\begin{pmatrix} 3(1-\beta) \\ \beta \\ \beta \\ \beta \end{pmatrix}. \tag{5.6.8}$$

The Jukes–Cantor Distance

Equations (5.6.6) and (5.6.8) provide two different results for the same quantity. Comparing them yields the equation

$$\frac{4}{3}\beta = 1 - \lambda_2^N = 1 - \left(1 - \frac{4}{3}\alpha\right)^N, \tag{5.6.9}$$

which predicts the fraction of sites with changes in terms of the mutation rate and the number of generations. We can solve this equation for N, with the elegant result

$$N = \frac{\ln\left(1 - \frac{4}{3}\beta\right)}{\ln\left(1 - \frac{4}{3}\alpha\right)}, \tag{5.6.10}$$

from which we have

$$d = \alpha\frac{\ln\left(1 - \frac{4}{3}\beta\right)}{\ln\left(1 - \frac{4}{3}\alpha\right)} = \frac{\alpha}{\ln\left(1 - \frac{4}{3}\alpha\right)}\ln\left(1 - \frac{4}{3}\beta\right). \tag{5.6.11}$$

This result still appears to depend on α, which is difficult to measure. In practice, this dependence is meaningless. Given the realistic assumption that α is very small, we can approximate[26] the Jukes–Cantor distance as

$$d = -\frac{3}{4}\ln\left(1 - \frac{4}{3}\beta\right). \tag{5.6.12}$$

This simple result is a reasonable approximation of the amount of genetic change corresponding to a particular net substitution probability β. The properties of this function match reasonable expectations.[27] It increases as β increases, with $d \approx \beta$ if β is small and $d \to \infty$ as $\beta \to 3/4$.[28]

Problems

5.6.1 (a) Derive (5.6.5) and (5.6.6).

(b) Use linear approximation to derive (5.6.12) from (5.6.11).

5.6.2(a) The Jukes–Cantor phylogenetic distance function

$$d = -\frac{3}{4}\ln\left(1 - \frac{4}{3}\beta\right).$$

[26] Problem 5.6.1b.

[27] Problem 5.6.2.

[28] Note that $\beta = 3/4$ means that the system has reached equilibrium; theoretically this requires infinite time.

can be rearranged to calculate the probability of a nucleotide difference β for a site in terms of the total number of mutations d. Plot that function.

(b) Use linear approximation to show that $\beta \approx d$ for small genome changes. Why does this make sense?

(c) Compute $\lim_{d\to\infty} \beta$. Explain the meaning of the result.

5.6.3* Let $\mathbf{M}$ be a 2×2 Markov chain matrix with entries a and b as shown below.

$$\mathbf{M} = \begin{pmatrix} & a \\ b & \end{pmatrix}.$$

(a) Fill in the blanks to complete the matrix.

(b) Show that $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector for $\lambda = 1$ if and only if the entries in each row of $\mathbf{M}$ sum to 1. What must be true about a and b in this case?

5.6.4 The Kimura model of genetic change assumes that the rates for the AG, GA, CT, and TC substitutions are faster than those for the other substitutions. (There is a biochemical basis for why this should be the case.)

(a) Construct the matrix $\mathbf{M}$ for the Kimura model, using α for the faster rate and γ for the slower rate.

(b) Show that $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ is an eigenvector for the Kimura model for $\lambda = 1$ and conclude that all nucleotides are equally likely.

5.6.5 The Felsenstein model of genetic change assumes that rates of change depend on the nucleotide being changed to, but not the nucleotide being changed from. (This allows for the model to predict different overall nucleotide frequencies.)

(a) Assume that other nucleotides change to A at rate a, G at rate g, and so on. Construct the matrix $\mathbf{M}$.

(b) Show that $\begin{pmatrix} a \\ g \\ c \\ t \end{pmatrix}$ is an eigenvector for the Felsenstein model for $\lambda = 1$.

Projects

This chapter has four projects.

A. Cheetahs are among the most endangered mammal species on earth. A matrix model can be used to seek ways to save the wild cheetah populations, as with the loggerhead sea turtle study of Sect. 5.5.

B. Aphids are among the most interesting biological organisms because of a life history that seems to have been designed for explosive population growth. A matrix model shows some unusual outcomes.

C. Teasel plants have a much more complicated life history than any animals. A model has been constructed for them, but the matrix has 14 nonzero entries, each representing a life history process. Experimentation with the matrix can identify which of the 14 processes are of only minor significance, leading to a simplified life history that tells a simpler biological story.

D. There is no better way to build intuition for modeling than to try to model a scenario in which you have to collect your own data in order to determine the life history and estimate parameter values. While we cannot do this with real populations in a mathematics class, we can do it with virtual populations that live in computer software. The last project of this chapter introduces you to "boxbugs," a virtual population that inhabits—you guessed it—a "bugbox."

Project 5A: Cheetah Conservation

Matrix population projection models have been used to study the extinction risk and possible conservation strategies for the endangered Serengeti cheetah (*Acinonyx jubatus*) [3, 10]. The model used for these studies divides cheetah populations into age classes 0–6, 6–12, 12–18, 18–24, 24–30, 30–36, 36–42, and 42+ (months). Here we consider a five-component version in which all of the reproductive age groups (2 years and up) are combined together. We assume a fertility of 1.277 female offspring per adult female per 6-month period. The survival probabilities for the five age groups in the model are s_1, 0.771, 0.771, 0.920, and 0.888, respectively. (We leave s_1 unspecified to allow for a variety of scenarios.) Assume all surviving individuals in the first four groups move to the next group in the following 6-month period, but survivors in the oldest group remain in that group.

(a) Write down the matrix $\mathbf{M}$ that represents the model.
(b) Use computer software to determine the largest eigenvalue for the case $s_1 = 0.081$, which is the value estimated in [3]. What does this result suggest about the survival chances of the population?
(c) Given arbitrary s_1, write the equations that the components of the eigenvector $\mathbf{v}$ must satisfy if the largest eigenvalue is 1, representing the case where the population is just barely viable. Assuming that the adult population is 1, the system can be solved to determine the other populations and the value of s_1 necessary for viability. Determine this value of s_1.
(d) Use computer software to plot the largest eigenvalue for $0.08 \leq s_1 \leq 0.2$.
(e) Discuss the outlook for the Serengeti cheetah in the wild, based on the results of your investigation of this model.

Project 5B: Pea Aphids

The pea aphid *Acyrthosiphon pisum* exhibits rapid population growth because of its unusual life history.[29] Asexual females hatch from eggs in the spring and find an annual host plant on which to found new colonies. For the remainder of the growing season, the colony consists of wingless asexual females that reproduce by cloning and are born live.[30] Sexual morphs are produced at the end of the growing season; they fly to trees, where they mate and lay eggs that will hatch into founders of new colonies in the next year. Experiments conducted by teams of undergraduates at the University of Nebraska–Lincoln measured the vital rates of aphids and aphid population growth [9].

(a) A stage-based model for aphid population dynamics needs six stages: first-, second-, third-, and fourth-instar nymphs, young adults, and mature adults. Aphids progress through each of the first five stages in 1–3 days. Under ideal conditions, experiments have obtained daily survival probabilities of 0.974, 0.952, 0.961, 0.930, 0.955, and 0.903 for the six stages, respectively. The probabilities of surviving and also advancing to the next stage in 1 day are 0.421, 0.714, 0.538, 0.379, and 0.455 for the immature stages. There is also a small probability of 0.034 for fourth-instar aphids

[29] This life history is common among aphid species.

[30] Pea aphids are born pregnant, so the amount of time required for a newborn aphid to become a reproductive adult can be as little as 8 days.

to become mature adults in 1 day. Young adults have an average of 2.106 offspring per day, and mature adults average 3.630 offspring per day. Use this data to construct a matrix for the aphid population model. Note that the survival probabilities include individuals who stay in the same stage for the whole day as well as individuals who move to the next stage.

(b) Use computer software to determine the long-term growth rate predicted by the model and the fraction of each stage in a population growing at that long-term rate. In particular, what fraction of the population are first-instar nymphs and what fraction are adults?

(c) Use the matrix model to run a simulation of aphid population growth, given a starting population of one reproductive adult and running for 15 days. What is the total population after 15 days?

(d) Use the 15-day total and the long-term growth rate to determine how long it will take for the theoretical population to reach 10,000.

(e) Plot N_t/N_{t-1} versus time (starting at $t = 0$), where N_t is the total population at time t.

(f) Offer a biological explanation for the shape of the curve in part (e). In particular, explain why the growth rate oscillates and why the amplitude of the oscillation decreases over time, and connect the behavior of the graph at the end of the 2 weeks with the result of part (b).[31]

Project 5C: Teasel Plants

The teasel plant (*Dipsacus sylvestris*) has a complicated life cycle consisting of dormant seeds, rosettes (which are vegetative but do not flower), and flowering plants. Rosettes have been subclassified as small, medium, and large by field biologists,[32] and the germination probability of dormant seeds decreases with age. Thus, it makes sense to consider a stage-structured model with two classes of dormant seeds, three classes of rosettes, and one class of flowering plants [1]. Using data obtained by Werner and Caswell [11], the matrix representing teasel populations has been reported as

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 322 \\ 0.966 & 0 & 0 & 0 & 0 & 0 \\ 0.013 & 0.010 & 0.125 & 0 & 0 & 3.45 \\ 0.007 & 0 & 0.125 & 0.238 & 0 & 30.2 \\ 0.008 & 0 & 0 & 0.245 & 0.167 & 0.862 \\ 0 & 0 & 0 & 0.023 & 0.750 & 0 \end{pmatrix}.$$

This model serves as a good example of how careful examination of data and results can yield useful biological information.

(a) Describe the various components of the life history of the teasel plant, as indicated by the pattern of nonzero entries in the matrix. [Hint: Draw a life history graph with a row of nodes for D1, R1, R2, R3, and F and a node for D2 below this row and between D1 and R1. Then draw arrows to represent each possible transition other than deaths. This means that there will be one arrow for each nonzero matrix entry. It is (barely) possible to locate these arrows in such a way that none of them cross. You can then describe the life history in terms of the graph.]

(b) Determine the long-term stable growth rate and the associated ratios of each group population to the adult population. (For a large matrix such as this one, you will want to use the combined MATLAB operation "max(eig(M))".)

[31] Experiments corresponding to this scenario consistently show the behavior predicted by the model.

[32] These distinctions are largely arbitrary. Since size is a continuous variable, we could just as easily use two or four classes of rosettes. There are *integral projection* models that deal with continuous size structure and discrete time, but these models are far beyond the scope of this book. They are also impractical, unless there is an enormous amount of data on the effect of rosette size on the future of the plant.

(c) Note that second-year dormant seeds have a very low probability of germinating and are only capable of producing small rosettes. Suppose their germination probability is changed to 0. How does that change the long-term growth rate?
(d) Based on the result of part (c), it is clear that second-year dormant seeds make no measurable contribution to the teasel population. Reformulate the matrix to omit this group. Note that the new matrix will be 5×5.
(e) One of the other stages of the teasel plant makes almost no difference to the population dynamics. Try to determine which one this is without doing any calculations. Then check your guess by systematically removing one group at a time. As in part (d), removing one group corresponds to removing one row and one column from the matrix.
(f) Having removed two of the six stages in the model without appreciably affecting the results, we now have a 4×4 matrix with 10 nonzero entries. Check the importance of each of these 10 processes by setting its matrix entry to 0 and finding the resulting growth rate. You should find that only three of the processes are so important that leaving them out drops the growth rate below 2. (Note: Some matrix entries should be moved rather than merely set to 0. For example, the entry in row 4, column 2 represents small rosettes becoming flowering plants. If that process is omitted, it makes sense that those same small rosettes would become large rosettes instead.)
(g) Set all of the entries that were not identified as critical in (f) to 0. This will not be enough to model the plant; however, if you use this as a starting point, you should be able to find one additional entry that makes the growth rate almost 2 using just four of the original 14 processes.
(h) Describe the simplified life history from (g) that accounts for more than 80% of the teasel plant growth.

Project 5D: Boxbugs

BUGBOX-population[33] is a virtual biology laboratory that gives students experience in observing biological systems, formulating mathematical models, and collecting data to determine model parameters. The BUGBOX is populated by organisms called "boxbugs," which have a life history that makes them more suited to population dynamics experiments than any real insects.[34] All boxbugs are female. The three life stages—larvae, pupae, and adults—are distinct in appearance. All stages are immobile; the fact that they neither move nor even rotate makes it easy to identify an individual as it moves through its life. There are four species, each with a more complicated life history than the previous one. Each can be modeled by a system of equations that predict the populations of the three stages at time $t + 1$ in terms of the populations at time t. The details of these equations must be determined by observation.

(a) Run experiments to determine the correct model for Species 1. The equations are very simple, with only one parameter. Use the letter f to represent this parameter.
(b) Determine the correct models for Species 2, 3, and 4, using r, b, and s to represent the new parameters in turn.
(c) Complete the model for Species 4 by estimating the values of the parameters.
(d) Run a computer simulation for Species 4 for 20 time steps, assuming an initial population of 10 adults.
(e) Plot the ratios L_{t+1}/L_t, P_{t+1}/P_t, and A_{t+1}/A_t together on a common set of axes, starting at $t = 2$.
(f) Plot the ratios L_t/A_t and P_t/A_t together on a common set of axes.
(g) Based on the plots of parts (e) and (f), describe the long-term behavior that the model predicts.
(h) Determine the long-term growth rate and population ratios.

[33] The data sets for this problem can be found at https://www.math.unl.edu/~gledder1/.

[34] This software was written when the author was co-teaching an interdisciplinary research course. The research involved population dynamics of aphids and coccinellids (ladybird beetles). Boxbug biology combines the biology of these two real insect types.

(i) Run several virtual experiments for 20 time steps. Is the model reasonably accurate for small numbers of time steps? How about large numbers?
(j) What biological feature of the "real" boxbug population is not built into the model?

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Nonlinear Dynamical Systems

6

In Chap. 4, we studied the dynamics of a single variable. Now we look at the dynamics of nonlinear systems. The methods available to us are analogous to the methods we used for single-variable dynamics, but with some important differences. The phase line for one variable scales up to the phase plane for two variables; however, there is no graphical method for discrete systems. The analytical method for determining stability with the derivative scales up to higher dimensions, but with technical complications that rapidly increase as the number of variables increases. When possible, we can greatly simplify the analysis of a model by using an appropriate approximation to reduce the number of dynamic equations.

Section 6.1 presents nullcline analysis, a powerful tool for understanding systems of two dynamic variables in continuous time. This section is about double the length of most of the other sections in the book, but there is no reasonable place to break the material. We develop an analytical (symbolic) method for determining stability of equilibria for continuous dynamical systems in Sects. 6.2 and 6.3. The first of these sections uses the standard method based on finding eigenvalues, while the second uses the more powerful method of applying the Routh–Hurwitz conditions directly to the entries in the matrix representing the linearized system. We then present a case study of onchocerciasis, a previously neglected parasitic tropical disease that has been the focus of an eradication campaign, with less than stellar success. The case study builds on the fundamental skills of modeling, stability analysis, and asymptotic simplification, and concludes with a problem on assessing the feasibility of eradication with current strategies. Other problems in this section continue development of an HIV model presented in earlier problem sets and explore a model for a portion of the human immune system. The chapter concludes with a section on linearized stability analysis of discrete systems using the Jury conditions. Both the algebraic work and the system behavior are more complicated than the corresponding continuous case; hopefully, this will drive home to readers the importance of using discrete models only when the biological setting requires them.

There are six projects in this chapter. Project 6A analyzes the dynamics of the SIR model with logistic growth. Project 6B considers an epidemiological model with isolation of some symptomatic individuals, showing some unusual behavior. Project 6C analyzes a predator–prey model capable of predicting the oscillatory solutions that have been observed in some biological systems. Project 6D examines the role of trained macrophages (a type of white blood cell) in the immune system. Projects 6E and 6F look at two discrete models, one for host–parasitoid dynamics and one for a stage-structured population destabilized by cannibalism.

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6.1 Phase Plane Analysis

After studying this section, you should be able to:

- Plot solution curves in the phase plane using data from simulations.
- Interpret phase portraits of two-dimensional dynamical systems.
- Sketch nullclines for two-dimensional dynamical systems.
- Use nullclines to draw conclusions about the stability of equilibrium points.

In this section, we generalize phase line analysis for single equations[1] to phase plane analysis for systems of the form

$$\frac{dx}{dt} = f(x, y)\,, \tag{6.1.1}$$

$$\frac{dy}{dt} = g(x, y)\,. \tag{6.1.2}$$

Note that the time does not appear explicitly in the functions that prescribe the rates of change. Systems such as these, where the rates of change depend only on the state of the system, are called *autonomous*. This situation generally occurs when there are no factors external to the mathematical system, such as temperature, that vary in time. The properties of such systems are similar to single-variable autonomous equations. In particular, many systems progress over time toward asymptotically stable equilibrium solutions.

6.1.1 Solution Curves in the Phase Plane

Understanding the behavior of second-order autonomous systems (6.1.1)–(6.1.2) is facilitated by plotting solutions as curves in the two-dimensional space of state variables. This two-dimensional space is called the *phase plane*. The resulting plot is called a *phase portrait*.

▶ **Example 6.1.1** Figure 6.1.1 shows the time series plot (a) and phase portrait (b) for the dimensionless Michaelis–Menten system[2]

$$s' = -s(1 - z) + hz\,, \quad s(0) = 1\,, \tag{6.1.3}$$

$$\epsilon z' = s(1 - z) - hz - rz\,, \quad z(0) = 0\,, \tag{6.1.4}$$

where h, r, and ϵ are all positive and the prime represents the time derivative. Each point in the phase portrait corresponds to the values of the state variables at some time; in this way, the plot indicates progression in time, even though the time variable does not appear as a coordinate. Phase portraits need not show the direction of forward time, as that can be inferred from the differential equations or time history. Here, the time history shows that s is always decreasing, so the solutions must move in the direction indicated by the arrow. Adding arrows is recommended as long as they don't clutter the plot. □

[1] Section 4.3.

[2] Section 3.8.

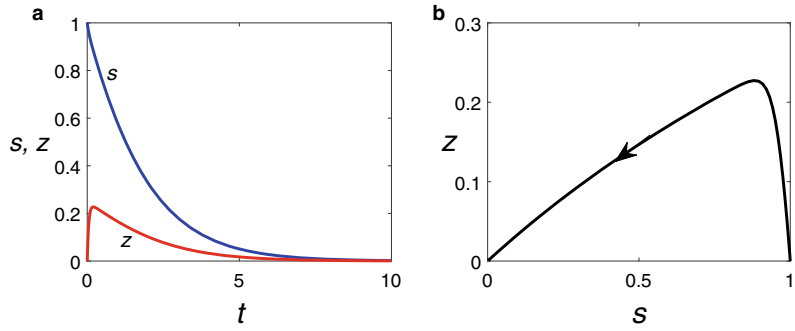


Fig. 6.1.1 The Michaelis–Menten system (6.1.3)–(6.1.4), with $h = 1$, $r = 2$, and $\epsilon = 0.2$, showing **a** the time series and **b** the phase portrait

Check Your Understanding 6.1.1:

(a) Identify the point in Fig. 6.1.1a that corresponds to the peak of the phase portrait in Fig. 6.1.1b.

(b) Does the solution move along the curve in Fig. 6.1.1b at constant speed?

6.1.2 Nullclines and Equilibria

In Fig. 6.1.1, we obtained a phase portrait by plotting a solution curve. The disadvantage of this method is that we have to select a specific set of parameter values for the simulation. Fortunately, it is also possible to get information about solutions by using graphical techniques that work in the general case. The technique is based on curves called *nullclines*.

Definition 6.1.1

A **nullcline** for a two-dimensional system is a curve in the phase plane comprising points at which one of the variables is not changing.

▶ **Example 6.1.2** In the dimensionless Michaelis–Menten system (6.1.3)–(6.1.4), the first variable s is unchanged when $s' = 0$, that is, when

$$-s(1 - z) + hz = 0\,.$$

This is the equation of the s-nullcline, which we can solve explicitly for z to get

$$z = \frac{s}{s + h}\,. \tag{6.1.5}$$

Similarly, the z-nullcline is given by

$$s(1 - z) - hz - rz\,,$$

or

$$z = \frac{s}{s + h + r}\,. \tag{6.1.6}$$

These nullclines are shown in Fig. 6.1.2. □

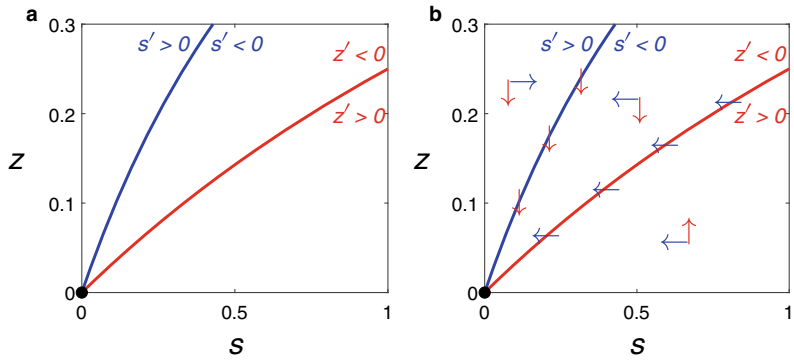


Fig. 6.1.2 The s (blue) and z (red) nullclines for the Michaelis–Menten model (using $h = 1$ and $r = 2$); **b** also shows direction arrows on the nullclines and in the three regions delineated by them. The equilibrium point at (0,0) is where the nullclines intersect. The arrows show directions determined by one of the differential equations: blue for the s equation and red for the z equation

Much of the analysis of autonomous systems is centered on finding the long-term fate of solutions, beginning with the identification of equilibrium points,[3] which can be found analytically or graphically.

Definition 6.1.2

Equilibrium points are points in a state space at which all derivatives are 0. Such points identify **equilibrium solutions**.

▶ **Example 6.1.3** Equilibria for the Michaelis–Menten system must satisfy the nullcline equations (6.1.5)–(6.1.6). Setting these formulas equal to each other yields the result $s = 0$ (given $r > 0$), whence also $z = 0$. Thus, the origin is the only equilibrium point for the system. Graphically, equilibria are at the intersection of nullclines of different "colors" (figuratively if not also literally), as shown in Fig. 6.1.2. □

The value of nullclines stems not only from their use in marking equilibria, but also because they partition the phase plane into regions where a variable is increasing and those where it is decreasing.

▶ **Example 6.1.4** The blue s-nullcline in Fig. 6.1.2 marks points where $s' = 0$; hence, it separates out regions where $s' > 0$ and $s' < 0$. These can be distinguished by looking at points where one variable is 0 or infinity; for example, the differential equation (6.1.3) yields $s' = hz \geq 0$ at $s = 0$ (keeping in mind that the variables and parameters are not negative). Thus, the left boundary of the plot is on the $s' > 0$ side of the nullcline. Similarly, the z-nullcline divides the plane into a different set of regions where $z' > 0$ and $z' < 0$. These can easily be distinguished by looking at the value of z' at either $s = 0$ (where z' is negative) or $z = 0$ (where z' is positive). □

Check Your Understanding 6.1.2:

Prepare a sketch similar to Fig. 6.1.2a for the system

$$x' = 1 - xy\,, \qquad y' = 1 - x\,,$$

and determine the equilibrium point(s).

[3] See also Sect. 4.3.

6.1.3 Nullcline Analysis

The full power of nullclines comes from combining them. Taken together, the nullclines divide the phase plane into regions in which the direction of change of each variable is known, and this information restricts the possible directions for the solution curves to one quadrant of the compass.

▶ **Example 6.1.5** Figure 6.1.2b shows direction marker pairs in each of the three regions created by the nullclines of the Michaelis–Menten system. The middle region, for example, is to the right of the blue s-nullcline and to the left of the red z-nullcline; hence, s and z are both decreasing. The direction marker pair shows that solution curves in this region must be moving to the "southwest." Similarly, solution curves move "southeast" in the left region and "northwest" in the right region. □

We've seen that the direction of a solution curve is restricted to one quadrant of the compass in each region marked out by nullclines. We can draw much stronger conclusions about the direction of solution curves on the nullclines themselves because one of the rates of change is known to be 0.

▶ **Example 6.1.6** In Fig. 6.1.2, the arrows must be vertical on the s-nullcline, since s is not changing. The direction is down, rather than up, in keeping with the downward components of the direction marker pairs in the regions on either side of the s-nullcline. Similarly, the arrows on the z-nullclines point to the left, consistent with the left-pointing arrows in the adjoining regions. □

Check Your Understanding 6.1.3:
Prepare a sketch similar to Fig. 6.1.2b for the system

$$x' = 1 - xy\,, \qquad y' = 1 - x\,.$$

No-Egress Regions

Having arrows on the nullclines themselves allows us to identify possible progressions of solution curves from one region to another. This is always helpful, but it is definitive when there is a region that cannot be exited by solution curves.

Definition 6.1.3

A **no-egress** region in the phase plane is a region whose boundaries consist of curves with arrows pointing inward or parallel.

▶ **Example 6.1.7** In the nullcline plot of Fig. 6.1.2, the arrows on both nullclines point toward the center region. This means that solution curves that enter the center region cannot leave. Ultimately, all such solution curves must approach the equilibrium point. Now consider a solution curve that begins at the biologically relevant initial point $(1, 0)$. The compass quadrants show that the curve must move up and to the left. Because of the shape of the nullcline itself, this solution curve must eventually cross the boundary into the center region. Once there, the solution curve must end at the origin. Note that the same argument must hold for any other initial point in the rightmost region. □

Check Your Understanding 6.1.4:

Describe the behavior of solution curves that begin in the leftmost region of Fig. 6.1.2.

Nullcline analysis of the Michaelis–Menten system leads to the conclusion that solution curves near the equilibrium point at the origin must move in its direction; hence, the origin is a locally asymptotically stable equilibrium point. Moreover, solution curves must ultimately enter the no-egress region from any starting point. The combination of forced entry into the region and forced progression through it to the origin identifies the equilibrium as globally asymptotically stable. We will see in Sects. 6.2 and 6.3 that analytical methods can usually determine the *local* stability of equilibrium solutions, but not the *global* stability.[4] Thus, we have obtained a very strong conclusion, with no calculation other than what was needed to plot the nullclines!

Check Your Understanding 6.1.5:

Use the sketch from Check Your Understanding 6.1.3 to draw conclusions about the stability of any equilibria or explain why that cannot be done.

Nullcline analysis does not always yield conclusions about global stability, or even local stability. It all depends on the information in the nullcline plot. No-egress boundaries combine with the compass quadrant in a no-egress region to significantly constrain the behavior of solution curves. Without a no-egress region, strong stability conclusions are not usually possible; however, even in the absence of such a region, nullcline analysis provides some useful information with minimal calculation and without requiring specific values for parameters.

Fast Variables

Sometimes the conclusions we can draw from nullclines are strengthened by making use of small parameters in a system of differential equations.

▶ **Example 6.1.8** The parameter ϵ in the dimensionless Michaelis–Menten system (6.1.3)–(6.1.4) represents the ratio of the initial enzyme concentration to the initial substrate concentration, and hence it is always small. Because this parameter multiplies dz/dt, it marks z as a "fast" variable. In Sect. 3.8, we used this observation to argue that z quickly approaches a point in time where the right side of the equation is small. In graphical terms, this means that solution curves quickly approach a z-nullcline. If the compass quadrant restrictions permit, the solution curves will subsequently follow closely to the nullcline of the fast equation. This information can be used to predict the shape of a solution curve in the Michaelis–Menten system to a considerable degree of detail.

Figure 6.1.3 illustrates this phenomenon by superimposing the solution curve on the nullcline plot. The initial point $(1, 0)$ is not near a z-nullcline, so the right side of the z equation is not near 0. Hence, z' must be large. Meanwhile, the s equation is not fast, so s' is never large. The inescapable conclusion is that the solution curve must move almost vertically in the phase plane until it gets close to the z nullcline, at which point the right side of the z equation becomes small and z' stops being large.

Once near the z nullcline, the solution curve must cross to the other side, because the arrows on that nullcline point to the left. The solution curve is subsequently constrained to follow the compass quadrant of the middle region, which means it must move down and to the left. While doing so, it must stay near the z nullcline; otherwise, z would begin to change rapidly. In the middle region, this rapid change in z would be downward, bringing the solution curve back to the nullcline.

[4] There are analytical methods that can sometimes be used to prove global stability, but they are far more sophisticated than anything in this book.

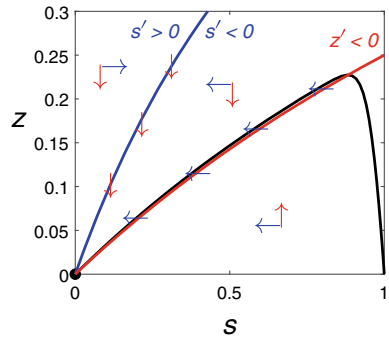


Fig. 6.1.3 The nullcline plot for the Michaelis–Menten model (using $h = 1$ and $r = 2$), along with the solution curve obtained with $\epsilon = 0.2$

The path of the solution curve is now clear. Once it enters the no-egress region, it must stay there, moving down and to the left. As it does so, it stays close to the z-nullcline, all the way to the origin. With a more realistic (smaller) value of ϵ, the solution will stay even closer to the z nullcline. □

6.1.4 Nullcline Analysis in General

The principal goal for the analysis of a dynamical system is to understand the phase portrait for the various ranges of parameter values. Nullcline analysis is a graphical tool for addressing this goal. Although the conclusions depend on the details of the system, the procedure is always as presented in the Michaelis–Menten example.

Algorithm 6.1.1
Nullcline analysis for a dynamical system of variables x and y in continuous time

1. Set $x' = 0$ to obtain an equation for x-nullcline. Plot the curve or curves for this equation and identify each of the regions marked out by the graph according to whether x is increasing or decreasing.
2. Repeat step 1 using the y equation.
3. Mark equilibria at intersections of nullclines of different "color." (Intersections of nullclines of the same type are not equilibria.)
4. Use the information from steps 1 and 2 to identify the appropriate compass quadrant for solution curves in each region delineated by the set of nullclines.
5. Use the compass quadrants to determine the direction of solution curves on each nullcline. Generally the direction reverses as the nullcline moves through an equilibrium point.
6. Identify any no-egress regions, if possible.
7. Use the compass quadrants and no-egress regions to determine the possible paths of solution curves in the phase plane. If either x or y is a fast variable, indicated by a small parameter in front of the derivative term in the dimensionless model, this means that solution curves move in the direction of the fast variable to a corresponding nullcline and remain near that nullcline, as long as doing so does not contradict the compass quadrant in the region.

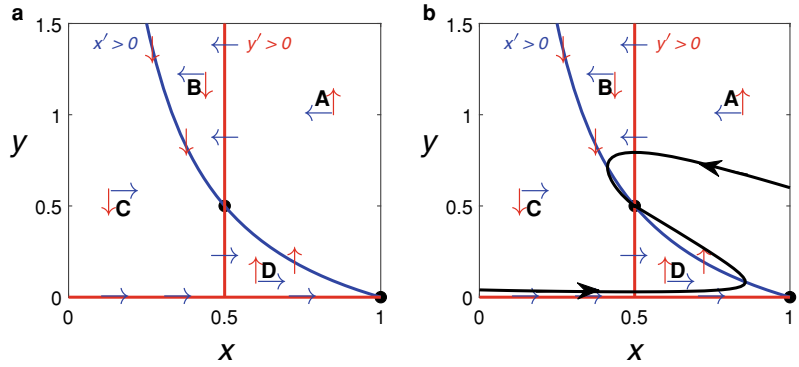


Fig. 6.1.4 The nullcline plot for Example 6.1.9. Two solution curves are shown: one begins at $x = 1$, $y = 0.7$ and moves through regions A, B, and C to the equilibrium point; the other begins at small values of x and y and moves through regions C, D, and A to the equilibrium point

The Michaelis–Menten model is a nice first example, but it is important to recognize that some of its features make it easier to analyze than the typical model.

1. The nullclines in the Michaelis–Menten model do not cross in the open first quadrant. This means that there are no equilibria with both variables positive. In general, there can be positive equilibria, and these might exist for certain ranges of parameter values and not for others.
2. Not all systems exhibit no-egress regions. Without such a region, the ultimate fate of solution curves can be unclear from the nullcline plot alone.
3. Not all systems have one fast and one slow variable. This means that there may not be solution curves that follow a nullcline closely.
4. For systems that do have a fast and a slow variable, it is not always possible for solutions to follow the fast nullcline. In the example system, the nullcline slopes down and to the left, a direction consistent with the arrows in the middle region. When the nullcline slope is not compatible with the direction arrows, the phase portrait becomes complicated.[5]

▶ **Example 6.1.9** Figure 6.1.4 shows the nullcline plot for the system

$$\frac{ds}{dt} = 1 - s - 2vs\,, \quad s \geq 0\,, \tag{6.1.7}$$

$$\frac{dv}{dt} = 2vs - v\,, \quad v \geq 0\,. \tag{6.1.8}$$

This nullcline plot lacks no-egress regions, so the analysis is more complicated than in the Michaelis–Menten model. Consider a solution curve that begins at a point $(1, y_0)$ in region A. From panel **a**, it is clear that the curve must move up and to the left, eventually crossing into region B. From there, it

[5] See Project 6C.

must move down and to the left into region C. But from region C there are two alternatives. Based on the nullclines alone, it appears that the solution curve will cross from region C into region D, from which it must return to region A. However, it is also possible that the solution curve in region C will move to the equilibrium point. Only by looking at the actual solution curve, added in panel **b**, can we tell which of these cases occurs.

While the ambiguity regarding solution curves in panel **a** makes stability of the equilibrium at $(0.5, 0.5)$ unclear from the nullclines alone, the same is not true for the equilibrium at $(1, 0)$. Solution curves in region D must move up into region A and continue to move up, either to the point $(0.5, 0.5)$ or into region B. Hence, the equilibrium $(1, 0)$ is clearly unstable. □

In general, there is usually a possibility of rotation when there are no no-egress regions, in which case the curves can move directly to an equilibrium, spiral either inward or outward, or form a closed loop. Without a no-egress region we generally cannot determine stability of all equilibria by the nullcline method. Unlike in the single-variable case, where the phase line method is always superior to linearized stability analysis, the graphical and analytical methods in the two-variable case are complementary, with each sometimes offering results that cannot be achieved by the other.

Check Your Understanding Answers

1. The peak is where z reaches its maximum, which occurs at a very small positive time. Thus, the solution moves from the initial point $(1, 0)$ to the peak very quickly; it then moves at a progressively slower pace along the curve from the peak toward the point $(0, 0)$. We know that the pace is slowing, because the time history graph for s gets flatter as time increases.
2. See Fig. 6.1.5a. There is a single equilibrium point at $(1, 0)$.
3. See Fig. 6.1.5b.
4. The presence of no-egress regions make clear that the equilibrium is unstable.
5. These solution curves must move down and to the right, cross the boundary into the center region, and continue to the origin.

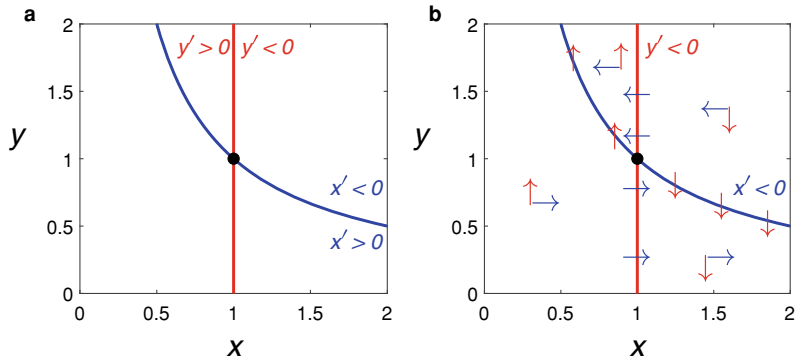


Fig. 6.1.5 Nullcline analysis for $x' = 1 - xy$, $y' = 1 - x$

Problems

The problems in this section call for nullcline plots, but parameters are given only in terms of inequality restrictions. In most cases, it is not difficult to sketch the nullclines by hand, but you can also get computer plots by choosing simple values that meet the requirements without being overly large or small. When preparing the plots, keep in mind that you should only show biologically relevant ranges for parameter values. For example, if a population n has been scaled by its maximum value, then only $0 \le n \le 1$ is relevant.

The inequalities that arise in the existence restrictions for equilibria also identify cases for the nullcline plots in these models. In general, linearized stability analysis[6] is required to fully identify cases.

6.1.1 This problem demonstrates the limitations of nullcline plots.

(a) Sketch the nullcline plot for the family of systems

$$x' = x - 1.2y\,,$$
$$y' = k(1.5x - y)\,,$$

where x and y are not required to be nonnegative. [Note that the nullcline plot is the same for all values of k.]

(b) Use computer simulations to plot some solution curves using the value $k = 0.8$. Is the origin stable or unstable?

(c) Repeat part (b) with $k = 1.2$.

(d) Can we conclude anything from the nullcline plot alone?

(This problem is continued in Problem 6.3.1.)

6.1.2* [Lead poisoning]
Prepare a nullcline plot for the lead poisoning model[7]

$$\epsilon\frac{dx}{dt} = q - (1+q)x + z\,,$$
$$\frac{dz}{dt} = x - z\,,$$

with $q = 7$. Draw conclusions about the stability of the equilibrium if possible.

(This problem is continued in Problem 6.2.10.)

6.1.3 (Continued from Problem 3.6.6.)

(a) Determine the equilibria for the SIR model with no demographics,

$$s' = -R_0 si\,,$$
$$i' = R_0 si - i\,,$$

being careful to indicate any restrictions on existence.

(b) Sketch the nullcline plot in the si-plane for the case $R_0 > 1$.

[6] See Sects. 6.2 and 6.3.

[7] Section 3.7.

(c) Add solution curves that begin with very small positive $i(0)$ and various $s(0)$ to the phase plane plot from part (b).

6.1.4 [Predator–prey and consumer–resource models] (Continued from Problem 3.6.9.)

(a) Determine the equilibria for the dimensionless Rosenzweig–MacArthur model with Holling type I predator response,

$$v' = v(1 - v - c)\,,$$
$$c' = mc(hv - 1)\,,$$

being careful to indicate any restrictions on existence.
(b) Sketch the nullcline plot in the vc-plane with $0 < h < 1$.
(c) Repeat part (b) for the case $h > 1$.
(d) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problem 6.2.3.)

6.1.5 [Self-limiting population] (Continued from Problem 4.3.6.)
Sketch the nullcline plot for the self-limiting population model

$$w' = ap\,,$$
$$p' = p(1 - p - w),$$

with $a = 0.5$. Add in solution curves for $p(0) = p_0$ and $w(0) = 0$, with several values of p_0 satisfying $0 < p_0 \leq 1$. The solution curves must be consistent with the restriction on $p + w$ obtained in Problem 4.3.6. Note that the stable equilibrium value for $p + w$ cannot be reached if p becomes 0, but in no case can it be exceeded.

6.1.6* [SIS disease with fixed birth rate] (Continued from Problem 3.9.4.)
With the rescaling $i = \delta y$, $\delta = \epsilon/d$, the SIS model with constant birth rate is

$$n' = \epsilon(1 - n - y)\,,$$
$$y' = \mathcal{R}_0 y(n - \mathcal{R}_0^{-1} - \delta y)\,,$$

where n is the scaled total population and y is a rescaled infectious population (so it can be any positive value).

(a) Determine the equilibrium points for the model, noting any restrictions on $\mathcal{R}_0$ for their existence.
(b) Sketch the nullcline plot in the ny-plane for the case $0 < R_0 < 1$.
(c) Repeat part (b) for the case $R_0 > 1$.
(d) The y-axis is not an n nullcline. Does this mean that the mathematical variable n can be negative? (If so, this would be a significant flaw in the model.)
(e) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problems 6.2.4 and 6.3.3.)

6.1.7 [SIR disease with fixed population] (Continued from Problem 3.9.1.)

(a) Determine the equilibria for the rescaled SIR model with fixed birth rate,

$$s' = \epsilon(1 - s - \mathcal{R}_0 sy)$$
$$y' = \mathcal{R}_0 sy - y\,,$$

where s and y are scaled susceptible and infectious class sizes and $\mathcal{R}_0$ is the basic reproductive number, being careful to indicate any restrictions on existence.
(b) Sketch the nullcline plot in the sy-plane for the case $0 < R_0 < 1$. Can the variable s become negative in this system?
(c) Repeat part (b) for the case $R_0 > 1$.
(d) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problems 6.2.5 and 6.3.4.)

6.1.8 [Plankton] (Continued from Problem 3.6.12.)

(a) Determine the equilibria for the plankton population model,

$$p' = p(1 - \alpha - p - z)\,,$$
$$z' = \delta z(p - \beta)\,,$$

where p and z are scaled phytoplankton and zooplankton biomasses, being careful to indicate any restrictions on existence.
(b) Sketch the nullcline plot in the pz-plane for the case $\alpha > 1$.
(c) Repeat part (b) for the case $\alpha < 1 < \alpha + \beta$.
(d) Repeat part (b) for the case $\alpha + \beta < 1$.
(e) Determine any conclusions that can be drawn from the nullcline plots. Of particular importance is the possible local extinction of the populations.

(This problem is continued in Problems 6.2.6 and 6.3.5.)

6.1.9* [SEIS disease with fixed population]
After scaling, the SEIS model with a fixed population takes the form

$$x' = bi(1 - x - i) - \nu x\,,$$

$$i' = \nu x - i\,,$$

where x and i are the fractions of the population that are latent (class E) and infectious, b is a transmission parameter that is approximately the basic reproduction number, and ν is approximately the ratio of mean infectious time to mean latent time. Take $\nu = 2$ for a convenient but realistic value. Plot nullclines for this model for the cases $b < 1$ and $b > 1$ and draw any possible conclusions about stability.

(This problem is continued in Problems 6.2.7 and 6.3.6.)

6.1.10 [Chemostat] (Continued from Problem 3.6.13.)

(a) Determine the equilibria for the dimensionless chemostat model,

$$r' = qr_0\left(1 - \frac{r}{r_0} - \frac{rc}{1+r}\right),$$
$$c' = c\left(\frac{r}{1+r} - q\right).$$

Also show that the equilibrium with $c > 0$ exists if and only if

$$q < \frac{r_0}{1+r_0}.$$

(b) Sketch the nullcline plot in the rc-plane for the case

$$\frac{r_0}{1+r_0} < q < 1.$$

(c) Repeat part (b) for the case

$$q < \frac{r_0}{1+r_0}.$$

(d) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problems 6.2.8 and 6.3.7.)

6.1.11 [Malaria] (Continued from Problem 3.4.17.)

(a) Sketch nullclines for the Ross malaria model

$$\frac{dx}{dt} = \alpha(1-x)y - x,$$
$$\frac{dy}{dt} = m[\beta x(1-y) - y].$$

Take $\beta = 1$ for convenience. You will need to consider the cases $\alpha > 1$ and $\alpha < 1$ separately.

(b) Use the nullcline plot to draw any possible conclusions about the stability of the equilibria.

(c) Sketch a solution curve starting at the point (0, 1). [Note that m is a large parameter.]

(This problem is continued in Problem 6.3.8.)

6.1.12 [SIS disease with logistic growth] (Continued from Problem 3.9.8.)
Consider the SIS model with logistic growth,

$$n' = \delta[n(1-n) - mi],$$
$$i' = \mathcal{R}_0 i\left(n - i - \mathcal{R}_0^{-1}\right).$$

This model is slightly modified from Problem 3.9.8 by defining $\delta = \epsilon\rho$ and $m = d/\delta$. These changes make the analysis slightly easier.

(a) Derive the equilibrium equation

$$n^*(1-n^*) = m(n^* - \mathcal{R}_0)$$

for any endemic disease equilibria ($i^* > 0$). Show that this equilibrium cannot exist if $\mathcal{R}_0 < 1$. [This is easy to do graphically.]

(b) Verify that the equilibrium value of n is decreased by the long-term presence of the disease. [Again, do this graphically.]

(c) Sketch the nullcline plot in the ni-plane for the case $0 < R_0 < 1$. The value $m = 2.4$ is convenient, as it yields an equilibrium n^* without square roots.

(d) Repeat part (c) for the case $R_0 > 1$.

(e) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problems 6.2.9 and 6.3.10.)

6.1.13 [SIS disease with logistic growth and standard incidence]
(Continued from Problem 3.9.9.)

(a) Determine the equilibrium points for the SIS model with logistic growth,

$$n' = \delta[n(1-n) - wi]\,,$$
$$i' = \mathcal{R}_0 i\left[1 - \mathcal{R}_0^{-1} - \frac{i}{n}\right],$$

where n and i are scaled variables that represent the total population and the infectious population, noting any restrictions on R_0 and w for their existence. You may find it convenient to replace R_0 with the parameter $\chi = 1 - R_0^{-1}$.

(b) Sketch the nullcline plot in the ni-plane for the case $0 < R_0 < 1$.

(c) Repeat part (b) for the case $R_0 > 1$, $w\chi < 1$. You may take $\mathcal{R}_0 = 2$, $w = 1$ for a specific example.

(d) Repeat part (b) for the case $R_0 > 1$, $w\chi > 1$. You may take $\mathcal{R}_0 = 2$, $w = 3$ for a specific example.

(e) Determine any conclusions that can be drawn from the nullcline plots.

(This problem is continued in Problem 6.2.2.)

6.1.14 [SIS disease with logistic growth and standard incidence] (Continued from Problem 3.9.10.)
The model

$$n' = \delta n(1 - n - wx)\,,$$
$$x' = \mathcal{R}_0 x(1 - \mathcal{R}_0^{-1} - x)$$

was obtained from that in Problems 3.9.9 and 6.1.13 by using the infectious population fraction $x = i/n$ to replace the infectious population i.

(a) Determine the equilibrium points, noting any restrictions on R_0 and w for their existence. You may find it convenient to replace R_0 with the parameter $\chi = 1 - R_0^{-1}$.

(b) Sketch the nullcline plot in the nx-plane for the case $0 < R_0 < 1$.

(c) Repeat part (b) for the case $R_0 > 1$, $w\chi < 1$. You may take $\mathcal{R}_0 = 2$, $w = 1$ for a specific example.

(d) Repeat part (b) for the case $R_0 > 1$, $w\chi > 1$. You may take $\mathcal{R}_0 = 2$, $w = 3$ for a specific example.

(e) Determine any conclusions that can be drawn from the nullcline plots regarding the stability of the equilibria.

(This problem is continued in Problem 6.2.1.)

6.1.15 [Immune system] (Continued from Problem 3.6.14.)
Some interesting behavior is exhibited by the very simple system

$$p' = p(1 - p - qm)\,,$$
$$m' = a[\delta(1 - m) - mp]\,,$$

where p is the scaled amount of pathogen in a patient's system, m is the scaled population of general-purpose macrophages,[8] a is the rate constant for macrophage loss, $a\delta$ is the rate constant for macrophage creation and natural death, and q is a measure of the strength of the macrophage against the pathogen. The parameter δ is small, as the creation of macrophages is slow compared to the rate at which they are lost in the battle against the pathogen. This model incorporates only one of the many components of the human immune system.

(a) Determine the equilibrium points, noting any restrictions on q and δ for their existence. In addition to general formulas, find the specific values of the equilibria with $\delta = 0.04$, $q = 2.35$.
(b) Prepare a nullcline plot in the pm-plane for the case $\delta = 0.04$, $q = 2.35$.
(c) Determine any conclusions that can be drawn from the nullcline plot regarding the stability of the equilibria for this case.
(d) Modify the MATLAB program *ODEsim.m* to run simulations for the model. Use it to compare two cases with $p(0) = 0.952$ and $p(0) = 0.953$, each using $\delta = 0.04$, $q = 2.35$, $a = 1$, $m(0) = 1$.
(e) Superimpose the nullclines onto a phase portrait of the solutions from the simulations in (d).[9]
(f) Discuss the results. What should we expect a realistic value of $p(0)$ to be, given that $p = 1$ would be the maximum capacity for the pathogen in a person without a functioning immune system?

(This problem is continued in Problems 6.2.11 and 6.3.9.)

6.2 Linearized Stability Analysis Using Eigenvalues

After studying this section, you should be able to:

- Compute the Jacobian to represent a nonlinear system near an equilibrium point.
- Use eigenvalues to determine the stability of two-component linear continuous dynamical systems.

In Sect. 4.4, we found that the local stability of a differential equation $x' = f(x)$ at an equilibrium point x^* depends on the value $f'(x^*)$. The idea of this analysis scales up to continuous dynamical systems with more than one component, but with one difficulty: we must find a higher dimensional analog for the quantity $f'(x^*)$.

[8] Macrophages are a type of white blood cell. See Problem 3.3.10 for more background.

[9] You should see very different outcomes for the two scenarios in spite of their being nearly identical. There is a curve in the phase plane called a *separatrix* that separates the plane into domains of attraction for the different stable equilibria. The initial conditions were chosen to give two solutions very close to the separatrix but on opposite sides.

6.2.1 Two-Component Linear Systems

Autonomous systems of two differential equations have the general form

$$\frac{dx}{dt} = f(x, y) \tag{6.2.1}$$

$$\frac{dy}{dt} = g(x, y)\,. \tag{6.2.2}$$

Eventually, we will want to analyze autonomous systems with nonlinear functions f and g; however, we begin with *linear* autonomous systems, which have a highly structured form.

Definition 6.2.1

An **autonomous linear system of two differential equations** is a system of the form

$$\frac{dx}{dt} = a_{11}x + a_{12}y + b_1\,,$$
$$\frac{dy}{dt} = a_{21}x + a_{22}y + b_2\,,$$

where the a_{ij} and b_j are all constants.

▶ **Example 6.2.1** The dimensionless two-component lead poisoning model,[10]

$$\epsilon x' = iq - (1+q)x + z\,, \tag{6.2.3}$$

$$z' = x - z\,, \tag{6.2.4}$$

where $q, \epsilon > 0$, and $i \in \{0, 1\}$, is an autonomous two-dimensional system of linear differential equations with

$$a_{11} = -\epsilon^{-1}(1+q), \quad a_{12} = \epsilon^{-1}, \quad a_{21} = 1, \quad a_{22} = -1, \qquad b_1 = iq, \quad b_2 = 0\,. \qquad \square$$

Autonomous linear systems are conveniently written in the matrix form

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{b} \tag{6.2.5}$$

by defining the vectors $\mathbf{x}$ and $\mathbf{b}$ and matrix $\mathbf{A}$ as

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}. \tag{6.2.6}$$

▶ **Example 6.2.2** The lead poisoning model of Example 6.2.1 has the form (6.2.5), with

$$\mathbf{A} = \begin{pmatrix} -\epsilon^{-1}(1+q) & \epsilon^{-1} \\ 1 & -1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} iq \\ 0 \end{pmatrix}. \qquad \square$$

[10] Section 3.7.

6.2.2 Eigenvalues and Stability

As long as $\det(\mathbf{A}) \neq 0$, there is always a single equilibrium point for (6.2.5), which is the unique solution of

$$\mathbf{Ax} + \mathbf{b} = \mathbf{0}\,. \tag{6.2.7}$$

Now suppose the matrix $\mathbf{A}$ has two real eigenvalues, λ_1 and λ_2. Then solutions of the linear equation

$$\mathbf{x}' = \mathbf{Ax} + \mathbf{b}$$

have the form

$$\mathbf{x} = \mathbf{x}^* + c_1\mathbf{v_1}e^{\lambda_1 t} + c_2\mathbf{v_2}e^{\lambda_2 t}\,,$$

where $\mathbf{x}^*$ is the solution of (6.2.7) and $\mathbf{v_k}$ is an eigenvector corresponding to the eigenvalue λ_k.[11] Asymptotic stability requires

$$\lim_{t\to\infty} \mathbf{x} = \mathbf{x}^*$$

for all solutions; hence, it occurs only when both eigenvalues are negative. The eigenvalues play the same role as $f'(x^*)$ in the scalar case.

The situation is more complicated than the scalar case, however. The eigenvalue equation is quadratic, so instead of a pair of unique real roots, there could be repeated real roots or a pair of complex roots. These cases make the solution formulas more complicated, but they do not affect stability. All real roots, repeated or distinct, yield solutions with the factor $e^{\lambda t}$.[12] Complex roots of the form $\lambda = \alpha \pm i\beta$ yield solutions with the factor $e^{\alpha t}$, where α is called the *real part* of λ.[13] This gives us the following general result.

Theorem 6.2.1 (Stability of Equilibria for Autonomous Linear Systems)
The equilibrium solution $\mathbf{x} = \mathbf{x}^*$ for the equation $\mathbf{x}' = \mathbf{Ax} + \mathbf{b}$, where $\det(\mathbf{A}) \neq 0$, is asymptotically stable if and only if all real eigenvalues are negative and all complex eigenvalues have a negative real part.

▶ **Example 6.2.3** For the model of Example 6.2.1, the equilibrium must satisfy

$$iq - (1+q)x + z = 0\,, \qquad x - z = 0\,.$$

Adding these equations together produces an equation with only the x variable:

$$iq - qx = 0\,.$$

Since $q > 0$, we have $z = x = i$ for all sets of parameter values. In the context of lead poisoning, the dimensionless variables x and z represent the amounts of lead in the blood and the bones; however, the

[11] Problem 6.2.14.

[12] Repeated roots have to contribute multiple solutions. If the eigenvalue equation has a factor $(\lambda - \lambda_1)^2$, then obviously one solution is $e^{\lambda_1 t}$. It can be shown that the other solution is $te^{\lambda_1 t}$. This is important if we need solution formulas, but irrelevant to the question of stability.

[13] Complex pairs of roots have to contribute two solutions. The eigenvalue pair $\alpha \pm i\beta$ corresponds to solutions $e^{\alpha t}\cos\beta t$ and $e^{\alpha t}\sin\beta t$. These solutions oscillate, with the rate of oscillation controlled by β; however, only the exponential factor affects stability.

z variable is scaled differently from x. With ϵ small, the model predicts an equilibrium state with a large amount of lead in the bones. This is why the solutions we saw in Sect. 3.7 reached equilibrium quickly for X but did not appear to reach equilibrium for Z at all. In order to reach a z value of approximately 1, it is necessary to wait for about $1/\epsilon$ times, however long it takes to reach an x value of approximately 1. In Fig. 3.7.2, X appears to reach equilibrium in about 100 days. With $\epsilon \approx 0.01$, it would take about 10,000 days (around 30 years!) for Z to reach equilibrium. Simulations are not a foolproof way to determine stability.

To determine the stability of the lead poisoning equilibrium analytically, we need only to find the eigenvalues of the matrix

$$\mathbf{A} = \begin{pmatrix} -\epsilon^{-1}(1+q) & \epsilon^{-1} \\ 1 & -1 \end{pmatrix}.$$

We have

$$\begin{aligned} 0 = \det(\mathbf{A} - \lambda \mathbf{I}) &= \begin{vmatrix} -\epsilon^{-1}(1+q) - \lambda & \epsilon^{-1} \\ 1 & -1-\lambda \end{vmatrix} \\ &= [-\epsilon^{-1}(1+q) - \lambda](-1-\lambda) - \epsilon^{-1} \\ &= \lambda^2 + [1 + \epsilon^{-1}(1+q)]\lambda + \epsilon^{-1} q \,. \end{aligned}$$

At this point, we can use the quadratic formula to calculate the eigenvalues for a particular set of parameter values. Whatever parameter values you pick, as long as $q > 0$ and $\epsilon > 0$, both eigenvalues will have negative real part. This will be established in Sect. 6.3. □

Check Your Understanding 6.2.1:

Determine the eigenvalues for the system of Example 6.2.3 for the typical case $q = 7.1, \epsilon = 0.009$.

One of the advantages of matrix notation is that theorems are often independent of the size of the system. Theorem 6.2.1 holds for systems of any dimension, not just dimension 2.

6.2.3 The Jacobian Matrix and Stability

Most meaningful systems in biology are nonlinear, so a method that works only for linear systems is of limited value. Fortunately, nonlinear systems can be linearized around an equilibrium point, with the results of the approximate linear system usually being valid for the original nonlinear system. The key to doing so is to find the appropriate matrix, which is called the *Jacobian*.

Definition 6.2.2

The **Jacobian** of an n-dimensional system of differential equations is the $n \times n$ matrix for which the entry in row i and column j is the partial derivative of the ith function with respect to the jth variable.

▶ **Example 6.2.4** The dimensionless Michaelis–Menten model[14] is

$$\frac{ds}{dt} = f(s,z) = -s(1-z) + hz\,,$$
$$\epsilon\frac{dz}{dt} = \epsilon g(s,z) = s(1-z) - hz - rz\,.$$

Thus,[15]

$$\frac{\partial f}{\partial s} = -1 + z\,, \quad \frac{\partial f}{\partial z} = s + h\,,$$
$$\frac{\partial g}{\partial s} = \epsilon^{-1}(1-z), \quad \frac{\partial g}{\partial z} = \epsilon^{-1}(-s-h-r)\,.$$

These derivatives combine to define the Jacobian matrix as

$$\mathbf{J}(s,z) = \begin{pmatrix} -1+z & s+h \\ \epsilon^{-1}(1-z) & -\epsilon^{-1}(s+h+r) \end{pmatrix}.$$

As we saw in our earlier analysis, the only equilibrium point is the origin. At this point,

$$\mathbf{J}(0,0) = \begin{pmatrix} -1 & h \\ \epsilon^{-1} & -\epsilon^{-1}(h+r) \end{pmatrix}. \tag{6.2.8}$$

Equation (6.2.8) defines the matrix for the linear approximation of the system near the equilibrium point $(0,0)$. □

Suppose (x^*, y^*) is an equilibrium point for a system of the form (6.2.1)–(6.2.2). If we zoom in on the equilibrium point in the phase plane, we approach a linear system with matrix (6.2.8).[16] We can determine the stability of the linearized system using Theorem 6.2.1. Because the behavior of solutions near an equilibrium point in the phase space doesn't change as we zoom in, it seems reasonable to expect that the nonlinear system and its linear approximation should have the same stability properties. This is generally true, although there are exceptions. The important result is summarized in the following theorem, which is written so as to apply to systems with any number of components:

Theorem 6.2.2 (Stability of Equilibria for Autonomous Nonlinear Systems)
Let $\mathbf{x}^*$ be an equilibrium point for a nonlinear autonomous system, where $\mathbf{x}$ is a vector containing the state variables of the system. If Theorem 6.2.1 applies to the linearized system having matrix $\mathbf{J}(\mathbf{x}^*)$, then the conclusion from that theorem applies to $\mathbf{x}^*$ in the corresponding nonlinear system.

▶ **Example 6.2.5** The matrix in Example 6.2.4 yields

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -1-\lambda & h \\ \epsilon^{-1} & -\epsilon^{-1}(h+r) - \lambda \end{pmatrix}.$$

[14] Section 3.8.

[15] See Appendix B for the definition of the partial derivative and methods for calculation.

[16] See Appendix F.

Setting the determinant equal to 0 results in the quadratic equation

$$\lambda^2 + [1 + \epsilon^{-1}(h + r)]\lambda + \epsilon^{-1} r = 0. \tag{6.2.9}$$

With any combination of parameters, both eigenvalues are negative or have negative real part. Thus, the origin for the linearized system is asymptotically stable by Theorem 6.2.1. By Theorem 6.2.2, this result extends to the equilibrium $(s, z) = (0, 0)$ for the original nonlinear system. □

Note that Theorem 6.2.2 does not apply in certain cases, such as when $\det \mathbf{A} > 0$, $\operatorname{tr} \mathbf{A} = 0$. This is not very common, but it is important to keep in mind that linearization does not always work.

Check Your Understanding Answers

1. $\lambda_1 = -0.876$, $\lambda_2 = -900$

Problems

Problem 1 is a good place to start, as it includes guidelines for best use of algebra, and the answer section shows an important intermediate result. Read Appendix G for a more general description of best practices in algebra.

6.2.1* [SIS disease with logistic growth and standard incidence]

(a) Determine the stability of all equilibria for the SIS model with logistic growth,

$$n' = \delta n(1 - n - wx)\,,$$
$$x' = \mathcal{R}_0 x(1 - \mathcal{R}_0^{-1} - x)\,,$$

where n and x are scaled variables that represent the total population and the infectious population fraction i/n, noting any restrictions on R_0 and w for their existence. Two algebraic strategies make this problem easy.

1. Don't multiply out products; instead take product rule derivatives. The first entry in the matrix (the partial derivative of the n' function with respect to n) is then

$$\delta(1 - n - wx) - \delta n$$

rather than

$$\delta(1 - 2n - wx)\,,$$

for example.

2. For the endemic disease equilibrium, use the equilibrium relations

$$1 - n^* - wx^* = 0\,, \qquad 1 - \mathcal{R}_0 - x^* = 0\,,$$

but do not substitute in the formulas for n^* or x^*.

(b) Explain why the algebraic advice given in (a) makes the problem easier.

6.2.2 [SIS disease with logistic growth and standard incidence]
(Continued from Problems 6.1.13 and 6.1.14.)
Discuss the results of Problem 6.2.1 with reference to Problems 6.1.13 and/or 6.1.14.

6.2.3 [Predator–prey and consumer–resource models] (Continued from Problem 6.1.4.)

(a) Determine the stability of the equilibria having $c = 0$ for the dimensionless Rosenzweig–MacArthur model with Holling type I predator response,

$$v' = v(1 - v - c)\,,$$
$$c' = mc(hv - 1)\,.$$

(b) Let $m = 0.1$ and $h = 2$ and let (v^*, c^*) be the equilibrium in which neither v nor c are 0. Determine the stability of the equilibrium.
(c) Repeat (b) with $h = 4$.
(d) Discuss the results with reference to Problems 3.6.9 and 6.1.4.

(This problem is continued in Problem 6.3.2.)

6.2.4* [SIS disease with fixed birth rate] (Continued from Problem 6.1.6.)

(a) Determine the stability of the disease-free equilibrium for the SIS model with constant birth rate,

$$n' = \epsilon(1 - n - y)\,,$$
$$y' = \mathcal{R}_0 y(n - \mathcal{R}_0^{-1} - \delta y)\,.$$

Note that the results could depend on the parameter values.
(b) Determine the stability for the endemic disease equilibrium with $\mathcal{R}_0 = 2$, $\delta = 0.1$, $\epsilon = 0.01$.
(c) Discuss the results with reference to Problems 3.9.4 and 6.1.6.

(This problem is continued in Problem 6.3.3.)

6.2.5 [SIR disease with fixed population] (Continued from Problem 6.1.7.)

(a) Determine the stability of the disease-free equilibrium for the SIR model with fixed birth rate

$$s' = \epsilon(1 - s - \mathcal{R}_0 sy)$$
$$y' = \mathcal{R}_0 sy - y\,.$$

Note that the results could depend on the parameter values.
(b) Determine the stability for the endemic disease equilibrium with $\mathcal{R}_0 = 4$, $\epsilon = 0.001$.
(c) Discuss the results with reference to Problems 3.9.1 and 6.1.7.

(This problem is continued in Problem 6.3.4.)

6.2.6 [Plankton] (Continued from Problem 6.1.8.)

(a) Determine the stability of the extinction equilibrium for the plankton population model

$$p' = p(1 - \alpha - p - z)\,,$$
$$z' = \delta z(p - \beta)\,.$$

(b) Repeat (a) for the equilibrium that has zooplankton extinction only. Keep in mind that some equilibria exist only for certain ranges of parameter values.
(c) Repeat (a) for the equilibrium in which neither population is extinct, using parameters $\alpha = 0.5$, $\beta = 0.25$, and $\delta = 0.1$.
(d) Discuss the results with reference to Problem 6.1.8.

(This problem is continued in Problem 6.3.5.)

6.2.7* [SEIS disease with fixed population] (Continued from Problem 6.1.9.)

(a) Determine the stability of the equilibria for the model

$$x' = bi(1 - x - i) - \nu x \,,$$

$$i' = \nu x - i \,,$$

with $b = 0.5$, $\nu = 2$.
(b) Repeat (a) with $b = 2$.
(c) Discuss the results with reference to Problem 6.1.9.

(This problem is continued in Problem 6.3.6.)

6.2.8 [Chemostat] (Continued from Problem 6.1.10.)

(a) Determine the stability of the $c = 0$ equilibrium for the chemostat model,

$$r' = qr_0\left(1 - \frac{r}{r_0} - \frac{rc}{1+r}\right) \,,$$
$$c' = c\left(\frac{r}{1+r} - q\right) \,.$$

(b) Discuss the results with reference to Problem 6.1.10.

(This problem is continued in Problem 6.3.7.)

6.2.9 [SIS disease with logistic growth] (Continued from Problem 6.1.12.)

(a) Determine the stability of the disease-free equilibrium for the SIS model with logistic growth,

$$n' = \delta[n(1-n) - mi] \,,$$
$$i' = \mathcal{R}_0 i(n - i - \mathcal{R}_0^{-1}) \,.$$

Note that the results could depend on the parameter values.
(b) Repeat for the endemic disease equilibrium, but take $\mathcal{R}_0 = 2$, $m = 2.4$, and $\delta = 0.001$.
(c) Discuss the results with reference to Problems 3.9.8 and 6.1.12.

(This problem is continued in Problem 6.3.10.)

6.2.10* [Lead Poisoning]
Determine the stability of the equilibrium solution of the three-component lead poisoning model

$$\begin{aligned}\frac{dX}{dT} &= R - (k_1 + k_3 + r_1)X + k_2 Y + k_4 Z\,,\\ \frac{dY}{dT} &= k_1 X - (k_2 + r_2)Y\,,\\ \frac{dZ}{dT} &= k_3 X - k_4 Z\end{aligned}$$

with parameters

$$R = 49.3\,, \quad r_1 = 0.0211\,, \quad r_2 = 0.0162\,,$$

$$k_1 = 0.0111\,, \quad k_2 = 0.0124\,, \quad k_3 = 0.0039\,, \quad k_4 = 0.000035\,.$$

(This problem is continued in Problem 6.3.11.)

6.2.11 [Immune system] (Continued from Problem 3.6.14.)
Determine the stability requirement for the disease-free equilibrium ($p = 0$) for the partial immune system model

$$\begin{aligned}p' &= p(1 - p - qm - sn)\,,\\ m' &= a[\delta(1 - m) - mp]\,,\\ n' &= b\left(\frac{p}{h + p} - n\right)\,,\end{aligned}$$

where p is a pathogen in a human host and m and n are populations of two different types of white blood cells.

(This problem is continued in Problem 6.3.9 and Project 6D.)

6.2.12 [SEIR disease with fixed population]
The SEIR model with fixed population is given in dimensionless form as

$$\begin{aligned}s' &= \epsilon(1 - s - bsy)\,,\\ x' &= bsy - \nu x\,,\\ y' &= \nu x - y\,,\end{aligned}$$

where x and y are the exposed and infectious populations, rescaled because they are both $O(\epsilon)$ at equilibrium.

(a) Find the Jacobian.
(b) Determine the stability of the disease-free equilibrium using the parameters $b = 5$, $\nu = 2$, and $\epsilon = 0.001$.
(c) Repeat (b) for the endemic disease equilibrium.

(This problem is continued in Problem 6.3.13.)

6.2.13 [SIR disease with fixed birth rate and loss of immunity] (Continued from Problem 3.9.7.)
Determine the existence and stability of the disease-free equilibrium for the SIR model with fixed birth rate and limited immunity, given in dimensionless form as

$$n' = \epsilon(1 - n - dy)\,,$$
$$s' = \epsilon[1 - s + \phi(n - s) - bsy]\,,$$
$$y' = bsy - y\,,$$

where n and s are the scaled total and susceptible populations and y is the rescaled infectious population. The parameter d is the fraction of people with the disease who die of it, b is the rescaled infection coefficient, and ϕ is the ratio of the mean lifespan to the mean duration of immunity, which could be larger than 1.[17]

(This problem is continued in Problem 6.3.14.)

6.2.14 Let $\mathbf{A}$ be a 2×2 matrix having eigenvectors $\mathbf{v_1}$ and $\mathbf{v_2}$ with corresponding eigenvalues λ_1 and λ_2. Let $\mathbf{b}$ be a two-dimensional vector and let x^* be the solution of the algebraic equation $\mathbf{Ax} = -\mathbf{b}$. Show that the function

$$\mathbf{x}(t) = \mathbf{x}^* + c_1\mathbf{v_1}e^{\lambda_1 t} + c_2\mathbf{v_2}e^{\lambda_2 t}$$

solves the differential equation

$$\mathbf{x}' = \mathbf{Ax} + \mathbf{b}\,.$$

6.3 Stability Analysis with the Routh–Hurwitz Conditions

After studying this section, you should be able to:

- Use the Routh–Hurwitz criteria to determine stability of equilibria for dynamical systems with two or three components.

So far, the general plan for analyzing a two-dimensional system like that of (6.2.3)–(6.2.4) involves two basic computational steps:

1. Rewrite the equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ as a polynomial equation for λ, which is called the *characteristic equation* of the matrix $\mathbf{A}$.
2. Check to see if all eigenvalues have a negative real part.

Unless the eigenvalues are obvious from the structure of the matrix, finding them involves a lot more work than actually needs to be done. This work is eliminated by employing some general mathematical results.

6.3.1 The Routh–Hurwitz Conditions for Two-Component Systems

We can simplify stability computations by making use of the relationship between the coefficients of the polynomial in step 1 of the plan and the sign of the eigenvalues in step 2. The following theorem supplies the key fact for a two-component system[18]:

[17] This model generalizes the SIS model, which has $\phi \to \infty$, as well as the SIR model, although deriving the SIS model from it is nontrivial because the scaling we used is based on the assumption that the duration of immunity is long compared to the duration of the disease.

[18] Problem 6.3.15.

Theorem 6.3.1 (Roots of Quadratic Polynomials)
Both roots of the polynomial equation

$$x^2 + bx + c = 0$$

have negative real parts if and only if

$$b, c > 0\,.$$

This theorem allows us to determine stability for a two-dimensional system without having to solve the characteristic equation.

▶ **Example 6.3.1** In Examples 6.2.3 and 6.2.5, we obtained characteristic equations in which all terms have the same algebraic sign. By Theorem 6.3.1, the roots of these equations have negative real parts, and thus the equilibria are stable. □

Theorem 6.3.1 is a nice improvement on the basic method because it eliminates step 2. We can also eliminate step 1 by connecting the coefficients of the polynomial that determines the eigenvalues directly to the entries in the matrix. Using this connection results in the Routh–Hurwitz[19] conditions, which provide stability criteria in terms of the entries in the matrix, thereby eliminating the need to actually compute $\det(\mathbf{A} - \lambda\mathbf{I})$.

Theorem 6.3.2 (Routh–Hurwitz Conditions for a System of Two Components)
Let

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The equilibrium solution (x^*, y^*) for a nonlinear system is asymptotically stable if

$$\operatorname{tr}\mathbf{A} < 0\,, \qquad \det\mathbf{A} > 0\,,$$

where $\mathbf{A}$ is the Jacobian of the system evaluated at the equilibrium point, $\det\mathbf{A} = ad - bc$, and the **trace** of the matrix is $\operatorname{tr}\mathbf{A} = a + d$. The equilibrium solution is unstable if $\det\mathbf{A} < 0$ or $\operatorname{tr}\mathbf{A} > 0$.

The derivation of the result in Theorem 6.3.2 is straightforward. Calculating the quantity $\det(\mathbf{A} - \lambda\mathbf{I})$ for the general matrix identifies the coefficients b and c as $-\operatorname{tr}\mathbf{A}$ and $\det\mathbf{A}$, respectively.[20]

▶ **Example 6.3.2** Given $q, \epsilon > 0$, the matrix

$$\mathbf{A} = \begin{pmatrix} -\epsilon^{-1}(1+q) & \epsilon^{-1} \\ 1 & -1 \end{pmatrix}$$

has

$$\operatorname{tr} A = -\epsilon^{-1}(1+q) - 1 < 0\,, \qquad \det A = \epsilon^{-1}q > 0\,;$$

[19] "Routh" rhymes with "mouth."
[20] Problem 6.3.16.

therefore, the equilibrium solution for the system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{b}$$

is asymptotically stable for any positive values of q and ϵ. □

Compare Example 6.3.2 with Example 6.2.3. Once the matrix is identified, the Routh–Hurwitz conditions require far less calculation than is required to find the eigenvalues.

6.3.2 The Routh–Hurwitz Conditions for Three-Component Systems

The difficulty of stability calculations increases rapidly as system size increases. Generally we must use the eigenvalue method in such cases, but we can only do specific parameter values using computer software rather than general cases. However, for three-component systems, it is often possible to obtain general results by using the corresponding Routh–Hurwitz conditions.

Theorem 6.3.3 (Routh–Hurwitz Conditions for a System of Three Components)
Let $\mathbf{A}$ be a 3×3 matrix. Let $\mathbf{A_k}$ be the 2×2 matrix obtained from $\mathbf{A}$ by deleting row k and column k. Define c_1, c_2, and c_3 by

$$c_1 = -\operatorname{tr}\mathbf{A}\,, \quad c_2 = \sum_{k=1}^{3} \det \mathbf{A_k}\,, \quad c_3 = -\det \mathbf{A}\,,$$

where $\mathbf{A}$ is the Jacobian evaluated at the equilibrium point and $\operatorname{tr}\mathbf{A}$ is the sum of the diagonal elements of $\mathbf{A}$. Then the equilibrium solution of the nonlinear system is asymptotically stable if all three coefficients are positive and $c_1c_2 > c_3$. The equilibrium solution is unstable if any of the coefficients is negative or $c_1c_2 < c_3$.

▶ **Example 6.3.3** The matrix A for the original lead poisoning model (3.7.1)–(3.7.3) is

$$A = \begin{pmatrix} -(k_1 + k_3 + r_1) & k_2 & k_4 \\ k_1 & -(k_2 + r_2) & 0 \\ k_3 & 0 & -k_4 \end{pmatrix}. \tag{6.3.1}$$

Using the notation of Theorem 6.3.3, we have

$$\begin{aligned} c_1 &= (k_1 + k_3 + r_1) + (k_2 + r_2) + k_4 > 0\,, \\ c_2 &= [k_4(k_2 + r_2)] + [k_4(k_1 + k_3 + r_1) - k_3k_4] + [(k_2 + r_2)(k_1 + k_3 + r_1) - k_1k_2] \\ &= k_4(k_2 + r_2) + k_4(k_1 + r_1) + k_2(k_3 + r_1) + r_2(k_1 + k_3 + r_1) > 0\,, \\ c_3 &= -[-k_4(k_2 + r_2)(k_1 + k_3 + r_1) + k_1k_2k_4 + k_3k_4(k_2 + r_2)] \\ &= k_4(k_2 + r_2)(k_1 + r_1) - k_1k_2k_4 = k_4(k_2r_1 + k_1r_2 + r_1r_2) > 0\,. \end{aligned}$$

Finally, we can use the observations $c_1 > k_4$ and $c_2 > (k_2r_1 + k_1r_2 + r_1r_2)$ to obtain $c_1c_2 > c_3$. The Routh–Hurwitz conditions are satisfied, and the equilibrium solution is asymptotically stable for any parameter values. □

If your eyes glaze over when you see the calculations of Example 6.3.3, there are several points to keep in mind. First, it is much easier to apply the Routh–Hurwitz conditions when at least some of the parameter values are known. Second, the calculations would have been far less tedious if we had scaled the model before doing the stability calculations. The dimensionless three-component model has a Jacobian with three parameters rather than six.[21] Even more importantly, identification of dimensionless parameters as small makes it much easier to check the Routh–Hurwitz conditions.[22]

For nonlinear problems, it is important to do the right algebraic steps at the right time. We illustrate some of the principles in a specific example.[23]

▶ **Example 6.3.4** Consider the one-parameter family of models with one predator and two prey that is given by the system

$$x' = x(1-x) - xz\,, \tag{6.3.2}$$
$$y' = ry(1-y) - yz\,, \tag{6.3.3}$$
$$z' = z(2x+2y-1)\,, \tag{6.3.4}$$

with $r > 0$. The Jacobian for this system is

$$\mathbf{J}(x,y,z) = \begin{pmatrix} 1-2x-z & 0 & -x \\ 0 & r-2ry-z & -y \\ 2z & 2z & 2x+2y-1 \end{pmatrix}. \tag{6.3.5}$$

Note that the equilibria satisfy

$$x = 0 \quad \text{or} \quad 1-x-z = 0\,, \tag{6.3.6}$$
$$y = 0 \quad \text{or} \quad r-ry-z = 0\,, \tag{6.3.7}$$
$$z = 0 \quad \text{or} \quad 2x+2y-1 = 0\,. \tag{6.3.8}$$

We can therefore look for a variety of equilibria, conveniently designated by identifying which components are nonzero. There is an extinction equilibrium Φ in which all variables are 0, an X equilibrium in which $y = 0$ and $z = 0$, and so on. Not all of these exist, for example, a Z equilibrium would have $x = 0$, $y = 0$, and $2x + 2y - 1 = 0$, which is impossible. Here we consider the XYZ equilibrium, which exists for some values of the parameter r. We leave the formulas for this equilibrium and the analysis of other equilibria for Problem 6.3.12. Here we use Equations (6.3.6)–(6.3.8) to analyze the XYZ equilibrium without using formulas for the equilibrium point components. To do this, observe that the first entry in the Jacobian can be rewritten as $(1-x-z)-x$. By (6.3.6), the portion in the parentheses is 0. Similar simplifications apply to the other diagonal entries, reducing the Jacobian to

$$\mathbf{J}(XYZ) = \begin{pmatrix} -x & 0 & -x \\ 0 & -ry & -y \\ 2z & 2z & 0 \end{pmatrix}.$$

Using the notation of Theorem 6.3.3, we have

$$c_1 = x + ry > 0\,, \qquad c_2 = rxy + 2yz + 2xz > 0\,, \qquad c_3 = 2rxyz + 2xyz > 0\,.$$

[21] Problem 6.3.11.

[22] Problem 6.3.11.

[23] See Appendix G for detailed guidelines and Problem 6.2.1 for a simple example.

It remains only to show $c_1c_2 > c_3$. Note that the first term in c_3 is the product of ry and $2xz$, while the second is the product of x and $2yz$. Thus, we can write

$$c_1c_2 > (x + ry) \cdot (2yz + 2xz) > x \cdot 2yz + ry \cdot 2xz = c_3.$$

Problems

6.3.1* (Continued from Problem 6.1.1.)

(a) Determine the stability of the origin for the system

$$x' = x - 1.2y\,,$$
$$y' = k(1.5x - y)\,.$$

Note that the answer depends on k.
(b) Compare the results of part (a) with Problem 6.1.1.

6.3.2 [Predator–prey and consumer–resource models] (Continued from Problems 3.6.9, 6.1.4, and 6.2.3.)

(a) Determine the stability of the equilibria having $c = 0$ for the dimensionless Rosenzweig–MacArthur model with Holling type I predator response,

$$v' = v(1 - v - c)\,,$$
$$c' = mc(hv - 1)\,.$$

(b) Let (v^*, c^*) be the equilibrium in which neither v nor c are 0. Use the equations that these quantities must satisfy to reduce the Jacobian to

$$J(v^*, c^*) = \begin{pmatrix} -v^* & -v^* \\ mhc^* & 0 \end{pmatrix}$$

and use this result to determine the stability of the equilibrium.
(c) Discuss the results with reference to Problems 3.6.9, 6.1.4, and 6.2.3.
(d) Explain the prediction the model makes for the effect of h on the behavior of the system.

6.3.3* [SIS disease with fixed birth rate] (Continued from Problems 6.1.6 and 6.2.4.)

(a) Compute the Jacobian for the endemic disease equilibrium of the SIS model with constant birth rate,

$$n' = \epsilon(1 - n - mi)\,,$$
$$i' = \mathcal{R}_0 i(n - i - \mathcal{R}_0^{-1})\,.$$

In so doing, you will find it most convenient to retain the variable i^* rather than to replace it with a solution formula.
(b) Determine the stability of the endemic disease equilibrium. Note that the results could depend on the parameter values.
(c) Discuss the results with reference to Problems 3.9.4, 6.1.6, and 6.2.4.

6.3.4 [SIR disease with fixed population] (Continued from Problems 6.1.7 and 6.2.5.)

(a) Determine the stability of the endemic disease equilibrium ($i^* > 0$) for the SIR model with fixed birth rate

$$s' = \epsilon(1 - s - \mathcal{R}_0 sy)$$
$$y' = \mathcal{R}_0 sy - y\,.$$

Keep in mind that both existence and stability can depend on the parameter values.

(b) Discuss the results with reference to Problems 3.9.1, 6.1.7, and 6.2.5.

6.3.5 [Plankton] (Continued from Problems 6.1.8 and 6.2.6.)

(a) Determine the stability of the coexistence equilibrium for the plankton population model

$$p' = p(1 - \alpha - p - z)\,,$$
$$z' = \delta z(p - \beta)\,.$$

(b) Discuss the results with reference to Problems 6.1.8 and 6.2.6.

(c) Explain the prediction the model makes for the effect of α and β on the behavior of the system. It will be helpful to make a plot in the first quadrant of the $\alpha\beta$ plane that identifies the regions in which each of the equilibria is stable.

6.3.6* [SEIS disease with fixed population] (Continued from Problems 6.1.9 and 6.2.7.)

(a) Use linearized stability analysis to determine the stability of the equilibria for the SEIS model with a fixed population,

$$x' = bi(1 - x - i) - \nu x\,,$$

$$i' = \nu x - i\,.$$

[When computing the Jacobian, write x^* in terms of i^* but don't substitute the formula for i^*. With proper algebraic simplification, the entry in row 1, column 2 of the Jacobian for the endemic disease equilibrium is simply a constant plus a constant times i^*. Once written in this form, the Routh–Hurwitz criteria resolve easily.]

(b) Compare the results with those of Problems 6.1.9 and 6.2.7.

6.3.7 [Chemostat] (Continued from Problems 6.1.10 and 6.2.8.)

(a) Determine the stability of the coexistence equilibrium for the chemostat model,

$$r' = qr_0\left(1 - \frac{r}{r_0} - \frac{rc}{1+r}\right)\,,$$
$$c' = c\left(\frac{r}{1+r} - q\right)\,,$$

being careful to account for the parameter requirements for existence. [This requires very little algebraic simplification, as we need only the signs of the trace and the determinant.]

(c) Sketch the region in the $r_0 q$ parameter space for which the consumer population has a stable positive value.

(d) Discuss the results with reference to Problems 6.1.10 and 6.2.8.

6.3.8 [Malaria] (Continued from Problem 6.1.11.)

(a) Determine the stability of the equilibria for the scaled Ross malaria model

$$\frac{dx}{dt} = \alpha(1-x)y - x\,,$$

$$\frac{dy}{dt} = m[\beta x(1-y) - y]\,.$$

Pay attention to restrictions on the existence of equilibria.

(b) Discuss the results with reference to Problem 6.1.11.

6.3.9 [Immune system] (Continued from Problems 6.1.15 and 6.2.11.)

(a) Find the Jacobian for the partial immune system model

$$p' = p(1-p-qm-sn)\,,$$
$$m' = a[\delta(1-m) - mp]\,.$$

(b) Use the Routh–Hurwitz conditions to show that the stability condition for the endemic disease equilibrium is $p^* > (1-\delta)/2$.

(c) Prepare a bifurcation plot of the disease-free and endemic disease equilibrium pathogen populations as a function of q with $\delta = 0.04$. Use a solid curve for ranges where each equilibrium is stable and a dashed curve for ranges where each equilibrium is unstable.

(d) Discuss the results with reference to Problem 6.1.15.

(This problem is continued in Project 6D.)

6.3.10 [SIS disease with logistic growth] (Continued from Problem 6.2.9.)

(a) Compute the Jacobian for the endemic disease equilibrium of the SIS model with constant birth rate,

$$n' = \delta[n(1-n) - mi]\,,$$
$$i' = \mathcal{R}_0 i(n - i - \mathcal{R}_0^{-1})\,.$$

In so doing, you will find it most convenient to retain the variables i^* and n^* rather than to replace them with solution formulas. However, make sure you use the equilibrium relation $n^* - i^* - \mathcal{R}_0^{-1} = 0$ to simplify when needed.

(b) Compute the trace and determinant of the Jacobian. Note that neither is obviously positive or negative.

(c) The parameter δ can be considered to be asymptotically small. This fact settles the question of whether the trace is positive or negative.

(d) Combine the equilibrium relations into a single relation for n^*. Rewrite this relation in the form $P(n^*) = 0$, where P is a quadratic polynomial.

(e) Use calculus to find the vertex of the parabola. Explain why there must be one positive value for n^* and it must be to the right of the vertex.

(f) Use the result of (e) to complete the determination of stability for the endemic disease equilibrium. Keep in mind that there is already a requirement that must be satisfied for the equilibrium to exist.
(g) Discuss the results with reference to Problems 3.9.8, 6.1.12, and 6.2.9.

6.3.11* [Lead Poisoning] (Continued from Problem 6.2.10.)
Had we scaled the three-component lead poisoning model before doing the calculations of Example 6.3.3, we would have obtained the matrix

$$A = \begin{pmatrix} -(1+a_1+b_1) & a_2 & \epsilon \\ a_1 & -(a_2+b_2) & 0 \\ 1 & 0 & -\epsilon \end{pmatrix},$$

where $a_1 = k_1/k_3$, $a_2 = k_2/k_3$, $b_1 = r_1/k_3$, $b_2 = r_2/k_3$, $\epsilon = k_4/k_3$, and ϵ is expected to be small.

(a) Approximate the parameter c_2 for the Routh–Hurwitz conditions by omitting any terms with the factor ϵ. Show with minimal calculation that $c_2 > 0$.
(b) Show that all terms in c_3 have a common factor of ϵ. Explain why it is unnecessary in this case to show that $c_1 c_2 > c_3$ for any realistic parameter values if the other conditions are satisfied.[24]
(c) Complete the demonstration of stability by showing $c_1 > 0$ and $c_3 > 0$.

6.3.12 [Predator–prey and consumer–resource models]
For the model of Example 6.3.4,

(a) Obtain a complete description of the stable equilibria for different ranges of r. To do this, you will need to identify all possible equilibria, paying careful attention to the ranges of r that they require, and determine their stability in terms of r. You should find that there is always a unique stable equilibrium.
(b) Note that r represents the relative growth rate advantage of the species y as compared to x. Explain why the results of parts (a) make sense biologically.
(c) What biologically plausible outcomes are not predicted by this over-simplified model?

6.3.13 [SEIR disease with fixed population] (Continued from Problem 6.2.12.)

(a) Determine the existence and stability of the equilibria for the SEIR model with fixed population, given in dimensionless form as

$$\begin{aligned} s' &= \epsilon(1 - s - bsy)\,, \\ x' &= bsy - \nu x\,, \\ y' &= \nu x - y\,, \end{aligned}$$

where x and y are the exposed and infectious populations, rescaled because they are both $O(\epsilon)$ at equilibrium.
(b) Discuss the results with reference to Problem 6.2.12.

6.3.14 [SIR disease with fixed birth rate and loss of immunity] (Continued from Problem 6.2.13.)

(a) Determine the existence and stability of the equilibria for the SIR model with fixed birth rate and limited immunity, given in dimensionless form as

[24] Calculation of c_3 is still messy, but the two hardest parts of the stability demonstration are considerably simplified by the assumption that ϵ is small.

$$n' = \epsilon(1 - n - dy)\,,$$
$$s' = \epsilon[1 - s + \phi(n - s) - bsy]\,,$$
$$y' = bsy - y\,,$$

where n and s are the scaled total and susceptible populations and y is the rescaled infectious population. The parameter d is the fraction of people with the disease who die of it, b is the rescaled infection coefficient, and ϕ is the ratio of the mean lifespan to the mean duration of immunity, which could be larger than 1. As part of the existence requirements, you must be concerned about the possibility that the disease renders the population unviable.

(b) Discuss the results with reference to Problem 6.2.13.

6.3.15 For the polynomial equation $x^2 + bx + c = 0$, determine the regions in the bc-plane where:

(a) The roots are real.
(b) The roots are real and differ in sign.
(c) The roots are real and positive.
(d) The roots are real and negative.
(e) The roots are complex and have negative real parts.
(f) The roots are complex and have positive real parts.
(g) Sketch a graph in the bc-plane showing the regions indicated in parts (b)–(f).

6.3.16 Use Theorem 6.3.1 to derive Theorem 6.3.2.

6.4 Case Study: Onchocerciasis

Onchocerciasis[25] is a parasitic disease endemic in parts of Africa. It is caused by a worm that has a complicated life cycle spent partly in a human host and partly in a species of black flies. While the need to incorporate the black fly vector makes the model more complicated, the within host dynamics is relatively simple: SEIS for humans and SI for flies [5]. Humans become infected when the parasite larvae enter the skin through the saliva of a biting fly. The latent stage lasts about a year, during which the parasite moves through larval stages to become an adult. The adults produce juveniles called microfilaria during the infectious human stage, and these migrate to the skin where they are taken up by biting flies. Adult worms can live 10–12 years, so this disease has an unusually long infectious period as well as the long incubation period. If a human host is able to clear all the adult worms before being reinfected, (s)he moves back to the susceptible class. Infected flies remain infected for the duration of their short lives of about 1 month.

In Problem 3.8.7, we considered the possibility of reducing the number of variables in a malaria model by thinking of the mosquito dynamics as instantaneous. This was not successful because the lifespan of the mosquito was a little too large compared to other time scales in the problem. In onchocerciasis, the 1-month fly lifespan is quite a bit shorter than the 1-year incubation period, so we will see that thinking of the fly dynamics as instantaneous works well, thereby making the analysis of long-term behavior much less difficult. Rather than just make this a mathematical assumption, we'll properly justify it through the use of scaling and asymptotic arguments, supplemented by simulation results.

[25] on-ko-sir-KI-a-sis.

6.4.1 Model Development

It is common practice to use the same letters for the human and vector infectious classes and distinguish multiple hosts through subscripts. For improved readability, we instead use U and V for the uninfected and infected fly vectors, as shown in Fig. 6.4.1. Note the assumption that there is no additional mortality from the disease. Onchocerciasis is seldom fatal, but it can cause blindness, hence its colloquial name "river blindness." The disease is the subject of an eradication program sponsored by the Carter Center, using a medication (ivermectin, which is used as a heartworm preventive for dogs and cats) that prevents the production of microfilaria. There is, however, both empirical and theoretical evidence that questions the value of this attempt at eradication [3, 5].

From the diagram in Fig. 6.4.1, with T as dimensional time, we obtain the system of equations

$$\frac{dS}{dT} = \mu(N - S) - \beta SV + \gamma I\,, \tag{6.4.1}$$

$$\frac{dE}{dT} = \beta SV - (\eta + \mu)E\,, \tag{6.4.2}$$

$$\frac{dI}{dT} = \eta E - (\gamma + \mu)I\,, \tag{6.4.3}$$

$$\frac{dU}{dT} = d(F - U) - \alpha UI\,, \tag{6.4.4}$$

$$\frac{dV}{dT} = \alpha UI - dV\,, \tag{6.4.5}$$

$$S + E + I = N\,, \qquad U + V = F\,. \tag{6.4.6}$$

The model can be written with just two differential equations for the humans and one for the flies by using the algebraic equations (6.4.6) to eliminate two of the variables. It is most convenient to eliminate S and U, leaving the model as

$$\frac{dE}{dT} = \beta(N - E - I)V - (\eta + \mu)E\,, \tag{6.4.7}$$

$$\frac{dI}{dT} = \eta E - (\gamma + \mu)I\,, \tag{6.4.8}$$

$$\frac{dV}{dT} = \alpha(F - V)I - dV\,. \tag{6.4.9}$$

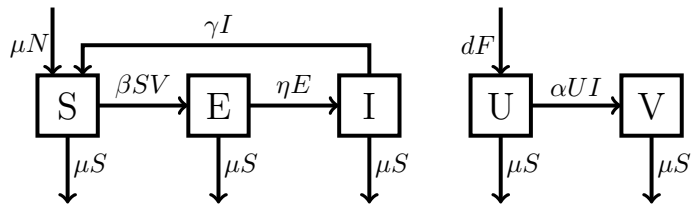


Fig. 6.4.1 The SEIS-UV onchocerciasis model. Note $N = S + E + I$ and $F = U + V$

6.4.2 Preparation for Analysis

The three-dimensional model has eight parameters, which we can reduce to five by nondimensionalization. Careful scaling will provide additional benefits that make the algebraic work of the Routh–Hurwitz stability analysis just barely feasible.

The obvious choices for scales work well here, that is, N for the human populations, F for the fly population, and the infectious duration $1/(\gamma + \mu)$ for the time. This means that dimensionless time $t = 1$ will represent approximately 10 years.

Using x, i, and v for the scaled versions of E, I, and V, the scaled system is

$$x' = bv(1 - x - i) - \nu x \,, \tag{6.4.10}$$

$$i' = \nu(1 - \epsilon)x - i \,, \tag{6.4.11}$$

$$\delta v' = a(1 - v)i - v \,, \tag{6.4.12}$$

where the dimensionless parameters are

$$b = \frac{\beta F}{\gamma + \mu} \,, \quad a = \frac{\alpha N}{d} \,, \quad \nu = \frac{\eta + \mu}{\gamma + \mu} \,, \quad \epsilon = \frac{\mu}{\eta + \mu} \,, \quad \delta = \frac{\gamma + \mu}{d} \,. \tag{6.4.13}$$

Each of these parameters has a straightforward epidemiological interpretation: b and a are the effective transmission coefficients from human to fly and fly to human, ν is the ratio of infectious duration to latent duration, ϵ is the ratio of latent duration to human lifespan, and δ is the ratio of fly lifespan to infectious duration. We can expect that ab will be the basic reproduction number, and thus somewhere in the range 1–5, while the information given earlier about time scales suggests estimates of 10 for ν, 0.02 or less for ϵ, and less than 0.01 for δ.

The analysis of the onchocerciasis model will make heavy use of the smallness of the parameters ϵ and δ. The first appears only in the combination $1 - \epsilon$. Given that all models are approximations and some parameters are known only to one significant figure or worse, we are fully justified in setting $\epsilon = 0$,[26] which will simplify the algebraic work noticeably even though it does not simplify the problem.

The role of δ in the model is much more significant than that of ϵ, so assuming it to be small will have a much larger impact but also requires more caution. The assumption that δ is asymptotically small marks v as a fast variable[27]; this means that we expect the solution to quickly approach a state in which the right-hand side of (6.4.12) is asymptotically small. Hence, the assumption $\delta \ll 1$ yields a two-component model consisting of (6.4.10) and (6.4.11), with $\epsilon = 0$, along with the algebraic equation

$$a(1 - v)i - v = 0 \,. \tag{6.4.14}$$

Analysis of this two-component model will be far easier than analysis of the full three-component model.

Ultimately, the question of whether the two-component model is an adequate alternative to the full model should be determined by numerical simulations, which we consider later. For now, a comparison with the malaria model of Problem 3.8.7 is worthwhile. That model had a parameter $1/m$ analogous to our δ, but its value was about 0.1. Simulations showed that the quasi-steady approximation of replacing the differential equation with an algebraic equation was not very good. The much longer infectious duration of onchocerciasis makes the corresponding time scale parameter much smaller, which we hope will be enough to justify the simplification.

[26] While we could formally demonstrate this in conjunction with real data using AIC (Sect. 2.4), it is clearly acceptable modeling practice to make this decision based on general principles, in the absence of hard data.

[27] Section 6.1.

6.4.3 Analysis of the Three-Component System

The analysis of the three-component system is somewhat tedious, but not overly difficult. The key is to make liberal use of the equilibrium equations but to avoid substituting equilibrium formulas.[28] We begin with the Jacobian,

$$\mathbf{J} = \begin{pmatrix} -(\nu + bv) & -bv & b(1-x-i) \\ \nu & -1 & 0 \\ 0 & \delta^{-1}a(1-v) & -\delta^{-1}(1+ai) \end{pmatrix}. \tag{6.4.15}$$

At the disease-free equilibrium, where each of the state variables is 0, the Jacobian reduces to

$$\mathbf{J}_{DF} = \begin{pmatrix} -\nu & 0 & b \\ \nu & -1 & 0 \\ 0 & \delta^{-1}a & -\delta^{-1} \end{pmatrix}. \tag{6.4.16}$$

The Routh–Hurwitz coefficients are then

$$c_1 = 1 + \nu + \delta^{-1} > \delta^{-1} > 0\,,$$

$$c_2 = \nu + \delta^{-1}\nu + \delta^{-1} \gg \nu > 0\,,$$

$$c_3 = \delta^{-1}\nu(1-ab)\,.$$

Stability therefore requires $ab < 1$. The final criterion is easily satisfied, as

$$c_1 c_2 \gg \delta^{-1}\nu > c_3.$$

The stability calculation for the endemic disease equilibrium works out easily with just the right algebra choices. The key is to combine the equilibrium relations to get the identity

$$ab(1-x-i) = a\frac{\nu x}{v} = \frac{ai}{v} = 1 + ai\,. \tag{6.4.17}$$

With this identity applied to the Jacobian entry $b(1-x-i)$ in row 1, column 3, the Jacobian is

$$\mathbf{J}_{ED} = \begin{pmatrix} -(\nu + bv) & -bv & a^{-1}(1+ai) \\ \nu & -1 & 0 \\ 0 & \delta^{-1}a(1-v) & -\delta^{-1}(1+ai) \end{pmatrix}. \tag{6.4.18}$$

While there is still a lot of messy algebra involved in the calculation of the Routh–Hurwitz coefficients,[29] it is relatively straightforward to use those results to verify that the endemic disease equilibrium is asymptotically stable whenever it exists.

6.4.4 The Endemic Disease Equilibrium

It may seem backward to do the stability calculation before calculating the equilibria; however, this is a good way to keep yourself from using the equilibrium results unless absolutely necessary. We do

[28] See Appendix F.

[29] Problem 6.4.2.

eventually want to find the equilibria, which are the solutions of the system

$$bv(1-x-i) = \nu x\,, \quad i = \nu x\,, \quad v = ai(1-v)\,. \tag{6.4.19}$$

We can solve the second and third of these equations to obtain

$$v^* = \frac{ai^*}{1+ai^*}\,, \qquad x^* = \nu^{-1} i^*\,. \tag{6.4.20}$$

Substituting these results into the first equilibrium relation and using (6.4.17) quickly yields the result

$$i^* = \frac{1-\mathcal{R}_0^{-1}}{1+b^{-1}+\nu^{-1}}\,, \qquad \mathcal{R}_0 \equiv ab\,; \tag{6.4.21}$$

hence, $\mathcal{R}_0 > 1$ is required for the existence of the endemic disease equilibrium.

6.4.5 Analysis of the Two-Component System

Compared to the three-component system, the two-component system requires more calculus and less algebra. In computing the Jacobian, it is easiest to leave the symbol v in the x' equation; however, this means that the product rule is needed to compute the derivative of x' with respect to i.[30] It makes the algebra easier to define a quantity

$$g = \frac{1}{1+ai}\,,$$

which allows us to write the derivative dv/di as

$$\frac{dv}{di} = \frac{a}{(1+ai)^2} = ag^2\,.$$

The Jacobian is then

$$\mathbf{J} = \begin{pmatrix} -(\nu+bv) & abg^2(1-x-i)-bv \\ \nu & -1 \end{pmatrix}. \tag{6.4.22}$$

At the equilibria, the Jacobian reduces to

$$\mathbf{J}_{DF} = \begin{pmatrix} -\nu & ab \\ \nu & -1 \end{pmatrix}, \qquad \mathbf{J}_{ED} = \begin{pmatrix} -(\nu+bv) & g-bv \\ \nu & -1 \end{pmatrix}, \tag{6.4.23}$$

where the subscripts identify the disease-free and endemic disease equilibria. The trace is clearly negative in both cases. The determinant for the disease-free case contains the factor $1-ab$, resulting in the stability requirement $ab < 1$, while the determinant in the endemic disease case is seen to be positive because $g < 1$.

6.4.6 Simulation

A worst-case environment for onchocerciasis has an equilibrium solution with about 50% of the human population and 30% of the fly population in the infectious stage [3], roughly corresponding

[30] Appendix B.

to the parameters $a = 0.9$, $b = 3.6$. Reasonable values for the other parameters are $\nu = 10$, $\epsilon = 0.02$, and $\delta = 0.008$.

We consider a scenario in which a small number of infected flies are introduced into a region with no prior exposure. Figure 6.4.2 shows the results. The left panel is the full three-component system (6.4.10)–(6.4.11), with no asymptotic simplifications. The right panel shows the solutions of both versions of the model for just the first 2 years. Up to about 4 months, there is a clear error in the variable v, as the two-component model misses the initial transient, during which the rate of change of v is order δ^{-1}. After that point, however, the solutions are distinguishable only because of the very limited range of y-axis values in the plot window. The persistent slight error after the initial transient is due to the approximation $\epsilon = 0$, which introduces a 2% error. As noted earlier, this small error is undoubtedly less important to an AIC score than the presence of one additional parameter. The assumption that v can be treated as a fast variable simplifies the analysis with no noticeable difference in results for the human population and a noticeable difference for the fly population for just the first few months of a long process.

Problems

6.4.1 The most common public health mitigation strategy for onchocerciasis is to administer ivermectin to infectious humans. This medication works by killing the microfilaria that are passed from humans to susceptible flies, but it does not actually treat the disease in humans. We can incorporate ivermectin distribution into the model as a reduction in one of the parameters.

(a) Calculate the stable equilibrium solutions for two possible public health campaigns:

1. Use ivermectin to reduce the production of microfilaria by 90%. (This is a very optimistic estimate of how much can be accomplished with widespread ivermectin treatment.) This will make a drastic change in one parameter.
2. Use some combination of physical barriers, chemical deterrents, and pesticides to reduce the rate of fly bites by 50%. (This is also a very optimistic estimate of how much can be accomplished with these measures.) This will make smaller changes in two parameters.

(b) Modify *ODEsim.m* to run the two-component onchocerciasis model. Check your work by reproducing the dash-dot curves of Fig. 6.4.2b.

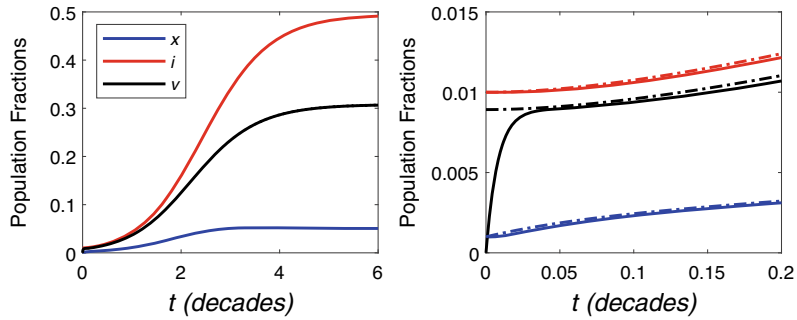


Fig. 6.4.2 A simulation of the onchocerciasis model, with $a = 0.9$, $b = 0.36$, $\nu = 10$, $\epsilon = 0.02$, $\delta = 0.008$; **b** shows a comparison of the two-component approximation (dash-dot) with the full three-component system

(c) Run a simulation of the ivermectin eradication strategy (1.). For initial conditions, use the pre-treatment stable equilibrium solution. Run the simulation for as long as it takes to get the full benefit of the strategy, then rerun for a 20-year period (keep in mind the scale used for dimensionless time).
(d) Run a simulation for the biting mitigation strategy (2.) using the pre-treatment stable equilibrium as the initial state. As in (c), run long enough to get the full impact and then rerun for a 20-year period.
(e) Discuss the benefits and drawbacks of the two strategies for reducing onchocerciasis. Is one clearly better than the other? Is either adequate?
(f) If we want to make a drastic reduction in the number of humans harboring the onchocerciases parasites, what kind of treatment should we be trying to develop? This should be relatively straightforward in light of the disease facts presented in the section.

6.4.2 Show that the Routh–Hurwitz conditions are satisfied for the endemic disease equilibrium (6.4.18). It is relatively straightforward to show that all three coefficients are positive. To show $c_1c_2 > c_3$, you can use simple underestimates for c_1 and c_2. Keep the largest terms for c_1; then all of the *smaller* terms in c_2 are sufficient.

6.4.3 Do a nullcline analysis of the two-component onchocerciasis model, considering both cases $ab > 1$ and $ab < 1$. Verify that the results are consistent with the analytical results in the text.

6.4.4 [HIV] (Continued from Problems 3.3.9, 3.4.18, and 3.6.11.)
Determine the stability of the equilibria for the dimensionless three-component HIV model,

$$s' = 1 - s - bvs\,, \tag{6.4.24}$$
$$\delta i' = bvs - i\,, \tag{6.4.25}$$
$$\epsilon v' = i - v\,. \tag{6.4.26}$$

6.4.5 [HIV] (Continued from Problem 6.4.4.)

(a) Use the assumption that ϵ is a small parameter to reduce the three-component HIV model (6.4.24)–(6.4.26) to a two-dimensional model.
(b) Modify *ODEsim.m* to produce figures similar to Fig. 6.4.2 showing that the two-component model is a fully adequate substitute for the three-component model. In particular, how long does the initial transient in the fast variable last?
(c) Do nullcline analysis and stability analysis for the two-component model. Depending on which problems you have done from Sects. 6.1–6.3, you may be able to copy prior work, as the two-component HIV model is essentially equivalent to a model appearing in those problem sets.

6.5 Discrete Nonlinear Systems

After studying this section, you should be able to:

- Run simulations for discrete nonlinear systems.
- Analyze the stability of fixed points for discrete nonlinear systems.

To build insight into the theory of multi-dimensional discrete nonlinear systems, we begin with the general case of a population divided between two age classes, juveniles (J) and adults (A), with the

additional requirements that all surviving juveniles become adults in the next year and all adults die after reproducing. With these restrictive assumptions, the populations can be modeled by equations of the form

$$J_{t+1} = f(A_t),$$
$$A_{t+1} = g(J_t).$$

Because generations do not overlap, we can combine the two equations into a single one that spans a discrete interval of 2 years. We can then apply all of the methods for dealing with such equations from Chap. 4.

▶ **Example 6.5.1** Suppose reproduction by adults is subject to competition for resources, while survival of juveniles is not, with no adult survival and a 1-year juvenile stage. Using the Holling type 2 function for the birth rate, we have the model

$$J_{t+1} = \frac{f A_t}{b + A_t},$$
$$A_{t+1} = s J_t .$$

The model can be simplified by choosing dimensionless forms of the state variables:

$$J = fj, \qquad A = ba . \tag{6.5.1}$$

With these substitutions, the model becomes

$$j_{t+1} = \frac{a_t}{1 + a_t}, \tag{6.5.2}$$
$$a_{t+1} = r j_t , \tag{6.5.3}$$

where the combined parameter $r = sf/b$ represents a recruitment rate.

The model equations combine to make a 2-year discrete model

$$a_{t+2} = r j_{t+1} = r \frac{a_t}{1 + a_t}, \tag{6.5.4}$$

which we can think of as

$$a_{n+1} = \frac{r a_n}{1 + a_n} \equiv h(a_n), \tag{6.5.5}$$

where n is a time coordinate that measures 2-year intervals.

Fixed points satisfy $a_{n+1} = a_n = a^*$, with the resulting values

$$a^{*0} = 0, \quad a^{*1} = r - 1,$$

with $r > 1$ required for existence of a^{*1}.

A fixed point a^* is asymptotically stable if and only if $|h'(a^*)| < 1$.[31] We have

$$h'(a) = \frac{r}{(1 + a)^2};$$

[31] Theorem 4.4.1.

thus,

$$h'(0) = r, \qquad h'(r-1) = \frac{1}{r}\,.$$

The fixed point $a^{*0} = 0$ is stable if $r < 1$, while the fixed point $a^{*1} = r - 1$ is stable whenever it exists. Cobweb analysis confirms these conclusions and also demonstrates that the long-term behavior of the population always converges to the stable fixed point. □

Check Your Understanding 6.5.1:

Plot cobweb diagrams for the cases $r = 2$ and $r = 0.8$ for the model

$$a_{n+1} = \frac{ra_n}{1 + a_n}$$

and use them to confirm the conclusions of Example 6.5.1.

Having completed the analysis of the model in Example 6.5.1 using prior methods, we can now use that model as a test case in developing methods for more general discrete nonlinear systems.

6.5.1 Linearization for Discrete Nonlinear Systems

In Chap. 5, we saw that the quantity λ in the discrete model $N_{t+1} = \lambda N_t$ generalizes to the eigenvalues for a discrete linear system. We also saw in Sect. 6.2 that the stability of an equilibrium solution of a continuous nonlinear system is determined by the eigenvalues of the Jacobian matrix at the corresponding equilibrium point. The same connections hold for discrete nonlinear systems. Near a fixed point, a discrete nonlinear system can be approximated by a linear system represented by the Jacobian matrix.[32] The eigenvalues of the Jacobian determine the behavior near the corresponding fixed point, with $|\lambda| < 1$ for all λ required for stability. It is often more convenient to use the equivalent criterion $|\lambda|^2 < 1$ for all λ; this generalizes to the case of complex eigenvalues, with the magnitude of a complex number defined by $|a + ib|^2 = a^2 + b^2$.

▶ **Example 6.5.2** The Jacobian for the system of Example 6.5.1 is

$$J = \begin{pmatrix} 0 & \frac{1}{(1+a)^2} \\ r & 0 \end{pmatrix}.$$

Thus, the eigenvalues are given by

$$0 = \det \begin{pmatrix} -\lambda & \frac{1}{(1+a^*)^2} \\ r & -\lambda \end{pmatrix} = \lambda^2 - \frac{r}{(1+a^*)^2}\,.$$

The stability requirement $\lambda^2 < 1$ yields the inequality

$$(1 + a^*)^2 > r\,.$$

This requirement reduces to $r < 1$ for $a^{*0} = 0$ and is always satisfied for $a^{*1} = r - 1$, which are the same results we found in Example 6.5.1 using the method for single discrete equations. □

[32] Problem 6.5.7.

It often requires significant algebraic calculation to compute eigenvalues for a model with arbitrary parameters. Some of this calculation can be avoided by making use of the Jury conditions for stability, which are a set of inequalities written in terms of quantities calculated directly from the Jacobian matrix. These correspond to the Routh–Hurwitz conditions for continuous systems.[33]

Theorem 6.5.1 (Jury Conditions for a System of Two Components)
Let $\mathbf{J}$ be the Jacobian matrix that represents a discrete nonlinear system of two components near a fixed point $\mathbf{x}^*$. The fixed point is asymptotically stable if

$$|\operatorname{tr}\mathbf{J}| < 1 + \det\mathbf{J} < 2,$$

where $\operatorname{tr}\mathbf{J}$ is the sum of the main diagonal entries of the Jacobian. The fixed point is unstable if any one of the inequalities

$$\operatorname{tr}\mathbf{J} > 1 + \det\mathbf{J}, \qquad \operatorname{tr}\mathbf{J} < -1 - \det\mathbf{J}, \qquad \det\mathbf{J} > 1$$

is true.

▶ **Example 6.5.3** For the model of Examples 6.5.1 and 6.5.2, we have

$$\operatorname{tr}\mathbf{J} = 0, \qquad \det\mathbf{J} = -\frac{r}{(1+a^*)^2}.$$

The first of these results reduces the Jury conditions to $-1 < \det\mathbf{J} < 1$, and the second of this latter pair is automatically satisfied because the determinant is negative. The remaining condition, $\det\mathbf{J} > -1$, reduces to $(1+a^*)^2 > r$, which we previously obtained from the eigenvalue formula in Example 6.5.2. □

As with the Routh–Hurwitz conditions for continuous systems, there are Jury conditions for any size of system, but they get more complicated as the size increases. Here we present the Jury conditions for 3×3 matrices.

Theorem 6.5.2 (Jury Conditions for a System of Three Components)
Let $\mathbf{A}$ be the Jacobian matrix for a three-component discrete system at a fixed point. Let $\mathbf{A_k}$ be the 2×2 matrix obtained from $\mathbf{A}$ by deleting row k and column k. Define c_1, c_2, and c_3 by

$$c_1 = -\operatorname{tr}\mathbf{A}\,, \quad c_2 = \sum_{k=1}^{3} (\det\mathbf{A_k})\,, \quad c_3 = -\det\mathbf{A}\,,$$

where $\operatorname{tr}\mathbf{A}$ is the sum of the diagonal elements of $\mathbf{A}$, all evaluated at a fixed point x^*. Then x^* is asymptotically stable if

1. $1 + c_1 + c_2 + c_3 > 0$,
2. $1 - c_1 + c_2 - c_3 > 0$,

[33] Section 6.3.

3. $|c_2 - c_1 c_3| < 1 - c_3^2$.

The fixed point is unstable if any of the inequalities is reversed.

6.5.2 A Structured Population Model with One Nonlinearity

In Sect. 6.3, we examined three-component continuous models with multiple nonlinearities. Here we consider a three-component discrete model with only one nonlinear equation. This is enough nonlinearity to produce some very complicated behavior.

Assume a life history encompassing three yearly stages, with the two older stages occupying the same ecological niche, but only the oldest stage reproducing. With density-independent survival and density-dependent fertility, similar to the two-component model of Example 6.5.1, we have the model

$$L_{t+1} = \frac{f A_t}{b + Y_t + A_t}, \tag{6.5.6}$$

$$Y_{t+1} = s_1 L_t, \tag{6.5.7}$$

$$A_{t+1} = s_2 Y_t, \tag{6.5.8}$$

where $f, b > 0$ and $0 < s_1, s_2 < 1$. This system can be rewritten in dimensionless form[34] as

$$\ell_{t+1} = \frac{2a_t}{2 + y_t + a_t}, \tag{6.5.9}$$

$$y_{t+1} = r\ell_t, \tag{6.5.10}$$

$$a_{t+1} = s y_t. \tag{6.5.11}$$

Figure 6.5.1 shows plots of y for four simulations using the common value $s = 0.6$ and r values of 9, 14, 15, and 60. The complicated patterns in these plots look more like computer art than scientific results.[35] They show that determining fixed points and stability from looking at simulations is not always possible. Here, it seems clear that the scenarios of panels **c** and **d** do not have a stable fixed point, but stability in panels **a** and **b** is unclear.

How were the parameter sets for the plots of Fig. 6.5.1 chosen so as to illustrate these strange behaviors? The r values were picked by trial and error with an eye toward interesting patterns. The value of s was not chosen by trial and error. Most values of s do not exhibit unusual patterns, but $s = 0.6$ is in a narrow range of values identified by a thorough analysis of the model, which we begin in the remainder of this section and finish in the exercises.[36]

6.5.2.1 Analysis of the Model

The Jacobian for the model (6.5.9)–(6.5.11) is

[34] Using $L = f\ell/2$, $Y = by/2$, $A = ba/2$, $r = fs_1/b$, and $s = s_2$.

[35] The patterns are even more complicated than they first appear. If you look for periodicity in the $r = 15$ case, you find what appears to be a cycle of period 171, but in fact there is a slight drifting away from a true periodic solution.

[36] Stumbling across a good set of parameter values by chance can be like throwing a dart at a fog-obscured board and managing to hit the bullseye. Finding good parameter values after the analysis is like walking up to the board and stabbing the bullseye with the dart.

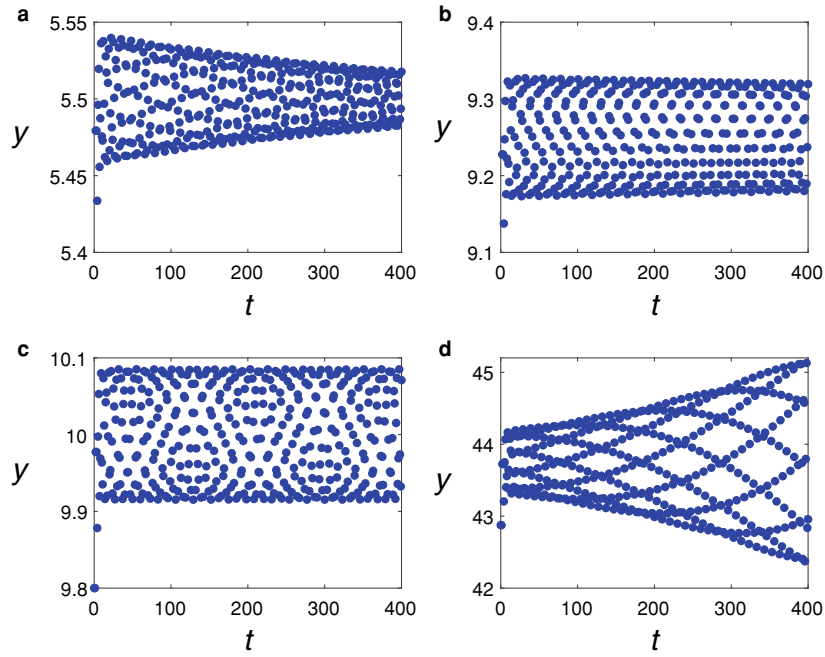


Fig. 6.5.1 y values for the model of (6.5.9)–(6.5.11), using $s = 0.6$, with (a) $r = 9$, (b) $r = 14$, (c) $r = 15$, (d) $r = 60$. In each case, the initial condition was 0.99 times the positive fixed point

$$J = \begin{pmatrix} 0 & J_{12} & J_{13} \\ r & 0 & 0 \\ 0 & s & 0 \end{pmatrix},$$

where

$$J_{12} = \frac{-2a}{(2+y+a)^2}, \qquad J_{13} = \frac{2(2+y)}{(2+y+a)^2}. \tag{6.5.12}$$

With c_1, c_2, c_3 defined as in Theorem 6.5.2, we have

$$c_1 = 0\,, \quad c_2 = -rJ_{12} \geq 0\,, \quad c_3 = -D = -rsJ_{13} < 0\,. \tag{6.5.13}$$

Using the determinant $D = -c_3$ in place of c_3, the Jury conditions become

$$1 + c_2 > D\,, \qquad 1 + c_2 + D > 0\,, \qquad c_2 < 1 - D^2\,. \tag{6.5.14}$$

The second of the conditions is automatically satisfied. The third requires $D < 1$, which guarantees that the first condition is satisfied as well (given $c_2 \geq 0$ and $D > 0$). Hence, stability hinges on the third condition. The subsequent analysis has to be done for each fixed point.

1. There is an obvious fixed point in which all components are 0. For this point,

$$c_2 = 0, \qquad D = rs.$$

This extinction point is stable if $rs < 1$.

The quantity rs has a clear biological interpretation. If we start with a tiny larva population ℓ_0, along with $y_0 = 0$ and $a_0 = 0$, we then get $\ell_1 = 0, y_1 = r\ell_0, a_1 = 0$, and $\ell_2 = 0, y_2 = 0, a_2 = rs\ell_0$, followed by

$$\ell_3 = \frac{2rs\ell_0}{(2 + rs\ell_0)} < rs\ell_0 \,.$$

In the limit of a small starting population, we have $\ell_3/\ell_0 \to rs$. Thus, rs represents the maximum possible population growth rate for a 3-year period. Of course the population dies out if this quantity is less than 1.

2. Now assume a non-extinction fixed point. These must satisfy

$$\ell^* = \frac{2a^*}{2 + y^* + a^*} \,, \qquad y^* = r\ell^* \,, \qquad a^* = rs\ell^* \,.$$

Substituting the second and third of these equations into the first and eliminating a factor of $\ell^* > 0$, the first equation becomes

$$1 = \frac{2rs}{2 + (s+1)r\ell^*} \,.$$

With a little bit of algebra, we eventually obtain

$$\ell^* = \frac{2(rs-1)}{r(1+s)} \,, \qquad y^* = \frac{2(rs-1)}{1+s} \,, \qquad a^* = sy^* \,. \tag{6.5.15}$$

As expected, this fixed point is only possible if $rs > 1$.

At this point, the algebraic steps needed to complete the analysis are quite obscure. We begin by using (6.5.15) to obtain the result

$$2 + y^* + a^* = 2 + (1+s)y^* = 2rs \,.$$

This combines with (6.5.12) and (6.5.13) to yield

$$c_2 = \frac{y^*}{2rs} \,, \qquad D = \frac{2 + y^*}{2rs} \,. \tag{6.5.16}$$

At this point, some messy algebra yields the convenient results

$$1 - D = \frac{y^*}{2r} \,, \qquad 1 + D = 2 - (1 - D) = 2 - \frac{y^*}{2r} \,. \tag{6.5.17}$$

By rewriting the stability requirement (the third condition of (6.5.14)) as

$$c_2 < 1 - D^2 = (1 - D)(1 + D) \,,$$

and substituting from (6.5.17), we get one factor of y^* to cancel, leaving the inequality

$$\frac{1}{s} < 2 - \frac{y^*}{2r} \,.$$

Substituting the formula for y^* and doing more algebra[37] eventually results in the inequality

$$rs^2 + (r+1)s - r > 0. \tag{6.5.18}$$

This is the simplest result we can obtain for the final stability criterion. Some additional characterization of stability will be worked out in Problem 6.5.5.

6.5.3 Choosing a Discrete or Continuous Model

This section reinforces lessons from Chap. 4. Discrete models have two significant drawbacks compared to continuous models:

1. Discrete models are harder to analyze. Cobweb analysis is much weaker than phase line analysis, and there is no graphical technique for discrete systems. While the Jury conditions do not seem at first glance to be significantly different from the Routh–Hurwitz conditions, in practice the calculations are usually much messier, particularly for three-component systems.
2. Discrete models have mathematical complications caused by their inability to respond quickly to dynamic changes, leading to overcompensation and instability. Using a discrete model in a setting where a continuous model could be used instead can result in strange behavior that is due to the choice of model rather than the properties of the setting. For another example of complicated behavior in a discrete model, see Ledder et al., 2020 [4].

Given the drawbacks of discrete models, it is reasonable to ask if there are any cases in which they are justified. Discrete models should only be used when the choice is mandated by synchrony of life events, such as population growth of a species that has an annual reproductive event or an annual harvest (as was the case in [4]). Discrete models should not be used when life stages are of the same duration but not synchronous, nor should they be used merely because data is collected at discrete points in time. Epidemiological models, for example, should be continuous rather than discrete in almost all cases.

Check Your Understanding Answers

1. The cobweb plots appear in Fig. 6.5.2. Note that n measures 2-year intervals.

Problems

6.5.1 Derive Equation (6.5.18).

6.5.2 Use (6.5.18) to determine the stability of the equilibria in the four cases of Fig. 6.5.1.

6.5.3 The full model (6.5.2)–(6.5.3) and the corresponding 2-year, one-component model (6.5.4) have some subtle differences.

(a) Plot solutions for (6.5.4) using each of the initial conditions $a_0 = 0.1$ and $a_0 = 2$. Use the 1-year time coordinate t rather than the 2-year time coordinate n.

[37] Problem 6.5.1.

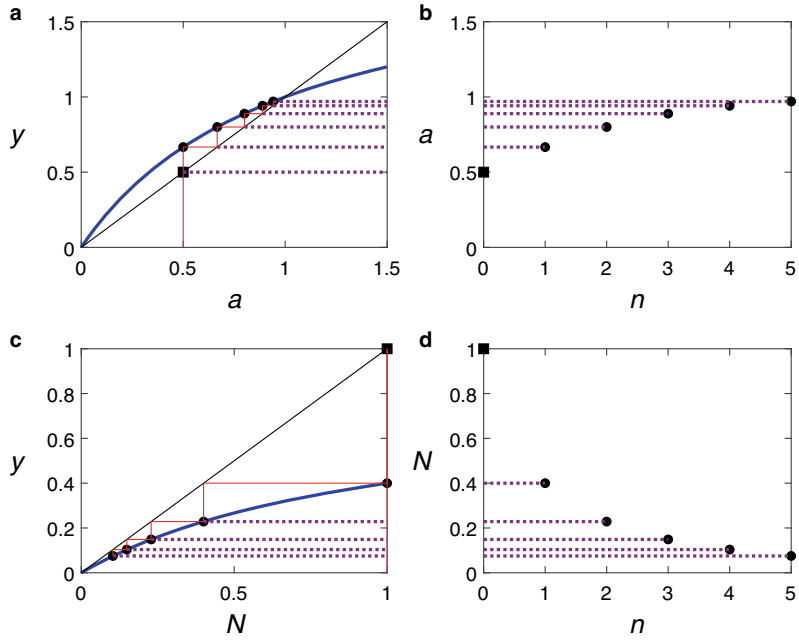


Fig. 6.5.2 Cobweb plots for Check Your Understanding 6.5.1: $r = 2.0$ (top); $r = 0.8$ (bottom)

(b) Plot solutions for (6.5.2)–(6.5.3) using the initial condition $(y_0, a_0) = (1, 0.1)$.
(c) Explain the connection between the solutions of (a) and (b).

6.5.4*

(a) Run numerical simulations for the model (6.5.9)–(6.5.11) with parameter values $r = 2.05$ and $s = 0.48$ and initial conditions $\ell_0 = 0.01$, $y_0 = 0.02$, and $a_0 = 0.01$. Plot graphs of y for 1,000 time steps and for 10,000 time steps. [Do not connect the points!] Do you think there is a stable fixed point?
(b) Repeat with $s = 0.495$.
(c) Repeat with $s = 0.51$.
(d) Do the results of these simulations agree with the analysis of Sec. 6.5.2.1 Explain.

6.5.5 This problem completes the general analysis begun in Sec. 6.5.2.1.

(a) Create a parameter space plot for the system by plotting the existence and stability boundaries for the extinction and coexistence fixed points in the sr-plane. Identify each region created by these boundaries according to which, if any, of the fixed points are stable.
(b) Observe that there is a value $s = s_2$ for which there is always one asymptotically stable solution whenever $s > s_2$. Find both the exact value and a numerical approximation.
(c) Find the point (s_1, r_1) where the two stability boundary curves intersect.
(d) Suppose $s < s_1$. As r increases, identify how the long-term behavior of the system changes as r increases.
(e) Repeat (d) for $s_1 < s < s_2$.
(f) Repeat (d) for $s > s_2$.

6.5.6* Parasitoids[38] are animals whose life history combines a free-living stage and a parasitic stage. Many wasps and flies, for example, lay eggs in a caterpillar or other insect host. When the eggs hatch, the wasp larvae consume the host from the inside. These animals are of significant interest in ecology because they are common in nature and because they can be useful as bio-control agents. In many cases, parasitoids have a synchronized life cycle, with one generation per year; in these cases, a discrete nonlinear model is most appropriate. The simplest example is the Nicholson–Bailey model, given in dimensionless form as

$$h_{t+1} = Rh_t e^{-p_t},$$
$$p_{t+1} = h_t[1 - e^{-p_t}],$$

where h and p are the dimensionless host and parasitoid populations.

(a) What is the significance of R? [Hint: Look at the simplified model for the case where there are no parasitoids.]
(b) Calculate the fixed points.
(c) Determine the range of R for which the fixed point that has no parasitoids is stable.
(d) Use Theorem 6.5.1 to reduce the stability conditions for the fixed point with $p^* > 0$ to a single inequality of the form $g(R) < 1$ for some function g.
(e) Show that $g(1) = 1$ and that g is an increasing function of R when $R > 1$.
(f) Use the result of (e) to identify the range of R for which the corresponding fixed point is stable.
(g) Run a simulation using $R = 1.1$, $h_0 = 1$, $p_0 = 0.4$, and a total of 120 years.
(h) Are the theoretical and simulation results consistent? Are they biologically realistic?

(This problem is continued in Project 6E.)

6.5.7 Suppose (X^*, Y^*) is a fixed point for a system

$$X_{n+1} = f(X_n, Y_n), \qquad Y_{n+1} = g(X_n, Y_n).$$

Near the fixed point, we can replace X and Y by

$$X = X^* + \epsilon x, \qquad Y = Y^* + \epsilon y.$$

Use the method of Appendix F to linearize the system near the fixed point. Conclude that the system is approximately

$$\mathbf{z}_{n+1} = \mathbf{J}(X^*, Y^*)\mathbf{z}_n,$$

where $\mathbf{z}$ is the vector whose components are x and y and $\mathbf{J}(X^*, Y^*)$ is the Jacobian matrix evaluated at the fixed point.

6.6 Projects

Project 6A: The SIR Model with Logistic Growth

The SIR model with logistic growth was introduced in Sect. 3.9 and scaled to facilitate comparison with the fixed birth rate SIR model. A slightly different scaling is more convenient for analysis of equilibria:

[38] PAIR-uh-si-toid.

$$n' = \epsilon[n(1-n) - dy]\,, \tag{6.6.1}$$

$$s' = \epsilon[n(1-n) + m(n-s) - \mathcal{R}_0 sy\,, \tag{6.6.2}$$

$$y' = y(\mathcal{R}_0 s - 1)\,, \tag{6.6.3}$$

where n and s are the total and susceptible populations relative to the population carrying capacity, y is the rescaled infectious population, $\mathcal{R}_0$ is the basic reproductive number, ϵ is the ratio of the disease time scale to the population growth time scale, m is the ratio of the natural death rate to the population growth rate, and d is the mortality fraction for the disease. In this project, we do a complete stability analysis, assuming only that ϵ is asymptotically small, that is, terms with more factors of ϵ are omitted when they are added to terms with fewer factors of ϵ.

1. Preliminary Analysis

   a. Identify the three equilibria for this model: an extinction equilibrium, a disease-free equilibrium, and an endemic disease equilibrium. For the latter, do not solve for the equilibrium values, just derive the results

$$y^* = n^*(1-n^*) + m(n^* - \mathcal{R}_0^{-1})\,, \tag{6.6.4}$$

$$n^*(1-n^*) = \frac{md}{1-d}(n^* - \mathcal{R}_0^{-1})\,. \tag{6.6.5}$$

   b. Keeping in mind $d < 1$, use a graph to show that there is always a unique solution to the n^* equation, provided $\mathcal{R}_0 > 1$.
   c. Compute the Jacobian.
   d. Show that the disease-free equilibrium is stable when $\mathcal{R}_0 < 1$ and that the extinction equilibrium is never stable.
   e. Show that the Jacobian for the endemic equilibrium is

$$J = \begin{pmatrix} \epsilon(1-2n^*) & 0 & -\epsilon d \\ \epsilon(1-2n^*+m) & -\epsilon(\mathcal{R}_0 y^* + m) & -\epsilon \\ 0 & \mathcal{R}_0 y^* & 0 \end{pmatrix}. \tag{6.6.6}$$

2. Routh–Hurwitz Conditions for the Endemic Disease Equilibrium

   a. Compute c_2, using the asymptotic limit $\epsilon \ll 1$. The answer is very simple (but not 0!).
   b. Derive the formula

$$c_3 = \epsilon^2 \mathcal{R}_0 y^*[(2n-1) + d(1-2n+m)]. \tag{6.6.7}$$

   c. Showing $c_3 > 0$ is not easy. You have to show that the quantity in the square brackets is positive. One way to do this is to use (6.6.5) to replace md, then factor out $1-d > 0$ and the denominator factor of $n^* - \mathcal{R}_0^{-1}$. Simplify what remains to get the expression $n^{*2} - 2\mathcal{R}_0^{-1} n^* + \mathcal{R}_0^{-1}$. If you add and subtract $\mathcal{R}_0^{-2}$, you can partition the resulting expression into the sum of a perfect square plus a term that is clearly positive.
   d. Show that $y^* > n^*(1-n^*)$ and use this result to obtain the inequality $\mathcal{R}_0 y^* > 1 - n^*$.
   e. Use the result from d to show that $c_1 > \epsilon(n+m) > 0$.
   f. The results obtained for c_1 and c_2 confirm $c_1 c_2 > C = \epsilon^2 \mathcal{R}_0 y^*(n^* + m)$. Use the formula for c_3 to show that $c_3 < C$. Conclude that the final stability requirement $c_1 c_2 > c_3$ is satisfied.

3. Some Additional Modeling

   Take $\mathcal{R}_0 = 3$ and $m = 0.5$ and plot bifurcation diagrams of n^* and y^* versus the mortality parameter d. Set an upper bound of $d = 0.5$, which is of course a huge mortality rate for the disease. Discuss the results. (For interpretation, keep in mind that the actual equilibrium infectious population is ϵy, with $\epsilon \approx 0.001$. So y^* is approximately the number of infectious individuals per thousand while $1 - n^*$ is the fraction by which the carrying capacity of the environment has been reduced by the disease.)

Project 6B: A Disease Model with Isolation

Consider an SIR disease with fixed birth rate, standard incidence, and isolation of infectious individuals. Isolation can be built into a model in various ways; here, we assume a spontaneous transition process[39] that siphons off infectious individuals into a class Q. Individuals in this class have no social contact until the recovery process moves them into class R. With these assumptions, the model is

$$S' = \mu N - BS\frac{I}{N-Q} - \mu S\,, \tag{6.6.8}$$

$$I' = BS\frac{I}{N-Q} - (\alpha + \gamma + \mu)I\,, \tag{6.6.9}$$

$$Q' = \alpha I - (\eta + \mu)Q\,, \tag{6.6.10}$$

$$R' = \gamma I + \eta Q - \mu R\,, \tag{6.6.11}$$

where $N = S + I + Q + R$ and $S + I + R = N - Q$ is the population that is available for contacts. The goals are to find the stable equilibria for various ranges of parameter values and to determine the impact of isolation on the population dynamics.

1. Preliminary Work

   a. Show that N is constant. We therefore need only three of the differential equations.
   b. The key to making this problem as easy as possible is to define a variable $A = N - Q$ and use it along with S and I, rather than using either Q or R as the third variable. Write down the system of differential equations for S, I, and A, replacing N and Q as needed.
   c. Define parameters

$$b = \frac{B}{\alpha + \gamma + \mu}\,, \qquad \epsilon = \frac{\mu}{\alpha + \gamma + \mu} \ll 1\,, \tag{6.6.12}$$

$$\rho = \frac{\alpha}{\alpha + \gamma + \mu} < 1\,, \qquad \nu = \frac{\eta + \mu}{\alpha}\,. \tag{6.6.13}$$

   Scale the problem using N as the population scale and $1/(\alpha + \gamma + \mu)$ as the time scale.
   d. The s equation shows that the infectious class should be $O(\epsilon)$. Use the rescaling $i = \epsilon y$ to get a properly scaled model with variables s, y, and a. Replace ϵ with the modified small parameter $\delta = \epsilon/\nu$. This makes the algebra slightly easier.

[39] Section 3.1.

2. Stability Analysis

   a. Show that there is a unique endemic disease equilibrium when $b > 1$. Calculate y^*, s^*, and a^* for this equilibrium. Also define the quantity $\phi = y^*/a^*$, which will be useful for the stability analysis. All of these quantities can be expressed in terms of just two parameters: b and δ.
   b. Compute the Jacobian, but don't make any substitutions for equilibrium values.
   c. Show that the disease-free equilibrium is asymptotically stable if and only if $b < 1$.
   d. Simplify the Jacobian at the endemic disease equilibrium by using the equilibrium equation $a^* = bs^*$ and replacing y^*/a^* with ϕ. Don't use the formula for ϕ until absolutely necessary. The quantity $1 + b\phi$ will appear in one of the matrix entries, but none of the other entries will have any sums.
   e. Calculate c_1, c_2, and c_3 for the Routh–Hurwitz conditions. Show that all three are positive. (Use what we already know about some of the parameter values.)
   f. Reduce the condition $c_1c_2 > c_3$ to

$$\nu\rho(1 + b\phi)C_1 + \delta b\phi(1 + b\phi - \rho\phi) > \rho^2\phi.$$

   g. Show that this final criterion is satisfied whenever $\nu > 1$ but may not be satisfied if ν is small enough.
   h. For the case where $c_1c_2 < c_3$, what do you think happens as $t \to \infty$? [Hint: Sherlock Holmes says that when you eliminate everything that is impossible, the only thing left has to be true, no matter how improbable. There are only so many possible behaviors for a system of three variables. After you have ruled out the most likely of these, what else is left, and which of these is most likely?]
   i. Set $\eta = \gamma = 0.1$, $\mu = 1/20000$, and $B = 1$ and consider $\alpha = 0.3$ and $\alpha = 0.5$. Check the Routh–Hurwitz conditions for both of these cases.
   j. Check your results by running numerical simulations for the two cases above. Try initial conditions $s = s^*$, $y(0) = 0.1$, and $a = 1$. Pay particular attention to the values of y^*.

3. The Primary Impact of Isolation of the Sick

   In 2a, we calculated y^*, which means we have a formula for i^* in terms of ϵ, δ, and b. To a first approximation, this result is

$$i^* = \epsilon(1 - b^{-1}).$$

   The question of interest is how much this quantity is changed by the isolation rate constant α. To address this question, we define i_0 to be the equilibrium value of i if $\alpha = 0$, with all other parameters the same. Then the quantity $z = i^*/i_0$ measures the ratio of the infectious population fraction to the default $\alpha = 0$ case. We must keep in mind that $i^* = 0$ whenever $b \leq 1$.

   a. Define $\mathcal{R}_0 = B/(\gamma + \mu)$ and $\epsilon_0 = \mu/(\gamma + \mu)$. Determine i_0 in terms of $\mathcal{R}_0$ and ϵ. Approximate the result by omitting any terms of order ϵ that are being added to a term that is not small.
   b. Define $q = \alpha/(\gamma + \mu)$. Show that $\epsilon = \epsilon_0/(1 + q)$ and obtain a similar formula for b in terms of q.
   c. Use the approximate results for i^* and i_0 to determine $z = i^*/i_0$ as a function of $\mathcal{R}_0$ and q.
   d. Plot z versus q for the cases of $\mathcal{R}_0 = 2, 3, 4$.
   e. Discuss the results.

Project 6C: Limit Cycles in a Predator–Prey Model

One of the interesting phenomena that has been observed in predator–prey systems is oscillatory patterns that appear to persist over time. These are often connected with the Lotka–Volterra model; however, this is a confusion of the neutrally (not asymptotically) stable behavior of the LV model with the more appropriate stable limit cycle behavior.[40] The simplest predator–prey model capable of producing a limit cycle is obtained by assuming logistic growth in the prey, natural death for the predator, predator growth proportional to consumption, and Holling type 2 predation dynamics. These assumptions yield the model

$$x' = x\left(1 - \frac{x}{k} - \frac{y}{1+x}\right), \tag{6.6.14}$$

$$y' = \epsilon y\left(\frac{hx}{1+x} - 1\right), \tag{6.6.15}$$

where x is the prey biomass scaled by the semisaturation level in the predation term, y is predator biomass scaled by the amount that could be achieved at maximum prey growth, k is the scaled prey carrying capacity, h is the predator hunting efficiency, and ϵ is the ratio of prey lifespan to predator lifespan, usually somewhat small.

1. Simulations

   a. Modify *ODEsim.m* to run simulations for the predator–prey model. Use $\epsilon = 0.05$ and $k = 2$ for all simulations.
   b. Run a simulation with $h = 1.4$ using initial conditions $x(0) = 1$, $y(0) = 1$. Try other initial conditions as well.
   c. Run a simulation with $h = 2$ using initial conditions $x(0) = 1$, $y(0) = 0.1$. Try other initial conditions as well.
   d. Repeat part 1c with $h = 4$.
   f. Describe the effect of increasing h, given $k = 2$.

   Note that simulations give you examples of what can happen, but only analysis can give you a complete picture.

2. Stability Analysis

   a. Compute the Jacobian for the system.
   b. Determine the equilibria (there are 3) and note restrictions on their existence.
   c. Determine the stability of the full extinction equilibrium.
   d. Determine the stability of the predator extinction equilibrium. The result will be an inequality with h and k.
   e. Repeat part 2d for the coexistence equilibrium.
   f. Plot the stability boundaries from parts 2d and 2e in the kh-plane. Label each region according to which equilibrium is stable, if any.
   g. Are the results of the stability analysis consistent with the simulations from part 1? [If not, recheck everything!]

[40] Neutrally stable periodic solutions have an amplitude determined by the initial conditions, whereas the amplitude of a limit cycle is determined entirely by the system parameters.

3. Nullcline Analysis

   a. Sketch the nullclines and direction arrows for the case $k = 2, h = 1.4$.
   b. Repeat part 3a for the case $k = 2, h = 2$.
   c. Repeat part 3a for the case $k = 2, h = 4$.
   d. Discuss the results with reference to the simulations and stability results.

4. Explaining the Limit Cycles

   a. Prepare a plot in the xy-phase plane that combines the nullclines for the $k = 2, h = 4$ case with the corresponding simulation results.

   b. The limit cycle can be thought of as having four phases. Describe and explain what happens as the solution curve moves through the nullcline plot. [Review the discussion of fast and slow variables in Sect. 6.1.]
   c. Explain why the limit cycle did not occur with $k = 2, h = 2$.
   d. Suppose you are explaining the phenomena of limit cycles to a biologist. What can you say about the biological characteristics of scenarios that have limit cycles as compared to those that don't? [Think about the biological significance of a large value of h.]

Project 6D: The Role of Trained Macrophages in the Immune System

In Problems 6.2.11 and 6.3.9, we studied the immune system model

$$\begin{aligned} p' &= p(1 - p - qm - sn)\,, \\ m' &= a[\delta(1 - m) - mp]\,, \\ n' &= b\left(\frac{p}{h + p} - n\right)\,, \end{aligned}$$

where p is a pathogen in a human host, m is the population of generalist macrophages that are immediately available but are lost in the battle with the pathogen, and n is the population of specialist macrophages that are available only after being "trained" to recognize the specific pathogen. There were two key findings:

1. The disease-free equilibrium is stable whenever $q > 1$ and unstable whenever $q < 1$, regardless of the value of s.
2. There remains a stable endemic disease equilibrium for q values between 1 and a large threshold value.

Given that the presence of the trained macrophages does not affect stability, these results raise the question, "What benefits do they actually provide to the system?" The purpose of this project is to address this question.

1. Weak Generalist Response ($q < 1$)

   a. Explain why the results of Problem 6.2.11 mean that the pathogen persists over time.

b. Explain why pathogen persistence at a level that is not asymptotically small means that we can expect m to be on the order of the small parameter δ at equilibrium.

c. Rescale the model for the case where p is not small by making the substitution $m = \delta y$.

d. Obtain an approximate solution for the endemic disease equilibrium by neglecting any terms in the equilibrium equation for p that contain factors of either h or δ. This should result in a very simple expression for p^*.

e. Explain why the result of (d) is meaningful only if $s < 1$ as well as $q < 1$. What does this imply must happen if $s > 1$, regardless of q?

f. Confirm the results of (d) and (e) for the specific case $q = 0.5$, $\delta = h = 0.04$ by plotting p^* versus s for $0 < s < 2$. Note that the results should not exactly match the approximation because $\delta = 0.04$ is not exactly the same as $\delta \to 0$. Nevertheless, you should see that the approximation is qualitatively correct.

g. Run simulations with $q = 0.5$, $s = 0.5$, $a = 1$, $b = 0.1$, $\delta = h = 0.04$, $m(0) = 1$, $n(0) = 0$, $p(0) = 0.01$. Check to make sure the results are consistent with what you have learned about the system.

h. Run simulations with $q = 0.5$, $s = 1.5$, $a = 1$, $b = 0.1$, $\delta = h = 0.04$, $m(0) = 1$, $n(0) = 0$, $p(0) = 0.01$. Check to make sure the results are consistent with what you have learned about the system.

2. Moderate Generalist Response $[1 < q < (1 + \delta)^2/(4\delta)]$

When $q > 1$, the disease-free equilibrium is stable, but if there are any endemic disease equilibria, the largest of them will be stable as well. Although initial pathogen load is usually low enough to be in the domain of attraction for the disease-free state, it is interesting to ask about the condition necessary for there to be no endemic disease state. The goal of this part of the project is to produce a plot in the qs-parameter space that shows this condition. To keep the algebra manageable, we take $h = \delta$.

a. Assuming q in the indicated range, simplify the equation for the endemic disease equilibria with $h = \delta$. The equation should be of the form "product of two linear factors containing p and δ equals linear factor containing q and s as well as p and δ."

b. Sketch the quadratic polynomial from (a) and several possible linear functions with different q and s. Positive equilibria p^* are at intersections of these curves.

c. Identify the additional relationship that the quadratic and linear curves must satisfy at the critical case that serves as the boundary between having one or more positive equilibria and having none. [See Sect. 4.5.6.]

d. There are now two equations for the critical case, with variables p, q, and s. The goal is to obtain a relationship between q and s. The easiest way to do this is to eliminate s from the system, solve the resulting equation for p in terms of q, and then use an already known relationship to determine s in terms of q.

e. Plot the critical relationship in the qs-plane. Identify the region where there is no disease-free equilibrium.

3. Summary

Review the results from parts 1 and 2 and earlier problems and write a summary describing how the system of generalist and specialist macrophages works to control pathogens. In particular, note when one of the two components is doing nearly all the work and when they are working together.

Project 6E: Modeling Host–Parasitoid Dynamics

As described in Problem 6.5.6, parasitoids have an interesting life history that offers a significant challenge to modelers. The Nicholson–Bailey model predicts that host–parasitoid dynamics is always chaotic, yet that does not appear to be the case. In this project, we work toward a host–parasitoid model that can predict a greater variety of outcomes. We do this by retaining the general structure of the Nicholson–Bailey model while modifying some of the details.

1. Development of A General Model

   a. Assume that the host population dynamics is given by $H_{t+1} = R_0 H_t$ in the absence of parasitoids, where $R_0 > 1$ is a growth factor comparable to the basic reproduction number for a disease. If $f(aP)$ is the fraction of hosts not parasitized when the parasitoid population is P (with a a constant having the dimension 1/parasitoid), write down the corresponding equation for H_{t+1}.
   b. Assume that each parasitized host results in c parasitoids in the next generation. Write down the equation for the dynamics of the parasitoid population P.
   c. Scale the model by multiplying the H equation by ac, multiplying the P equation by a, and using the substitutions $p = aP$ and $h = acH$.
   d. Explain why the model makes sense only if $f(0) = 1$ and $f' < 0$.
   e. Confirm that the Nicholson–Bailey model of Problem 6.5.6 is an example of our more general model.

2. Preliminary Analysis

   a. Compute the Jacobian matrix for a fixed point $(h, p) = (h^*, p^*)$ with both components positive.
   b. The h equation yields a simple equation for the parasitoid population at a fixed point in terms of the function f. Use this equation to eliminate f from the Jacobian. Note that it still contains f'.
   c. Compute tr $\mathbf{J}$, and det $\mathbf{J}$.
   d. Use the property $f' < 0$ to determine the signs of tr $\mathbf{J}$ and det $\mathbf{J}$.
   e. Use the results of (d) and the assumption $R_0 > 1$ to show that the first of the two conditions in Theorem 6.5.1 is always satisfied. Hence, stability is determined by the requirement det $\mathbf{J} < 1$.
   f. Use the other fixed point equation to eliminate h from the formula for the determinant. Conclude that the fixed point determined by $f(p^*) = 1/R_0$ is stable if and only if

$$-R_0 p^* f'(p^*) < 1 - R_0^{-1}. \tag{6.6.16}$$

   Obviously we can go no further without a specific function for f.[41]

3. A More General Attack Function

   The function f in the Nicholson–Bailey model corresponds to the assumption of a Poisson distribution for the number of parasitoid attacks in a given amount of time. The assumption of a negative binomial distribution yields a different function f:

$$f(p) = \left(1 + \frac{p}{k}\right)^{-k}.$$

[41] Note that $f' < 0$ and $R_0 > 1$, so both quantities in the inequality are positive.

Table 6.6.1 Reported parameter values for the *Tribolium* model

Reference	b	s	m	α	β	γ
Cushing et al. [1]	11.6772	0.4871	0.1108	0.0093	0.0110	0.0178
Dennis et al. [2]	10.45	0.8000	0.007629	0.01731	0.01310	0.004619

The parameter $k > 0$ measures the similarity to the Poisson distribution, with the latter achieved in the limit $k \to \infty$. This suggests that we could see different behavior with smaller values of k.

a. Calculate the fixed point p^* for $k = 0.5$ and use (6.6.16) to determine its stability.
b. Run a simulation for the model using $k = 0.5$, $R_0 = 1.1$, $h_0 = 1$, $p_0 = 0.4$, and a total of 120 years.
c. Are the results of the simulations consistent with the theoretical results?

Project 6F: A Discrete Model with Chaotic Dynamics

The flour beetle *Tribolium confusum* is often used as a model insect species for both theory and experiment. The best known mathematical model for this population is a discrete stage-structured model with a time step of 2 weeks [1, 2]. Flour beetles are larvae for about 2 weeks; then they go through three life stages (nonfeeding larvae, pupae, and callow adults) in approximately 2 weeks before becoming adults that can live for several years.[42] The model assumes that adults lay an average of b eggs, but these numbers lead to far fewer larvae because larvae and adults eat eggs. A fraction s of larvae survive to become "pupae," the pupae are either eaten by adults or survive to become adults, and a fraction m of adults die in each 2-week period. The full model is

$$\begin{aligned} L_{t+1} &= bA_t e^{-\alpha L_t - \beta A_t}, \\ P_{t+1} &= sL_t, \\ A_{t+1} &= P_t e^{-\gamma A_t} + (1-m)A_t. \end{aligned}$$

Reported parameter values are given in Table 6.6.1.[43] The value of γ can be manipulated experimentally by removing additional pupae by hand at each census.

The full model is rather complicated, so we make a few minor simplifications. Cannibalism of pupae by adults does not add anything different to the outcomes, so we set $\gamma = 0$. Also, there is little difference between α and β values, so we make these the same. Finally, nondimensionalization would remove α from the model. Thus, we consider

[42] It is common in practice to use discrete models for cases such as this, where the stage durations are approximately comparable. As noted in Sect. 6.5.3, this is not a good modeling decision. It is not clear whether the complex behavior of the model represents complex biological behavior or is merely an artifact of the use of a discrete model. Nevertheless, the model is very interesting from a mathematical perspective.

[43] The reported values have as many as four significant figures, suggesting a high degree of precision. However, such apparent precision is unwarranted in light of the limited accuracy of the values; for example, the reported values for s differ by a factor of almost 2. One should not put too much faith in reported parameter values in ecology or epidemiology; in particular, it is misleading to use values that appear to indicate a high degree of precision. One should be particularly careful of theoretical results that rely on a narrow range of parameter values.

$$L_{t+1} = bA_t e^{-(L_t+A_t)}, \tag{6.6.17}$$

$$P_{t+1} = sL_t, \tag{6.6.18}$$

$$A_{t+1} = P_t + (1-m)A_t. \tag{6.6.19}$$

When numerical values are needed, we'll choose $s = 0.5$ and $m = 0.1$ as reasonable values while varying b. The only effect of our taking $\gamma = 0$ is that much higher values of b will be needed for instability.

1. Determine the Jacobian matrix.
2. Use the Jury conditions to show that the extinction fixed point is stable whenever $bs < m$. This condition is not met, given $b > 1$ and $s > m$.
3. Show that the Jacobian for the fixed point with positive values can be simplified to

$$\mathbf{J} = \begin{pmatrix} -L^* & 0 & \frac{m}{s} - L^* \\ s & 0 & 0 \\ 0 & 1 & 1-m \end{pmatrix}.$$

4. Show that the first of the Jury conditions is always satisfied.
5. Show that the critical stability condition with $s = 0.5$ and $m = 0.1$ is $L^* < 5/6$.
6. Solve the fixed point equations to obtain a formula for L^* in terms of m, s, and $R = bs/m$.
7. Combine the results of 5 and 6 to determine the critical stability condition for b.
8. Run simulations showing the behavior of the system with $b = 20, 80, 300, 1000$. In each case, use starting values that are 95% of the fixed point values.
9. Compare the results of the simulations and the stability analysis. Are they consistent?

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Correction to: Mathematical Modeling for Epidemiology and Ecology

Correction to:
G. Ledder, *Mathematical Modeling for Epidemiology and Ecology*, Springer Undergraduate Texts in Mathematics and Technology, https://doi.org/10.1007/978-3-031-09454-5

The original version of the book was inadvertently published with incorrect URL link. The incorrect URL link:

"http://www.math.unl.edu/∼gledder1/MMEE/, http://www.springer.com/978-1-4614-7275-9" in the pages 17, 18, 28, 46, 60, 76, 81, 95, and 256 has been updated as "https://www.math.unl.edu/∼gledder1/".

The book has been updated with the changes.

The updated version of the book can be found at https://doi.org/10.1007/978-3-031-09454-5

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Using MATLAB and Octave

A

MATLAB is commercial software for scientific computation, published by a company called MathWorks. It lacks the capacity of Python or C++ for object-oriented programming, but this capacity is only necessary for making professional user-friendly tools run entirely from a graphical user interface or for managing large amounts of data. On the other hand, MATLAB is much better designed for computation than computer algebra systems such as Maple and Mathematica. It has a CAS add-on package, but it remains primarily a platform for scientific computation.

In the first edition, I used R rather than MATLAB for two reasons. First, R is more common in biology than MATLAB. Second, at the time there was no suitable free substitute for MATLAB. The second issue is no longer relevant (see below), while the first seems less important to me now. Readers whose background is in mathematics rather than biology will not have any experience with R, and MATLAB has proven easy to learn for my colleagues who already know R. The main drawbacks of R are its primitive graphical capabilities and the challenges to debugging posed by the difficulty of getting intermediate results to print.

For this book, I have used only basic MATLAB, with no special packages. In particular, I have shunned any symbolic computational tools in accordance with my experience, which shows that careful algebraic computation informed by human judgment is superior to brute-force symbolic computation based on algorithms. The partnership of humans with computers is of value primarily because of complementary skills: computers have speed and accuracy that humans lack, while humans have ingenuity that computers cannot match.

For those readers who object to commercial software, either for financial or philosophical reasons, there is a largely suitable alternative to MATLAB in the Octave Online platform, which is part of the broad GNU community of free software. Octave is able to run MATLAB programs, but with some caveats:

1. Not all Octave numerical routines are of equal quality with their MATLAB counterparts. In particular, the differential equation routines are not as good, although comparable results can be obtained by carefully lowering error tolerances.
2. Graphics statements in Octave are not always comparable to those of MATLAB. Some merely need minor adjustments, such as to marker sizes when plotting points. Others have less functionality, such as the Octave function *hist* as compared to MATLAB's *histogram*.
3. Octave Online is obviously slower than MATLAB running on a personal computer. This is unnoticeable for most of the computation in this book. The computationally intensive *MMfit.m* in Sect. 2.5 is limited in Octave, and long simulations with stiff differential equations are likely to be timed out unless you pay for a subscription to premium service.

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The second of these issues has been largely addressed by comments in my programs that suggest modifications to use with Octave Online to make plots that look more like the plots obtained when running the programs in MATLAB.

MATLAB Files

MATLAB files fall into three categories.

1. Script files contain instructions that are executed in sequence, as though the statements had been entered into the console. All variable values are retained in current memory after the script is finished.
2. Standard function files begin with a *function* statement of the form

   ```
   function output = funcname(input)
   ```

   where `function` is part of the statement structure, "funcname" is the name of the function (which should be the same as the name of the file), "input" is a list of variables to be supplied when the function is called, and "output" is a list of variables, enclosed in square brackets if there is more than one, defined by the function and returned as results. Functions of this type must be called from the console window, a script, or another function; in other words, they cannot be run directly from the editor. Any variables defined in the function and not returned as output are not saved in current memory.
3. There is a hybrid type I call a "funscript" that has the syntax "`function result = funcname()`", where the optional name "result" is a single output quantity. These files are technically functions, with locally defined variables not saved in current memory. However, they function as scripts in that they can be run directly from the editor. Funscript files can have function definitions following the main function as well as functions that are internal to the main function.[1]

It is customary practice to use only lower case letters for the names of MATLAB functions. I always use an initial capital in the name of any script or funscript, so the initial capital identifies files that can be run directly from the editor. Where possible, functions I have written for use in scripts or funscripts appear in the main file rather than in a separate function file. The exceptions are the functions *hpsr*, *findmin*, *seir*, and *seaihrd*, which appear in separate files because they are used by more than one script. Those scripts with names that begin with HPSR, SEIR, or SEAIHRD include calls to the corresponding external function. *findmin* is used in *Semilin.m*.

Programs that include function definitions have been written as funscripts in the interest of Octave compatibility, while those that do not have been written as scripts.

Documentation for software products generally suffers from being too complex; that is, many entries include examples that can only be understood by people who are experts on everything except what is in that entry. This kind of documentation shows the full power of the item being discussed, but it is not very helpful to a beginner who knows only the basics. It would be better for a documentation entry to include the simplest possible examples for that item. One can find tutorials that are better, but these are necessarily of broad scope and written for a general audience, which means that they include many topics the reader of this book will not need while missing a few topics that are needed. The alternative

[1] After years of resisting customer requests, MATLAB now allows script files to contain function definitions *after* the final statement of the script. Octave still does not allow this, although it does allow function definitions in a script if they occur *before* the function is used. As of March 2022, if a program with an internal function is to be compatible with both MATLAB and Octave, it must be in the form of a funscript.

is a targeted tutorial that focuses on what is needed for a particular suite of programs. This appendix is intended for that purpose.

The tutorial presented here is divided into three lessons. In lesson 1, we examine each line of code in the program *Vaccination.m*, which is the simplest of the programs used in this book. That way, we start with a limited number of features but also see them used to make a complete program. Lesson 2 examines statements in the program *DiscreteVar.m* that are not covered by lesson 1. In lesson 3, we present a few additional items that do not appear in either *Vaccination.m* or *DiscreteVar.m*.

The best way to learn computer programming is by a combination of imitation and experimentation. Hardly ever does anyone write a computer program from scratch, the way you would write a paper. Instead, you piece together modified elements of programs that you have already used, adding new elements only when needed. Programs written for this book have a generic structure that facilitates this approach.

The MATLAB funscript *Vaccination.m*

It is good programming practice to impose stylistic conventions on your programs to make it easier to understand and reuse programs written years before, an example being the naming convention, which makes it immediately obvious that *seir.m* is a function program and that *SEIR_onesim.m* is a script that uses that function.

MATLAB syntax allows a lot of flexibility for how to structure programs. This flexibility can be used to write programs that can be read by people other than the author or read by the author after several years. Any line in MATLAB that starts with the percent symbol is a comment, which the MATLAB interpreter ignores. Any line that starts with `%%` rather than a single `%` additionally marks the beginning of a new section.

Vaccination.m uses `%%` comments to organize the statements into five sections: `DATA`, `INITIALIZATION`, `COMPUTATION and PLOTS`, `OUTPUT`, and `END`. The primary purpose of this structure is to group statements according to how likely they are to need a change when the program is used. The `DATA` section defines parameters and the function used to calculate the rate of change needed for a differential equation. The `INITIALIZATION` section contains statements that set up plot structures. In a more complicated program, there will also be statements here that create data structures. The main computational work is in the `COMPUTATION and PLOTS` section. The `OUTPUT` section contains some additional statements that modify the appearance of the plot and the `END` section is there merely to set off the statement needed to end a file written with function syntax. We can run new experiments and even change the mathematical model merely by modifying the small `DATA` section. Where possible, it is best to make the other sections of a program generic so that there is a large section of program code that never needs to change.

The program file *Vaccination.m* contains 57 lines, but only 19 of these comprise the program code. These statements fall into three broad classes, with 7 assignment statements, 6 control statements, and 6 graphical statements.

Simple Assignment Statements

Assignment statements have the generic structure “name=value.” In the simplest of these, the values are just numbers or vectors, as in the statements

```
phi = 0.02;
dWdt = -phi*WW;
V = a-W;
```

The semicolon at the end of a statement prevents the result from being printed in the console window. These are often omitted when debugging so that the programmer can see intermediate results.

Data Structures

Most programs require data structures, so it is important to understand how these are defined and accessed. Two of the assignment statements in *Vaccination.m* define vector data structures:

```
avals = [0.6,0.8];
interval = [0 100];
```

These statements use square brackets to create a list that also functions as a row vector. The elements can be separated by commas or just blank space. It is common to use commas for lists that might have more than two elements and to use blank spaces for lists of two elements that serve to define intervals, but either style is fine. Separation with semicolons would make the list function as a column vector rather than a row vector.

The statement

```
a = avals(i);
```

selects the element in position `i` from the vector `avals` and assigns its value to the name `a`. Of course this only works if `i` is an integer in the range from 1 to the number of elements in the vector.

Note that square brackets are used to create lists while parentheses are used to select values (and also to replace individual values). Beginners often have trouble remembering this apparently contradictory notation.

Function Calls

The statement

```
[t,W] = ode45(@rates,interval,a);
```

is a call to the (built-in) MATLAB function `ode45`, which implements the rk4 method for approximating solutions of differential equations.[2] This function requires three input parameters. The first is the name of the function that calculates the rate of change; note that it is written as `@rates` rather than `rates`. The second is the time interval over which the solution is to be calculated and the third is an initial condition for a single dependent variable or a list of initial conditions for a vector of dependent variables. It is good practice to define these quantities in the DATA section where possible. If the text `[0 100]` is typed here rather than `interval`, it will eliminate the need for the statement in line 21, but it will bury information that might need to be changed for a different example. This will be harder to do if the information is not located in the DATA section.[3]

User-Defined Functions

The function `rates` is a user-defined function. These have the same syntax when they are in a program as when they are the content of a function file, that is,

```
function output = funcname(input)
```

where "input" is a list of parameters to be passed into the function and "output" is either a list in square brackets or a single quantity (which may itself be a list). The structure of the input and output

[2] see Appendix D.

[3] Similarly, the information in lines 43 and 45 could have been buried inside lines 44 and 46 rather than being defined separately, but this is bad practice. There are no prizes awarded for writing the most compact code. The only prizes are for writing code that works and for writing code that can easily be modified.

quantities must match whatever program code is used for the function call. In the case of `rates`, the requirements are determined by `ode45`: the input quantities must be the independent variable and a vector of the dependent variable values (just a single value for a single dependent variable) and the output quantity must be a vector of the rates of change for the dependent variables. Function syntax requires that each function be closed with an `end` statement.

Note that the entire program is written as a user-defined function, starting with the function statement on line 1 and ending with the final end statement in line 57.

The `for` Loop

The four statements

```
a = avals(i);
[t,W] = ode45(@rates,interval,a);
V = a-W;
plot(t,V,'LineWidth',1.4);
```

need to be executed once for each desired value of the parameter a, which are the values in `avals`. This is accomplished efficiently using a `for` loop, which has syntax

```
for index=range
    statements
end
```

The reserved words `for` and `end` are required, "index" is the name of the variable whose value update marks passage through the loop, "range" is a set of values taken by the index variable, and "statements" are the statements that are repeated in each passage of the loop.

In the `for` loop of the example, the index variable `i` takes the values 1 and 2 in consecutive runs through the loop, since `avals` has two entries. Each time through the loop, the program identifies the appropriate a value from the list, runs the differential equation solver with that value as the initial condition, calculates the desired output variable V from the differential equation variable W, and then uses the results from the solver to plot a curve.

Graphics Statements

The graphics output is produced by 6 statements in the program. Three of these are used to set up the graph:

- `clf`
  clears the previous graph from the current figure window (use `figure` to create a new window instead),
- `hold on`
  directs MATLAB to add new curves to the current plot rather than deleting previous ones, and
- `box on`
  formats the axes as a box rather than just the usual axes (this is a matter of taste).

A `plot` statement creates either a plot of points, a dot-to-dot curve, or both. In the statement

```
plot(t,V,'LineWidth',1.4)
```

- The first argument specifies the list of values for the horizontal coordinate.
- The second argument specifies the list of values for the vertical coordinate.
- There is an optional third item to specify the marker or curve style and the color. This argument is missing here, so the plot will be a dot-to-dot curve using the next color in MATLAB's default color palette.

- The final item in the list is an option–value pair that specifies the line width, overriding the default. I find the default line width to be a little thinner than I like, but this is a matter of taste.

The final two graphics statements label the axes. The default font is somewhat small, so I always specify the font size as an option–value pair. The default rotation is to orient the vertical axis label so that it reads from bottom to top rather than left to right. For labels that are short, such as a single letter or subscripted letter, it is best to set the rotation to 0, which puts the axis label in the more readable left to right form.

The MATLAB script *DiscreteVar.m*

The program file *DiscreteVar.m* is only slightly more complicated than *Vaccination.m*, but it does contain a few additional features.

More Assignment Statements and Display Statements

The values in assignment statements can be strings as well as numbers, such as in the statement

```
varname = '\it{N}';
```

As before, the semicolon at the end of the statement prevents the result from being printed in the console window. One should only allow results to be printed if they are desired end results or if they are temporarily needed for debugging. I prefer having statements that print results come in the `OUTPUT` section. Here, the statement

```
N
```

merely displays the result calculated earlier.

One-line Function Definition

The model being used for the computation in *DiscreteVar.m* consists of a formula that specifies a value of the dependent variable N in terms of the previous value of N. This means that we need to use the same formula repeatedly, but with different input values. The one-line function statement

```
seqfnc = @(N) N+R*N.*(1-N/K);
```

accomplishes this purpose using an input variable `N` along with the previously defined parameter values `R` and `K`. The parameters are global, meaning that their values are taken from previous assignments, while the argument `N` is local, meaning that its value is independent of the use of the same symbol in the main program. The program statement `z = seqfnc(y)` would define the value of `z` by using the formula in `seqfnc` with the value of `y` being used as `N` in the calculation. Here, we have defined $K = 1000$ and $R = 1.5$. So `z = seqfnc(20)` would calculate the value for `z` as 20+1.5*20*(1-20/1000), which is 49.4.

Note the use of `.*` for multiplication. The usual multiplication symbol `*` is reserved in MATLAB for vector multiplication, whereas `.*` is for listwise multiplication. In this particular formula, we use `/` for division because `K` is always a scalar quantity. In contrast, the input variable `N` could be a vector quantity. For example, `z = seqfnc([0.5, 0.8])` would define `z` to be a vector of two components, each computed from `seqfnc`, one with `N`=0.5 and the other with `N`=0.8. This only works if we use `.*` to indicate that a scalar calculation is to be performed for each element in a vector. For example, suppose `K` is 1 and `N` is [0.5, 0.8]. Then `1-N/K` is [0.5, 0.2] and `N.*(1-N/K)` is [0.25, 0.16] (obtained by multiplying the corresponding pairs, 0.5*0.5 and 0.8*0.2). In contrast, the formula

`N*(1-N/K)` (without the extra "`.`") would be interpreted by MATLAB as an attempt to multiply a pair of 2×1 vectors, which is not defined. When in doubt, one should use `.*` for multiplication, `./` for division, and `.^` for raising to a power.

We could have used standard function definition syntax rather than that of the inline function, but that would have required the program to take the form of a funscript rather than a simple script.

Data Structures

Unlike *Vaccination.m*, most programs require user-defined data structures that are more complicated than pre-defined lists, so it is important to understand how these are defined and accessed.

```
N = N0*ones(1,T+1);
```

creates a matrix with 1 row and T+1 columns and assigns the value 1 to each position. Since it has just one row, the quantity `N0*ones(1,T+1)` serves as a row vector of T+1 copies of the value N0. There is a similar function `zeros` that is particularly helpful in creating data structures in which many of the entries are initially 0; the nonzero values can then be substituted for the initial 0 values later.

```
N(i+1) = seqfnc(N(i));
```

overwrites the entry in position i+1 of the list `N` with a value calculated from a formula that uses the previous entry in `N` as the input value. It was convenient to assign the value N0 to all of the list positions initially, given that this statement overrides all of these assignments except the one to the first position in the list.

Note again that square brackets are used to create lists while parentheses are used to select values and to replace individual values.

The `for` Loop

Note that the value of `N` at time `i` is stored as `N(i+1)`, not `N(i)`; this is necessary because the first space in the list is needed for `N` at time 0. We always need to be cognizant of this fact when we write loops and access data structures.

`for` loops can be embedded in other `for` loops, giving additional programming power. Here, the `COMPUTATION` section consists of a loop that runs through the range of values in `N0vals`. Each time through the loop, the program pulls the initial value `N0` from its list, uses that value to initialize the data structure, uses an internal `for` loop to compute the values of the sequence, and plots the graph of the results. In this way, we efficiently obtain a plot with `N` different data sets.

More Graphics Details

As noted earlier, the `plot` statement creates either a plot of points, a dot-to-dot curve, or both, with a dot-to-dot curve as the default. In the statement (note that the lists for the horizontal and vertical coordinates must contain the same number of entries)

```
plot(0:T,N,'.','MarkerSize',12)
```

the third item specifies dots for the points without specifying the color. You can use a variety of options that prescribe colors, symbols, and line styles; these include the standard color symbols `b` for blue, `k` for black, `r` for red, and `g` for green. I find the standard green to be too light, so I replace it with a dark green color that I specify with a statement in the `DATA` section.[4] Specifiers such as "`.`" in the program statement that specify a point style also suppress the default dot-to-dot curve, but you can use specifiers for both if that is what you want.

[4] See *DiscreteVars.m* for an example.

Additional specifications to a plot statement can be added using an option–value pair, as is done here with the marker size. Marker sizes do not translate from MATLAB to Octave, but these can easily be changed as directed in the program comments to give graphs that look comparable.

Additional MATLAB Programming Features

The small programs we have discussed so far do not include all of the MATLAB features needed for the programs in this book. In particular, there is some data structure functionality we have not seen, as well as statements that produce multiple panels in a single figure and control structures that only execute statements when some condition is met. We also need to say a little bit more about user-defined functions.

Data Structures

The primary data structures in MATLAB are vectors and matrices. These can be created using template functions such as `ones` and `zeros` and then subsequently filled in item by item using loops. They can also be created directly using square brackets, with spaces or commas separating elements within a row and semicolons separating rows. An example is the statement

```
M = [0,104,160;0.01,0,0;0,0.3,0]
```

that creates a matrix with three rows and three columns in *LinSys.m*. It is often convenient to specify a range of values, usually integers. For example, `t = 0:4` is the same as `t = [0 1 2 3 4]`.

We've seen how to select individual elements from vectors. The same syntax works for multiple elements and for matrices, but with some added features. The code `S = results(:,1)` that appears in *seir.m* is used to select the entire first column from the matrix `results` and save it as `S`. Similar statements can select other portions of a matrix; for example, `results(3:end,1:2)` would select the portion of matrix that is in the first two columns and all but the first two rows.

Multi-panel Figures

We often want to prepare figures with multiple panels. An example of this appears in *DiscreteVars.m*. The graphics statements in the `INITIALIZATION` section are

```
clf
for k=1:2
    subplot(1,2,k)
    hold on
    box on
    ...
    end
end
```

These statements generally serve the same purpose as the simpler versions in scripts with a single plot. The figure will have one row and two columns, with the first panel accessed using `subplot(1,2,1)` and the second using `subplot(1,2,2)`. The code section combines the setup requirements into a `for` loop—note that the clear function statement is for the whole figure while the hold and box statements are for each subpanel.

The `if` Statement

Sometimes it is necessary for a program to execute a block of instructions only when certain conditions are met. This is generally accomplished using the `if` statement. An example is in *DiscreteVars.m*. This

funscript produces a plot with one row and two columns. It is easy to resize windows in MATLAB, but less so in Octave. For whatever reason, one might want to use the statement `axis square` to force equal lengths of the two axes. For maximum flexibility, the `DATA` section of this program includes a statement that defines a variable `sq`. If can be left at the default value of 0 or set to some positive value. The code

```
if sq>0
    axis square
end
```

appears inside the loop (above) that sets up the subpanel plots. The `if` statement means that the axes will be square if `sq` has been set to a positive number, but not if it has been left at 0.

Note that the condition could have been written so that it would be triggered if `sq~=0` or `sq==1`, the first condition meaning that `sq` is anything but 0, while the second requiring it to be exactly 1. The double equal sign is required.[5]

The `switch` Statement

DiscreteVars.m contains another conditional control structure that can be very useful: the `switch` statement. Note the syntax of this structure:

```
switch size
    case 2
        ...
    case 3
        ...
end
```

The initial `switch` statement identifies the name of a variable. This is followed by `case` statements that indicate particular values of that variable, each of which has one or more embedded statements that are executed if the switch variable has that particular value. Since there are only two cases in this example, these portions of the code could have been written using the form

```
if size==2
   ...
else
   ...
end
```

I prefer the symmetry of the switch–case construction.[6]

More About Functions

When a lot of coding is required for a project, it is often best to split off the heavy computation as one or more separate functions. Such functions can be put in their own function file if the computation results are going to be used in multiple ways. This is the idea behind the SEIR suite of programs, which consists of a function program, *seir.m*, and three driver scripts: *SEIR_onesim.m*, *SEIR_comparison.m*,

[5] In spite of our claims that we mathematicians love rigor, we can be very sloppy about notation. We use the equal sign in mathematics for two completely different purposes—to assign a value to a name and to assert equality between two things that have been previously defined independently. Computers are strict literalists, so this ambiguity is forbidden in the writing of programming languages. Some of the difficulties students have in conceptual understanding of mathematics could be alleviated if we insisted on the same level of literalism in our mathematical notation.

[6] The use of embedded conditionals if, elseif, else is required in Python, which does not have a switch statement.

and *SEIR_paramstudy.m*. In other instances, such as *CobwebPlotter.m*, the function (in this case called *cobweb*) appears within the larger funscript file.

In many cases, a function is used simply to split off a portion of a large code for easier understanding and testing. More critical are cases where a function definition is required as part of a problem statement. Like *Vaccination.m*, the program *ODEsim.m* requires a function to represent the differential equation(s), which means that it needs to be changed each time you want to reuse the program for a different system of differential equations. The function appears in the demonstration version of this program as

```
function dydt = rates(ttt,yyy)
    % Unpack variables
    s = yyy(1);
    i = yyy(2);
    % Calculate derivatives
    sp = -b*s*i;
    ip = b*s*i-i;
    % Assemble vector derivative
    dydt = [sp;ip];
end
```

Since the function is used as input to the built-in MATLAB function *ode45*, it has to have a specific structure: two arguments, one for the independent variable and one for a column vector of dependent variables. There are ways to pass in parameters, but it is easier to allow the parameters to be defined globally; in this example, `b` was defined in a prior statement. The risk of this method is that the input variable names must not conflict with names used in the main code. I use `ttt` and `yyy` to represent t and y with names that are certainly not being used elsewhere.

The function definition could have been written as a single line of code inside the `function–end` delimiters:

```
dydt = [-b*yyy(1)*yyy(2);b*yyy(1)*yyy(2)-yyy(2)];
```

This is not wrong, but it is a bad practice because writing things in ways that are hard to read means that it is also harder to catch errors.[7] I break up functions for differential equations into three sections, as done here. The argument `yyy` is a column vector that contains one entry for each of the variables in the system. Splitting these out into the individual components makes the subsequent derivative formulas more human-readable. Similarly, it is more human-readable if the derivatives are calculated separately before being assembled into a vector. The semicolons in the last statement make `dydt` a column vector, as required.

List of Programs

There are a total of 28 programs associated with this book. Here is a brief listing of what they do, in the order in which they appear.

1. *DecaySim.m* (Sect. 1.2)
   This funscript runs a probability-based simulation of natural decay.
2. *hpsr.m* (Sect. 1.5)
   This function does the computation for an agent-based model for a disease simulation.
3. *HPSRsim.m* (Sect. 1.5)
   This script uses *hpsr* to plot the results of one simulation.

[7] As previously noted, there are no prizes for writing a program using the fewest lines of code. The only prize worth having is the one for writing an easily modified code with no errors.

4. *HPSRavg.m* (Sect. 1.5)
   This script uses *hpsr* to determine the average results of multiple simulations.
5. *TransitionSim.m* (Project 1E)
   This funscript generalizes *DecaySim* to incorporate multiphase transitions.
6. *LeastSq_1var.m* (Sect. 2.1)
   This script fits the model $y = mx$ to data.
7. *LeastSq.m* (Sect. 2.2)
   This script fits the model $y = b + mx$ to data.
8. *findmin.m* (Appendix C and Sect. 2.3)
   This function uses a variant of the bisection method to find the minimizer for a function of one variable.
9. *Semilin.m* (Sect. 2.3)
   This funscript fits any semilinear model $y = Af(x; p)$ to data, using *findmin.m*. The specific function is coded by the user.
10. *PolyLS.m* (Sect. 2.4)
    This script fits a polynomial of any degree to data.
11. *MMfit.m* (Sect. 2.5)
    This funscript compares the semilinear and several linearization methods for fitting the Michaelis–Menten function to simulated or real data.
12. *Vaccination.m* (Chap. 3)
    This funscript does the computation and plots results for a single differential equation. The specific system must be coded by the user. While the original version uses a naive model for vaccination, the program is easily modified to solve a more sophisticated vaccination model or any other model with a single differential equation.
13. *ODEsim.m* (Chaps. 3, 4, and 6)
    This funscript does the computation and plots results for a system of differential equations. The specific system must be coded by the user.
14. *seir.m* (Sect. 3.4)
    This function does the computation for an SEIR disease model. It reports state variables for each day of a scenario.
15. *SEIR_onesim.m* (Sect. 3.4)
    This script uses *seir* to plot the results of one simulation.
16. *SEIR_comparison.m* (Sect. 3.4)
    This script uses *seir* to plot the results of multiple simulations.
17. *SEIR_paramstudy.m* (Sect. 3.4)
    This script uses *seir* to plot graphs showing how results depend on a parameter.
18. *seirabm.m* (Sect. 3.4)
    This function runs a simulation of an agent-based SEIR model. It reports state variables for each day of a scenario.
19. *R0fit.m* (Sect. 3.4)
    This funscript uses *findmin* with *seir* to determine the best fit $\mathcal{R}_0$ value for a dataset produced by *seirabm.m*. A graph shows the best fit model results along with the data.
20. *seaihrd.m* (Sect. 3.5)
    This function does the computation for the SEAIHRD disease model used for the March 2020 COVID-19 scenario. It reports state variables for each day of a scenario.
21. *SEAIHRD_onesim.m* (Sect. 3.5)
    This script uses *seaihrd* to plot the results of one simulation.
22. *SEAIHRD_comparison.m* (Sect. 3.5)
    This script uses *seaihrd* to plot the results of multiple simulations.

23. *SEAIHRD_paramstudy.m* (Sect. 3.5)
    This script uses *seaihrd* to plot graphs showing how results depend on a parameter.
24. *DiscreteVar.m* (Sect. 4.1)
    This script runs a simulation for a discrete dynamic variable.
25. *CobwebPlotter.m* (Sect. 4.2)
    This funscript runs a simulation for a discrete dynamic variable and returns a cobweb plot along with an optional plot of the dynamic variable.
26. *DiscreteVars.m* (Sect. 5.1)
    This script runs a simulation for a discrete linear system of two or more components without using eigenvalues.
27. *LynSys.m* (Sect. 5.3)
    This script runs a simulation for a discrete linear system of two or more components, and computes eigenvalues to determine the long-term behavior.
28. *DiscreteSys.m* (Sect. 6.5)
    This funscript does the computation and plots results for a discrete-time system of dynamic variables. The specific system must be coded by the user.

Derivatives and Differentiation

B

The *derivative* is one of the central concepts of calculus. A full mathematical presentation of the derivative is outside the scope of this book. Instead, we focus on the derivative concept and derivative computation.

The Derivative Concept

The text material for the derivative in a standard calculus book includes the motivation for the derivative, its definition, rules for computing derivatives, and a variety of applications of the derivative.[8] While all of this material is important in calculus, all we really need for dynamical systems analysis are the geometric concept of the derivative and derivative computation rules. Here is a concise summary of the geometric concept.[9]

Figure B.1 illustrates a pair of functions, $x(t)$ and $y(t)$, that might represent the population size of a growing community. The function x shows *linear* growth, as would happen when individuals join the community at a fixed rate from the outside. That rate is the change in x divided by the change in t, which is the slope of the line in the graph. We can calculate the slope using algebra by picking two points, calculating the differences in their x and t values, and taking the ratio; here,

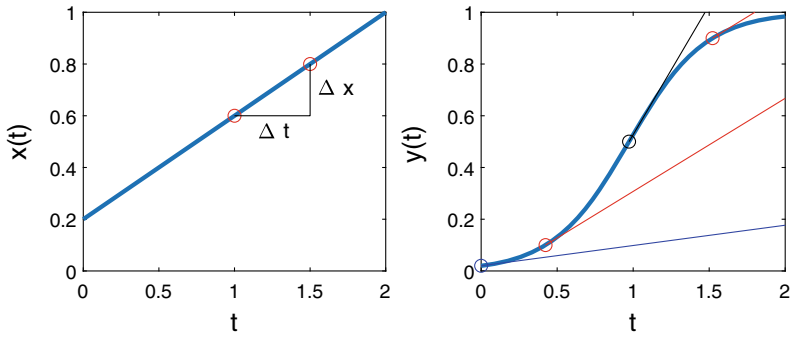


Fig. B.1 A linear function $x(t)$ and a nonlinear function $y(t)$ (the thick curves), showing the slope and instantaneous slope (the thin lines) of the two functions at various values of t

[8] This material is adapted from [2].

[9] See [1] for a more complete presentation.

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$$\text{slope} = \frac{\Delta x}{\Delta t} = \frac{0.8 - 0.6}{1.5 - 1} = 0.4. \tag{B.1}$$

The function y shows what is called *logistic* growth, which means that the growth is limited by the availability of resources. In this instance, the population grows by virtue of having more births than deaths, with no migration. It grows slowly at first because there are few potential parents, then increases more rapidly, and then levels off as the availability of resources or space becomes critical.

It still makes sense to talk about the slope of the graph of y, which we can think of as the slope of the line tangent to the graph at a particular point. A tangent line has only one known point, so we can't calculate its slope with standard algebra; techniques for determining the slopes of tangent lines are developed in calculus. For our purposes, we can assign a symbol to represent the slope at a point t and focus on its interpretation. The symbol we use is dy/dt, which is similar to the notation $\Delta y/\Delta t$ for the slope of a line. The key difference is that we have to interpret dy/dt as a single symbol, not a quotient of two different things.

Definition B.1

For a function $y(t)$ that has a smooth graph, the **derivative** dy/dt at any given point is the slope of the tangent line at that point, which represents the instantaneous rate of change of the quantity y.[10]

Applications of the derivative follow largely from the fact that the tangent line to a graph of a function serves as the linear approximation of the function.

While some important function properties are *global*, meaning that they are based on an extended graph of the function, others are *local*, meaning that they are seen by zooming in on a point.

1. Particularly important local properties are the asymptotic stability of an equilibrium point for a continuous dynamical system and a fixed point for a discrete dynamical system. As local properties, their determination requires calculation of derivatives; hence, some facility in computing derivatives is necessary in mathematical biology.
2. Mathematical models often take the form of differential equations, in which the rate of change of one or more variables are given as functions of the values of those variables rather than the independent variable for the system (time). Differential equations can sometimes be solved by methods of calculus, but most of the important properties of these systems can be determined by applying the concept of the derivative as a rate of change.

There are two different notations for the derivative of a quantity y that depends on a quantity t through a formula $y = f(t)$. The notation dy/dt is based on thinking of the ty system as containing an independent variable t and a dependent variable y, as we did in introducing the derivative concept. The notation $f'(t)$ is based on thinking of $f(t)$ as a mathematical function, in which the identity of the independent variable is deemphasized. When computing derivatives, it is usually more convenient to think of formulas as representing functions rather than quantities. In actual practice, the two notations are mingled indiscriminately, with the hybrid notation $y'(t)$ also in use, particularly when the formal relationship between y and t is determined by a problem rather than a function f.

[10] Mathematicians will object that this is not a mathematical definition, and of course that is correct. The definition offered here is the fundamental semantic definition; the mathematical definition is what is needed so that the mathematically defined derivative matches the semantic definition.

Derivative Computation

The interested reader can find the formal mathematical definition of the derivative in any calculus book. Fortunately, the mathematical consequences of this formal definition provide a complete algorithm for calculating derivatives without having to understand the definition. This algorithm is based on mathematical formulas at two levels of organization:

- A small number of specific derivative formulas for elementary functions;
- Some very general rules for reducing complex derivative problems to elementary ones.

Elementary Derivative Formulas

Table B.1 summarizes the elementary derivative formulas that we need to be able to find derivatives for all polynomial, rational, exponential, and logarithmic functions, as well as more complicated functions composed of these elements. A more complete derivative table includes trigonometric functions and their inverses, but these are not needed in this book.

General Derivative Rules

The task of symbolic differentiation is greatly facilitated by the use of general rules that allow us to reduce differentiation of various algebraic structures to differentiation of elementary functions. We consider these in turn.

Let f and g be differentiable functions and let a and b be real constants.

Linearity Rule:

$$[af(x) + bg(x)]' = af'(x) + bg'(x). \tag{B.2}$$

▶ **Example B.1** Let $f(x) = 3x^2 + 4\sqrt{x}$. Using the linearity rule, we can reduce the task of calculating f' to application of the power rule from Table B.1:

$$f' = 3\left(x^2\right)' + 4\left(x^{1/2}\right)' = 3 \cdot 2x + 4 \cdot \frac{1}{2}x^{-1/2} = 6x + \frac{2}{\sqrt{x}}.$$

□

▶ **Example B.2** Let $f(x) = 2^x + \ln(3x)$. This derivative can also be computed using the linearity rule and basic rules. However, some initial algebraic manipulation of the exponential and logarithmic functions is necessary. We have

$$2^x = e^{x \ln 2}, \qquad \ln(3x) = \ln 3 + \ln x.$$

Table B.1 Elementary derivative formulas

$f(x)$	$f'(x)$
$x^p \quad [p \neq 0]$	px^{p-1}
e^{ax}	ae^{ax}
$\ln x$	$\frac{1}{x}$

Thus,

$$f(x) = e^{x \ln 2} + \ln 3 + \ln x.$$

The first term is of the form e^{ax}, with $a = \ln 2$. The second term is a constant; hence, its derivative is 0. Therefore, we have

$$f' = \left(e^{x \ln 2}\right)' + (\ln 3)' + (\ln x)' = (\ln 2)e^{x \ln 2} + 0 + \frac{1}{x} = (\ln 2)2^x + \frac{1}{x}.$$

□

Product Rule:

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x). \tag{B.3}$$

▶ **Example B.3** Let $f(x) = x^2 e^{3x}$. This function is a product of two elementary functions listed in Table B.1. Hence, we apply the product rule and obtain

$$\begin{aligned}\left[x^2 e^{3x}\right]' &= (x^2)' e^{3x} + x^2 (e^{3x})' = 2x e^{3x} + x^2 (3e^{3x}) \\ &= (2x + 3x^2) e^{3x}.\end{aligned}$$

□

Quotient Rule:

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \quad g \neq 0. \tag{B.4}$$

▶ **Example B.4** Let

$$f(x) = \frac{x}{a + x}.$$

This function is a quotient, with a parameter a. Using the quotient rule, we have

$$\left[\frac{x}{a+x}\right]' = \frac{(x)'(a+x) - x(a+x)'}{(a+x)^2} = \frac{(a+x) - x \cdot 1}{(a+x)^2} = \frac{a}{(a+x)^2}.$$

□

▶ **Example B.5** Let

$$f(x) = \frac{x}{y(x) + x}.$$

This function is similar to that in Example B.4, except that it has a function $y(x)$ in place of the parameter a. The quotient rule still applies:

$$\left[\frac{x}{y(x)+x}\right]' = \frac{(x)'(y+x)-x(y(x)+x)'}{(y+x)^2} = \frac{(y+x)-x(y'+1)}{(y+x)^2}$$
$$= \frac{y+x-xy'-x}{(y+x)^2} = \frac{y-xy'}{(y+x)^2}.$$

Note that it is helpful to use the full notation $y(x)$ wherever it is important to remember that y is a function of x rather than a parameter, while using the abbreviated notation y wherever it does not matter that y is a function. Note also that we cannot compute y', since y is not given. In such cases, it is acceptable to leave y' in the answer. □

Chain Rule:

$$[f(g(x))]' = f'(g(x))g'(x). \tag{B.5}$$

The chain rule is often confusing to students. It is helpful to think of it in a verbal form rather than the abstract mathematical form:

The derivative of $f(g(x))$ is f' evaluated at $g(x)$ multiplied by the derivative of $g(x)$.

It is helpful to focus on the specific function f, while continuing to think of g in general terms.

▶ **Example B.6** Let $f(x) = (1+e^{2x})^3$. This function is a composition of the form $(g(x))^3$, so we apply the chain rule. This rule says, "The derivative of $(g(x))^3$ is $3(g(x))^2$ multiplied by the derivative of $g(x)$." Hence,

$$[(1+e^{2x})^3]' = 3(1+e^{2x})^2 \cdot \left(1+e^{2x}\right)'$$
$$= 3(1+e^{2x})^2 \cdot 2e^{2x}$$
$$= 6e^{2x}(1+e^{2x})^2.$$

□

Partial Derivatives

We can define derivatives for functions of more than one variable by treating all but one of the variables as parameters. We also modify the notation.[11]

Partial derivative of *f(x,y)* with respect to *x*: *the formula obtained by treating y as a parameter and differentiating with respect to x. This quantity is denoted by* $\dfrac{\partial f}{\partial x}$ *or by* f_x.

[11] In some contexts, it is important to have a partial derivative notation that is distinguishable from ordinary derivative notation.

Note that the prime notation cannot be used for partial derivatives; the subscript notation is an acceptable alternative, provided the context makes clear that the subscript is indicating an independent variable for partial differentiation.

The partial derivative $f_x(x, y_0)$, where y_0 is a fixed value of y, can be interpreted as the tangent slope for a graph of the single variable function $g(x) = f(x, y_0)$. In this book, the sole use of partial derivatives is to construct Jacobian matrices for stability analysis, so we will not need to be concerned with interpretation.

▶ **Example B.7** Let

$$f(x, y) = \frac{sxy}{a + x}, \qquad s, a > 0.$$

To compute the partial derivative of f with respect to x, we treat y (as well as s and a) as a parameter:

$$f_x(x, y) = sy \cdot \frac{\partial}{\partial x}\left(\frac{x}{a + x}\right) = sy\frac{a}{(a + x)^2} = \frac{asy}{(a + x)^2},$$

where we have used the result of Example B.4. Similarly,

$$f_y(x, y) = \frac{sx}{a + x} \cdot \frac{\partial}{\partial y}(y) = \frac{sx}{a + x}.$$

□

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Nonlinear Optimization

C

Nonlinear optimization is concerned with the question of how to find the value of one or more independent variables that maximizes or minimizes the value of a dependent variable. It is a large enough subject for a 3-credit upper division course, and a thorough treatment requires background in linear algebra and advanced calculus. In this book, we consider only a small class of problems associated with fitting parameters in a model to data. That empirical modeling topic appears in Chap. 2; here, we consider only the underlying mathematics problem.

Given a function F of one variable that is concave up, find the point p^* that yields the minimum value of F.

This class of problems is quite limited, first by the restriction to one independent variable and second by the requirement that F be concave up, as in Fig. 2.3.2. The second requirement will nearly always be met in a curve fitting problem, as the results are invariably worse when the variable is farther from its optimal value. These properties guarantee that the minimizing value of p will occur at a unique point where $F'(p) = 0$.

A fully analytical solution is possible only if the function F' is known and the equation $F' = 0$ is solvable.[12] This is the case when F is the residual sum of squares for a model $y(x) = b + mx$, with b known,[13] but it is not generally the case for any other curve-fitting problems. When doing least squares for a nonlinear model or a numerical simulation, we must apply a numerical method, which we now consider.

C.1 A Guess and Check Method

For a one-off problem, it takes less time to determine the solution to an adequate degree of precision by a guess and check method than a formal numerical method. One first identifies an interval that contains the minimizer and then calculates the function values for a set of points that spans the interval. The smallest function value identifies the minimizer to a level of precision determined by the spacing of the points. This method was used in Example 2.3.3.

[12] Optimization problems in calculus books meet these requirements, although their graphs are not necessarily concave up.

[13] Sections 2.1 and 2.2.

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C.2 The Bisection/Quintsection Methods

If you are going to be solving a lot of minimization problems for functions $F(p)$, then it is worth writing a computational function to implement a variant of the bisection method. There are much better methods available, but the better methods all have subtle issues that can pose difficulties, while these n-section methods always work if the initial requirements are met.

The standard bisection method is for solving an equation $f(x) = 0$. The method requires prior knowledge of an interval $[x_1, x_2]$ guaranteed to contain a root. This is accomplished by finding values x_1 and x_2 that have function values of opposite sign, that is, $f(x_1)f(x_2) < 0$. In each step of this iterative method, we test the midpoint $x_m = (x_1 + x_2)/2$ and use this new point to replace either x_1 or x_2 so as to maintain the condition of opposite sign function values. The bracketing interval shrinks by half with each step, so the precision improves by a factor of roughly 10 for three steps. The process stops when the interval width falls within some predetermined precision requirement. Convergence is slow but guaranteed, and with today's computational power this will almost certainly take only a small amount of time for all but the most complicated functions.

We can adapt the bisection method for the minimization of a function $F(p)$ that is always concave up. Suppose we know the minimizer of F lies in the interval $[p_{min}, p_{max}]$.[14] Let n be an integer that represents the number of subintervals into which the bracketing interval is to be divided[15] and let $\Delta p = (p_{max} - p_{min})/n$. Then define $p_0, p_1, \ldots p_n$ by $p_i = p_{min} + i\,\Delta p$. Compute the function values at the new intermediate points and let I be the subscript for the smallest of these. Then set $p_{min} = p_{I-1}$ and $p_{max} = p_{I+1}$ when $0 < I < n$, instead using $p_{min} = p_0$ if $I = 0$ or $p_{max} = p_n$ if $I = n$.[16] This single iteration reduces the width of the bracketing interval by a factor of (at least) $n/2$. Additional iterations are continued until a predetermined precision requirement is met.

The author's MATLAB function *findmin* implements the "quintsection" method ($n = 10$) as described here, with provision for finding a suitable bracketing interval if the interval supplied to the function is not bracketing.

[14] A good implement of this algorithm, such as the author's *findmin.m*, can shift the search interval if the initial interval does not include the minimizer.

[15] $n = 4$ is most mathematically efficient, while $n = 10$ is more intuitive and works fine as long as the computation is not too slow. Because each iteration cuts the interval by a factor of $n/2$, we could use the term "bisection" for $n = 4$ and "quintsection" for $n = 10$.

[16] If the original interval does indeed bracket the minimizer, then the concave up property of F makes it unlikely that $I = 0$ or $I = n$, but this cannot be ruled out.

A Runge–Kutta Method for Numerical Solution of Differential Equations

D

It is easy to program simulations for discrete dynamical systems because the equations of the model are all that is needed. For continuous dynamical systems, we need numerical methods that use a discrete approximation of the system of differential equations to calculate a set of approximate values x_j of the dependent variable[17] at a collection of times t_j. Thus, they are based on discrete approximations to a continuous equation.

The most obvious way to discretize a differential equation $x' = f(t, x)$ is to evaluate the differential equation at the point (t, x) using a forward difference approximation for the derivative:

$$\frac{x_{j+1} - x_j}{h_j} = f(t_j, x_j), \qquad h_j = t_{j+1} - t_j,$$

which leads to the Euler approximation

$$x_{j+1} = x_j + h_j f(t_j, x_j).$$

The drawback of Euler's method is the use of the value $f(t_j, x_j)$ to represent dx/dt over the entire interval $[t_j, t_{j+1}]$. As the actual solution curve moves through this interval, its slope changes and the approximation becomes less accurate. A better approximation for the average rate of change over the interval could in theory be obtained by averaging $f(t_j, x_j)$ and $f(t_{j+1}, x_{j+1})$, the latter being the slope at the end of the interval. This *trapezoidal* method does not work easily in practice because the value x_{j+1} is not known. However, it can be adapted into the *modified Euler* method, in which the quantity $f(t_{j+1}, x_{j+1})$ needed for the slope at the end of the time step is calculated using the Euler approximation for x_{j+1} rather than the unknown actual value.

The Euler, trapezoidal, and modified Euler methods are all examples of the broad class of Runge–Kutta[18] methods. These work by defining approximations for slopes in the given interval and averaging them together to obtain the slope to be used to compute x_{j+1}. All Runge–Kutta methods can be written in the generic form

$$x_{j+1} = x_j + mh_j, \qquad h_j = t_{j+1} - t_j, \tag{D.1}$$

[17] For convenience, we are assuming a single differential equation. Numerical methods for single differential equations also work for systems.

[18] RUN-ga KUT-ta.

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where m is an approximate average slope of the solution curve on the interval $[t_j, t_{j+1}]$. The panoply of Runge–Kutta methods is obtained by using different definitions for m_j. So far, we have seen three of these definitions.

1. Euler's method has
$$m = f(t_j, x_j).$$
2. The trapezoidal method has
$$m = \frac{f(t_j, x_j) + f(t_{j+1}, x_{j+1})}{2}.$$
3. The modified Euler method is conveniently defined by separating the formulas for computing slopes from the formula for averaging them:
$$\begin{aligned} m_1 &= f(t_j, x_j)\,, \\ m_2 &= f(t_j + h, x_j + m_1 h)\,, \\ m &= \frac{m_1 + m_2}{2}. \end{aligned}$$

The trapezoidal method is used only for differential equations with a property called "stiffness" that makes them difficult to solve numerically.[19] This is because the value of m must itself be calculated numerically.

D.1 Accuracy of Runge–Kutta Methods

To precisely determine the error in a numerical approximation, we would need to know the exact solution, which is only possible for a few cases. Instead, the error can be estimated by using a better numerical method as a surrogate for the exact solution; for example, the modified Euler approximation could be used to estimate the error in Euler's method.

It is helpful to have some idea of the error in a method without reference to a particular differential equation. This can be done by looking at how the error changes when the step size h is changed. With Euler's method, any test problem can show you that if you make h small enough to start, cutting h by a further factor of 2 reduces the error by a factor of 2. Try the same experiment with the modified Euler method, and you will see that reducing a small value of h by a factor of 2 cuts the error by a factor of 4. In general, cutting a small h by a factor of 2 for any method reduces the error by a factor of 2^p, where p is said to be the "order" of the method. This raises the question of whether we can find methods of orders higher than 2. The answer is that careful definitions of more complicated methods can achieve as high an order as you want, but eventually there is little gained by increasing the order further.

D.2 The Runge–Kutta rk4 Method

Much numerical experimentation has identified order 4 as the ideal compromise between accuracy and simplicity. With a method of order 4, cutting h in half reduces the error by a factor of 16, so you don't have to decrease h by an enormous amount to get very high accuracy. For example, cutting h by a factor of 10 decreases the error by a factor of 10^4. Only the most tricky problems will require a very small step size with a method of order 4.

[19] See any book on numerical solution of differential equations.

It is possible to achieve a method of order 4 using a weighted average of four approximations for the slope m, but to do it you have to choose just the right approximation formulas and averaging weights. The result is the explicit RK method of order 4, often abbreviated as "rk4," which serves as the base method for more sophisticated professional numerical solvers, such as MATLAB's *ode45*. In some circumstances, such as when you want to generate simulation data at a fixed set of times, it is more efficient to use one's own rk4 function than to use a professional solver. To that end, we present here the rk4 algorithm, details of which can be found in any book on numerical analysis of ordinary differential equations, such as that by Iserles [1].

Algorithm D.1 *Runge–Kutta method of order 4 for* $x' = f(t, x)$.
Given (t_j, x_j) *and* $h = t_{j+1} - t_j$, x_{j+1} *is given by*

$$\begin{aligned} m_1 &= f(t_j, x_j)\,, \\ m_2 &= f(t_j + 0.5h, x_j + 0.5m_1h)\,, \\ m_3 &= f(t_j + 0.5h, x_j + 0.5m_2h)\,, \\ m_4 &= f(t_j + h, x_j + m_3h)\,, \\ m &= \frac{m_1 + 2m_2 + 2m_3 + m_4}{6}\,, \\ x_{j+1} &= x_j + mh. \end{aligned}$$

Note that m_1 is the slope at the left end of the interval, m_2 and m_3 are approximations for the slope at the midpoint of the interval, and m_4 is an approximation for the slope at the right end of the interval. The overall slope m is a weighted average, based on Simpson's rule for numerical integration, which you can find in any calculus book.

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Scales and Dimensionless Parameters

E

There are a number of advantages to writing models in dimensionless form; some of these follow simply from any nondimensionalization, while others follow only when the scales for nondimensionalization are properly chosen. Scaling is standard practice in modeling of physical systems but is far less common in biology. The main text focuses on the mechanics of nondimensionalization and the analysis of dimensionless models rather than the art of scaling. Here we present some of the basic ideas in the context of a simple endemic disease model from Sect. 3.9. (See [6] for a more thorough presentation.)

Definition E.1

Nondimensionalization is the systematic replacement of dimensional quantities in a model with dimensionless quantities.

The primary benefit of nondimensionalization is to reduce the number of parameters that need to be considered for analysis.

Definition E.2

Scaling is nondimensionalization using reference values that are representative of the values the variables will actually have.

In addition to the benefits of nondimensionalization, scaling puts a model into a form where asymptotic arguments can be used either to simplify the analysis of the model or to obtain a simpler model that approximates the original one.

There are two aspects to the art of scaling in mathematical biology: choosing scales and choosing dimensionless parameters. These are somewhat connected because dimensionless parameters can always be interpreted as ratios of two competing scales. Most written treatments of scaling focus exclusively on the choices of scales, but in biology it is often the case that the choice of dimensionless parameters is equally important.

We focus here on a specific example, the endemic SIR model with fixed birth rate (3.9.7)–(3.9.9):

$$N' = \Lambda - \mu N - \alpha I \,. \tag{E.1}$$

$$S' = \Lambda - \beta SI - \mu S \,, \tag{E.2}$$

$$I' = \beta SI - (\gamma + \alpha + \mu) I \,. \tag{E.3}$$

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The variables N, S, and I represent the total population and the numbers of susceptible and infectious individuals, respectively. The model includes processes for transmission (βSI), recovery (γI), disease-induced death (αI), natural mortality (μS, μI, μN), and a fixed birth rate (Λ).

Reference Populations

If there is no disease-induced death, the equation for total population becomes

$$N' = \Lambda - \mu N\,.$$

This equation has a stable equilibrium population of $K = \Lambda/\mu$. This is clearly the right scale for the total population N and the susceptible population S. The dimensionless populations n and s will then represent fractions of the disease-free total population.

The choice of scale for I is problematic. On the one hand, it would be natural for i to be given as a fraction of the disease-free population. On the other hand, after the disease becomes endemic, the current fraction of infectious individuals stays small, as seen in Fig. 3.9.2. Best practice is to choose the equilibrium population K, but to be prepared to rescale when it is time to do analysis.

Reference Times

There are a lot of parameters and combinations that have the dimension of 1/time. These generally fall into two broad categories: fast rates and slow rates. Each of them is associated with a characteristic time, given by the reciprocal of the rate.

Fast rates are those that are associated with standard disease processes.

1. The combined term $(\gamma + \alpha + \mu)I$ has a fast rate constant $(\gamma + \alpha + \mu)$. The reciprocal of this rate constant is the time
$$T_i = \frac{1}{\gamma + \alpha + \mu}\,,$$
which represents the mean amount of time an individual spends in the infectious class. (We are using capital T to reserve t for dimensionless time.)
2. The individual term γI has a fast rate constant γ. Its reciprocal $1/\gamma$ is the mean infectious duration for those individuals who recover from the disease.
3. The term βSI is a standard disease process. From the I equation, the transmission rate per unit infective is βS. While this is not constant, we can replace S with its scale K to obtain the fast rate constant βK. The reciprocal of this quantity represents the mean transmission time in a fully susceptible population.

Slow rates are those that are associated with demographic processes.

1. The terms μS, μI, and μN have a slow rate constant μ. The reciprocal of this rate constant is the time
$$T_\mu = \frac{1}{\mu}\,,$$
which represents the mean lifespan of an individual, independent of disease-induced death.
2. The term Λ is a demographic process. It does not have the right dimension to be a rate constant. Instead, we need to think of the quantity as $(\Lambda/S)S$, with rate parameter Λ/S. This gives us a rate constant $\Lambda/K = \mu$. For this simple model, the birth and death processes have the same rate constant.

There is one additional rate constant to be considered, coming from the term αI. While this term represents a fast *process*, α is not a fast *rate* if the disease mortality is low. In general, a rate is fast if it is comparable to the standard fast rate $\gamma + \alpha + \mu$. This may or may not be true for α.

Time Scale Options

In endemic disease models, it is very helpful to identify one fast time and one slow time as potential time scales. In this model, the mean lifespan T_μ is the only candidate for a slow time scale. For the fast time scale, we can quickly set aside the transmission time $1/\beta K$ on the grounds that it would be best to have a time scale that does not depend on a population scale. The choice between T_i and $1/\gamma$ is more subtle. On asymptotic grounds, they are equivalent, since their ratio is not likely to be much different from 1. There are subtle differences that make it clear why T_i is the better choice. These will be mentioned as they arise.

Once the top candidates for a fast and a slow time scale are identified, the final choice between them is not critical. Simulations show behaviors that occur on both time scales, as we saw in Fig. 3.9.2. While the slow time is theoretically better for stability analysis, the mathematical analysis is the same in either case and the algebraic form using the fast time is slightly more convenient. This additional algebraic convenience is in most cases the only real difference; hence, in this book we choose the fast time for epidemiology models.

Dimensionless Parameters

It is not technically necessary to identify dimensionless parameters before the nondimensionalization step because dimensionless groupings emerge from that process. Nevertheless, there are multiple sets of parameters that can be chosen, and it is generally best to try to identify good choices of parameters at the outset. A key idea that facilitates this process is that dimensionless parameters can always be thought of as ratios between competing scales.

In our model (E.1)–(E.3), we will need three dimensionless parameters. We know this because the model has a total of five parameters, but we will lose one parameter by scaling the populations and another by scaling time.

Three possible dimensionless parameters can be identified before nondimensionalization.

1. The basic reproduction number is the mean number of infections caused by one primary infective in a fully susceptible population.[20] Here, one infective produces infections in a fully susceptible population at rate βK for an average time period of T_i; hence, the basic reproductive number is

$$\mathcal{R}_0 = \frac{\beta K}{\gamma + \alpha + \mu} \,. \tag{E.4}$$

   Note that this is the ratio of the transmission rate for a fully susceptible population to the infectious duration rate.
2. The ratio of the fast time T_i to the slow time T_μ is a critical parameter that we need to facilitate asymptotic analysis[21]:

$$\epsilon = \frac{T_i}{T_\mu} = \frac{\mu}{\gamma + \alpha + \mu} \ll 1 \,. \tag{E.5}$$

3. The mortality fraction of the disease is given by the rate at which people die from the disease relative to the total rate at which people leave class I; in other words,

[20] Section 3.4.

[21] It is traditional in modeling to reserve the symbol ϵ for a parameter assumed to be small.

$$d = \frac{\alpha}{\gamma + \alpha + \mu}\,. \tag{E.6}$$

This parameter is easier to measure than any of its component parameters and runs from 0 up to perhaps 0.8, which might have been the mortality fraction for the bubonic plague when that disease was most rampant.

The notation $\ll$ is used in asymptotic analysis to indicate a parameter that is assumed to be arbitrarily small at such time as this information is useful. This is not quite the same thing as taking a limit as $\epsilon \to 0$, which assumes a parameter is arbitrarily small at the cost of losing valuable information. The assumption $\epsilon \ll 1$ is well justified. Given a life expectancy of about 70 years, a disease duration of 3.5 weeks would yield a value $\epsilon \approx 0.001$. Most diseases have shorter durations, yielding an even smaller value for ϵ.[22]

Let's revisit why T_i is a better choice of time scale than $1/\gamma$. The most important reason is that T_i is the time that we need for the basic reproduction number. With time scale $1/\gamma$, we could have taken $\epsilon = \mu/\gamma$ and $d = \alpha/\gamma$ instead of the definitions we picked for them in light of the choice of T_i. However, the scaling process would have given us the combined dimensionless grouping $\beta K/\gamma$ instead of $\beta K/(\gamma + \alpha + \mu)$. We could have made a parameter $b = \beta K/\gamma$ to fit the equation, but b would not be the basic reproductive number. If we had wanted to use $\mathcal{R}_0$ as one of the parameters, then we would have had to write $\beta K/\gamma$ as $\mathcal{R}_0/(1 + d + \epsilon)$, which would complicate the subsequent algebra.

Rescaling

With K as the scale for all populations and T_i as the time scale, we obtain the dimensionless model

$$n' = \epsilon(1 - n) - di\,, \tag{E.7}$$

$$s' = \epsilon(1 - s) - \mathcal{R}_0 si\,, \tag{E.8}$$

$$i' = \mathcal{R}_0 si - i\,, \tag{E.9}$$

where the prime symbol now refers to the t derivative.

At first glance, the scaled model (E.7)–(E.9) looks great. It has three parameters and simple equations. However, a closer look from an asymptotic perspective reveals a problem. The idea of asymptotics is that if all the "smallness" is in the *parameters*, you can immediately tell which terms in an equation are the most important. The purpose of scaling is to remove any "smallness" from *variables*. We did not succeed in doing this. The system should make sense in the long term if we set $\epsilon \to 0$. But doing so gives us

$$n' = -di$$

for the long-term behavior of the population. This would eventually yield a negative population unless $i \to 0$. However, the whole point of approximation is to keep the most important terms in an equation. The term di is not more important than $\epsilon(1 - n)$ if $i = 0$!

The only possible escape from this dilemma is that i must be small as the dynamics approaches the endemic phase. This is a wonderful insight, an important mathematical result that we got before even starting the analysis. As a disease reaches the endemic stage, the fraction of infectious people becomes small, no matter how contagious the disease. Indeed, the author grew up before the development of the measles vaccine. Nearly the entire population in those days consisted of people who had already

[22] In most models, taking a parameter to be small when its value is on the order of 0.01 is well justified by simulations.

recovered from measles, with most people getting it as a young child. At any given time, the fraction of the population that had measles was tiny in spite of its being the most contagious of human diseases.[23]

What this analysis tells us is that the system (E.7)–(E.9) is not properly scaled because there is some "smallness" in the variable i. This is an important issue, but one that is easy to correct. The two terms in the n equation should be of comparable importance, so i is of the *same* degree of smallness as ϵ. We can factor the smallness out of i by using the new variable y defined by

$$i = \epsilon y. \tag{E.10}$$

Since i must be on the order of ϵ in the endemic phase, y must be order 1. With this change, the model becomes

$$n' = \epsilon(1 - n - dy). \tag{E.11}$$

$$s' = \epsilon(1 - s - \mathcal{R}_0 sy), \tag{E.12}$$

$$y' = y(\mathcal{R}_0 s - 1). \tag{E.13}$$

When we look at this new system, we see two important features that were missing from (E.7)–(E.9). First, the parameter ϵ has factored out of the s and n equations. It is now strictly a"time-scale" parameter, meaning that it identifies s and n as slow variables compared to i, and that it will be absent from the equilibrium equations (obtained by setting the derivatives equal to 0). Second, the absence of any terms with factors of ϵ inside the parentheses means that all of the processes we incorporated into the model—transmission, removal, disease-related death, and demographics—will play a meaningful role in the endemic phase. Had some terms remained small compared to others, we could have argued that the processes they represent are unimportant.

Conclusions

We can extract some general principles from the example of this section.

1. Nondimensionalization simplifies a model by reducing the number of parameters. Additionally, it is sometimes easier to estimate a dimensionless parameter than its component dimensional factors.
2. It is best to choose primary long-time and short-time scales and create one dimensionless parameter as the ratio of these fundamental scales.
3. Scaling, along with a judicious choice of dimensionless variables, can simplify the analysis of long-term behavior by removing one or more parameters from the equilibrium relations. For this to work, the dependent variables must be scaled according to their sizes near the equilibrium state.

[23] The omicron strain of COVID-19 has a basic reproduction number approaching that of measles. Unlike measles, for which immunity is generally lifelong, COVID-19 immunity is short-lived. It will not become a childhood disease like measles. Instead, those who do not get annual vaccinations with updated vaccines can expect to get COVID-19 roughly once per year.

Approximating a Nonlinear System at an Equilibrium Point

F

Suppose we have a two-dimensional nonlinear system of the form

$$\frac{dx}{dt} = f(x, y) \tag{F.1}$$

$$\frac{dy}{dt} = g(x, y)\,, \tag{F.2}$$

and we would like to approximate this system near an equilibrium point where, of course,

$$f(x^*, y^*) = 0\,, \qquad g(x^*, y^*) = 0\,. \tag{F.3}$$

In general, the linear approximation of a function $f(x, y)$ near a point (x^*, y^*) is given by[24]

$$f(x, y) \approx f(x^*, y^*) + \frac{\partial f}{\partial x}(x^*, y^*)(x - x^*) + \frac{\partial f}{\partial y}(x^*, y^*)(y - y^*)\,.$$

However, the first term is 0 for an equilibrium point; hence, the linear approximation formula is

$$f(x, y) \approx \frac{\partial f}{\partial x}(x^*, y^*)(x - x^*) + \frac{\partial f}{\partial y}(x^*, y^*)(y - y^*)\,.$$

Replacing the functions f and g with their linear approximations yields the system

$$x' \approx \frac{\partial f}{\partial x}(x^*, y^*)(x - x^*) + \frac{\partial f}{\partial y}(x^*, y^*)(y - y^*)\,, \tag{F.4}$$

$$y' \approx \frac{\partial g}{\partial x}(x^*, y^*)(x - x^*) + \frac{\partial g}{\partial y}(x^*, y^*)(y - y^*)\,. \tag{F.5}$$

Now define new variables u and v by

$$u = x - x^*\,, \qquad v = y - y^*\,. \tag{F.6}$$

Note that $du/dt = dx/dt$ because x^* is a constant. Thus, we can rewrite (F.4) and (F.5) as

[24] Linear approximation for functions of more than one variable requires one term for each independent variable, using partial derivatives rather than ordinary derivatives. See Appendix B.

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$$u' = \frac{\partial f}{\partial x}(x^*, y^*)u + \frac{\partial f}{\partial y}(x^*, y^*)v \ ,$$
$$v' = \frac{\partial g}{\partial x}(x^*, y^*)u + \frac{\partial g}{\partial y}(x^*, y^*)v \ .$$

In matrix-vector form, this is

$$\mathbf{u}' = \mathbf{J}(x^*, y^*)\,\mathbf{u} \ , \qquad \mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \ , \tag{F.7}$$

where

$$\mathbf{u} = \begin{pmatrix} u \\ v \end{pmatrix} \ .$$

Thus, the origin in the linear system (F.7), is equivalent to the original equilibrium point in the nonlinear system. In most cases (as covered by the stability theorems), the stability of the origin in that linear system carries over to the stability of the original equilibrium point in the nonlinear system.

An analogous argument holds for systems of more than two components.

Best Practices in the Use of Algebra

G

Best practices in application of algebra can make a huge difference, following a general principle:

> Do algebra with a scalpel, not a meat cleaver.

Whenever you do an algebraic manipulation, ask two questions: (1) Is it legal? and (2) Did it help? In general, do legal manipulations that help, but don't do legal manipulations that don't help. Of course it is not always obvious whether a manipulations makes things better or worse. There are no universal rules for simplification; context is critical.

Much of the application of algebra in mathematical biology occurs in the context of stability calculations (Sects. 6.2 and 6.3). Here are two algebraic guidelines that help in this context.

1. When computing Jacobians, don't multiply out products in the original functions; instead take product rule derivatives and do not combine terms. The reason is that equilibrium points for a system involving an equation $x' = F(x, y)G(x, y)$ must have either $F = 0$ or $G = 0$. If you write the row 1 column 1 entry in the Jacobian (the partial derivative of FG with respect to x) as $FG_x + GF_x$, then you are guaranteed that each equilibrium point will have one of the two terms be 0. If you combine $FG_x + GF_x$ into something that looks simpler as a full formula, you will have to undo some of that algebraic work when you evaluate at an equilibrium point.
2. Individual equilibrium relations are simple, while the equilibrium point formulas obtained from them can be very complicated. Use the equilibrium relations whenever they seem to simplify the Jacobian. Don't use complicated equilibrium formulas unless you have no other way to move forward.

Multiple examples of the use of these guidelines occur in Sects. 6.2 and 6.3.

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Hints and Answers to Selected Problems

Chapter 1

Section 1.1

1.1.2 $t_z = \dfrac{\ln 650 - \ln z}{0.3}$

1.1.6 (a) $\left(\dfrac{b}{2}, -\dfrac{b^2}{4}\right)$

1.1.11

(a) $y = Ae^{-1.2}$
(b) $y = 650 + Ae^{-1.2}$
(c) $A = \frac{650}{1-e^{-1.2}} \approx 930, \quad y_{\min} = Ae^{-1.2} \approx 280$

1.1.12 (c) Hint: The units of Q are the units of q divided by the units of p.

1.1.14

(a) $y(t) = e^{-mt}, \quad s = e^{-m}$
(b) $n = -\dfrac{\ln s}{s - s^2}$. $n > 0$ follows from $0 < s < 1$.

Section 1.3

1.3.3 Needing at least three tries is equivalent to having two consecutive failures; thus, the probability is $b(0; 2, 0.3) = 0.49$.

1.3.7 $E_4(1) = 1 - e^{-4} = 0.9817$

1.3.11 (a) Means are all 5; standard deviations are 2.5, 1.25, and 0.625.

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Section 1.4

1.4.1

(a) $(X, Y) = (0, 0)$ and $(X, Y) = m/cq, r/q$. There is no possibility that the predators go extinct while the prey survives.
(b) $x \to \infty$

1.4.3

(a) $q^2, \quad 2pq, \quad p^2$
(b) 0.36, 0.41

Chapter 2

Section 2.1

2.1.5 The slopes are $m_1 = 1 - 0.4c$ and $m_2 = 1 - 0.2c$. Errors at the edges of the graph matter more than errors near the middle.

2.1.8 $m^* = 0.271$, RSS $= 1469$. Note that the residual sum of squares is much larger than we got in Example 2.1.2. This does not mean the fit is less satisfactory; horizontal distances between points are larger than vertical distances because the data covers a wider range of x values than y values.

Section 2.2

2.2.2

(a) $C = -15.6 + 4.219t$
(b) There is a lot of scatter in the data, which limits our confidence in the choice of a linear model.

2.2.4 $N = 6.01e^{0.400t}$

Section 2.3

2.3.1 $N = 6.18e^{0.390t}$

2.3.4 $y = \dfrac{0.9982}{1 + 0.0275x}$, RSS $= 43$, as compared to RSS $= 63$ for $y = 4.014x^{0.409}$.

Section 2.4

2.4.2

(a) $y = -0.11667x + 0.04x^2 - 0.00083x^3$
(c) The cubic polynomial is too "curvy" for a good fit.

2.4.7

1. $y = -15.60 + 4.219t$, RSS $= 368$, AIC $= 33.7$
2. $y = -9.08 + 3.342t + 0.02508t^2$, RSS $= 361$, AIC $= 35.6$
3. $y = 15.5e^{0.0695t}$, RSS $= 581$, AIC $= 36.9$
4. $y = 12.0e^{0.0817t}$, RSS $= 742$, AIC $= 38.6$

The linear function is clearly best, but the fit is not very good.

Section 2.5

2.5.3 The AIC difference is 3.6 for Lineweaver–Burk and just 0.02 for Haney–Woolf. The Haney–Woolf linearization performs so well here that one has to wonder whether the data was picked over, as seems to have been done in Mendel's famous paper[25] Even if that was the case, the Haney–Woolf method seldom has an AIC difference above 0.3 with real or simulated data.

Chapter 3

Section 3.1

3.1.6 The average of t is $1/\gamma$.

Section 3.2

3.2.5 Hint: Separately show that each side of the differential equation is $\beta\,(y_0 - 1)\,e^{-\beta t} I^2$, where $y_0 = 1/I_0$ slightly simplifies the notation.

3.2.8 Hint: If we define $u(t) = 4 + 96e^{-t} = y^{-1}$, the right side of the differential equation becomes $(uy)y - 4y^2 = (u - 4)y^2$.

3.2.10 The solution with the given parameter values is the same function as the one in Problem 3.2.8.

Section 3.3

3.3.3

$$\begin{aligned}
\frac{dS}{dt} &= -\beta SI + \sigma R\,,\\
\frac{dE}{dt} &= \beta SI - \eta E\,,\\
\frac{dI}{dt} &= \eta E - \gamma I\,,\\
\frac{dR}{dt} &= \gamma I - \sigma R\,.
\end{aligned}$$

[25] See Fairbanks DJ and B Rytting. Mendelian controversies: A botanical and historical review. *American Journal of Botany* **88**, 737–752 (2001).

3.3.6

$$\begin{aligned}
\frac{dP}{dt} &= -\beta P I - \Phi(W,t)P\,,\\
\frac{dU}{dt} &= -\beta U I\,,\\
\frac{dE}{dt} &= \beta(P+U)I - \eta E\,,\\
\frac{dI}{dt} &= \eta E - \gamma I\,,\\
\frac{dR}{dt} &= \Phi(W,t)P + \gamma I\,.
\end{aligned}$$

The vaccination rate function is

$$\Phi(W,t) = \frac{\phi g(t) W}{K^2 + W^2}\,, \qquad g(t) = \min\left(\frac{t}{\tau}, 1\right)\,.$$

Section 3.4

3.4.16 $\mathcal{R}_0 = \dfrac{b^2 pqV}{\mu\gamma H}$. Note that the disease benefits from having a larger mosquito population and a smaller human population. This is because the mosquito biting rate does not depend on the size of the human population.

3.4.17

(a) The populations stabilize at about $t = 8$ with roughly 60% of humans and 15% of mosquitos infected.
(b) With a 20% reduction in biting rate, it takes almost twice as long for the populations to stabilize, and the values they reach are roughly 30% less for humans and 40% less for mosquitoes.

3.4.10 Hint: Rewrite the assumptions so that they are about I and E rather than $\ln I$ and $\ln E$. Use the E assumption to rewrite both differential equations in terms of I alone. Then use the I assumption to get equations with $\lambda + \eta$ and $\lambda + \gamma$ on one side. Combine these together to get the desired result.

3.4.13 The maximum infectious population with $\mathcal{R}_0 = 5$ is about 48%. The corresponding value for an SEIR model would be lower, but still quite large.

Section 3.6

3.6.3 $Y = e^{-T}$, where $Y = y/y_0$ and $T = kt$

3.6.6 $s' = -\mathcal{R}_0 si\,,\ i' = \mathcal{R}_0 si - i$

3.6.9 $v' = v(1 - v - c)\,,\ c' = mc(hv - 1)$

Section 3.7

3.7.1 $r = k_1 + r_1 + \dfrac{k_1 k_2}{k_1 + r_2}$

Section 3.8

3.8.7 The value $m = 10$ is not large enough for the simplification to work well. The plot of x ends up in the same place, but it arrives about 1 time unit earlier in the 1-variable reduction than in the full system. The simplification is much better with $m = 100$.

Section 3.9

3.9.1 The equations for s and i are identical; only the n equation is different. The disease will suppress the overall population without changing the proportions.

3.9.4 $i' = \mathcal{R}_0 i(n - i) - i$. The graphs are almost the same for the first 20 days, but then they are quite different. For the short-term 60-day plot, the total population continues to fall, the infectious population falls more slowly than in the SIR case, and the susceptible population levels off at about 0.2. Repeat spikes occur more quickly, but they have all but stopped at 10 years. The overall population drops to just 0.2, with most people susceptible.

3.9.5 The equilibrium value $n \approx \mathcal{R}_0^{-1}$ means that the infectiousness of the disease is a far more important factor in determining the suppression of the population than the mortality. With $x \approx \delta(\mathcal{R}_0 - 1)$, both are important, but a high death rate means small δ, which means only a small fraction are infected at the same time. If we find a population of animals with a low infection rate, this does not mean that the disease has a small impact. Of course these results are specifically for diseases that do not confer immunity.

Chapter 4

Section 4.1

4.1.2

(a) $N \to 1000$
(b) $N \to 1000$
(c) The system has a stable 2-cycle.
(d) The system has a stable 4-cycle.
(e) The system has a stable 8-cycle.
(f) The system is chaotic.
(g) The system is chaotic, but quite different from (f). In particular, there is a point in time with 6 consecutive small values.

4.1.4

(a) $S - 1 + \dfrac{A}{1 + N_t}$
(b) Adults do not survive.

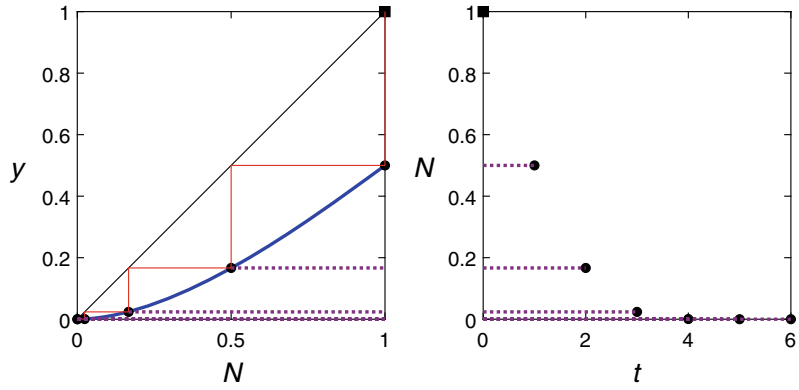


Fig. A.1 Problem 4.2.1

(c) $N^* = \dfrac{A-(1-S)}{1-S}$, provided $A+S>1$.
(d) The plot viewing window should be $0 \le S \le 1$ and $0 \le A \le \hat{A}$, where $\hat{A}$ is somewhere around 3–5.
(e) $N \to 0$, consistent with predictions.
(f) $N \to 2$, consistent with predictions.
(g) $N \to 0.6$, consistent with predictions.

Section 4.2

4.2.1 See Fig. A.1.

4.2.4

(a) $N \to 0$, consistent with predictions.
(b) $N \to 2$, consistent with predictions.
(c) $N \to 0.6$, consistent with predictions. See Fig. A.2.

Section 4.3

4.3.1 17.5% of the population is unvaccinated at the 20-week mark.

4.3.2 0 and K are stable, while T is unstable.

4.3.3 0 is stable if $E>1$ and unstable if $E<1$. $E-1$ is stable whenever it exists; that is, when $E<1$.

4.3.10 (c) The plot of E^* vs A seems to show that there are two values of E^* for some values of A. There are two ways to show that one part of the graph is meaningful and the other part is not. (1) Think about whether the optimal effort should increase or decrease as the cost goes up. (2) Plot R vs E for a value of A that seems to give two answers.

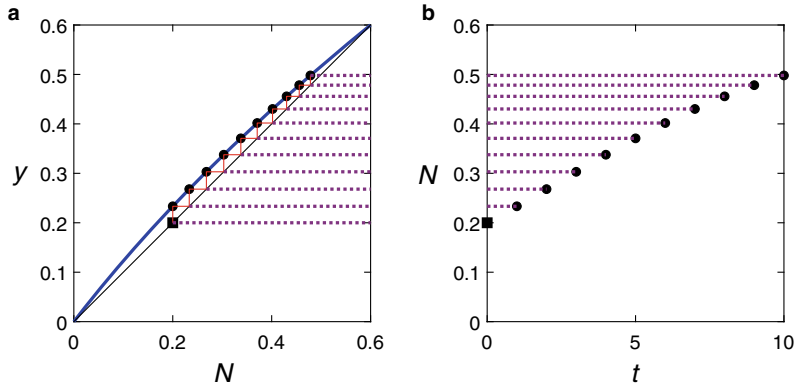


Fig. A.2 Problem 4.2.4

Section 4.4

4.4.2 $g'(0) = 8$, so 0 is unstable; $g'(\ln 8) = -1.079$, so $\ln 8$ is also unstable.

4.4.7

(a) 0 and 961
(b) $F'(0) = 1.7$ and $F'(961) = -2.045$, so both points are unstable.

4.4.8

(a) 0 is stable if $S + A < 1$; $\dfrac{A - (1 - S)}{1 - S}$ is stable if $S + A > 1$.
(b) The results of different methods are fully consistent.

4.4.11 The calculations are easier if you apply the product rule to the factored form of f and don't combine terms. This yields

$$f' = -\left(1 - \frac{N}{T}\right)\left(1 - \frac{N}{K}\right) + \frac{N}{T}\left(1 - \frac{N}{K}\right) + \frac{N}{K}\left(1 - \frac{N}{T}\right).$$

This looks messy, but two of the three terms vanish for each of the equilibria. We get $f'(0) = -1$, $f'(T) = 1 - T/K > 0$, and $f'(K) = 1 - K/T < 0$. Thus, 0 and K are stable and T is unstable.

4.4.14 The nonzero equilibrium $x^* = 1/r - 1$ is unstable whenever it exists.

4.4.15 When you take the derivative, use the product rule and don't combine terms. This gets you very quickly to

$$f'(v^*) = v^*\left(\frac{c}{(1 + v^*)^2} - \frac{1}{8}\right)$$

for any positive equilibria. Then use the equilibrium relation $v' = 0$ to eliminate c from the formula. If you factor out $1/8(1 + v^*)$ from the resulting formula, you quickly get a critical value of v^* that marks the smallest stable value. This will mark one portion of the graph of v^* vs c as stable and the other as unstable.

Section 4.5

4.5.1

(a) 0 and 8.9
(b) 0, 0.7, 2.0, and 7.3
(c) 0 and 0.36

Chapter 5

Section 5.1

5.1.1 $\lambda = 1.1$, $j = 22$

5.1.5 $s + rf = 1$

Section 5.2

5.2.1 About 17%

Section 5.3

5.3.1 4

5.3.5 $\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t$, where $\mathbf{x} = \begin{pmatrix} J \\ A \end{pmatrix}$ and $\mathbf{M} = \begin{pmatrix} 0 & f \\ r & b \end{pmatrix}$. $\det \mathbf{M} = -rf$

5.3.7

(a) $\lambda = 3, -1$
(b) The ratios of the components of the vectors must be $1:3$ and $-1:1$, respectively.

Section 5.4

5.4.1 $\lambda_1 = 4$. Eigenvectors have ratio 3:2.

5.4.3 The dominant eigenvalue is $\lambda = 3$. Eigenvectors have ratios $(ac + b):2c:2$.

Section 5.6

5.6.3

(a) There is only one way to complete the matrix that is consistent with the requirements for Markov chain matrices.

Chapter 6

Section 6.1

6.1.2 The nullclines are straight lines that intersect at the equilibrium point (1, 1). The arrows indicate counterclockwise rotation, but it is unclear whether the equilibrium is stable or unstable.

6.1.6 The $\mathcal{R}_0 > 1$ case is illustrated in Fig. A.3a with $rr = 2$, $\delta = 0.1$. The disease-free equilibrium at (1, 0) is unstable, while the stability of the endemic disease equilibrium at (0.55, 0.444) is unclear.

6.1.9 The $b < 1$ case is illustrated in Fig. A.3b with $b = 0.5$, $\nu = 2$. A no-egress region guarantees the stability (global as well as local) of the disease-free equilibrium at (1, 0). There are no other equilibria.

Section 6.2

6.2.1

(a) The Jacobian for the endemic disease equilibrium is

$$J = \begin{pmatrix} -\delta n^* & -\delta w n^* \\ 0 & -\mathcal{R}_0 x^* \end{pmatrix}.$$

Thus, both eigenvalues are negative whenever the equilibrium exists.

(b) No matter how messy the formulas for n^* and x^*, all we needed to know is that the quantities are positive.

6.2.4

(a) The disease-free equilibrium has eigenvalues $\mathcal{R}_0 - 1$ and $-\epsilon$, so it is stable when $\mathcal{R}_0 < 1$. The nullcline plot gives the additional information that the stability is global as well as local.

(b) Given the example parameter values, the eigenvalues for the endemic disease equilibrium are complex with real part -0.0505; hence, the equilibrium is locally asymptotically stable. This is more information than could be obtained from the nullcline plot, but it is still only an example. The general case is addressed in Problem 6.3.3 with the help of an additional tool covered in Sect. 6.3.

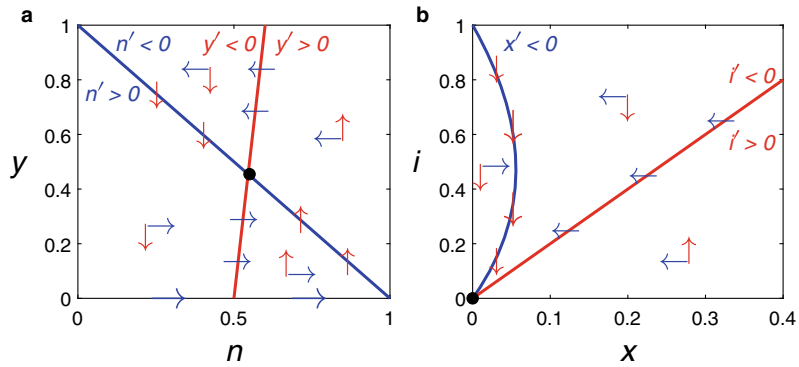


Fig. A.3 **a** Problem 6.1.6c; **b** Problem 6.1.9, $b < 1$

6.2.7

(a) With $b = 0.5$, $\nu = 2$, there is only a disease-free equilibrium. It's eigenvalues are -0.382 and -2.62; hence, it is locally asymptotically stable. The nullclines (Fig. A.3b) provide the additional information that the stability is global.
(b) With $b = 2$, the disease-free equilibrium has an eigenvalue of 0.562, so it is unstable. The eigenvalues for the endemic disease equilibrium are $-2/3$ and -3, so that equilibrium is stable. This is more information than could be obtained from the nullcline plot, but it is still only an example. The general case is addressed in Problem 6.3.6 with the help of an additional tool covered in Sect. 6.3.

6.2.10 The eigenvalues are -0.00003, -0.02, and -0.0447, so the equilibrium is stable.

Section 6.3

6.3.1

(a) The origin is stable for $k > 1$ and unstable for $k < 1$.
(b) The nullcline plot is unable to distinguish these cases.

6.3.3

(a) The Jacobian for the endemic disease equilibrium is

$$J = \begin{pmatrix} -\epsilon & -\epsilon \\ \mathcal{R}_0 y^* & -\mathcal{R}_0 \delta y^* \end{pmatrix}.$$

(b) Stable because the trace is negative and the determinant positive whenever the EDE exists.

6.3.6

(a) The disease-free equilibrium is stable whenever $b < 1$, while the endemic disease equilibrium is stable whenever it exists, that is, when $b > 1$.

6.3.11

(a) $c_2 \approx (1 + a_1 + b_1)(a_2 + b_2) - a_1 a_2 = (1 + b_1)a_2 + (1 + a_1 + b_1)b_2 > 0$
(b) Under the assumption that ϵ is arbitrarily small, we have $c_3 \to 0$ while $c_1 c_2 > 0$. Hence, $c_1 c_2 > c_3$ for ϵ small enough, provided all factors are positive.

Section 6.5

6.5.4

(a) The solution moves to the extinction fixed point.
(b) The solution appears to be cycling around the positive fixed point.
(c) The positive fixed point might be stable, but it is hard to tell.
(d) All results are consistent with the analysis, which shows that the positive fixed point in (c) is stable.

6.5.6

(b) $(0, 0)$ and $\left(\dfrac{R \ln R}{R-1}, \ln R\right)$, with the latter only if $R > 1$.
(c) $R < 1$
(d) $g(R) = R^{-1} + \ln R$
(f) The coexistence fixed point is never stable.
(g) While instability is a biological possibility, it seems unlikely that there cannot be a stable scenario.

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